On Statistical Methods for Genetics

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Fisher, Ewens and the Bayesian Programme

Nicholas G. Polson

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For Warren J. Ewens,
whose questions began these chapters

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Preface

This book grew out of a correspondence. Over the course of 2026 Warren Ewens sent me a series of questions, some about the Fundamental Theorem of Natural Selection, on which he had just written what he called the end of a story, and more about the sampling formula that carries his name. What happens to the formula when a Poisson Dirichlet subdivision is sampled with a nonuniform density? How much of that survives in the two parameter family? What is the mean number of singleton colours in a paintbox? Does a derivation from de Finetti's theorem say anything about probabilities in Kingman's coalescent? Each question became a note, most of the notes became papers, and the papers turned out to be chapters of a single argument.

The argument is that the classical questions of population genetics are statistical questions, and that the statistical tools developed in recent decades (global local shrinkage priors, Bayesian nonparametrics, data augmentation, and generative methods for inference without a likelihood) answer some of them in ways that were not available to Fisher, Haldane, Wright, Kimura or Ewens himself. In the other direction, several objects that statisticians now use routinely, from the Dirichlet process to the horseshoe prior, turn out to be objects that population geneticists defined first, for reasons of their own, or to share their marginal laws exactly.

The book has three parts, framed by a historical introduction, a chapter of mathematical preliminaries, and a closing chapter on open problems. Chapter 1 tells the history of the subject and introduces its basic models. Chapter 2, *A Mathematical Toolkit*, collects in one place the probability and statistics that the later chapters draw on repeatedly, so that those chapters can get to their arguments quickly. Part I (Chapters 3, 4 and 5) is about selection and Fisher's additive model; Part II (Chapters 6, 7, 8 and 9) about the Ewens sampling formula and the sampling theory of neutral alleles; Part III (Chapters 10 and 11) about inference for the coalescent when the likelihood cannot be computed. Chapter 12 collects the open problems, stated where possible as precise questions. Each chapter has its own summary, notation and references, so that it can be read independently. The cross references between chapters are there for the reader who wants to follow a thread.

Two features are meant to make the book usable for teaching and for self study. Every chapter contains worked examples, set apart from the text, in which the main results are carried out on cases small enough to check by hand or with a few lines of code, and most chapters contain at least one analysis of real genetic or ecological data. And every chapter ends with a set of exercises, mixing proofs,

calculations and short computations, each with a hint.

The chapters need not be read in order. Some suggested paths:

- A course on the sampling formula: Chapters 2, 6, 7 and 8, adding Chapter 9 for a longer course.
- A course on selection and the Fundamental Theorem: Chapters 2, 3, 4 and 5.
- A course on likelihood free inference: Chapters 2, 6, 10 and 11.
- A short introduction for a reader new to the subject: Chapter 1, then Chapter 2, then the first three sections of Chapter 6, and finally Chapter 12.

Chapter 1 can serve as background reading for any of these.

The prerequisites are a working knowledge of probability and of Bayesian inference at the level of a first graduate course. No previous acquaintance with population genetics is assumed; Chapter 1 supplies the history and the basic models, Chapter 2 the common mathematical tools, and each later chapter defines what it uses. The numerical results and worked examples in every chapter were produced by short Python programs, and the figures can be reproduced from them.

Acknowledgements

My first and largest debt is to Warren Ewens. This book exists because of our correspondence: a long run of emails in which he asked the questions that started almost every chapter, read the answers as they arrived, pointed out what was wrong or unclear in them, and then, more often than not, asked the next question. I am grateful for every one of those emails, for the generosity with which he shared his preprints and his memory of how the subject developed, and for the motivating questions themselves, which were always better than the answers I first gave. His work over more than half a century defines much of the territory the book covers, and Chapter 1 tries to give some account of it. The book is dedicated to him.

I am also grateful to Vadim Sokolov for many discussions, in particular of double descent and of overparameterised models, and to the Booth School of Business at the University of Chicago for its support. The errors that remain are mine.

Nicholas G. Polson
Chicago, October 2026

CHAPTER 1

A Short History of Population Genetics

Population genetics is among the oldest mathematical theories in biology and one of the two parents of modern statistics. This chapter traces the subject from Darwin's difficulty with blending inheritance, through Mendel, the quarrel between the biometricians and the Mendelians, and the synthesis of Fisher, Haldane and Wright, to Kimura's diffusions and the neutral theory, the sampling theory of Ewens and Kingman, the coalescent, and the genomic era of biobanks and polygenic traits. Along the way six worked examples carry out, on numbers small enough to check by hand, the calculation that defined each period: Mendel's $3:1$ *ratio and its chi square test, the Hardy Weinberg law, Fisher's decomposition of the genetic variance, Haldane's* $2s$*, the Ewens sampling formula and the expected times of the coalescent. The emphasis throughout is on the statistical questions each period left open, because those are the questions the remaining chapters of this book take up.*

1.1 Two subjects with one founder

It is unusual for a science and its methodology to be founded by the same person in the same paper, but that is roughly what happened to population genetics and statistics. In 1918 R. A. Fisher published "The correlation between relatives on the supposition of Mendelian inheritance" (Fisher, 1918). The paper showed that the correlations Karl Pearson and his school had measured between relatives were exactly what Mendel's discrete factors would produce if a trait were governed by many of them, and in order to show it Fisher introduced the word *variance* and the decomposition of a total variance into additive, dominance and epistatic components. The analysis of variance, which Fisher later developed at Rothamsted for agricultural experiments, was born as a statement about genes.

The intertwining did not stop there. Fisher's work on maximum likelihood was motivated in part by the estimation of linkage, Haldane wrote one of the first papers on what we would now call a default prior, with genetic linkage as his leading example (Haldane, 1932b), the Dirichlet process of Bayesian nonparametrics turned out to be the de Finetti measure behind a sampling formula derived for alleles, and approximate Bayesian computation was invented by population geneticists who could simulate their models but could not write down their likelihoods. A book on statistical methods for genetics is therefore also, in some measure, a book on the history of statistical thought.

This chapter sketches that history, with an eye on the problems it left unsolved. It is not a substitute for the full histories of the subject, and it makes no attempt at completeness. Its purpose is to put the later chapters in context and to introduce, in their historical order, the handful of models on which everything else rests. For that reason each period is accompanied by a worked example, set apart from the narrative, in which the central calculation of the period is carried out on small numbers. The mathematical tools that these examples only touch, and that the later chapters use in earnest, are collected in Chapter 2.

1.2 Darwin's difficulty and the biometricians

The problem that population genetics was invented to solve was a problem about inheritance, and it was posed by Darwin's critics before anyone had a theory of heredity adequate to answer it. Darwin's *Origin of Species* (1859) offered natural selection as the mechanism of evolution but had no adequate account of inheritance. The prevailing assumption was blending: the offspring is an average of its parents. In 1867 the engineer Fleeming Jenkin pointed out, in a review that Darwin took seriously, that blending halves the heritable variation in every generation (Jenkin, 1867). If X and Y are the values of two unrelated parents with common variance σ^2, a blended offspring $(X+Y)/2$ has variance $\sigma^2/2$. A favourable variant arising in one individual would be diluted away long before selection could act on it, and variation itself would collapse unless something replenished it at an implausible rate. The objection is, at bottom, a statement about a variance, and its resolution would require a theory in which variance is conserved.

Francis Galton approached heredity from the opposite direction, by measurement. His studies of stature in parents and children led to the discovery of regression towards the mean (Galton, 1886) and to the correlation coefficient, and his "law of ancestral heredity" attempted to apportion an individual's character among the contributions of its ancestors. Karl Pearson and W. F. R. Weldon built on Galton to found biometry: the statistical study of continuous variation in natural populations, with the journal *Biometrika* established in 1901 as its organ. For the biometricians evolution was gradual, variation was continuous, and the correct description of heredity was a set of correlations between relatives.

1.3 Mendel and the rediscovery

The answer to Jenkin had in fact been published a year before his review appeared, in a journal almost nobody read. Gregor Mendel's experiments on the garden pea, reported to the Natural History Society of Brünn in 1865 and published the following year (Mendel, 1866), proposed something entirely different from blending: heredity is carried by discrete factors that come in pairs, segregate cleanly at the formation of gametes, and recombine at fertilisation without blending. The ratios $3:1$ and $9:3:3:1$ were the visible consequence. The paper was largely ignored until 1900, when it was rediscovered more or less simultaneously by Hugo

de Vries, Carl Correns and Erich von Tschermak. William Bateson became its most energetic advocate in the English speaking world, coined the word *genetics*, and saw in Mendelism a theory of discontinuous evolution by large steps.

EXAMPLE 1.1 (Mendel's 3 : 1 ratio and a chi square check). Cross a pure line of round seeded peas with a pure line of wrinkled seeded peas. Every first generation plant carries one factor of each kind, say Rr, and since round is dominant every seed it bears on the parent plant is round. Self fertilise the first generation. Each parent passes R or r with probability $1/2$, independently, so a second generation seed is RR, Rr or rr with probabilities $1/4$, $1/2$ and $1/4$, and is round unless it is rr. The predicted proportion of round seeds is therefore $3/4$.

Mendel counted 5474 round and 1850 wrinkled seeds, 7324 in all, a ratio of 2.96 to one. Under the hypothesis the expected counts are $\frac{3}{4} \times 7324 = 5493$ and $\frac{1}{4} \times 7324 = 1831$, so the observed counts are each 19 away from expectation, and Pearson's statistic is

$$X^2 = \frac{19^2}{5493} + \frac{19^2}{1831} = 0.066 + 0.197 = 0.263.$$

On one degree of freedom this has upper tail probability 0.61. Equivalently, the number of round seeds is binomial with standard deviation $\sqrt{7324 \times \frac{3}{4} \times \frac{1}{4}} = 37.1$, and the observed shortfall of 19 is half a standard deviation. The data agree with the 3 : 1 law about as well as one could ask.

Too well, in Fisher's later opinion. Pooling the chi square statistics over all of Mendel's experiments, Fisher (1936) found that the aggregate agreement was closer than chance would ordinarily allow, and suggested that the data had been adjusted, perhaps by an assistant who knew what was expected. The debate over Fisher's charge has never been fully settled. Its interest for us is that the most famous statistical analysis of Mendel's data is a test of goodness of fit in the *lower* tail: a test of whether the data were too good to be true.

What followed the rediscovery was one of the more bitter disputes in the history of science. The Mendelians held that discrete factors explained heredity and that continuous variation was largely environmental noise; the biometricians held that the correlations between relatives, which were real and measurable, could not be explained by Mendel's factors. Both sides were partly right, and the reconciliation was in plain sight. G. Udny Yule argued as early as 1902 that many Mendelian factors acting together would produce continuous variation and the correlations Pearson observed (Yule, 1902), and Pearson himself in 1904 derived some of the consequences of Mendelian inheritance for correlations, though he concluded that the agreement was imperfect.

A second early result was more decisive for the mathematics. In 1908 G. H. Hardy, responding to a question from the geneticist R. C. Punnett, showed that under random mating a dominant allele does not increase in frequency merely because it is dominant: genotype frequencies settle in one generation at $p^2 : 2pq : q^2$

and allele frequencies stay fixed in the absence of other forces (Hardy, 1908). Wilhelm Weinberg had derived the same result independently in the same year (Weinberg, 1908). The Hardy Weinberg law answered Jenkin. Under particulate inheritance variation is conserved, so selection has something persistent to act upon. It also established the null model against which every later force (selection, mutation, drift, migration, nonrandom mating) would be measured, and the manifold of product measures it defines will reappear, in a geometrical guise, in Chapter 3.

EXAMPLE 1.2 (The Hardy Weinberg law in numbers). Suppose a population has genotype frequencies $AA : Aa : aa = 0.6 : 0.2 : 0.2$, which are far from Hardy Weinberg proportions. The frequency of A is $p = 0.6 + \frac{1}{2}(0.2) = 0.7$, because each AA individual carries two copies and each Aa one. Under random mating a zygote is formed by two independent draws from the gamete pool, each A with probability 0.7, so the next generation has frequencies

$$p^2 = 0.49, \qquad 2pq = 0.42, \qquad q^2 = 0.09.$$

The allele frequency of this generation is $0.49 + 0.21 = 0.7$ again, so the generation after has the same genotype frequencies, and so on for ever. Equilibrium is reached in a single generation, and nothing in the calculation depends on which allele is dominant. That was Hardy's answer to Punnett: a dominant allele at frequency 0.3 stays at 0.3.

The same arithmetic run backwards is how the law is most often used. If a recessive condition affects one birth in 2500, then $q^2 = 1/2500$ gives $q = 0.02$, and the frequency of carriers is $2pq = 2(0.98)(0.02) = 0.0392$, about one person in 25.5. There are 98 carriers for every affected individual, which is why selection against a rare recessive is so slow: almost all copies of the allele are hidden in heterozygotes.

Finally, a test. In a hypothetical sample of $N = 500$ individuals with counts 180, 220 and 100, the estimated frequency is $\hat{p} = (2 \times 180 + 220)/1000 = 0.58$, the expected counts are 168.2, 243.6 and 88.2, and $X^2 = 4.69$ on one degree of freedom ($P = 0.03$). The departure has a single interpretable parameter. Writing the observed heterozygote frequency as $2\hat{p}\hat{q}(1 - \hat{F})$ gives $\hat{F} = 0.097$, a deficit of heterozygotes of about ten percent, and one can check that $X^2 = N\hat{F}^2$ exactly (Exercise 4). Wright's inbreeding coefficient, met below, is this F.

1.4 Fisher, Haldane and Wright

The reconciliation of Mendelism with the biometrical facts and with Darwinian selection was the work of three people in roughly fifteen years. Each brought a different kind of mathematics: Fisher the analysis of variance and the diffusion equation, Haldane the deterministic recursion and the branching process, Wright path analysis and the inbreeding coefficient.

Fisher's 1918 paper settled the biometrical question by showing that a large number of Mendelian factors with small effects produces a normally distributed

trait with exactly the correlations between relatives that the biometricians had measured, provided one separates the additive component of the genetic variance from the dominance and epistatic components. The separation is a least squares regression, and it is worth seeing once at a single locus.

EXAMPLE 1.3 (Fisher's decomposition at one locus). Let the genotypes A_1A_1, A_1A_2 and A_2A_2 have genotypic values a, d and $-a$, and let A_1 have frequency p with $q = 1 - p$, in Hardy Weinberg proportions. Regress the genotypic value on the number $x \in \{2, 1, 0\}$ of A_1 alleles, weighting each genotype by its frequency. The least squares slope is Fisher's *average effect of a gene substitution*,

$$\alpha = a + d(q - p), \tag{1.1}$$

and the variance of the fitted values (the *breeding values*) and of the residuals (the *dominance deviations*) are

$$V_A = 2pq\,\alpha^2, \qquad V_D = (2pqd)^2. \tag{1.2}$$

Because the residuals of a least squares fit are orthogonal to the fitted values, the genotypic variance splits exactly as $V_G = V_A + V_D$.

Take $a = 1$, $d = 0.5$ and $p = 0.3$. The genotype frequencies are 0.09, 0.42 and 0.49, the mean genotypic value is $a(p - q) + 2pqd = -0.19$, and a direct calculation gives $V_G = 0.6489$. Formula (1.1) gives $\alpha = 1 + 0.5(0.4) = 1.2$, which is also the slope a weighted regression returns. The breeding values are 1.49, 0.29 and -0.91, the dominance deviations are -0.49, 0.21 and -0.09, and

$$V_A = 2(0.3)(0.7)(1.2)^2 = 0.6048, \qquad V_D = \{2(0.3)(0.7)(0.5)\}^2 = 0.0441,$$

which sum to 0.6489 as they must. Most of the variance is additive even though the locus shows substantial dominance, as is usual when dominance is partial.

Fisher's point about relatives now follows. A parent passes one allele to its offspring, so the covariance between parent and offspring is $V_A/2$, and the covariance between full sibs, who share both alleles with probability 1/4, is $V_A/2 + V_D/4$. Adding an environmental variance $V_E = 1$ gives a phenotypic variance $V_P = 1.6489$, a heritability $V_A/V_P = 0.37$, a parent and offspring correlation of 0.183 and a full sib correlation of 0.190. These are the kinds of quantities the biometricians measured, and the Mendelian factors produce them without any blending.

In 1922 Fisher introduced the diffusion approximation to the stochastic fluctuation of gene frequencies (Fisher, 1922), and in *The Genetical Theory of Natural Selection* (Fisher, 1930) he set out a comprehensive theory of evolution in Mendelian populations. Its centrepiece was the Fundamental Theorem of Natural Selection: the rate of increase in fitness of any organism at any time is equal to its genetic variance in fitness at that time. Fisher compared it to the second law of thermodynamics. As Chapters 3 and 4 explain, its meaning remained

disputed for half a century, and the dispute was resolved only by recognising that Fisher's average effects are the coefficients of a least squares projection, exactly the projection of Example 1.3, but taken in an inner product that changes as the population evolves.

J. B. S. Haldane, in a series of papers beginning in 1924 under the title "A mathematical theory of natural and artificial selection" (Haldane, 1924), worked out the deterministic dynamics of selection for a great variety of genetic systems, and in the fifth paper of the series (Haldane, 1927) showed that a new advantageous mutant with selective advantage s is fixed with probability approximately $2s$. His book *The Causes of Evolution* (Haldane, 1932a) brought the results to a wide audience. Haldane also thought carefully about inference: his 1932 note on inverse probability proposed the improper prior proportional to $\{\theta(1-\theta)\}^{-1}$ for a proportion (Haldane, 1932b), a prior that reappears in Chapter 3 as the limiting member of a ladder whose middle rung is the horseshoe.

EXAMPLE 1.4 (Haldane's $2s$ and Kimura's formula). Haldane's argument is a branching process. A single new mutant in a large population is carried by a heterozygote whose expected number of offspring carrying the mutant is $1+s$. If that number is Poisson with mean $1+s$, the probability π that the mutant's line survives for ever solves

$$1-\pi = \exp\{-(1+s)\pi\}, \tag{1.3}$$

since the line dies out exactly when each of the offspring lines does. Expanding the exponential to second order gives $\pi \approx (1+s)\pi - \frac{1}{2}(1+s)^2\pi^2$, so $\pi \approx 2s/(1+s)^2 \approx 2s$. More precisely $\pi = 2s - \frac{8}{3}s^2 + O(s^3)$ (Exercise 7). Survival of the line in an infinite population is the same thing as fixation in a large one, because a line that has become common is no longer at risk from chance.

Kimura's diffusion formula (Kimura, 1962) removes the assumption of an infinite population. For a diploid population of size N with genic selection (fitnesses 1, $1+s$ and $1+2s$) the probability that an allele at initial frequency p is eventually fixed is

$$u(p) = \frac{1-e^{-4Nsp}}{1-e^{-4Ns}}, \qquad u\Big(\frac{1}{2N}\Big) = \frac{1-e^{-2s}}{1-e^{-4Ns}}. \tag{1.4}$$

When $4Ns$ is large the denominator is essentially one and $u \approx 1-e^{-2s} \approx 2s$, which is Haldane's answer. When $s \to 0$ the formula tends to $1/(2N)$, the neutral answer: each of the $2N$ copies present now is equally likely to be the ancestor of the whole population in the future.

Table 1.1 compares the three approximations with the exact fixation probability of the Wright Fisher chain, computed by solving the linear equations for absorption of the 1001 state Markov chain with $N = 500$. Haldane's $2s$ is good when s is small and $4Ns$ is large, is thirteen percent too small at $4Ns = 2$, and is useless when $4Ns$ is well below one, where drift rather than selection decides the mutant's fate;

there the probability of fixation of a slightly advantageous mutant is barely more than that of a neutral one, the observation that underlies the nearly neutral theory. Kimura's formula is accurate throughout, with an error of order s^2 when s is not small.

TABLE 1.1. Probability of fixation of a single new mutant in a diploid population of size $N = 500$ with genic selection coefficient s. Haldane: $2s$; branching: root of (1.3); Kimura: formula (1.4); exact: Wright Fisher chain.

s	$4Ns$	Haldane	Branching	Kimura	Exact
0.05	100	0.10000	0.09370	0.09516	0.09364
0.01	20	0.02000	0.01974	0.01980	0.01972
0.001	2	0.00200	0.00200	0.00231	0.00231
0.0001	0.2	0.00020	0.00020	0.00110	0.00110
0	0	0	0	0.00100	0.00100

Sewall Wright emphasised what the other two treated as secondary: the random fluctuation of gene frequencies in finite populations, inbreeding and population structure, and the interaction of genes. His coefficients of inbreeding and relationship (Wright, 1922) and his long paper "Evolution in Mendelian populations" (Wright, 1931) made genetic drift a central evolutionary force, and his shifting balance theory and adaptive landscapes framed a debate with Fisher about the relative importance of drift and selection that was never fully settled. Wright's F statistics (Wright, 1951) remain the standard language for population structure; the F of Example 1.2 is the simplest of them.

The mathematical theory of Fisher, Haldane and Wright was absorbed into biology in the 1930s and 1940s by Dobzhansky, Mayr, Huxley, Simpson and others, in what became known as the Modern Synthesis. On the mathematical side the most important continuation was Gustave Malécot's probabilistic treatment of identity by descent (Malécot, 1948), which looked backwards in time from the sample to its ancestors and in that sense anticipated the coalescent by three decades.

1.5 Diffusions, drift and the neutral theory

After the war the stochastic theory was put on a rigorous footing, and the question of how much variation natural populations carry was, for the first time, put to direct measurement. Motoo Kimura, working first with James Crow, solved the diffusion equations for random drift (Kimura, 1955), derived the probability of fixation of a mutant with arbitrary dominance and selection (Kimura, 1962), and turned the diffusion approximation into the main computational tool of the subject. P. A. P. Moran introduced the overlapping generation model that carries his name (Moran, 1958), in which many diffusion results are exact. Kimura also wrote the first serious critique of the Fundamental Theorem, showing in 1958 that Fisher's

statement does not hold as a statement about the total change in mean fitness (Kimura, 1958).

In 1966 the technique of protein gel electrophoresis allowed, for the first time, a direct estimate of how much genetic variation natural populations actually contain. Lewontin and Hubby in *Drosophila* (Lewontin and Hubby, 1966) and Harris in humans (Harris, 1966) found far more polymorphism than the classical view expected. If all of it were maintained by balancing selection the genetic load would be intolerable. Kimura's answer (Kimura, 1968), and independently that of King and Jukes (King and Jukes, 1969), was the neutral theory of molecular evolution: most molecular variation is selectively neutral, and its frequencies are governed by mutation and drift. The last row of Table 1.1 contains the theory's most useful consequence. A neutral mutant is fixed with probability $1/(2N)$, and $2N\mu$ new mutants arise in each generation, so neutral substitutions accumulate at rate $2N\mu \times 1/(2N) = \mu$, independent of the population size. That is the molecular clock. The neutralist and selectionist controversy that followed was in large part a statistical controversy, because it turned on what patterns of variation in a sample could distinguish neutral from selected alleles.

To make the neutral theory testable one needs mutation models. Kimura and Crow's infinite alleles model (Kimura and Crow, 1964) assumes that every mutation creates an allele never seen before, which is the right model for electrophoretic data, where one can tell alleles apart but not how different they are. Kimura's infinite sites model (Kimura, 1969) assumes that every mutation strikes a new nucleotide site, which is the right model for DNA sequence data, where one can count segregating sites. These two models are the two worked examples of Chapter 10.

1.6 The sampling theory of selectively neutral alleles

The neutral theory made a quantitative prediction about samples, and the next step was to work it out. The question was this: given a sample of n genes from a population at equilibrium under the infinite alleles model, what is the probability of the observed allelic configuration? Warren Ewens answered it in 1972 (Ewens, 1972). If a_j is the number of alleles represented exactly j times in the sample, so that $\sum_j j a_j = n$, then

$$\mathbb{P}(a_1, \ldots, a_n) \;=\; \frac{n!}{\theta(\theta+1)\cdots(\theta+n-1)} \prod_{j=1}^{n} \frac{(\theta/j)^{a_j}}{a_j!}, \tag{1.5}$$

where $\theta = 4N\mu$ is the scaled mutation rate. Karlin and McGregor gave a second proof the same year (Karlin and McGregor, 1972). The formula has a striking statistical property, which Ewens emphasised from the start: the number of distinct alleles $K_n = \sum_j a_j$ is sufficient for θ, so that the configuration of allele counts, given K_n, carries no information about the mutation rate and can be used to test neutrality without reference to it. It follows that the natural estimator of θ, Ewens's moment

estimator, depends on the data only through K_n, and that its variance decreases only like $1/\log n$.

EXAMPLE 1.5 (The sampling formula for five genes). With $n = 5$ there are seven possible allelic configurations, one for each partition of 5. For $\theta = 1$ the leading factor of (1.5) is $5!/(1 \cdot 2 \cdot 3 \cdot 4 \cdot 5) = 1$, and each probability is a product of the factors $(1/j)^{a_j}/a_j!$. For the configuration with one allele seen four times and one seen once ($a_4 = a_1 = 1$) it is $\frac{1}{4} \times 1 = \frac{1}{4}$; for one allele seen three times and one twice it is $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$; and so on. Table 1.2 lists all seven for $\theta = 1$ and $\theta = 2$. Each column sums to one, and a larger mutation rate moves probability from configurations with few alleles to configurations with many.

The sufficiency of K_5 can be seen in the table. Two configurations have $K_5 = 2$, namely $4 + 1$ and $3 + 2$. For $\theta = 1$ their probabilities are $\frac{1}{4}$ and $\frac{1}{6}$, for $\theta = 2$ they are $\frac{1}{6}$ and $\frac{1}{9}$, and in both cases the conditional probabilities given $K_5 = 2$ are $\frac{3}{5}$ and $\frac{2}{5}$. In general the factor $\theta^{K_n}/\{\theta(\theta+1)\cdots(\theta+n-1)\}$ is the only place θ enters, so the conditional law of the configuration given K_n is free of θ, and the law of K_n itself is

$$\mathbb{P}(K_n = k) = \frac{|s(n,k)|\,\theta^k}{\theta(\theta+1)\cdots(\theta+n-1)},$$

where $|s(n,k)|$ is an unsigned Stirling number of the first kind. For $n = 5$ these are 24, 50, 35, 10 and 1, giving $\mathbb{P}(K_5 = k) = \frac{1}{5}, \frac{5}{12}, \frac{7}{24}, \frac{1}{12}, \frac{1}{120}$ for $\theta = 1$ and $k = 1, \ldots, 5$. The mean is $\mathbb{E}K_n = \sum_{i=0}^{n-1} \theta/(\theta+i)$, which is $137/60 \approx 2.28$ for $\theta = 1$ and 2.9 for $\theta = 2$. Since $\mathbb{E}K_n \approx \theta \log n$ for large n, doubling the sample adds only about $\theta \log 2$ alleles, and that is the source of the slow $1/\log n$ rate.

TABLE 1.2. The Ewens sampling formula for a sample of $n = 5$ genes. Each configuration is written as a partition of 5 into allele counts.

Configuration	K_5	$\theta = 1$	$\theta = 2$
5	1	$1/5 = 0.200$	$1/15 = 0.067$
$4+1$	2	$1/4 = 0.250$	$1/6 = 0.167$
$3+2$	2	$1/6 = 0.167$	$1/9 = 0.111$
$3+1+1$	3	$1/6 = 0.167$	$2/9 = 0.222$
$2+2+1$	3	$1/8 = 0.125$	$1/6 = 0.167$
$2+1+1+1$	4	$1/12 = 0.083$	$2/9 = 0.222$
$1+1+1+1+1$	5	$1/120 = 0.008$	$2/45 = 0.044$

Formula (1.5) turned out to be far more than a result about alleles. Its connection with the species abundance problem had in effect been glimpsed by Fisher himself, whose logarithmic series for the number of species represented by j individuals (Fisher et al., 1943) is the large sample limit of the frequency spectrum implied by (1.5). Watterson studied the corresponding sufficiency and estimation questions for the infinite sites model (Watterson, 1975). Kingman, in a series of papers from 1975, showed that the Ewens formula is the canonical example of an exchangeable

random partition, that every such partition arises by sampling from a random discrete distribution (his "paintbox" representation (Kingman, 1978)), and that for the Ewens formula that distribution is the Poisson Dirichlet law. Meanwhile, and at first quite independently, Ferguson had introduced the Dirichlet process as a prior for Bayesian nonparametric inference (Ferguson, 1973), and Antoniak had shown that the partition induced by a sample from it is exactly Ewens's (Antoniak, 1974). Hoppe's urn (Hoppe, 1984), the Chinese restaurant process, and eventually the two parameter family of Pitman and Yor (Pitman and Yor, 1997) made the sampling formula a central object of modern probability and of machine learning. Hoppe's urn also gives the quickest proof of the mean in Example 1.5: the $(i+1)$th gene is a new allele with probability $\theta/(\theta+i)$, whatever happened before.

Part II of this book is about this formula: its derivation from de Finetti's theorem (Chapter 6), what happens to it when the sample is ascertained rather than drawn at random (Chapter 7), and how to do Bayesian inference for its parameter (Chapters 8 and 9).

1.7 The coalescent

Kingman's second great contribution looked at the same problem from the other end, and it turned the prospective theory of Fisher and Wright into a retrospective one. Trace the ancestry of a sample of n genes backwards in time in a large population: each pair of lineages coalesces into a common ancestor at rate one on the appropriate time scale, so that while there are k lineages the waiting time to the next coalescence is exponential with rate $\binom{k}{2}$. This is the coalescent (Kingman, 1982). Superimposing neutral mutation on the coalescent tree as a Poisson process reproduces the Ewens formula under the infinite alleles model and the distribution of segregating sites under the infinite sites model, and it makes simulation of neutral samples almost trivial.

EXAMPLE 1.6 (Coalescence times for a sample of five). Measure time in units of $2N$ generations, so that two lineages in a diploid population of size N find a common ancestor at rate one. Let T_k be the time during which the sample has exactly k ancestral lineages. Since each of the $\binom{k}{2}$ pairs coalesces at rate one, T_k is exponential with rate $\binom{k}{2}$ and mean $2/\{k(k-1)\}$. For $n=5$,

$$\mathbb{E}T_5 = \tfrac{1}{10}, \qquad \mathbb{E}T_4 = \tfrac{1}{6}, \qquad \mathbb{E}T_3 = \tfrac{1}{3}, \qquad \mathbb{E}T_2 = 1.$$

The expected time to the most recent common ancestor is the sum, $\mathbb{E}T_{\mathrm{MRCA}} = 2(1-1/n) = 1.6$, with standard deviation $\{\sum_k (\mathbb{E}T_k)^2\}^{1/2} = 1.07$. The last two lineages account for $1/1.6 = 62.5$ percent of the expected depth of the tree, and the depth for a sample of five is already 80 percent of the depth 2 for an infinitely large sample. A small sample sees most of the history of the population; a large sample mainly adds short branches near the tips.

The total branch length is $L = \sum_k kT_k$, with mean $\mathbb{E}L = \sum_{k=2}^{5} 2/(k-1) = 2h_5 = 25/6$, where $h_n = \sum_{j=1}^{n-1} 1/j$ and $h_5 = 25/12$. Mutations fall on the branches at rate

$\theta/2$, so the expected number of segregating sites is $\mathbb{E}S = \theta h_5 = 2.083\,\theta$; this is the basis of Watterson's estimator $\hat{\theta}_W = S/h_n$ (Watterson, 1975). With $N = 10{,}000$ the expected depth of the tree is $1.6 \times 20{,}000 = 32{,}000$ generations. A simulation of 200,000 coalescent trees with $n = 5$ gives a mean depth of 1.596 and a mean length of 4.156, in agreement with the exact values 1.6 and 4.167.

Hudson extended the coalescent to recombination (Hudson, 1983), Tajima used it to study nucleotide diversity (Tajima, 1983) and later proposed the neutrality test that bears his name (Tajima, 1989), and Griffiths and Marjoram gave the ancestral recombination graph its modern form (Griffiths and Marjoram, 1996). Selection was brought in by the ancestral selection graph of Krone and Neuhauser (Krone and Neuhauser, 1997), and populations with high fecundity and skewed offspring distributions by the multiple merger, or Λ, coalescents (Pitman, 1999).

The coalescent turned population genetics into a subject of computational statistics. The likelihood of a sample is an integral over all genealogies consistent with it, which is intractable for any realistic model, but samples are easy to simulate. Two lines of response emerged. One evaluated the likelihood by importance sampling or Markov chain Monte Carlo over genealogies, beginning with Griffiths and Tavaré (Griffiths and Tavaré, 1994) and Felsenstein's group. The other abandoned the likelihood: Tavaré, Balding, Griffiths and Donnelly (Tavaré et al., 1997) and Pritchard and colleagues (Pritchard et al., 1999) proposed accepting simulated parameter values whose simulated data resembled the observed data, and Beaumont, Zhang and Balding (Beaumont et al., 2002) named the method approximate Bayesian computation. Its two acknowledged defects, the choice of summary statistics and the choice of a tolerance, are the subject of Part III (Chapters 10 and 11).

1.8 Warren Ewens and the statistical reading of population genetics

One career runs through almost every theme of this chapter, and through every chapter of this book. Warren Ewens trained in Australia as a student of P. A. P. Moran, whose *Statistical Processes of Evolutionary Theory* (Moran, 1962) was the first systematic account of the stochastic theory after Fisher, Wright and Malécot. From the beginning Ewens approached population genetics as a statistician: not only what a model predicts, but what a sample drawn from it can tell us, and with what precision. That emphasis is the thread this book tries to follow.

The sampling formula and its statistics. The formula (1.5) of 1972 is usually remembered as a probability result, but the paper in which it appeared (Ewens, 1972) is a paper about inference. It observes that the number of alleles K_n is sufficient for θ, so that the allelic configuration given K_n is free of the mutation rate and can be used to test the neutral model without estimating anything; it derives the estimator of θ as the root of $\mathbb{E}(K_n) = k$, where k is the observed number of alleles; and it notes that the precision of that estimator improves only like $1/\log n$, so that very large samples buy surprisingly little. Each of these

observations reappears below: the conditional test as the Ewens and Watterson test of neutrality (Watterson, 1978), the sufficiency of K_n as the benchmark that the learned summaries of Chapter 10 must rediscover, and the logarithmic information as the length of the coalescent tree in Chapter 6. Simple consequences of the formula often have long histories in his hands; the mean number of singletons in a sample, for instance, is

$$\mathbb{E}(A_1) = \frac{n\theta}{\theta + n - 1},$$

a consequence of the sampling formula.

The textbook. Mathematical Population Genetics (Ewens, 1979), revised and much expanded in its second edition (Ewens, 2004), set out the diffusion theory, the sampling theory and the coalescent in a single rigorous framework. For two generations of probabilists and statisticians it has been the route into the subject, and much of the notation of this book is its notation.

Effective population size. The parameter N in every formula of the theory is an idealisation, and real populations are not Wright Fisher populations. Ewens's analysis of the effective population size (Ewens, 1982) asked which features of a real population an effective size can and cannot summarise, and whether different definitions (inbreeding, variance, eigenvalue) agree. His recent critical assessment with Hössjer (Ewens and Hössjer, 2026) returns to the question with the historical record in view, and is one of the starting points of Chapter 5.

From population genetics to genetic epidemiology. With Spielman and McGinnis, Ewens introduced the transmission disequilibrium test (Spielman et al., 1993), which tests for linkage between a marker and a disease locus in the presence of association by comparing the alleles a heterozygous parent transmits to an affected child with those not transmitted. Because each parent serves as its own control, the test is immune to the population stratification that confounds a case control comparison. It became one of the standard designs of family based association studies, and its logic (condition on what is fixed, compare what is random) is exactly the logic of the sufficiency arguments of the 1972 paper. With Grant he also wrote one of the first statistical textbooks for bioinformatics (Ewens and Grant, 2001).

The Fundamental Theorem. For more than thirty years Ewens has argued that Fisher's theorem, read as Fisher intended, is a statement about the partial change in mean fitness due to changes in gene frequency, and that so read it is exact (Ewens, 1989; Ewens and Lessard, 2015). His most recent account (Ewens, 2024) sets out the history and calls itself the end of a story. Part I of this book begins where that account ends: if the average effects are a least squares projection, then estimating the theorem's terms from data is a many parameter regression, and the statistical questions take over.

A ubiquitous formula. Half a century after it appeared, the sampling formula is used in ecology, in Bayesian nonparametrics, in combinatorics, in the analysis of random permutations and random integers, and in machine learning, often by people unaware of its origin in the electrophoresis data of the 1960s (Arratia et al., 2003; Crane, 2016). The questions that began the chapters of this book (what happens when the paintbox is sampled with a nonuniform density, how much of that survives in the two parameter family, what a derivation from de Finetti's theorem says about the coalescent) were his questions, asked in correspondence, and they show that the formula still has things to teach.

1.9 The genomic era

The last forty years have changed the data far more than the models. DNA sequence variation in a natural population was first surveyed by Kreitman in 1983, at the alcohol dehydrogenase locus of *Drosophila* (Kreitman, 1983). Two decades later the human genome had been sequenced (International Human Genome Sequencing Consortium, 2001), the HapMap had catalogued common variation and its patterns of linkage disequilibrium (International HapMap Consortium, 2005), and genome wide association studies had begun to map the loci underlying common diseases, beginning in earnest with the Wellcome Trust Case Control Consortium (Wellcome Trust Case Control Consortium, 2007). The statistical situation changed completely. A typical study measures hundreds of thousands or millions of variants on a sample that, until recently, numbered in the thousands: many more parameters than observations.

Three findings of this era bear directly on Part I of the book. First, the associated variants explained only a small fraction of the heritability estimated from twin and family studies, the problem of "missing heritability" (Manolio et al., 2009). Second, Yang and colleagues showed that much of the missing heritability is recovered when all common variants are fitted jointly as random effects (Yang et al., 2010), so that complex traits are highly polygenic, with very many loci of very small effect. Third, Boyle, Li and Pritchard argued that the polygenicity is so extreme that essentially every gene expressed in a relevant tissue affects the trait, the omnigenic model (Boyle et al., 2017). These findings put a question at the centre of the subject that is, in statistical language, a question about the prior: is the vector of genetic effects sparse, dense, or something in between? Ten years of association studies (Visscher et al., 2017) have not settled it. Fisher's infinitesimal model of 1918 corresponds to a Gaussian prior, the classical picture of a few genes of large effect to a sparse prior, and the horseshoe of Chapters 3, 4 and 5 to a family that lets the data choose.

At the same time the scale of the data has grown by orders of magnitude. Biobanks genotype hundreds of thousands of people, catalogues of rare variation are built from more than a hundred thousand exomes and genomes, ancient DNA has turned the history of human populations into a time series (Green et al., 2010), and the genealogies of samples as large as entire biobanks can now be simulated

and stored as tree sequences (Kelleher et al., 2016). In this regime the classical approximations begin to fail: when the sample is comparable in size to the effective population, the Kingman coalescent is no longer an adequate model of the sample's genealogy (Wakeley and Takahashi, 2003; Bhaskar et al., 2014), and the rare variant end of the frequency spectrum, which is exactly what the Ewens and Pitman formulae describe, becomes the most informative part of the data. Example 1.6 shows why: the extra lineages of a very large sample live on short branches near the present, which is where the rare variants are and where the assumptions of the coalescent are weakest.

1.10 The plan of this book

The chapters that follow began as answers to questions. Most were posed by Warren Ewens in correspondence, a few by reading his recent work on the Fundamental Theorem and on the effective population size, and the remainder by following the answers where they led. After this chapter and a chapter of mathematical preliminaries they are arranged in three parts.

Chapter 2, *A Mathematical Toolkit*, collects the probability and statistics that the later chapters use repeatedly, such as exchangeability and de Finetti's theorem, the Dirichlet and Poisson Dirichlet laws, diffusion approximations and shrinkage priors. A reader who knows this material can skip it and return to it as needed.

Part I, *Selection and the Additive Model*, concerns Fisher's Fundamental Theorem. Chapter 3 explains why the theorem as Fisher stated it is false, why the error went undetected for so long, and what replaces it: the average effects are a projection onto a fixed subspace in a moving inner product, and the neglected term vanishes exactly when the population stays on the manifold of Hardy Weinberg and linkage equilibrium. It then asks what data can say about the question, which leads to Jeffreys' Bayes factors, Haldane's prior and the horseshoe. Chapter 4 takes Ewens's corrected reading as given and shows that the remaining content of the theorem is a regression problem with more loci than individuals, so that the plug in estimate of the additive genetic variance is badly inflated and a horseshoe prior repairs it. Chapter 5 generalises the point to many parameter models in population genetics and proves that the global local shrinkage priors of sparse Bayesian statistics and the two parameter Poisson Dirichlet family are linked by an exact identity of marginal laws (the shrinkage weight has the law of the first stick breaking weight, and the frequency spectrum intensities agree), so that the sparsity level is Pitman's α; whether the full partition laws agree is left open.

Part II, *The Sampling Theory of Neutral Alleles*, concerns the Ewens sampling formula. Chapter 6 derives it from de Finetti's theorem in the elementary form of Heath and Sudderth, identifies its parameter from the empirical distribution function, connects the information about θ to the length of the coalescent tree, and reanalyses Corbet's Malayan butterflies. Chapter 7 studies ascertainment, the sampling of a Poisson Dirichlet subdivision with a nonuniform density, in the two parameter family, and proves that when the discount is positive no

sampling density that charges every interval carries the size biased sticks to those of another member of the family. Chapter 8 gives the Bayesian account of Fisher's species model and Good's predictive rule through Pólya inverse gamma augmentation. Chapter 9 develops rational conjugacy, the theory of Pochhammer priors for concentration parameters that enter a likelihood through ratios of Gamma functions.

Part III, *Inference Without a Likelihood*, concerns the coalescent. Chapter 10 replaces approximate Bayesian computation by a learned transport map fitted by quantile regression, and shows that sufficiency itself can be tested by fitting: the method rediscovers Ewens's theorem that K_n is sufficient, and proves that Watterson's S is not. Chapter 11 studies generative Bayesian computation for the Kingman coalescent, gives an exact law for the number of segregating sites, and establishes a ceiling on what permutation invariant architectures can learn.

Chapter 12 closes the book with the open problems that the preceding chapters leave, stated where possible as precise questions, and with some thoughts on where the subject is going.

Each chapter is self contained, with its own notation and references, its own worked examples, and a set of exercises at the end. The reader who wants a single thread through the book might read Chapter 2 and then Chapter 6, since the sampling formula and the de Finetti representation behind it recur in almost every chapter, then Chapter 3, and then follow the cross references. The preface suggests shorter paths for readers with a particular interest in selection, in the sampling formula, or in likelihood free inference.

1.11 Exercises

1. Mendel also scored seed colour in the second generation and counted 6022 yellow and 2001 green seeds. Test the $3:1$ hypothesis with Pearson's statistic and with the binomial normal approximation, and compare the agreement with that of Example 1.1. *Hint.* The expected counts are 6017.25 and 2005.75, and $X^2 = 0.015$.

2. In a dihybrid cross Mendel counted 315 round yellow, 101 wrinkled yellow, 108 round green and 32 wrinkled green seeds. Test the $9:3:3:1$ hypothesis. How many degrees of freedom does the test have, and why? *Hint.* $X^2 = 0.47$ on three degrees of freedom, with upper tail probability about 0.93.

3. Prove that under random mating, with no selection, mutation or migration and nonoverlapping generations, an autosomal locus with any number of alleles reaches Hardy Weinberg proportions after one generation. For two loci with recombination fraction r, show that the linkage disequilibrium coefficient is multiplied by $1 - r$ in each generation. *Hint.* A gamete is a recombinant with probability r, and then its two alleles come from independent parental haplotypes.

4. For a biallelic locus with N genotyped individuals, define $\hat{F}$ by writing the observed heterozygote frequency as $2\hat{p}\hat{q}(1-\hat{F})$. Show that Pearson's statistic for Hardy Weinberg proportions is exactly $X^2 = N\hat{F}^2$. *Hint.* All three observed minus expected counts equal $\pm N\hat{F}\hat{p}\hat{q}$ up to a factor of two.

5. Derive (1.1) and (1.2) by weighted least squares, and show that $V_G = V_A + V_D$. *Hint.* The regressor x has mean $2p$ and variance $2pq$ under Hardy Weinberg proportions.

6. With complete dominance ($d = a$) show that $V_A = 8pq^3a^2$, where q is the frequency of the recessive allele. Conclude that the additive variance, and hence by the Fundamental Theorem the response to selection, vanishes like q^3 as a recessive allele becomes rare. Relate this to the carrier calculation of Example 1.2. *Hint.* $1 + q - p = 2q$.

7. Show that the positive root of (1.3) satisfies $\pi = 2s - \frac{8}{3}s^2 + O(s^3)$, and check the value 0.01974 in Table 1.1. *Hint.* Substitute $\pi = c_1 s + c_2 s^2$ and match coefficients of s^2 and s^3; the next term is $\frac{28}{9}s^3$, which is needed to reproduce 0.01974 to five places.

8. Show that Kimura's $u(p)$ in (1.4) tends to p as $s \to 0$ and to $1 - e^{-2s}$ as $N \to \infty$ with $p = 1/(2N)$. For which values of Ns is the relative error of Haldane's $2s$ below ten percent? *Hint.* $2s/u(1/2N) \approx (1+s)(1-e^{-4Ns})$ for small s.

9. Verify the column $\theta = 2$ of Table 1.2, and show from (1.5) that the conditional law of the configuration given $K_n = k$ does not depend on θ. Compute $\mathbb{P}(K_5 = k)$ for $\theta = 2$. *Hint.* The answer is $\frac{1}{15}, \frac{5}{18}, \frac{7}{18}, \frac{2}{9}, \frac{2}{45}$.

10. For the coalescent with n lineages show that $\mathbb{E}T_{\mathrm{MRCA}} = 2(1 - 1/n)$, $\mathbb{E}L = 2h_n$ and $\operatorname{Var} S = \theta h_n + \theta^2 \sum_{j=1}^{n-1} j^{-2}$. Write a short simulation that checks the values 1.6 and 25/6 of Example 1.6, and the value $\operatorname{Var} S = 46.0$ for $n = 5$ and $\theta = 5$. *Hint.* T_k are independent exponentials, and given L the count S is Poisson with mean $\theta L/2$.

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CHAPTER 2

A Mathematical Toolkit

The later chapters of this book draw on four bodies of mathematics that are rarely found between the same covers: the classical theory of genotype frequencies and quantitative genetics, the stochastic theory of drift in finite populations, the theory of exchangeable random partitions that grew out of the Ewens sampling formula, and a set of Bayesian and likelihood free computational devices. This chapter collects what is needed, in the notation used later, with definitions, short proofs and at least one worked numerical example for each topic. We cover Hardy Weinberg proportions and linkage disequilibrium; average effects as a weighted least squares projection and the additive genetic variance; the Wright Fisher and Moran models, the diffusion approximation and Kimura's fixation probability; exchangeability, de Finetti's theorem, the Pólya urn, the Dirichlet distribution and process, stick breaking and the two parameter Poisson Dirichlet family; the Ewens and Pitman sampling formulae, the number of alleles and Fisher's logarithmic series; the Kingman coalescent with Watterson's estimator and the site frequency spectrum; global local shrinkage, the horseshoe and several data augmentation identities; and rejection approximate Bayesian computation, the check loss and the noise outsourcing lemma. Every number in the chapter has been checked by exact computation or simulation.

2.1 How to use this chapter

This chapter is a reference, not an argument. Each of the later chapters is self contained and introduces its own notation, but each also assumes some background that a statistician may lack on the genetic side or a geneticist may lack on the probabilistic side. The sections below supply that background in the order in which the book uses it. Section 2.2 is needed for Part I (Chapters 3, 4 and 5); Sections 2.4 and 2.5 for Part II (Chapters 6 to 9); Sections 2.3 and 2.6 for Part III (Chapters 10 and 11); and Sections 2.7 and 2.8 throughout.

A reader who knows a topic can skip its section; nothing later depends on the proofs here, only on the statements. We have nevertheless given proofs, or sketches that a reader can complete, because several of the identities (the projection formula for average effects, the spectrum identity $\mathbb{E}\xi_i = \theta/i$, the Beta augmentation of a ratio of Gamma functions) are used later in ways that only make sense once one has seen why they are true. Worked examples use small numbers that can be checked by hand; the scripts that check them are in the directory `code/toolkit` that accompanies the book.

Two conventions hold throughout. The rising factorial is $(x)_m = x(x+1)\cdots(x+m-1)$, with $(x)_0 = 1$, so that $(x)_m = \Gamma(x+m)/\Gamma(x)$; the book also writes it $x^{(m)}$. Population sizes are counted in genes: a population of N genes is a haploid population of N individuals or a diploid population of $N_d = N/2$ individuals, and we say so whenever the distinction matters. In this chapter, therefore, the scaled mutation rate is $\theta = 2N\mu$ and coalescent time is measured in units of N generations. Most other chapters (Chapters 1, 3, 5, 8, 9, 10 and 11) instead write N for the number of diploid individuals, or for the effective size N_e, so that their $\theta = 4N\mu$ and their time unit of $2N$ generations are the same quantities as $\theta = 2N\mu$ and N generations here; the θ itself, and every formula stated in terms of θ, is unchanged.

2.2 Genotypes, gametes and the additive model

This section sets out the static theory of a Mendelian population: how genotype frequencies relate to gene frequencies, how two loci are associated, and how a genotypic value is split into an additive part and a remainder. These are the objects of Fisher's Fundamental Theorem in Part I, and the projection of Section 2.2.3 is the lemma on which Chapters 3 to 5 are built.

2.2.1 Hardy Weinberg proportions

Consider an autosomal locus with alleles $A_1, \ldots, A_k$ in a diploid population. Write P_{ij} for the frequency of the unordered genotype A_iA_j, so that $\sum_{i\le j} P_{ij} = 1$, and $p_i = P_{ii} + \frac{1}{2}\sum_{j\neq i} P_{ij}$ for the frequency of A_i among genes.

DEFINITION 2.1 (Hardy Weinberg proportions). A genotype distribution is in Hardy Weinberg proportions if $P_{ii} = p_i^2$ and $P_{ij} = 2p_ip_j$ for $i < j$; equivalently, if the two genes of a random individual are independent draws from $(p_1, \ldots, p_k)$.

The intuition is that Mendelian segregation hands each offspring one gene drawn at random from each parent, so if parents pair at random the offspring's two genes are independent draws from the gene pool. Nothing in the argument depends on the parental genotype frequencies.

THEOREM 2.2 (Hardy and Weinberg). *Suppose generations are discrete, mating is at random, the gene frequencies are the same in the two sexes, there is no selection, mutation or migration, and the population is infinite. Then whatever the parental genotype frequencies, the offspring are in Hardy Weinberg proportions with the parental gene frequencies p_i, and the population remains so in every later generation.*

Proof. A gamete from a random parent carries A_i with probability $P_{ii} + \frac{1}{2}\sum_{j\neq i} P_{ij} = p_i$, by Mendel's first law. Under random mating the two gametes forming a zygote come from independent parents, so the ordered pair of genes is (A_i, A_j) with probability p_ip_j, giving $P'_{ii} = p_i^2$ and $P'_{ij} = 2p_ip_j$. The gene frequencies of the

offspring are $p_i^2 + \tfrac{1}{2}\sum_{j\neq i} 2p_ip_j = p_i$, so the next generation is formed from the same p_i and the argument repeats. □

The theorem is a statement about geometry as much as about genetics. The genotype frequencies live in a simplex of dimension $k(k+1)/2 - 1$; the Hardy Weinberg distributions form a smooth submanifold of dimension $k-1$, parametrised by the gene frequencies, consisting of the symmetric product measures $p \otimes p$. Random mating projects any point of the simplex onto this manifold in one generation along the fibres of constant gene frequency. In Chapter 3 the Fundamental Theorem is shown to hold exactly when the population moves tangentially to the corresponding manifold for many loci, and a transverse coordinate at one biallelic locus can be taken to be the inbreeding coefficient F (that chapter uses Lindley's log cross product ratio, a function of F and p), defined by

$$P_{11} = p^2 + Fpq, \qquad P_{12} = 2pq(1-F), \qquad P_{22} = q^2 + Fpq, \qquad q = 1 - p. \tag{2.1}$$

Every biallelic genotype distribution with $0 < p < 1$ can be written this way for a unique F, which measures the excess of homozygotes; Hardy Weinberg is $F = 0$.

EXAMPLE 2.3 (Ford's scarlet tiger moth). Ford's sample of the scarlet tiger moth *Panaxia dominula* from Cothill, a standard textbook data set (Ford, 1971), contained 1469 of the common homozygote, 138 heterozygotes and 5 of the rare homozygote, $n = 1612$ in all. The common allele has frequency $\hat p = (2\cdot 1469 + 138)/3224 = 0.9541$. The Hardy Weinberg expectations are $n\hat p^2 = 1467.4$, $2n\hat p\hat q = 141.2$ and $n\hat q^2 = 3.4$, and Pearson's statistic is $\sum (O-E)^2/E = 0.83$ on one degree of freedom (p value 0.36). The estimated inbreeding coefficient is $\hat F = 1 - (138/1612)/(2\hat p\hat q) = 0.023$. The data sit on the Hardy Weinberg manifold to within sampling error.

2.2.2 *Two loci and linkage disequilibrium*

With two biallelic loci, alleles A, a and B, b, a gamete is one of four haplotypes AB, Ab, aB, ab with frequencies $x_{AB}, x_{Ab}, x_{aB}, x_{ab}$. The marginal frequencies are $p_A = x_{AB} + x_{Ab}$ and $p_B = x_{AB} + x_{aB}$.

DEFINITION 2.4 (Linkage disequilibrium). The coefficient of linkage disequilibrium is $D = x_{AB} - p_Ap_B$. The population is in linkage equilibrium if $D = 0$, that is, if the haplotype distribution is the product of its marginals.

Expanding $x_{AB}(1 - x_{AB} - x_{Ab} - x_{aB}) - x_{Ab}x_{aB} = x_{AB} - (x_{AB}+x_{Ab})(x_{AB}+x_{aB})$ gives the symmetric form

$$D = x_{AB}\,x_{ab} - x_{Ab}\,x_{aB}, \tag{2.2}$$

the determinant of the 2×2 table of haplotype frequencies. The haplotype simplex has dimension three, the product measures form a surface of dimension two, and D is the coordinate transverse to it. Two normalised versions are common: $r^2 = D^2/(p_Aq_Ap_Bq_B)$, the squared correlation between the allelic indicators, and $D' = D/D_{\max}$ with $D_{\max} = \min(p_Aq_B, q_Ap_B)$ when $D > 0$.

The next result says that recombination drives D to zero geometrically without changing the gene frequencies.

PROPOSITION 2.5 (Decay of disequilibrium). *Under random mating with recombination fraction c between the loci, and no other forces, p_A and p_B are constant and $D_{t+1} = (1-c)D_t$, so $D_t = (1-c)^t D_0$.*

Proof. A gamete produced by an individual is a parental haplotype with probability $1-c$ and a recombinant with probability c. A parental haplotype is AB with probability x_{AB}. A recombinant takes its A locus allele from one of the parent's two haplotypes and its B locus allele from the other; under random mating these are independent draws from the haplotype pool, so the recombinant is AB with probability $p_A p_B$. Hence $x'_{AB} = (1-c)x_{AB} + c p_A p_B$, and subtracting $p_A p_B$ gives $D' = (1-c)D$. The marginals are unchanged because recombination only reshuffles alleles between loci. □

EXAMPLE 2.6 (Disequilibrium and its decay). Take $x_{AB} = 0.5$, $x_{Ab} = x_{aB} = 0.1$, $x_{ab} = 0.3$. Then $p_A = p_B = 0.6$, $D = 0.5 - 0.36 = 0.14$, and (2.2) gives the same value $0.15 - 0.01$. The correlation is $r^2 = 0.14^2/0.24^2 = 0.340$ and $D' = 0.14/0.24 = 0.583$. With $c = 0.1$ the disequilibrium after 1, 5, 10 and 20 generations is 0.126, 0.083, 0.049 and 0.017; iterating the haplotype recursion directly reproduces these values. Loosely linked loci forget their association quickly; tightly linked loci, with c near zero, preserve it for many generations, which is why disequilibrium maps the genome.

2.2.3 *Average effects as a projection*

Fisher's average effect is the central quantity of Part I, and it is defined most cleanly as a regression coefficient (Fisher, 1918, 1941). Take one biallelic locus, alleles A_1 and A_2 with frequencies p and $q = 1-p$, genotype frequencies P_g for $g \in \{11, 12, 22\}$, and genotypic values G_{11}, G_{12}, G_{22} (the mean phenotype, or the fitness, of each genotype). Let $X_g \in \{2, 1, 0\}$ count the copies of A_1 and $\mu = \sum_g P_g G_g$. Work in the space of functions on the three genotypes with the inner product $\langle f, h \rangle_P = \sum_g P_g f_g h_g$.

DEFINITION 2.7 (Average excess, average effect, breeding value). The average excess of A_1 is the mean genotypic value of the carriers of a randomly chosen A_1 gene, less μ:

$$e_1 = \frac{P_{11}G_{11} + \tfrac{1}{2}P_{12}G_{12}}{p} - \mu, \qquad e_2 = \frac{P_{22}G_{22} + \tfrac{1}{2}P_{12}G_{12}}{q} - \mu.$$

The average effect α of substituting A_1 for A_2 is the slope of the weighted least squares regression of G on X with weights P:

$$(\mu_0, \alpha) = \arg\min_{(m,b)} \sum_g P_g \, (G_g - m - bX_g)^2.$$

The breeding value is the fitted value centred at the mean, $A_g = \alpha(X_g - 2p)$, and the dominance deviation is the residual $\delta_g = G_g - \mu - A_g$.

So the breeding value is the orthogonal projection of $G - \mu$ onto the span of $X - 2p$ in the inner product $\langle \cdot, \cdot \rangle_P$, and the dominance deviation is orthogonal to both constants and X. The point that Chapter 3 turns on is already visible: the subspace (functions linear in gene content) is fixed, but the inner product moves with the genotype frequencies.

THEOREM 2.8 (Average effect and average excess). *For any genotype frequencies of the form* (2.1),

$$pe_1 + qe_2 = 0, \qquad \alpha = \frac{e_1 - e_2}{1 + F}. \tag{2.3}$$

Under Hardy Weinberg proportions, $\alpha = e_1 - e_2 = a + d(q - p)$, *where* $a = (G_{11} - G_{22})/2$ *and* $d = G_{12} - (G_{11} + G_{22})/2$, *and the variance of* G *splits orthogonally as*

$$\sigma_G^2 = \sigma_A^2 + \sigma_D^2, \qquad \sigma_A^2 = 2pq\,\alpha^2, \qquad \sigma_D^2 = (2pqd)^2. \tag{2.4}$$

Proof. First, $pe_1 + qe_2 = P_{11}G_{11} + P_{12}G_{12} + P_{22}G_{22} - (p + q)\mu = 0$. The weighted least squares slope is $\mathrm{Cov}_P(X, G)/\mathrm{Var}_P(X)$. Since $p = P_{11} + \frac{1}{2}P_{12}$,

$$\mathrm{Cov}_P(X, G) = \sum_g P_g X_g (G_g - \mu) = 2P_{11}(G_{11} - \mu) + P_{12}(G_{12} - \mu) = 2pe_1.$$

From $pe_1 = -qe_2$ we get $e_1 - e_2 = e_1(1 + p/q) = e_1/q$, so $\mathrm{Cov}_P(X, G) = 2pq(e_1 - e_2)$. Under (2.1), $\mathbb{E}_P X^2 = 4(p^2 + Fpq) + 2pq(1 - F)$ and $\mathbb{E}_P X = 2p$, so $\mathrm{Var}_P(X) = 2pq(1 + F)$, which proves (2.3). At $F = 0$, $e_1 = pG_{11} + qG_{12} - \mu$ and $e_2 = pG_{12} + qG_{22} - \mu$; writing $G_{11} = m + a$, $G_{12} = m + d$, $G_{22} = m - a$ gives $e_1 - e_2 = p(a - d) + q(d + a) = a + d(q - p)$. The decomposition (2.4) is Pythagoras in $\langle \cdot, \cdot \rangle_P$, since δ is orthogonal to A; $\sigma_A^2 = \alpha^2 \mathrm{Var}_P(X) = 2pq\alpha^2$. Finally a direct computation gives $\delta_{11} = -2q^2 d$, $\delta_{12} = 2pqd$, $\delta_{22} = -2p^2 d$, whence $\sigma_D^2 = 4p^2q^2d^2(q^2 + 2pq + p^2) = (2pqd)^2$. □

The theorem says that average excess and average effect coincide exactly when the population is in Hardy Weinberg proportions, and differ by the factor $1 + F$ otherwise. Fisher's 1941 paper drew this distinction; Chapter 3 shows that the gap between the two is the one dimensional shadow of the term Fisher's theorem neglects.

EXAMPLE 2.9 (One locus with partial dominance). Let $p = 0.4$ and $(G_{11}, G_{12}, G_{22}) = (10, 8, 4)$, so $a = 3$ and $d = 1$. Under Hardy Weinberg the genotype frequencies are $(0.16, 0.48, 0.36)$ and $\mu = 6.88$. The average excesses are $e_1 = 0.4 \cdot 10 + 0.6 \cdot 8 - 6.88 = 1.92$ and $e_2 = 0.4 \cdot 8 + 0.6 \cdot 4 - 6.88 = -1.28$, and indeed $0.4 \cdot 1.92 - 0.6 \cdot 1.28 = 0$. The average effect is $\alpha = 3.2 = a + d(q - p) = 3 + 0.2$, and the weighted regression of G on X returns intercept 4.32 and slope 3.2. The breeding values

are $(3.84, 0.64, -2.56)$ and the dominance deviations $(-0.72, 0.48, -0.32)$. The variances are

$$\sigma_G^2 = 5.1456, \qquad \sigma_A^2 = 2(0.24)(3.2)^2 = 4.9152, \qquad \sigma_D^2 = (0.48)^2 = 0.2304,$$

and they add. The proportion of genotypic variance that is additive is 0.955: with only mild dominance, almost all of the variance is transmissible. If instead the population carries excess homozygosity $F = 0.2$ at the same p, the genotype frequencies become $(0.208, 0.384, 0.408)$, the weighted regression slope falls to $\alpha = 3.133$, while $e_1 - e_2 = 3.76 = 1.2\,\alpha$, exactly as (2.3) predicts.

With many loci the same construction applies with X replaced by the vector of gene contents at all loci, and σ_A^2 is then the variance of the projection onto the span of those contents. When the loci are in Hardy Weinberg and linkage equilibrium the gene contents at different loci are uncorrelated, the projection splits locus by locus, and σ_A^2 is the sum of the single locus values. Otherwise the regression must be fitted jointly, and with more loci than individuals it is the overparametrised regression of Chapters 4 and 5.

2.3 Drift: the Wright Fisher and Moran models

This section introduces the two standard models of a finite population, shows how their gene frequency fluctuates, and derives the probability that a new allele is fixed. These results are the forward time counterpart of the coalescent of Section 2.6, and they supply the scaled parameters such as θ that appear in every later chapter. The standard reference for this material, and for much of Sections 2.5 and 2.6, is Ewens (2004).

2.3.1 The two models

DEFINITION 2.10 (Wright Fisher model). A population of N genes, i of type A, produces the next generation by N independent draws from a gamete pool in which A has frequency $\psi(i/N)$. Thus $X_{t+1} \mid X_t = i \sim \mathrm{Bin}(N, \psi(i/N))$. Without selection $\psi(x) = x$; with genic selection of strength s in favour of A, $\psi(x) = x(1+s)/(1+sx)$.

DEFINITION 2.11 (Moran model). A population of N genes changes by one event at a time. At each event an individual is chosen to reproduce, with probability proportional to fitness ($1+s$ for A, 1 for a), and an individual chosen uniformly at random dies and is replaced by the offspring. There are N events per generation.

The Wright Fisher model has nonoverlapping generations and is the natural discrete model; the Moran model has overlapping generations and is a birth and death chain, which makes many of its quantities exactly computable. For a diploid population of N_d individuals one takes $N = 2N_d$ genes.

PROPOSITION 2.12 (Neutral drift). *In the neutral Wright Fisher model the frequency $x_t = X_t/N$ is a martingale, the probability that A is eventually fixed is x_0, and the expected heterozygosity $H_t = \mathbb{E}[2x_t(1-x_t)]$ satisfies $H_t = (1-1/N)^t H_0$.*

Proof. $\mathbb{E}[x_{t+1} \mid x_t] = x_t$ by the binomial mean. The chain is absorbed at 0 or 1 in finite time almost surely, and bounded martingales converge, so optional stopping gives $x_0 = \mathbb{E}x_\infty = \mathbb{P}(\text{fixation})$. For the heterozygosity, $\mathbb{E}[x_{t+1}^2 \mid x_t] = x_t^2 + x_t(1-x_t)/N$, so $\mathbb{E}[x_{t+1}(1-x_{t+1}) \mid x_t] = x_t(1-x_t)(1-1/N)$; take expectations. □

Heterozygosity is lost at rate $1/N$ per generation, so drift acts on a time scale of N generations. This is the reason population genetic results are stated in units of N (or $2N_d$) generations and in scaled parameters such as Ns and $N\mu$.

EXAMPLE 2.13 (Heterozygosity in a population of twenty genes). With $N = 20$ and $x_0 = 1/2$, $H_0 = 0.5$ and the proposition predicts $H_1 = 0.475$, $H_5 = 0.3869$ and $H_{10} = 0.2994$. Propagating the exact 21×21 transition matrix gives the same three numbers to five decimals. After ten generations, a population of twenty genes has lost forty percent of its variation to chance alone.

2.3.2 The diffusion approximation

In one Wright Fisher generation with weak selection the change $\Delta x = x_{t+1} - x_t$ has

$$\mathbb{E}[\Delta x \mid x] = \psi(x) - x = sx(1-x) + O(s^2),$$
$$\operatorname{Var}[\Delta x \mid x] = \frac{\psi(x)\{1-\psi(x)\}}{N} = \frac{x(1-x)}{N} + O\Big(\frac{s}{N}\Big).$$

Both moments are of order $1/N$ when $s = O(1/N)$, and higher moments are of smaller order. As $N \to \infty$ with Ns fixed, the process $x_{\lfloor Nt \rfloor}$ converges to the diffusion on $[0,1]$ with generator

$$\mathcal{L}f(x) = \tfrac{1}{2}x(1-x)f''(x) + \sigma x(1-x)f'(x), \qquad \sigma = Ns, \tag{2.5}$$

time being measured in units of N generations. This is the approximation Fisher introduced in 1922 and Kimura made the main computational tool of the subject (Kimura, 1962). For the Moran model the variance per generation is $2x(1-x)/N$ because each of the N events changes the count by exactly one with probability $2x(1-x)$ and otherwise leaves it unchanged; the same diffusion arises with N replaced by $N/2$. The Moran model drifts twice as fast.

Quantities defined by absorption satisfy ordinary differential equations in the starting point. If $u(x)$ is the probability of fixation from frequency x and $T(x)$ the expected time to absorption, then

$$\mathcal{L}u = 0, \quad u(0) = 0,\ u(1) = 1; \qquad \mathcal{L}T = -1, \quad T(0) = T(1) = 0. \tag{2.6}$$

These are obtained by conditioning on the first small step of the process, exactly as for a random walk.

2.3.3 Fixation probabilities

The intuition behind the next theorem is that selection pushes the frequency up at a rate proportional to Ns, while drift spreads it at a rate independent of s; the fixation probability balances the two, and depends on s and N only through their product.

THEOREM 2.14 (Kimura's formula). *For the diffusion* (2.5)*, the probability of fixation from frequency x is*

$$u(x) = \frac{1 - e^{-2Nsx}}{1 - e^{-2Ns}}. \tag{2.7}$$

For a new mutant, $x = 1/N$, and $Ns \gg 1$ with s small, $u \approx 1 - e^{-2s} \approx 2s$.

Proof. The equation $\frac{1}{2}x(1-x)u'' + Nsx(1-x)u' = 0$ reduces on $(0,1)$ to $u'' = -2Nsu'$, so $u'(x) = Ce^{-2Nsx}$ and $u(x) = C'(1 - e^{-2Nsx})$. The boundary condition $u(1) = 1$ fixes C'. For $x = 1/N$ the numerator is $1 - e^{-2s}$ and the denominator tends to one when Ns is large. □

For a diploid population of N_d individuals in which each copy of the favoured allele adds s to fitness, $N = 2N_d$ and (2.7) becomes the familiar $u(x) = (1 - e^{-4N_d sx})/(1 - e^{-4N_d s})$, while a new mutant at $x = 1/(2N_d)$ is fixed with probability close to Haldane's $2s$. Even a strongly favoured mutant is usually lost: with $s = 0.01$ it survives with probability about two percent.

For the Moran model there is an exact answer. The number i of A genes moves up by one at rate proportional to $(1+s)i(N-i)$ and down by one at rate proportional to $i(N-i)$, so it is a gambler's ruin chain with ratio of down to up probabilities r^{-1}, $r = 1 + s$, and

$$u_i^{\text{Moran}} = \frac{1 - r^{-i}}{1 - r^{-N}}. \tag{2.8}$$

EXAMPLE 2.15 (Exact against diffusion for small populations). Table 2.1 compares the exact fixation probability of a single new mutant in the Wright Fisher model, obtained by solving the linear system for absorption probabilities of the $(N+1)$ state chain, with Kimura's formula at $x = 1/N$, and lists the exact Moran value (2.8). Kimura's formula is accurate to within one percent when $Ns < 1$ and to within three and a half percent when $s = 0.1$, even for ten genes. The Moran values are smaller, close to s rather than $2s$ for large Ns, because the Moran model drifts twice as fast; Kimura's formula with N replaced by $N/2$ matches them to within five percent in every case of the table. The exact neutral Wright Fisher answer is $1/N$, as Proposition 2.12 requires. For the mean absorption time from $x = 1/2$, the solution of $\mathcal{L}T = -1$ is $T(x) = -2\{x \log x + (1-x)\log(1-x)\}$ in units of N generations, so $T(1/2) = 2N \log 2 = 1.386N$ generations; the exact Wright Fisher values for $N = 10$, 20 and 50 are 12.6, 26.2 and 67.5 generations against 13.9, 27.7 and 69.3 from the diffusion.

TABLE 2.1. Fixation probability of a single new mutant in a population of N genes.

N	s	exact Wright Fisher	Kimura (2.7)	relative error	exact Moran
10	0.01	0.1092	0.1092	0.0%	0.1045
10	0.05	0.1494	0.1505	0.8%	0.1233
10	0.10	0.2046	0.2096	2.5%	0.1480
20	0.01	0.0600	0.0601	0.1%	0.0549
20	0.05	0.1086	0.1101	1.3%	0.0764
20	0.10	0.1787	0.1847	3.3%	0.1068
50	0.01	0.0313	0.0313	0.2%	0.0253
50	0.05	0.0941	0.0958	1.8%	0.0522
50	0.10	0.1756	0.1813	3.3%	0.0917

2.4 Exchangeability and random discrete distributions

Part II of the book rests on one idea: a sample of genes whose order carries no information is exchangeable, and an exchangeable sample behaves as if drawn independently from an unknown random distribution. This section states that idea precisely and builds, step by step, the random distributions that the Ewens and Pitman formulae sample from.

2.4.1 De Finetti's theorem

DEFINITION 2.16 (Exchangeability). A sequence $X_1, X_2, \dots$ of random variables is exchangeable if for every n and every permutation π of $\{1, \dots, n\}$, $(X_{\pi(1)}, \dots, X_{\pi(n)})$ has the same law as $(X_1, \dots, X_n)$.

Independent identically distributed sequences are exchangeable, and so are mixtures of them. De Finetti's theorem says that there is nothing else.

THEOREM 2.17 (de Finetti; Hewitt and Savage). *Let $X_1, X_2, \dots$ be an exchangeable sequence taking values in a Polish space $\mathcal{X}$. Then there is a unique probability measure Π on the space of probability measures on $\mathcal{X}$ such that for all n and measurable $A_1, \dots, A_n$,*

$$\mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \int \prod_{i=1}^{n} P(A_i)\, \Pi(\mathrm{d}P).$$

Moreover the empirical distribution $F_n = n^{-1} \sum_{i \le n} \delta_{X_i}$ converges weakly almost surely to a random measure P with law Π, and given P the X_i are independent with law P.

Sketch for binary sequences. Let $S_n = X_1 + \cdots + X_n$. Given $S_n = m$, exchangeability makes every arrangement of the m ones equally likely, so $X_1, \dots, X_k$ are the first k draws without replacement from an urn with m ones and $n - m$ zeros. Hence for a fixed pattern with r ones among the first k,

$$\mathbb{P}(\text{pattern}) = \sum_{m} \mathbb{P}(S_n = m) \frac{(m)_{r\downarrow} (n-m)_{(k-r)\downarrow}}{(n)_{k\downarrow}},$$

where $(x)_{j\downarrow} = x(x-1)\cdots(x-j+1)$. Drawing with and without replacement differ by at most $k(k-1)/(2n)$ in total variation, so the right side is $\int p^r(1-p)^{k-r}\,\mathrm{d}\Pi_n(p) + O(k^2/n)$ with Π_n the law of S_n/n. By Helly's selection theorem some subsequence Π_{n_j} converges weakly to a measure Π on $[0,1]$, and the continuous integrand passes to the limit. Uniqueness follows because a measure on $[0,1]$ is determined by its moments. □

The proof is the elementary one of Heath and Sudderth (1976); Chapter 6 gives it in full for a general space, and Aldous (1985) surveys the subject. The bound $k(k-1)/(2n)$ is also the content of the finite form of the theorem (Diaconis and Freedman, 1980): the first k of n exchangeable variables are within $k(k-1)/(2n)$ in total variation of a mixture of independent sequences. In genetics the random measure P is the vector of allele frequencies in the population and Π is its distribution; the theorem says that any model for an unordered sample is a prior on population frequencies.

2.4.2 *The Pólya urn*

DEFINITION 2.18 (Pólya urn). An urn contains $b > 0$ black and $w > 0$ white balls, where b and w need not be integers. A ball is drawn at random, its colour recorded, and it is returned together with one more ball of the same colour. Let $X_i = 1$ if the ith draw is black.

PROPOSITION 2.19 (The urn is a Beta mixture). *Any sequence of n draws containing r blacks has probability $(b)_r(w)_{n-r}/(b+w)_n$, which equals $\int_0^1 p^r(1-p)^{n-r}\mathrm{Be}(p;b,w)\,\mathrm{d}p$. Hence the draws are exchangeable, the fraction of black balls converges almost surely to $V \sim \mathrm{Be}(b,w)$, and given V the draws are independent Bernoulli(V).*

Proof. At the jth black draw the urn holds $b+j-1$ black balls, and at the ith draw overall it holds $b+w+i-1$ balls in all; multiplying the conditional probabilities gives the first expression, which does not depend on the order of the draws. The Beta integral $\int p^{b+r-1}(1-p)^{w+n-r-1}\,\mathrm{d}p/B(b,w) = B(b+r,w+n-r)/B(b,w)$ equals the same ratio of rising factorials. The remaining claims follow from Theorem 2.17, the mixing measure being identified by its moments. □

EXAMPLE 2.20 (Two balls). With $b = w = 1$, the sequences BBW, BWB and WBB each have probability $\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{1}{4} = \frac{1}{12} = \int_0^1 p^2(1-p)\,\mathrm{d}p$. The number of blacks in n draws is uniform on $\{0,1,\dots,n\}$; for $n = 5$ each value has probability exactly $1/6$. With $b = 2$, $w = 1$ the limiting fraction is $\mathrm{Be}(2,1)$, with mean $2/3$ and variance $1/18 = 0.0556$; four thousand simulated urns run for two thousand draws give 0.665 and 0.056. The random limit is the urn's analogue of an unknown allele frequency.

2.4.3 *The Dirichlet distribution*

DEFINITION 2.21 (Dirichlet distribution). For $a_1, \dots, a_k > 0$ and $a = \sum_i a_i$, the Dirichlet distribution $\mathrm{Dir}(a_1, \dots, a_k)$ on the simplex $\{P_i \geq 0, \sum P_i = 1\}$ has density proportional to $\prod_i P_i^{a_i - 1}$ with respect to Lebesgue measure on the first $k-1$ coordinates; the normalising constant is $\Gamma(a) / \prod_i \Gamma(a_i)$. For $k = 2$ it is the Beta distribution.

PROPOSITION 2.22 (Properties of the Dirichlet). *(i) If $G_i \sim \mathrm{Ga}(a_i, 1)$ independently and $G = \sum G_i$, then $(G_1/G, \dots, G_k/G) \sim \mathrm{Dir}(a)$, independent of $G \sim \mathrm{Ga}(a, 1)$.*

(ii) Aggregation: merging cells adds parameters, e.g. $(P_1 + P_2, P_3, \dots, P_k) \sim \mathrm{Dir}(a_1 + a_2, a_3, \dots, a_k)$.

(iii) Moments: $\mathbb{E} P_i = a_i / a$ and $\mathrm{Var}\, P_i = a_i(a - a_i)/\{a^2(a+1)\}$.

(iv) Conjugacy: if $P \sim \mathrm{Dir}(a)$ and, given P, the counts $(n_1, \dots, n_k)$ are multinomial, then $P \mid n \sim \mathrm{Dir}(a_1 + n_1, \dots, a_k + n_k)$.

Proof. (i) is the change of variables $(G_1, \dots, G_k) \mapsto (G, G_1/G, \dots, G_{k-1}/G)$, whose Jacobian is G^{k-1}; the joint density factorises. (ii) follows from (i) because $G_1 + G_2 \sim \mathrm{Ga}(a_1 + a_2, 1)$. (iii) follows from (ii) with the Beta moments. (iv) is immediate from the densities. □

The gamma representation (i) is the finite form of a fact used repeatedly in Part II: the Dirichlet process is a normalised gamma process, and the Ewens formula is the sampling law of normalised gamma jumps. For example, $\mathrm{Dir}(1, 2, 3)$ gives $P_1 + P_2 \sim \mathrm{Be}(3, 3)$, with mean $1/2$ and variance $1/28 = 0.0357$; two hundred thousand simulated vectors give 0.4999 and 0.0357.

2.4.4 *The Dirichlet process*

DEFINITION 2.23 (Dirichlet process). Let $\theta > 0$ and let G_0 be a probability measure on $\mathcal{X}$. A random probability measure G is a Dirichlet process $\mathrm{DP}(\theta, G_0)$ if for every finite measurable partition $B_1, \dots, B_k$ of $\mathcal{X}$, $(G(B_1), \dots, G(B_k)) \sim \mathrm{Dir}(\theta G_0(B_1), \dots, \theta G_0(B_k))$ (Ferguson, 1973).

The aggregation property makes these finite dimensional laws consistent, so the process exists. The base measure G_0 is the mean, $\mathbb{E} G(B) = G_0(B)$, and the concentration θ controls the spread, $\mathrm{Var}\, G(B) = G_0(B)\{1 - G_0(B)\}/(\theta + 1)$. Three facts follow from conjugacy (Proposition 2.22(iv)) applied to every partition at once. If $X_1, \dots, X_n \mid G$ are independent with law G, then

$$G \mid X_1, \dots, X_n \sim \mathrm{DP}\Big(\theta + n,\ \frac{\theta G_0 + \sum_{i \leq n} \delta_{X_i}}{\theta + n}\Big), \tag{2.9}$$

and so the predictive law, integrating out G, is the generalised Pólya urn of Blackwell and MacQueen (1973),

$$X_{n+1} \mid X_1, \dots, X_n \;\sim\; \frac{\theta}{\theta+n}\, G_0 + \frac{1}{\theta+n} \sum_{i \le n} \delta_{X_i}. \tag{2.10}$$

With probability $\theta/(\theta+n)$ the next draw is new, from G_0; otherwise it copies a uniformly chosen earlier draw. If G_0 has no atoms, ties among the X_i occur only by copying, which forces G to be discrete almost surely. Chapter 6 shows the converse: if the predictive rule is affine in the empirical distribution, exchangeability forces exactly the weights in (2.10).

2.4.5 Stick breaking and the Poisson Dirichlet family

The atoms and weights of a Dirichlet process can be generated explicitly by breaking a stick.

DEFINITION 2.24 (Two parameter stick breaking). Let $0 \le \alpha < 1$ and $\theta > -\alpha$. Let $W_1, W_2, \dots$ be independent with $W_j \sim \mathrm{Be}(1-\alpha, \theta + j\alpha)$ and set

$$P_1 = W_1, \qquad P_j = W_j \prod_{i<j} (1 - W_i), \quad j \ge 2.$$

The law of $(P_1, P_2, \dots)$ is $\mathrm{GEM}(\alpha, \theta)$, and the law of its decreasing rearrangement is the two parameter Poisson Dirichlet distribution $\mathrm{PD}(\alpha, \theta)$ (Pitman and Yor, 1997). The one parameter case $\alpha = 0$ is Kingman's $\mathrm{PD}(\theta)$.

The stick of length one is broken at a random fraction W_1, the remainder at a fraction W_2, and so on; the pieces sum to one almost surely. Sethuraman (1994) proved that if $(P_j) \sim \mathrm{GEM}(0, \theta)$ and Y_j are independent draws from G_0, then $G = \sum_j P_j \delta_{Y_j} \sim \mathrm{DP}(\theta, G_0)$. The GEM order is the size biased order: P_1 is the frequency of the type of a randomly chosen individual, P_2 that of the first individual not of that type, and so on. In genetic language, $\mathrm{PD}(\theta)$ is the distribution of the ranked allele frequencies in a neutral population at equilibrium under the infinite alleles model.

Because the W_j are independent, the expected weights are explicit:

$$\mathbb{E} P_j = \frac{1-\alpha}{\theta + 1 + (j-1)\alpha} \prod_{i<j} \frac{\theta + i\alpha}{\theta + 1 + (i-1)\alpha}, \qquad \mathbb{E}\Big[\sum_{i>k} P_i\Big] = \prod_{i \le k} \frac{\theta + i\alpha}{\theta + 1 + (i-1)\alpha}. \tag{2.11}$$

When $\alpha = 0$ the expected tail is $\{\theta/(\theta+1)\}^k$, geometric; when $\alpha > 0$ each factor is $1 - (1-\alpha)/\{\theta + 1 + (i-1)\alpha\}$ and the tail decays only like $k^{-(1-\alpha)/\alpha}$, a power law. The discount α is the parameter that controls how heavy the tail of rare types is, and in Chapter 5 it is identified with the sparsity level of a shrinkage prior.

EXAMPLE 2.25 (Geometric against power law tails). For $\mathrm{GEM}(0, 5)$, $\mathbb{E} P_1 = 1/6 = 0.167$ and $\mathbb{E} P_2 = 0.139$, and the expected mass beyond the first k sticks falls below

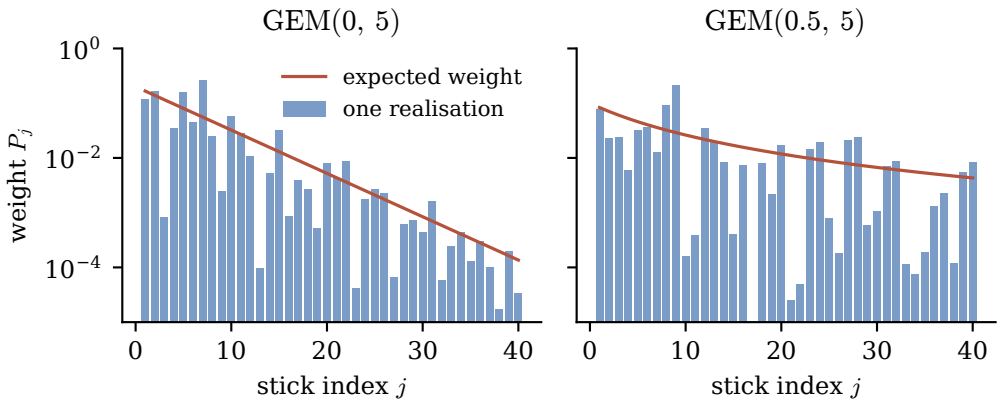


FIGURE 2.1. Stick breaking weights P_j on a logarithmic scale for GEM$(0,5)$ (left) and GEM$(0.5,5)$ (right): one realisation (bars) and the expected weights (2.11) (line). With $\alpha = 0$ the weights fall geometrically; with $\alpha = 1/2$ they fall like a power of j.

one percent at $k = 26$. For GEM$(0.5,5)$, $\mathbb{E}P_1 = 0.083$ and $\mathbb{E}P_2 = 0.071$, but the expected mass beyond the first 25 sticks is still 0.306, and it does not fall below one percent until $k = 1089$. Figure 2.1 shows one realisation of each with the expected weights (2.11). The two parameter family puts appreciable mass on a very long tail of rare types, which is what species abundance and rare variant data often require.

The last object of this section connects random distributions to random partitions. A random partition Π of $\{1,2,\dots\}$ is exchangeable if its law is invariant under finite permutations of the integers. Given a sequence $s_1 \geq s_2 \geq \cdots \geq 0$ with $\sum_j s_j \leq 1$, the paintbox partition is formed by painting each integer independently with colour j with probability s_j, or with a unique colour of its own with probability $1 - \sum_j s_j$, and grouping integers of the same colour.

THEOREM 2.26 (Kingman's paintbox). *Every exchangeable random partition of $\{1,2,\dots\}$ is a mixture of paintbox partitions: there is a random ranked sequence $S = (S_1, S_2, \dots)$ such that, given S, the partition is the paintbox of S. The S_j are the limiting relative frequencies of the blocks, ranked. The Ewens partition of Section 2.5 corresponds to $S \sim \mathrm{PD}(0,\theta)$ and Pitman's to $S \sim \mathrm{PD}(\alpha,\theta)$ (Kingman, 1978; Pitman, 1995).*

The paintbox plays for partitions the role de Finetti's theorem plays for sequences. In Chapter 7 the paint is applied with a nonuniform density, which models ascertainment.

2.5 The Ewens and Pitman sampling formulae

This section derives the sampling formula for a neutral sample under the infinite alleles model from a single predictive rule, records the distribution and moments

of the number of alleles, and extends both to the two parameter family. These are the objects of Part II.

2.5.1 *The Chinese restaurant and Hoppe's urn*

Apply the predictive rule (2.10) with a nonatomic G_0 and record only which draws coincide. Customers enter a restaurant one at a time. The first sits at a table. When m customers are seated at k tables with $m_1, \dots, m_k$ at each, the next customer

$$\begin{aligned} &\text{starts a new table with probability } \frac{\theta}{\theta+m}, \\ &\text{joins table } j \text{ with probability } \frac{m_j}{\theta+m}. \end{aligned} \tag{2.12}$$

Hoppe (1984) gave the same scheme as an urn: a black ball of weight θ and coloured balls of weight one; drawing black adds a ball of a new colour, drawing a colour adds another of that colour. In genetic terms a table is an allele, the black ball is mutation, and $\theta = 2N\mu$ for a population of N genes ($4N_d\mu$ for N_d diploids).

Write a_j for the number of alleles represented exactly j times among n genes, so that $\sum_j j a_j = n$, and $K_n = \sum_j a_j$ for the number of distinct alleles.

THEOREM 2.27 (Ewens sampling formula). *Under* (2.12),

$$\mathbb{P}(a_1, \dots, a_n) = \frac{n!}{(\theta)_n} \prod_{j=1}^{n} \frac{(\theta/j)^{a_j}}{a_j!}. \tag{2.13}$$

Proof. Fix a seating sequence ending with k tables of sizes $m_1, \dots, m_k$. Its probability is a product of n factors whose denominators are $\theta, \theta+1, \dots, \theta+n-1$. The numerators are θ for each of the k new tables and $1, 2, \dots, m_j - 1$ for the later arrivals at table j, so the sequence has probability $\theta^k \prod_j (m_j-1)!/(\theta)_n$, whatever the order of arrivals. The number of ways to partition n labelled genes into blocks with a_j blocks of size j is $n!/\prod_j (j!)^{a_j} a_j!$. Multiplying, and using $(j-1)!/j! = 1/j$, gives (2.13). □

Ewens (1972) derived (2.13) from the Wright Fisher model with infinite alleles mutation, Antoniak (1974) found it as the partition of a Dirichlet process sample, and Section 2.6 derives it a third way, from the coalescent.

2.5.2 *The number of alleles*

The $(m+1)$st gene is a new allele with probability $\theta/(\theta+m)$, independently of everything before, so

$$K_n = \sum_{m=0}^{n-1} \xi_m, \qquad \xi_m \sim \text{Bernoulli}\Big(\frac{\theta}{\theta+m}\Big) \text{ independent}, \tag{2.14}$$

TABLE 2.2. Probabilities of the five allelic configurations of $n = 4$ genes.

block sizes	(a_1, a_2, a_3, a_4)	K_4	Ewens $\theta = 1$	Ewens $\theta = 2$	Pitman $(\frac{1}{2}, 1)$
4	$(0,0,0,1)$	1	$1/4$	$1/10$	$5/64$
3,1	$(1,0,1,0)$	2	$1/3$	$4/15$	$3/16$
2,2	$(0,2,0,0)$	2	$1/8$	$1/10$	$3/64$
2,1,1	$(2,1,0,0)$	3	$1/4$	$2/5$	$3/8$
1,1,1,1	$(4,0,0,0)$	4	$1/24$	$2/15$	$5/16$
$\mathbb{E}K_4$			$25/12$	$77/30$	$187/64$

and therefore

$$\mathbb{E}K_n = \sum_{m=0}^{n-1} \frac{\theta}{\theta + m}, \qquad \operatorname{Var} K_n = \sum_{m=0}^{n-1} \frac{\theta m}{(\theta + m)^2}, \qquad \mathbb{P}(K_n = k) = \frac{|s(n,k)|\,\theta^k}{(\theta)_n}, \tag{2.15}$$

where $|s(n,k)|$ is the unsigned Stirling number of the first kind, the coefficient of θ^k in $(\theta)_n$. For large n, $\mathbb{E}K_n = \theta\{\psi(\theta + n) - \psi(\theta)\} \approx \theta \log(1 + n/\theta)$, where ψ is the digamma function, and $K_n / \log n \to \theta$ almost surely (Korwar and Hollander, 1973).

The factorisation of (2.13) into $\theta^{K_n}/(\theta)_n$ times a function of a alone shows that K_n is sufficient for θ: the configuration given K_n carries no information about the mutation rate. The maximum likelihood estimator solves $K_n = \mathbb{E}_\theta K_n$, and since $\operatorname{Var} K_n \approx \theta \log n$ the Fisher information about θ is only about $\log n / \theta$. Chapters 6, 8 and 11 return to this logarithmic information from three directions.

EXAMPLE 2.28 (All partitions of four genes). There are five allelic configurations of $n = 4$ genes. Table 2.2 lists their probabilities under (2.13) for $\theta = 1$ and $\theta = 2$, and under Pitman's formula of Section 2.5.3 for $\alpha = 1/2$, $\theta = 1$; each column sums to one. At $\theta = 1$ the probabilities are $(6, 8, 3, 6, 1)/24$, the proportions of permutations of four objects with each cycle type, because the Ewens formula at $\theta = 1$ is the law of the cycle lengths of a uniform random permutation. For $\theta = 2$, the Stirling numbers $|s(4,k)| = 6, 11, 6, 1$ give $\mathbb{P}(K_4 = k) = (0.1, 0.3667, 0.4, 0.1333)$ for $k = 1, \ldots, 4$; four hundred thousand simulated restaurants give $(0.1002, 0.3657, 0.4013, 0.1328)$. The moments from (2.15) are $\mathbb{E}K_4 = 77/30 = 2.567$ and $\operatorname{Var} K_4 = 641/900 = 0.712$, and summing over the table reproduces both exactly. At $\theta = 1$, $\mathbb{E}K_4 = 25/12$ and $\operatorname{Var} K_4 = 95/144$.

2.5.3 Pitman's two parameter formula

The two parameter generalisation changes the restaurant so that large tables are slightly less attractive and new tables slightly more so the more tables there are.

With $0 \le \alpha < 1$ and $\theta > -\alpha$, when m customers occupy k tables the next customer

$$\begin{aligned} &\text{starts a new table with probability } \frac{\theta + k\alpha}{\theta + m}, \\ &\text{joins table } j \text{ with probability } \frac{m_j - \alpha}{\theta + m}. \end{aligned} \tag{2.16}$$

The same counting as in Theorem 2.27 shows that a particular partition of $\{1, \dots, n\}$ into blocks of sizes $m_1, \dots, m_k$ has probability

$$p(m_1, \dots, m_k) = \frac{\prod_{i=1}^{k-1}(\theta + i\alpha)}{(\theta+1)_{n-1}} \prod_{j=1}^{k} (1-\alpha)_{m_j - 1}, \tag{2.17}$$

Pitman's exchangeable partition probability function (Pitman, 1995, 2006). It is symmetric in the block sizes, so the partition is exchangeable; its paintbox is $\mathrm{PD}(\alpha, \theta)$; and at $\alpha = 0$ it reduces to the Ewens formula. The number of blocks has mean

$$\mathbb{E}K_n = \frac{\theta}{\alpha}\left\{ \frac{(\theta+\alpha)_n}{(\theta)_n} - 1 \right\}, \qquad \alpha > 0, \tag{2.18}$$

which grows like n^α rather than $\log n$. Indeed K_n / n^α converges almost surely to a strictly positive random variable, Pitman's α diversity. In Table 2.2, (2.18) with $\alpha = 1/2$, $\theta = 1$, $n = 4$ gives $187/64 = 2.92$, larger than either Ewens column, and the configuration with four singletons is now the second most probable.

2.5.4 The logarithmic series and Fisher's alpha

Fisher's problem was ecological: a sample of n individuals contains S species, of which a_j are represented by exactly j individuals. If species abundances are a Poisson process with gamma intensity and the number of specimens of each species is Poisson, the expected number of species seen j times is a negative binomial series, and in the limit of a gamma shape tending to zero it becomes Fisher's logarithmic series (Fisher et al., 1943)

$$\mathbb{E}a_j = \alpha_F \frac{x^j}{j}, \qquad x = \frac{n}{n + \alpha_F}, \qquad \mathbb{E}S = \alpha_F \log\left(1 + \frac{n}{\alpha_F}\right), \tag{2.19}$$

with Fisher's diversity index α_F (not the discount α of the previous subsection). Under the Ewens formula the expected multiplicities are

$$\mathbb{E}a_j = \frac{\theta}{j} \prod_{i=0}^{j-1} \frac{n-i}{\theta + n - 1 - i} \;\longrightarrow\; \frac{\theta}{j} \qquad (n \to \infty,\ j \text{ fixed}), \tag{2.20}$$

which is (2.19) with $\alpha_F = \theta$ and $x \to 1$. Fisher's index is the Ewens parameter: the logarithmic series is the large sample frequency spectrum of the Ewens formula, and $\mathbb{E}S$ in (2.19) is the approximation $\mathbb{E}K_n \approx \theta \log(1 + n/\theta)$.

EXAMPLE 2.29 (Williams' Macrolepidoptera). Williams' Rothamsted light trap catch of moths, analysed in Fisher et al. (1943), contained $n = 15{,}609$ individuals of $S = 240$ species. Solving $S = \alpha_F \log(1 + n/\alpha_F)$ gives Fisher's $\alpha_F = 40.25$, and $x = 0.99743$. The Ewens maximum likelihood estimate, solving $240 = \theta\{\psi(\theta + n) - \psi(\theta)\}$, is $\hat{\theta} = 40.15$ with standard error 2.84, the value used in Chapter 8. The expected numbers of species seen once, twice, three and ten times are 40.05, 19.97, 13.28 and 3.92 under (2.20) at $\hat{\theta}$, and 40.14, 20.02, 13.31 and 3.92 under the logarithmic series: the two are indistinguishable at this sample size. Note that the cruder approximation $\theta \log n = 388$ is far from 240; the logarithmic growth of K_n sets in only when n is large compared with θ, and here n/θ is about 390.

2.6 The Kingman coalescent

The coalescent describes the ancestry of a sample rather than the future of a population. It is the model behind Part III, and it gives a third derivation of the Ewens formula. This section derives it from the Wright Fisher model, computes the moments of its height and length, and superimposes the two mutation models of the neutral theory.

2.6.1 Coalescence times

In the neutral Wright Fisher model with N genes, two genes have the same parent in the previous generation with probability $1/N$. The number of generations back to their common ancestor is therefore geometric with mean N, and measured in units of N generations it converges to an exponential with mean one. With k genes, the probability that some pair shares a parent is $\binom{k}{2}/N + O(N^{-2})$ and the probability that three or more do, or two pairs at once, is $O(N^{-2})$. Taking the limit gives Kingman's definition (Kingman, 1982).

DEFINITION 2.30 (Kingman coalescent). Starting from n lineages, while there are $k \geq 2$ lineages each pair merges at rate one, independently. Thus the time T_k during which there are exactly k lineages is exponential with rate $\binom{k}{2}$, the T_k are independent, and at the end of T_k a uniformly chosen pair merges. Time is measured in units of N generations ($2N_d$ for a diploid population).

The height of the tree, the time to the most recent common ancestor, is $H_n = \sum_{k=2}^{n} T_k$ and the total branch length is $L_n = \sum_{k=2}^{n} kT_k$. Since $\mathbb{E}T_k = 2/\{k(k-1)\}$ and $\operatorname{Var} T_k = 4/\{k(k-1)\}^2$, and $2/\{k(k-1)\} = 2/(k-1) - 2/k$ telescopes,

$$\begin{gathered} \mathbb{E}H_n = 2\Big(1 - \frac{1}{n}\Big), \qquad \mathbb{E}L_n = 2h_n, \qquad \operatorname{Var} L_n = 4b_n, \\ h_n = \sum_{k=1}^{n-1} \frac{1}{k}, \qquad b_n = \sum_{k=1}^{n-1} \frac{1}{k^2}. \end{gathered} \tag{2.21}$$

Two features deserve comment. The expected height is bounded by two whatever n, and more than half of it, on average, is the time T_2 when only two lineages remain:

adding genes to a sample adds short twigs near the present, not depth. The length grows only like $2\log n$, and its variance stays bounded, so a large sample says little more about the deep tree than a small one. For $n = 10$, $\mathbb{E}H_{10} = 1.8$, $\mathbb{E}L_{10} = 5.66$ and $\operatorname{Var} L_{10} = 6.16$; for $n = 100$, $\mathbb{E}H_{100} = 1.98$, $\mathbb{E}L_{100} = 10.35$ and $\operatorname{Var} L_{100} = 6.54$.

2.6.2 *Mutation on the tree*

Neutral mutations do not affect the genealogy, so they can be added afterwards. If each gene mutates with probability μ per generation, then in units of N generations mutations fall on each lineage as a Poisson process of rate $\theta/2$, with $\theta = 2N\mu$ ($4N_d\mu$ for diploids).

Under the infinite alleles model each mutation creates a new allele. Looking back from the present, kill a lineage when it meets a mutation, since the allele it carries is then determined. While k lineages remain, coalescence occurs at rate $k(k-1)/2$ and mutation at rate $k\theta/2$, so the next event is a mutation with probability $\theta/(\theta + k - 1)$. Reading the history forwards, each added gene is a new allele with probability $\theta/(\theta + k - 1)$ and otherwise copies a uniformly chosen existing gene: this is the Hoppe urn (2.12), and the sample's allelic partition follows the Ewens formula (2.13).

Under the infinite sites model each mutation strikes a new nucleotide site, so every mutation on the tree produces one segregating site, and the derived allele at that site is carried by exactly the sampled genes below the mutated branch. Let S_n be the number of segregating sites and ξ_i the number of sites at which the derived allele is carried by exactly i of the n genes, $1 \le i \le n-1$. The vector $(\xi_1, \dots, \xi_{n-1})$ is the unfolded site frequency spectrum and $S_n = \sum_i \xi_i$. Figure 2.2 shows one realisation.

2.6.3 *Watterson's estimator and the site frequency spectrum*

THEOREM 2.31 (Segregating sites). *Under the infinite sites model,* $S_n \mid L_n \sim \mathrm{Po}(\theta L_n/2)$, *and*

$$\mathbb{E}S_n = \theta h_n, \qquad \operatorname{Var} S_n = \theta h_n + \theta^2 b_n. \tag{2.22}$$

Watterson's estimator $\hat\theta_W = S_n/h_n$ *is unbiased, with variance* $\theta/h_n + \theta^2 b_n/h_n^2$ *(Watterson, 1975).*

Proof. Given the tree, mutations form a Poisson process of rate $\theta/2$ on a set of total length L_n. Then $\mathbb{E}S_n = \frac{\theta}{2}\mathbb{E}L_n$ and $\operatorname{Var} S_n = \mathbb{E}\operatorname{Var}(S_n \mid L_n) + \operatorname{Var}\mathbb{E}(S_n \mid L_n) = \frac{\theta}{2}\mathbb{E}L_n + \frac{\theta^2}{4}\operatorname{Var} L_n$; substitute (2.21). □

Since $h_n \sim \log n$ while $b_n \to \pi^2/6$, the variance of $\hat\theta_W$ falls only like $\theta/\log n$, and for moderate samples and large θ it is dominated by the second term, $\theta^2 b_n/h_n^2$, which comes from the randomness of the single genealogy that all the sites share. Chapter 10 proves that S_n, unlike K_n in the infinite alleles model, is not sufficient for θ, and Chapter 11 gives the exact law of S_n as a sum of independent geometric variables. Information beyond S_n lies in the spectrum.

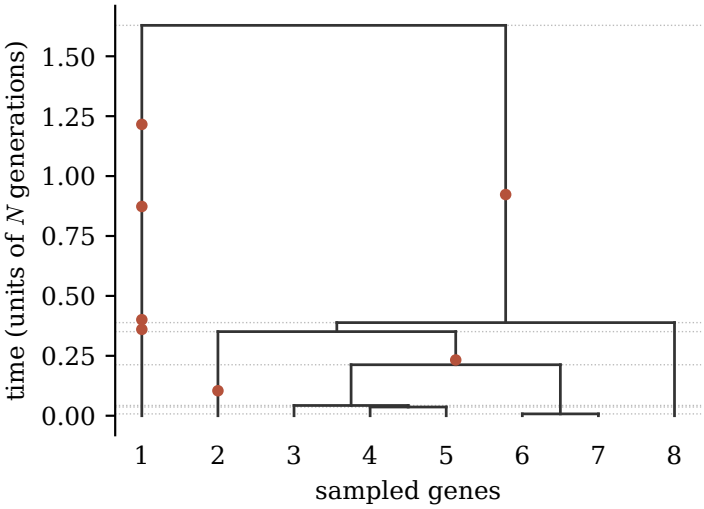


FIGURE 2.2. A realisation of the Kingman coalescent for $n = 8$ genes with infinite sites mutations at $\theta = 3$ (dots). Dotted lines mark the coalescence times. This tree has height 1.63, total length 4.30 and $S = 7$ segregating sites, with spectrum $\xi_1 = 5$, $\xi_5 = 1$, $\xi_7 = 1$ and the remaining $\xi_i = 0$; the expected values are $\mathbb{E}H_8 = 1.75$, $\mathbb{E}L_8 = 5.19$ and $\mathbb{E}S_8 = 7.78$.

THEOREM 2.32 (Expected spectrum). *Under the infinite sites model,* $\mathbb{E}\xi_i = \theta/i$ *for* $1 \le i \le n-1$.

Proof. Let L_i be the total length of branches that subtend exactly i sampled genes, so $\mathbb{E}\xi_i = \frac{\theta}{2}\mathbb{E}L_i$. During the epoch with k lineages, the numbers of sampled genes descending from the k lineages form a random composition of n into k positive parts. By the Pólya urn argument of Proposition 2.19 applied to the forward history, with k colours each starting with one ball, every composition is equally likely, so a given lineage subtends exactly i genes with probability $\binom{n-i-1}{k-2}/\binom{n-1}{k-1}$. Hence

$$\mathbb{E}L_i = \sum_{k=2}^{n-i+1} k\,\mathbb{E}T_k \,\frac{\binom{n-i-1}{k-2}}{\binom{n-1}{k-1}} = \sum_{k=2}^{n-i+1} \frac{2}{k-1}\,\frac{\binom{n-i-1}{k-2}}{\binom{n-1}{k-1}}.$$

Writing $j = k-2$ and expanding the binomials, the summand is $2\,(n-i-1)!\,(i-1)!\,\binom{n-2-j}{i-1}/(n-1)!$, and the hockey stick identity $\sum_{j=0}^{n-i-1}\binom{n-2-j}{i-1} = \binom{n-1}{i}$ gives $\mathbb{E}L_i = 2\,(n-i-1)!\,(i-1)!\,\binom{n-1}{i}/(n-1)! = 2/i$. □

The spectrum is a decreasing harmonic sequence: singletons are the most common class, and the expected number of sites at frequency i/n is inversely proportional to i. It is the infinite sites counterpart of (2.20), and the departures of observed spectra from θ/i are the basis of the neutrality tests of Tajima and of Fu and Li (Fu, 1995). Summing over i recovers $\mathbb{E}S_n = \theta h_n$.

EXAMPLE 2.33 (Five genes). For $n = 5$ and $\theta = 2$ the theory gives $\mathbb{E}(\xi_1, \xi_2, \xi_3, \xi_4) = (2, 1, 0.667, 0.5)$, $\mathbb{E}S_5 = 2h_5 = 4.167$, $\operatorname{Var} S_5 = 2h_5 + 4b_5 = 9.861$, $\mathbb{E}H_5 = 1.6$ and $\mathbb{E}L_5 = 4.167$. One hundred thousand simulated coalescent trees give spectrum means $(2.010, 0.998, 0.669, 0.501)$, $\mathbb{E}S = 4.178$, $\operatorname{Var} S = 9.854$, $\mathbb{E}H = 1.603$ and $\mathbb{E}L = 4.175$. The probability that the sample is monomorphic is $\prod_{k=2}^{5}(k-1)/(k-1+\theta) = 1/15 = 0.0667$, against 0.0669 in simulation; this product form is the first instance of the geometric decomposition of Chapter 11.

EXAMPLE 2.34 (Kreitman's alcohol dehydrogenase sample). Kreitman's survey of the *Adh* locus of *Drosophila melanogaster* (Kreitman, 1983) sequenced $n = 11$ alleles and found $S = 43$ segregating sites. With $h_{11} = 2.929$ and $b_{11} = 1.550$, Watterson's estimate for the region is $\hat{\theta}_W = 43/2.929 = 14.7$. At that value the standard deviation of S is $\sqrt{14.7 \cdot 2.929 + 14.7^2 \cdot 1.550} = 19.4$, so the standard error of $\hat{\theta}_W$ is about 6.6: the coefficient of variation is forty five percent, and almost all of it comes from the term $\theta^2 b_n$, the variability of the genealogy itself. Sequencing more alleles would barely help; sequencing more independent loci would. Under neutrality the expected spectrum at $\hat{\theta}_W$ is $14.7/i$, that is 14.7 singletons, 7.3 doubletons and so on down to 1.5 sites at frequency $10/11$.

2.7 Bayesian tools

This section collects the Bayesian devices used in the later chapters: scale mixtures of normals, the horseshoe prior and its shrinkage weight, and three data augmentation identities that turn awkward likelihoods into conditionally conjugate ones. It closes with a paragraph on the large sample behaviour of posteriors.

2.7.1 Scale mixtures of normals

A random variable β is a scale mixture of normals if $\beta = \sqrt{V}\, Z$ with $Z \sim \mathcal{N}(0,1)$ independent of a positive V, so that $p(\beta) = \int \mathcal{N}(\beta; 0, v)\, p(v)\, \mathrm{d}v$. The class is closed under the operations Bayesian computation needs: given V everything is Gaussian, and V is updated by a one dimensional draw. Two classical members are the Student t_ν, for which $1/V \sim \mathrm{Ga}(\nu/2, \nu/2)$, and the Laplace distribution with scale b, for which V is exponential with mean $2b^2$ (Andrews and Mallows, 1974). Numerically, at $\beta = 1.3$ the t_3 density 0.150389 and the Laplace density 0.136266 (with $b = 1$) are both reproduced to six decimals by integrating the normal density against the mixing law.

2.7.2 Global local shrinkage and the horseshoe

In a sparse normal means problem, $y_i \mid \beta_i \sim \mathcal{N}(\beta_i, \sigma^2)$ for $i = 1, \ldots, p$, and most β_i are near zero while a few are large. A global local prior is a scale mixture with one scale shared by all coordinates and one particular to each:

$$\beta_i \mid \lambda_i, \tau \sim \mathcal{N}(0, \lambda_i^2 \tau^2 \sigma^2), \qquad \lambda_i \sim \pi(\lambda_i), \qquad \tau \sim \pi(\tau). \tag{2.23}$$

The global scale τ adapts to the overall level of sparsity; the local scales λ_i let individual signals escape it. The horseshoe of Carvalho et al. (2010) takes $\lambda_i \sim \mathrm{C}^+(0,1)$, the half Cauchy distribution with density $2/\{\pi(1+\lambda^2)\}$ on $(0,\infty)$.

Given the scales the posterior mean is linear, and the shrinkage weight

$$\kappa_i = \frac{1}{1+\lambda_i^2\tau^2} \in (0,1), \qquad \mathbb{E}[\beta_i \mid y_i, \lambda_i, \tau] = (1-\kappa_i)\, y_i, \tag{2.24}$$

is the fraction of the observation that is shrunk away. Integrating over λ_i, $\mathbb{E}[\beta_i \mid y_i, \tau] = \{1 - \mathbb{E}(\kappa_i \mid y_i, \tau)\} y_i$, so the whole behaviour of the prior is carried by the law of κ_i.

PROPOSITION 2.35 (The horseshoe weight). *With $\tau = \sigma = 1$ and $\lambda \sim \mathrm{C}^+(0,1)$, the weight $\kappa = 1/(1+\lambda^2)$ has the* $\mathrm{Be}(1/2, 1/2)$ *distribution, and its posterior given y has density proportional to $(1-\kappa)^{-1/2} e^{-\kappa y^2/2}$ on $(0,1)$. For general τ, $\mathbb{P}(\kappa < 1/2) = (2/\pi)\arctan\tau$.*

Proof. Since $\lambda = \{(1-\kappa)/\kappa\}^{1/2}$, $|\,\mathrm{d}\lambda/\,\mathrm{d}\kappa| = \frac{1}{2}\kappa^{-3/2}(1-\kappa)^{-1/2}$ and $1+\lambda^2 = 1/\kappa$, the density of κ is $\{2\kappa/\pi\} \cdot \frac{1}{2}\kappa^{-3/2}(1-\kappa)^{-1/2} = \kappa^{-1/2}(1-\kappa)^{-1/2}/\pi$, the $\mathrm{Be}(1/2,1/2)$ density. Given κ, $y \sim \mathcal{N}(0, 1+\lambda^2) = \mathcal{N}(0, 1/\kappa)$, so the likelihood of κ is $\kappa^{1/2} e^{-\kappa y^2/2}$; multiply by the prior. Finally $\kappa < 1/2$ if and only if $\lambda\tau > 1$, which has probability $1 - (2/\pi)\arctan(1/\tau) = (2/\pi)\arctan\tau$. □

The $\mathrm{Be}(1/2,1/2)$ density is unbounded at both ends, the horseshoe shape that names the prior: it expects each coordinate to be either almost entirely noise (κ near one) or almost entirely signal (κ near zero). Chapter 3 reads Fisher's additive model as the hypothesis $\kappa = 1$, and Chapter 5 identifies κ with the first stick of a GEM sequence. The marginal prior density of β has no closed form but satisfies, with $c_0 = (2\pi^3)^{-1/2}$,

$$\tfrac{c_0}{2}\log\Big(1+\frac{4}{\beta^2}\Big) < p(\beta) < c_0 \log\Big(1+\frac{2}{\beta^2}\Big), \tag{2.25}$$

so it has a logarithmic pole at zero and Cauchy tails (Carvalho et al., 2010). At $\beta = 0.1$, 1 and 3 the density is 0.603, 0.117 and 0.0237, inside the bounds $(0.381, 0.673)$, $(0.102, 0.140)$ and $(0.0233, 0.0255)$.

EXAMPLE 2.36 (How the horseshoe shrinks). With $\tau = \sigma = 1$, the posterior means of κ and of β for several observations are

y	0	0.5	1	2	3	5	10
$\mathbb{E}(\kappa \mid y)$	0.667	0.655	0.620	0.469	0.263	0.084	0.020
$\mathbb{E}(\beta \mid y)$	0	0.172	0.380	1.063	2.210	4.579	9.798

At $y = 0$, $\mathbb{E}(\kappa \mid y) = B(2, \frac{1}{2})/B(1, \frac{1}{2}) = 2/3$ exactly. Small observations are shrunk by about two thirds, large ones are left almost alone: the bias at $y = 10$ is 0.20, and it tends to zero as y grows. A normal prior with the same shrinkage at the origin

would shrink every observation by the same fraction. With a small global scale the prior becomes more sceptical: at $\tau = 0.1$ the prior probability that $\kappa < 1/2$ is only 0.063.

2.7.3 *Beta augmentation of Gamma ratios*

Concentration parameters such as the Ewens θ enter likelihoods through ratios $\Gamma(\theta)/\Gamma(\theta+n) = 1/(\theta)_n$, which block conjugate updating. The Beta integral removes the block:

$$\frac{\Gamma(\theta)}{\Gamma(\theta+n)} = \frac{1}{\Gamma(n)} \int_0^1 \eta^{\theta-1}(1-\eta)^{n-1}\, \mathrm{d}\eta, \tag{2.26}$$

which is $B(\theta, n) = \Gamma(\theta)\Gamma(n)/\Gamma(\theta+n)$ rearranged. For the Ewens likelihood $\theta^K/(\theta)_n$ with a $\mathrm{Ga}(a_0, b_0)$ prior, introduce η with joint density proportional to $\theta^{K+a_0-1}e^{-b_0\theta}\eta^{\theta-1}(1-\eta)^{n-1}$. Both conditionals are standard:

$$\eta \mid \theta \sim \mathrm{Be}(\theta, n), \qquad \theta \mid \eta \sim \mathrm{Ga}(K + a_0,\ b_0 - \log\eta), \tag{2.27}$$

a variant of the sampler of Escobar and West (1995), who augment with $\mathrm{Be}(\theta+1, n)$ and obtain a two component Gamma mixture for θ. Chapter 8 uses this variant throughout and Chapter 9 develops the general theory of priors for such ratios.

EXAMPLE 2.37 (A small Ewens posterior). At $\theta = 2.5$ and $n = 10$ both sides of (2.26) equal 9.714×10^{-9}. Take a sample of $n = 30$ genes with $K = 8$ alleles and the prior $\mathrm{Ga}(1, 0.2)$, an exponential with mean five. Fifty eight thousand iterations of (2.27) give posterior mean 3.61 and standard deviation 1.61; numerical integration of the exact posterior $\propto \theta^8 e^{-0.2\theta}/(\theta)_{30}$ gives 3.616 and 1.592. The same posterior is the target of the likelihood free methods of Section 2.8.

2.7.4 *Pólya Gamma and Pólya inverse gamma*

Two further identities are stated here for reference; both are proved in the papers cited.

A random variable ω has the Pólya Gamma distribution $\mathrm{PG}(b, c)$, $b > 0$, $c \in \mathbb{R}$, if

$$\omega \overset{d}{=} \frac{1}{2\pi^2} \sum_{k=1}^{\infty} \frac{g_k}{(k-1/2)^2 + c^2/(4\pi^2)}, \qquad g_k \sim \mathrm{Ga}(b, 1) \text{ independent.}$$

Polson et al. (2013) prove that for $a \in \mathbb{R}$ and $h = a - b/2$,

$$\frac{(e^\psi)^a}{(1+e^\psi)^b} = 2^{-b} e^{h\psi} \int_0^\infty e^{-\omega\psi^2/2}\, p(\omega \mid b, 0)\, \mathrm{d}\omega, \tag{2.28}$$

that $\mathbb{E}[e^{-t\omega}] = \cosh^{-b}(\sqrt{t/2})$ for $\omega \sim \mathrm{PG}(b, 0)$, that $\mathbb{E}\omega = \{b/(2c)\}\tanh(c/2)$ for $\omega \sim \mathrm{PG}(b, c)$, and that $\omega \mid \psi \sim \mathrm{PG}(b, \psi)$ under the joint density implied by (2.28). The logistic and negative binomial likelihoods are of the form on the left with ψ linear in the regression coefficients, so given ω the likelihood is Gaussian in ψ.

As checks, the product $\prod_k \{1 + t/(2\pi^2(k-1/2)^2)\}^{-b}$ truncated at two hundred thousand terms equals $\cosh^{-1}(\sqrt{0.35}) = 0.847340$ at $b = 1$, $t = 0.7$, and the series mean of $\mathrm{PG}(1, 1.5)$ is $0.21172 = \tanh(0.75)/3$.

The Pólya inverse gamma law of He et al. (2019) handles a parameter that appears in a reciprocal Gamma function. A positive random variable X_0 has the $\mathrm{PIG}(0)$ law if

$$\mathbb{E}\left[e^{-s^2 X_0}\right] = \frac{e^{-\gamma s}}{\Gamma(1+s)}, \qquad s > 0, \tag{2.29}$$

with γ the Euler Mascheroni constant, so that $1/\Gamma(1+s) = e^{\gamma s}\mathbb{E}[e^{-s^2 X_0}]$ is Gaussian in s given X_0. The law is an infinite convolution of scaled inverse gamma variables and has mean $\pi^2/12$. Chapter 8 uses (2.26) for Fisher's richness parameter and (2.29) for his shape parameter.

2.7.5 Bernstein and von Mises

In a regular parametric model with a prior that is continuous and positive at the true value θ_0, the posterior is asymptotically normal, centred at the maximum likelihood estimator, with variance the inverse of the Fisher information: the total variation distance between $\pi(\theta \mid y_{1:n})$ and $\mathcal{N}(\hat{\theta}_n, I_n(\hat{\theta}_n)^{-1})$ tends to zero in probability (van der Vaart, 1998). Bayesian and likelihood intervals then agree to first order, and the prior is forgotten. What matters in genetics is the rate at which I_n grows. For the Ewens parameter it grows like $\log n/\theta$, not n, so the effective sample size is the number of alleles K_n, and the approximation is accurate only when K_n is large; for the small and hyperdiverse samples of Chapter 8, and for the many parameter regressions of Part I where the number of parameters grows with n, the theorem does not apply and the prior matters.

2.8 Likelihood free inference

Coalescent models are easy to simulate and hard to evaluate: the likelihood of a sample is an integral over all genealogies. This section describes the two responses used in Part III, rejection approximate Bayesian computation and quantile regression on simulations, and the lemma that justifies the second.

2.8.1 Rejection ABC

Let $\pi(\theta)$ be a prior, $p(y \mid \theta)$ a model that can be simulated, S a summary statistic, ρ a distance and $\epsilon \geq 0$ a tolerance. Rejection approximate Bayesian computation (Tavaré et al., 1997; Pritchard et al., 1999; Beaumont et al., 2002) repeats: draw $\theta \sim \pi$; draw $y \sim p(\cdot \mid \theta)$; keep θ if $\rho(S(y), S(y_{\mathrm{obs}})) \leq \epsilon$.

PROPOSITION 2.38 (What rejection ABC computes). *The accepted values are independent draws from $\pi\{\theta \mid \rho(S(Y), S(y_{\mathrm{obs}})) \leq \epsilon\}$. If S is discrete and $\epsilon = 0$ this is $\pi(\theta \mid S(Y) = S(y_{\mathrm{obs}}))$, and if moreover S is sufficient it is the exact posterior $\pi(\theta \mid y_{\mathrm{obs}})$.*

Proof. The pair (θ, Y) is drawn from the joint law, and accepting on an event E of positive probability yields the conditional law of θ given E. With $\epsilon = 0$ the event is $\{S(Y) = s_{\text{obs}}\}$, and sufficiency means $\pi(\theta \mid y)$ depends on y only through $S(y)$. □

The two defects of the method are visible in the statement: the analyst must choose S, and unless S is sufficient the target is not the posterior; and the analyst must choose ϵ, and a positive ϵ conditions on a larger event, which inflates the spread.

EXAMPLE 2.39 (ABC for the Ewens parameter). Return to $n = 30$, $K = 8$ and the prior $\mathrm{Ga}(1, 0.2)$ of Example 2.37. Since K_n is sufficient, rejection with $S = K_n$ and $\epsilon = 0$ is exact. Of two hundred thousand simulations, using the Bernoulli representation (2.14), 7.0 percent are accepted; the accepted values have mean 3.61 and standard deviation 1.60, and 5, 50 and 95 percent quantiles 1.55, 3.34 and 6.65, against the exact 3.616, 1.592 and 1.555, 3.345, 6.599. Widening the tolerance to $|K - 8| \leq 2$ raises the acceptance rate to 34 percent but inflates the standard deviation to 1.89, an error that more simulation does not remove.

2.8.2 The check loss and quantile regression

For $u \in (0,1)$ the check loss is $\rho_u(z) = z\{u - \mathbf{1}(z < 0)\}$: it charges $u|z|$ for an underestimate and $(1-u)|z|$ for an overestimate. Quantile regression (Koenker and Bassett, 1978) minimises it.

LEMMA 2.40 (The check loss identifies quantiles). *Let X be a real random variable with $\mathbb{E}|X| < \infty$ and distribution function F. Then $c \mapsto \mathbb{E}\rho_u(X - c)$ is convex and minimised exactly at the u quantiles of X, that is at the c with $F(c-) \leq u \leq F(c)$. Consequently, for a pair (θ, Y) with $\mathbb{E}|\theta| < \infty$, the minimiser over measurable functions h of $\mathbb{E}\rho_u(\theta - h(Y))$ is the conditional quantile $h(y) = F^{-1}_{\theta|Y=y}(u)$ for almost every y.*

Proof. Convexity holds because ρ_u is convex. Writing $\rho_u(z) = uz^+ + (1-u)z^-$ and differentiating under the expectation, the right derivative of $c \mapsto \mathbb{E}\rho_u(X - c)$ is $-u\mathbb{P}(X > c) + (1-u)\mathbb{P}(X \leq c) = F(c) - u$, and the left derivative is $F(c-) - u$. A convex function is minimised where the left derivative is at most zero and the right derivative at least zero. For the conditional statement, apply the first part to the conditional law of θ given $Y = y$ and integrate over y. □

Fitting h for one u gives one conditional quantile; letting u be an input, as Chapter 10 does, gives the whole conditional quantile function, and so the whole posterior, from simulated pairs (θ_i, y_i) alone. Least squares returns one number per y; the check loss, integrated over u, returns a law.

EXAMPLE 2.41 (Posterior quantiles by regression). On one thousand and one standard normal draws, minimising $\sum \rho_u(z_i - c)$ over c returns the empirical u quantile exactly for $u = 0.1$, 0.5 and 0.9. For the Ewens problem, simulate six thousand pairs (θ_i, K_i) from the prior and the model with $n = 30$, and fit for each u

the linear quantile regression of $\log\theta$ on $(1, \log K, \log^2 K)$ by linear programming; since log is increasing, quantiles of $\log\theta$ exponentiate to quantiles of θ. At $K = 3, 8$ and 15 the fitted 5, 50 and 95 percent quantiles are

	$K = 3$	$K = 8$	$K = 15$
$u = 0.05$, fitted (exact)	0.208 (0.239)	1.564 (1.555)	5.174 (5.266)
$u = 0.50$, fitted (exact)	0.815 (0.839)	3.402 (3.345)	9.621 (9.573)
$u = 0.95$, fitted (exact)	2.158 (2.217)	6.820 (6.599)	16.954 (16.758)

One regression per quantile level, trained once, answers every data set, with no tolerance and no rejection; the error, at most about three percent except in the lower tail, is an estimation error that shrinks with the simulation budget and the flexibility of the basis.

2.8.3 Noise outsourcing

The last result says that the posterior is not merely approximable by a function of the data and an independent uniform variable, but is one.

THEOREM 2.42 (Noise outsourcing). *Let (θ, Y) be a random element of $\Theta \times \mathcal{Y}$, with Θ standard Borel. There are a measurable map $H : (0,1) \times \mathcal{Y} \to \Theta$ and a random variable $U \sim \mathrm{Unif}(0,1)$ independent of Y such that $\theta = H(U, Y)$ almost surely. When $\Theta \subseteq \mathbb{R}$ one may take $H(u, y) = F^{-1}_{\theta|Y=y}(u)$ (Kallenberg, 2002, Theorem 6.10).*

Sketch. Take $\Theta = \mathbb{R}$; a standard Borel space is Borel isomorphic to a Borel subset of $\mathbb{R}$, which reduces the general case to this one. Let F_y be a regular version of the conditional distribution function of θ given $Y = y$, and let $V \sim \mathrm{Unif}(0,1)$ be independent of (θ, Y). Define $U = F_Y(\theta-) + V\{F_Y(\theta) - F_Y(\theta-)\}$, which randomises within any atom. Given $Y = y$, U is uniform on $(0,1)$ (the probability integral transform, randomised at atoms), so U is uniform and independent of Y. And $F_y^{-1}(U) = \theta$ almost surely, because U lies in $[F_y(\theta-), F_y(\theta)]$ and the quantile function maps that interval back to θ except on a null set. □

Read together with Lemma 2.40, the theorem says that the posterior is the map $u \mapsto H(u, y_{\mathrm{obs}})$, that this map minimises an expected check loss over simulated pairs, and that it can therefore be learned by regression. Chapter 10 builds the method on this observation and shows that a learned map with a low dimensional bottleneck that attains the Bayes risk certifies a sufficient statistic, and Chapter 11 applies it to the coalescent.

2.9 Exercises

1. Show that for a biallelic locus with genotype frequencies (P_{11}, P_{12}, P_{22}) one generation of random mating gives Hardy Weinberg proportions, and that the inbreeding coefficient F of (2.1) satisfies $F = (P_{11}P_{22} - P_{12}^2/4)/(pq)$. *Hint.* Expand $P_{11}P_{22}$ using (2.1).

2. With three alleles at frequencies $(0.5, 0.3, 0.2)$, compute the Hardy Weinberg genotype frequencies and the expected heterozygosity $1 - \sum_i p_i^2$. What is the dimension of the Hardy Weinberg manifold inside the simplex of genotype frequencies? *Hint.* The heterozygosity is 0.62; count free parameters on each side.

3. Prove that $|D| \le \frac{1}{4}$ for any two biallelic loci, and find the haplotype frequencies that attain it. *Hint.* $D = x_{AB}x_{ab} - x_{Ab}x_{aB} \le x_{AB}x_{ab}$ and $x_{AB} + x_{ab} \le 1$.

4. In Example 2.9, let p vary in $(0,1)$ with the genotypic values fixed. Plot $\sigma_A^2(p)$ and $\sigma_D^2(p)$, and show that σ_A^2 is largest at $p = (7 - \sqrt{17})/8 \approx 0.360$, where it equals 4.957, while σ_D^2 is largest at $p = 1/2$. *Hint.* $\sigma_A^2 = 8p(1-p)(2-p)^2$, whose derivative is $8(2-p)(4p^2 - 7p + 2)$.

5. For the neutral Wright Fisher model with N genes, show that $\mathbb{E}[x_t(1-x_t)] = (1 - 1/N)^t x_0(1-x_0)$ and deduce that the probability that the population is still polymorphic at time t is at least $4(1-1/N)^t x_0(1-x_0)$. For $N = 20$ and $x_0 = 1/2$, how long is the population guaranteed to stay polymorphic with probability at least one half? *Hint.* $x(1-x) \le \frac{1}{4}$ on the event of polymorphism and is zero otherwise.

6. Solve $\mathcal{L}T = -1$ with $T(0) = T(1) = 0$ for the neutral diffusion and show that the mean absorption time of a new mutant, $x = 1/N$, is about $2(1 + \log N)$ generations, while conditional on fixation it is about $2N$ generations. *Hint.* For the conditional time solve $\mathcal{L}(uT^*) = -u$ with $u(x) = x$.

7. Using Proposition 2.22(i), show that if $P \sim \mathrm{Dir}(a_1, \dots, a_k)$ then $P_1/(P_1 + P_2) \sim \mathrm{Be}(a_1, a_2)$ independently of $P_1 + P_2$. Check numerically for $a = (1, 2, 3)$. *Hint.* The ratio involves only G_1 and G_2.

8. Prove (2.11) and deduce that for $\alpha = 0$ the number of sticks needed to leave expected mass below ϵ is about $\log(1/\epsilon)/\log(1 + 1/\theta)$. Compare with the value 26 found in Example 2.25. *Hint.* Independence of the W_j gives $\mathbb{E}\prod(1 - W_j) = \prod \mathbb{E}(1 - W_j)$.

9. Enumerate the seven configurations of $n = 5$ genes, compute their Ewens probabilities at $\theta = 1$, and check that they are the cycle type proportions of the symmetric group on five letters. *Hint.* The numbers of permutations are $1, 10, 15, 20, 20, 30, 24$.

10. Show that $\mathbb{E}K_n$ in (2.18) tends to the Ewens value $\sum_{m<n} \theta/(\theta + m)$ as $\alpha \to 0$. *Hint.* Write $(\theta + \alpha)_n/(\theta)_n = \prod_m (1 + \alpha/(\theta + m))$ and expand to first order in α.

11. Show that $\mathbb{E}T_2/\mathbb{E}H_n \to 1/2$ as $n \to \infty$, and that $\mathrm{Var}\, H_n \to \frac{4}{3}\pi^2 - 12 \approx 1.16$. *Hint.* $\mathrm{Var}\, H_n = \sum_k 4/\{k(k-1)\}^2$; use partial fractions.

12. Write a rejection ABC sampler for θ from Watterson's S with $n = 11$, $S = 43$ and a flat prior on $(0, 60)$, and compare the accepted values with the exact posterior

obtained from the geometric representation of Chapter 11. *Hint.* Simulate the T_k of Definition 2.30, form L_n, and then draw $S \sim \text{Po}(\theta L_n/2)$.

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PART I

Selection and the Additive Model

CHAPTER 3

The Fundamental Theorem of Natural Selection: From Deterministic to Stochastic, Fisher to Bayes

Fisher's Fundamental Theorem of Natural Selection asserts that the additive genetic variance is the total rate of change of mean fitness. The assertion is false, as Kimura showed in 1958 and as Ewens (2024) sets out definitively. This chapter asks what the falsehood costs, why sixty six years of correction failed to displace it, and what should replace it.

The deterministic half rests on a lemma. The average effects are the coefficients of an orthogonal projection of fitness onto a fixed *subspace taken in a* moving *inner product, so Fisher computed the rate of change of $\Pi_t m$ as though Π_t were constant. With δ the residuals and σ_A^2 denoting Fisher's term $2\sum_u \dot{p}_u \beta_u$, $d\bar{m}/dt = \sigma_A^2 + \sum_g \dot{P}_g \delta_g$ holds for arbitrary paths and for the whole genome, and the remainder vanishes for every assignment of fitnesses exactly when the association coordinates are stationary, as they are when the distribution moves within the manifold of gene frequency product measures: Fisher's claim is that a population never departs from Hardy Weinberg proportions and linkage equilibrium. Whenever the distribution moves inside an exponential family whose main effects are the gene contents, the remainder factorizes over the transverse coordinates alone; at a biallelic locus that sum has one term and equals $D\,\mathcal{I}(\alpha)\,\dot{\alpha}$ in Lindley's (1988) coordinates, the fitness curvature times the Fisher information about the disequilibrium coefficient times its velocity, whence Fisher's 1941 constant genic environment condition is necessary as well as sufficient there. A two locus system has $\sigma_A^2 \equiv 0$ while mean fitness falls twenty percent, and at quasi linkage equilibrium the relative error is $\epsilon\sqrt{p_A q_A p_B q_B}$, independent of the recombination rate: recombination destroys the disequilibrium and leaves the error untouched.*

The stochastic half asks what data can say. Fisher's claim is the hypothesis $\kappa_\ell = 1$ on a shrinkage weight. Jeffreys's Bayes factor shows it can be refuted but never confirmed, and detection at one locus needs $n \approx 50\, s^{-4}$ zygotes per genotype. Global local shrinkage estimates the distribution of κ and finds Fisher right at almost every locus, wrong in aggregate by the entire quantity. The horseshoe places Jeffreys's binomial prior on κ, with Haldane's as the improper limit; Haldane's 1932 spike and slab on Primula *linkage anticipated the machinery inside genetics; and the Ewens Sampling Formula is itself a marginal likelihood, Kingman's paintbox playing the role of de Finetti's theorem. We close with the consequences for practice and the one measurement that would settle the question.*

3.1 Introduction

3.1.1 A history of the theorem, and of priors in genetics

The Fundamental Theorem is usually told as a story about one man's cryptic sentence. It is better told as a story about two traditions that grew up in the same rooms, in the same journals, at the same time, and never spoke. This section sets out that history, because the rereading offered in this chapter is not an importation of modern statistics into an old problem: almost every ingredient was present by 1960, and several of them were supplied by geneticists.

Two traditions in one room, 1918 to 1936

Fisher (1918) introduced the decomposition of variance into additive and residual parts, and with it the average effect of a gene. Fisher (1930a) stated the Fundamental Theorem. In the same years he was campaigning against inverse probability: Fisher (1930b) in the *Proceedings of the Cambridge Philosophical Society*, and again in Fisher (1932), arguing that likelihood should replace it.

Haldane (1932) replied in the same journal, and constructively. He objected that Bayes's theorem as then used assumes all values near the observed one are equally probable a priori, and asked what better assumption could be made. Two answers came out of that note. The first is the prior that bears his name, $\pi(\theta) \propto \{\theta(1-\theta)\}^{-1}$, motivated by the fact that mutation rates and recombination fractions range over orders of magnitude, so that a flat prior asserts $\mathbb{P}(\theta < 10^{-6}) = 10^{-6}$, which nobody believes about a mutation rate. The second, and for our purposes the more remarkable, is a mixture of a point mass and a continuous density fitted to linkage data in *Primula sinensis*: a spike and slab, more than half a century before the term existed. Its prior mass of $11/12$ on free recombination was a chromosome count, a physical fact about the plant, checkable with a microscope. Whatever one thinks of subjective priors, that is not one.

Wrinch and Jeffreys had already argued, in a series of three papers beginning in 1919 (see Etz and Wagenmakers, 2017, and Wrinch and Jeffreys, 1921), that a general law deserves prior mass, and Jeffreys (1935, 1936) turned the argument into significance tests, introducing the notation K and the convention of equal prior odds. Etz and Wagenmakers (2017) argue that Jeffreys knew Haldane's note and may have been prompted by it.

So by 1936 the following existed: a variance decomposition, a theorem built on it, a spike and slab prior on a recombination fraction, a general theory of sharp null testing, and a public dispute between the author of the first two and the authors of the last two. The dispute was about foundations. Nobody applied the last two to the first two.

The error, found and not taken, 1958 to 1972

Kimura (1958) computed the neglected term. By Crow's (1987) account Fisher knew of the paper, possibly reviewed it, and regarded his own statement as complete and not in need of clarification.

In the following year the literature acquired what we call the decoy. Mulholland and Smith (1959), Scheuer and Mandel (1959) and Kingman (1961) proved that at one locus with constant fitnesses mean fitness cannot decrease. That is a true theorem, easy to verify, and about a different proposition from Fisher's. It has been mistaken for the Fundamental Theorem ever since, and because it is true in exactly the case that everyone checks, it has functioned as a standing confirmation of a claim it does not support.

Price (1972) gave the partial change reading. Ewens (1989) made it precise and Lessard (1997) proved the various expressions for the remainder equal. Ewens (2024) now regards the partial change concept as unhelpful, on the ground that no biological interpretation was ever found for it; Remark 3.6 argues that this is what one should expect, since the quantity is the tangential component of a motion and therefore a property of an approximation rather than of a population.

Bayesian objects built by geneticists who did not name them, 1972 to 1988

The same decades in which the theorem was being corrected saw population genetics construct, in its own vocabulary, a substantial part of Bayesian nonparametrics.

Ewens (1972) derived the sampling formula for the infinitely many alleles model. In modern terms it is the marginal likelihood obtained by integrating a Poisson Dirichlet prior against a multinomial sampling model and observing only the allelic partition; the estimator of θ from the number of distinct alleles is type II maximum likelihood, which is to say empirical Bayes. Kingman (1975, 1978) supplied the representation theorem: every exchangeable random partition is a mixture over paintboxes, so exchangeability does not permit a prior but forces one, exactly as de Finetti's theorem does for sequences. Antoniak (1974) and Ferguson (1973) approached the same object from the other side. Hoppe (1984) exhibited the sampling formula as a Pólya urn, which is to say a posterior predictive rule, and Donnelly (1986) developed the urn alongside partition structures and the ages of alleles. Section 3.14 develops this reading.

Lindley (1988) contributed the piece that this chapter needs most. Working on Hardy Weinberg equilibrium, he took as his disequilibrium parameter the logarithm of the cross product ratio of genotype frequencies, on the ground that a condition on the simplex should be stated invariantly. Section 3.4 shows that this parameter is the unique transverse coordinate to the manifold on which Fisher's theorem is exact, that it is the two way interaction term of a log linear model, and that the neglected term of the Fundamental Theorem factorizes exactly through its

time derivative. Lindley was not writing about Fisher's theorem, and so far as we can tell nobody noticed that he had supplied its missing coordinate.

He also observed, in passing, that the Bayes factor can favour Hardy Weinberg equilibrium on data that classical tests reject. That asymmetry between refutation and confirmation of a sharp null is the phenomenon that Section 3.12 finds governing the Fundamental Theorem as well.

Why the error resisted detection

The historical puzzle is not detection but propagation: the error was found in 1958 and again four times since, and the claim is still repeated. Section 3.7 gives six mechanisms, and they are worth previewing here because they motivate everything in Part II.

The remainder is one order higher in the selection coefficient than the term Fisher kept, so the relative error is $O(s)$ and amounts, for the fitness shape of Example 3.5.2, to one part in 360 at a one percent selection coefficient. To see it at a single locus one would need $n^* \approx 50\, s^{-4}$ zygotes per genotype, five billion at that coefficient, because the quantity to be measured is the square of a small contrast and squaring a noisy small number is a very noisy business. In the setting where the theorem is usually taught, one locus in continuous time with Hardy Weinberg proportions imposed, the remainder is identically zero, so the check confirms it. The proposition most people actually test is the decoy of 1959, which is true. The neglected term has never been computable at whole genome scale, so nobody has inspected it and judged it small. Across loci the dominance contribution is small and partly cancels, and at experimental sample sizes the estimator's own plug in bias exceeds Fisher's error, so his is not even the largest error present.

The conclusion to draw from this is not that the error is unimportant. It is that the per locus question is unanswerable, and that is a statement about which estimand one should be pursuing.

From an unanswerable question to a hierarchical one

If δ_ℓ cannot be estimated at any locus, the response is to change the estimand from δ_ℓ to a functional of the distribution G from which the δ_ℓ are drawn. Information about G accumulates in the number of loci rather than in the depth at any one, and in our simulations the relative error of the genome wide nonadditive total falls from 19% at 500 loci to 3% at twenty thousand, while no individual locus is ever resolved.

That move requires no Bayesian commitment. It rests on three independent legs, of which only one is Bayesian: de Finetti, since exchangeable loci are conditionally independent draws from a common G, so the hierarchical model is what exchangeability means rather than an assumption added to it; Stein, since for three or more parallel problems the unshrunk estimator is inadmissible, so refusing to pool is dominated rather than cautious; and Robbins, since the compound

decision rule built from the empirical distribution attains the risk of the best rule one could have chosen knowing G.

The genetics literature made this move without waiting for the argument. Smith and Stephens (1995, 1996) gave Bayesian inference for multipoint recombination fractions, which set the decay rate of the very transverse coordinate whose motion is the remainder. Pritchard, Stephens and Donnelly (2000) built STRUCTURE on Dirichlet priors and it became one of the most used methods in the field. Li and Stephens (2003) introduced a copying model that is structurally a prior over haplotypes and now underlies essentially every imputation pipeline, so that any modern genomic data set has had a prior applied to it before anyone sees the genotypes. Carbonetto and Stephens (2012), Zhou, Carbonetto and Stephens (2013) and Stephens (2017) supplied spike and slab variable selection at association scale, the sparse plus polygenic model, and nonparametric estimation of the effect distribution respectively. The last of these is the operational form of what this chapter proposes.

The rereading

Placed in that history, Fisher's assertion reads differently. He faced a problem with $2^L - L - 1$ interaction parameters and no prospect of estimating them, and closed it by fiat: every nonadditive deviation is exactly zero. In the notation of Section 3.10 that is $\kappa_\ell = 1$ at every locus, a point mass on a shrinkage weight, asserted before any data. Read as a prior it is defensible and, as Section 3.13 shows, close to right: the posterior mean of κ across loci runs from 0.86 to 0.98. Read as a theorem it is false, and the falsehood is the subject of Part I.

The Bayesian rereading thus does three things at once. It names the assumption Fisher made implicitly; it locates the assumption in a family whose other members are Bayes Laplace and Haldane, with the horseshoe as the Jeffreys member (Section 3.11.2); and it replaces the point mass with an estimated sparsity level, which is a quantity a genome can be asked about. The apparatus for asking is Lindley's, Jeffreys's and the hierarchical machinery of the last three decades, and the chronology below records that almost all of it was built by people one step from the problem.

Year	Event	Reading offered here
1918	Fisher: variance decomposition	the projection Π_t
1930	Fisher: the Fundamental Theorem	the tangential component only
1930, 1932	Fisher against inverse probability	
1932	Haldane: prior, and spike and slab on ρ	the first mixture prior, on the decay rate of the transverse coordinate
1935, 1936	Jeffreys: significance tests, K	the test of $\kappa_\ell = 1$
1941	Fisher: constant genic environment	$\dot{\alpha} = 0$; necessary and sufficient
1947	Haldane and Smith: linkage	Bayesian inference in human genetics
1958	Kimura: the neglected term	$\mathcal{R} \neq 0$
1959	Mulholland Smith; Scheuer Mandel	the decoy theorem
1961	Kingman: the same inequality	
1972	Ewens: the sampling formula	a marginal likelihood
1972	Price: the partial change	the tangential component named
1975, 1978	Kingman: paintbox	de Finetti for partitions
1984	Hoppe: the Pólya urn	the posterior predictive rule
1986	Donnelly: Pólya urns	
1988	Lindley: Hardy Weinberg disequilibrium	the transverse coordinate α
1989	Ewens: partial rate of change	
1995	Smith and Stephens: recombination fractions	inference for the decay rate of the transverse coordinate
1997	Lessard: the forms agree	
2000	Pritchard, Stephens, Donnelly: STRUCTURE	Dirichlet priors go mainstream
2003	Li and Stephens: the copying model	a prior inside every genome
2010	Carvalho, Polson, Scott: horseshoe	the Jeffreys member of the κ family
2012, 2013	Carbonetto, Zhou, Stephens	scalable sparse and sparse plus dense models
2017	Stephens: adaptive shrinkage	nonparametric estimation of G
2019, 2024	Lessard and Ewens; Ewens	the verdict

3.1.2 The claim, and the error in one line

Fisher (1930a, 1958) states the Fundamental Theorem of Natural Selection as: *the rate of increase in fitness of any organism at any time is equal to its genetic variance in fitness at that time*. In modern notation, with $\bar{m}$ the mean Malthusian parameter and σ_A^2 the additive genetic variance,

$$\frac{d\bar{m}}{dt} \stackrel{?}{=} \sigma_A^2. \tag{3.1}$$

The error is a product rule. Mean fitness satisfies $\bar{m} = 2\sum_u p_u \beta_u$, where p_u are allele frequencies and β_u are average effects. Differentiating along a path on which

the gene frequencies change by selection,

$$\frac{d\bar{m}}{dt} = \underbrace{2\sum_u \dot{p}_u \beta_u}_{\sigma_A^2} + \underbrace{2\sum_u p_u \dot{\beta}_u}_{\mathcal{R}}, \tag{3.2}$$

and Fisher kept the first term. That is the whole of it. No probability is needed to see it or to state it. Kimura (1958) gave an equivalent expression for $\mathcal{R}$; Price (1972) and Ewens (1989) clarified the interpretation; Lessard (1997) proved the various forms of $\mathcal{R}$ equal; Ewens (2024) concludes that from start to finish the theorem was not correct and should be banned from the literature.

This chapter takes that verdict as settled and asks four questions it leaves open. How much does the error matter in a given population? Why did sixty six years of correction fail to displace the claim? How much of Fisher's instinct survives? And what should replace it?

3.1.3 The intuition: a quantity effect and a price effect

Everything below unpacks one picture, so it is worth stating before any notation.

Read $\bar{m} = 2\sum_u p_u \beta_u$ as a weighted average with two moving parts. The frequencies p_u say how much of each allele the population holds. The average effects β_u say what each allele is worth. This is the structure of an index number, in which a total moves both because quantities change and because prices change, and the change decomposes into a *quantity effect* and a *price effect*. Irving Fisher (1922) was working out that arithmetic in the same decade, and the moral transfers: one may not compute the change in a total by moving the quantities while holding the prices, absent a reason to believe the prices did not move.

In these terms σ_A^2 *is* the quantity effect, exactly and always. That is a real achievement and nothing here disputes it: the additive genetic variance is a complete and exact accounting of the contribution of gene frequency change. What Fisher dropped is the price effect, and the price effect is not zero, because the worth of an allele is not a property of the allele. It is the average fitness that allele confers *in the genetic background it currently finds itself in*, averaged over the partners it meets. Selection changes that background. So the prices move whenever the quantities do, and for the same reason.

This also explains why the mistake is natural. Calling β_u the *average effect of the allele* makes it sound like a mass or a charge, an attribute the allele carries about with it. Read that way, $\bar{m} = 2\sum p_u \beta_u$ is a weighted sum of constants with moving weights, and differentiating only the weights is not an oversight but the obvious thing to do. Ewens (2024) puts the point exactly: that the β values are not fixed constants but functions of the frequencies is an important fact not mentioned in 1930, 1941 or 1958.

3.1.4 What the neglected term contains

The price effect has a concrete biological identity worth naming early.

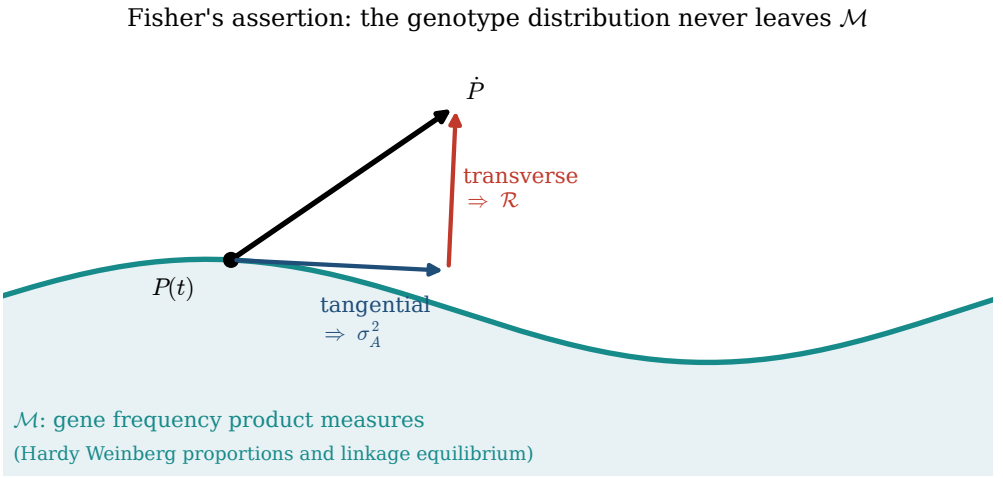


FIGURE 3.1. The geometry of the Fundamental Theorem. The genotype distribution $P(t)$ moves in the simplex. Its velocity decomposes into a component tangential to the manifold $\mathcal{M}$ of gene frequency product measures, which the additive genetic variance registers exactly, and a transverse component, which it cannot see. Fisher's assertion is that the transverse component is absent.

Selection acts on genotypes; inheritance transmits genes. Each generation selection builds up *associations* among genes, departures from Hardy Weinberg proportions within a locus and linkage disequilibrium between loci, while segregation and recombination tear them down. Gene frequencies are one bookkeeping of the state; the associations are the rest of it. Since β is by construction the best description of fitness expressible as a sum over single genes, the residual δ is exactly the part of fitness living in the associations, which is what dominance and epistasis *are*. When the associations move, mean fitness moves in a way no gene frequency records.

Section 3.3 makes this exact. Writing $\mathcal{R} = \sum_g \dot{P}_g \delta_g$, the remainder vanishes precisely when the association coordinates are stationary, in particular when the genotype distribution moves within the manifold of gene frequency product measures, and is otherwise the transverse motion weighted by δ (Figure 3.1). Section 3.5.3 gives the pure case: two loci, fitness high exactly when they disagree, no gene frequency ever moving, σ_A^2 identically zero, and mean fitness falling twenty percent as the disequilibrium decays.

3.1.5 Two questions, and the division of the chapter

Two quite different questions arise, and confusing them has caused much of the trouble.

The first is deterministic. The Fundamental Theorem belongs to the deterministic half of population genetics: infinitely large populations, no stochastic

element, no data. In that world one asks whether $\mathcal{R} = 0$. The answer is no, and no data are consulted to reach it. Part I is written in that idiom and uses statistical vocabulary only to say how its quantities could be measured.

The second is stochastic, and belongs beside the Ewens Sampling Formula rather than beside the Fundamental Theorem. Given genomic data, what can be said about the size of the deviations δ across a genome? This is not an interpretation of Fisher. It is the question a person with a data set has, and it is the one on which the practical status of σ_A^2 turns. Part II addresses it.

The two come apart cleanly. Part I would be unchanged if every genome were measured without error. Part II would be unchanged if Fisher had never written anything.

3.1.6 Contributions

1. *A moving projection lemma* (Lemma 3.2): for arbitrary genotype frequency paths, $d\bar{m}/dt = \sigma_A^2 + \sum_{u,v} \dot{P}_{uv}\delta_{uv}$, with σ_A^2 denoting Fisher's term $2\sum_u \dot{p}_u\beta_u$ and with no assumption of Hardy Weinberg proportions. The remainder is the pairing of the least squares residuals with the logarithmic velocity of the genotype frequencies.

2. *A whole genome version* (Theorem 3.5), with a necessary and sufficient criterion: $\mathcal{R} = 0$ for every assignment of fitnesses if and only if $\partial_t \log P$ lies in the span of the gene contents, that is, if and only if the association coordinates are stationary, as they are when the distribution moves within the manifold of product measures.

3. *An identification of the transverse coordinate* with the disequilibrium coefficient of Lindley (1988), which Section 3.4 shows to be the two way interaction term of the log linear model on the genotype table, with the multilocus transverse coordinates being the higher order interactions.

4. *An exact factorization of the remainder over the transverse coordinates* (Theorem 3.8): whenever the genotype distribution moves inside an exponential family whose main effects are the gene contents, $\mathcal{R} = \sum_A \langle \delta, u_A \rangle \dot{\theta}_A$, the main effect directions contributing exactly zero. The number of terms is the codimension of $\mathcal{M}$, so the counting of Section 3.17 is a counting of terms. At a biallelic locus the sum has one term and evaluates to $\mathcal{R} = D\,\mathcal{I}(\alpha)\,\dot{\alpha}$, the fitness curvature times the Fisher information about Lindley's coefficient times its velocity, with the average effects cancelling identically (Corollary 3.10). The same calculation gives the nonadditive variance at arbitrary genotype frequencies in closed form, $\sigma_D^2 = D^2\mathcal{I}(\alpha)$, which reduces to the classical $(2pqd)^2$ under Hardy Weinberg proportions (Proposition 3.11). A consequence is that Fisher's 1941 constant genic environment condition is necessary as well as sufficient for $\mathcal{R} = 0$ at a biallelic locus, sharpening the standard statement and supplying the statistical meaning that Ewens (2024) reports as missing.

5. *A closed form for the remainder* in discrete time at one locus with arbitrarily many alleles (Theorem 3.16), together with the exact two allele relative error $Dpq/(2\bar{w})$, which shows the error is $O(s)$ rather than the $O(s^2)$ one might guess (Corollary 3.17).
6. *A dynamical calculation* (Section 3.6): at quasi linkage equilibrium the relative error is $\epsilon\sqrt{p_A q_A p_B q_B}$, independent of the recombination rate and of the selection intensity. Recombination suppresses the disequilibrium by a factor of ten across the range examined and suppresses the error not at all.
7. *A quantitative account of nondetection*: $n^* \approx 50\, s^{-4}$ zygotes per genotype are needed to see the error at one locus, five billion at a one percent selection coefficient.
8. *A reformulation of Fisher's claim as* $\kappa_\ell = 1$, a point mass on a shrinkage weight, and its examination by Jeffreys's Bayes factor and by global local shrinkage.
9. *Two identifications in the Bayesian apparatus*: the horseshoe places Jeffreys's binomial prior on the shrinkage weight, with Haldane's as the improper limit of the same Beta family; and the Ewens Sampling Formula is the marginal likelihood of a Dirichlet process prior, with Kingman's paintbox theorem in the role of de Finetti's.
10. *A prescription* for genomic practice, and an identification of the one measurement that would settle the empirical question.

Plan. Section 3.1.1 gives the history and the rereading in outline; the rest of the introduction states the error and the intuition. Part I is then deterministic: the projection setup (Section 3.2), the lemma and its whole genome form (Section 3.3), the Lindley coordinates and the exact factorization of the remainder (Section 3.4), the discrete closed form and two examples (Section 3.5), the quasi linkage equilibrium dynamics (Section 3.6), the account of nondetection (Section 3.7) and a note on terminology (Section 3.8). Part II is stochastic: the bridge (Section 3.9), the κ formulation (Section 3.10), Haldane (Section 3.11), the Jeffreys test (Section 3.12), the horseshoe (Section 3.13) and the sampling formula (Section 3.14). Part III draws the practical consequences and gives a prescription. Part IV places the argument in the existing literature and states a research programme. Closing sections collect proofs, numerical verification, notation and exercises.

Because the argument of Part I is algebraic, it can be followed on small tables with a pencil, and the chapter does this repeatedly. One locus table, a 2×2 genotype array with excess homozygotes, is carried through Examples 3.1, 3.4 and 3.12; a two locus haplotype table off linkage equilibrium is worked in Example 3.15; the MN blood group data supply a real Hardy Weinberg calculation in Example 3.13; and Part II works a Bayes factor and a horseshoe fit through by hand in Examples 3.18 and 3.19. Every number in them was checked by computer.

PART I THE DETERMINISTIC THEORY

3.2 A fixed subspace and a moving metric

Everything in Part I concerns a deterministic system. The theory has no samples, no observations and no random variables; where Part I mentions them, it is only to say how the quantities of the theory could be measured.

Fix a locus with alleles $A_1, \dots, A_k$. Let $P_{uv}(t) > 0$ be the ordered genotype frequencies at time t, with $\sum_{u,v} P_{uv} = 1$ and $p_u = \sum_v P_{uv}$. *No Hardy Weinberg assumption is made.* Let m_{uv} be the Malthusian parameters, fixed real numbers, and $\bar{m}(t) = \sum_{u,v} P_{uv}(t) m_{uv}$.

Regard m as a fixed vector in $\mathbb{R}^{k\times k}$. Let

$$V \;=\; \{\, f_{uv} = c_u + c_v \;:\; c \in \mathbb{R}^k \,\}$$

be the subspace of additively separable arrays. V is a *fixed* linear subspace: it does not depend on t, on P, or on m. Equip $\mathbb{R}^{k\times k}$ with the time varying inner product

$$\langle f, g\rangle_t \;=\; \sum_{u,v} P_{uv}(t)\, f_{uv} g_{uv},$$

and let Π_t denote orthogonal projection onto V in this inner product.

Ewens's β definition of the average effects is exactly the statement that $\Pi_t m$ has coefficients β: minimizing $\sum_{u,v} P_{uv}(m_{uv} - \beta^*_u - \beta^*_v)^2$ is minimizing $\|m - f\|_t$ over $f \in V$. Write

$$A(t) \;=\; \Pi_t m, \qquad A_{uv} = \beta_u + \beta_v, \qquad \delta \;=\; m - A(t), \qquad \delta_{uv} = m_{uv} - \beta_u - \beta_v.$$

The stationarity conditions are

$$\sum_v P_{uv}\, \delta_{uv} = 0 \quad \text{for every } u, \qquad \text{hence} \qquad \sum_{u,v} P_{uv}\, \delta_{uv} = 0. \tag{3.3}$$

The subspace V is fixed and the vector m is fixed. What moves is the inner product, and therefore the projection. This is the entire source of the difficulty, and it is a fact of linear algebra rather than of statistics.

Three things follow, and none is stated in most textbook treatments. First, σ^2_A is an explained sum of squares, Fisher's own 1918 decomposition applied to fitness, and it inherits every property of such a decomposition, including dependence on the distribution of the predictors. Second, since the weights are the current genotype frequencies, $\sigma^2_A = \sigma^2_A(P)$ is a functional of the state and not a fixed genetic parameter: two populations with identical genotypic fitnesses have different additive genetic variances and different average effects, which is Ewens's point that the β move with the p; see also Lee and Chow (2013). Third, and this is the crux, the residuals are orthogonal to gene content *under $P(t)$ and under no other distribution*. Condition (3.3) is a property of the fit, not of the genotypes, and treating σ^2_A as the total change silently invokes it at a later time, where it is false.

The normal equations at a biallelic locus. It helps to see the least squares problem written out once in full. With two alleles, put $x = P_{12} = P_{21}$ for each ordered heterozygote, so that the unordered heterozygote frequency is $2x$. The two conditions (3.3) read

$$\begin{aligned} (2P_{11} + x)\,\beta_1 + x\,\beta_2 &= P_{11}m_{11} + x\,m_{12}, \\ x\,\beta_1 + (2P_{22} + x)\,\beta_2 &= P_{22}m_{22} + x\,m_{12}. \end{aligned} \tag{3.4}$$

The determinant is $(2P_{11} + x)(2P_{22} + x) - x^2 = 4P_{11}P_{22} + 2x(P_{11} + P_{22}) > 0$, so the average effects exist and are unique whenever at least two genotype classes are present. Nothing in (3.4) refers to Hardy Weinberg proportions; when they hold, the system reduces to the familiar regression of genotypic value on gene count. The three variance components are

$$\sigma_G^2 = \sum_{u,v} P_{uv}(m_{uv} - \bar{m})^2, \qquad \sigma_A^2 = \sum_{u,v} P_{uv}(A_{uv} - \bar{m})^2, \qquad \sigma_D^2 = \sum_{u,v} P_{uv}\delta_{uv}^2,$$

and $\sigma_G^2 = \sigma_A^2 + \sigma_D^2$ by Pythagoras in $\langle \cdot, \cdot \rangle_t$, the cross term $\langle A - \bar{m}, \delta \rangle_t$ vanishing because $A - \bar{m} \in V$ and $\delta \perp V$.

EXAMPLE 3.1 (Average effects by weighted least squares). Take a population with an excess of homozygotes, as inbreeding or population subdivision would produce, and an overdominant fitness:

genotype	A_1A_1	A_1A_2	A_2A_2
frequency (unordered)	0.30	0.30	0.40
Malthusian parameter m	0.10	0.30	-0.20

So $P_{11} = 0.30$, $x = 0.15$, $P_{22} = 0.40$, the allele frequency is $p_1 = 0.30 + 0.15 = 0.45$, and Hardy Weinberg proportions at that frequency would be $(0.2025, 0.4950, 0.3025)$: the population is far off the surface $\mathcal{M}$. Mean fitness is $\bar{m} = 0.03 + 0.09 - 0.08 = 0.04$ and the fitness curvature is $D = m_{11} - 2m_{12} + m_{22} = -0.7$.

The normal equations (3.4) are

$$0.75\,\beta_1 + 0.15\,\beta_2 = 0.075, \qquad 0.15\,\beta_1 + 0.95\,\beta_2 = -0.035,$$

with determinant 0.69, so

$$\beta_1 = \frac{0.0765}{0.69} = \frac{51}{460} = 0.110870, \qquad \beta_2 = -\frac{0.0375}{0.69} = -\frac{25}{460} = -0.054348.$$

The fitted array and the residuals are

	A_1A_1	A_1A_2	A_2A_2
$A_{uv} = \beta_u + \beta_v$	0.221739	0.056522	-0.108696
$\delta_{uv} = m_{uv} - A_{uv}$	-0.121739	0.243478	-0.091304

and one checks (3.3) directly: $0.30(-0.121739) + 0.15(0.243478) = 0$ and $0.15(0.243478) + 0.40(-0.091304) = 0$. The variance components are

$$\sigma_G^2 = 0.3(0.06)^2 + 0.3(0.26)^2 + 0.4(0.24)^2 = 0.0444,$$

$\sigma_A^2 = 0.018835$ and $\sigma_D^2 = 0.025565$, which add up as they should. The additive part is 42 percent of the genetic variance.

Now hold the fitnesses and the allele frequency fixed and move the population onto $\mathcal{M}$, to $(0.2025, 0.4950, 0.3025)$. The average effects become $\beta = (0.155875, -0.029125)$, the additive variance falls to 0.016941 and the nonadditive variance rises to 0.030016. The same alleles with the same genotypic fitnesses have different average effects and different variance components in the two populations. This is the second observation above in numbers: β and σ_A^2 are functionals of the state.

3.3 The moving projection lemma

3.3.1 The lemma

LEMMA 3.2 (Moving projection). *For arbitrary smooth genotype frequency paths $P(t)$, with σ_A^2 here denoting Fisher's term $2\sum_u \dot{p}_u \beta_u$ (it equals the additive variance along the selection paths of Example 3.4, but not for every path),*

$$\frac{d\bar{m}}{dt} = \underbrace{\sum_{u,v} \dot{P}_{uv}\, A_{uv}}_{=\ \sigma_A^2} + \underbrace{\sum_{u,v} \dot{P}_{uv}\, \delta_{uv}}_{=:\ \mathcal{R}}, \tag{3.5}$$

and since $\sum_{u,v} P_{uv}\delta_{uv} = 0$ the remainder may be written $\mathcal{R} = \langle \delta,\ \dot{P}/P \rangle_t$, the pairing of the least squares deviations with the logarithmic velocity of the genotype frequencies.

Proof. $\bar{m} = \sum_{u,v} P_{uv} m_{uv}$ with m fixed, so $d\bar{m}/dt = \sum_{u,v} \dot{P}_{uv} m_{uv}$. Split $m = A + \delta$. For the first piece,

$$\sum_{u,v} \dot{P}_{uv}(\beta_u + \beta_v) = 2\sum_u \beta_u \sum_v \dot{P}_{uv} = 2\sum_u \beta_u \dot{p}_u = \sigma_A^2,$$

which is Ewens's Equation (2). The second piece is $\mathcal{R}$ by definition. The final form follows from $\dot{P}_{uv} = P_{uv} \cdot (\dot{P}_{uv}/P_{uv})$. □

Nothing in the proof uses Hardy Weinberg, weak selection, one locus rather than many, or any property of m. The identity is exact for arbitrary paths. Fisher asserted $d\bar{m}/dt = \sigma_A^2$, that is $\mathcal{R} = 0$; equivalently he differentiated $\bar{m} = 2\sum_u p_u \beta_u$ and dropped $2\sum_u p_u \dot{\beta}_u$. Lemma 3.2 says the same thing from the other side: he computed the rate of change of $\Pi_t m$ as though Π_t were constant.

REMARK 3.3 (Where the dot sits). Ewens's Equation (15) writes the remainder as $2\sum_u p_u\dot{\beta}_u$ and (3.5) writes it as $\sum_{u,v}\dot{P}_{uv}\delta_{uv}$. These are the same number, and the difference is only which slot the product rule is applied in.

Since $\sum_{u,v}P_{uv}\delta_{uv} = 0$, mean fitness may be written against either the fitted vector or the original one, $\bar{m} = \sum_{u,v}P_{uv}A_{uv} = \sum_{u,v}P_{uv}m_{uv}$. Differentiating the first gives two terms because both factors move,

$$\frac{d\bar{m}}{dt} = \sum_{u,v}\dot{P}_{uv}A_{uv} + \sum_{u,v}P_{uv}\dot{A}_{uv} = \sigma_A^2 + 2\sum_u p_u\dot{\beta}_u,$$

which is Ewens's form. Differentiating the second gives only one term, because m is a fixed vector of constants and P is the only thing that depends on t; splitting $m = A + \delta$ afterwards gives (3.5). The first terms agree, so the second terms do too.

We use the $\dot{P}$ form for two reasons. It differentiates only what actually moves: the Malthusian parameters are constants and the β are derived quantities, functions of P, so a dot on β is bookkeeping at one remove and a dot on P is bookkeeping at the source. And, more importantly, the $\dot{P}$ form exhibits the remainder as an inner product between δ and the velocity, which gives the vanishing criterion of Section 3.3.3 immediately. Kimura's form, Ewens's Equation (7), is a third repackaging of the same quantity, in which the transverse coordinate $P_{uv}/(p_u p_v)$ is made explicit; Lessard (1997) proved his form equal to Ewens's, and (3.5) sits between the two. Section 3.30 reports a numerical check in which the three expressions agree to ten significant figures on a randomly chosen non Hardy Weinberg path.

3.3.2 The geometry, and Kimura's coordinate

Lemma 3.2 makes the condition for Fisher's claim geometric rather than mysterious. Let

$$\mathcal{M} \;=\; \left\{\, P : P_{uv} = p_u p_v \text{ for some } p \,\right\}$$

be the Hardy Weinberg surface inside the simplex, a $(k-1)$ dimensional submanifold. Suppose $P(t) \in \mathcal{M}$ and $\dot{P}$ is tangential to $\mathcal{M}$, so that $\dot{P}_{uv} = \dot{p}_u p_v + p_u \dot{p}_v$. Then

$$\mathcal{R} = \sum_{u,v}(\dot{p}_u p_v + p_u\dot{p}_v)\delta_{uv} = 2\sum_u \dot{p}_u \sum_v p_v\delta_{uv} = 0$$

by (3.3). So the remainder measures exactly the component of $\dot{P}$ transverse to the Hardy Weinberg surface, weighted by the deviations δ.

> *Fisher's assertion is that the genotype frequencies move along the Hardy Weinberg surface.* They do not: at one locus because selection on diploids perturbs the genotype proportions, and across the genome because linkage disequilibrium is a transverse coordinate of exactly this kind.

This also explains why the theorem looks true in the simplest checks. The projection of the motion onto $\mathcal{M}$ is all that σ_A^2 ever sees, and in a one locus model with Hardy Weinberg proportions maintained by fiat there is nothing else.

Kimura's (1958) expression, Ewens's Equation (7), is

$$\frac{d\bar{m}}{dt} = \sigma_A^2 + \sum_{u,v} p_u p_v \, \delta_{uv} \, \frac{d}{dt}\left(\frac{P_{uv}}{p_u p_v} \right),$$

which Lessard (1997) showed equals the $2\sum p_u \dot{\beta}_u$ form. Lemma 3.2 gives a third expression for the same quantity and makes the common content visible: all three pair the deviations δ with a measure of how the genotype frequencies move off $\mathcal{M}$. Kimura's $P_{uv}/(p_u p_v)$ is the natural transverse coordinate, and its time derivative is what he pairs with δ. Section 3.4 identifies the invariant version of that coordinate and computes the pairing in closed form.

The lemma is easiest to absorb by holding the state fixed and varying the velocity. Three velocities are biologically natural, and between them they show the remainder taking every role from nothing to everything.

EXAMPLE 3.4 (Three velocities through one state). Keep the population of Example 3.1.

Viability selection without mating, $\dot{P}_{uv} = P_{uv}(m_{uv} - \bar{m})$, is what happens within a generation, or in continuous time when genotype proportions are not reset by mating. In unordered form $\dot{P} = (0.018,\, 0.078,\, -0.096)$ and $\dot{p}_1 = 0.018 + 0.039 = 0.057$. Then

$$\frac{d\bar{m}}{dt} = \sum_{u,v} \dot{P}_{uv} m_{uv} = 0.0444,$$

$$2\sum_u \dot{p}_u \beta_u = 2(0.057)(0.110870 + 0.054348) = 0.018835,$$

and the remainder is $\mathcal{R} = 0.0444 - 0.018835 = 0.025565$. These are exactly σ_G^2, σ_A^2 and σ_D^2 of Example 3.1, and not by coincidence: with this velocity $\sum \dot{P} A = \mathrm{Cov}_P(m, A) = \sigma_A^2$ and $\mathcal{R} = \mathrm{Cov}_P(m, \delta) = \sigma_D^2$. Under pure selection the total change is the total genetic variance and Fisher's term misses the whole of the nonadditive part, here 58 percent of it. Differentiating the normal equations numerically along this path gives $\dot{\beta} = (0.006087, 0.018261)$, so Ewens's form is $2(0.45 \times 0.006087 + 0.55 \times 0.018261) = 0.025565$, and Kimura's form returns the same number: the three expressions of Remark 3.3 agree to every printed figure.

Selection on the breeding value alone, $\dot{P}_{uv} = P_{uv}(A_{uv} - \bar{m})$, has $\partial_t \log P \in V$, so the motion is tangent to the leaf through P on which the association is constant, and $\mathcal{R} = \langle \delta, A - \bar{m} \rangle_t = 0$. Here $d\bar{m}/dt = \mathrm{Cov}_P(m, A) = \sigma_A^2 = 0.018835$ and Fisher's statement is exact.

Random mating with no selection, $\dot{P}_{uv} = p_u p_v - P_{uv}$, relaxes the population toward $\mathcal{M}$. In unordered form $\dot{P} = (-0.0975,\, 0.1950,\, -0.0975)$, so $\dot{p}_1 = 0$ and the term Fisher kept is zero. Yet

$$\frac{d\bar{m}}{dt} = -0.0975(0.10) + 0.1950(0.30) - 0.0975(-0.20) = 0.06825,$$

all of it remainder: mean fitness rises because heterozygotes, the fittest class, are being manufactured by mating, and no gene frequency records it. This is the one locus analogue of the two locus system of Section 3.5.3, where it is recombination rather than mating that moves the population across the surface.

A caution about what tangent means off the surface. The velocity $\dot{P}_{uv} = \dot{p}_u p_v + p_u \dot{p}_v$, which is tangent to $\mathcal{M}$ at points *of* $\mathcal{M}$, is not tangent to anything useful here: with $\dot{p}_1 = 0.057$ it gives $\mathcal{R} = 0.002255 \neq 0$. Away from $\mathcal{M}$ the criterion is the one of Theorem 3.5, $\partial_t \log P \in V$.

3.3.3 The whole genome

Ewens (2024) insists, rightly, that the Fundamental Theorem is a whole genome claim which a one locus analysis cannot assess. Lemma 3.2 generalizes without extra work, and the geometric criterion becomes sharper rather than weaker.

Index the whole genome autosomal genotypes by g, with frequencies $P_g(t)$ and fixed Malthusian parameters m_g. For each allele A_{ij} let $X_{ij}(g) \in \{0, 1, 2\}$ be its copy number in genotype g, and let

$$V \;=\; \Big\{\, f_g = \textstyle\sum_{i,j} c_{ij} X_{ij}(g) \;:\; c \in \mathbb{R}^{\cdot} \,\Big\}$$

be the span of the gene contents. Since $\sum_{i,j} X_{ij}(g)$ is twice the number of loci for every g, the constants lie in V. Ewens's Equation (10) defines the whole genome average effects by minimizing $\sum_g P_g \big(m_g - \sum_{(g)} \beta_{ij}\big)^2$, which is exactly the $L^2(P_t)$ projection of m onto V, with stationarity conditions $\langle \delta, X_{ij} \rangle_t = 0$ for every allele.

THEOREM 3.5 (Whole genome). *For arbitrary genotype frequency paths $P(t)$, with no assumption of Hardy Weinberg proportions or of linkage equilibrium,*

$$\frac{d\bar{m}}{dt} \;=\; \sigma_A^2 \;+\; \mathcal{R}, \qquad \sigma_A^2 = 2 \sum_{i,j} \dot{p}_{ij} \beta_{ij}, \qquad \mathcal{R} = \sum_g \dot{P}_g \, \delta_g = \big\langle\, \delta,\; \partial_t \log P \,\big\rangle_t. \quad (3.6)$$

Moreover $\mathcal{R} = 0$ whenever $\partial_t \log P \in V$, and if $\mathcal{R} = 0$ for every assignment of Malthusian parameters m then necessarily $\partial_t \log P \in V$.

Proof. $d\bar{m}/dt = \sum_g \dot{P}_g m_g$ and $m = A + \delta$. For the first piece, $A_g = \sum_{i,j} \beta_{ij} X_{ij}(g)$, so

$$\sum_g \dot{P}_g A_g = \sum_{i,j} \beta_{ij} \frac{d}{dt} \sum_g P_g X_{ij}(g) = \sum_{i,j} \beta_{ij} \frac{d}{dt} \big(2 p_{ij}\big) = 2 \sum_{i,j} \beta_{ij} \dot{p}_{ij} = \sigma_A^2$$

by Ewens's Equation (11), the middle step being the definition of an allele frequency as half the mean copy number. The second piece is $\mathcal{R}$, and $\dot{P}_g = P_g \, \partial_t \log P_g$ gives the pairing form. If $\partial_t \log P \in V$ then $\mathcal{R} = 0$ because $\delta \perp V$. Conversely, as m ranges over $\mathbb{R}^g$ the residual δ ranges over all of $V^\perp$, so $\mathcal{R} = 0$ for every m forces $\partial_t \log P \perp V^\perp$, that is $\partial_t \log P \in V$. □

The criterion has a clean reading. The set

$$\mathcal{M} \;=\; \Big\{\, P :\; \log P_g = \textstyle\sum_{i,j} c_{ij} X_{ij}(g) - \psi(c) \,\Big\}$$

of distributions whose logarithm is additive in gene content is exactly the set of product measures over genes, that is the genotype distributions in Hardy Weinberg proportions *and* in linkage equilibrium. It is an exponential family, and V is its tangent space in exponential coordinates. So Theorem 3.5 says:

> *The additive genetic variance registers only the component of the motion tangent to $\mathcal{M}$.* Every departure from Hardy Weinberg proportions and every change in linkage disequilibrium is transverse to $\mathcal{M}$ and invisible to σ_A^2.

The one locus statement is the case of a single transverse coordinate; across the genome there are as many transverse directions as there are dominance and association parameters, which is why the failure grows worse rather than better as loci are added (Section 3.17). Theorem 3.8 makes the dependence on each of them explicit. The pairing $\langle \delta, \partial_t \log P \rangle$ in (3.6) is the natural duality between a mixture coordinate residual and an exponential coordinate velocity, so the structure is the dually flat one of information geometry (Amari, 1985). We mention this only because it makes clear that nothing probabilistic is involved: the objects are a fixed vector, a fixed subspace and a path of measures.

3.3.4 Two remarks on interpretation

REMARK 3.6 (Why the partial change has no biological interpretation). Ewens (1989) introduced the partial rate of change, showed $\partial \bar{m}/\partial t = \sigma_A^2$, and Ewens (2024) now writes that the concept did not lead to a useful discussion because no biological interpretation for it could be found. Lemma 3.2 identifies the partial change as the tangential component of the motion, which is to say the increment of a fitted projection with its coefficients frozen at their current values. That is a property of an approximation procedure applied to a population, not a property of the population. There is no reason for such a quantity to have a biological meaning, and the long failure to find one is what one should expect. The revised judgement is right, and right for a structural reason rather than as a matter of taste.

REMARK 3.7 (The Price equation route). The same conclusion in one line. Apply the Price equation to the breeding value rather than to fitness. Mendelian segregation under random mating transmits breeding value without bias, so the transmission term vanishes and

$$\bar{w}\,\Delta\bar{A} = \mathrm{Cov}_P(w, A) = \mathrm{Cov}_P(A + \varepsilon, A) = \mathrm{Var}_P(A) = \sigma_A^2,$$

the middle equality being orthogonality of least squares residuals. Correctly stated, then, the theorem concerns the mean *breeding value* of fitness and not mean fitness.

The two coincide at time t, because the residual has mean zero there, but their increments do not. That single substitution is the entire error, and it is worth putting to a biologist in exactly those words. Ewens (1989, 1992) and Lessard (1997) establish the corresponding statement without random mating, with arbitrarily many loci and under fertility selection, the average effects then being defined by a linear system rather than by a Hardy Weinberg weighted regression; see Castilloux and Lessard (1995) and Lessard and Castilloux (1995).

3.4 Lindley coordinates and an exact form for the remainder

Section 3.3.3 shows that $\mathcal{R}$ measures transverse motion. This section identifies the transverse coordinate, shows that the identification is forced rather than conventional, and computes $\mathcal{R}$ in closed form in terms of it. The coordinate is Lindley's (1988).

3.4.1 The coordinate is essentially unique

At a biallelic locus the genotype simplex is two dimensional and $\mathcal{M}$, parametrized by the allele frequency, is one dimensional. The codimension is therefore exactly one: there is a single transverse direction, and a single scalar coordinate on it. Any two such coordinates are functions of one another, so the only question is which representative to use.

Lindley (1988) answers it. Writing P_{11}, P_{12}, P_{22} for the unordered genotype frequencies, he takes

$$\alpha \;=\; \log \frac{P_{11}P_{22}}{\left(\frac{1}{2}P_{12}\right)^2} \;=\; \log P_{11} - 2\log \tfrac{1}{2}P_{12} + \log P_{22}, \tag{3.7}$$

so that Hardy Weinberg equilibrium is $\alpha = 0$, equivalently $P_{12}^2/(P_{11}P_{22}) = 4$. His argument for the logarithmic scale is invariance: a condition on the simplex should be stated in a form that does not depend on the arbitrary choice of coordinates used to describe the simplex, and the log cross product ratio is the invariant such statement. The same argument was later formalized in the log ratio transformations of compositional data analysis (Egozcue et al., 2003).

Three features of (3.7) are worth separating.

It is the canonical parameter of an exponential family. By Theorem 3.5, $\mathcal{M}$ is the set of distributions whose logarithm is additive in gene content. On a 2×2 genotype table that is the log linear model with main effects only, and α is precisely its two way interaction term. So the coordinate transverse to $\mathcal{M}$ is the interaction parameter of a log linear model, and $\mathcal{M}$ is the subfamily obtained by setting it to zero.

It generalizes. With L biallelic loci the gamete simplex has dimension $2^L - 1$ and $\mathcal{M}$ has dimension L, so there are $2^L - 1 - L$ transverse coordinates: exactly the higher order interaction terms of the log linear model on the 2^L table. Lindley's α is the same kind of coordinate for the 2×2 table of the two gene copies within

a locus, and for $L = 2$ the single between locus coordinate is the log odds ratio λ of Section 3.4.4. Within locus departures from Hardy Weinberg and between locus linkage disequilibria are the two kinds, and the counting of Section 3.17 is the counting of interaction terms. We should flag that Lindley treated the single locus case, and that the multilocus extension, though natural, is ours rather than his.

It is a second difference, like D. Comparing (3.7) with $D = m_{11} - 2m_{12} + m_{22}$, the disequilibrium coefficient is the second difference of the log genotype *frequencies* and the fitness curvature is the second difference of the genotype *fitnesses*. The two objects that meet in the remainder are the same contrast applied to the two halves of the model.

3.4.2 The remainder factorizes over the transverse coordinates

The third observation is not a coincidence. Whenever the genotype distribution moves inside an exponential family whose main effects are the gene contents, the remainder factorizes over the interaction coordinates alone, and the contribution of the main effects is exactly zero.

THEOREM 3.8 (Exponential family factorization). *Let the genotype distribution move within the family*

$$\log P_g(t) \;=\; \sum_j c_j(t)\, X_j(g) \;+\; \sum_A \theta_A(t)\, u_A(g) \;-\; \psi\big(c(t),\theta(t)\big),$$

where $\{X_j\}$ spans V, the gene contents, and $\{u_A\}$ completes the basis. Then

$$\Big\langle \delta,\, \frac{\partial \log P}{\partial c_j} \Big\rangle_t \;=\; 0 \quad \textit{for every main effect } j, \qquad \textit{and} \qquad \mathcal{R} \;=\; \sum_A \big\langle \delta,\, u_A \big\rangle_t \,\dot\theta_A. \tag{3.8}$$

Proof. In an exponential family with natural parameter η_k and sufficient statistic T_k, $\partial \log P_g / \partial \eta_k = T_k(g) - \mathbb{E}_P T_k$. Hence

$$\Big\langle \delta, \frac{\partial \log P}{\partial \eta_k} \Big\rangle_t = \sum_g P_g\, \delta_g\, T_k(g) - \mathbb{E}_P[T_k] \sum_g P_g \delta_g = \mathrm{Cov}_P(\delta, T_k) = \langle \delta, T_k \rangle_t,$$

using $\mathbb{E}_P \delta = 0$ from (3.3). For a main effect, $T_k = X_j$ and $\langle \delta, X_j \rangle_t = 0$ by the same stationarity conditions, which is the first claim. The second follows by the chain rule, $\partial_t \log P = \sum_j (\partial \log P / \partial c_j)\dot c_j + \sum_A (\partial \log P / \partial \theta_A)\dot\theta_A$, paired with δ as in Theorem 3.5. □

Three things are worth reading off (3.8).

The remainder depends on the velocities of the interaction coordinates and on nothing else. The allele frequencies may move as fast as they like without contributing a single term, which is Theorem 3.5 in differential form: $\mathcal{R} = 0$ when $\dot\theta = 0$, that is when the motion is confined to a leaf of constant θ, and to $\mathcal{M}$ itself when $\theta = 0$.

The number of terms in the sum is the codimension of $\mathcal{M}$, namely $2^L - 1 - L$ on the gamete simplex of L biallelic loci. So the counting of Section 3.17 is a counting of the terms in (3.8), and each transverse coordinate contributes its velocity weighted by the covariance between the nonadditive deviation and the corresponding interaction statistic.

The weights $\langle \delta, u_A \rangle_t$ are covariances between the residual and the interaction statistics, so a transverse coordinate contributes only insofar as the fitness deviation is correlated with the association it measures. An epistatic interaction that no fitness deviation tracks moves the population without moving mean fitness.

3.4.3 The biallelic case in closed form

At one locus with two alleles there is a single interaction coordinate, and evaluating (3.8) gives the whole remainder explicitly. One preliminary fact is needed first, because the quantity that appears in the answer has a statistical identity worth fixing exactly.

LEMMA 3.9 (Information about Lindley's coefficient). *In the trinomial model for unordered genotype counts, parametrized by (p, α) with p the allele frequency and α as in* (3.7)*, the parameters are orthogonal and the Fisher information about α per observation is*

$$\mathcal{I}(\alpha) \;=\; \Big(\frac{1}{P_{11}} + \frac{4}{P_{12}} + \frac{1}{P_{22}}\Big)^{-1}, \tag{3.9}$$

where P_{12} is the unordered heterozygote frequency. Under Hardy Weinberg proportions $\mathcal{I}(\alpha) = p^2 q^2$.

Proof. Write $c = (1, -2, 1)$, so $\alpha = 2\log 2 + \sum_g c_g \log P_g$, and $h_g = c_g / P_g$. Holding p fixed, the heterozygote frequency is the only free coordinate: $P = (p - x,\, 2x,\, 1 - p - x)$ with $x = \tfrac{1}{2} P_{12}$. Then $\partial\alpha/\partial x = -1/P_{11} - 2/x - 1/P_{22} = -1/\mathcal{I}(\alpha)$, so $\partial P/\partial\alpha = -\mathcal{I}(\alpha)\,(-1, 2, -1) = \mathcal{I}(\alpha)\, c$, and

$$\sum_g \frac{(\partial P_g/\partial\alpha)^2}{P_g} = \mathcal{I}(\alpha)^2 \sum_g \frac{c_g^2}{P_g} = \mathcal{I}(\alpha)^2 \cdot \frac{1}{\mathcal{I}(\alpha)} = \mathcal{I}(\alpha).$$

For orthogonality, the cross term is $\sum_g (\partial P_g/\partial p)(\partial P_g/\partial\alpha)/P_g = \mathcal{I}(\alpha) \sum_g (\partial P_g/\partial p) h_g = \mathcal{I}(\alpha)\, \partial\alpha/\partial p$, the derivative being taken at fixed α, where it is zero. Under Hardy Weinberg $1/p^2 + 4/(2pq) + 1/q^2 = (q + p)^2/(p^2 q^2)$. □

Equivalently, $1/\mathcal{I}(\alpha)$ is the asymptotic variance per observation of the plug in estimate $\hat\alpha = 2\log 2 + \sum_g c_g \log \hat P_g$, by the delta method: $n \operatorname{Var}(\hat\alpha) \to \sum_g c_g^2/P_g - (\sum_g c_g)^2 = 1/\mathcal{I}(\alpha)$.

COROLLARY 3.10 (Remainder in Lindley coordinates). *At a biallelic locus, parametrize the genotype simplex by (p, α). Then*

$$\mathcal{R} \;=\; D\,\mathcal{I}(\alpha)\,\dot\alpha, \qquad D = m_{11} - 2m_{12} + m_{22}, \tag{3.10}$$

with $\mathcal{I}(\alpha)$ the Fisher information (3.9)*.*

Proof. In the ordered array with off diagonal $x = \frac{1}{2}P_{12}$ we have $P_{11} = p - x$, $P_{22} = 1 - p - x$, and by the proof of Lemma 3.9, $x' = \partial x/\partial\alpha = -\mathcal{I}(\alpha)$ at fixed p. So $\partial P/\partial\alpha$ has ordered entries $(-x', x', x', -x')$ and

$$\Big\langle \delta, \frac{\partial \log P}{\partial \alpha} \Big\rangle_t = \sum_{u,v} \delta_{uv} \frac{\partial P_{uv}}{\partial \alpha} = x'\big(2\delta_{12} - \delta_{11} - \delta_{22}\big)$$
$$= -x'\big(\delta_{11} - 2\delta_{12} + \delta_{22}\big) = -x'D = D\,\mathcal{I}(\alpha),$$

the average effects cancelling identically because $\delta_{uv} = m_{uv} - \beta_u - \beta_v$ and the second difference annihilates any additively separable array. By Theorem 3.8 the p direction contributes nothing, so $\mathcal{R}$ is this weight times $\dot\alpha$. □

There is a second proof, shorter and more informative, because at a biallelic locus the residual itself can be written down. The idea is that $V^\perp$ is one dimensional, so the residual is a known vector times one number, and the second difference reads off the number.

PROPOSITION 3.11 (The residual in closed form). *At a biallelic locus with all three genotype classes present, let $h = (1/P_{11}, -2/P_{12}, 1/P_{22})$ on the unordered genotypes A_1A_1, A_1A_2, A_2A_2. Then h spans the orthogonal complement of V in $L^2(P)$, and*

$$\delta \;=\; D\,\mathcal{I}(\alpha)\,h, \qquad \sigma_D^2 \;=\; D^2\,\mathcal{I}(\alpha), \qquad \sigma_A^2 \;=\; \sigma_G^2 - D^2\,\mathcal{I}(\alpha), \tag{3.11}$$

and for every velocity $\mathcal{R} = \langle \delta, \dot P/P\rangle_t = D\,\mathcal{I}(\alpha)\,\dot\alpha$. Under Hardy Weinberg proportions $\sigma_D^2 = (pqD)^2$, which is the classical dominance variance $(2pqd)^2$ in the parametrization $m = (\mu + a, \mu + d, \mu - a)$, where $D = -2d$.

Proof. Functions of the unordered genotype are functions of the gene count $n \in \{2, 1, 0\}$, a three dimensional space in which $V = \operatorname{span}\{1, n\}$ has dimension two. The vector h is orthogonal to both: $\sum_g P_g h_g = 1 - 2 + 1 = 0$ and $\sum_g P_g h_g n_g = 2 - 2 + 0 = 0$. So $V^\perp$ is the line through h and $\delta = ch$ for some scalar c. Apply the second difference $\Delta^2 f = f(2) - 2f(1) + f(0)$, which annihilates V: $\Delta^2\delta = \Delta^2 m = D$, while $\Delta^2 h = 1/P_{11} + 4/P_{12} + 1/P_{22} = 1/\mathcal{I}(\alpha)$. Hence $c = D\,\mathcal{I}(\alpha)$. Then $\sigma_D^2 = c^2 \sum_g P_g h_g^2 = c^2/\mathcal{I}(\alpha) = D^2\mathcal{I}(\alpha)$, and $\sigma_A^2 = \sigma_G^2 - \sigma_D^2$ by Pythagoras. Finally $\langle \delta, \dot P/P\rangle_t = c \sum_g \dot P_g h_g = c\,(\dot P_{11}/P_{11} - 2\dot P_{12}/P_{12} + \dot P_{22}/P_{22}) = c\,\dot\alpha$. The Hardy Weinberg case follows from Lemma 3.9. □

The proposition says something the textbook formulae conceal. The dominance variance at arbitrary genotype frequencies is the squared fitness curvature times the Fisher information about the departure from Hardy Weinberg, and the familiar $(2pqd)^2$ is the special case in which that information equals p^2q^2. Section 3.30 reports numerical confirmation of Theorem 3.8 on randomly generated exponential families and of (3.10) across randomly drawn states and velocities.

EXAMPLE 3.12 (Lindley's coordinate on the worked table). Return to Example 3.1, with $(P_{11}, P_{12}, P_{22}) = (0.30, 0.30, 0.40)$ and $D = -0.7$. Lindley's coefficient is

$$\alpha = \log \frac{0.30 \times 0.40}{0.15^2} = \log 5.3333 = 1.6740,$$

well away from the Hardy Weinberg value 0; the information is

$$\mathcal{I}(\alpha) = \Big(\frac{1}{0.30} + \frac{4}{0.30} + \frac{1}{0.40}\Big)^{-1} = \frac{1}{19.1667} = 0.052174,$$

somewhat below the value $p^2q^2 = 0.061256$ it would take on $\mathcal{M}$ at $p = 0.45$. Then $h = (3.3333, -6.6667, 2.5)$ and $D\,\mathcal{I}(\alpha) = -0.036522$, so (3.11) gives $\delta = (-0.121739,\ 0.243478,\ -0.091304)$, the residuals of Example 3.1 obtained without solving any normal equations, and $\sigma_D^2 = 0.49 \times 0.052174 = 0.025565$.

For the velocities of Example 3.4: under selection without mating $\dot\alpha = \Delta^2(m - \bar m) = D = -0.7$, so $\mathcal{R} = (-0.7)(0.052174)(-0.7) = 0.025565$; under selection on the breeding value $\dot\alpha = \Delta^2 A = 0$ and $\mathcal{R} = 0$; under random mating

$$\dot\alpha = \frac{-0.0975}{0.30} - 2\,\frac{0.1950}{0.30} + \frac{-0.0975}{0.40} = -1.86875,$$
$$\mathcal{R} = (-0.7)(0.052174)(-1.86875) = 0.06825.$$

Each agrees with the direct computation. The fitness enters only through D; the state only through $\mathcal{I}(\alpha)$; the motion only through $\dot\alpha$.

EXAMPLE 3.13 (Lindley's coefficient on the MN blood groups). A classic sample of 1000 people in England typed for the MN blood group, reproduced in Hartl and Clark (2007), gave 298 MM, 489 MN and 213 NN. The allele frequency is $\hat p_M = (2 \times 298 + 489)/2000 = 0.5425$ and the plug in coefficient is

$$\hat\alpha = \log \frac{0.298 \times 0.213}{0.2445^2} = 0.0600.$$

By Lemma 3.9 its standard error is $\{n\,\mathcal{I}(\hat\alpha)\}^{-1/2} = (16.2305/1000)^{1/2} = 0.127$, so $\hat\alpha$ is half a standard error from zero; the Pearson statistic is 0.22 on one degree of freedom. The estimated information $\mathcal{I}(\hat\alpha) = 0.06161$ is indistinguishable from $\hat p^2\hat q^2 = 0.06160$.

A Bayes factor in Lindley's spirit can be had in closed form. Under the alternative put a uniform Dirichlet prior on the genotype frequencies; under Hardy Weinberg put a uniform prior on p. With counts n_{11}, n_{12}, n_{22} summing to n,

$$K = \frac{2^{n_{12}}\, B(2n_{11} + n_{12} + 1,\ n_{12} + 2n_{22} + 1)\,(n+2)!}{2\; n_{11}!\, n_{12}!\, n_{22}!},$$

which is $K = 11.3$ for these data: strong support, on Jeffreys's scale, for Hardy Weinberg proportions, and about as much as a thousand individuals can supply under a sharp null.

What this does and does not say about the Fundamental Theorem should be kept straight. It says the *state* is close to $\mathcal{M}$, so that $\mathcal{I}(\alpha) \approx p^2q^2$. It says nothing about the *velocity* $\dot\alpha$, which is what $\mathcal{R}$ depends on. To estimate $\dot\alpha$ one needs two samples separated in time, and the standard error of a difference of two independent $\hat\alpha$ at this frequency is $\sqrt{2} \times 0.127/\sqrt{n/1000}$. Detecting a change of 0.1 in α at two standard errors therefore needs about 13,000 individuals in each sample. That is large but not absurd, which is the point of the prescription in Section 3.20: unlike the variance components, the transverse velocity is within reach of an ordinary survey.

3.4.4 *Two loci off linkage equilibrium*

The same construction works for any table whose transverse space is one dimensional. The simplest multilocus case is a haploid population, or the gametic phase of a diploid one, with two biallelic loci. Order the haplotypes AB, Ab, aB, ab with frequencies $x_1, \dots, x_4$, let $n_A, n_B \in \{0,1\}$ be the gene contents, so $V = \operatorname{span}\{1, n_A, n_B\}$, and define

$$\lambda = \log \frac{x_1 x_4}{x_2 x_3}, \qquad E = m_1 - m_2 - m_3 + m_4,$$

$$\mathcal{I}(\lambda) = \Big(\sum_{i=1}^{4} \frac{1}{x_i} \Big)^{-1}, \qquad D_{AB} = x_1 x_4 - x_2 x_3.$$

Here λ is the log odds ratio of the haplotype table, the canonical measure of linkage disequilibrium, E is the interaction contrast of fitness, and $\mathcal{I}(\lambda)$ is the Fisher information about λ per haplotype, by the argument of Lemma 3.9.

PROPOSITION 3.14 (Two locus residual). *With $h = (1/x_1, -1/x_2, -1/x_3, 1/x_4)$ one has $h \perp V$ in $L^2(x)$, and*

$$\delta = E\,\mathcal{I}(\lambda)\,h, \qquad \sigma^2_{AB} = E^2\,\mathcal{I}(\lambda), \qquad \mathcal{R} = E\,\mathcal{I}(\lambda)\,\dot\lambda,$$

where $\sigma^2_{AB} = \sum_i x_i \delta_i^2$ is the epistatic variance. In particular, recombination at rate r in continuous time, $\dot x = -rD_{AB}(1,-1,-1,1)$, moves no allele frequency and changes mean fitness at the rate $\mathcal{R} = -r\,D_{AB}\,E$, exactly.

Proof. $\sum x_i h_i = 1 - 1 - 1 + 1 = 0$, $\sum x_i h_i n_A = 1 - 1 = 0$ and $\sum x_i h_i n_B = 1 - 1 = 0$, and V has codimension one in $\mathbb{R}^4$, so $\delta = ch$. The interaction contrast annihilates V, so $E = c\sum_i 1/x_i$ and $c = E\,\mathcal{I}(\lambda)$. The variance and the remainder follow as in Proposition 3.11, with $\sum_i \dot x_i h_i = \dot\lambda$. For recombination, $\sum_i \dot x_i h_i = -rD_{AB}\sum_i 1/x_i$, so $\mathcal{R} = -rD_{AB}E$. □

EXAMPLE 3.15 (Tangential and transverse parts with two loci). Take haplotype frequencies and Malthusian parameters

haplotype	AB	Ab	aB	ab
frequency x	0.40	0.10	0.20	0.30
Malthusian parameter m	0.20	0.00	0.10	0.40

So $p_A = 0.5$, $p_B = 0.6$ and $D_{AB} = 0.12 - 0.02 = 0.10$; linkage equilibrium would put 0.30 rather than 0.40 on AB. Mean fitness is $\bar{m} = 0.22$ and the interaction contrast is $E = 0.5$.

Least squares. Fitting $m \approx b_0 + b_A n_A + b_B n_B$ with weights x gives the normal equations

$$\begin{pmatrix} 1 & 0.5 & 0.6 \\ 0.5 & 0.5 & 0.4 \\ 0.6 & 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} b_0 \\ b_A \\ b_B \end{pmatrix} = \begin{pmatrix} 0.22 \\ 0.08 \\ 0.10 \end{pmatrix}, \qquad (b_0, b_A, b_B) = (0.32,\ -0.08,\ -0.10).$$

The breeding values are $A = (0.14, 0.24, 0.22, 0.32)$ and the residuals $\delta = (0.06, -0.24, -0.12, 0.08)$. Directly, $\sigma_G^2 = 0.0176$, $\sigma_A^2 = 0.0056$ and $\sigma_{AB}^2 = 0.0120$. Proposition 3.14 gives the same residuals at once: $\sum 1/x_i = 20.8333$, $\mathcal{I}(\lambda) = 0.048$, $E\,\mathcal{I}(\lambda) = 0.024$, and $0.024 \times (2.5, -10, -5, 3.3333) = (0.06, -0.24, -0.12, 0.08)$; and $\sigma_{AB}^2 = 0.25 \times 0.048 = 0.012$.

Both average effects are negative, although the best haplotype ab and the second best AB share no allele. At the same allele frequencies in linkage equilibrium the coefficients would be $(0.25, -0.10, -0.05)$, with $\sigma_A^2 = 0.0031$ and $\sigma_{AB}^2 = 0.015$: the average effect of B halves when nothing but the association changes.

Selection. Under $\dot{x} = x(m - \bar{m})$ the velocity is $(-0.008, -0.022, -0.024, 0.054)$, so $\dot{p}_A = -0.030$ and $\dot{p}_B = -0.032$. The logarithmic velocity $m - \bar{m}$ splits orthogonally into a tangential part, the centred breeding value, and a transverse part, the residual:

	AB	Ab	aB	ab
$\partial_t \log x = m - \bar{m}$	-0.02	-0.22	-0.12	0.18
tangential, $A - \bar{m}$	-0.08	0.02	0.00	0.10
transverse, δ	0.06	-0.24	-0.12	0.08

The term Fisher keeps is $b_A \dot{p}_A + b_B \dot{p}_B = 0.0024 + 0.0032 = 0.0056 = \sigma_A^2$ (no factor two, the organism being haploid). The remainder is $\mathcal{R} = \sum \dot{x}_i \delta_i = 0.0120 = \sigma_{AB}^2$, and since $\dot{\lambda} = \sum_i \dot{x}_i h_i = E = 0.5$, also $\mathcal{R} = 0.5 \times 0.048 \times 0.5$. Fisher's term is 32 percent of the true rate 0.0176.

Recombination. Add recombination at rate $r = 0.5$. Its velocity is $-0.05 \times (1, -1, -1, 1)$, it moves neither allele frequency, and it contributes $\mathcal{R} = -rD_{AB}E = -0.025$ to the rate of change of mean fitness. With selection and recombination together, $d\bar{m}/dt = 0.0176 - 0.025 = -0.0074$: mean fitness *falls*, while Fisher's term, still $+0.0056$, says it rises. The additive variance has the wrong sign for the total, and every unit of the discrepancy is carried by the motion of λ.

3.4.5 What the factorization says

Equation (3.10) decomposes the remainder into three interpretable factors, and each carries a separate lesson.

D: the average effects do not appear. The fitness input is the raw second difference of the Malthusian parameters. The β values, which are the objects that caused all the trouble, cancel identically. So the remainder is not a subtle functional of the fitted model; it is the plain curvature of the fitness function, and the only reason it was ever hard to see is that the term it belongs to was dropped.

$\dot{\alpha}$: Fisher's condition is necessary as well as sufficient. Fisher's (1941) constant genic environment, in the form Ewens (2024) gives as Equation (9) and read as $\sum_u p_u \dot{\beta}_u = 0$ rather than as constancy of every β_u, is exactly $\dot{\alpha} = 0$. Equation (3.10) shows that at a biallelic locus this is not merely sufficient for $\mathcal{R} = 0$ but, whenever $D \neq 0$, necessary. The theorem is exactly true when Lindley's disequilibrium coefficient is stationary and at no other time. That sharpens the standard statement, which gives sufficiency only, and it makes precise what Ewens and Lessard describe as the circularity: the condition does not constrain the biology, it names the conclusion.

$\mathcal{I}(\alpha)$: the remainder is weighted by the information about α. The middle factor is the Fisher information about the very coordinate whose motion drives the remainder. So the remainder is large where α is well determined and small where it is poorly determined. $\mathcal{I}(\alpha)$ tends to zero as any genotype class becomes rare, so $\mathcal{R} \to 0$ at the boundary of the simplex for any bounded $\dot{\alpha}$; near $\mathcal{M}$ it equals $p^2q^2 \leq 1/16$, so that $|\mathcal{R}| \leq |D\,\dot{\alpha}|/16$. The remainder is a phenomenon of intermediate frequencies, which is where segregating variation lives, and it is exactly there that α is easiest to measure. The reasons the error went unseen, collected in Section 3.7, therefore do not include any difficulty in estimating α. They lie in D and $\dot{\alpha}$ each being of the order of the selection coefficient, and in the fact that nobody tried to measure $\dot{\alpha}$ at all.

3.4.6 Consequences for measurement

Corollary 3.10 converts the central question of this chapter into an estimation problem with existing tooling, and this is the concrete content behind prescription item 9 of Section 3.20.

To know whether $\mathcal{R}$ matters in a given population one needs D, which is a fitness contrast, and $\dot{\alpha}$, which is the rate of change of a function of genotype counts. The second is the hard part only in the sense that nobody reports it. It is exactly what Lindley (1988) built inference for, what Shoemaker, Painter and Weir (1998) extended by arguing that no population is ever exactly in equilibrium so that the useful question is the magnitude of the departure, and what Wakefield (2010) scaled to genome wide panels. Two time points of genotype counts at a segregating locus suffice for a first estimate of $\dot{\alpha}$, with a standard error computable from Lemma 3.9 as in Example 3.13; the linkage disequilibrium spectrum, and the log odds ratio λ of Proposition 3.14, supply the between locus analogue.

So the proposition that has been argued about since 1930 has, at a biallelic locus, the form *a measurable coefficient does not move*. Nobody needs to settle it by argument.

3.5 Discrete time: a closed form, and two examples

Ewens (2024) is right that Fisher's claim is continuous time and whole genome, and that a one locus discrete time analysis cannot settle it. What follows is offered only because it is the one place where $\mathcal{R}$ can be written down explicitly and inspected.

3.5.1 The closed form and its order

Take discrete nonoverlapping generations, one locus, random mating, constant genotypic fitnesses w_{ij}, so $P_t(ij) = p_i p_j$, and write $\bar{w} = \sum p_i p_j w_{ij}$, $w_{i\cdot} = \sum_j p_j w_{ij}$, $\alpha_i = w_{i\cdot} - \bar{w}$, $\Delta p_i = p_i \alpha_i / \bar{w}$.

THEOREM 3.16 (Exact remainder, one locus, k alleles).

$$\Delta\bar{w} \;=\; \frac{\sigma_A^2}{\bar{w}} \;+\; \frac{1}{\bar{w}^2}\sum_{i,j}\delta_{ij}\,(p_i\alpha_i)(p_j\alpha_j), \qquad \delta_{ij} = w_{ij} - \bar{w} - \alpha_i - \alpha_j.$$

With two alleles and $D = w_{11} - 2w_{12} + w_{22}$ this is $\Delta\bar{w} = \sigma_A^2/\bar{w} + D(\Delta p)^2$.

Proof. $\bar{w}(p)$ is an exact quadratic form in p, so $\Delta\bar{w} = \sum_i 2(\bar{w} + \alpha_i)\Delta p_i + \sum_{i,j} w_{ij}\Delta p_i \Delta p_j$. The first sum is $\sigma_A^2/\bar{w}$ since $\sum_i p_i\alpha_i = 0$. In the second, put $u_i = p_i\alpha_i$ and substitute $w_{ij} = \bar{w} + \alpha_i + \alpha_j + \delta_{ij}$; the terms in $\bar{w}$ and α drop because $\sum_i u_i = 0$, leaving $\bar{w}^{-2}\sum_{i,j}\delta_{ij}u_i u_j$. □

This is Ewens's Equation (16) with the neglected term supplied. In the standard parametrization $w_{11} = \mu + a$, $w_{12} = \mu + d$, $w_{22} = \mu - a$ one has $D = -2d$ and $\mathcal{R} = -2d(\Delta p)^2$, so Fisher's total change claim is exactly the hypothesis $d = 0$: a sharp point null on a single interpretable parameter. That observation is the hinge on which Part II turns.

Write $w_{ij} = 1 + s\,a_{ij}$. Then $D = O(s)$ and $\Delta p = O(s)$, so

$$\mathcal{R} = O(s^3), \qquad \sigma_A^2/\bar{w} = O(s^2), \qquad \frac{|\mathcal{R}|}{\sigma_A^2/\bar{w}} = O(s). \tag{3.12}$$

The relative error is first order in the selection coefficient, not second. With two alleles it can be computed exactly, and the answer is simpler than either of its ingredients.

COROLLARY 3.17 (Exact relative error, two alleles). *At a biallelic locus under the conditions of Theorem 3.16, with $q = 1 - p$,*

$$\frac{\mathcal{R}}{\sigma_A^2/\bar{w}} \;=\; \frac{D\,pq}{2\,\bar{w}}, \qquad \mathcal{R} = D(\Delta p)^2.$$

Proof. Since $p\alpha_1 + q\alpha_2 = 0$, $\alpha_2 = -p\alpha_1/q$ and $\sigma_A^2 = 2(p\alpha_1^2 + q\alpha_2^2) = 2p\alpha_1^2(1 + p/q) = 2p\alpha_1^2/q$. With $\Delta p = p\alpha_1/\bar{w}$,

$$\frac{D(\Delta p)^2}{\sigma_A^2/\bar{w}} = \frac{D\,p^2\alpha_1^2/\bar{w}^2}{2p\alpha_1^2/(q\bar{w})} = \frac{Dpq}{2\bar{w}}.$$

□

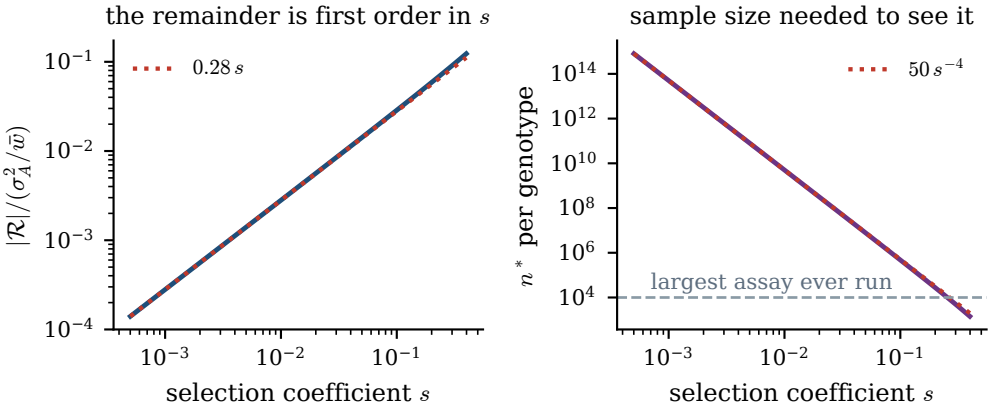


FIGURE 3.2. Left: the relative error of Fisher's term is first order in the selection coefficient, $|\mathcal{R}|/(\sigma_A^2/\bar{w}) = 0.28\,s$ over nearly three decades. Right: the sample size n^* at which the neglected term first exceeds the sampling noise in the term retained, $n^* \approx 50\,s^{-4}$. At a one percent selection coefficient this is five billion zygotes per genotype.

The average excess α_1, which carries all the dependence on the additive part of fitness, cancels. The relative error of Fisher's term at one locus depends only on the curvature D, the heterozygosity and mean fitness. With $w_{ij} = 1 + s\,a_{ij}$ it is $s\,|D_a|\,pq/(2\bar{w})$, where $D_a = a_{11} - 2a_{12} + a_{22}$, and since $pq \le \frac{1}{4}$ it never exceeds $s|D_a|/(8\bar{w})$. For the shape of Example 3.5.2 below, $D_a = -8/3$ and $pq = 0.21$, so the relative error is $0.28\,s/\bar{w}$ exactly, tending to $0.28\,s$ as $s \to 0$ (Figure 3.2, left panel). At a one percent selection coefficient the theorem is wrong by about one part in 360.

Two further consequences. The claim is false as an identity, since $\mathcal{R}$ vanishes for all p only when $\delta \equiv 0$, which is the purely additive model. And the remainder is not sign definite: $\mathcal{R} < 0$ under overdominance and $\mathcal{R} > 0$ under underdominance. At one locus with constant fitnesses $\Delta\bar{w}$ is nonetheless nonnegative, but that is a separate theorem, due to Mulholland and Smith (1959), Scheuer and Mandel (1959) and Kingman (1961), and it is a one locus phenomenon. Conflating it with the Fundamental Theorem has done real damage, since it is true in precisely the case everybody checks; Section 3.7 returns to this.

3.5.2 Example A: one locus, moderate selection

Take $w_{11} = 0.9$, $w_{12} = 1.2$, $w_{22} = 0.7$ and $p_1 = 0.3$, so $D = -0.8$ and the heterozygote is fittest. Then $\bar{w} = 0.9280$, $w_{1\cdot} = 1.1100$, $w_{2\cdot} = 0.8500$, $\alpha_1 = 0.1820$, $\alpha_2 = -0.0780$ and $\Delta p = 0.058836$.

Quantity	Value
Total genetic variance σ_G^2	0.056616
Additive part σ_A^2	0.028392
Nonadditive part σ_D^2	0.028224
Fisher's term $\sigma_A^2/\bar{w}$	0.030595
Remainder $\mathcal{R} = D(\Delta p)^2$	-0.002769
Total $\Delta\bar{w}$	0.027825

The additive part is 50.1 percent of the genetic variance, yet the remainder is only 9.05 percent of Fisher's term, so his term overstates the true change by 9.95 percent. The 9.05 percent is Corollary 3.17 in action: $|D|pq/(2\bar{w}) = 0.8 \times 0.21/1.856 = 0.0905$. By hand, $\Delta p = p\alpha_1/\bar{w} = 0.3 \times 0.182/0.928 = 0.058836$ and $\mathcal{R} = -0.8 \times 0.058836^2 = -0.002769$, while $\sigma_A^2 = 2(0.3 \times 0.182^2 + 0.7 \times 0.078^2) = 0.028392$. Two percentages are in play throughout and it is worth fixing the convention: we report $|\mathcal{R}|/(\sigma_A^2/\bar{w})$, the remainder as a fraction of the term Fisher kept, because that is what compares directly with the relative standard error of $\hat{\sigma}_A^2$ in Section 3.15.

The error is small because of (3.12): the remainder is one order higher in s than the term Fisher kept, while the quality of the fit is not. Anyone expecting a fit that explains half the variance to produce a fifty percent error will find ten percent and conclude the theorem is fine. At $s = 0.03$ the same configuration errs by less than one percent, and at $s = 0.003$ by about one part in a thousand. What one sees is reassuring and misleading.

3.5.3 Example B: two loci, σ_A^2 identically zero while mean fitness falls

This is the example to put in front of a sceptic, because it removes any possibility of treating the discrepancy as a higher order nuisance.

Two loci with alleles A/a and B/b, gametes ordered AB, Ab, aB, ab with frequencies x, free recombination $r = 1/2$, and

$$w(AaBb) = w(AAbb) = w(aaBB) = 1.2, \qquad \text{every other genotype} = 0.6,$$

with no position effect, so the coupling and repulsion double heterozygotes share the value 1.2. Fitness is a pure exclusive or: high when the two loci disagree. Start at $x = (0.05, 0.45, 0.45, 0.05)$, with the standard recursion

$$\bar{w}\,x_1' = x_1(Wx)_1 - r\,w_{14}D, \qquad \bar{w}\,x_2' = x_2(Wx)_2 + r\,w_{14}D, \qquad D = x_1x_4 - x_2x_3,$$

and symmetrically for x_3', x_4'. Here $W = (w_{ij})$ is the 4×4 array of fitnesses of the genotype formed by gametes i and j, so $w_{14} = 1.2$ is the fitness of the coupling double heterozygote AB/ab, and $(Wx)_i = \sum_j w_{ij}x_j$ is the marginal fitness of gamete i.

Two facts make this example clean, and both follow from one symmetry. Let σ exchange A with a and B with b simultaneously. It maps the gametes $AB \leftrightarrow ab$

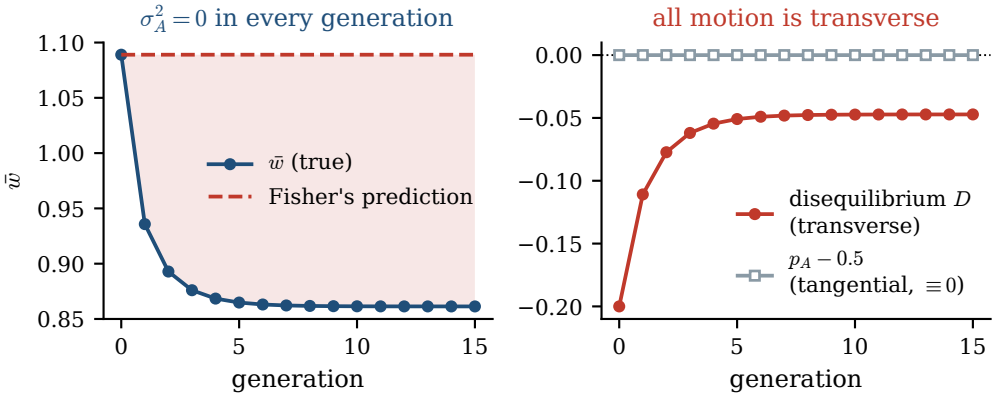


FIGURE 3.3. The exclusive or system. Mean fitness falls twenty percent (left, blue circles) while Fisher's prediction is a flat dashed line (red), since $\sigma_A^2 = 0$ in every generation. No allele frequency moves at any point (right, grey open squares); the whole change is carried by the decay of linkage disequilibrium (right, red circles), a coordinate transverse to $\mathcal{M}$.

and $Ab \leftrightarrow aB$, it preserves the fitness (whether the two loci disagree is unchanged), and it preserves the state whenever $x_1 = x_4$ and $x_2 = x_3$. The recursion commutes with σ, so these equalities hold in every generation, and then $p_A = x_1 + x_2 = \frac{1}{2}$ and likewise $p_B = \frac{1}{2}$: the allele frequencies never move. Second, the weighted regression of fitness on the gene counts n_A, n_B is unique, and σ sends $n_A \mapsto 2 - n_A$, $n_B \mapsto 2 - n_B$ while leaving fitness and weights unchanged. So if (b_0, b_A, b_B) is the least squares fit then so is $(b_0 + 2b_A + 2b_B, -b_A, -b_B)$, and uniqueness forces $b_A = b_B = 0$. Every average effect is zero, $\sigma_A^2 = 0$ in every generation, and Fisher's quantity predicts $\Delta\bar{w} = 0$ forever.

Generation	$x_1 = x_4$	$x_2 = x_3$	$\bar{w}$	D	$\Delta\bar{w}$
0	0.05000	0.45000	1.089000	−0.20000	−0.153210
1	0.13912	0.36088	0.935790	−0.11088	−0.042901
2	0.17270	0.32730	0.892889	−0.07730	−0.016889
3	0.18804	0.31196	0.876001	−0.06196	−0.007558

Mean fitness falls monotonically from 1.089 towards 0.86, about twenty percent, driven entirely by the decay of linkage disequilibrium under recombination, with no allele frequency changing at any point. In the language of Theorem 3.5 the tangential component is identically zero and the whole change is transverse.

The point is not that Fisher's term is inexact. In this system it carries no information at all about the total change: it is zero while the truth is large and negative. This is the exact complement of the Lessard and Ewens (2019) example, in which the second term does involve gene frequency change, defeating the reading under which it is evolutionarily uninteresting; here it involves no gene frequency

change at all and yet is the entire story. Between the two, both halves of that proposed interpretation fail. Decreases of mean fitness of this kind go back to Kojima and Kelleher (1961) and Moran (1964).

3.6 Dynamics: how far does a selected path leave the manifold?

Theorem 3.5 says $\mathcal{R}$ measures motion transverse to $\mathcal{M}$, and Section 3.5.3 shows the transverse motion can be everything. Neither says how far an actual selected path departs. That is a dynamical question. This section attempts it in the two locus case, and the answer is not what one would guess.

3.6.1 Setup and two findings

Two loci, additive across loci with coefficient s at each and a pure epistatic interaction of size ϵ, so that a genotype carrying n_A and n_B copies has fitness

$$w \;=\; 1 + s(n_A - 1) + s(n_B - 1) + \epsilon\,(n_A - 1)(n_B - 1),$$

with recombination rate r and no dominance within either locus, so that all nonadditivity is epistatic. Start at linkage equilibrium and iterate. When $r \gg s$ the disequilibrium equilibrates long before the allele frequencies move appreciably: the classical quasi linkage equilibrium regime.

The disequilibrium follows the classical formula. Measured at quasi equilibrium,

$$D^* \;=\; \frac{\epsilon\; p_A q_A\; p_B q_B}{r}, \tag{3.13}$$

confirmed numerically to within half a percent at $r = 0.5$ and $r = 0.25$ across ϵ from 0.005 to 0.02. Tighter linkage means proportionally more disequilibrium, as one expects, and the left panel of Figure 3.4 shows the $1/r$ decline over a decade of r.

The relative error does not depend on r. The natural defence of Fisher at this point is that recombination breaks up the associations, so that under free recombination the epistatic remainder should be negligible. It is not.

At quasi linkage equilibrium,

$$\frac{|\mathcal{R}|}{\sigma_A^2/\bar{w}} \;\approx\; \epsilon\,\sqrt{p_A q_A\; p_B q_B}, \tag{3.14}$$

with no dependence on the recombination rate r and none on the selection intensity s.

Across a grid with s spanning sixteenfold (0.0005 to 0.008), ϵ fourfold and r from 0.5 to 0.2, the ratio $|\mathcal{R}|/(\sigma_A^2/\bar{w}) \div \epsilon$ stayed between 0.224 and 0.255 against the predicted $p(1-p) = 0.25$ at $p = \frac{1}{2}$. Varying the allele frequency from 0.17 to 0.84 reproduced the $p_A q_A$ factor with ratios 1.002, 1.003, 0.992, 0.998 and 0.998.

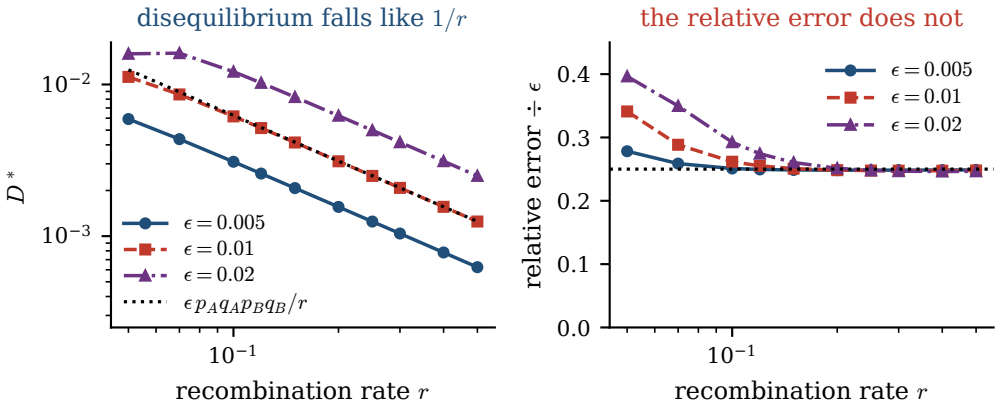


FIGURE 3.4. Quasi linkage equilibrium, $s = 0.001$. Left: the disequilibrium falls like $1/r$, matching (3.13) (dotted). Right: the relative error of Fisher's term, divided by ϵ, is flat at $p_A q_A = 0.25$ over the range where the timescales separate. Recombination removes the disequilibrium without removing the error. The rise at $r \lesssim 0.1$ is the quasi equilibrium approximation failing.

3.6.2 *Two consequences, and caveats*

For the mathematics, (3.14) completes the pattern of (3.12). At one locus the relative error is $0.28\,s$, first order in the selection coefficient, because the dominance curvature there scales with s. Between loci it is $\epsilon\sqrt{p_A q_A p_B q_B}$, first order in the epistatic coefficient and *independent of the selection intensity*. The two contributions therefore behave differently as selection weakens: the dominance remainder vanishes with s and the epistatic remainder does not. In the weak selection limit where population genetics operates, the epistatic contribution is the one that survives.

For the argument, this removes the most natural defence of the theorem. One expects free recombination to protect Fisher by destroying the associations that carry the nonadditive fitness, and the left panel of Figure 3.4 shows that it does destroy them, by a factor of ten across the range plotted. The right panel shows that this buys nothing. The disequilibrium falls like $1/r$ and the error stays put, because the same recombination that suppresses D also suppresses the rate at which the population is dragged across the surface, and the two effects cancel. There is no recombination rate at which the theorem becomes safe.

Caveats. Equation (3.14) is verified numerically rather than derived, and we would want the derivation before relying on it. The factor $\sqrt{p_A q_A p_B q_B}$ is confirmed at $p_A = p_B$ and in three of four asymmetric configurations tried; the fourth, at $p_A \approx 0.3$ and $p_B \approx 0.5$, did not settle to the predicted value and we have not diagnosed why, though a sign change in $\mathcal{R}$ nearby would explain it. The model has no dominance within loci by construction, so this isolates the epistatic contribution

rather than describing a realistic architecture. And quasi linkage equilibrium requires $r \gg s$ and $r \gg \epsilon$; the departure at small r in Figure 3.4 comes from those conditions failing, not the formula.

3.7 Why the error was not detected

The question needs rephrasing before it can be answered, because the error *was* detected. Kimura found it in 1958, and by Crow's (1987) account Fisher knew and regarded his own statement as complete and not in need of clarification. Price found it again in 1972, Ewens in 1989 and 1992, Lessard in 1997, Lessard and Ewens in 2019; Edwards (2002) and Ewens (2004, Ch. 6) survey the position. So the puzzle is not detection. It is why sixty six years of correct mathematics failed to displace the claim. The answer, we think, is that the error has essentially no observable consequence in any regime anyone has worked in, and six separate mechanisms conspire to make that so.

1. The remainder is one order smaller in the selection intensity. By (3.12) the relative error is $O(s)$, and for the shape of Example 3.5.2 it is $0.28\,s$ across nearly three decades:

s	$\sigma_A^2 / \bar{w}$	$\mathcal{R}$	relative error
0.3	3.06×10^{-2}	-2.77×10^{-3}	9.05%
0.1	3.23×10^{-3}	-9.27×10^{-5}	2.87%
0.03	2.86×10^{-4}	-2.42×10^{-6}	0.85%
0.01	3.16×10^{-5}	-8.88×10^{-8}	0.28%

Selection coefficients at segregating loci are small. At one percent the theorem is wrong by about one part in 360, which no biologist would notice and none ought to care about.

2. Detecting it at one locus needs a sample of order s^{-4}. This is the decisive number. The relative standard error of $\hat{\sigma}_A^2$ scales as $1/(s\sqrt{n})$, because the estimator is a quadratic functional of a contrast of size $O(s)$ estimated with $O(n^{-1/2})$ noise, while the relative bias scales as s. Setting them equal gives

$$n^* \;\approx\; 50\, s^{-4} \quad \text{zygotes per genotype.} \tag{3.15}$$

s	relative bias	n^* per genotype
0.3	9.05%	5.0×10^3
0.1	2.87%	4.8×10^5
0.03	0.85%	6.1×10^7
0.01	0.28%	5.0×10^9
0.003	0.084%	6.2×10^{11}
0.001	0.028%	5.0×10^{13}

At a one percent selection coefficient one would need five billion zygotes of each genotype before the neglected term rose above the sampling noise in the term that was kept. No viability experiment in the history of the subject has come within five orders of magnitude of that. Fisher's error is not merely undetected: at realistic selection intensities it is undetectable, and the fourth power scaling means it will stay so (Figure 3.2, right).

3. The standard teaching setting makes the claim exactly true. By Section 3.3, $\mathcal{R}$ vanishes when the motion is tangential to the Hardy Weinberg surface. In continuous time at one locus with Hardy Weinberg proportions imposed rather than derived, which is how the case is almost always presented, the tangency is exact and $\mathcal{R} \equiv 0$ identically; we verified this numerically for two to five alleles. So a reader who checks the theorem in the setting where it is usually taught confirms it, correctly, and concludes that the general statement is safe. The error appears only when the genotype distribution is allowed off the surface, and the standard presentation forbids that by construction.

4. The proposition people actually test is a different one, and it is true. Ask a population geneticist to check the Fundamental Theorem and they will check whether mean fitness increases. At one locus with constant fitnesses it does, by the theorem of Mulholland and Smith (1959), Scheuer and Mandel (1959) and Kingman (1961), and the check succeeds every time. Producing a failure requires two loci, recombination and epistasis together, as in Section 3.5.3 and in Kojima and Kelleher (1961) and Moran (1964). Almost nobody's working example has all three. The monotonicity theorem thus acts as a decoy: it is true, easy to verify, constantly mistaken for the Fundamental Theorem, and verifying it feels like verifying Fisher.

5. The neglected term has never been computable. Ewens (2024) writes that it is impossible to find explicit expressions for the whole genome average effects, depending as they do on epistatic effects between loci and on linkage disequilibrium values. Then $2\sum_{i,j} p_{ij}\dot{\beta}_{ij}$ has never been evaluated for any real system, and could not have been. This is worth separating from the others: it is not that anyone looked at the second term and judged it small. Nobody has ever been in a position to look at it.

6. Across loci the dominance contribution is small, and partly cancels. A genome simulation with $L = 20{,}000$ segregating deleterious alleles, $w = (1, 1-hs, 1-s)$, selection coefficients averaging one percent, gives a whole genome relative error from $\mathcal{R}$ of 0.126% under directional dominance $h \sim U(0.1, 0.4)$, where every $D_\ell = s_\ell(2h_\ell - 1)$ is negative so nothing cancels, and 0.059% under symmetric $h \sim U(0, 1)$, where about half cancels. The cancellation is far weaker than the $\sqrt{L}$ a central limit heuristic would suggest, because the sum is dominated by a handful of loci, which is the sparsity of Section 3.13 showing up again.

That last paragraph is a caveat against Part II as much as an explanation, and we state it as such: for *dominance* at ordinary segregating loci under weak

selection, the whole genome error really is of order a tenth of a percent, and Fisher's approximation is excellent. What that calculation excludes is exactly what Section 3.5.3 and Section 3.6 contain. Linkage disequilibrium and epistasis are transverse coordinates too, they are not bounded by any $O(s^3)$ argument, and by (3.14) they are not damped by recombination either.

Two further reasons that are not mathematical. The theorem never made a prediction. It was offered as foundational reassurance that Darwinian selection survives Mendelian segregation, not as a tool with downstream consumers, so no calculation ever broke in a way that would have propagated back to it. A statement that is never used cannot be falsified by use. And the 1958 text is visibly corrupt, with many printing errors and a plausible miscommunication with the publisher, which gives every charitable reader licence to assume the intended statement was the correct one. Section 3.8 adds a third.

3.8 A note on terminology

One further reason the error survived is grammatical, and it is worth separating from the mathematics.

Consider what has to be true for the error to be easy to make and hard to notice. Lemma 3.2 is a product rule. Nobody forgets the product rule. What one forgets is that something is a function of t at all. And the phrase *average effect of an allele* is a noun phrase naming a property of an allele. It is grammatically an attribute, like a mass or a charge. Read that way, $\bar{m} = 2\sum_u p_u \beta_u$ is a weighted sum of constants with moving weights, and differentiating only the weights is not an oversight but the obvious thing to do.

Names like projection, fitted value, or least squares fit are *relational*: each names something as the output of an operation applied to a state, and each therefore carries its own time dependence along with it. *Average effect* does not. That asymmetry is not cosmetic. A noun that conceals a functional dependence is exactly the kind of thing that survives sixty six years of correct mathematics, and if the aim is to stop the error being repeated, it is worth naming as one of the mechanisms by which it propagated.

PART II THE STOCHASTIC THEORY

3.9 The bridge: a different question

Part I settles a question about a deterministic system: is $\mathcal{R} = 0$? The answer is no, and no data were consulted. That question is closed.

There is a second question, and it belongs on the stochastic side of population genetics, next to the Ewens Sampling Formula rather than next to the Fundamental Theorem. Suppose one has genomic data. What can be said about the size of the

deviations δ across the genome? This is not an interpretation of Fisher. It is the question a person with a data set has, and it is the one on which the practical status of σ_A^2 turns. Part I says σ_A^2 is not the whole of $d\bar{m}/dt$. Part II asks how much of it σ_A^2 is, in a real genome, and what evidence could settle that.

It is worth being explicit about why the two questions do not collapse into one. Ewens's argument solves a *testing* problem: the hypothesis is $\mathcal{R} = 0$, the loss is zero one, and the answer is reject. A reader with a genome solves an *estimation* problem: the estimand is $d\bar{m}/dt$, the action is to report a number, and the loss is squared error, so that an unbiased estimator $\hat{\sigma}_A^2$ of σ_A^2 has risk

$$\mathbb{E}\big(\hat{\sigma}_A^2 - d\bar{m}/dt\big)^2 \;=\; \mathrm{Var}(\hat{\sigma}_A^2) \;+\; \mathcal{R}^2.$$

The deterministic argument fixes only that the second term is positive. It says nothing about its size, nothing about the first term, and nothing about whether some other action has lower risk. These are different problems and the answer to the first does not determine the answer to the second.

3.10 Fisher's $\kappa = 1$ assumption

At locus ℓ let δ_ℓ denote the nonadditive deviation and let y_ℓ be a standardized estimate of it from data, scaled so that pure noise would be standard normal. Model

$$\begin{gathered}\delta_\ell \mid \lambda_\ell, \tau \sim N(0, \tau^2\lambda_\ell^2), \qquad \kappa_\ell = \frac{1}{1+\tau^2\lambda_\ell^2} \in [0,1], \\ \mathbb{E}[\delta_\ell \mid y_\ell] = (1 - \mathbb{E}[\kappa_\ell \mid y_\ell])\, y_\ell.\end{gathered} \tag{3.16}$$

So $\kappa_\ell = 1$ shrinks the locus to exact additivity and $\kappa_\ell = 0$ retains the deviation in full. In this notation:

Fisher's assumption is $\kappa_\ell = 1$ at every locus, asserted before any data.

Not that deviations are usually small, nor that they average out, but a point mass at $\kappa = 1$ imposed across the whole genome. By Theorem 3.16 the two allele version is the point null $d = 0$, so the claim is a sharp hypothesis on a single interpretable parameter, and testable in the framework Jeffreys built. Two tests follow, doing different jobs: Jeffreys's Bayes factor at one locus (Section 3.12) and global local shrinkage across the genome (Section 3.13).

3.11 Haldane 1932: genetics, priors, and Fisher

Before the machinery, a historical section, because what follows is less anachronistic than it looks. It was invented by a population geneticist of Fisher's own generation, in the same journal, within two years of Fisher's own attack on inverse probability, and it was invented to solve a linkage problem.

3.11.1 What Haldane did

Haldane (1932), *A note on inverse probability*, opens by objecting that Bayes's theorem assumes all values in the neighbourhood of the observed one are equally probable a priori, and proposes to ask what more reasonable assumptions could be made. Two things come out of the note.

The first is the prior that bears his name, $\pi(\theta) \propto \{\theta(1-\theta)\}^{-1}$, or $\mathrm{Beta}(0,0)$. His motivation was genetic: mutation rates and rare recombination fractions range over orders of magnitude, and a flat prior on θ asserts $\mathbb{P}(\theta < 10^{-6}) = 10^{-6}$, which is not a statement anyone would make about a mutation rate. Haldane's prior is flat in $\log\{\theta/(1-\theta)\}$, so it is indifferent to the order of magnitude rather than to the value.

The second, and for present purposes the more striking, is that his worked example is a mixture of a point mass and a continuous density: a spike and slab, more than half a century before the term. The example is linkage mapping in the Chinese primrose. Four hundred meioses, one hundred and sixty crossovers, recombination fraction ρ. Since *Primula sinensis* has twelve chromosomes of roughly equal length and unlinked loci recombine freely, he places a point mass of $11/12$ on $\rho = 1/2$, the two loci lying on different chromosomes, and spreads the remaining $1/12$ over the linked case.

Haldane's prior was cytology, not taste. The detail worth dwelling on is where the $11/12$ came from. It is a chromosome count: a physical fact about the organism, checkable by anyone with a microscope, and nothing about it is subjective or convenient. That is the model answer to the standard objection against priors, and it was produced inside genetics in the year Fisher was arguing that inverse probability was illegitimate. Etz and Wagenmakers (2017) argue that the conceptual step of assigning prior mass to a general law belongs to Wrinch and Jeffreys, but that Haldane's concrete mixture was an important development, that Jeffreys knew of it, and that it may have pushed him toward the implementation of 1935. Vogl (2017) should be read alongside: he argues that Haldane used a flat prior in this very example, against his own stated principle.

3.11.2 Why Haldane's prior is not Jeffreys's, and where both reappear

Both are uniform, on different scales, and the scale is the entire content. Writing $I(\theta)$ for the Fisher information of one Bernoulli trial,

$$\text{Bayes Laplace: } \pi \propto 1, \qquad \text{Jeffreys: } \pi \propto \sqrt{I(\theta)} = \{\theta(1-\theta)\}^{-1/2},$$
$$\text{Haldane: } \pi \propto I(\theta) = \{\theta(1-\theta)\}^{-1}.$$

Jeffreys takes the square root because he wants the volume element of the Fisher Rao metric, so his prior is flat in statistical distance and invariant under reparametrization. Haldane takes the information itself, which is flat in the natural parameter of the exponential family, the log odds. Haldane's posterior is

$\text{Beta}(y, n-y)$, so its mean is exactly y/n and there is no shrinkage at all, at the cost of impropriety when $y = 0$ or $y = n$; Jeffreys gives $\text{Beta}(y + \frac{1}{2}, n - y + \frac{1}{2})$, always proper, shrinking by half an observation at each end.

This is not a digression, because the identical ladder governs the shrinkage weights of (3.16). With $\lambda_\ell \sim C^+(0,1)$ and $\tau = 1$,

$$\kappa \;\sim\; \text{Beta}(\tfrac{1}{2}, \tfrac{1}{2}),$$

which is exactly Jeffreys's prior for a binomial proportion. Simulated quantiles of κ at levels $0.1, 0.25, 0.5, 0.75, 0.9$ are $0.0245, 0.1463, 0.5002, 0.8550, 0.9756$ against $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ values $0.0245, 0.1464, 0.5000, 0.8536, 0.9755$.

The reason this is the natural place for the invariance argument is that Jeffreys's rule applied to the *location* is useless here. For $y \sim N(\theta, 1)$ we have $I(\theta) = 1$, so the prior is flat, $\mathbb{E}[\theta \mid y] = y$, and there is no shrinkage; in three or more dimensions it is also inadmissible, and Jeffreys knew his rule misbehaved in multiparameter problems, which is why he modified it to $1/\sigma$ for the normal mean and variance. The horseshoe applies the argument instead to the shrinkage weight κ, which is a proportion: the fraction of an observation attributed to noise. And the Jeffreys prior for a proportion is $\text{Beta}(\frac{1}{2}, \frac{1}{2})$.

Within the generalized beta family (Armagan, Dunson and Clyde, 2011) the three classical binomial priors reappear as three shrinkage rules. The Strawderman and Berger prior, which places $\text{Beta}(\frac{1}{2}, 1)$ on κ (Carvalho, Polson and Scott, 2010), is listed for comparison; it has a pole at $\kappa = 0$ only, so it protects large signals but has no prong at exact additivity:

prior on κ	$\mathbb{P}(\kappa < 0.1)$	$\mathbb{P}(\kappa > 0.9)$	name
$\text{Beta}(1,1)$	0.100	0.100	Bayes Laplace; the uniform shrinkage prior
$\text{Beta}(\frac{1}{2}, 1)$	0.316	0.051	Strawderman (1971) and Berger
$\text{Beta}(\frac{1}{2}, \frac{1}{2})$	0.205	0.205	*Jeffreys; the horseshoe*
$\text{Beta}(0,0)$	$\to \frac{1}{2}$	$\to \frac{1}{2}$	*Haldane*; improper

The induced tails behave accordingly: $\mathbb{P}(|\theta| > 10)$ is 0.010, 0.050 and 0.376 for $\text{Beta}(1,1)$, $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ and $\text{Beta}(0.05, 0.05)$. The horseshoe is the Jeffreys member of the family and Haldane's is its improper limit, in which large signals are not shrunk at all, exactly as his binomial posterior returns the maximum likelihood estimate untouched. Figure 3.5 plots the three densities.

One caveat, since the correspondence is an interpretation rather than a mechanical derivation. If one applies Jeffreys's rule literally to the marginal scale model $y \mid \kappa \sim N(0, 1/\kappa)$, the log likelihood is $\frac{1}{2}\log\kappa - \frac{1}{2}\kappa y^2$, so $I(\kappa) = 1/(2\kappa^2)$ and $\sqrt{I} \propto \kappa^{-1}$, which is improper and sits on the Haldane side of the horseshoe rather than at it. The $\text{Beta}(\frac{1}{2}, \frac{1}{2})$ is justified by reading κ as an exclusion probability, not by running the rule on the scale model, and the two readings disagree in the direction of Haldane. The genuinely multivariate content is not in the $\text{Beta}(\frac{1}{2}, \frac{1}{2})$: it

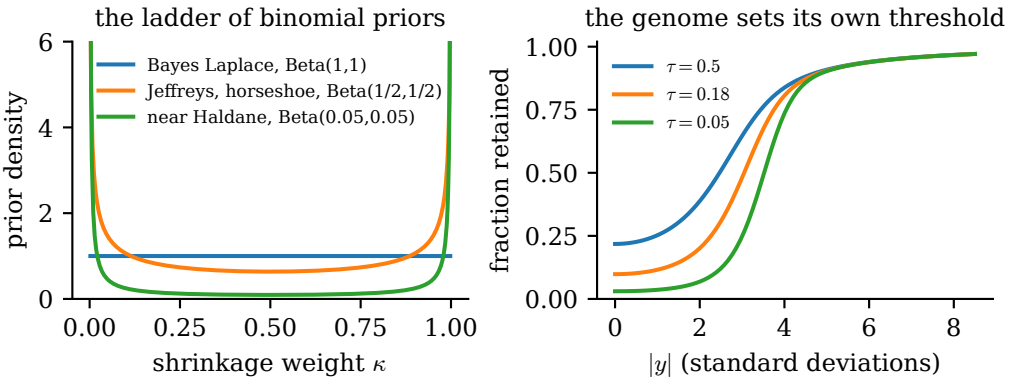


FIGURE 3.5. Left: the ladder of priors on the shrinkage weight κ reproduces the classical ladder of binomial priors, with the horseshoe as the Jeffreys member and Haldane's as the improper limit. Right: the resulting shrinkage rule at three global scales. Small deviations are erased and large ones pass through essentially untouched whatever the scale.

is in the shared global scale τ, which lets the coordinates borrow strength. Jeffreys's rule has no such mechanism, which is precisely where it fails in high dimension.

3.11.3 The coincidence worth noticing

Haldane's spike and slab was built to answer the question: *are these two loci linked?* It places its point mass on $\rho = 1/2$, free recombination, and its continuous part on $\rho < 1/2$.

That is the same question as the one in Part I. By Theorem 3.5 the remainder $\mathcal{R}$ vanishes when the motion is tangential to $\mathcal{M}$, and linkage disequilibrium is exactly a transverse coordinate; free recombination between unlinked loci is the case in which the transverse coordinate decays fastest, and linkage is what allows it to persist. So the first mixture prior in the history of statistics was written down to test the presence of the very structure that makes Fisher's assertion false, by a man working in the same field. Nobody, so far as we can discover, ever joined the two.

The chronology is worth stating plainly. Fisher (1930b) attacked inverse probability in the *Proceedings of the Cambridge Philosophical Society*. Haldane (1932) replied constructively in the same journal with a better prior and a mixture. Fisher (1932) replied in turn, in the same volume. Jeffreys (1935, 1936) then built the significance tests. *The Genetical Theory* had appeared in 1930 and the second edition, with the sentence at issue, in 1958. So the tools of Section 3.12 were assembled, in Fisher's own journal and partly by a fellow population geneticist, in the decades between the two editions of the book, and Fisher spent that period arguing that the tools were illegitimate.

3.12 The Jeffreys test: one locus, sharply

3.12.1 The sharp null, and Jeffreys's test

By Theorem 3.16, $\kappa = 1$ at a biallelic locus is the point null $d = 0$, with μ, a and the error scale as nuisance parameters common to both hypotheses and so assignable improper invariant priors without penalty. This is precisely the structure Jeffreys built his significance tests around: a sharp null asserting that a single new parameter is absent, a diffuse alternative, and nuisance parameters shared. Jeffreys (1935) treated the first cases; Jeffreys (1936) introduced the notation K for the ratio of posterior to prior odds on the null, with equal prior probabilities as a working convention he was careful not to call a belief.

There is something fitting in the arrangement. Fisher's own 1918 paper supplies the variance decomposition, Fisher's own theorem supplies the null, and the test is the one built by the contemporary who spent a decade disputing his foundations.

If genotypic fitnesses are estimated from replicated assays with a common error variance, the relevant statistic is the contrast

$$\hat{D} = \hat{w}_{11} - 2\hat{w}_{12} + \hat{w}_{22}, \qquad \operatorname{se}(\hat{D})^2 = \sigma^2\Big(\tfrac{1}{n_{11}} + \tfrac{4}{n_{12}} + \tfrac{1}{n_{22}}\Big), \qquad t = \hat{D}/\operatorname{se}(\hat{D}),$$

on ν degrees of freedom. With a Cauchy prior on the standardized dominance effect, Jeffreys's approximation (Jeffreys, 1961; see Ly, Verhagen and Wagenmakers, 2016, for a modern account of these default tests) is

$$K \;\approx\; \sqrt{\tfrac{\pi\nu}{2}}\left(1 + \tfrac{t^2}{\nu}\right)^{-(\nu-1)/2}. \tag{3.17}$$

Both facts that matter are visible already. The dependence on t is exponential, so evidence against additivity accumulates fast. The dependence on ν is only $\sqrt{\nu}$, so evidence in favour of additivity accumulates as the square root of the sample size. *A sharp null of this kind can be destroyed but never established.* At $\nu = 100$, $t = 2$ gives $K = 1.8$, no evidence at all despite a conventionally significant t; $t = 3.34$ gives $K = 0.067$; $t = 4.72$ gives $K = 5.9 \times 10^{-4}$.

3.12.2 An exact binomial version, with no tuning constant

Because a reader may object to the Cauchy scale, here is a version with no free constants. In a viability experiment with n zygotes of each genotype and $s_g \sim \mathrm{Bin}(n, \theta_g)$, θ_g proportional to fitness, put independent uniform priors on θ_{11}, θ_{22} under both hypotheses and set

$$H_0 : \theta_{12} = \tfrac{1}{2}(\theta_{11} + \theta_{22}), \qquad H_1 : \theta_{12} \sim U(0,1) \text{ independently.}$$

Then K is a ratio of a two dimensional integral to a product of Beta functions, with no tuning constant. Table 3.1 evaluates it at expected data for the Example 3.5.2 fitnesses rescaled to $\theta = (0.45, 0.60, 0.35)$ so that $d = 0.20$; for a small dominance $\theta = (0.45, 0.42, 0.35)$ so that $d = 0.02$; and for the null itself, $\theta = (0.45, 0.40, 0.35)$.

	$d = 0.20$		$d = 0.02$	$d = 0$
n per genotype	K	$1/K$	K	K
25	9.4×10^{-1}	1.1	3.4	3.5
50	3.4×10^{-1}	2.9	4.7	4.8
100	3.2×10^{-2}	31	6.4	6.7
200	2.1×10^{-4}	4.9×10^{3}	8.5	9.5
400	6.0×10^{-9}	1.7×10^{8}	10.7	13.4
1,000	8.4×10^{-23}	1.2×10^{22}	12.1	21.1
4,000	9.3×10^{-93}	1.1×10^{92}	4.6	42.1
20,000	$< 10^{-300}$	$> 10^{300}$	1.5×10^{-3}	94.2
100,000	$< 10^{-300}$	$> 10^{300}$	$< 10^{-10}$	210.6

TABLE 3.1. Jeffreys's K in favour of $\kappa = 1$, evaluated at expected data. $K < 1$ is evidence against Fisher's assumption. Computed by windowed quadrature about the posterior mode; see Section 3.30.

The table uses expected data. It is worth doing one calculation with counts of the kind an experiment actually produces, to see what the numbers look like in the hand.

EXAMPLE 3.18 (A Bayes factor from a viability assay). Suppose 500 eggs of each genotype are set up and the survivors to adulthood counted, as in a classical *Drosophila* viability assay. The counts below are illustrative, not data; they are chosen to be typical of such assays. Three outcomes are compared, differing only in the heterozygote:

outcome	s_{11}	s_{12}	s_{22}	$\hat{d}$	$\hat{D}/\operatorname{se}(\hat{D})$	two sided p	K
(i) overdominant	331	372	302	0.111	-4.49	7×10^{-6}	1.1×10^{-3}
(ii) borderline	331	340	302	0.047	-1.82	0.069	3.0
(iii) additive	331	318	302	0.003	-0.11	0.91	15.1

Here $\hat{\theta}_g = s_g/500$, $\hat{d} = \hat{\theta}_{12} - \frac{1}{2}(\hat{\theta}_{11} + \hat{\theta}_{22})$, $\hat{D} = \hat{\theta}_{11} - 2\hat{\theta}_{12} + \hat{\theta}_{22} = -2\hat{d}$, and $\operatorname{se}(\hat{D})^2 = \sum_g c_g^2 \hat{\theta}_g(1 - \hat{\theta}_g)/500$ with $c = (1, -2, 1)$. For outcome (i), $\hat{\theta} = (0.662, 0.744, 0.604)$, $\hat{D} = -0.222$ and $\operatorname{se}(\hat{D})^2 = (0.2238 + 4 \times 0.1905 + 0.2392)/500 = 0.00245$, so the standardized contrast is -4.49.

K is the exact Bayes factor of Table 3.1. Under H_1 the marginal likelihood is $\prod_g B(s_g + 1,\, 500 - s_g + 1)$; under H_0 it is the two dimensional integral

$$\int_0^1 \int_0^1 \theta_{11}^{s_{11}} (1 - \theta_{11})^{500 - s_{11}}\, \theta_{22}^{s_{22}} (1 - \theta_{22})^{500 - s_{22}}\, \bar{\theta}^{\,s_{12}} (1 - \bar{\theta})^{500 - s_{12}}\, d\theta_{11}\, d\theta_{22},$$

where $\bar{\theta} = \frac{1}{2}(\theta_{11} + \theta_{22})$; a grid of 1600^2 points centred on the mode evaluates it without difficulty.

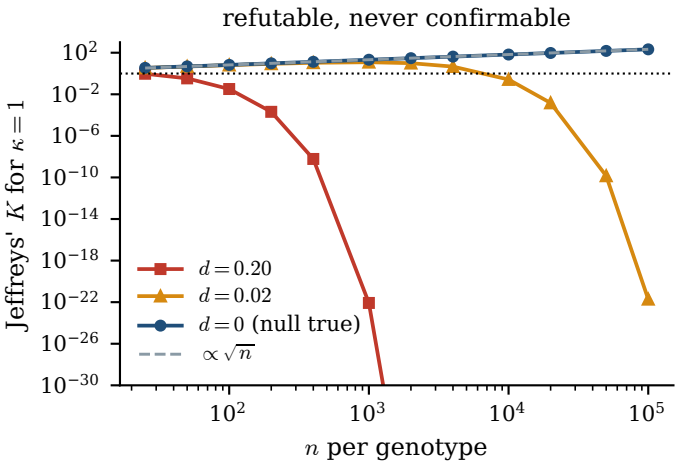


FIGURE 3.6. The asymmetry of a sharp null. Under the null (blue circles) the Bayes factor grows only as $\sqrt{n}$, reaching 211 at a hundred thousand zygotes per genotype. Under a visible dominance effect (red squares) it collapses geometrically. A small deviation (amber triangles) rises to apparent strong support for Fisher before collapsing at sample sizes no experiment reaches.

Three things can be read off. Outcome (i) is decisive against additivity: posterior odds of about 900 to one against $\kappa = 1$ from equal prior odds. Outcome (iii) is as favourable to Fisher as data with these outer counts can be; perfectly additive data, with $s_{12} = 316.5$, would give $K = 15.2$, so 15 is the ceiling for outer counts like these at 500 per genotype and roughly matches the $\sqrt{n}$ growth of the $d = 0$ column of Table 3.1. Outcome (ii) is the Lindley (1957) phenomenon in miniature: a contrast nearly significant at the five percent level, which a classical analyst would report as suggestive evidence of dominance, gives odds of three to one *in favour* of additivity. The classical and Bayesian readings of the same assay point in opposite directions, and only the second has the asymmetry of Section 3.12 built in.

3.12.3 Three readings

The asymmetry. Where the null is exactly true, K grows like $\sqrt{n}$: quadruple the sample size to double the evidence, and $K = 211$ at a hundred thousand zygotes per genotype, a size no viability experiment has attained. Where dominance is present, K collapses geometrically. Lindley (1988) reported the same asymmetry in the Hardy Weinberg problem, where the Bayes factor can favour equilibrium on data that classical tests reject. Fisher's assumption can be refuted but not confirmed, so a century of not having refuted it is close to no evidence at all.

The turnover. The $d = 0.02$ column is the most instructive. A dominance effect one tenth the size of Example 3.5.2's drives K up to 12 at $n = 1000$, which on Jeffreys's scale is strong support for $\kappa = 1$, before collapsing at $n = 20{,}000$. At every sample

size a real experiment uses, a small but strictly nonzero deviation masquerades as exact additivity and is actively rewarded by the evidence. Fisher's claim is false in the manner hardest to detect: not by a wide margin where anyone would notice, but by a small margin everywhere, accumulated over the genome. This is the Lindley (1957) phenomenon in its natural habitat.

The genome wide conjunction. Jeffreys (1936) opens by observing that independent batches of relevant evidence multiply in their effect on the probability of the null. Fisher's assertion requires $\kappa_\ell = 1$ at every locus at once, so if $\mathbb{P}(\kappa_\ell = 1 \mid \text{data}) = \pi$ independently at each of L loci then the posterior probability of Fisher's claim is π^L.

Per locus posterior null probability π	$L = 10^3$	$L = 10^4$
0.90	1.7×10^{-46}	$< 10^{-300}$
0.95	5.3×10^{-23}	1.7×10^{-223}
0.99	4.3×10^{-5}	2.2×10^{-44}
0.999	0.37	4.5×10^{-5}

Even at $\pi = 0.99$ per locus, ten thousand loci give 2.2×10^{-44}. Fisher's claim does not fail because some locus refutes it. It fails because it is a conjunction of thousands of sharp nulls, and conjunctions of sharp nulls have vanishing posterior probability under any evidence a finite experiment supplies. Wakefield (2010) makes the corresponding point for genome wide Hardy Weinberg testing. Note that this argument is genome wide by construction, which meets the objection that a one locus analysis cannot settle a whole genome claim.

3.12.4 An Ockham reading: Mercury and the additive model

The asymmetry of Section 3.12 has a well known reading in the philosophy of science, and setting the two side by side sharpens what is at stake.

Jefferys and Berger (1992) give the canonical illustration.[1] General relativity predicts an anomalous advance of Mercury's perihelion of 43 arcseconds per century with no adjustable parameter; the Newtonian alternatives of the period, an intramercurial planet or a modified inverse square exponent or an oblate sun, could accommodate that figure but could equally have accommodated many others. The measured value is 43.11 ± 0.21 arcseconds per century. Because the sharp theory concentrated its predictive mass where the alternative spread it, confirmation buys an Ockham factor, the ratio of prior predictive densities at the observed value. This is Jeffreys's simplicity postulate made quantitative, and it is the reason a parameter free theory is worth more when it survives.

[1] William H. Jefferys, the astronomer, is not Harold Jeffreys, whose tests Section 3.12 uses. The near homonymy is unfortunate and is not ours.

Fisher's claim is a Mercury type theory. It is worth being explicit that Fisher is on the favoured side of this argument. His assertion is the maximally sharp one available: a parameter free identity, $\kappa_\ell = 1$ at every locus, against an alternative carrying a free sparsity level. Of the two hypotheses in Section 3.10 it is Fisher's that Ockham's razor rewards, and Section 3.18 says why the reward is deserved rather than accidental: faced with $2^L - L - 1$ interaction parameters, the additive model is not a convenient simplification but the only well posed one available in 1930.

The reward is of the same size, and that is the problem. A sharp null against a diffuse alternative of prior width Δ, measured with standard error σ, earns roughly $\Delta/(\sigma\sqrt{2\pi})$ when confirmed. For Mercury, with $\sigma = 0.21$ arcseconds and a prior range of tens of arcseconds for an unexplained residual, that is a factor of order 10^2. For the Fundamental Theorem, the $d = 0$ column of Table 3.1 gives $K/\sqrt{n}$ falling from 0.69 at $n = 25$ to 0.666 from $n = 4000$ on, the Laplace limit $\{2\pi(v_{12} + \frac{1}{4}v_{11} + \frac{1}{4}v_{22})\}^{-1/2}$ with $v_g = \theta_g(1-\theta_g)$ at these null frequencies; so $K \approx 0.67\sqrt{n}$ under the true null, and at a hundred thousand zygotes per genotype the factor is 211: the same order.

So the difference between the two cases is not the size of the available reward. It is that Mercury needed the reward once and Fisher needs it L times. With $\pi = K/(1+K)$ the per locus posterior probability, the conjunction has probability $\pi^L \approx \exp(-L/K)$, so confirming it with posterior probability one half requires $K \gtrsim L/\log 2$, that is

$$n \gtrsim \left(\frac{L}{c\log 2}\right)^2, \qquad c \approx 0.67,$$

which is 4.6×10^6 at a thousand loci and 4.6×10^8 at ten thousand. At genome scale that is about an order of magnitude below the $n^* \approx 5 \times 10^9$ of (3.15) required to *refute* the claim at a one percent selection coefficient. The sample size at which the genome wide conjunction could first be believed and the sample size at which its falsehood would first become visible are of comparable magnitude, and neither is attainable.

What this adds. Three things. First, it corrects a natural misreading of Section 3.12: the conjunction π^L is not intrinsically hostile to a universal claim. Had Fisher been right, the same conjunction would have delivered K^L in his favour, which is an enormous number. The conjunction is hostile only because it magnifies whichever way the per locus evidence points, and the per locus evidence points against him.

Second, it locates precisely where Fisher's position differs from Einstein's. Both advanced a parameter free prediction. Einstein's concerned a single observable measured to half a percent; Fisher's concerns a latent parameter at each of 10^4 loci, none of which any experiment reports directly. A theory earns its Ockham reward only where its prediction is checkable, and the Fundamental Theorem has never been anywhere that its prediction is checkable.

Third, it explains why the argument of this chapter has to be genome wide rather than locus by locus. If the reward and the penalty both scale with L, then the estimand that a genome can speak to is a functional of the distribution of κ across loci, not the value of κ_ℓ at any one. That is the move of Section 3.13, and the Ockham accounting is a second route to it.

3.12.5 The Ockham factor in genetics, and in the horseshoe

It is worth asking whether genetics has an equivalent of the Ockham factor. It does, in at least two places, and in one of them the calculation has been performed daily since 1955 by people who would decline the label.

Morton's LOD three. The threshold for declaring linkage in human genetics is a LOD score of 3, that is a likelihood ratio of 1000 to one. The number does not come from a significance level. Morton (1955) derived it from prior odds: two loci chosen at random from the genome are unlikely to be linked closely enough to detect, with prior odds of roughly 1 in 50, so a likelihood ratio of 1000 leaves posterior odds of about 20 to one, or a posterior probability near 0.95. That is precisely the Ockham accounting of Section 3.12.4, with the prior odds supplied by genome length rather than by opinion, and it is the same calculation Haldane (1932) made from a chromosome count twenty three years earlier. The most widely used threshold in human genetics is a Bayes factor calibrated against a prior derived from cytology.

The genome wide threshold. The modern successor is the association threshold of 5×10^{-8}, which is a multiplicity correction with the same Bayesian reading: the prior odds that a given variant is causal are of order 10^{-5} to 10^{-6}, so a comparable Bayes factor is needed before the posterior odds become respectable. Wakefield (2007) makes the translation explicit and Stephens and Balding (2009) argue on exactly these grounds that association studies should report Bayes factors rather than p values. Churchill and Doerge's (1994) permutation threshold, discussed in Section 3.13, is the frequentist construction of the same quantity.

MacKay's formulation, and what κ is. MacKay (1992, 2003) gives the Ockham factor its standard modern form. The evidence for a model factorizes approximately as the best fit likelihood times the ratio of the posterior accessible volume of parameter space to the prior accessible volume, and that ratio is the penalty a model pays for having had parameters free to adjust. A model with no adjustable parameter pays nothing; a model whose posterior has collapsed to a small fraction of its prior range pays a great deal.

Applied to a single locus in the model of Section 3.10, the calculation is exact and identifies κ. With $y_\ell \sim N(\delta_\ell, 1)$ and $\delta_\ell \mid \kappa_\ell \sim N\big(0, (1-\kappa_\ell)/\kappa_\ell\big)$, the posterior is $N\big((1-\kappa_\ell)y_\ell,\, 1-\kappa_\ell\big)$, so

$$\text{Ockham factor} \;=\; \frac{\text{posterior width}}{\text{prior width}} \;=\; \frac{\sqrt{1-\kappa_\ell}}{\sqrt{(1-\kappa_\ell)/\kappa_\ell}} \;=\; \sqrt{\kappa_\ell}. \tag{3.18}$$

The shrinkage weight is the squared Ockham factor. That gives Section 3.10 a third reading alongside the two already offered. κ_ℓ is the fraction of the estimate discarded; it is the exclusion probability on which Jeffreys's prior is placed (Section 3.11.2); and it is the square of the volume ratio by which locus ℓ is penalized for having had an interaction parameter free to move.

In that third reading Fisher's assertion becomes: *the Ockham factor is one at every locus in the genome*. No interaction parameter was ever free, so none is penalized, and the additive model pays nothing anywhere. Table 3.2 is then an estimate of the Ockham factor genome wide, and its finding is that the mean of κ lies between 0.86 and 0.98, so the Ockham factor at that mean shrinkage, $\sqrt{\bar{\kappa}}$, lies between 0.93 and 0.99: Fisher's model pays almost nothing at almost every locus, and pays heavily at a few. The global scale $\hat{\tau}$ is the genome's own estimate of how much it should pay in total, which is why Section 3.20 asks for it to be reported.

3.12.6 The scale question

One difficulty must be recorded. Additivity is not scale invariant, so $\kappa = 1$ is not well posed until one says on which scale. Fisher's claim concerns $m = \log w$; the discrete recursion concerns w. Lindley's argument, transposed from frequencies to fitnesses, selects $\log w_{11} - 2\log w_{12} + \log w_{22} = 0$, that is multiplicative fitness, equivalently additive Malthusian parameters, which is Fisher's own scale.

This sharpens rather than repairs the failure. Put $w_{11} = c^2$, $w_{12} = c$, $w_{22} = 1$, so the invariant null holds exactly. Then

$$D = w_{11} - 2w_{12} + w_{22} = (c-1)^2 > 0 \quad \text{unless } c = 1,$$

so $\mathcal{R} > 0$ strictly and Fisher's term now *understates* the change. By Corollary 3.17 the shortfall is $(c-1)^2 pq/(2\bar{w})$: at $c = 1.5$, $p_1 = 0.3$ one has $\bar{w} = 1.3225$ and a shortfall of $0.25 \times 0.21/2.645 = 2.0$ percent, and at $c = 1.25$ it is 0.6 percent. The two nulls are mutually exclusive except under no selection. Jeffreys, whose programme was organized around invariance, would have raised this before any other objection.

3.13 The horseshoe test: the genome, softly

The sharp test cannot confirm. But Fisher's claim is not really about any single locus; it is about the whole distribution of κ across the genome, and that distribution *is* estimable, because information about it accumulates in the number of loci as well as in the sample size per locus. Replace the point mass by $\lambda_\ell \sim C^+(0,1)$, which by (3.16) gives κ_ℓ, at $\tau = 1$, a $\mathrm{Beta}(1/2, 1/2)$ prior with unbounded density at both endpoints, and estimate the global scale τ from all loci jointly. This is the horseshoe of Carvalho, Polson and Scott (2010). Van der Pas, Kleijn and van der Vaart (2014) showed that its posterior contracts around a sparse mean vector at the minimax rate, up to a constant, when τ is set at about the fraction of nonzero means.

3.13.1 The horseshoe in genetic terms

Before the formulae, the picture, since the procedure is easier to justify to a geneticist than its name suggests.

What κ_ℓ means at a locus. At locus ℓ one has an estimated nonadditive deviation y_ℓ, standardized so pure noise would be standard normal. Part of it is real and part measurement error, and one does not know the split. The weight κ_ℓ is the fraction discarded: $\kappa_\ell = 1$ throws the whole thing away and calls the locus additive, $\kappa_\ell = 0$ keeps it entire. So $\mathbb{E}[\kappa_\ell \mid y_\ell]$ answers the question a geneticist actually asks: how much of this locus's apparent nonadditivity should I believe?

Why a prior with mass at both ends. One wants to reach either verdict at any locus, additive or not, without knowing in advance which loci are which. A prior concentrated in the middle, say $\mathrm{Beta}(5,5)$, shrinks every locus by about half, blurring the genuine epistasis and retaining half the noise. What is wanted is weight at the two ends and little between, so the data can push each locus to one verdict rather than leaving everything at a compromise. $\mathrm{Beta}(\frac{1}{2},\frac{1}{2})$ has unbounded density at 0 and 1 and a minimum at $\frac{1}{2}$; drawn, it is a horseshoe, and the two prongs are the two verdicts. Fisher's assumption keeps one prong.

What τ means: a threshold the genome sets for itself. The global scale τ is one number estimated from all loci jointly, and it is best read as an answer to *how additive is this genome*. It fixes the effective threshold, in standard deviations, above which a locus is allowed to keep its deviation:

true fraction of loci nonadditive	fitted $\hat{\tau}$	threshold at which half the estimate survives
0.10	0.164	3.09σ
0.05	0.072	3.40σ
0.02	0.027	3.71σ
0.01	0.016	3.87σ

The sparser the truth, the higher the bar, and nobody chose the bar. This is the service a permutation threshold performs in QTL mapping (Churchill and Doerge, 1994): the analyst does not set the significance level, the genome wide distribution of the statistic sets it. The difference is that the horseshoe does it inside the estimator rather than as a separate calibration step, and it returns a graded weight rather than a verdict.

A real signal is never penalized for the company it keeps. The rule $\mathbb{E}[\delta_\ell \mid y_\ell] = (1 - \mathbb{E}[\kappa_\ell \mid y_\ell])y_\ell$ is a soft threshold. The fraction of the estimate retained is:

$\lvert y\rvert$	$\tau = 0.5$	$\tau = 0.18$	$\tau = 0.05$
0	0.218	0.098	0.030
2	0.388	0.201	0.069
3	0.646	0.481	0.248
4	0.838	0.806	0.735
6	0.940	0.939	0.939
8	0.967	0.967	0.967

Read the bottom rows. At six standard deviations the retained fraction is 0.94 whatever the global scale, and at eight it is 0.967: a genuinely large epistatic deviation passes through untouched even in a genome judged almost entirely additive. That is what the two prongs buy, and it distinguishes this from ridge or lasso, which shrink a large coefficient by an amount growing with the penalty, so a strong signal is punished for sitting among null ones. At two standard deviations, where the winner's curse lives, between 61 and 93 percent is erased. (These entries, and the thresholds above, are computed from (3.19) below; the fitted scales are those of Table 3.2.)

What it is not. The horseshoe is not a claim about biology and does not assert that genomes are sparse in their epistasis. It is a statement about what one may conclude when interaction parameters outnumber observations, and it degrades gracefully if the assumption is wrong: a genome with pervasive epistasis of a few standard errors per locus returns a large $\hat{\tau}$, a low threshold and little shrinkage. (Pervasive deviations each well below their standard errors are the exception, and Section 3.24 takes them up.) Fisher's $\kappa = 1$ has no such escape, returning zero whatever the genome contains.

How the numbers are computed. Everything in this section follows from a single one dimensional density. With $y_\ell \mid \delta_\ell \sim N(\delta_\ell, 1)$ and the prior (3.16), integrating out δ_ℓ gives $y_\ell \mid \kappa_\ell \sim N(0, 1/\kappa_\ell)$, and combining with the induced prior on κ_ℓ (Section 3.29) gives

$$p(\kappa \mid y, \tau) \;\propto\; \frac{e^{-\kappa y^2/2}\,(1-\kappa)^{-1/2}}{1-(1-\tau^2)\kappa}, \qquad 0 < \kappa < 1. \tag{3.19}$$

Given κ, $\delta \mid y, \kappa \sim N\big((1-\kappa)y,\, 1-\kappa\big)$. So $\mathbb{E}[\delta \mid y] = (1 - \mathbb{E}[\kappa \mid y])\, y$, and $\mathbb{E}[\delta^2 \mid y] = \mathbb{E}[1-\kappa \mid y] + y^2\, \mathbb{E}[(1-\kappa)^2 \mid y]$, each a ratio of two one dimensional integrals; the substitution $\kappa = 1 - u^2$ removes the singularity at $\kappa = 1$ and an ordinary quadrature suffices. The global scale is fitted by maximizing the marginal likelihood $\prod_\ell \int_0^1 N(y_\ell \mid 0, 1/\kappa)\, p(\kappa \mid \tau)\, d\kappa$ over τ. Two checks are available by hand: at $\tau = 1$ and $y = 0$, (3.19) is the Beta$(1, \frac{1}{2})$ density, so $\mathbb{E}[\kappa \mid y = 0] = 2/3$; and as $\lvert y\rvert \to \infty$ the factor $e^{-\kappa y^2/2}$ drives κ to zero whatever τ is, which is the tail robustness above.

EXAMPLE 3.19 (The horseshoe on a panel of twelve loci). Suppose a dominance scan of twelve loci returns the standardized contrasts in the first column below.

The values are illustrative: two loci look strongly nonadditive, two are borderline and the rest are of the size noise produces. The table gives the posterior shrinkage weight and the posterior mean deviation under two global scales: $\tau = 0.2$, as if borrowed from a genome wide fit, and $\hat{\tau} = 1.02$, the marginal maximum likelihood estimate from these twelve loci alone.

	$\tau = 0.2$		$\hat{\tau} = 1.02$	
y_ℓ	$\mathbb{E}[\kappa_\ell \mid y]$	$\mathbb{E}[\delta_\ell \mid y]$	$\mathbb{E}[\kappa_\ell \mid y]$	$\mathbb{E}[\delta_\ell \mid y]$
6.0	0.061	5.632	0.057	5.656
4.5	0.127	3.929	0.105	4.026
−2.7	0.610	−1.053	0.318	−1.840
2.1	0.766	0.492	0.445	1.166
1.6	0.834	0.266	0.539	0.738
−1.2	0.864	−0.163	0.595	−0.486
0.9	0.878	0.110	0.626	0.337
−0.8	0.881	−0.095	0.634	−0.293
0.6	0.886	0.068	0.647	0.212
0.4	0.890	0.044	0.656	0.137
−0.3	0.891	−0.033	0.659	−0.102
0.1	0.892	0.011	0.663	0.034

Four lessons. First, the two large contrasts are treated almost identically under the two scales: the locus at 6.0 keeps 94 percent of its estimate either way. Second, the global scale decides the fate of the middle of the table: the contrast at -2.7 keeps 39 percent under $\tau = 0.2$ and 68 percent under $\hat{\tau}$. Third, the small contrasts are shrunk toward zero but not to it, since the horseshoe has no point mass; under $\tau = 0.2$ their weights sit near 0.89, close to Fisher's $\kappa = 1$ but not at it. Fourth, the estimate of the nonadditive total depends on using the right functional. Under $\tau = 0.2$ the posterior mean of $\sum_\ell \delta_\ell^2$ is $\sum_\ell \mathbb{E}[\delta_\ell^2 \mid y] = 54.2$, whereas squaring the posterior means gives $\sum_\ell (\mathbb{E}[\delta_\ell \mid y])^2 = 48.6$; under $\hat{\tau}$ the two are 62.0 and 54.0. The unbiased moment estimate $\sum_\ell y_\ell^2 - 12$ is 62.0. Fisher's answer is 0.

A last caution, which the genome sized calculations below remove. Twelve loci say little about τ: the marginal log likelihood is -27.87 at $\hat{\tau}$, -28.38 at $\tau = 0.5$ and -29.99 at $\tau = 0.2$, a likelihood ratio of about eight across a fivefold range of scales. The point of pooling thousands of loci is to pin this one number down.

3.13.2 What the genome says

Table 3.2 reports simulations with $L = 5000$ loci, a fraction carrying real nonadditive deviations, one standardized estimate per locus, and τ estimated by marginal maximum likelihood. The target is the total nonadditive contribution $\sum_\ell \delta_\ell^2$, which is what Fisher's second term is built out of.

Three readings, and they are the substance of the empirical test.

design		estimates of $\sum_\ell \delta_\ell^2$					diagnostics		
sparsity	sd	truth	Fisher	unshrunk	plug in	horseshoe	$\hat{\tau}$	mean $\mathbb{E}[\kappa \mid y]$	top 5% share
0.10	3.0	4766	0	4971	3877	5046	0.164	0.859	0.966
0.05	3.0	1998	0	1952	1477	2085	0.072	0.932	0.988
0.02	2.5	541	0	779	374	644	0.027	0.974	0.993
0.01	4.0	623	0	486	396	555	0.016	0.985	0.998

TABLE 3.2. Global local shrinkage across $L = 5000$ loci, one simulated genome per row: a fraction of loci (sparsity) carries $\delta_\ell \sim N(0, \text{sd}^2)$, the rest $\delta_\ell = 0$, and $y_\ell \sim N(\delta_\ell, 1)$. Unshrunk is $\sum_\ell y_\ell^2 - L$; plug in is $\sum_\ell (\mathbb{E}[\delta_\ell \mid y])^2$; horseshoe is the posterior mean $\sum_\ell \mathbb{E}[\delta_\ell^2 \mid y]$ at the marginal maximum likelihood $\hat{\tau}$. The top five percent share is that of the plug in total. Fisher's $\kappa = 1$ forces the estimate to zero in every row.

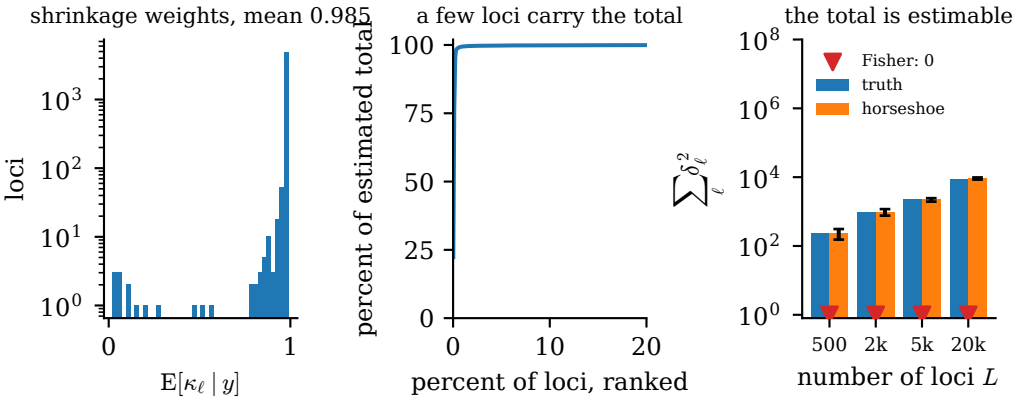


FIGURE 3.7. Genome scale behaviour. Left: the posterior shrinkage weights pile up near one, so Fisher is nearly right at nearly every locus, but the distribution is not the point mass he assumed. Centre: the estimated nonadditive total is concentrated in a tiny minority of loci. Right: the aggregate against L, averaged over replicate genomes, with bars of one Monte Carlo standard deviation; Fisher's answer is zero in every case.

Fisher is right about most of the genome. The posterior mean of κ runs from 0.86 to 0.98. Between eighty six and ninety eight percent of the shrinkage he imposed by assumption is shrinkage the data endorse. His instinct about the primacy of the additive component is not merely defensible; it is quantitatively confirmed.

The deviations are sparse, not absent. The top five percent of loci carry between 96.6 and 99.8 percent of the estimated nonadditive signal. This is the structure a $\text{Beta}(1/2, 1/2)$ prior on κ is designed to represent and a point mass cannot. It is also the reason the error is invisible locus by locus: at a randomly chosen locus Fisher is very nearly right.

The aggregate is wrong by the whole quantity. Fisher's answer for $\sum_\ell \delta_\ell^2$ is zero, against truths between 540 and 4800. The horseshoe recovers the totals to within 4 to 19 percent. Refusing to shrink is not the better alternative: the unshrunk moment

estimate is unbiased on average but accumulates the noise of every null locus, and in this table it errs by -22 to $+44$ percent. Nor is shrinking and then squaring: the plug in total is low in every row, by 19 to 36 percent, for the reason explained in Section 3.13.3. The choice is between a point mass and a prior that lets the loci speak separately, used with the right functional.

A precedent is worth noting. Shoemaker, Painter and Weir (1998) argued that no population is ever exactly in Hardy Weinberg equilibrium and that the useful question is the magnitude of the departure rather than its existence. Table 3.2 is that argument applied to δ rather than to genotype proportions.

3.13.3 Standard errors

Since Part I contains no standard errors by construction, the question arises only here. It divides into a one locus sampling question and a genome wide one, and the two answer differently.

One locus: the standard error goes to zero and the error does not. In a designed viability assay with n zygotes of each genotype and $s_g \sim \mathrm{Bin}(n, \theta_g)$, the allele frequency is fixed by the experimenter and only θ is estimated. With $\alpha = p\theta_{11} + (q-p)\theta_{12} - q\theta_{22}$ and $\sigma_A^2 = 2pq\alpha^2$, the delta method gives

$$\mathrm{Var}\left(\hat{\sigma}_A^2\right) \;\approx\; \frac{16p^2q^2\alpha^2}{n}\Big[p^2\theta_{11}\bar{\theta}_{11} + (q-p)^2\theta_{12}\bar{\theta}_{12} + q^2\theta_{22}\bar{\theta}_{22}\Big], \qquad \bar{\theta} = 1-\theta. \tag{3.20}$$

For Example 3.5.2 rescaled to $\theta = (0.45, 0.60, 0.35)$ with $p = 0.3$, this is a relative standard error of $638\%/\sqrt{n}$, confirmed by simulation to two figures: additive genetic variance is a quadratic functional of a small contrast and correspondingly hard to pin down. Against it sits the relative bias of 9.05%, which does not depend on n.

n per genotype	rel. s.e. (delta)	rel. s.e. (simulated)	rel. bias	s.e./bias
100	63.8%	64.6%	9.05%	7.05
1,000	20.2%	20.1%	9.05%	2.23
5,000	9.0%	9.1%	9.05%	1.00
10,000	6.4%	6.3%	9.05%	0.71
100,000	2.0%	2.1%	9.05%	0.22

The mean squared error of $\hat{\sigma}_A^2/\bar{w}$ for $\Delta\bar{w}$ is, for large n, $\mathrm{Var} + \mathcal{R}^2$, the first term $O(1/n)$ and the second fixed, with crossover at $n^* \approx 4975$. Below it the experiment is too noisy for anyone to have noticed; above it further data buys nothing, since what remains is bias. Everybody worked to the left of n^*, and no standard error reveals a bias.

Genome wide: the aggregate is estimable, its interval needs checking. The quantity Fisher got wrong is the total $\sum_\ell \delta_\ell^2$, and at genome scale it behaves better than any single locus. The table summarizes replicate genomes with five percent of loci

nonadditive, $\delta_\ell \sim N(0, 9)$ there, at four genome sizes (40, 40, 30 and 20 replicates); truth and posterior mean are averages over replicates, the Monte Carlo standard deviation is that of the error (posterior mean minus truth), and coverage is that of the nominal 95 percent interval, posterior mean plus or minus 1.96 posterior standard deviations.

L	truth	posterior mean	MC s.d. of error	rel. MC s.d.	posterior s.d.	plug in bias	coverage
500	233	232	39	19.2%	32	−31%	0.85
2,000	949	970	71	7.8%	65	−27%	0.90
5,000	2188	2198	102	4.8%	99	−29%	0.97
20,000	8876	9150	260	3.0%	202	−27%	0.60

Three things follow.

First, the good news, and it is the reverse of the Jeffreys asymmetry: no single locus can confirm $\kappa_\ell = 1$ at any sample size, but the aggregate is estimated to within three percent at twenty thousand loci, the relative error falling roughly like $L^{-1/2}$. Information about the total accumulates in L even though information about any sharp null does not.

Second, the functional matters more than the prior. The posterior mean of $\sum_\ell \delta_\ell^2$ is

$$\sum_\ell \mathbb{E}[\delta_\ell^2 \mid y] = \sum_\ell (\mathbb{E}[\delta_\ell \mid y])^2 + \sum_\ell \mathrm{Var}(\delta_\ell \mid y),$$

and with it the horseshoe is close to unbiased at every genome size, within three percent on average. Squaring the posterior means and summing, the plug in that one is tempted to compute, drops the variance term and is biased low by 27 to 31 percent at every L. The reason is structural: a shrinkage estimator moves exactly the loci near its threshold from the mean into the variance, and those are the loci that make up much of a quadratic total. So the plug in is a lower bound on the nonadditive total, and the posterior mean is an estimate of it. Either way Fisher's answer of zero is not in contention.

Third, the credible interval for the total cannot be taken on trust. Its coverage moves between 60 and 97 percent across the four genome sizes, and at twenty thousand loci a relative bias of three percent is already larger than the posterior standard deviation. This is the phenomenon described by van der Pas, Szabó and van der Vaart (2017): horseshoe credible sets with an adaptively chosen sparsity level are honest only under a restriction on excessive bias, and a quadratic functional dominated by signals near the threshold is close to the excluded case. The comparison with the unshrunk moment estimate $\sum_\ell y_\ell^2 - L$ is also instructive: here the noise variance is known exactly, the moment estimate is unbiased, and its relative standard deviation (22.5, 8.4, 5.6 and 2.3 percent) is comparable to the horseshoe's, so for the total alone shrinkage buys little when the noise model is exact. What shrinkage buys is the locus level picture of Table 3.2, which the moment estimate cannot give, and robustness when per locus standard errors must themselves be estimated. One should therefore report the horseshoe estimate of

the nonadditive total as a point estimate with a Monte Carlo or bootstrap standard error, and quote its credible interval only after checking its coverage by simulation.

3.14 The sampling theory of selectively neutral alleles

This section makes explicit a structure that is present in Ewens (1972) and is usually left unnamed.

3.14.1 The sampling formula is a marginal likelihood, and the prior is forced

The infinitely many alleles model with scaled mutation rate θ has a hierarchy with three layers, whether or not one calls it that. The unobserved population allele frequencies, in decreasing order, follow the Poisson Dirichlet law with parameter θ, which is the law of the ranked atoms of a Dirichlet process. Given those frequencies, a sample of n genes is drawn independently. Only the allelic partition of the sample is recorded: a_j is the number of alleles represented exactly j times among the n genes, so that $\sum_j j a_j = n$.

Integrating the second layer against the first and recording only the partition gives exactly

$$\mathbb{P}(a_1, \dots, a_n) \;=\; \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{\theta^{a_j}}{j^{a_j} a_j!}, \qquad \theta_{(n)} = \theta(\theta+1)\cdots(\theta+n-1),$$

the Ewens Sampling Formula. In the vocabulary of Section 3.10: Poisson Dirichlet is the prior, i.i.d. sampling is the likelihood, the partition is the data, θ is the hyperparameter, and the sampling formula is the *marginal likelihood* obtained by integrating the prior out. Antoniak (1974) and Kingman (1975) set this out; Ferguson's Dirichlet process (1973) is the same object approached from the other side.

Kingman's paintbox is de Finetti's theorem for partitions. The reason this is not an accident of algebra is a representation theorem.

De Finetti's theorem says that an infinite exchangeable sequence is a mixture of independent identically distributed sequences, and that the mixing measure is unique. The content is that exchangeability does not permit a prior; it *forces* one. A modeller who assumes his observations are exchangeable has thereby assumed a prior, and the only remaining choice is whether to write it down.

Kingman's paintbox theorem is the exact counterpart for exchangeable random partitions of the integers. Every such partition is a mixture over frequency vectors $(p_1, p_2, \dots)$ with $\sum p_i \le 1$, and conditionally on the frequencies the partition arises by independent sampling from them, with the residual mass producing singletons. The paintbox measure is the de Finetti measure of the problem.

Apply it. Ewens (1972) assumes selectively neutral alleles with every mutation novel, that the sampled genes are exchangeable, and that only the partition is observed. Kingman's theorem then guarantees the formula is a mixture, and

	Ewens (1972)	global local shrinkage
latent layer	population allele frequencies	per locus deviations δ_ℓ
prior	Poisson Dirichlet(θ)	scale mixture with global scale τ
hyperparameter	$\theta = 4Nu$	τ
data	allelic partition of a sample	standardized contrasts y_ℓ
marginal likelihood	the sampling formula	$\prod_\ell \int N(y_\ell \mid \delta)\pi(\delta \mid \tau)\, d\delta$
hyperparameter estimate	K_n sufficient, solve $\mathbb{E}[K_n] = k$	marginal maximum likelihood
representation theorem	Kingman paintbox	de Finetti

TABLE 3.3. The hierarchy of Ewens (1972) and the hierarchy of global local shrinkage, term by term.

identifies the mixing measure as Poisson Dirichlet(θ). The prior was not chosen and could not have been avoided: its existence is implied by the exchangeability and its form by the neutral model, both assumed on biological grounds, in the same way that Haldane's 11/12 was implied by the chromosome count.

3.14.2 The estimator of θ is empirical Bayes

The θ dependence of the sampling formula is $\theta^{K_n}/\theta_{(n)}$, where K_n is the number of distinct alleles, so K_n is sufficient for θ and the maximum likelihood estimate solves

$$\sum_{i=0}^{n-1} \frac{\theta}{\theta + i} \;=\; k.$$

We checked this against simulated Chinese restaurant samples: at $n = 100$, $\theta = 5$ the formula gives $\mathbb{E}[K_n] = 15.72$ against a simulated 15.76, and a single sample with $k = 15$ returns $\hat{\theta} = 4.66$.

That procedure is maximization of a marginal likelihood over a hyperparameter with the latent layer integrated out. It is type II maximum likelihood, which is to say empirical Bayes, and it is the identical operation by which $\hat{\tau}$ is obtained in Section 3.13. The two hierarchies line up term for term, as Table 3.3 shows.

3.14.3 Four priors already in use

The sampling formula is the clearest instance but not the only one. Since an anti Bayesian position is rarely a position without priors, it is worth listing the ones actually in play in this subject. All four are justified by mechanism rather than taste, which is what makes them invisible.

The additive subspace. The sharpest, because it sits inside the Fundamental Theorem itself. The average effects are defined by minimizing $\sum_{u,v} P_{uv}(m_{uv} - b_u - b_v)^2$, a projection onto the space of functions additive in gene content. Nothing forces that subspace: out of 2^L dimensions it selects $L + 1$ and discards every interaction coordinate before any data are seen. The justification is biological, namely that genes are transmitted singly and segregation dismantles the rest. That is a good reason and it is precisely a prior justified by mechanism. Fisher's error was to promote it to a theorem, not to have selected it, but selected it was.

Exchangeability and the neutral model, which force Poisson Dirichlet. As above.

The measure hiding in generically nonzero. The claim that the second term of (3.2) is not in general zero is, as mathematics, unimpeachable: the zero set of a nontrivial analytic function has Lebesgue measure zero. But the step from *the exceptions have measure zero* to *therefore this matters for real populations* requires that populations do not sit in or near the exceptional set, and that is a claim about where organisms live rather than a theorem. It is not idle: the exceptional set contains the one locus continuous time case with Hardy Weinberg proportions maintained, where $\mathcal{R} \equiv 0$ exactly, and $\mathcal{R}$ stays small, only $0.28\,s$ of Fisher's term, in a large neighbourhood of it.

Conditioning on K rather than integrating θ out. The Ewens Watterson test uses the sampling formula conditional on the number of alleles, because that conditional is free of θ. We verified it: at $n = 8$ and $K = 3$ the conditional distribution over partitions is $(0.255864, 0.307036, 0.255864, 0.095949, 0.085288)$ at $\theta = 0.5$, at $\theta = 2$ and at $\theta = 10$, identical to six figures. That is disposal of a nuisance parameter by conditioning on a sufficient statistic; a Bayesian disposes of the same parameter by integration. The two routes differ in tactics and both require the same three layer hierarchy to be in place first.

The pattern is that an anti Bayesian practice is rarely one without priors, but one whose priors are so well motivated that nobody calls them priors: genes transmitted singly, mutations always novel, samples exchangeable, chromosomes countable. These are Haldane's kind, derived from mechanism and checkable against the organism. Worth adding is where no prior is used at all, and rightly: the refutation itself requires no measure on anything, which is why Part I is written without one.

3.14.4 Pooling is not a Bayesian imposition

Equation (3.15) says the sample sizes needed to estimate small selective differences locus by locus are impossible. One might conclude that the topic goes nowhere. The conclusion does not follow, and the reason is the point of a hierarchical model. Pooling across loci rests on three independent legs, only one of which is Bayesian.

De Finetti. If, before seeing the data, one has no reason to single out any particular locus, the per locus deviations are exchangeable, and the representation theorem says they are conditionally independent draws from a

common distribution G. The hierarchical model is not an assumption added on top; it is what exchangeability means. The same move was accepted in 1972 for genes.

Stein. For three or more parallel estimation problems (James and Stein, 1961), the unshrunk estimator is inadmissible under total squared error: the James Stein rule, shrinking toward a fixed common centre, strictly dominates it, uniformly, for every parameter value. That is a frequentist theorem requiring no prior and no belief. It says that refusing to pool is not a neutral or cautious choice but a dominated one.

Robbins (1956). In a compound decision problem with many parallel instances, the empirical distribution of the observations can be used to construct a decision rule attaining, as the number of instances grows, the risk of the best rule one could have chosen knowing G. The prior is estimated, not asserted, and the justification is frequentist risk rather than coherence.

Together these say that when the per locus problem is unanswerable, the estimand should be changed from δ_ℓ to a functional of G, and information about G accumulates in the number of loci rather than in the depth at any one. The impossibility of the per locus problem is the premise of the argument, not an objection to it.

PART III CONSEQUENCES FOR PRACTICE

3.15 Two biases, and why they get confused

Two entirely different things are called bias in this subject, and the confusion between them is the practical reason the situation has stayed muddled. Keep them apart by asking what would happen with infinite data.

The structural bias $\mathcal{R}$. Using $\sigma_A^2/\bar{w}$ for $\Delta\bar{w}$ is wrong by $\mathcal{R}$ even when P and m are known exactly. It is not a statistical bias but the mathematical error of Part I, and no quantity of data reduces it: 9.05% of Fisher's term at any n.

The plug in bias. $\sigma_A^2 = 2pq\alpha^2$ is *quadratic* in the parameters, and squaring a noisy estimate inflates it, since $\mathbb{E}[\hat{\alpha}^2] = \alpha^2 + \mathrm{Var}(\hat{\alpha})$. Even the deterministic σ_A^2 is therefore overestimated by the obvious estimator, by $2pq\,\mathrm{Var}(\hat{\alpha})$, a relative $10.19/n$ here, which does vanish with data. The two biases coincide at $n = 113$.

n per genotype	plug in bias	structural bias	rel. s.e.	largest term
25	40.8%	9.05%	127.7%	noise
50	20.4%	9.05%	90.3%	noise
113	9.0%	9.05%	60.0%	noise
500	2.0%	9.05%	28.5%	noise
4,975	0.2%	9.05%	9.0%	structural
20,000	0.05%	9.05%	4.5%	structural

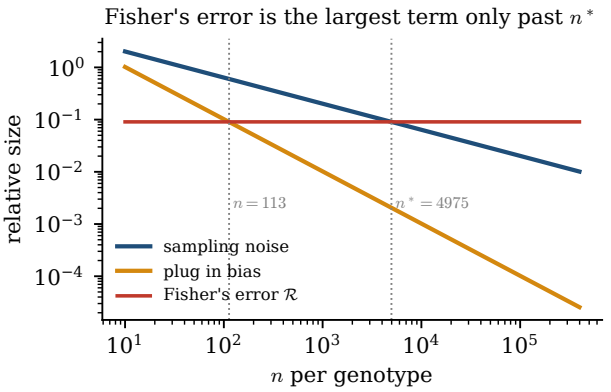


FIGURE 3.8. Three sources of error in estimating $\Delta\bar{w}$ from a viability assay. Sampling noise falls like $n^{-1/2}$, the plug in bias like n^{-1}, and Fisher's structural error not at all. Below $n = 113$ the estimator's own bias exceeds Fisher's; only past $n^* = 4975$ does Fisher's error become the largest term, and that is at $s = 0.3$.

Read the last column. At every sample size a real experiment has used, the dominant term is noise rather than either bias; below a hundred zygotes per genotype the estimator's own upward bias exceeds Fisher's error; and Fisher's error becomes the largest term only past five thousand, at $s = 0.3$, which is not a realistic selection coefficient. This is a further reason nobody noticed.

3.16 The delta method, and what it does not cover

The delta method is worth stating in full because its two outputs are exactly the two things being confused.

Let $\hat{\vartheta} = \vartheta + \varepsilon$ with $\mathbb{E}\varepsilon = 0$ and $\operatorname{Var}\varepsilon = \Sigma$, and let g be smooth. Expanding to second order and taking the variance uses the *first* derivative while taking the expectation uses the *second*:

$$\operatorname{Var}\big(g(\hat{\vartheta})\big) \approx \nabla g^{\top}\Sigma\nabla g, \qquad \mathbb{E}\big[g(\hat{\vartheta})\big] - g(\vartheta) \approx \tfrac{1}{2}\operatorname{tr}\big(\nabla^2 g\,\Sigma\big).$$

For $g = \sigma_A^2 = 2pq\alpha^2$ with $\alpha = c^{\top}\theta$, $c = (p, q-p, -q)$, one has $\nabla g = 4pq\alpha\, c$ and $\nabla^2 g = 4pq\, cc^{\top}$, giving $\operatorname{Var}(\hat{\sigma}_A^2) \approx 16p^2q^2\alpha^2\, c^{\top}\Sigma c$ and bias $\approx 2pq\, c^{\top}\Sigma c$, the two numbers of Section 3.15.

Now the point. Both $\mathcal{R}$ and the delta method bias are second order Taylor terms, they look alike, and they are different objects. $\mathcal{R}$ is second order in a *deterministic* expansion of $\bar{w}(p)$ in Δp: the population moving. The delta method bias is second order in a *stochastic* expansion in $\hat{\theta} - \theta$: the sample not being the population. Compute a delta method standard error, or bootstrap one, and the second is addressed while the first is untouched. The ritual of uncertainty quantification is complete and the error survives it.

3.17 Why the failure needs two or more loci

Theorem 3.5 says $\mathcal{R} = 0$ when the motion is tangential to $\mathcal{M}$. Whether the population can move transversally is a question about which coordinates *persist*.

At one locus with random mating and discrete generations, selection does push the adult genotype proportions off Hardy Weinberg, but random mating restores them exactly among next generation's zygotes. The transverse coordinate exists within a generation and is annihilated at the start of the next: no memory, not a state variable, unable to accumulate. In continuous time with Hardy Weinberg maintained, $\mathcal{R} \equiv 0$ identically. So the entire persistent state is the vector of allele frequencies, every direction of motion is tangential, and Fisher's claim is not approximately right but right.

With two loci this changes qualitatively. Recombination does not set the linkage disequilibrium D to zero in one generation; it multiplies it by roughly $(1 - r)$. So D is a genuine state variable with its own dynamics and its own memory, selection can drive it, and epistasis means it carries fitness. Section 3.5.3 is the extreme case; Section 3.6 the generic one.

The counting makes it stark. Under random mating the persistent state of an L locus biallelic system is the gamete frequency vector, a simplex of dimension $2^L - 1$, of which exactly L coordinates are allele frequencies.

loci L	dimension of state	visible to σ_A^2	fraction visible
1	1	1	100%
2	3	2	66.7%
3	7	3	42.9%
5	31	5	16.1%
10	1,023	10	0.98%
20	1,048,575	20	0.0019%

> The additive genetic variance tracks L of the $2^L - 1$ dimensions of the state. Fisher's assertion is that the population never moves in the other $2^L - 1 - L$.

At $L = 1$ that assertion is a theorem. At $L = 2$ it is a substantive and false claim about one third of the state space being inert. At $L = 20$ it is a claim about all but two thousandths of one percent. This is why the one locus case is not a special case of the general one but its opposite, and it is the sharpest form of the insistence that a one locus analysis cannot settle a whole genome claim (Figure 3.9, left).

3.18 Sparsity is not a convenience

The same counting explains why sparsity, which can look like a modelling fashion, is forced.

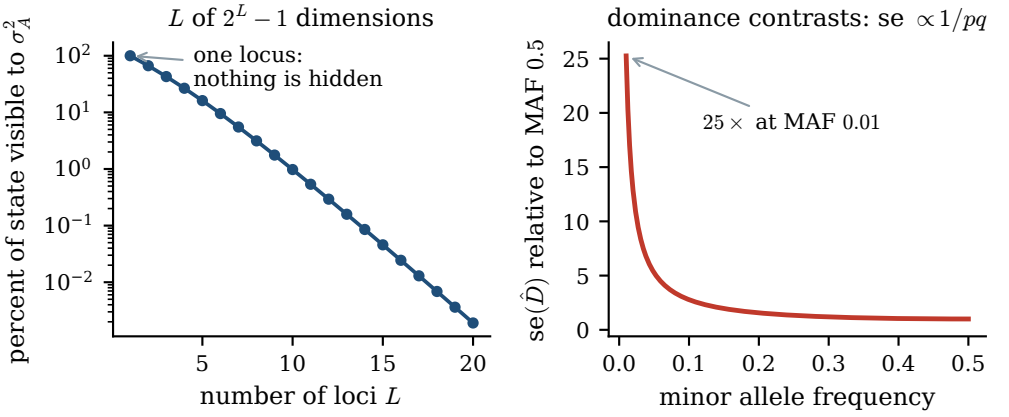


FIGURE 3.9. Left: the fraction of the persistent state that the additive genetic variance can see, L of $2^L - 1$ dimensions. Right: dominance contrasts are strongly heteroskedastic, with $\text{se}(\hat{D}) \propto 1/pq$, so the standard errors vary by a factor of twenty five across the allele frequency spectrum. Any genome wide analysis must be built around per locus standard errors.

The additive model on L biallelic loci has $L + 1$ parameters. The saturated model on gametes has 2^L. So there are $2^L - L - 1$ interaction parameters: about a billion at $L = 30$, and beyond any conceivable data by $L = 60$. No experiment estimates them. No experiment ever will.

Some assumption restricting the interactions is therefore not optional; the problem is not well posed without one. It cannot be a smoothness bound, since epistasis is not smooth in gene content. A magnitude bound, such as a dense variance component spreading a small effect over every interaction, does make the problem answerable, but only in aggregate: it estimates how large the interactions are in total and singles out none of them. If the interactions that matter are to be found, what remains is an assumption about how *many* of them matter. That is sparsity, and it enters not because it is elegant but because it is the only class of assumption under which an exponentially overparameterized problem says which interactions matter. The two are combined in Section 3.22.

Seen this way, Fisher and the horseshoe are doing the same thing and differ in one respect:

	Fisher (1930a, 1958)	global local shrinkage
assumption	$\kappa_\ell = 1$ for all ℓ	$\kappa_\ell \sim \text{Beta}(1/2, 1/2)$ at $\tau = 1$
interactions retained	none	a data determined few
sparsity level	fixed at zero, a priori	estimated, as $\hat{\tau}$
refutable	no	yes

Fisher closed an exponentially overparameterized problem by fiat in 1930. That was reasonable and, by Table 3.2, close to right; his mistake was to write the fiat as a theorem. The modern move is the same instinct made falsifiable, and the number carrying the content is $\hat{\tau}$, an estimate of how additive a genome is.

Sparsity also explains the one favourable result of Section 3.13.3: because a few loci carry almost the whole nonadditive total, estimating a few things well suffices for the aggregate, whose relative error therefore falls like $L^{-1/2}$ even though no single locus is ever resolved.

3.19 Why more parameters make it worse

Five effects compound, and they push in the same direction.

The invisible subspace grows exponentially: L tangential coordinates against $2^L - 1 - L$ transverse ones.

Conjunctions of sharp nulls die: Fisher's claim has posterior probability π^L, which at $\pi = 0.99$ and $L = 10^4$ is 2.2×10^{-44}. Adding loci strictly hurts a universal claim, however well each locus behaves.

Plug in bias in quadratic functionals scales with the parameter count: the delta method bias is $\frac{1}{2}\operatorname{tr}(\nabla^2 g\,\Sigma)$, a trace over the parameter space, so with p parameters and n observations it is $O(p/n)$ rather than $O(1/n)$. In the modern regime where p is comparable to n it does not vanish at all.

Selection on the estimate inflates it: if loci are chosen for attention because $\hat{\delta}_\ell$ was large, the retained values are biased upward. Shrinkage corrects this automatically; plug in estimation does not.

The uncertainty quantification degrades exactly where it is needed: Section 3.13.3 found the nominal 95 percent credible interval for $\sum_\ell \delta_\ell^2$ covering as little as 60 percent of the time.

3.20 A prescription

What we would ask of a genomics study that uses any of this, ordered by how much difference each makes.

1. *Name the estimand.* State whether the reported quantity is σ_A^2, the quantity effect, or $d\bar{m}/dt$, the total. They are different numbers and the literature routinely uses the language of the second while computing the first. The phrase *rate of increase in fitness* should not appear over a variance component.
2. *Write $\kappa = 1$ down as an assumption when you use it.* A study reporting σ_A^2 and concluding something about the change in mean fitness has assumed every nonadditive deviation in the genome is exactly zero: a strong assumption, almost certainly false, belonging in the assumptions rather than in the arithmetic.

3. *Do not let a standard error stand in for a check on the formula.* By Section 3.16 the delta method and the bootstrap quantify noise in the inputs and are silent about $\mathcal{R}$. Report both, or state which has not been addressed.

4. *Estimate the sparsity level and report it.* $\hat{\tau}$, or better the whole estimated G, is the interesting number and is almost never reported. The choice is not between assuming all interactions vanish and estimating all of them, neither being possible, but between fixing the sparsity level and estimating it.

5. *Use per locus standard errors.* Dominance contrasts have $\mathrm{se}(\hat{D}) \propto 1/pq$, a twenty five fold spread from minor allele frequency 0.5 to 0.01 (Figure 3.9, right). A homoskedastic analysis is not an approximation here, it is a misspecification.

6. *Use the correct functional.* $\sum_\ell (\mathbb{E}[\delta_\ell \mid y])^2$ is not the posterior mean of $\sum_\ell \delta_\ell^2$, which requires the $\sum_\ell \mathrm{Var}(\delta_\ell \mid y)$ term.

7. *Do not quote shrinkage credible intervals for quadratic functionals unchecked.* Report a point estimate with a Monte Carlo or bootstrap standard error, and check any interval's coverage by simulation.

8. *Design across loci, not within them.* Detection at a single locus needs $n \approx 50\, s^{-4}$, which is unattainable, while information about the distribution of κ accumulates in L. A genome wide design with modest per locus depth therefore beats a deep design at few loci, which inverts the usual instinct.

9. *Test the tangency, which is measurable.* Whether $\mathcal{R}$ is large is a question about whether the association coordinates are moving, and those are observable: the Hardy Weinberg disequilibrium coefficient in Lindley's parametrization and the linkage disequilibrium spectrum. A study reporting both stationary has earned Fisher's assumption for its population. A study reporting neither has assumed it.

Item 9 is the one we would press hardest, because it converts a foundational dispute into an empirical question. By Corollary 3.10 the constant genic environment condition is exactly $\dot{\alpha} = 0$, necessary as well as sufficient at a biallelic locus. Nobody need argue about whether the Fundamental Theorem holds in general. One can look.

PART IV CONTEXT, PROGRAMME, AND LIMITS

3.21 Where this programme already lives

Read alone, Part II sounds like a proposal to build something new. It is not. Almost every component exists, most of it built by Matthew Stephens and Peter Carbonetto,

and the lineage runs back through Donnelly and Pritchard to the 1972 sampling formula.

Hoppe (1984) showed that the sampling formula arises from a Pólya urn scheme, and Donnelly (1986) built on it: each new gene is a novel allele with probability proportional to θ, or a copy of an existing one with probability proportional to its current count. That urn is the Chinese restaurant process and the rule is the posterior predictive of a Dirichlet process. Donnelly and Tavaré (1986) worked out the corresponding genealogical structure. So the Bayesian skeleton of Section 3.14 was articulated by a population geneticist a decade before the machine learning literature adopted the same objects.

Fourteen years later, Pritchard, Stephens and Donnelly (2000) built STRUCTURE on Dirichlet priors over allele frequencies and admixture proportions. Stephens and Donnelly (2000) gave importance sampling for coalescent likelihoods; Stephens, Smith and Donnelly (2001) used an approximate coalescent prior for haplotype reconstruction. None of this was framed as an argument about Bayes. It was framed as population genetics, and the priors were the biology.

A last observation on the same theme. Li and Stephens (2003) introduced a hidden Markov copying model in which each new haplotype is an imperfect mosaic of those already seen: an approximation to the coalescent and, structurally, a prior over haplotypes, on which essentially every modern imputation, phasing and local ancestry pipeline is built. Any contemporary genomic data set has therefore had a prior applied before anyone sees the genotypes, and the prior does real work, since it is what fills in the untyped variants. One may decline to be a Bayesian in one's own analysis; one cannot decline the prior embedded in one's data.

3.22 The empirical Bayes programme, and two corrections

The argument of Part II reduces to: do not fix the sparsity level, estimate it. That is the organizing idea of Stephens's methodological work, executed more carefully than here.

Adaptive shrinkage (Stephens, 2017) takes a vector of estimates with standard errors, posits a unimodal prior g on the underlying effects, estimates g by nonparametric maximum likelihood, and returns shrunken estimates and local false sign rates. The prior is not chosen at all; it is a nonparametric object fitted to the data.

Multivariate adaptive shrinkage (Urbut, Wang, Carbonetto and Stephens, 2019) extends this to many conditions, learning the covariance structure of effects rather than assuming it. In the present problem the analogue would be learning how nonadditive deviations covary across environments or populations, which is a question nobody has asked of dominance deviations.

Variable selection (Servin and Stephens, 2007; Carbonetto and Stephens, 2012) puts the spike and slab into association analysis with the mixing weight estimated

from data, and *SuSiE* (Wang, Sarkar, Carbonetto and Stephens, 2020) makes the fine mapping version tractable.

3.22.1 Comparison with the horseshoe

Both are normal scale mixtures; they differ in what they do with the mixing measure.

	horseshoe	adaptive shrinkage
mixing measure	half Cauchy, fixed shape	unimodal, estimated by NPMLE
free parameters	one (τ)	mixture weights on a scale grid
point mass at zero	impossible	available
noise variance	common	per observation se_ℓ
computation	MCMC or EB search	convex, seconds
tails	Cauchy; never overshrinks	set by the largest grid scale
headline output	posterior mean	posterior mean and local false sign rate

Correction one. For the estimand that matters, a functional of G, a nonparametric estimate of G is the better instrument. Our $\hat{\tau}$ is a one parameter summary of what adaptive shrinkage estimates as a whole distribution, and heteroskedasticity settles the matter: dominance contrasts have se $\propto 1/pq$, so per locus standard errors vary twenty five fold across the allele frequency spectrum, and the horseshoe as used here has no place to put them.

Correction two. Zhou, Carbonetto and Stephens (2013) introduced the Bayesian sparse linear mixed model, combining a sparse spike and slab component with a dense polygenic one. For this problem that is probably the right model class and a pure sparsity prior is not. The biology suggests two contributions: dominance deviations at ordinary segregating loci, numerous and individually tiny, since inbreeding depression testifies to directional dominance almost everywhere and Section 3.7 put their whole genome contribution at about a tenth of a percent spread thin; and epistatic contributions, which by Section 3.6 are not damped by recombination and are plausibly concentrated. That is a dense background plus a sparse foreground. We would now propose the two component model, and expect the dense component to absorb the dominance and the sparse component to carry the epistasis. That is a testable prediction.

Where the horseshoe wins. Tail robustness. Because the mixing measure has genuinely Cauchy tails, a large signal is never overshrunk: 0.94 retained at six standard deviations and 0.967 at eight, identical across τ. A finite scale grid inherits whatever tail its largest component provides.

An awkwardness worth naming. Fisher's hypothesis is a point mass at $\kappa = 1$, and the horseshoe is continuous, so $\mathbb{P}(\delta = 0) = 0$: it cannot represent the hypothesis being tested. That is why Section 3.12 used a spike and slab and Section 3.13 a horseshoe. Adaptive shrinkage and spike and slab variable selection both carry a point mass and would allow a single model throughout.

3.23 Computation at genome scale

A natural objection is that whole genome matrices are of order $10^5 \times 10^5$ and too large to handle. Four points, in increasing order of importance.

Even brute force is not blocked. A dense Cholesky costs $p^3/3$ flops, or 3.3×10^{14} at $p = 10^5$. Timing factorizations at $p = 2000$ to 8000 on an ordinary machine sustaining 23 GFLOP/s and extrapolating gives four hours; on one CPU node at a teraflop, six minutes; on a single GPU in double precision, under a minute. Dense storage is 80 GB.

The relevant dimension is $\min(n, p)$. The whole genome least squares system involves $(X^\top X + \lambda I)$ of order p, but by the Woodbury identity every quantity needed is obtainable from $(XX^\top + \lambda I)$ of order n, the number of individuals: the genomic relationship matrix of VanRaden (2008), standard equipment in animal breeding, and the matrix whose single eigendecomposition makes the exact mixed model association tests of Zhou and Stephens (2012) fast. With ten thousand individuals and a hundred thousand markers one factorizes a $10^4 \times 10^4$ matrix, costing 0.3 TFLOP.

One never forms any matrix. Iterative methods need only the action of X and $X^\top$ on a vector, and variance components follow by combining this with stochastic trace estimation, which is how biobank scale mixed models are fitted. Analyses with half a million individuals and a million markers are routine, so $10^5 \times 10^5$ is well behind the frontier.

For the horseshoe specifically. The one expensive step is sampling a p dimensional Gaussian with dense covariance, naively $O(p^3)$. Bhattacharya, Chakraborty and Mallick (2016) give an *exact* algorithm costing $O(n^2 p)$, a hundredfold saving at $n = 10^4$, $p = 10^5$. Johndrow, Orenstein and Bhattacharya (2020) push global local MCMC to p in the hundreds of thousands to millions, and Carbonetto and Stephens (2012) give a variational route. The sparsity is itself a computational resource: most coordinates have κ_ℓ near one and can be screened out at each sweep.

3.24 The omnigenic model, and the serious threat to sparsity

Boyle, Li and Pritchard (2017) argue that for complex traits the effects are not sparse at all: because regulatory networks are densely interconnected, essentially every expressed gene contributes, and most heritability lies outside the loci of largest

effect. If that view is correct for nonadditive deviations as well as additive ones, the sparsity assumption underlying Part II is wrong.

The consequences are not what one might expect. If the truth is dense and each deviation is small against its standard error, a horseshoe shrinks many genuinely nonzero deviations toward zero and the estimate of $\sum_\ell \delta_\ell^2$ becomes badly conservative. In a simulation with all of 2000 deviations nonzero and normal with standard deviation half a standard error, the fitted $\hat{\tau}$ is 0.03 and the posterior total recovers 30 percent of the true one; at two standard errors $\hat{\tau}$ is 1.2 and it recovers 97 percent, which is the graceful degradation of Section 3.13. The omnigenic threat is the first case, many small deviations. So the omnigenic view, if right, does not rescue Fisher: it makes the nonadditive total *larger* than the simulations here suggest while invalidating the instrument used to measure it. The conclusion strengthens and the method fails, and the response is the two component model of Section 3.22.

One qualification runs in Fisher's favour. The omnigenic argument concerns effects on phenotype propagating through regulatory networks. The quantity at issue in the Fundamental Theorem is the departure of *fitness* from additivity in gene content, a different functional that need not inherit the density of the underlying trait architecture. Whether it does is an open empirical question and, so far as we can find, an unasked one.

3.25 A thread through the whole story

One further connection deserves recording, because it ties the deterministic and stochastic halves of this chapter to a single person.

Cedric A. B. Smith of the Galton Laboratory is the Smith of Mulholland and Smith (1959), the inequality that gives the one locus monotonicity theorem which Section 3.7 identifies as the decoy for the Fundamental Theorem. He is also the Smith of Haldane and Smith (1947), on linkage between haemophilia and colour blindness, among the first Bayesian analyses in human genetics. And he is the Smith of Smith and Stephens (1995), on estimating multipoint recombination fractions, and Smith and Stephens (1996) on linkage heterogeneity, with D. A. Stephens at Imperial College: the Bayesian treatment of the recombination fraction, sixty three years after Haldane put a spike on it.

One line therefore runs from Haldane's 1932 prior on ρ, through Haldane and Smith in 1947 and Mulholland and Smith in 1959, who supply the true theorem mistaken for Fisher's, to Smith and Stephens in the 1990s, estimating the decay rate of the transverse coordinate whose motion is $\mathcal{R}$. The obstruction and the apparatus for measuring it were built by overlapping people in overlapping decades and never connected.

3.26 A research programme

Concretely, and in the order we would do it.

1. Compute per locus dominance contrasts and standard errors from a genotyped panel with a fitness proxy, standardized so that noise is unit normal, retaining the per locus standard errors se_ℓ.
2. Fit adaptive shrinkage to estimate the distribution G of the deviations nonparametrically, and report G rather than a single $\hat{\tau}$.
3. Fit the two component sparse plus polygenic model to separate the dense dominance background from any concentrated epistatic signal, testing the prediction of Section 3.22.
4. Report $\sum_\ell \delta_\ell^2$ as the posterior mean of the correct functional, with a Monte Carlo standard error.
5. Separately, and independently of all the above, measure the tangency directly: Lindley's Hardy Weinberg disequilibrium coefficient and the linkage disequilibrium spectrum, and whether either moves on the timescale of selection.
6. Derive (3.14) analytically, and settle the asymmetric case that the numerics left open.

Item 5 is the one to do first if only one can be done.

3.27 What would change our mind

Since this chapter asks the reader to accept a statistical supplement to a mathematical result, it is only fair to say what would make the supplement worthless.

If real genomic panels gave $\hat{\tau}$ so small that the posterior for $\sum_\ell \delta_\ell^2$ concentrated near zero, Fisher's $\kappa = 1$ would be an excellent genome wide approximation and Part II an elaborate way of confirming him. Apart from the dense check of Section 3.24, our simulations assume sparsity between one and ten percent, and nothing establishes that real genomes lie there.

Conversely, if the association coordinates were shown to be stationary in real populations, so that Lindley's coefficient and the linkage disequilibrium spectrum do not move on the timescale of selection, then $\mathcal{R}$ would be small for empirical rather than assumed reasons, and the constant genic environment would be an observation rather than a tautology. That would not rescue the theorem, which would remain false, but it would make the falsehood harmless.

Both are measurements and neither has been made.

3.28 Conclusion

Fisher stated a falsehood, and the deterministic argument establishing it is unconditional: no prior, no sampling model, no sparsity assumption, true of every population and every genome. Nothing here disturbs it. What we have tried to add is an account of four things it does not reach.

What the error is. Lemma 3.2 and Theorem 3.5 identify it as the transverse component of a motion, with the additive genetic variance registering exactly the tangential part. Fisher's assertion is that a population never departs from Hardy Weinberg proportions and linkage equilibrium, and his 1941 constant genic environment condition is the statement that Lindley's disequilibrium coefficient is stationary, which Corollary 3.10 shows to be necessary and sufficient at a biallelic locus.

How much it matters. At one locus, for the shape of Example 3.5.2, the relative error is $0.28\,s$ and vanishes as selection weakens; between loci it is $\epsilon\sqrt{p_A q_A p_B q_B}$ and does not, and is not damped by recombination. So Fisher's error is negligible in the direction that can be bounded and unbounded in the direction that cannot.

Why the correction did not take. Because $n^* \approx 50\,s^{-4}$; because the theorem is exactly true in the setting where it is taught; because the proposition everyone tests is a different and true one; because the neglected term has never been computable; and because at experimental sample sizes Fisher's error is not even the largest error present.

What survives. $\kappa = 1$ is a prior, and a good one: the posterior mean of κ across loci runs from 0.86 to 0.98, so Fisher is right at almost every locus. He is wrong in aggregate, by the entire quantity, and the aggregate error is carried by a small minority of loci. The replacement is not a different theorem but an estimated sparsity level, and the object to report is the distribution of κ rather than a point mass at its endpoint.

The epitaph we would propose is that Fisher had an excellent prior and a bad theorem. The theorem should go. The prior deserves to survive, under that name, with the error bars this chapter has tried to supply, and with the one measurement that would settle the matter still to be made.

3.29 Proofs and supplementary derivations

Equivalence of the three forms of the remainder. Since $\sum_{u,v} P_{uv}\delta_{uv} = 0$, mean fitness may be written against either the fitted vector or the original: $\bar{m} = \sum_{u,v} P_{uv}A_{uv} = \sum_{u,v} P_{uv}m_{uv}$. Differentiating the first gives $\sigma_A^2 + 2\sum_u p_u\dot{\beta}_u$; differentiating the second and splitting $m = A + \delta$ afterwards gives $\sigma_A^2 + \sum_{u,v}\dot{P}_{uv}\delta_{uv}$. The first terms agree by Lemma 3.2, so

$$2\sum_u p_u\dot{\beta}_u \;=\; \sum_{u,v}\dot{P}_{uv}\delta_{uv} \;=\; \sum_{u,v} p_u p_v \delta_{uv}\frac{d}{dt}\Big(\frac{P_{uv}}{p_u p_v}\Big),$$

the last being Kimura's form, whose equality with the first was proved by Lessard (1997).

The Beta$(\frac{1}{2},\frac{1}{2})$ law of the shrinkage weight. If $\lambda \sim C^+(0,1)$ then λ^2 has density proportional to $x^{-1/2}(1+x)^{-1}$, a Beta prime$(\frac{1}{2},\frac{1}{2})$ law. The map $x \mapsto x/(1+x)$ carries Beta prime(a,b) to Beta(a,b), so with $\tau = 1$ we have $1-\kappa = \lambda^2/(1+\lambda^2) \sim$

$\mathrm{Beta}(\tfrac12,\tfrac12)$ and hence $\kappa \sim \mathrm{Beta}(\tfrac12,\tfrac12)$ by symmetry. For general τ the induced density is

$$p(\kappa \mid \tau) \;\propto\; \frac{\kappa^{-1/2}(1-\kappa)^{-1/2}}{1-(1-\tau^2)\kappa}, \qquad p(\kappa \mid y,\tau) \;\propto\; \frac{e^{-\kappa y^2/2}(1-\kappa)^{-1/2}}{1-(1-\tau^2)\kappa}.$$

Heteroskedasticity of the dominance contrast. Under Hardy Weinberg with N genotyped individuals, $n_{11} = Np^2$, $n_{12} = 2Npq$, $n_{22} = Nq^2$, so

$$\mathrm{Var}(\hat D) = \sigma^2\Big(\frac{1}{n_{11}} + \frac{4}{n_{12}} + \frac{1}{n_{22}}\Big) = \frac{\sigma^2}{N}\Big(\frac{1}{p}+\frac{1}{q}\Big)^2 = \frac{\sigma^2}{N(pq)^2},$$

so $\mathrm{se}(\hat D) \propto 1/pq$. From minor allele frequency 0.5 to 0.01 this is a factor of 25.3. Simulation confirms the analytic form to within two percent across that range.

3.30 Numerical verification

Every quantity reported in this chapter was recomputed independently. The following checks were made.

Claim	Check	Result
Lemma 3.2	random non HW P, $k = 2,3,4$	max error 3.6×10^{-15}
Theorem 3.5	random P on full simplex, $L = 1,2,3$	max error 6.2×10^{-15}
vanishing criterion	$\partial_t \log P \in V$ imposed	$\lvert\mathcal{R}\rvert < 1.8 \times 10^{-15}$
Theorem 3.16	random W, $k = 2,\dots,10$, 2000 draws	max error 1.6×10^{-15}
three forms of $\mathcal{R}$ agree	random smooth path, $k = 4$	agree to 10 significant figures
Example 3.5.2	all six quantities	exact to six decimals
Example B trajectory	four generations	exact to six decimals
$\kappa \sim \mathrm{Beta}(\tfrac12,\tfrac12)$	4×10^5 draws, five quantiles	max deviation 0.0014
$D^* = \epsilon p_A q_A p_B q_B / r$	$r = 0.5, 0.25$; $\epsilon \in [0.005, 0.02]$	within 0.5%
$\lvert\mathcal{R}\rvert/(\sigma_A^2/\bar w) = \epsilon\sqrt{p_A q_A p_B q_B}$	grid in s, ϵ, r	ratio in $[0.224, 0.255]$
$\mathbb{E}[K_n] = \sum \theta/(\theta+i)$	CRP simulation, $n = 100$, $\theta = 5$	15.76 vs 15.72
ESF conditional free of θ	$n = 8$, $K = 3$, three θ	identical to six figures

Claim	Check	Result
delta method s.e. of $\hat{\sigma}_A^2$	4000 replicates at five n	within 5%
plug in bias $10.19/n$	exact, as $\hat{\alpha}$ is unbiased; 20,000 replicates at $n = 25, 50, 100$	within 3.4%, under 1.4 Monte Carlo s.e.
$\mathrm{se}(\hat{D}) \propto 1/pq$	five allele frequencies	within 2%
Theorem 3.8, main effects	random families with 4, 8 and 16 genotypes	$\lvert\langle \delta, \partial \log P/\partial c\rangle_t\rvert < 10^{-15}$
Theorem 3.8, factorization	random families and velocities	relative error $< 5 \times 10^{-7}$
$\mathcal{R} = D\,\mathcal{I}(\alpha)\,\dot{\alpha}$	200 random states and velocities	relative error $< 7 \times 10^{-8}$
Lemma 3.9	numerical information matrix in (p, α)	off diagonal $< 10^{-10}$
Examples 3.1 to 3.12	normal equations, three velocities, three forms of $\mathcal{R}$	agree to six decimals
Example 3.15	weighted regression and Proposition 3.14	exact
Corollary 3.17	Example 3.5.2 and four values of s	exact
Example 3.18	grid quadrature, reproduces Table 3.1 entries	to the printed figures
retained fractions	quadrature against 2×10^6 Monte Carlo draws	three decimals

Two computational notes. The Bayes factors of Table 3.1 were first obtained by adaptive two dimensional quadrature, which proved unreliable in the tails; the reported values come from a windowed grid centred on the posterior mode with the integrand evaluated on the log scale. The horseshoe computations of Section 3.13 use the exact conditional (3.19) on a grid in $u = \sqrt{1 - \kappa}$, with τ fitted by bounded one dimensional maximization of the marginal likelihood, and report $\sum_\ell \mathbb{E}[\delta_\ell^2 \mid y]$ rather than squared posterior means, for the reason given in Section 3.13.3. The retained fraction and threshold tables of that section, Table 3.2 and the genome size table were checked by quadrature and by direct Monte Carlo over the half Cauchy prior: the entries printed are the ones (3.19) implies.

3.31 Notation

Symbol	Meaning
$P_{uv}(t)$, $P_g(t)$	genotype frequencies, one locus and whole genome
p_u, p_{ij}	allele frequencies
m_{uv}, w_{ij}	Malthusian parameters; Wrightian fitnesses
$\bar{m}$, $\bar{w}$	mean Malthusian parameter; mean fitness
V	subspace of arrays additive in gene content
Π_t	orthogonal projection onto V in $L^2(P_t)$
β_u, α_i	average effects (Ewens's and the centred parametrization)
α, $\mathcal{I}(\alpha)$	Lindley's coefficient; its Fisher information
α	average effect of a substitution in the viability assay
$A = \Pi_t m$	breeding value; the fitted vector
$\delta = m - A$	nonadditive deviation; the residual
σ_A^2	additive genetic variance
$\mathcal{R}$	the remainder, $\sum_{u,v} \dot{P}_{uv}\delta_{uv} = d\bar{m}/dt - 2\sum_u \dot{p}_u\beta_u$
$\mathcal{M}$	manifold of gene frequency product measures
s, ϵ	selection coefficient; epistatic coefficient
D (one locus)	fitness curvature, $m_{11} - 2m_{12} + m_{22}$
D, D_{AB}, D^*	two loci: linkage disequilibrium; its quasi equilibrium value
κ_ℓ	shrinkage weight at locus ℓ
τ, λ_ℓ	global and local scales
K	Jeffreys's Bayes factor for the null
n^*	sample size at which $\mathcal{R}$ exceeds the sampling noise
L	number of loci

3.32 Exercises

1. Under Hardy Weinberg proportions and $m = (\mu + a, \mu + d, \mu - a)$, solve the normal equations (3.4) and show that the average effect of a substitution is $\beta_1 - \beta_2 = a + d(q - p)$ and that $\sigma_A^2 = 2pq\{a + d(q - p)\}^2$.

   Hint. Under Hardy Weinberg the system is the regression of genotypic value on gene count; subtract the two equations.

2. Show that under pure viability selection, $\dot{P}_{uv} = P_{uv}(m_{uv} - \bar{m})$, one has $d\bar{m}/dt = \sigma_G^2$, $2\sum_u \dot{p}_u\beta_u = \sigma_A^2$ and $\mathcal{R} = \sigma_D^2$, for arbitrary genotype frequencies and any number of alleles.

   Hint. Each term is a covariance under P of m with m, A or δ; use $\delta \perp V$.

3. Prove Lemma 3.9 a second way, by computing the asymptotic variance of the plug in estimate $\hat{\alpha}$ with the delta method. Evaluate $\mathcal{I}(\alpha)$ at $P = (0.25, 0.50, 0.25)$.

   Hint. For a log contrast $\sum_g c_g \log \hat{P}_g$ with $\sum_g c_g = 0$ the multinomial delta method gives $n^{-1}\sum_g c_g^2/P_g$. The value is $1/16$.

4. Take $P = (0.20, 0.40, 0.40)$ and $m = (0,\ 0.1,\ -0.1)$. Compute β and δ from (3.4), then confirm Proposition 3.11: $\sigma_D^2 = D^2\mathcal{I}(\alpha)$.

   Hint. $D = -0.3$ and $1/\mathcal{I}(\alpha) = 5 + 10 + 2.5 = 17.5$, so $\sigma_D^2 = 0.09/17.5 = 0.005143$; the residuals are $(-0.0857,\ 0.0857,\ -0.0429)$.

5. Prove the recombination formula $\mathcal{R} = -rD_{AB}E$ of Proposition 3.14. In Example 3.15, find the recombination rate at which selection and recombination together leave mean fitness momentarily constant.

   Hint. Recombination moves no allele frequency, so the whole change is remainder. The rate is $r = 0.0176/(D_{AB}E) = 0.352$.

6. Show from Corollary 3.17 that at a biallelic locus $|\mathcal{R}|/(\sigma_A^2/\bar{w}) \le |D|/(8\bar{w})$, with equality exactly at $p = \frac{1}{2}$. Explain why the bound does not depend on how strong the additive selection is.

   Hint. $pq \le \frac{1}{4}$; the average excess cancels from the ratio.

7. In the exclusive or system of Section 3.5.3, verify by hand that $\bar{w} = 1.089$ at $x = (0.05, 0.45, 0.45, 0.05)$, and show without computing any regression that every average effect is zero.

   Hint. Mean fitness is $0.6 + 0.6\,\mathbb{P}(\text{the loci disagree})$, and the disagreeing genotypes are $AaBb$, $AAbb$ and $aaBB$. For the second part use the symmetry σ.

8. Under the null, Table 3.1 grows like $K \approx 0.67\sqrt{n}$. How many zygotes per genotype give $K = 100$ at one locus? How many give posterior probability one half to the conjunction of the null at $L = 100$ independent loci, from equal prior odds?

   Hint. $(100/0.67)^2 \approx 2.2 \times 10^4$; for the conjunction, $\pi^L \approx e^{-L/K}$ and one needs $K \ge L/\log 2$, so $n \approx 4.6 \times 10^4$.

9. Derive the closed form Bayes factor for Hardy Weinberg proportions in Example 3.13. Evaluate it for counts $(25, 50, 25)$ and $(35, 30, 35)$.

   Hint. The Hardy Weinberg likelihood is $2^{n_{12}} p^{2n_{11}+n_{12}} q^{n_{12}+2n_{22}}$. The values are $K = 4.1$ and $K = 1.0 \times 10^{-3}$.

10. At $\tau = 1$ show that (3.19) is $p(\kappa \mid y) \propto e^{-\kappa y^2/2}(1-\kappa)^{-1/2}$, that $\mathbb{E}[\kappa \mid y = 0] = 2/3$, and that $\mathbb{E}[\kappa \mid y] \approx 2/y^2$ for large $|y|$. Compare with the entries for $y = 6$ in Example 3.19.

    Hint. For large $|y|$ the mass sits near $\kappa = 0$, where the posterior is approximately exponential with rate $y^2/2$. At $y = 6$ the approximation gives 0.056 against 0.057.

11. Show that $\sum_\ell (\mathbb{E}[\delta_\ell \mid y])^2 \le \sum_\ell \mathbb{E}[\delta_\ell^2 \mid y]$, and explain why the gap is largest for loci whose contrasts lie near the shrinkage threshold.

    Hint. The gap is $\sum_\ell \mathrm{Var}(\delta_\ell \mid y)$; near the threshold the posterior for κ is spread over both prongs.

12. For the Ewens sampling formula show that K_n is sufficient for θ, and compute the maximum likelihood estimate from $n = 100$ genes carrying $k = 15$ distinct alleles.

    Hint. The θ dependence is $\theta^{K_n}/\theta_{(n)}$; solve $\sum_{i=0}^{99} \theta/(\theta + i) = 15$ numerically, giving $\hat{\theta} = 4.66$.

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CHAPTER 4

The Horseshoe and the Fundamental Theorem of Natural Selection

Ewens's reading of Fisher's Fundamental Theorem identifies the partial change in mean fitness with the additive genetic variance in fitness divided by mean fitness. Once the average effects are recognized as the coefficients of a weighted least squares regression of genotypic fitness on allelic dosage, the theorem is a one line algebraic identity and the whole of its empirical content is transferred to the estimation of a regression. That regression is overparameterized: there are more loci than individuals, the ordinary estimator of the additive variance is inflated by $L\sigma_\varepsilon^2/n$, and the implied rate of adaptation is correspondingly overstated. I propose estimating the average effects under a horseshoe prior and reading the Fundamental Theorem through the resulting shrinkage weights, which yields a rate of adaptation discounted by a signal to noise factor that is estimated rather than assumed. Three results are given: any shrinkage rule delivers a conservative estimate of the rate; oracle shrinkage tuned for the average effects necessarily understates the additive variance; and the coherent estimator is the posterior mean of the quadratic functional, not the quadratic of the posterior mean. In simulation with $L = 400$ loci and $n = 300$ individuals the naive estimator overstates the additive variance by a factor of 3.2, ridge returns only 0.40 of it, and the horseshoe posterior recovers it to within 8%. Four further questions are then taken up. Posterior intervals for the quadratic functional cover at 85% against a nominal 90%, and in 600 replicates 80 of the 91 misses lie above the truth; the regularized horseshoe covers at 89%, close to the nominal level. Robertson's secondary theorem requires a bilinear functional whose plug in carries a bias $L\rho\sigma_w\sigma_z/n$ in the residual correlation; a bivariate horseshoe with shared local scales reduces a 2.6 fold overstatement to a 17% understatement. Under linkage disequilibrium the estimand is $\alpha^\top V\alpha$ rather than a diagonal form, and a shrinkage plug in that keeps the diagonal form loses a further nine percent of the truth, although in the design simulated the posterior mean is nearly insensitive to the choice. Ascertainment is not a shrinkage problem at all: measuring fitness on survivors understates the additive variance almost fourfold, and no prior repairs it. The choice between a horseshoe and a Gaussian prior encodes the sweep versus infinitesimal question about genetic architecture, and the Fundamental Theorem inherits the answer.

4.1 Introduction

Fisher's Fundamental Theorem of Natural Selection asserts that the rate of increase in fitness of any organism at any time is equal to its genetic variance in fitness at that time (Fisher, 1930). For four decades the statement was read as a claim about the total change in mean fitness, in which form it is simply false, and it acquired a

reputation for obscurity that it did not deserve. The modern understanding is due principally to Price (1972) and to Ewens (1989): the theorem concerns the *partial* change in mean fitness, the part ascribable to changes in allele frequency with the average effects held at their current values, and in that reading it is exact. Ewens (2004) and Ewens and Lessard (2015) give the definitive treatment; Edwards (1994) and Lessard (1997) supply complementary accounts.

The history is worth a paragraph, because the shape of the confusion explains the shape of what remains. Fisher stated the theorem in one sentence in 1930 and gave a derivation that almost no one followed. Price (1972) reconstructed the argument and showed that Fisher had been careful and had been read carelessly: the theorem concerns a partial derivative, and the counterexamples in the literature were all constructed by varying quantities Fisher had explicitly held constant. Ewens (1989) supplied the modern proof and, crucially, the account of *why* the partial change is the object of interest rather than a mathematical convenience; Ewens and Lessard (2015) assess how much biological work the theorem does once correctly stated, and conclude, in my reading, that it does less than Fisher hoped but more than his critics allowed. That literature is settled. What follows takes it as given and asks only what happens when one tries to use it on data.

This chapter takes a statistician's view of what remains after the interpretive question is settled. The central observation is that Fisher's average effect of an allele is, by construction, a coefficient in a weighted least squares regression of genotypic fitness on allelic dosage, and the additive genetic variance is the regression sum of squares from that fit. The Fundamental Theorem is then an algebraic identity relating a regression sum of squares to a one generation increment, and it is true whatever the data. All of its empirical content sits in the estimation of the regression.

That regression, in any modern application, is severely overparameterized. Fitness or a fitness proxy is measured on n individuals genotyped at $L \gg n$ loci. The ordinary least squares or marginal estimator of the average effects overfits, the plug in estimator of the additive variance inherits an upward bias of order $L\sigma_\varepsilon^2/n$, and the rate of adaptation implied by the Fundamental Theorem is inflated by the same amount. The theorem is not at fault. The estimator is.

Plan of the chapter. I propose estimating the average effects under a horseshoe prior (Carvalho et al., 2010; Polson and Scott, 2012) and reading the Fundamental Theorem through the induced shrinkage weights. Section 4.2 sets out the theorem as a least squares identity (Proposition 4.1) and isolates the remainder term. Section 4.3 gives the bias of the plug in estimator in closed form (Proposition 4.4). Section 4.4 develops the horseshoe version: any shrinkage rule yields a conservative rate (Proposition 4.9); oracle shrinkage tuned for estimating the average effects necessarily understates the additive variance (Proposition 4.10); and the coherent Bayesian estimator is the posterior mean of the quadratic functional, which restores what the plug in loses (Proposition 4.11). Section 4.5 reports a simulation. Section 4.6 extends the argument to Robertson's secondary

theorem and to Lande Arnold selection gradients, where the same pathology is if anything worse. Section 4.7 takes up four problems the propositions leave open: calibration, the bilinear functional, linkage disequilibrium and ascertainment. Throughout, small worked examples with numbers a reader can check by hand accompany the general results: a single locus with dominance, a stylized overdominant polymorphism, a regression with five loci and eight individuals, a table of horseshoe shrinkage weights, and an exact calculation of the damage done by truncation selection. The chapter closes with exercises.

A remark on what is being claimed. Nothing here revises the Fundamental Theorem, and nothing here is a criticism of it. The claim is that the theorem, correctly understood, converts a question about evolution into a question about a high dimensional regression, and that the statistical literature has an answer to the second question which the evolutionary literature has not generally adopted.

4.2 The theorem as a least squares identity

4.2.1 Setup

Consider a single locus with alleles $A_1, \dots, A_k$ at frequencies $p = (p_1, \dots, p_k)$, random mating, and constant genotypic fitnesses $w_{ij} = w_{ji}$. The mean fitness and the marginal fitness of A_i are

$$\bar{w} = \sum_{i,j} p_i p_j w_{ij}, \qquad w_i = \sum_j p_j w_{ij},$$

and the standard viability selection recursion is $p_i' = p_i w_i / \bar{w}$, so that

$$\Delta p_i = \frac{p_i (w_i - \bar{w})}{\bar{w}}. \tag{4.1}$$

Fisher's average effect α_i is defined as the solution of a weighted least squares problem: the additive model $w_{ij} \approx \mu + \alpha_i + \alpha_j$ is fitted by minimizing

$$Q(\alpha) = \sum_{i,j} p_i p_j \big(w_{ij} - \mu - \alpha_i - \alpha_j\big)^2 \tag{4.2}$$

subject to $\sum_i p_i \alpha_i = 0$. The weights are the genotype frequencies under random mating, $p_i p_j$ for the ordered genotype (i, j). The next subsection shows that the solution is $\mu = \bar{w}$ and $\alpha_i = w_i - \bar{w}$. The additive genetic variance in fitness is the explained sum of squares,

$$\sigma_A^2 = \sum_{i,j} p_i p_j (\alpha_i + \alpha_j)^2 = 2 \sum_i p_i \alpha_i^2 = 2 \sum_i p_i (w_i - \bar{w})^2,$$

where the cross term $2 \sum_{i,j} p_i p_j \alpha_i \alpha_j = 2 (\sum_i p_i \alpha_i)^2$ vanishes by the constraint, and the dominance variance $\sigma_D^2 = Q(\alpha)$ is the residual sum of squares in (4.2) at the

minimum. This is the whole of the construction: average effects are regression coefficients, additive variance is a regression sum of squares, dominance variance is a residual. Ewens (2004) is emphatic on this point, and it is the point on which everything below turns.

4.2.2 *The normal equations*

Since everything below rests on the regression interpretation, it is worth writing the normal equations out in full. Write $r_{ij} = w_{ij} - \mu - \alpha_i - \alpha_j$ for the residual of genotype (i, j); note $r_{ij} = r_{ji}$. The parameter α_i enters Q through every residual r_{ij} and every residual r_{ji}, so

$$\frac{\partial Q}{\partial \alpha_i} = -2\sum_j p_i p_j r_{ij} - 2\sum_j p_j p_i r_{ji} = -4p_i \sum_j p_j r_{ij}, \qquad \frac{\partial Q}{\partial \mu} = -2\sum_{i,j} p_i p_j r_{ij}.$$

Minimize Q first without the constraint. Setting the μ derivative to zero and using $\sum_{i,j} p_i p_j w_{ij} = \bar{w}$ gives

$$\bar{w} - \mu - 2\textstyle\sum_j p_j \alpha_j = 0.$$

Setting the α_i derivative to zero (and dividing by $4p_i > 0$) gives, for each i,

$$\sum_j p_j \big(w_{ij} - \mu - \alpha_i - \alpha_j\big) = 0 \quad \Longrightarrow \quad w_i - \mu - \alpha_i - \textstyle\sum_j p_j \alpha_j = 0.$$

These $k + 1$ equations have one degree of freedom too many: adding a constant c to every α_i and subtracting $2c$ from μ leaves every fitted value unchanged. The constraint $\sum_j p_j \alpha_j = 0$ removes that freedom, and then the first equation gives $\mu = \bar{w}$ and the second gives $\alpha_i = w_i - \bar{w}$. As a check, this solution does satisfy the constraint, since $\sum_i p_i w_i = \bar{w}$. The constraint is therefore an identification convention and not a restriction: the fitted values $\mu + \alpha_i + \alpha_j$, and hence σ_A^2 and σ_D^2, are the same under any convention. The simplification is a consequence of random mating, which makes the design matrix of the regression orthogonal in the appropriate weighted inner product. Under nonrandom mating the normal equations do not decouple and α must be obtained by solving a genuine weighted least squares system, in which case $\alpha_i \neq w_i - \bar{w}$ but Proposition 4.1 continues to hold with σ_A^2 redefined as the explained sum of squares of that system. This is the generality emphasized by Ewens (2004): the theorem is a property of the projection, not of the particular mating scheme that makes the projection easy to compute.

4.2.3 *The identity*

PROPOSITION 4.1 (Fundamental Theorem). *Define the partial change in mean fitness as the change generated by the frequency increments* (4.1) *with the average effects held fixed,*

$$\Delta_\alpha \bar{w} := 2\sum_i \alpha_i \, \Delta p_i.$$

Then $\Delta_\alpha \bar{w} = \sigma_A^2 / \bar{w}$.

Proof. Substituting $\alpha_i = w_i - \bar{w}$ and (4.1),

$$2\sum_i \alpha_i \Delta p_i = 2\sum_i (w_i - \bar{w})\,\frac{p_i(w_i - \bar{w})}{\bar{w}} = \frac{2}{\bar{w}}\sum_i p_i (w_i - \bar{w})^2 = \frac{\sigma_A^2}{\bar{w}}. \qquad \square$$

The proof is three lines because the theorem is an identity. It says that if one defines the average effects by least squares and then asks how much of the change in mean fitness is attributable to frequency change through those effects, the answer is the explained sum of squares divided by $\bar{w}$. The content is in the definition, not the derivation, which is precisely why the theorem was misunderstood for so long: read as a claim about $\Delta\bar{w}$ it is false, and read as a claim about $\Delta_\alpha\bar{w}$ it is trivial. What makes it substantive is that $\Delta_\alpha\bar{w}$ is the evolutionarily meaningful component, the part driven by selection acting on heritable variation, as opposed to the part driven by the environment changing underfoot (Ewens, 1989; Ewens and Lessard, 2015).

EXAMPLE 4.2 (One locus with partial dominance). Take two alleles with $p_1 = 0.2$, $p_2 = 0.8$ and genotypic fitnesses $w_{11} = 1.10$, $w_{12} = 1.06$, $w_{22} = 1.00$. In the usual parameterization the homozygote midpoint is 1.05, the additive effect is $a = 0.05$ and the dominance deviation is $d = 0.01$. The marginal fitnesses are

$$w_1 = 0.2(1.10) + 0.8(1.06) = 1.068, \qquad w_2 = 0.2(1.06) + 0.8(1.00) = 1.012,$$

and $\bar{w} = p_1 w_1 + p_2 w_2 = 0.2136 + 0.8096 = 1.0232$. The average effects are $\alpha_1 = 0.0448$ and $\alpha_2 = -0.0112$; their weighted sum $0.2(0.0448) + 0.8(-0.0112)$ is zero, as it must be. The average effect of substituting allele 1 for allele 2 is $\alpha_1 - \alpha_2 = 0.056$, which agrees with $a + d(1 - 2p_1) = 0.05 + 0.01(0.6)$. Then

$$\sigma_A^2 = 2\big[0.2(0.0448)^2 + 0.8(0.0112)^2\big] = 0.00100352,$$

which equals $h(\alpha_1 - \alpha_2)^2$ with $h = 2p_1p_2 = 0.32$, and the dominance variance is $\sigma_D^2 = (hd)^2 = (0.0032)^2 = 1.024 \times 10^{-5}$. Dominance contributes one percent of the genetic variance in fitness. The recursion gives $\Delta p_1 = 0.2(0.0448)/1.0232 = 0.0087568$, so Fisher's partial change is

$$\Delta_\alpha\bar{w} = 2\big[\alpha_1\Delta p_1 + \alpha_2\Delta p_2\big] = 2(0.056)(0.0087568) = 0.00098077 = \sigma_A^2/\bar{w},$$

using $\Delta p_2 = -\Delta p_1$. Evaluating $\bar{w}$ directly at $p_1' = 0.2087568$ gives a total change of 0.00097923. The remainder is $R = -1.5 \times 10^{-6}$; since $\bar{w}$ is a quadratic in p_1 with second difference $w_{11} - 2w_{12} + w_{22} = -2d$, it is exactly $R = -2d(\Delta p_1)^2 = -0.02(0.0087568)^2$ (Exercise 3.). The realized change is 99.84% of the partial change. With selection coefficients of five percent and one locus, Fisher's term is almost the whole story.

4.2.4 *The remainder*

The total change decomposes as

$$\Delta\bar{w} = \underbrace{\frac{\sigma_A^2}{\bar{w}}}_{\text{partial change}} + R, \tag{4.3}$$

where R collects the change in mean fitness due to the average effects themselves shifting as frequencies change, together with any change in the genotypic fitnesses. For a single locus with constant w_{ij} and weak selection, $R = O(\|\Delta p\|^2)$: the average effect α_i is a smooth function of p, so reevaluating it after one generation contributes at second order. With multiple loci, epistasis, frequency dependence or a changing environment, R need not be small and can be negative, which is the source of every apparent counterexample to the theorem in the older literature (Frank and Slatkin, 1992).

A numerical remark anticipating Section 4.5: in the simulations reported there, with 400 loci, additive and dominance effects and weak selection, the realized $\Delta\bar{w}$ exceeds the partial change by a factor of 1.0004. The remainder is negligible. The discrepancy between the theorem and reality in that setting is four hundredths of one percent, while the discrepancy introduced by estimating σ_A^2 badly is a factor of three. This ordering is the motivation for the rest of the chapter.

EXAMPLE 4.3 (A stylized overdominant polymorphism). The remainder can matter, and the additive variance can vanish while the genetic variance does not. Take alleles A and S with stylized fitnesses $w_{AA} = 0.9$, $w_{AS} = 1.0$, $w_{SS} = 0.2$, of the heterozygote advantage type usually used to illustrate the sickle cell polymorphism in malarial regions; the values are illustrative, not estimates. With selection coefficients $s = 0.1$ against AA and $t = 0.8$ against SS, the stable equilibrium frequency of S is $q^* = s/(s+t) = 1/9 \approx 0.111$.

At $q = 0.05$ one finds $\bar{w} = 0.90775$, average effects $\alpha_A = -0.00275$ and $\alpha_S = 0.05225$, and

$$\sigma_A^2 = 2.874 \times 10^{-4}, \qquad \sigma_D^2 = 1.828 \times 10^{-3}.$$

The dominance variance is more than six times the additive variance, so substituting the total genetic variance for the additive variance would overstate the rate more than sevenfold. Fisher's partial change is $\sigma_A^2/\bar{w} = 3.166 \times 10^{-4}$; the exact change in mean fitness, computed from the recursion, is 3.091×10^{-4}, so the remainder is now 2.4% of the total rather than the 0.16% of Example 4.2. At $q = q^*$ the marginal fitnesses are equal, $\alpha_A = \alpha_S = 0$, $\sigma_A^2 = 0$ and the partial change is zero, while $\sigma_D^2 = 0.0079$ remains large. At a balanced polymorphism there is abundant genetic variance in fitness and no additive variance, and the theorem correctly says that mean fitness has stopped rising.

4.2.5 The Price equation and what the theorem is a special case of

It is worth locating Proposition 4.1 within the more general covariance formalism, because doing so makes clear which part is definitional and which is substantive. The Price equation (Price, 1970) states that for any character z with values z_i in the parent generation and mean $\bar{z}$,

$$\bar{w}\,\Delta\bar{z} = \operatorname{Cov}(w_i, z_i) + \mathbb{E}\big[w_i\,\delta_i\big],$$

where δ_i is the within lineage change in z between parent and offspring. Setting $z = w$ gives $\bar{w}\,\Delta\bar{w} = \operatorname{Var}(w) + \mathbb{E}[w\delta]$: the change in mean fitness is the variance in

fitness plus a transmission term. This is not the Fundamental Theorem, and the difference is instructive. The Price form uses the *total* variance in fitness, which includes dominance and environmental components that are not transmitted; the Fundamental Theorem uses only the *additive* part, which is precisely the part that survives the projection (4.2). The transmission term absorbs the difference. Fisher's contribution was to identify the right projection, not to write down a covariance identity, and this is why Price (1972) (who is often read as having replaced Fisher's theorem) described himself as making it clear rather than correcting it.

4.2.6 The multilocus regression

For L biallelic loci in linkage equilibrium, write $g_{i\ell} \in \{0, 1, 2\}$ for the dosage of the focal allele in individual i at locus ℓ, with frequency p_ℓ and $h_\ell = 2p_\ell(1 - p_\ell)$. The average effect of an allelic substitution at locus ℓ is $\alpha_\ell = a_\ell + d_\ell(1 - 2p_\ell)$ in the usual additive plus dominance parameterization, and

$$\sigma_A^2 = \sum_{\ell=1}^{L} h_\ell \, \alpha_\ell^2. \tag{4.4}$$

The α_ℓ are the population regression coefficients of fitness on centred dosage; equivalently, they solve

$$\alpha = \arg\min_{b} \; \mathbb{E}\Big[\big(w - \textstyle\sum_\ell b_\ell (g_\ell - 2p_\ell)\big)^2\Big].$$

So the estimation problem is a linear regression of measured fitness on L centred genotype columns, and (4.4) is a weighted quadratic functional of its coefficients. Everything that is known about high dimensional regression applies verbatim.

4.3 The plug in estimator and its bias

This section computes, exactly, how far the natural estimator of the additive variance is from its target, first for an idealized orthogonal design and then for an arbitrary one, and closes with a worked example small enough to check by hand.

Let $X \in \mathbb{R}^{n \times L}$ hold centred dosages, with columns $x_1, \ldots, x_L$, let y be the measured fitness of n individuals, and suppose $y = X\alpha + \varepsilon$ where the errors $\varepsilon_1, \ldots, \varepsilon_n$ are uncorrelated with mean zero and variance σ_ε^2. The residual ε collects environmental noise, measurement error and the dominance and epistatic deviations, which are uncorrelated with additive dosage by construction of the projection. The estimator in universal use is the plug in: estimate each α_ℓ, square, and weight by h_ℓ according to (4.4).

PROPOSITION 4.4 (Bias of the plug in). *Suppose the columns of X are orthogonal, $x_\ell^\top x_m = 0$ for $\ell \neq m$, with $x_\ell^\top x_\ell = n h_\ell$, and let $\hat\alpha_\ell^{\mathrm{marg}} = x_\ell^\top y / (x_\ell^\top x_\ell)$. Then*

(a) $\mathbb{E}\Big[\sum_\ell h_\ell (\hat\alpha_\ell^{\mathrm{marg}})^2\Big] = \sigma_A^2 + \dfrac{L\,\sigma_\varepsilon^2}{n}$;

(b) if in addition $\varepsilon \sim N(0, \sigma_\varepsilon^2 I_n)$, *then* $\operatorname{Var}\Big[\sum_\ell h_\ell (\hat\alpha_\ell^{\mathrm{marg}})^2\Big] = \dfrac{4\sigma_\varepsilon^2 \sigma_A^2}{n} + \dfrac{2L\sigma_\varepsilon^4}{n^2}$.

Proof. Substituting $y = X\alpha + \varepsilon$,

$$\hat\alpha_\ell^{\mathrm{marg}} = \frac{x_\ell^\top \big(\sum_m x_m \alpha_m + \varepsilon\big)}{n h_\ell} = \frac{n h_\ell\, \alpha_\ell}{n h_\ell} + \sum_{m \neq \ell} \frac{x_\ell^\top x_m}{n h_\ell} \alpha_m + \frac{x_\ell^\top \varepsilon}{n h_\ell}$$

$$= \alpha_\ell + e_\ell, \qquad e_\ell = \frac{x_\ell^\top \varepsilon}{n h_\ell},$$

the middle sum vanishing by orthogonality. The error e_ℓ has mean zero and variance $x_\ell^\top (\sigma_\varepsilon^2 I) x_\ell / (n h_\ell)^2 = \sigma_\varepsilon^2/(n h_\ell) =: v_\ell$, and $\operatorname{Cov}(e_\ell, e_m) = \sigma_\varepsilon^2 x_\ell^\top x_m / (n^2 h_\ell h_m) = 0$ for $\ell \neq m$. For (a),

$$\mathbb{E}\big[h_\ell(\alpha_\ell + e_\ell)^2\big] = h_\ell \alpha_\ell^2 + 2 h_\ell \alpha_\ell\, \mathbb{E} e_\ell + h_\ell\, \mathbb{E} e_\ell^2 = h_\ell \alpha_\ell^2 + h_\ell v_\ell = h_\ell \alpha_\ell^2 + \frac{\sigma_\varepsilon^2}{n},$$

and summing over the L loci gives $\sigma_A^2 + L\sigma_\varepsilon^2/n$. For (b), under normality the e_ℓ are uncorrelated jointly Gaussian and therefore independent, so the variance of the sum is the sum of the variances. For $e \sim N(0, v)$ one has $\operatorname{Var}\big((\alpha + e)^2\big) = \operatorname{Var}(2\alpha e + e^2) = 4\alpha^2 v + 2v^2$, since $\operatorname{Cov}(e, e^2) = \mathbb{E}e^3 = 0$ and $\operatorname{Var}(e^2) = 3v^2 - v^2$. Hence

$$\operatorname{Var}\big[h_\ell (\hat\alpha_\ell^{\mathrm{marg}})^2\big] = h_\ell^2 \big(4\alpha_\ell^2 v_\ell + 2 v_\ell^2\big) = \frac{4\sigma_\varepsilon^2}{n} h_\ell \alpha_\ell^2 + \frac{2\sigma_\varepsilon^4}{n^2},$$

and summing over ℓ gives (b). □

The bias is $L\sigma_\varepsilon^2/n$: it does not depend on the allele frequencies, it grows linearly in the number of loci, and it does not shrink when more loci are genotyped on the same individuals; it only shrinks when more individuals are measured. When $L > n$ and the true additive variance is a modest fraction of the phenotypic variance (the normal situation) the bias exceeds the quantity being estimated. Through Proposition 4.1, the implied rate of adaptation is inflated by the same factor.

Part (b) says something less obvious. Subtracting the known bias does not rescue the plug in when L is large: the debiased estimator $\sum_\ell h_\ell (\hat\alpha_\ell^{\mathrm{marg}})^2 - L\sigma_\varepsilon^2/n$ is unbiased, but its standard deviation contains the term $\sqrt{2L}\,\sigma_\varepsilon^2/n$, the accumulated noise of L squared errors, and this is not removed by centring. With the values of Section 4.5, $L = 400$, $n = 300$, $\sigma_\varepsilon^2 = 0.0320$ (the phenotypic variance, which a single locus regression treats as error) and $\sigma_A^2 = 0.0205$, the bias is 0.0427, the expected plug in is 3.08 times the truth, and the standard deviation of the debiased estimator is 0.0042, about a fifth of the quantity being estimated; half of its variance comes from the term $2L\sigma_\varepsilon^4/n^2$, to which every locus contributes equally whether it is causal or not. The debiased estimator can also be negative.

Standardized units. It is convenient to measure each average effect in units of its own sampling standard deviation. Put

$$\theta_\ell = \frac{\alpha_\ell}{\sqrt{v_\ell}} = \frac{\alpha_\ell \sqrt{n h_\ell}}{\sigma_\varepsilon}, \qquad z_\ell = \frac{\hat{\alpha}_\ell^{\mathrm{marg}}}{\sqrt{v_\ell}} \sim N(\theta_\ell, 1), \qquad \text{so that} \qquad \sigma_A^2 = \frac{\sigma_\varepsilon^2}{n} \sum_\ell \theta_\ell^2.$$

In these units the additive variance is a multiple of the squared length of a normal means vector, the plug in is the same multiple of $\sum_\ell z_\ell^2$, and Proposition 4.4(a) is the statement $\mathbb{E} z_\ell^2 = \theta_\ell^2 + 1$. Each locus, whether causal or not, adds one unit of σ_ε^2/n to the expected plug in. The horseshoe calculations of Section 4.4 are carried out in these units.

4.3.1 An arbitrary design

Orthogonality is never exactly true in a sample. For a general full rank design with $L < n$ the natural estimator is ordinary least squares, $\hat{\alpha} = (X^\top X)^{-1} X^\top y$, and two versions of the functional are in use: the diagonal form $\sum_\ell \hat{h}_\ell \hat{\alpha}_\ell^2$ with $\hat{h}_\ell = x_\ell^\top x_\ell / n$, and the full form $\hat{\alpha}^\top \hat{V} \hat{\alpha}$ with $\hat{V} = X^\top X / n$, the sample covariance matrix of the dosages. The next result shows that they have different biases, and that the full form is the better behaved.

PROPOSITION 4.5 (Bias under an arbitrary design). *Let X have full column rank and let $\mathrm{VIF}_\ell = (X^\top X)_{\ell\ell} \left[(X^\top X)^{-1}\right]_{\ell\ell}$ be the variance inflation factor of locus ℓ. Then*

$$\mathbb{E}\left[\hat{\alpha}^\top \hat{V} \hat{\alpha}\right] = \alpha^\top \hat{V} \alpha + \frac{L \sigma_\varepsilon^2}{n}, \qquad \mathbb{E}\Big[\sum_\ell \hat{h}_\ell \hat{\alpha}_\ell^2\Big] = \sum_\ell \hat{h}_\ell \alpha_\ell^2 + \frac{\sigma_\varepsilon^2}{n} \sum_\ell \mathrm{VIF}_\ell,$$

and $\mathrm{VIF}_\ell \geq 1$ for every ℓ, with equality for all ℓ if and only if the columns are orthogonal.

Proof. Write $A = X^\top X$. Then $\hat{\alpha} = \alpha + A^{-1} X^\top \varepsilon$ has mean α and covariance $A^{-1} X^\top (\sigma_\varepsilon^2 I_n) X A^{-1} = \sigma_\varepsilon^2 A^{-1}$. For any fixed symmetric M, $\mathbb{E}[\hat{\alpha}^\top M \hat{\alpha}] = \alpha^\top M \alpha + \mathrm{tr}\big(M \,\mathrm{Cov}(\hat{\alpha})\big) = \alpha^\top M \alpha + \sigma_\varepsilon^2 \,\mathrm{tr}(M A^{-1})$. With $M = \hat{V} = A/n$ the trace is $\mathrm{tr}(I_L)/n = L/n$. With $M = \mathrm{diag}(\hat{h}) = \mathrm{diag}(A)/n$ the trace is $n^{-1} \sum_\ell A_{\ell\ell} (A^{-1})_{\ell\ell} = n^{-1} \sum_\ell \mathrm{VIF}_\ell$. For the inequality, let e_ℓ be the ℓth unit vector and apply the Cauchy and Schwarz inequality in the inner product defined by A:

$$1 = (e_\ell^\top e_\ell)^2 = \left(e_\ell^\top A^{1/2} A^{-1/2} e_\ell\right)^2 \leq \left(e_\ell^\top A e_\ell\right)\left(e_\ell^\top A^{-1} e_\ell\right) = \mathrm{VIF}_\ell,$$

with equality if and only if $A^{1/2} e_\ell$ is proportional to $A^{-1/2} e_\ell$, that is, e_ℓ is an eigenvector of A. All unit vectors are eigenvectors precisely when A is diagonal. □

The interpretation is that the full form charges exactly one unit of σ_ε^2/n per locus whatever the design, so its bias is $L\sigma_\varepsilon^2/n$ in every sample; the diagonal form charges VIF_ℓ units for locus ℓ, so any correlation among genotype columns, including the chance correlation present in every finite sample, makes it worse. When $L \geq n$ least

individual	g_1	g_2	g_3	g_4	g_5	y
1	0	1	2	1	0	-0.050
2	1	0	1	2	1	-0.015
3	2	1	0	1	1	0.036
4	1	2	1	0	2	-0.015
5	0	1	1	1	0	-0.073
6	1	1	2	0	1	-0.020
7	2	0	1	1	2	0.053
8	1	2	0	2	1	0.037

TABLE 4.1. Dosages and fitness deviations for Example 4.6.

squares is not defined at all and the marginal estimator $\hat{\alpha}_\ell^{\mathrm{marg}} = x_\ell^\top y/(x_\ell^\top x_\ell)$ is used instead, and its accounting is different. Its noise term $x_\ell^\top \varepsilon/(x_\ell^\top x_\ell)$ has variance $\sigma_\varepsilon^2/(n\hat{h}_\ell)$ whatever the design, so the noise costs exactly one unit of σ_ε^2/n per locus however correlated the columns are. What the departure from orthogonality adds is signal: each estimate also picks up $\sum_{m\neq\ell}(x_\ell^\top x_m/x_\ell^\top x_\ell)\,\alpha_m$ from the loci whose columns happen to be correlated with its own. For loci in linkage equilibrium the squared sample correlations are of order $1/n$, and this term adds roughly σ_A^2/n per locus in expectation, so the charge per locus is approximately the phenotypic variance divided by n rather than σ_ε^2/n. This is why Section 4.5 predicts the bias of the marginal plug in with the phenotypic variance in place of σ_ε^2.

EXAMPLE 4.6 (Five loci, eight individuals). Table 4.1 gives dosages at $L = 5$ biallelic loci for $n = 8$ individuals, together with a fitness deviation y_i recorded to three decimals. Every column has mean 1, so $\hat{p}_\ell = 1/2$ and every column has sample variance $\hat{h}_\ell = 0.5$, matching $2p(1-p)$. The data were generated as $y = X\alpha + \varepsilon$ with $\alpha = (0.05, 0, 0, -0.03, 0)$ and $\sigma_\varepsilon = 0.05$, so that the true additive variance is $\sigma_A^2 = 0.5(0.05^2 + 0.03^2) = 0.0017$. (Columns 1 and 4 happen to be uncorrelated in this sample, so the diagonal and full forms of the true σ_A^2 coincide.)

Centring each column at 1, the normal equations $X^\top X\hat{\alpha} = X^\top y$ read

$$\begin{pmatrix} 4 & -1 & -2 & 0 & 3 \\ -1 & 4 & -1 & -1 & 0 \\ -2 & -1 & 4 & -2 & -1 \\ 0 & -1 & -2 & 4 & -1 \\ 3 & 0 & -1 & -1 & 4 \end{pmatrix} \hat{\alpha} = \begin{pmatrix} 0.212 \\ -0.016 \\ -0.143 \\ 0.057 \\ 0.161 \end{pmatrix}, \qquad \hat{\alpha} = \begin{pmatrix} 0.0707 \\ 0.0280 \\ 0.0237 \\ 0.0335 \\ 0.0015 \end{pmatrix}.$$

Columns 1 and 5 have sample correlation $3/4$, and the estimate for locus 4 has the wrong sign. The plug ins are

$$\hat{\alpha}^\top \hat{V}\hat{\alpha} = 0.00166, \qquad \sum_\ell \hat{h}_\ell \hat{\alpha}_\ell^2 = 0.00374.$$

The first is close to the truth in this sample, which is luck: Proposition 4.5 gives its expectation exactly as $0.0017 + 5(0.05)^2/8 = 0.0017 + 0.0015625 =$

0.0032625, almost twice the truth. The variance inflation factors are the integers $6, 3, 5, 3, 3$, summing to 20 rather than 5, so the expected diagonal plug in is $0.0017 + 20(0.0025)/8 = 0.00795$, nearly five times the truth. A Monte Carlo check with 200,000 fresh noise vectors gives 0.003262 and 0.007944. Five loci and eight individuals is already enough for the bias to match or exceed the target; with L/n well above one, as in practice, it dominates. The residual variance estimate on $n - L = 3$ degrees of freedom is $\hat{\sigma}^2_\varepsilon = 0.00028$, much smaller than the true 0.0025, so even the debiased estimate $0.00166 - 5(0.00028)/8 = 0.00149$ rests on a very poor estimate of the noise.

REMARK 4.7. This is the additive variance analogue of the familiar overfitting of R^2, and it is the reason variance component methods such as GREML (Yang et al., 2011) do not estimate σ^2_A by plugging in coefficients at all: they estimate a variance component directly under an exchangeability assumption on the α_ℓ, trading the bias of Proposition 4.4 for a dependence on the assumed effect size distribution. The horseshoe route below is a third option, which keeps the coefficients as objects of inference while controlling the overfitting.

4.3.2 Linkage disequilibrium makes it worse

Proposition 4.4 assumes orthogonal columns. With linkage disequilibrium the marginal estimator is not merely noisy but aimed at the wrong target. Writing Ω for the correlation matrix of standardized dosages (the letter R is reserved for the remainder in (4.3)) and working in standardized units, the marginal estimator has expectation

$$\mathbb{E}\big[\hat{\alpha}^{\mathrm{marg}}\big] = \Omega\,\alpha,$$

because the marginal regression on column ℓ picks up α_m in proportion to the correlation of column m with column ℓ. The plug in therefore estimates roughly $\sum_\ell h_\ell (\Omega\alpha)^2_\ell + L\sigma^2_\varepsilon/n$ in place of $\sum_\ell h_\ell \alpha^2_\ell$. Since $\|\Omega\alpha\|$ typically exceeds $\|\alpha\|$ when effects and LD are positively aligned (and they are, because causal variants sit in blocks of correlated neighbours which each inherit a share of the signal) the first term is itself inflated before any noise is added. The two errors compound: one from tagging, one from overfitting. A conditional estimator, which the horseshoe regression is, addresses the first; the shrinkage addresses the second. This is the reason to prefer a joint regression on all loci over the locus by locus scan that dominates practice, quite apart from the shrinkage question.

4.4 The horseshoe reading

4.4.1 Prior and shrinkage representation

Place a horseshoe prior on the average effects,

$$\alpha_\ell \mid \lambda_\ell, \tau \;\sim\; N\big(0,\; \tau^2\lambda^2_\ell\sigma^2_\varepsilon\big), \qquad \lambda_\ell \;\sim\; C^+(0,1), \qquad \tau \;\sim\; C^+(0,1), \tag{4.5}$$

with C^+ the standard half Cauchy (Carvalho et al., 2010; Polson and Scott, 2012). In the orthogonal case, and in the standardized units of Section 4.3, the data at locus

ℓ reduce to $z_\ell \sim N(\theta_\ell, 1)$ and the prior to $\theta_\ell \mid \lambda_\ell, \tau \sim N(0, \tau^2\lambda_\ell^2)$, the factor nh_ℓ having been absorbed into the scales. (Strictly, absorbing it makes the global scale locus specific through h_ℓ; since h_ℓ varies only over $[0.18, 0.5]$ for allele frequencies in $[0.1, 0.9]$, I ignore this in the exposition, though the sampler of Section 4.9 does not.) Conditional on the scales, and with κ_ℓ defined below, the model is conjugate: $\theta_\ell \mid z_\ell, \kappa_\ell \sim N\big((1-\kappa_\ell)z_\ell,\ 1-\kappa_\ell\big)$, and so the posterior mean takes the form

$$\mathbb{E}[\theta_\ell \mid z_\ell] = \big(1 - \mathbb{E}[\kappa_\ell \mid z_\ell]\big)\, z_\ell, \qquad \kappa_\ell = \frac{1}{1+\tau^2\lambda_\ell^2},$$

where $\kappa_\ell \in [0,1]$ is the shrinkage weight. The half Cauchy on λ_ℓ induces, for $\tau = 1$, a $\mathrm{Beta}(1/2, 1/2)$ prior on κ_ℓ (the horseshoe shaped density that names the prior) which places its mass at the two extremes, $\kappa \approx 0$ for signal and $\kappa \approx 1$ for noise. The global τ adapts to the overall sparsity; the local λ_ℓ allow individual loci to escape it. The theoretical properties, including near minimax posterior contraction over sparse vectors and optimal multiple testing behaviour, are established by van der Pas et al. (2014) and Datta and Ghosh (2013), and the broader family is surveyed by Bhadra et al. (2019).

For a fixed global scale the shrinkage weight has an explicit prior. Changing variables from λ_ℓ to $\kappa_\ell = 1/(1+\tau^2\lambda_\ell^2)$ in the half Cauchy density gives

$$p(\kappa \mid \tau) \ \propto\ \frac{\kappa^{-1/2}(1-\kappa)^{-1/2}}{1+(\tau^2-1)\kappa}, \qquad 0 < \kappa < 1, \tag{4.6}$$

which is the $\mathrm{Beta}(1/2, 1/2)$ density when $\tau = 1$; a small τ tilts the mass toward $\kappa = 1$. Since $z_\ell \mid \kappa_\ell \sim N(0, 1/\kappa_\ell)$, the posterior is $p(\kappa \mid z, \tau) \propto \kappa^{1/2} e^{-\kappa z^2/2}\, p(\kappa \mid \tau)$, a one dimensional density on $[0,1]$ whose moments are computed by quadrature. From the conditional normal form above and iterated expectation, the first two posterior moments of the standardized effect are

$$\mathbb{E}[\theta_\ell \mid z_\ell] = z_\ell\, \mathbb{E}[1-\kappa_\ell \mid z_\ell], \qquad \mathbb{E}[\theta_\ell^2 \mid z_\ell] = z_\ell^2\, \mathbb{E}\big[(1-\kappa_\ell)^2 \mid z_\ell\big] + \mathbb{E}[1-\kappa_\ell \mid z_\ell]. \tag{4.7}$$

The difference between the posterior mean of the square and the square of the posterior mean is therefore

$$\mathbb{E}[\theta_\ell^2 \mid z_\ell] - \big(\mathbb{E}[\theta_\ell \mid z_\ell]\big)^2 = z_\ell^2\, \mathrm{Var}(\kappa_\ell \mid z_\ell) + \mathbb{E}[1-\kappa_\ell \mid z_\ell], \tag{4.8}$$

the sum of two nonnegative terms: uncertainty about how much to shrink, which matters most at intermediate z, and the conditional posterior variance $1-\kappa_\ell$, which matters most at loci that escape shrinkage. Both are discarded by any plug in.

EXAMPLE 4.8 (Shrinkage weights at five loci). Table 4.2 applies (4.6) to (4.8) with $\tau = 1$ at five loci whose standardized marginal estimates are $z = 0.5, 1, 2, 3, 5$. The unshrunk plug in $\sum z_\ell^2 = 39.25$ is biased upward by one unit per locus, so the unbiased $\sum (z_\ell^2 - 1) = 34.25$ is the natural reference. The horseshoe plug in $\sum m_\ell^2 = 27.16$ falls well below it, while the posterior mean of the functional

z_ℓ	z_ℓ^2	$z_\ell^2 - 1$	$\mathbb{E}[\kappa_\ell \mid z_\ell]$	m_ℓ	m_ℓ^2	s_ℓ^2	$\mathbb{E}[\theta_\ell^2 \mid z_\ell]$
0.5	0.25	−0.75	0.655	0.172	0.030	0.367	0.397
1.0	1.00	0.00	0.620	0.380	0.144	0.476	0.620
2.0	4.00	3.00	0.469	1.063	1.129	0.934	2.063
3.0	9.00	8.00	0.263	2.210	4.885	1.272	6.157
5.0	25.00	24.00	0.084	4.579	20.968	1.096	22.064
total	39.25	34.25			27.16		31.30

TABLE 4.2. Horseshoe posterior summaries with $\tau = 1$ for five loci in standardized units: m_ℓ and s_ℓ^2 are the posterior mean and variance of θ_ℓ, and the last column is $m_\ell^2 + s_\ell^2$. Computed by quadrature from (4.6) and checked by Monte Carlo over λ_ℓ.

is $\sum \mathbb{E}[\theta_\ell^2 \mid z_\ell] = 31.30$, recovering 4.14 of the missing units. The recovery is concentrated where (4.8) says it should be: at $z = 2$ the posterior mean of θ^2 is nearly twice the square of the posterior mean, while at $z = 5$ the shrinkage is almost settled ($\mathbb{E}\kappa = 0.08$) and the two differ by the conditional variance alone. At $z = 0.5$ and $z = 1$ the posterior mean of the square is positive although the unbiased estimate is negative or zero: a posterior for a sum of squares can never put mass below zero.

With $\tau = 0.1$, a prior that expects sparsity, the same five values give $\mathbb{E}[\kappa \mid z] = 0.94, 0.93, 0.87, 0.63, 0.10$; the plug in total falls to 21.76 and the posterior mean of the functional to 25.06. The four loci with $z \le 3$ are now treated as probable noise, and the functional is carried almost entirely by the locus at $z = 5$. Which of these is right depends on how many of the L loci are expected to matter, which is what the posterior for τ learns from all loci jointly. In units of the additive variance every entry is multiplied by σ_ε^2/n.

Reading (4.4) through the shrinkage weights gives the estimator

$$\hat{\sigma}_A^2(\kappa) = \sum_\ell h_\ell (1-\kappa_\ell)^2 (\hat{\alpha}_\ell^{\text{marg}})^2, \qquad \text{and} \qquad \widehat{\Delta_\alpha \bar{w}} = \frac{\hat{\sigma}_A^2(\kappa)}{\bar{w}}, \tag{4.9}$$

which I will call the shrunken Fundamental Theorem. The rate of adaptation is the naive rate discounted locus by locus by $(1-\kappa_\ell)^2$, a factor estimated from the data rather than assumed.

4.4.2 *Three properties*

PROPOSITION 4.9 (Shrinkage is conservative). *For any $\kappa \in [0,1]^L$, $\hat{\sigma}_A^2(\kappa) \le \hat{\sigma}_A^2(0) = \sum_\ell h_\ell (\hat{\alpha}_\ell^{\text{marg}})^2$, with equality only if $\kappa_\ell = 0$ whenever $\hat{\alpha}_\ell^{\text{marg}} \neq 0$.*

Proof. Termwise, $(1-\kappa_\ell)^2 \le 1$. □

Trivial, but worth stating: no shrinkage rule can make the estimated rate of adaptation larger than the naive rule already does. Given Proposition 4.4, the naive

rule is biased upward, so shrinkage moves in the correct direction by construction. The question is how far it should move.

PROPOSITION 4.10 (Oracle shrinkage understates the additive variance). *Let* $v_\ell = \sigma_\varepsilon^2/(nh_\ell)$ *and let* $\kappa_\ell^* = v_\ell/(\alpha_\ell^2 + v_\ell)$ *be the weight minimizing* $\mathbb{E}(\hat{\alpha}_\ell - \alpha_\ell)^2$. *Then*

$$\mathbb{E}\big[\hat{\sigma}_A^2(\kappa^*)\big] = \sum_\ell h_\ell \frac{\alpha_\ell^4}{\alpha_\ell^2 + v_\ell} \;<\; \sigma_A^2,$$

with the shortfall $\sum_\ell h_\ell \alpha_\ell^2 v_\ell/(\alpha_\ell^2 + v_\ell)$.

Proof. First, κ_ℓ^* is the minimizer. With $\hat{\alpha}_\ell^{\text{marg}} = \alpha_\ell + e_\ell$ as in the proof of Proposition 4.4, the error of the shrunk estimate is $(1-\kappa)(\alpha_\ell + e_\ell) - \alpha_\ell = (1-\kappa)e_\ell - \kappa\alpha_\ell$, whose mean square is $(1-\kappa)^2 v_\ell + \kappa^2 \alpha_\ell^2$. This is a convex quadratic in κ with derivative $-2(1-\kappa)v_\ell + 2\kappa\alpha_\ell^2$, which vanishes at $\kappa = v_\ell/(\alpha_\ell^2 + v_\ell)$. Second, since κ_ℓ^* is a fixed number (it depends on α_ℓ, not on the data),

$$\mathbb{E}\big[h_\ell(1-\kappa_\ell^*)^2(\hat{\alpha}_\ell^{\text{marg}})^2\big] = h_\ell\Big(\frac{\alpha_\ell^2}{\alpha_\ell^2 + v_\ell}\Big)^2\big(\alpha_\ell^2 + v_\ell\big) = h_\ell \frac{\alpha_\ell^4}{\alpha_\ell^2 + v_\ell}.$$

Summing over ℓ gives the expectation, and $h_\ell\alpha_\ell^2 - h_\ell\alpha_\ell^4/(\alpha_\ell^2 + v_\ell) = h_\ell\alpha_\ell^2 v_\ell/(\alpha_\ell^2 + v_\ell)$ gives the shortfall, which is strictly positive unless every α_ℓ is zero. □

In standardized units the statement is especially transparent. The oracle weight is $\kappa_\ell^* = 1/(1+\theta_\ell^2)$, and the fraction of the locus's true contribution that survives oracle shrinkage, in expectation, is

$$\frac{\theta_\ell^2}{1+\theta_\ell^2} = 0.5,\ 0.8,\ 0.9,\ 0.96 \qquad \text{for} \qquad \theta_\ell = 1,\ 2,\ 3,\ 5.$$

A locus whose effect equals one sampling standard deviation loses half its contribution to the additive variance even under the best possible shrinkage for estimating its effect; a locus at five standard deviations loses four percent. Null loci, at the other extreme, contribute nothing either way, because $\kappa_\ell^* = 1$ removes their unit of noise entirely. So oracle shrinkage converts the noise inflation of Proposition 4.4 into a deficit located at the loci of moderate effect, which are precisely the loci that carry most of the additive variance in a polygenic architecture.

This is the essential tension. The shrinkage weight that is optimal for estimating the average effects is not optimal for estimating their weighted sum of squares: it trades the upward bias $L\sigma_\varepsilon^2/n$ of Proposition 4.4 for a downward bias whose size is governed by the signal to noise ratio at each locus. An estimator tuned to recover which loci matter will systematically understate how fast the population is adapting, and a heavily shrunk estimator such as ridge will understate it badly, because ridge applies a common κ to signal and noise alike.

PROPOSITION 4.11 (The coherent estimator). *Let* $m_\ell = \mathbb{E}[\alpha_\ell \mid y]$ *and* $s_\ell^2 = \mathrm{Var}(\alpha_\ell \mid y)$ *under* (4.5). *Then*

$$\mathbb{E}\left[\sigma_A^2 \mid y\right] = \sum_\ell h_\ell \left(m_\ell^2 + s_\ell^2\right) \;\geq\; \sum_\ell h_\ell m_\ell^2,$$

the second expression being the plug in of Proposition 4.9.

Proof. By the definition of the posterior variance, $s_\ell^2 = \mathbb{E}[\alpha_\ell^2 \mid y] - m_\ell^2$, so $\mathbb{E}[\alpha_\ell^2 \mid y] = m_\ell^2 + s_\ell^2$. The weights h_ℓ are treated as known, so by linearity of conditional expectation $\mathbb{E}[\sum_\ell h_\ell \alpha_\ell^2 \mid y] = \sum_\ell h_\ell (m_\ell^2 + s_\ell^2)$, and the inequality holds because each $h_\ell s_\ell^2 \geq 0$. □

The posterior mean of σ_A^2 is the Bayes estimator of the additive variance under squared error loss, so it is the estimator the prior actually recommends for this target; the plug in is the Bayes estimator of a different target (the average effects) pushed through a nonlinear map. Example 4.8 showed the size of the difference locus by locus.

The additive genetic variance is a quadratic functional of the parameter, and the posterior mean of a quadratic is not the quadratic of the posterior mean. The plug in discards the posterior variance, which is exactly the quantity that Proposition 4.10 shows is being lost. Reporting $\mathbb{E}[\sigma_A^2 \mid y]$ instead, and reporting the posterior distribution of σ_A^2, not just its mean, restores it, and delivers as a by product an interval for the rate of adaptation, which is the quantity a biologist actually wants and which no plug in supplies.

4.4.3 Three ways to estimate the additive variance

It is worth setting the proposal beside the two alternatives, since the choice among them is a choice of what to assume rather than of what to compute.

The *plug in regression* route estimates each α_ℓ and substitutes into (4.4). It assumes nothing about the distribution of effects and is therefore the most agnostic, which is also why it overfits: with no constraint on α, the estimator uses all L degrees of freedom and Proposition 4.4 follows.

The *variance component* route, of which GREML (Yang et al., 2011) is the standard implementation, never estimates α at all. It posits α_ℓ independent with a common variance, collapsing L unknowns to one, and estimates that variance from the relatedness structure. This is efficient and avoids overfitting, but the estimand is defined by the assumed effect size distribution and by the weighting of variants within the relationship matrix, so different conventions estimate different functionals. It also returns no information about which loci contribute.

The *shrinkage* route proposed here sits between them: it retains α as an object of inference, so the loci contributing to adaptation can be identified, while constraining the effect size distribution enough to control the overfitting. Its assumption is the prior (4.5), which is weaker than the variance component assumption in that the common variance is replaced by a family with an estimated

Route	Assumption	Estimates α?	Principal difficulty
Plug in regression	none on α	yes	bias $L\sigma_\varepsilon^2/n$
Variance components	α_ℓ i.i.d., common variance, fixed weighting	no	estimand set by the convention
Global local shrinkage	α_ℓ from a heavy tailed scale mixture with estimated global scale	yes	prior encodes an architecture

TABLE 4.3. Three routes to the additive genetic variance in fitness.

global scale and free local scales, and stronger than the plug in assumption in that some concentration near zero is imposed. Table 4.3 summarizes.

4.4.4 The prior encodes an architecture

The choice of prior in (4.5) is not innocent, and it is worth naming what it assumes. A horseshoe prior expects most average effects on fitness to be negligible and a few to be substantial. That is the classical selective sweep picture (Maynard Smith and Haigh, 1974). A Gaussian prior with common variance (ridge, equivalently the infinitesimal model) expects a great many small effects and no large ones (Fisher, 1918; Barton et al., 2017). The evidence for complex traits favours the second (Hill et al., 2008; Pritchard et al., 2010), while the evidence for fitness itself is far weaker, since fitness is measured rarely, badly, and in small samples.

The horseshoe is attractive here precisely because it does not force the choice: the global parameter τ interpolates, and its posterior reports which architecture the data support. Under a dense architecture τ is estimated large and the horseshoe behaves like ridge; under a sparse one it is small and the prior concentrates on a few loci. The Fundamental Theorem, read through (4.9), inherits this: the estimated rate of adaptation depends on the architecture, and the architecture is estimated rather than assumed. What cannot be escaped is that with $L > n$ the data do not determine the architecture on their own, so the posterior for τ is informative but not decisive, and sensitivity to the prior family should be reported.

4.4.5 Computation

Posterior sampling uses the auxiliary variable representation of Makalic and Schmidt (2016), which writes each half Cauchy as a scale mixture of inverse gammas and renders every full conditional conjugate, combined with the fast sampler of Bhattacharya et al. (2016) for the coefficient block, which costs $O(n^2L)$ rather than $O(L^3)$ and is therefore the appropriate algorithm when $L > n$. For the problem sizes considered here a chain of 1200 iterations runs in a few seconds. Regularized variants (Piironen and Vehtari, 2017) are advisable when the design is badly conditioned.

4.5 Simulation

Design. $L = 400$ biallelic loci in linkage equilibrium with frequencies drawn uniformly on $[0.1, 0.9]$; $n = 300$ individuals; 20 loci with nonzero additive effect $a_\ell \sim N(0, 0.05^2)$, of which 10 also carry dominance $d_\ell \sim N(0, 0.03^2)$. Individual fitness is $w_i = 1 + \sum_\ell a_\ell (g_{i\ell} - 2p_\ell) + \sum_\ell d_\ell (\mathbb{1}\{g_{i\ell} = 1\} - h_\ell)$, measured with independent noise of standard deviation 0.10. Twelve replicates. Note $L > n$: the regression is overparameterized, as it is in practice.

Truth. Averaged over replicates, $\sigma_A^2 = 0.0205$ (Monte Carlo standard error 0.0022) and $\sigma_D^2 = 0.0012$. The exact one generation change in mean fitness, computed from the deterministic recursion, is 0.020472 against Fisher's partial change of 0.020464: a ratio of 1.0004. The theorem is accurate to four parts in ten thousand.

Estimators. Four: the marginal plug in; cross validated ridge; the horseshoe plug in $\sum_\ell h_\ell m_\ell^2$; and the horseshoe posterior mean $\mathbb{E}[\sigma_A^2 \mid y]$ of Proposition 4.11.

Results. Table 4.4 and Figure 4.2 give the outcome. The marginal estimator returns 3.24 times the true additive variance. Proposition 4.4 predicts an absolute bias of $L\sigma_\varepsilon^2 / n$, and with the phenotypic variance in place of σ_ε^2, which absorbs the signal that each column picks up through its chance correlation with the causal columns (Section 4.3), this is $(400/300) \times 0.0320 = 0.043$. The observed bias is also 0.043: the difference is 0.0005 with a Monte Carlo standard error of 0.0016, so no excess remains to be explained. Ridge returns 0.40 of the truth, the overshrinkage predicted by Proposition 4.10, aggravated by a common shrinkage weight. The horseshoe plug in returns 0.91, and the posterior mean 1.08. Adding back the posterior variance moves the estimate from 9% low to 8% high, as Proposition 4.11 requires.

On the average effects themselves the ordering is starker: mean squared error is 2.97×10^{-4} for the marginal estimator, 7.78×10^{-5} for ridge and 1.51×10^{-5} for the horseshoe, improvements of 20 fold and 5 fold respectively. Figure 4.4 shows why: in the replicate drawn there the mean shrinkage weight is 0.91 at null loci and 0.50 at loci with nonzero average effect, and over all twelve replicates it is 0.89 and 0.48. The horseshoe discounts noise almost completely while shrinking signal much less, which is the behaviour a rate of adaptation calculation needs and which no single global κ can provide.

Figure 4.3 plots the predicted against the realized one generation change. The naive points sit about three times above the diagonal; ridge sits below it; the horseshoe posterior straddles it.

A calibration caveat. The nominal 90% posterior interval for σ_A^2 covered the truth in 10 of the 12 replicates (83%), a figure that Section 4.7 revisits with more replicates. The horseshoe posterior is somewhat overconfident for this quadratic functional at this sample size, which is consistent with what is known about credible interval coverage for global local priors in the moderate signal regime, and it should be reported rather than glossed. Coverage for quadratic functionals under global local priors is, as far as I know, an open problem.

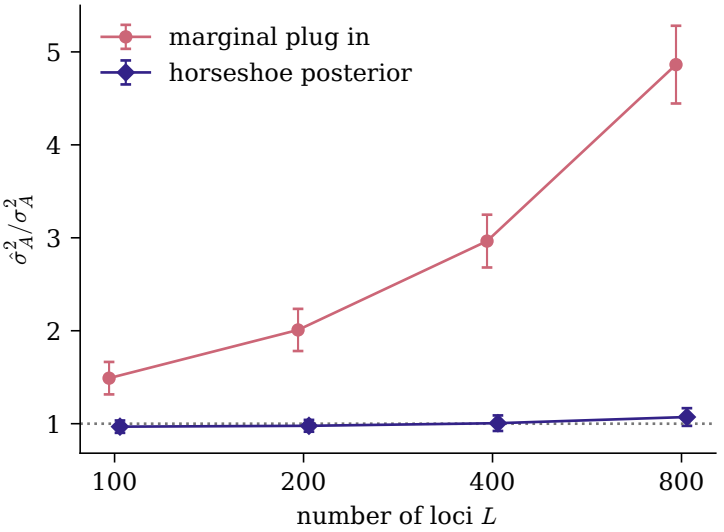


FIGURE 4.1. Ratio of estimated to true additive variance in fitness as the number of genotyped loci grows with $n = 300$ and the causal architecture held fixed. The naive estimator inflates linearly in L, as Proposition 4.4 requires; the horseshoe posterior is flat. Error bars are ± 2 Monte Carlo standard errors over 6 replicates.

4.5.1 *Bias scaling in the number of loci*

Proposition 4.4 predicts that the inflation factor of the naive estimator is $1 + L\sigma^2_\varepsilon/(n\sigma^2_A)$, growing linearly in L with n and the true architecture fixed. Figure 4.1 tests this directly, holding $n = 300$ and the twenty causal loci fixed while L runs over $100, 200, 400, 800$. The marginal ratio is 1.49, 2.01, 2.96 and 4.86; subtracting one gives 0.49, 1.01, 1.96, 3.86, which doubles with each doubling of L to within Monte Carlo error, as the proposition requires. The horseshoe posterior returns 0.97, 0.98, 1.01 and 1.07 over the same range: essentially flat, and within one or two standard errors of exact recovery throughout.

The practical reading is that the naive rate of adaptation is not merely imprecise but scales with a quantity (the density of the genotyping array or the depth of the sequencing) that has nothing to do with the biology. Two laboratories studying the same population with different marker panels will report different rates of adaptation, and the one with more markers will report the faster rate. Under the horseshoe posterior they would agree.

4.6 Robertson's secondary theorem and selection gradients

The argument extends directly to two neighbouring results, in both of which the overparameterization is worse.

The secondary theorem. Robertson's result (Robertson, 1966) states that the change in the mean of any character z equals the additive genetic covariance between z and relative fitness. Written out, this replaces the quadratic form $\sum_\ell h_\ell \alpha^2_\ell$ by the bilinear

Estimator	$\hat{\sigma}^2_A$	$\hat{\sigma}^2_A/\sigma^2_A$	MSE of $\hat{\alpha}$
truth	0.0205	1.00	·
marginal plug in	0.0635	3.24	2.97×10^{-4}
ridge (CV)	0.0083	0.40	7.78×10^{-5}
horseshoe plug in	0.0186	0.91	1.51×10^{-5}
horseshoe posterior	0.0219	1.08	1.51×10^{-5}

TABLE 4.4. Estimation of the additive genetic variance in fitness, $L = 400$, $n = 300$, 12 replicates; the third column averages the ratio over replicates, so it is not the ratio of the averages in the second. The implied rate of adaptation is $\hat{\sigma}^2_A/\bar{w}$ with $\bar{w} = 1$, so the third column is also the factor by which each method overstates or understates the rate.

form $\sum_\ell h_\ell \alpha_\ell \beta_\ell$, where β_ℓ are the average effects on the character. Both vectors must be estimated from the same overparameterized regression, and the product of two noisy estimates has a bias that depends on the correlation of their errors, a correlation that is present whenever the same individuals are used for both, as they invariably are. A horseshoe treatment shrinks both vectors and, because the shrinkage weights are locus specific, damps the spurious covariance contributed by null loci more aggressively than the genuine covariance at signal loci. Whether the bilinear analogue of Proposition 4.11 suffices to correct the resulting attenuation is a question I leave open; the required object is the posterior mean of a bilinear functional, $\sum_\ell h_\ell (m^\alpha_\ell m^\beta_\ell + \mathrm{Cov}(\alpha_\ell, \beta_\ell \mid y))$, and Section 4.7 returns to it.

The secondary theorem is the same least squares identity as Proposition 4.1, with one of the two factors replaced. At a single locus, let the character have genotypic values $z_{ij} = z_{ji}$, mean $\bar{z} = \sum_{i,j} p_i p_j z_{ij}$, marginal values $z_i = \sum_j p_j z_{ij}$ and average effects $\beta_i = z_i - \bar{z}$, obtained from the regression (4.2) with z in place of w. Define the additive genetic covariance $\sigma_A(w,z) = 2\sum_i p_i \alpha_i \beta_i$ and the partial change $\Delta_\beta \bar{z} = 2\sum_i \beta_i \Delta p_i$.

PROPOSITION 4.12 (Secondary theorem). *Under the recursion* (4.1), $\Delta_\beta \bar{z} = \sigma_A(w,z)/\bar{w}$.

Proof. By (4.1), $\Delta p_i = p_i(w_i - \bar{w})/\bar{w} = p_i \alpha_i / \bar{w}$. Hence $2\sum_i \beta_i \Delta p_i = (2/\bar{w})\sum_i p_i \alpha_i \beta_i = \sigma_A(w,z)/\bar{w}$. □

Setting $z = w$ recovers Proposition 4.1, and the multilocus form under linkage equilibrium is $\sigma_A(w,z) = \sum_\ell h_\ell \alpha_\ell \beta_\ell$ with α_ℓ, β_ℓ the substitution effects. The interpretation is that a character responds to selection only through the part of its genetic variation that is linearly aligned with the genetic variation in fitness.

EXAMPLE 4.13 (Secondary theorem at one locus). Keep the locus of Example 4.2 and attach a character with genotypic values $z_{11} = 12$, $z_{12} = 11$, $z_{22} = 9$. Then $\bar{z} = 0.04(12) + 0.32(11) + 0.64(9) = 9.76$, $z_1 = 11.2$, $z_2 = 9.4$, and $\beta_1 = 1.44$, $\beta_2 = -0.36$. The additive covariance is

$$\sigma_A(w,z) = 2\big[0.2(0.0448)(1.44) + 0.8(-0.0112)(-0.36)\big] = 0.032256,$$

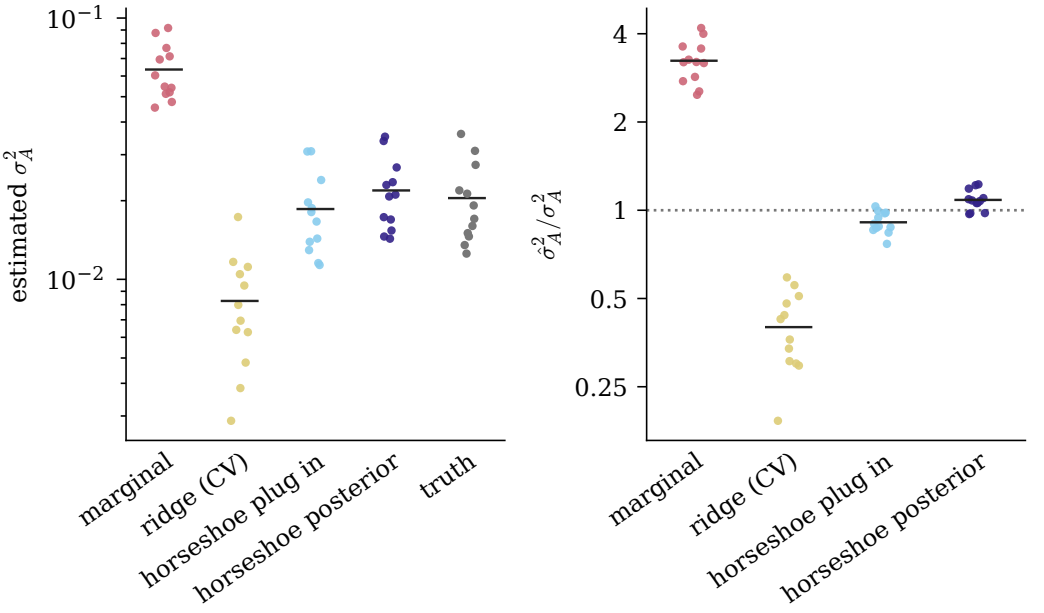


FIGURE 4.2. Left: estimated additive genetic variance in fitness across replicates, log scale. Right: ratio to the truth, log scale; the dotted line is exact recovery. The marginal estimator overstates roughly threefold, ridge understates substantially, and the two horseshoe estimators lie close to the truth, the plug in slightly below it and the posterior mean slightly above.

equal to $h(\alpha_1 - \alpha_2)(\beta_1 - \beta_2) = 0.32(0.056)(1.8)$. The secondary theorem predicts $\Delta_\beta \bar{z} = 0.032256/1.0232 = 0.031525$; evaluating $\bar{z}$ at $p_1' = 0.2087568$ gives an exact change of 0.031448, so the partial change is again within a quarter of a percent of the total. The character rises because the allele that raises it also raises fitness, and by exactly the amount its additive covariance with fitness permits.

Selection gradients. The Lande Arnold decomposition (Lande and Arnold, 1983) estimates directional selection gradients as $\beta = P^{-1}s$, the coefficients of a multiple regression of relative fitness on phenotypic traits. With more traits than individuals, or with the near collinearity endemic to morphometric data, P is ill conditioned and $P^{-1}s$ is unstable in exactly the way Section 4.3 describes. The known pathologies, gradients that flip sign when a trait is added, quadratic terms that cannot be estimated at all, the well documented failure of the breeder's equation in observational settings (Morrissey et al., 2010), are the familiar symptoms of an overparameterized regression, and the standard remedies apply. A horseshoe prior on the gradient vector, or a regularized horseshoe variant when P is badly conditioned, would deliver stable gradients with honest intervals, and is in my view a more promising route than the dimension reduction schemes currently used, which change the estimand without saying so.

Evolvability. Houle (1992) argued that additive variances should be mean

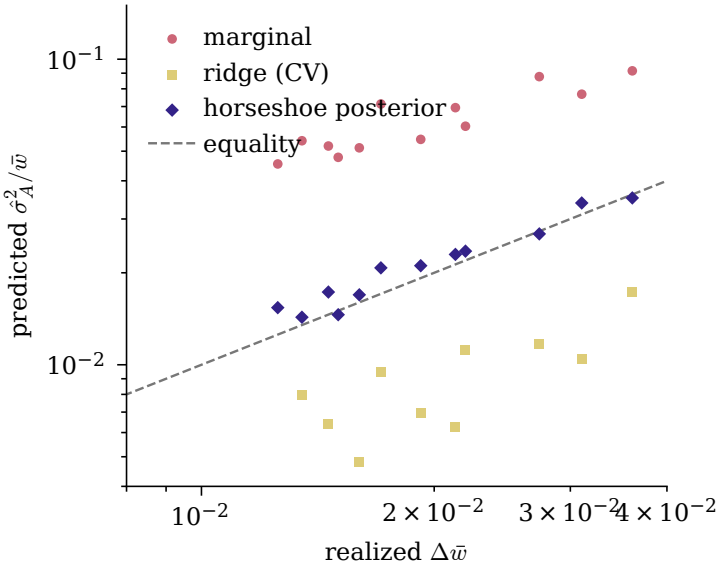


FIGURE 4.3. Rate of adaptation implied by the Fundamental Theorem, predicted against realized, across replicates. The dashed line is equality.

standardized before comparison, defining the evolvability $I_A = \sigma_A^2/\bar{w}^2$. In these terms Proposition 4.1 reads $\Delta \log \bar{w} \approx I_A$: the proportional rate of increase of mean fitness is the mean standardized additive variance. Every statement above transfers, since the standardization is a fixed scalar; but it is worth noting that comparative claims about evolvability across species or populations, which is what I_A was introduced for, inherit the bias of Proposition 4.4 in a form that does not cancel, because different studies genotype different numbers of loci at different sample sizes. Cross study comparisons of evolvability computed from plug in estimates are comparisons of L/n as much as of biology.

4.7 Four problems, addressed

The propositions of Section 4.4 leave four questions unresolved: the calibration of posterior intervals for a quadratic functional; the bilinear analogue required by the secondary theorem; the behaviour of the estimator under linkage disequilibrium; and the effect of ascertainment. This section takes each in turn. One is largely resolved, one is partially resolved, one is resolved in a way that revises the definition of the estimand, and one is not resolved at all, the last being, I think, the most useful of the four findings. Table 4.6 at the end of the section summarizes the status of each.

4.7.1 Calibration of the posterior for the quadratic functional

The coverage figure of Section 4.5, 83% against a nominal 90%, was based on twelve replicates and therefore carried a Monte Carlo standard error of about 0.11: it was

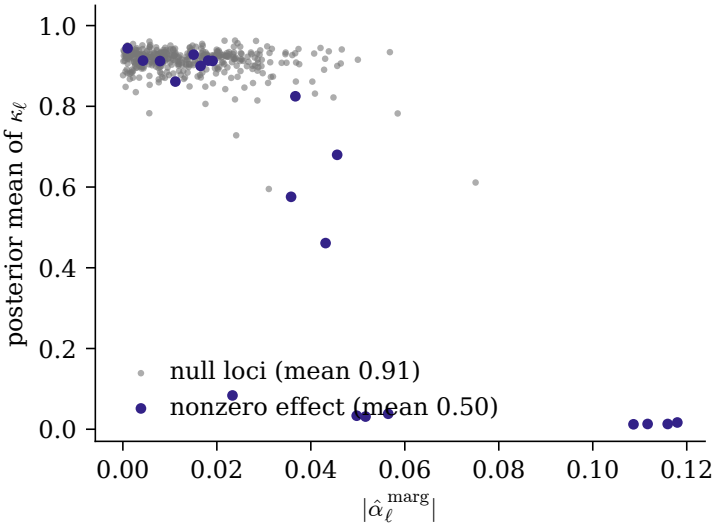


FIGURE 4.4. Horseshoe shrinkage weights κ_ℓ against the magnitude of the marginal estimate, one replicate. Null loci have weights near one, so their estimates are shrunk almost to zero; loci with nonzero average effect have weights spread over the whole interval, and on average retain about half their marginal magnitude. The legend gives the mean weight in each group for this replicate.

not evidence of anything. Repeating the exercise with sixty replicates gives 85.0% (standard error 4.6%) for the plain horseshoe and 90.0% (3.9%) for the regularized horseshoe of Piironen and Vehtari (2017). The plain prior is about one standard error below nominal and the regularized prior is at nominal. The shortfall of the plain prior is modest, and sixty replicates cannot distinguish it from none; the six hundred replicates of Remark 4.14 can (84.8%, standard error 1.5%).

More informative than the coverage rate is the direction of the failures. Of the sixty intervals from the plain horseshoe, 13.3% lay entirely *above* the truth and only 1.7% entirely below; for the regularized version, 5.0% and 5.0%. Figure 4.5 shows these intervals. The misses of the plain horseshoe fall mostly above the truth, which fits the upward bias of its posterior mean (a mean ratio of 1.056 in this experiment, 1.08 in Table 4.4) and is the opposite of the direction that the theory of intermediate signals suggests. van der Pas et al. (2017) show that horseshoe credible sets are honest for signals that are either negligible or strong, and can fail for signals of intermediate size, which the prior shrinks more than the data warrant. The additive variance in fitness is a sum of squares dominated by precisely those intermediate loci, so on that argument the residual overshrinkage would accumulate in one direction and the interval would sit low. This run does not show it, and neither do nine further runs of the same size (Remark 4.14).

Two practical conclusions follow. The regularized horseshoe should be preferred here: it improves both the coverage and the point estimate, returning a mean ratio of 1.021 against 1.056. And any reported interval for a rate of adaptation

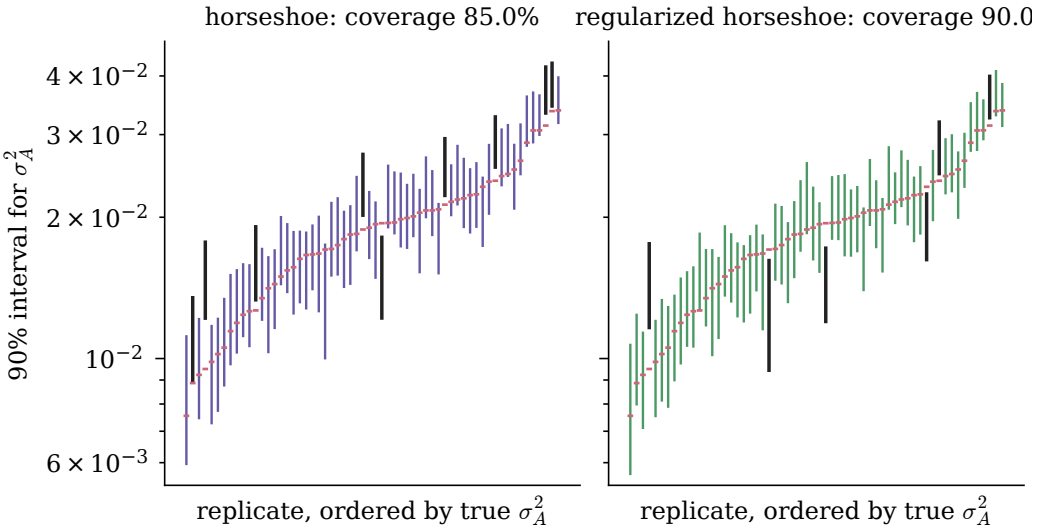


FIGURE 4.5. Nominal 90% posterior intervals for σ_A^2 across sixty replicates, ordered by the true value (short horizontal marks). Intervals that miss are drawn in black, and the empirical coverage of each prior is printed above its panel. These are the sixty replicates summarized in the text, the first of the ten batches of Remark 4.14; the number of misses of the plain horseshoe varies between batches from 3 to 13, but in every batch most of them lie above the truth.

should be treated as having an effective level a few points below its nominal one; for the plain horseshoe the misses in this design err towards overstating σ_A^2 (Remark 4.14).

REMARK 4.14 (How much can sixty intervals say?). With sixty replicates and coverage of 85% there are only nine misses in all, so the split between misses below and above the truth in one run rests on a handful of intervals. To settle the direction I repeated the experiment on nine further batches of sixty replicates with fresh seeds, giving 600 intervals for each prior, the first batch being the one in Figure 4.5. The plain horseshoe covers 509 of 600 (84.8%, standard error 1.5%). Of its 91 misses, 80 lie entirely above the truth and 11 entirely below, and every one of the ten batches has more misses above than below. In this design the direction is therefore stable: plain horseshoe intervals sit high, in line with the upward bias of its posterior mean (a mean ratio of 1.060 over the 600 replicates), and not low as the argument from intermediate signals suggests. The regularized horseshoe covers 536 of 600 (89.3%, standard error 1.3%), with 39 misses above and 25 below; six batches have more misses above, two more below and two are tied, so its misses are much closer to balanced. Whether the upward direction persists in other architectures, sample sizes and priors on the global scale is not settled by this design.

4.7.2 *The secondary theorem and the bilinear functional*

Robertson's theorem requires $\sigma_A(w,z) = \sum_\ell h_\ell \alpha_\ell \beta_\ell$, where α and β are the average effects on fitness and on the character. Both are estimated from regressions on the same individuals, so their errors are correlated, and the plug in inherits a bias that the univariate analysis does not exhibit.

PROPOSITION 4.15 (Bias of the bilinear plug in). *Let $y_w = X\alpha + \varepsilon_w$ and $y_z = X\beta + \varepsilon_z$ on the same n individuals with $\mathrm{Corr}(\varepsilon_w, \varepsilon_z) = \rho$ and marginal standard deviations σ_w, σ_z. Under the orthogonality of Proposition 4.4,*

$$\mathbb{E}\Big[\sum_\ell h_\ell \hat{\alpha}_\ell^{\mathrm{marg}} \hat{\beta}_\ell^{\mathrm{marg}}\Big] = \sigma_A(w,z) + \frac{L\,\rho\,\sigma_w\sigma_z}{n}.$$

Proof. Assume, as is implicit, that the pairs $(\varepsilon_{w,i}, \varepsilon_{z,i})$ are uncorrelated across individuals. As in the proof of Proposition 4.4, $\hat{\alpha}_\ell^{\mathrm{marg}} = \alpha_\ell + e_\ell$ and $\hat{\beta}_\ell^{\mathrm{marg}} = \beta_\ell + f_\ell$ with $e_\ell = x_\ell^\top \varepsilon_w/(nh_\ell)$ and $f_\ell = x_\ell^\top \varepsilon_z/(nh_\ell)$, both of mean zero. Then

$$\mathbb{E}\big[\hat{\alpha}_\ell^{\mathrm{marg}} \hat{\beta}_\ell^{\mathrm{marg}}\big] = \alpha_\ell\beta_\ell + \mathbb{E}[e_\ell f_\ell] = \alpha_\ell\beta_\ell + \frac{x_\ell^\top \mathrm{Cov}(\varepsilon_w, \varepsilon_z)\, x_\ell}{(nh_\ell)^2} = \alpha_\ell\beta_\ell + \frac{\rho\sigma_w\sigma_z\, nh_\ell}{(nh_\ell)^2},$$

since $\mathrm{Cov}(\varepsilon_w, \varepsilon_z) = \rho\sigma_w\sigma_z I_n$. Multiplying by h_ℓ gives $h_\ell\alpha_\ell\beta_\ell + \rho\sigma_w\sigma_z/n$, and summing over the L loci gives the result. □

For a sense of scale, take $L = 300$ loci, $n = 250$ individuals, a residual correlation $\rho = 0.5$, fitness residual standard deviation $\sigma_w = 0.1$ and a character measured in units of its residual standard deviation, $\sigma_z = 1$. The bias is $300(0.5)(0.1)(1)/250 = 0.06$. Unlike the quadratic case the bias can have either sign, since it takes the sign of ρ, so the plug in can manufacture a predicted response of the character in a direction the genetics does not support. A Monte Carlo check of the proposition on an orthogonal design with $L = 40$, $n = 64$ reproduces the predicted mean to three significant figures.

The structure mirrors Proposition 4.4 with σ_ε^2 replaced by the residual covariance, and the bias vanishes only if the two residuals are uncorrelated, which they are not, since environmental effects on fitness and on a character measured in the same organism are correlated by construction, and since each residual also contains the signal of the loci not being tested, which are shared. In practice the effective ρ differs from the environmental correlation for this second reason, by an amount that depends on the genetic correlation between fitness and the character.

The coherent estimator is the posterior mean of the bilinear functional,

$$\mathbb{E}\big[\sigma_A(w,z) \mid y\big] = \sum_\ell h_\ell \Big(m_\ell^\alpha m_\ell^\beta + \mathrm{Cov}(\alpha_\ell, \beta_\ell \mid y)\Big), \tag{4.10}$$

which requires a joint posterior over the two effect vectors. I use a bivariate horseshoe in which the two locus specific effects share a local scale,

$$(\alpha_\ell, \beta_\ell)^\top \mid \lambda_\ell, \tau \;\sim\; N_2\big(0,\; \tau^2\lambda_\ell^2\, \mathrm{diag}(c_w, c_z)\big), \qquad \lambda_\ell \sim C^+(0,1),$$

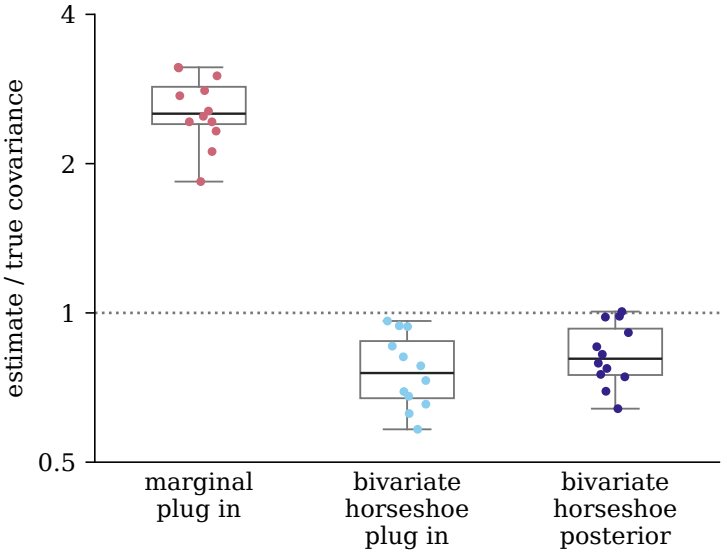


FIGURE 4.6. Estimation of the additive genetic covariance between fitness and a character, twelve replicates, log scale. The marginal plug in overstates by a factor of about two and a half; both bivariate horseshoe estimators understate moderately, the posterior form less than the plug in.

with $c_w = \sigma_w^2(1-\rho^2)$ and $c_z = \sigma_z^2(1-\rho^2)$ the conditional residual variances, and the residual covariance Σ_ε also estimated. Sharing λ_ℓ encodes the substantive belief that a locus is either relevant to both fitness and the character or to neither, which is the sparsity pattern the secondary theorem is about. Sampling proceeds by two conditional draws per sweep, each an ordinary horseshoe update on a residual adjusted response, so the fast sampler of Section 4.4 applies unchanged; details are in Section 4.9.

In simulation ($L = 300$, $n = 250$, twenty loci affecting both, residual correlation 0.5, twelve replicates) the marginal plug in returns 2.58 times the true genetic covariance. The bivariate horseshoe plug in returns 0.77 and the posterior form (4.10) returns 0.83: the posterior covariance term supplies part of what the plug in loses, as in the univariate case, but not all of it. Figure 4.6 shows the distributions. A residual attenuation of about 17% remains, and I do not have a correction for it; the shrinkage of two correlated vectors interacts in a way the univariate argument of Proposition 4.10 does not capture. What can be said is that the bilinear estimator is wrong by seventeen percent in the conservative direction where the standard estimator is wrong by a factor of two and a half in the anticonservative one.

4.7.3 *Linkage disequilibrium changes the functional, not just the estimator*

The formula $\sigma_A^2 = \sum_\ell h_\ell \alpha_\ell^2$ of (4.4) assumes linkage equilibrium. In general the additive genetic variance is the variance of the breeding value $\sum_\ell \alpha_\ell g_\ell$, so

$$\sigma_A^2 = \alpha^\top V \alpha, \qquad V = \operatorname{Var}(g), \tag{4.11}$$

with V reducing to $\mathrm{diag}(h)$ only under equilibrium. This is not a refinement of the estimator; it is a different estimand, and using the diagonal form when V is not diagonal discards the covariance between the breeding value contributions of linked loci. Propositions 4.4 and 4.11 generalize cleanly.

PROPOSITION 4.16 (General quadratic form). *Under* (4.11)*, the ordinary least squares plug in satisfies*

$$\mathbb{E}\big[\hat{\alpha}^\top V \hat{\alpha}\big] = \sigma_A^2 + \sigma_\varepsilon^2 \,\mathrm{tr}\big(V (X^\top X)^{-1}\big),$$

and the posterior mean of the functional is $\mathbb{E}[\sigma_A^2 \mid y] = m^\top V m + \mathrm{tr}\big(V \,\mathrm{Cov}(\alpha \mid y)\big)$.

Proof. $\mathbb{E}[\hat{\alpha}^\top V \hat{\alpha}] = \alpha^\top V \alpha + \mathrm{tr}(V \,\mathrm{Var}(\hat{\alpha}))$ with $\mathrm{Var}(\hat{\alpha}) = \sigma_\varepsilon^2 (X^\top X)^{-1}$; the posterior statement is the same identity applied to the posterior distribution. □

Under orthogonality with $V = \mathrm{diag}(h)$ and $X^\top X = n\,\mathrm{diag}(h)$, the trace is L/n and Proposition 4.4 is recovered; with V replaced by the sample covariance $X^\top X/n$ the trace is L/n for every design, which is Proposition 4.5; so (4.11) is the general statement of which the earlier result is the equilibrium special case.

Simulation with block linkage disequilibrium (blocks of ten, within block latent correlation 0.8, which gives a mean sample correlation of 0.58 between dosages in the same block, $L = 400$, $n = 300$, sixty replicates) bears out the first half of this argument and qualifies the second. The marginal plug in with diagonal weights returns 6.22 times the truth, far worse than the 3.24 of the equilibrium case, because LD tagging and overfitting compound as Section 4.3 argues. The horseshoe plug in with diagonal weights returns 0.80, and the same plug in with the full form $m^\top V m$ returns 0.90: shrinkage removes the overfitting, and the diagonal form then loses a further nine percent of the truth (a paired difference of 0.09, Monte Carlo standard error 0.02), because the posterior mean spreads each causal effect over the loci that tag it and the diagonal form discards their positive cross products. The posterior mean of the functional returns 1.02 with diagonal weights and 1.00 with the full form $\alpha^\top V \alpha$, so for the coherent estimator the choice of functional matters little here. That is a property of this design rather than a general reassurance: the twenty causal loci rarely share a block, so the diagonal form of the true effects averages 1.01 times $\alpha^\top V \alpha$, whereas when causal loci share blocks the cross terms $2\sum_{\ell<m} V_{\ell m}\alpha_\ell \alpha_m$ need not vanish in a given population and the diagonal form is wrong at the truth itself. The lesson is that under LD the estimand is $\alpha^\top V \alpha$, and that a plug in which fixes the estimator but keeps diagonal weights adds a systematic error of the same order as the overshrinkage of the plug in itself.

4.7.4 Ascertainment is not a shrinkage problem

Fitness is not measured on a random sample. It is measured on the individuals who survived long enough to be measured, which is selection on the very variable being analysed. To gauge the effect I repeated the base simulation with $n = 400$ (sixty replicates), then discarded the lower half of the population by realized fitness and reestimated from the survivors alone.

The result is stark. On the full sample the horseshoe posterior returns 1.03 times the true additive variance (Monte Carlo standard error 0.01). On the truncated sample it returns 0.27 (standard error 0.02), and rescaling by the ratio of genotypic variances between the two samples (the obvious first aid correction) barely moves it, to 0.28. Truncating on the dependent variable attenuates every regression coefficient, and squaring the attenuated coefficients attenuates the additive variance roughly quadratically. An almost fourfold understatement swamps the twofold to sixfold overstatement that shrinkage was introduced to fix, and it runs in the opposite direction, so the two errors partially cancel in a way that is entirely accidental and cannot be relied upon.

This is a negative result and I state it plainly: no shrinkage prior addresses this. Ascertainment is a bias in the estimand, not variance in the estimator, and the two are orthogonal problems. The remedy has to come from the likelihood (a truncated or selection corrected model of the sampling, in the tradition of Heckman type corrections, with the selection rule modelled explicitly) and only then does the shrinkage question arise, on the corrected likelihood. Since almost all fitness data in natural populations are ascertained in some such way, I would put this ahead of everything else in Section 4.4 as the practically binding constraint on estimating rates of adaptation. It is also, unlike the others, not a statistical problem that can be solved by a better prior.

An exact calculation for truncation selection

The size of the ascertainment effect can be computed exactly in a Gaussian version of the problem, and the calculation shows both why the damage is so large and why the cure has to come from the likelihood. Let the centred dosage vector $x \in \mathbb{R}^L$ have covariance V, let $y = x^\top \alpha + \varepsilon$ with ε independent of x and of variance σ_ε^2, and write $\sigma_A^2 = \alpha^\top V \alpha$, $V_y = \sigma_A^2 + \sigma_\varepsilon^2$ and $R^2 = \sigma_A^2 / V_y$ for the proportion of the measured variance that is additive. Suppose (x, y) is jointly Gaussian, which is accurate for the breeding value when many loci each contribute a little, and that only individuals with $y > c$ are measured, a fraction q of the population. Put $c' = (c - \mathbb{E}y)/\sqrt{V_y}$, $\lambda = \phi(c')/q$ with ϕ the standard normal density, and

$$\delta = \lambda(\lambda - c') \in (0, 1),$$

the proportional reduction in the variance of a standard normal truncated below at c'.

PROPOSITION 4.17 (Truncation on measured fitness). *Among the measured individuals, the regression coefficients of y on x are*

$$\alpha_S = \frac{1 - \delta}{1 - \delta R^2}\, \alpha,$$

and the additive variance computed from them is

$$\alpha_S^\top V_S\, \alpha_S = \frac{(1-\delta)^2}{1 - \delta R^2}\, \sigma_A^2 \qquad \text{or} \qquad \alpha_S^\top V \alpha_S = \Big(\frac{1-\delta}{1-\delta R^2}\Big)^2 \sigma_A^2,$$

according as the genotypic covariance V_S of the survivors or the covariance V of the whole population is used.

Proof. Because (x, y) is jointly Gaussian, $x = \mathbb{E}x + b\,(y - \mathbb{E}y) + u$ with $b = \mathrm{Cov}(x, y)/V_y = V\alpha/V_y$ and u Gaussian, independent of y, with covariance $V - bb^\top V_y$. Selection on y alone leaves the distribution of u unchanged and u independent of y. Among survivors, therefore, $\mathrm{Var}(y \mid S) = V_y(1 - \delta)$ by the truncated normal variance formula, and

$$\mathrm{Cov}(x, y \mid S) = b\,V_y(1 - \delta) = (1 - \delta)V\alpha,$$
$$V_S = \mathrm{Var}(x \mid S) = V - bb^\top V_y + bb^\top V_y(1 - \delta) = V - \frac{\delta}{V_y} V\alpha\alpha^\top V.$$

The survivor regression coefficient is $\alpha_S = V_S^{-1}\mathrm{Cov}(x, y \mid S)$. To invert V_S on the vector $V\alpha$, note that

$$\Big(V - \frac{\delta}{V_y} V\alpha\alpha^\top V\Big)\alpha = V\alpha\Big(1 - \frac{\delta\,\alpha^\top V\alpha}{V_y}\Big) = (1 - \delta R^2)\,V\alpha,$$

so $V_S^{-1}V\alpha = \alpha/(1 - \delta R^2)$ and $\alpha_S = (1 - \delta)\alpha/(1 - \delta R^2)$. Finally $\alpha^\top V_S\alpha = \sigma_A^2 - \delta\sigma_A^4/V_y = \sigma_A^2(1 - \delta R^2)$, which, multiplied by the square of the scalar factor, gives the first form; the second is immediate. □

Every coefficient is attenuated by the same factor, so truncation does not distort which loci matter, only how much. This is why no prior can detect it: to a regression fitted on the survivors, a population with effects α seen through truncation looks exactly like an untruncated population with the smaller effects α_S, and nothing in a prior on effect sizes can tell the two apart. The additive variance is attenuated by roughly the square of the factor, and rescaling to the population genotypic variance (the first aid correction mentioned above) changes only the denominator: it replaces $(1 - \delta)^2/(1 - \delta R^2)$ by $(1 - \delta)^2/(1 - \delta R^2)^2$ and leaves the dominant $(1 - \delta)^2$ untouched.

EXAMPLE 4.18 (Discarding the less fit half). If the lower half is discarded, $q = 1/2$, $c' = 0$, $\lambda = \phi(0)/0.5 = 0.7979$ and $\delta = \lambda^2 = 2/\pi = 0.6366$. With $R^2 = 0.5$ the coefficients shrink by the factor $0.3634/0.6817 = 0.533$; the additive variance computed on survivors is 0.194 of the truth and the rescaled version 0.284. For the setting of Section 4.5, where $\sigma_A^2 = 0.0205$ out of a phenotypic variance of 0.0320, so $R^2 \approx 0.64$, the large sample values are 0.223 and 0.376. The simulated horseshoe values reported above, 0.27 and 0.28, use the diagonal form $\sum_\ell h_\ell\alpha_\ell^2$, with survivor and population dosage variances respectively. The diagonal form misses the linkage disequilibrium that truncation induces among the causal loci, which lowers the variance of the breeding value but barely changes the dosage variance of any single locus, so both values should be compared with the second form of the proposition, whose limit at the simulation's own $R^2 = 0.62$ is 0.36. They

fall below it, which is plausible for 200 survivors genotyped at 400 loci, far from the large sample limit, where the overfitting of Proposition 4.4 pushes upward and the prior's own shrinkage pushes downward; the simulation does not separate the two. With the full survivor covariance, the posterior mean of $\alpha^\top V_S \alpha$, the same fits return 0.22 (standard error 0.01), against a limit of 0.22 from the first form. That agreement should not be read as a confirmation of the limit: the proposition puts the diagonal form above the full form by the factor $1/(1-\delta R^2) = 1.65$, whereas in the fits the ratio of the two averages is 1.21, so the finite sample effects move the two forms differently. What the simulation does confirm is the size of the loss, roughly three quarters of the additive variance whichever form is used. Table 4.5 gives the factors over a range of truncation fractions and heritabilities of fitness. Even mild ascertainment, losing the least fit tenth, costs roughly forty to fifty percent of the additive variance computed on the survivors.

As a check, a simulation with 200 independent biallelic loci and 400,000 individuals, of whom the fitter half are kept, gives with $R^2 = 0.5$ a coefficient factor of 0.531, a survivor ratio of 0.195 and a rescaled ratio of 0.283, against theoretical values 0.533, 0.194 and 0.284; the binomial dosages are not Gaussian, but the breeding value summed over many loci is close enough.

The proposition also shows what a likelihood correction looks like. If the truncation fraction is known, δ is known, $V_y = \mathrm{Var}(y \mid S)/(1-\delta)$ is estimable from the survivors, and the first identity of the proposition can be solved for the target:

$$\sigma_A^2 = \frac{s_S}{(1-\delta)^2 + \delta\, s_S / V_y}, \qquad s_S = \alpha_S^\top V_S\, \alpha_S.$$

In the same simulation this returns 1.003 times the truth with $R^2 = 0.5$ and 1.008 with $R^2 = 0.65$. The correction works because the selection rule was written into the model, as in the sample selection models of econometrics (Heckman, 1979). It depends entirely on knowing the selection rule; in a natural population where the measured survivors were selected on an unobserved combination of viability and fertility, the same algebra holds with an unknown δ, and the data on survivors can recover it only through strong assumptions about the shape of the fitness distribution.

4.8 Discussion

The Fundamental Theorem is exact and, in Ewens's formulation, clear. What it delivers, however, is a statement about a least squares quantity, and the difficulty of using it empirically is entirely the difficulty of estimating that quantity when there are more loci than individuals. The naive estimator overstates the rate of adaptation by a factor that grows linearly in the number of loci. Ridge type shrinkage overcorrects. A horseshoe prior, combined with the recognition that the additive variance is a quadratic functional requiring the posterior mean rather than the plug in, comes close to the truth in the simulations reported here.

		$R^2 = 0.1$		$R^2 = 0.5$		$R^2 = 0.65$	
kept, q	δ	survivors	rescaled	survivors	rescaled	survivors	rescaled
0.90	0.288	0.522	0.538	0.592	0.692	0.624	0.767
0.75	0.465	0.300	0.314	0.373	0.486	0.410	0.588
0.50	0.637	0.141	0.151	0.194	0.284	0.225	0.384
0.25	0.758	0.063	0.068	0.094	0.151	0.115	0.227

TABLE 4.5. Ratio of the additive variance estimated from survivors to the true additive variance under truncation selection on measured fitness, from Proposition 4.17. "Survivors" uses the survivors' genotypic covariance, "rescaled" the population covariance; R^2 is the additive proportion of the variance in measured fitness.

Problem	Status	Numerical summary
Coverage of σ_A^2 intervals	largely resolved; use the regularized horseshoe	85.0% plain, 90.0% regularized in sixty replicates (84.8% and 89.3% in 600), nominal 90%; plain misses mostly above the truth (80 of 91 in 600 replicates)
Bilinear functional	partially resolved; bivariate prior with shared local scale	marginal 2.58×; bivariate posterior 0.83×
Linkage disequilibrium	resolved by changing the estimand to $\alpha^\top V \alpha$	horseshoe plug in, diagonal 0.80×, full 0.90×; posterior mean 1.00×; marginal 6.22×
Ascertainment	not resolved; not a shrinkage problem	1.03× full sample, 0.27× on survivors

TABLE 4.6. Status of the four problems.

Three caveats deserve emphasis. First, the choice of prior encodes a hypothesis about genetic architecture, and with $L > n$ the data cannot fully adjudicate it; sensitivity across prior families should be reported alongside any estimated rate. Second, the coverage of posterior intervals for the quadratic functional is a few points below nominal, and in 600 replicates the misses of the plain horseshoe fall mostly above the truth (Remark 4.14), so a reported interval for a rate of adaptation should be read as somewhat narrower than its nominal level warrants; the regularized horseshoe should be the default for this reason. Third, and most seriously, real fitness data are ascertained. Section 4.7 shows that measuring fitness on survivors alone can understate the additive variance almost fourfold, that this is a bias in the estimand rather than variance in the estimator, and that no choice of prior repairs it. Until the selection mechanism is written into the likelihood, the shrinkage question is secondary to it.

The broader point is that a century of interpretive dispute about the Fundamental Theorem was, in the end, a dispute about what quantity Fisher

was regressing on what. Now that the question is settled, the remaining obstacle is not conceptual but statistical, and it is the same obstacle that afflicts every other high dimensional problem in genetics: too many parameters, too few individuals, and an estimator that mistakes noise for signal unless it is told not to.

4.9 Posterior sampling

The half Cauchy priors in (4.5) are written as inverse gamma scale mixtures following Makalic and Schmidt (2016): if $\lambda^2 \mid \nu \sim \mathrm{IG}(1/2, 1/\nu)$ and $\nu \sim \mathrm{IG}(1/2, 1)$, then $\lambda \sim C^+(0,1)$, and similarly for τ with auxiliary ξ. Every full conditional is then conjugate. With $D = \mathrm{diag}(\sigma_\varepsilon^2\tau^2\lambda_1^2, \ldots, \sigma_\varepsilon^2\tau^2\lambda_L^2)$ the sweep is

1. $\alpha \mid \cdot \;\sim\; N\big(A^{-1}\Phi^\top \tilde{y},\; A^{-1}\big)$ with $\Phi = X/\sigma_\varepsilon$, $\tilde{y} = y/\sigma_\varepsilon$, $A = \Phi^\top\Phi + D^{-1}$;
2. $\sigma_\varepsilon^2 \mid \cdot \;\sim\; \mathrm{IG}\big(\frac{n+L}{2},\; \frac{1}{2}\|y - X\alpha\|^2 + \frac{1}{2}\sum_\ell \alpha_\ell^2/(\tau^2\lambda_\ell^2)\big)$;
3. $\lambda_\ell^2 \mid \cdot \;\sim\; \mathrm{IG}\big(1,\; 1/\nu_\ell + \alpha_\ell^2/(2\tau^2\sigma_\varepsilon^2)\big)$ and $\nu_\ell \mid \cdot \sim \mathrm{IG}(1, 1 + 1/\lambda_\ell^2)$;
4. $\tau^2 \mid \cdot \;\sim\; \mathrm{IG}\big(\frac{L+1}{2},\; 1/\xi + \frac{1}{2\sigma_\varepsilon^2}\sum_\ell \alpha_\ell^2/\lambda_\ell^2\big)$ and $\xi \mid \cdot \sim \mathrm{IG}(1, 1 + 1/\tau^2)$.

Step 1 is executed by the algorithm of Bhattacharya et al. (2016), which avoids forming A: draw $u \sim N(0, D)$ and $\delta \sim N(0, I_n)$, set $v = \Phi u + \delta$, solve $(\Phi D \Phi^\top + I_n) w = \tilde{y} - v$, and return $\alpha = u + D\Phi^\top w$. The cost is $O(n^2 L)$ per iteration, so the sampler is cheapest exactly when $L > n$, which is the regime of interest. The bivariate sampler of Section 4.7 adds one step. Conditional on β, the fitness residual is regressed after adjusting for the character residual: with $r_z = y_z - X\beta$, the response $y_w - \rho(\sigma_w/\sigma_z) r_z$ follows an ordinary regression on X with error variance $\sigma_w^2(1-\rho^2)$, so step 1 applies verbatim; the roles are then exchanged to draw β. The shared local scale is updated from both coefficients, $\lambda_\ell^2 \mid \cdot \sim \mathrm{IG}\big(3/2,\; 1/\nu_\ell + (\alpha_\ell^2/c_w + \beta_\ell^2/c_z)/(2\tau^2)\big)$ with c_w, c_z the conditional variances, which is the two dimensional group horseshoe update. The residual covariance is refreshed from the current residuals each sweep.

All results in Section 4.5 use 1200 iterations with the first 400 discarded; the additive variance is computed on each retained draw as $\sum_\ell h_\ell \alpha_\ell^2$, giving the posterior distribution required by Proposition 4.11 directly rather than by a delta method approximation.

4.10 Exercises

1. Show that the total genetic variance in fitness at one locus decomposes as $\sum_{i,j} p_i p_j (w_{ij} - \bar{w})^2 = \sigma_A^2 + \sigma_D^2$, and check it on Example 4.2. *Hint.* The normal equations say the residuals r_{ij} are orthogonal, in the $p_i p_j$ weighted inner product, to the constant and to every $\alpha_i + \alpha_j$.

2. For two alleles with $w_{11} = m + a$, $w_{12} = m + d$, $w_{22} = m - a$ and $p = p_1$, show that $\alpha_1 - \alpha_2 = a + d(1 - 2p)$, $\sigma_A^2 = 2p(1-p)\big[a + d(1-2p)\big]^2$ and $\sigma_D^2 =$

$\big[2p(1-p)d\big]^2$. *Hint.* Compute $w_1 - w_2$ directly; for σ_D^2 evaluate the residual at the heterozygote and use Exercise 1..

3. With the parameterization of Exercise 2., show that the exact one generation change is $\Delta\bar{w} = 2(\alpha_1 - \alpha_2)\Delta p + (w_{11} - 2w_{12} + w_{22})(\Delta p)^2$, so that the remainder in (4.3) is $R = -2d(\Delta p)^2$. Verify $R = -7.5 \times 10^{-6}$ in Example 4.3 at $q = 0.05$. *Hint.* $\bar{w}$ is a quadratic in p, so its second order Taylor expansion is exact, and $d\bar{w}/dp = 2(w_1 - w_2)$.

4. In Example 4.3, show that $w_A = w_S$, and hence $\sigma_A^2 = 0$, exactly when $q = s/(s+t)$, and that $\sigma_D^2 = \big[2q^*(1-q^*)\big]^2(s+t)^2/4$ there. Evaluate it. *Hint.* In the notation of Exercise 2. the dominance deviation is $d = (s+t)/2$ relative to the homozygote midpoint.

5. Under the conditions of Proposition 4.4 with Gaussian errors and $\alpha = 0$, show that $(n/\sigma_\varepsilon^2)\sum_\ell h_\ell(\hat{\alpha}_\ell^{\mathrm{marg}})^2$ has a χ^2_L distribution, and deduce part (b) of the proposition in this case. *Hint.* In standardized units the statistic is $\sum_\ell z_\ell^2$ with independent standard normal z_ℓ.

6. For the design of Table 4.1, compute $(X^\top X)^{-1}$ and confirm that the variance inflation factors are $6, 3, 5, 3, 3$. Then draw 1000 noise vectors with $\sigma_\varepsilon = 0.05$ and check the two expectations in Example 4.6. *Hint.* The diagonal of $X^\top X$ is 4 throughout, so $\mathrm{VIF}_\ell = 4\,[(X^\top X)^{-1}]_{\ell\ell}$.

7. Derive the density (4.6) of the shrinkage weight from $\lambda \sim C^+(0,1)$, and show it reduces to $\mathrm{Beta}(1/2, 1/2)$ when $\tau = 1$. *Hint.* $\lambda = \tau^{-1}\sqrt{(1-\kappa)/\kappa}$ and $|d\lambda/d\kappa| = \tfrac{1}{2}\tau^{-1}\kappa^{-3/2}(1-\kappa)^{-1/2}$.

8. Prove (4.8) using the law of total variance applied to θ_ℓ given z_ℓ, conditioning on κ_ℓ. *Hint.* $\mathrm{Var}(\theta \mid z) = \mathbb{E}[\mathrm{Var}(\theta \mid z, \kappa) \mid z] + \mathrm{Var}(\mathbb{E}[\theta \mid z, \kappa] \mid z)$.

9. (Ridge as a common weight.) Suppose a single weight κ is applied to every locus, $\hat{\theta}_\ell = (1-\kappa)z_\ell$. Show that the weight minimizing $\mathbb{E}\sum_\ell(\hat{\theta}_\ell - \theta_\ell)^2$ is $\kappa^* = L/(\|\theta\|^2 + L)$ and that $\mathbb{E}\sum_\ell \hat{\theta}_\ell^2 = \|\theta\|^4/(\|\theta\|^2 + L)$ at this weight. Deduce that when $\|\theta\|^2 = L/2$ oracle ridge recovers one third of the additive variance, and compare with Proposition 4.10. *Hint.* Repeat the proof of Proposition 4.10 with the sums over loci taken before minimizing.

10. In the bivariate sampler of Section 4.9, show that conditional on β the adjusted response $y_w - \rho(\sigma_w/\sigma_z)(y_z - X\beta)$ equals $X\alpha$ plus Gaussian noise of variance $\sigma_w^2(1-\rho^2)$, independent across individuals. *Hint.* Use the conditional distribution of ε_w given ε_z for a bivariate normal pair.

11. Verify that $\delta = 2/\pi$ when the lower half is discarded, and use Proposition 4.17 to reproduce the row $q = 0.75$ of Table 4.5. What fraction must be kept for the survivors' additive variance to be at least 90% of the truth when $R^2 = 0.5$? *Hint.* $c' = \Phi^{-1}(1-q)$; solve $(1-\delta)^2/(1-\delta R^2) = 0.9$ for δ and then for q numerically.

12. (Computation.) Simulate $n = 200$ individuals at $L = 50, 100, 200, 400, 800$ independent loci with twenty causal loci of fixed effect, compute the marginal plug in, and plot its excess over σ_A^2 against L. Fit a line through the origin and compare its slope with σ_y^2/n. *Hint.* For the marginal estimator the effective error variance of each single locus regression is the phenotypic variance, which includes the signal of the other loci.

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CHAPTER 5

Many Parameter Models in Population Genetics

Population genetics has become a many parameter subject. Whole genome average effects, polygenic effect sizes and allelic frequency spectra all involve parameter counts that are large relative to any sample, and in Fisher's whole genome formulation the count is astronomically large. We make three points. First, Fisher's definition of the average effect, as restated by Ewens (2024), is a weighted least squares projection, and the plug in additive genetic variance obtained from a linear smoother H carries a noise term $\sigma^2 \operatorname{tr}(H^\top H)/N$, which for the unregularised fit of that projection is an inflation of exactly $p\sigma^2/N$. Second, the natural prior for such problems is a species sampling prior in the sense of Kingman (1975), of which the Ewens Sampling Formula is the canonical member. Third, the global local shrinkage priors of sparse Bayesian statistics and the two parameter Poisson Dirichlet family are linked by an exact identity of marginal laws: if $\kappa \sim \mathrm{Be}(a, b)$ is the shrinkage weight of a prior in the three parameter beta normal class, then κ has the law of the first stick breaking weight of $\mathrm{GEM}(1-a,\, a+b-1)$, and the mean intensity of the matched frequency spectrum is $\nu(x) = x^{-1} f_\kappa(x)$ exactly; the full partition laws are not shown to agree. Sparsity level is Pitman's α. We then prove the resulting rate directly for the shrinkage prior as used: at the canonical global scale $\tau_n = n^{-1/2}$ the expected effective model size is $\asymp n^{1-a}$, matching the Poisson Dirichlet discovery rate in exponent exactly and in constant to within $\Gamma(2-a) \in [0.8856, 1]$. The horseshoe and the Strawderman Berger prior therefore sit at $n^{1/2}$, while the uniform shrinkage prior and the other priors with $a = 1$ sit at $\log n$ (Laplace type priors, outside the beta normal class, stay bounded), and the conventional choice of global scale is the unique power of n at which the two sides agree for every a. We give proofs, worked one locus and two locus examples, a small design on which the optimism can be checked by hand, a worked Kingman correspondence at $n = 3$, a table of discovery counts for three members of the beta normal class, a conjugate augmentation for the mutation parameter with worked small sample and butterfly examples, and simulations. In an epistatic whole genome simulation with $L = 400$ loci the unregularised projection in its orthogonal design form inflates σ_A^2 by a factor of 3.09 at $N = 200$; a species sampling prior at the global scale the theory identifies, with no oracle input, never inflates it and reaches 0.98 of the truth by $N = 3200$.

5.1 Introduction

Ewens (2024) opens with the observation that the direction of research in population genetics theory is now, and correctly, retrospective, and that this is driven by the volume of genomic data available. That change of direction has a statistical consequence which has not been much discussed. Retrospective inference

from genomic data is inference in models whose parameter count is large relative to the sample, often very large, and sometimes astronomically large. The classical prospective theory was developed for models with a handful of parameters and its estimators inherit no protection against the difficulties that arise when parameter count grows with, or exceeds, the amount of data.

This chapter is about what to do in that regime, and about the fact that population genetics already contains, in the Ewens Sampling Formula, the prior distribution that the statistics literature spent the last twenty years rediscovering under a different name.

We organise the argument around three questions.

1. *Is the additive genetic variance safe to estimate by plugging in?* No. Fisher's definition of the average effect, restated by Ewens (2024, Equation 10), is a weighted least squares projection onto an additive subspace whose dimension is of the order of the number of alleles in the genome. Section 5.2 shows that fitting that projection produces an estimate of σ_A^2 exceeding the truth in expectation by $p\sigma^2/N$, where p is the number of fitted average effects. This is the optimism that inflates R^2 in any regression, and at genomic parameter counts it is not a small correction.

2. *What is the right prior?* Section 5.5 argues that the right prior is a species sampling prior. Kingman (1975) introduced the random discrete distributions obtained by normalising a subordinator and showed that sampling from one induces an exchangeable partition. The Ewens Sampling Formula is the partition induced by the gamma subordinator, and the global local shrinkage priors of Carvalho, Polson and Scott (2010) and Polson and Scott (2012) are built from subordinator increments, without the normalisation.

3. *How far does the correspondence go?* Section 5.6 gives our main results. Theorem 5.15 identifies the shrinkage weight of the three parameter beta normal family with the first stick breaking weight of a two parameter Poisson Dirichlet process. Theorem 5.18 shows the identification is not confined to one weight: the entire frequency spectrum intensity is the shrinkage density with its size bias removed. Corollary 5.22 converts this into discovery rates.

Section 5.3 explains how the resulting estimators relate to Fisher's own weighted least squares criterion, Section 5.7 treats inference for the mutation parameter, Section 5.9 reports simulations, Section 5.10 draws out the genetic consequences and Section 5.12 concludes. Proofs not given in the text are collected in Section 5.13, and Section 5.15 contains exercises.

5.1.1 Methodological contribution

The chapter draws on two literatures which do not usually meet, so we state at the outset what is on offer.

1. *An identification, Theorem 5.15.* The shrinkage weight of a global local prior in the three parameter beta normal class is the first stick breaking weight of a two parameter Poisson Dirichlet process, under the explicit bijection $(\alpha, \theta) = (1 - a,\, a + b - 1)$. The horseshoe is $\mathrm{PD}(1/2, 0)$, the Strawderman Berger prior is $\mathrm{PD}(1/2, 1/2)$, and the uniform shrinkage prior is the Ewens Sampling Formula at $\theta = 1$.

2. *An extension of that identification to the whole spectrum, Theorem 5.18.* The mean intensity of allele frequencies is $\nu(x) = x^{-1} f_\kappa(x)$, the shrinkage density with its size bias removed, from which Fisher's log series follows as the case $\kappa \sim \mathrm{Be}(1, \theta)$.

3. *A design rule, Corollary 5.22, proved directly for the shrinkage prior in Theorem 5.23.* The number of variants a prior will ever declare grows like n^{1-a}, and the effective model size of (5.13) at the canonical global scale $\tau_n = n^{-1/2}$ has the same exponent with a constant agreeing to within $\Gamma(2 - a) \in [0.8856, 1]$. Choosing a prior is therefore choosing a discovery rate, and the choice is a genetic statement about the polygenic architecture, not a computational convenience.

4. *A bias calculation for Fisher's own estimand, Proposition 5.5 and Corollary 5.6.* The plug in additive genetic variance obtained by fitting Equation (5.1) has expectation $\|HX\beta\|^2/N + \sigma^2 \mathrm{tr}(H^\top H)/N$, which for the unregularised projection is $\sigma_A^2 + p\sigma^2/N$. Separately, Proposition 5.3 shows that even the population estimand fails to decompose across loci under linkage disequilibrium, by the factor $c^\top R^{-1} c$ against $c^\top c$.

5. *A reading of the shrinkage estimator as penalised weighted least squares, Section 5.3.* The estimator we propose is Fisher's criterion with a quadratic penalty, and the penalty weights are drawn from the same family of random discrete distributions that supplies the least squares weights. This is the sense in which the proposal is internal to Fisher's formulation rather than imported.

6. *An assessment of the existing repairs for overparameterisation, Section 5.4.* Each buys identifiability with an assumption that is not testable against the data it is imposed on, minimum norm interpolation included, since minimum ℓ_2 norm is a Gaussian prior in disguise. What distinguishes the proposal here is not that it avoids an assumption but that its assumption is the neutral genetic model itself and is falsifiable through the frequency spectrum.

7. *A precise account of where the correspondence stops, Proposition 5.27.* Means agree; fluctuations do not. The coefficient of variation of the effective model size vanishes like $n^{-(1-a)/2}$ while that of K_n converges to the Mittag Leffler constant 0.7555 at $a = 1/2$.

8. *A conjugate augmentation for θ, Proposition 5.32.* The latent beta variable of Escobar and West, extended to many loci and combined with a heavy tailed

Pochhammer prior, replaces the customary point estimate with a posterior at negligible cost.

5.1.2 *Empirical findings*

Four numerical results are reported, all in Section 5.9.

1. *The predicted discovery rates are attained.* Simulating the matched partition directly, the horseshoe case has fitted log log slope 0.498 against the predicted $1/2$, while the two $a = 1$ cases show no power law and grow linearly in $\log n$ with slopes 1.005 and 2.835 against the predicted $\theta = 1$ and $\theta = 3$. At $n = 3 \times 10^4$ the horseshoe case has occupied 166 blocks on average and the $\mathrm{Be}(1, 1)$ case 10.6.

2. *The additive genetic variance is inflated threefold at realistic dimensions.* With $L = 400$ loci and epistasis present, the fitted projection, in its orthogonal design form, reports 3.09 times the true σ_A^2 at $N = 200$, falling to 1.14 at $N = 3200$, tracking $1 + p(\sigma_A^2 + \sigma^2)/(N\sigma_A^2)$. The species sampling prior reports 0.65 rising to 0.98 and never inflates.

3. *Rare variant prediction is essentially unbiased.* Estimating θ from K_n and predicting $\mathbb{E}K_{2n}$ gives mean relative errors between 4.3% and 9.3% across four settings, against a naive doubling rule which overstates K_{2n} by a factor of between 1.6 and 1.8.

4. *Uncertainty in θ is of the order of θ itself.* At $n = 100$ and $\theta = 5$ the median 95% interval is $[2.53, 8.52]$, a width of 120% of θ, with exact frequentist coverage 0.943. At $n = 1000$ it is still 83%, consistent with the $(\log n)^{-1/2}$ rate implied by Proposition 5.30.

5.1.3 *Relation to the companion literature*

Three recent papers of Ewens and of Ewens and Hössjer set the context, and this chapter is in large part a statistical commentary on them.

Ewens (2024) gives the definitive account of Fisher's Fundamental Theorem, concluding that the theorem as Fisher stated it is not correct and should be set aside. What survives, and what concerns us, is the apparatus Fisher built in order to state it: the average effect defined by weighted least squares, the additive genetic variance defined as the sum of squares that fit removes, and the observation in his Equation (15) that the rate of change of mean fitness exceeds the additive genetic variance by a term reflecting the movement of the average effects themselves. Section 5.2 treats his Equation (10) as the high dimensional regression it is, and Remark 5.8 argues that his Equation (15) and Efron's covariance penalty are needed for the same structural reason.

Hössjer and Ewens (2026) construct a self contained continuous time theory of allele frequency change, showing that the standard treatment borrows its generations concept from a discrete approximation and is therefore not self

contained at all. Their remedy, a time change defined from the age distribution of parents, has a direct consequence for us: it makes the mutation parameter convention dependent, which is the subject of Section 5.8. The methodological parallel is also worth noting. Their objection is that a continuous time theory should not borrow its time scale from a discrete approximation; ours is that a shrinkage prior should not borrow its sparsity level from a tuning constant.

Ewens and Hössjer (2026) show that there is no such thing as *the* effective population size, only the inbreeding, variance and eigenvalue notions, which coincide for the idealised monoecious model and diverge for two sex and subdivided populations, in their Table 1 by as much as fourteen per cent at small N_m. The same is true of *the* sparsity level. Theorem 5.15 says which two parameter family the alternatives inhabit and Corollary 5.22 says what distinguishes them, which is the closest analogue we can offer to their resolution.

5.2 Fisher's average effects as a many parameter regression

5.2.1 The estimand and the worked one locus case

This section takes Fisher's definition of the average effect at face value, as a weighted least squares problem, and asks what happens when it is fitted from a sample. The answer has two parts: a population level failure of additivity across loci, caused by linkage disequilibrium, and a sample level inflation of the fitted variance, caused by dimension.

Following Ewens (2024), list the whole genome autosomal genotypes as $g = 1, \ldots, G$, write P_g for the frequency of genotype g and m_g for its Malthusian parameter. The average effects $\{\beta_{ij}\}$ minimise

$$Q(\beta) \;=\; \sum_{g} P_g \Big(m_g - \sum_{(g)} \beta_{ij} \Big)^2, \tag{5.1}$$

the inner sum running over the genes present in genotype g, each counted once, twice or not at all. The additive genetic variance σ_A^2 is the sum of squares removed from the total variance of the Malthusian parameters by this fit.

Ewens (2024) notes that no explicit solution exists, the average effects depending on epistasis between loci and on linkage disequilibrium. We add that (5.1) is a population quantity, defined with the true P_g and the true m_g. Any empirical version replaces these with sample counterparts, and at that moment the problem becomes a regression with p parameters fitted from N observations. The statistical properties of the fitted σ_A^2 are then not those of the estimand.

A worked one locus case. To fix ideas, and to connect (5.1) to the average excess concept that Ewens (2024) discusses, take a single locus with alleles $A_1, \ldots, A_k$ at frequencies $p_1, \ldots, p_k$, random mating so that $P_{uv} = p_u p_v$, and Malthusian parameters m_{uv}.

PROPOSITION 5.1. *Under random mating the minimisers of* $\sum_u \sum_v P_{uv}(m_{uv} - \beta_u - \beta_v)^2$ *are*

$$\beta_u \;=\; m_{u\cdot} - \tfrac{1}{2}\bar{m}, \qquad m_{u\cdot} = \sum_v p_v m_{uv}, \qquad \bar{m} = \sum_u \sum_v p_u p_v m_{uv},$$

so that $\bar{m} = 2\sum_u p_u \beta_u$, *and*

$$\beta_u - \bar{\beta} \;=\; a_u \;:=\; m_{u\cdot} - \bar{m}, \qquad \sigma_A^2 \;=\; 2\sum_u p_u a_u^2,$$

where a_u *is the average excess of* A_u *and* $\bar{\beta} = \sum_u p_u \beta_u$.

Three consequences are worth stating. First, the identity $\bar{m} = 2\sum_u p_u \beta_u$ is Ewens (2024, Equation 3), here derived rather than assumed. Second, under random mating the average effect and the average excess coincide up to the additive constant $\bar{\beta}$, which is Fisher's classical result and explains why Ewens (2024) is able to move between the two. Third, combining $\sigma_A^2 = 2\sum_u p_u a_u^2$ with Fisher's $\mathrm{d}p_u/\,\mathrm{d}t = p_u a_u$ recovers Ewens (2024, Equation 2), namely $\sigma_A^2 = 2\sum_u (\,\mathrm{d}p_u/\,\mathrm{d}t)\beta_u$, since $\sum_u p_u a_u = 0$.

EXAMPLE 5.2 (Overdominance). Take $k = 2$, $m_{11} = m_{22} = 0$ and $m_{12} = m_{21} = s > 0$. Then $m_{1\cdot} = p_2 s$, $m_{2\cdot} = p_1 s$ and $\bar{m} = 2p_1 p_2 s$, so $a_1 = p_2 s(1 - 2p_1)$ and $a_2 = p_1 s(2p_1 - 1)$, giving

$$\sigma_A^2 \;=\; 2p_1 p_2 s^2 (1 - 2p_1)^2.$$

This vanishes at $p_1 = 1/2$, the equilibrium, even though the variance in Malthusian parameters across genotypes does not. With numbers: at $p_1 = 0.3$ and $s = 1$ the genotypic variance is $2p_1 p_2(1 - 2p_1 p_2) = 0.42 \times 0.58 = 0.2436$, while $\sigma_A^2 = 2(0.3)(0.7)(0.4)^2 = 0.0672$, so only 28 per cent of the genotypic variance is additive; at $p_1 = 0.45$ the additive share falls to 2 per cent, and at $p_1 = 1/2$ it is zero. The additive projection can be null while the underlying variation is large. Any estimator of σ_A^2 must therefore be able to report values near zero without being dominated by fitting noise, which is exactly what Corollary 5.6 says the unregularised fit cannot do: its expectation is bounded below by $p\sigma^2/N$ whatever the truth.

5.2.2 Why the whole genome is not the sum of its loci, and how large the failure is

Ewens (2024) records that a principal error of Fisher (1930) was to assume that the whole genome additive genetic variance is the sum of one locus values and that whole genome average effects are given by the one locus values. In regression notation the reason is immediate. Write the empirical design as $X = [X_1\, X_2\, \cdots\, X_L]$ with one block of columns per locus. The least squares fit satisfies $X^\top X\hat{\beta} = X^\top y$ and the explained sum of squares decomposes across blocks if and only if $X_\ell^\top X_{\ell'} = 0$ for $\ell \neq \ell'$. That orthogonality is exactly linkage equilibrium. Under linkage disequilibrium the cross blocks are nonzero, the fitted effect at one locus depends on the others, and the decomposition fails. Epistasis contributes separately, by

placing signal outside the column space of X altogether, where it is absorbed into the residual and therefore into σ^2. Ewens (2024) reaches the same conclusion from the genetics; we note only that it is a statement about a Gram matrix, and that its magnitude is therefore computable from data.

The Gram matrix statement of the previous paragraph can be made quantitative, and the magnitude is not small. Suppose the coded loci $z_1, \dots, z_L$ have been standardised to unit variance, write R for their correlation matrix, which is the matrix of linkage disequilibrium coefficients in standardised form, and write $c_\ell = \mathrm{Cov}(m, z_\ell)$.

PROPOSITION 5.3. *The whole genome additive genetic variance is $\sigma_A^2 = c^\top R^{-1} c$, whereas the sum of the one locus values is $\sum_\ell c_\ell^2 = c^\top c$. These agree for all c if and only if $R = I$, that is under complete linkage equilibrium.*

EXAMPLE 5.4 (Two loci). Take $L = 2$ with $c = (1, 1)^\top$ and R having off diagonal r. Then $c^\top R^{-1} c = 2/(1 + r)$ while $c^\top c = 2$. At $r = 1/2$ the correct value is $4/3$ and the sum of one locus values is 2, an overstatement of fifty per cent. At $r = -1/2$ the correct value is 4 and the sum of one locus values is again 2, an understatement by a factor of two.

Example 5.4 makes concrete the error that Ewens (2024) attributes to Fisher (1930), and shows that it is signed neither way: positive linkage disequilibrium between loci with effects of the same sign causes the one locus sum to overstate, and negative disequilibrium causes it to understate. There is therefore no direction in which one may safely err. It also shows that the error is not a small sample phenomenon; it is present in the population quantities themselves, and no amount of data removes it. What data can remove is the separate, additive contribution studied in the next subsection.

5.2.3 The optimism of the fitted additive genetic variance

Write the empirical version of (5.1) as

$$y = X\beta + \eta, \qquad \mathbb{E}\eta = 0, \quad \mathrm{Var}\eta = \sigma^2 I_N, \tag{5.2}$$

where X is $N \times p$ with standardised gene contents, y collects observed Malthusian parameters or phenotypes, and η absorbs epistatic and environmental contributions, that is everything orthogonal to the additive subspace. Let $\hat{y} = Hy$ for a linear smoother H and define the plug in additive genetic variance $\hat{\sigma}_A^2 = \|Hy\|^2 / N$.

PROPOSITION 5.5 (Optimism of a linear smoother). *Under* (5.2),

$$\mathbb{E}\,\hat{\sigma}_A^2 \;=\; \frac{\|HX\beta\|^2}{N} \;+\; \frac{\sigma^2 \,\mathrm{tr}(H^\top H)}{N}.$$

COROLLARY 5.6 (Inflation of Fisher's fitted projection). *If H is the orthogonal projection onto the column space of X and X has full column rank p, then* $\mathrm{tr}(H^\top H) = p$

and $HX\beta = X\beta$, *so*

$$\mathbb{E}\,\hat{\sigma}^2_A \;=\; \sigma^2_A + \frac{p\,\sigma^2}{N}, \qquad \sigma^2_A = \|X\beta\|^2/N.$$

The bias is additive, does not vanish as the signal strengthens, and is proportional to the ratio of parameter count to sample size. In a whole genome analysis with p of order 10^6 and N of order 10^5 the projection interpolates the data and, by Corollary 5.10, the bias is the whole residual variance σ^2, which exceeds σ^2_A whenever the heritability is below one half. Fisher's σ^2_A is a projection, and projections onto high dimensional subspaces capture noise in proportion to their dimension.

Proposition 5.5 also says what a good estimator must do. Shrinking H away from the projection reduces $\operatorname{tr}(H^\top H)$, the variance term, at the cost of attenuating $\|HX\beta\|^2$, the signal term. The whole question is which H, and the answer is supplied by a prior.

EXAMPLE 5.7 (The optimism on a design small enough to check by hand). Take $N = 6$ individuals typed at $p = 3$ biallelic loci, each genotype coded -1, 0 or 1 by the centred count of one allele, with design and true average effects

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \\ -1 & -1 & 0 \\ -1 & 0 & -1 \end{pmatrix}, \qquad \beta = \begin{pmatrix} 1 \\ 1/2 \\ 0 \end{pmatrix}, \qquad \sigma^2 = 1.$$

Every column sums to zero and $X^\top X = \operatorname{diag}(6,4,4)$, so the loci are in linkage equilibrium in this sample and nothing but dimension is at work. The true additive genetic variance is $\sigma^2_A = \|X\beta\|^2/N = (6\cdot 1 + 4\cdot \tfrac{1}{4})/6 = 7/6 \approx 1.167$, and the third locus is null.

For ridge smoothers $H_\lambda = X(X^\top X + \lambda I)^{-1}X^\top$ the matrix H_λ has eigenvalues $d_j/(d_j+\lambda)$ on the column space, where $d_j \in \{6,4,4\}$ are the eigenvalues of $X^\top X$, and zero elsewhere. Hence $\operatorname{tr}(H_\lambda^\top H_\lambda) = \sum_j d_j^2/(d_j+\lambda)^2$ and $\|H_\lambda X\beta\|^2 = \sum_j d_j\{d_j/(d_j+\lambda)\}^2\beta_j^2$, and Proposition 5.5 gives the three rows of the following table.

smoother	$\operatorname{tr} H$	$\operatorname{tr}(H^\top H)$	$\|HX\beta\|^2/N$	$\mathbb{E}\hat{\sigma}^2_A$	Monte Carlo
projection, $\lambda = 0$	3.000	3.000	1.167	1.667	1.664
ridge, $\lambda = 2$	2.083	1.451	0.637	0.878	0.878
ridge, $\lambda = 6$	1.300	0.570	0.277	0.372	0.372

For the projection the arithmetic is $7/6 + 3/6 = 5/3$: half a unit of residual variance has been booked as additive genetic variance, an inflation of 43 per cent with only

three parameters. At $\lambda = 2$ the shrinkage factors are $3/4$ on the first locus and $2/3$ on the other two, so the variance term falls to $1.451/6 = 0.242$, while the signal term falls from 1.167 to $(6 \cdot \frac{9}{16} + 4 \cdot \frac{4}{9} \cdot \frac{1}{4})/6 = 0.637$. The estimate is now biased downwards, and heavier shrinkage makes it more so. The last column averages $\|Hy\|^2/N$ over 4×10^5 simulated response vectors and agrees with the formula to within 0.003.

Two lessons generalise. First, the relevant trace is $\mathrm{tr}(H^\top H)$, not the degrees of freedom $\mathrm{tr}\, H$ that appear in Efron's covariance penalty; the two agree only for projections. Second, no fixed λ is right: a uniform penalty that is light enough to keep the first locus removes too little noise from the null third locus, and one heavy enough to silence the null locus destroys the signal. A penalty that varies by locus, light where the effect is large and heavy where it is null, is what Section 5.3 obtains from a prior.

REMARK 5.8 (A formal parallel with Ewens (2024), Equation 15). Ewens (2024) shows that $\mathrm{d}\bar{m}/\,\mathrm{d}t = \sigma_A^2 + 2\sum_i \sum_j p_{ij}\ \mathrm{d}\beta_{ij}/\,\mathrm{d}t$ and that the second term is generally nonzero because the average effects are functions of the gene frequencies and therefore move. The bias in Corollary 5.6 has the same origin in a different variable. In the statistical problem the fitted basis moves with the data rather than with time, and the correction, which in Efron's covariance penalty form is $2\sum_i \mathrm{Cov}(\hat{y}_i, y_i)/N = 2\,\mathrm{df}\,\sigma^2/N$, is the analogue of Fisher's second term. We do not claim a theorem. We observe that in both settings the naive identity fails because the coefficients being summed are not constants, and that in both settings the size of the failure is governed by how fast they move. Ewens (2024) concludes that the Fundamental Theorem, read as an equality, is not correct. The statistical reading is the same: the explained sum of squares is not the parameter it is taken for, and the discrepancy is the drift term.

5.3 Weighted least squares, and what a prior adds to it

Corollary 5.6 says the fitted projection must be regularised; it does not say how, and it does not say whether regularising is faithful to what Fisher was doing. This section addresses both. The claim is that the estimator we propose is not an alternative to Fisher's criterion but that criterion with one term added, and that the added term is built from the same probabilistic object as the criterion itself.

5.3.1 Fisher's criterion as a projection, and its empirical version

Equation (5.1) is weighted least squares, the weights being the genotype frequencies P_g. Geometrically, let $\mathcal{M}$ be the space of real functions on the genotype set equipped with the inner product

$$\langle f, h \rangle_P \;=\; \sum_g P_g f(g) h(g),$$

and let $\mathcal{A} \subset \mathcal{M}$ be the additive subspace spanned by the gene content functions, that is by the indicators counting how often each allele appears in a genotype.

Then (5.1) reads $Q(\beta) = \|m - \mathcal{A}\beta\|_P^2$, the average effects are the coordinates of the $L^2(P)$ orthogonal projection $\Pi_{\mathcal{A}} m$, and

$$\sigma_A^2 \;=\; \|\Pi_{\mathcal{A}} m\|_P^2 - \bar{m}^2.$$

The normal equations are $\mathcal{A}^\top W \mathcal{A}\beta = \mathcal{A}^\top W m$ with $W = \operatorname{diag}(P_g)$. Ewens (2024) observes that these cannot be solved explicitly for the whole genome because $\mathcal{A}^\top W m$ involves epistatic effects and $\mathcal{A}^\top W \mathcal{A}$ linkage disequilibrium values. Proposition 5.3 is the quantitative version of that observation: $\mathcal{A}^\top W \mathcal{A}$ is, after standardisation, the correlation matrix R, and $\sigma_A^2 = c^\top R^{-1} c$ rather than $c^\top c$.

The connection between this population criterion and the regression of Section 5.2 is exact, and it is the point at which sampling error enters.

PROPOSITION 5.9. *Let genotypes $g_1, \dots, g_N$ be drawn independently from P with observed values $y_i = m_{g_i} + \varepsilon_i$, where the genotype set is finite, the ε_i are independent and identically distributed with mean zero and variance σ^2 and independent of the g_i, and redundant gene content functions have been removed so that $\mathcal{A}^\top W \mathcal{A}$ is nonsingular. Then the ordinary least squares criterion $N^{-1} \sum_{i=1}^N (y_i - \sum_{(g_i)} \beta_{ij})^2$ is the weighted criterion* (5.1) *evaluated at the empirical measure $\hat{P}_N$, and converges almost surely to $Q(\beta) + \sigma^2$ uniformly on compacta. Its minimiser converges almost surely to Fisher's average effects.*

So sampling N individuals and running ordinary least squares *is* Fisher's weighted least squares with the weights estimated by their empirical frequencies. The estimator of the average effects is consistent. The plug in estimator of σ_A^2 is consistent for fixed p but not when p/N stays away from zero, and Corollary 5.6 says by how much: the projection is computed in $\hat{P}_N$ rather than in P, and a projection computed in the empirical measure captures $p\sigma^2/N$ of noise before it captures any signal.

5.3.2 *The shrinkage estimator is penalised weighted least squares*

Our proposal replaces (5.1) by

$$\hat{\beta} \;=\; \arg\min_{\beta} \Big\{ \underbrace{\sum_g \hat{P}_g \Big(m_g - \sum_{(g)} \beta_{ij} \Big)^2}_{\text{Fisher's criterion}} \;+\; \underbrace{\frac{\sigma^2}{N} \sum_j \frac{\beta_j^2}{s^2 \tau^2 \lambda_j^2}}_{\text{penalty}} \Big\}, \qquad s^2 = \frac{\sigma^2}{N}. \tag{5.3}$$

Nothing has been discarded: the first term is exactly (5.1) with the weights estimated and the Malthusian parameters replaced by their observed values. The scale s is the standard error of a least squares coefficient in an orthogonal design, and measuring the prior scale in units of s keeps τ dimensionless. The penalty is $2\sigma^2/N$ times the negative log density of the prior $\beta_j \mid \lambda_j, \tau \sim N(0, s^2\tau^2\lambda_j^2)$, and the first term is $2\sigma^2/N$ times the negative Gaussian log likelihood, so (5.3) is the posterior mode, which for a Gaussian prior is also the posterior mean. Given

the local scales λ, differentiating gives the solution $\hat{\beta} = (X^\top X + D_\lambda^{-1})^{-1} X^\top y$ with $D_\lambda = \mathrm{diag}(\tau^2 \lambda_j^2 / N)$, and the corresponding smoother is $H_\lambda = X(X^\top X + D_\lambda^{-1})^{-1} X^\top$. In the orthogonal case $X^\top X = NI$ this collapses to

$$\hat{\beta}_j \;=\; (1 - \kappa_j)\, z_j, \qquad \kappa_j = \frac{1}{1 + \tau^2 \lambda_j^2}, \qquad z_j = N^{-1} (X^\top y)_j, \tag{5.4}$$

so that κ_j, the object of Theorem 5.15, is precisely the fraction of the weighted least squares coefficient that the penalty removes. This is the operational meaning of everything in Sections 5.5 and 5.6: a statement about κ is a statement about how much of Fisher's fitted average effect survives.

By Proposition 5.5 the effect on the estimand is exactly what one wants. Since $\mathrm{tr}(H_\lambda^\top H_\lambda) = \sum_j (1 - \kappa_j)^2 \le p$, with equality only when nothing is shrunk, the variance term is reduced by the factor $p^{-1} \sum_j (1 - \kappa_j)^2$, the effective number of retained average effects $\sum_j (1 - \kappa_j)^2$ divided by the nominal number p. The signal term $\| H_\lambda X \beta \|^2 / N$ becomes $\sum_j (1 - \kappa_j)^2 \beta_j^2$ in place of $\sigma_A^2 = \sum_j \beta_j^2$. A good prior is one that makes κ_j near one where β_j is near zero and near zero elsewhere, which is the design requirement that the horseshoe was constructed to meet.

5.3.3 The weights and the penalty come from the same object

We can now say why the species sampling machinery is the natural source of the penalty rather than one arbitrary source among many.

In Fisher's criterion the weights P_g are genotype frequencies. They are a random discrete distribution, generated by mutation and drift, and the classical description of that distribution is the Ewens Sampling Formula, that is a normalised gamma subordinator in the sense of Kingman (1975). In (5.3) the penalty weights $1/(\tau^2 \lambda_j^2)$ are chosen by the analyst, and by Theorem 5.15 the standard choices correspond to stick breaking weights of a two parameter Poisson Dirichlet process, that is to normalised subordinators of the same family or, when $\alpha > 0$ and $\theta \neq 0$ as for the Strawderman Berger prior, to a normalised stable subordinator with the law of its total reweighted. Theorem 5.18 extends the correspondence to the mean intensity of the induced frequency spectrum.

So the object appearing on the left of (5.3) as observation weights and the object appearing on the right as penalty weights are drawn from one family. Fisher's formulation already contains that family; we are proposing to use it a second time, on the other side of the criterion. The alternative regularisations available in the literature, ridge, lasso, elastic net, are not wrong but are exogenous: they impose a geometry on the coefficients which has no counterpart in the genetics. A species sampling penalty imposes the geometry that the genotype frequencies already have.

5.3.4 Moving weights: iterative reweighting and Fisher's drift term

There is a final structural parallel, and it explains Remark 5.8 in the language of this section.

In (5.3) the penalty weights are not fixed. The local scales λ_j and the global scale τ are estimated, so $\hat{\beta}$ is the fixed point of an iteratively reweighted least squares recursion: given λ, solve for β; given β, update λ. The map H is then not linear in y, and the optimism in Proposition 5.5 must be computed for the converged smoother, whose effective degrees of freedom exceed the naive count $\sum_j (1 - \kappa_j)$ by an amount reflecting the estimation of λ.

Ewens (2024) faces the same structure in a different variable. His average effects are functions of the gene frequencies, the gene frequencies change under selection, and therefore the coefficients being summed in the Fundamental Theorem are not constants. His Equation (15) splits $\mathrm{d}\bar{m}/\,\mathrm{d}t$ into σ_A^2 plus a term $2\sum_i \sum_j p_{ij}\,\mathrm{d}\beta_{ij}/\,\mathrm{d}t$ which is generally nonzero, and his conclusion is that the Theorem read as an equality is false. In (5.3) the weights move with the data rather than with time, the corresponding correction is Efron's covariance penalty, and the corresponding conclusion is that the explained sum of squares is not the estimand it is taken for.

Neither statement is a corollary of the other; the parallel is structural rather than formal. But it identifies the same culprit in both settings, namely a projection whose basis is not held fixed.

5.4 Overparameterisation in genomic data

Before proposing anything we set out the difficulty in the form it actually takes in genomic data, and review what is currently done about it. We do this because the review is not flattering to any of the available methods, including ours, and it is better to say so at the start.

5.4.1 The problem

A modern genome wide study measures p variants on N individuals with p of order 10^6 to 10^7 and N of order 10^4 to 10^6. In Fisher's whole genome formulation (5.1) the parameter count is worse still, since the average effects are indexed by every allele at every locus and the genotype sum runs over a set of astronomical size. Four things go wrong at once.

Nonidentifiability. When $p > N$ the Gram matrix $X^\top X$ is singular, the normal equations have infinitely many solutions, and infinitely many vectors of average effects fit the data exactly. Fisher's estimand is still well defined at the population level; its empirical version is not defined at all without a further rule for choosing among the exact fits.

Optimism. Even when $p < N$, Corollary 5.6 shows the fitted additive genetic variance carries the additive bias $p\sigma^2/N$. At $p/N = 1/2$ that bias is half the residual variance, and it does not shrink as the signal strengthens.

Collinearity. Linkage disequilibrium makes columns of X locally near duplicates, so even where a solution is unique it is unstable, and by Proposition 5.3 the estimand itself fails to decompose across loci in either direction.

Selection effects. Variants are reported because they passed a threshold, so reported effect sizes are conditioned on their own magnitude, a bias which grows as p grows relative to the number of true signals.

5.4.2 *What is currently done, and what each method assumes*

method	the assumption that buys identifiability
single variant testing with multiplicity control	effects can be assessed one at a time; the joint model is never fitted, so linkage disequilibrium is handled by post hoc pruning rather than by the model
ridge, genomic best linear unbiased prediction, mixed model heritability	every variant contributes, with effects independent and normal of common variance; sparsity is excluded by assumption
lasso and elastic net	exact sparsity in the chosen basis, with a tuning constant selected by cross validation and no distributional statement about the result
spike and slab and finite variance class mixtures	a small number of effect size classes with prior mixing weights, both chosen by the analyst
regression of test statistics on linkage disequilibrium scores	a specified functional relation between the two, plus polygenicity and no confounding
minimum norm interpolation	nothing stated, and therefore something unstated: minimum ℓ_2 norm is a Gaussian prior in disguise

The pattern is the same in every row. Identifiability is not recovered from the data; it is purchased with an assumption, and the assumption is generally not testable against the data it is imposed on. It is fair to ask what makes our proposal different, and the honest answer has two parts.

The first is negative. We do not escape the pattern. The species sampling prior of Sections 5.5 and 5.6 is an assumption like any other, and if the allelic architecture is not of that form the resulting estimates will be wrong in ways the data will not reveal.

The second is positive, and it is the reason we think the proposal is worth making. Our assumption is not imported from the statistics of some other subject; it is the neutral population genetics model itself, the Ewens Sampling Formula, used on the other side of Fisher's criterion as explained in Section 5.3. And, unlike the assumptions in the table, it is checkable. By Theorem 5.15 the prior is indexed by (α, θ); by Theorem 5.18 those parameters determine the entire expected frequency spectrum through (5.14); and by Corollary 5.20 the growth of the singleton count identifies α. The assumption therefore makes a prediction about a quantity every genomic study already computes, and can be falsified by it. That is a weaker claim than solving the problem, and it is the strongest claim we are able to defend.

5.4.3 A remark on double descent

The behaviour of the overparameterised regime deserves separate comment, because it is frequently offered as evidence that the difficulty above has been overstated.

COROLLARY 5.10 (Saturation of the optimism). *Let H be the orthogonal projection onto the column space of X, of rank* $\min(p, N)$ *generically. Then Proposition 5.5 gives*

$$\mathbb{E}\,\hat{\sigma}_A^2 \;=\; \sigma_A^2 \;+\; \frac{\sigma^2 \min(p, N)}{N},$$

so the inflation grows linearly in p up to the interpolation threshold $p = N$ and is constant at σ^2 thereafter.

The right panel of Figure 5.1 confirms this. The design has $N = 200$ individuals, ten causal variants of unit effect and $\sigma^2 = 1$, so the true σ_A^2 averages 10.11 across the sixty replicates and the predicted rise from small p to the threshold is one unit. Because the same genotype matrix is used for every p within a replicate, the Monte Carlo standard error of each plotted point is only about 0.12, small against that rise. The plotted values climb from 10.20 at $p/N = 0.1$ to 11.10 at $p/N = 1$ and then stay at 11.10 for every larger p. Averaged over the points below the threshold they give 10.67 against a predicted 10.68, and over the points at or above it 11.10 against a predicted 11.11. The substantive point is that adding variants beyond the threshold neither helps nor hurts this estimand: once $p \geq N$ the projection is the identity and $\hat{\sigma}_A^2$ is simply $\|y\|^2/N$, in contrast to the prediction risk in the left panel.

The prediction risk behaves quite differently, and is the phenomenon known as double descent (Belkin et al., 2019; Hastie et al., 2022). The left panel of Figure 5.1 shows the out of sample risk of the minimum norm interpolator rising from 0.11 at $p/N = 0.1$ to about 9.8×10^3 at $p/N = 1$, then descending to 5.4 at $p/N = 1.6$, a regime in which more variants than individuals is better than an equal number.

The Bayesian reading of this, developed by Polson and Sokolov (2025), is the point we wish to make here, and it bears directly on the assessment in the previous subsection. The minimum norm solution is not assumption free. It is the posterior mean under a Gaussian prior in the limit of infinite prior variance, restricted to the row space of X, and the peak at $p = N$ is the divergence of that posterior as the smallest singular value approaches zero. The descent beyond the threshold occurs because adding columns to a wide matrix raises its smallest singular value, so the implicit prior is progressively better conditioned. Nothing has been obtained for free; an assumption has been made silently and then rewarded.

Two consequences. First, the benign behaviour of the heavily overparameterised regime is evidence about a particular implicit prior, not about overparameterisation as such. Second, and visible in the left panel, an explicit prior removes the pathology entirely: the ridge curve has no peak, and at $p/N = 4$ it is indistinguishable from the interpolator. If one is going to make

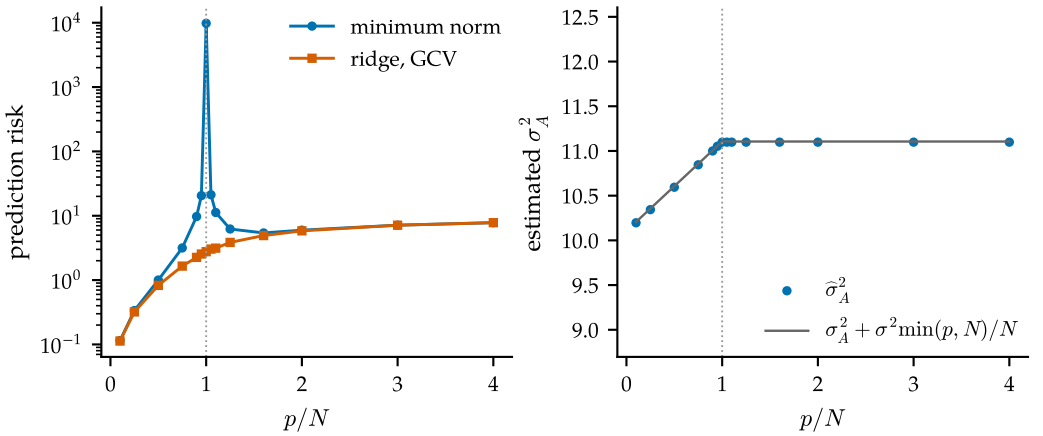


FIGURE 5.1. $N = 200$ individuals, ten causal variants of unit effect, sixty replicates. Left: out of sample prediction risk on a logarithmic scale. The minimum norm interpolator peaks at the interpolation threshold and descends beyond it; ridge with the scale chosen by generalised cross validation is monotone and never spikes; at $p/N = 4$ the two coincide at 7.8. Right: estimated σ_A^2 against $\sigma_A^2 + \sigma^2 \min(p, N)/N$, illustrating the saturation of Corollary 5.10. The per point Monte Carlo standard error is about 0.12; the group means are 10.67 below the threshold and 11.10 at or above it, against predictions of 10.68 and 11.11.

an assumption in order to proceed, and Section 5.4 argues that one must, the argument for stating it is that a stated assumption can be tuned, criticised and, in the case of Corollary 5.20, tested.

5.5 Species sampling priors

5.5.1 Kingman's correspondence

Let ρ be a Lévy density on $(0, \infty)$ with $\int_0^\infty \rho(x)\,\mathrm{d}x = \infty$ and $\int_0^\infty \min(1, x)\rho(x)\,\mathrm{d}x < \infty$. Let $J_1 \geq J_2 \geq \cdots$ be the ranked points of a Poisson process on $(0, \infty)$ with intensity ρ, put $T = \sum_i J_i$, finite almost surely by the second condition, and set

$$P_i \;=\; J_i/T, \qquad i = 1, 2, \ldots \tag{5.5}$$

Kingman (1975) showed that (5.5) produces a random discrete distribution and that drawing n exchangeable samples from (P_i) induces an exchangeable random partition of $\{1, \ldots, n\}$.

The two integrability conditions are worth separating, because they play opposite roles and both are needed. The second, $\int_0^\infty \min(1, x)\rho(x)\,\mathrm{d}x < \infty$, is what makes the normalisation in (5.5) possible at all: its part on $(0, 1)$ makes the small jumps summable and its part on $(1, \infty)$ makes the large jumps finitely many, and without it there is no total to divide by. The first, $\int_0^\infty \rho(x)\,\mathrm{d}x = \infty$, forces infinitely many atoms, since the number of points exceeding ε is Poisson with mean $\int_\varepsilon^\infty \rho(x)\,\mathrm{d}x$, which tends to infinity as $\varepsilon \downarrow 0$. It is the formal expression of

the statement that there are always more rare types waiting to be discovered, and every result in this chapter about discovery rates is a quantitative refinement of it.

Two cases are classical:

$$\rho(x) = \theta x^{-1} e^{-x} \implies (P_i) \sim \mathrm{PD}(0, \theta),$$
$$\rho(x) = \tfrac{\alpha}{\Gamma(1-\alpha)} x^{-1-\alpha} \implies (P_i) \sim \mathrm{PD}(\alpha, 0).$$

The first is the gamma subordinator, for which the Laplace exponent is $\psi(t) = \theta \log(1+t)$ and $T \sim \mathrm{Gamma}(\theta, 1)$; the normalised jumps give the Dirichlet process, and the total T is independent of the normalised sequence (P_i), a special property of the gamma case which is what makes the Ewens Sampling Formula so tractable. The second is the α stable subordinator, with $\psi(t) = t^\alpha$; here T has no finite mean and is not independent of (P_i), and that dependence is exactly what Proposition 5.27 later shows to be the source of the nonvanishing fluctuation in the number of discovered types.

The relation between the two cases is the whole story. Both densities behave like a power of x near the origin, x^{-1} in the first and $x^{-1-\alpha}$ in the second, and it is behaviour near the origin that controls how many small atoms exist and hence how many rare types a sample uncovers. The gamma case is the boundary $\alpha = 0$ multiplied by the exponential tilt e^{-x}, which is harmless at the origin but truncates the large jumps. Corollary 5.22 below is, in the end, a statement about what that boundary costs: the pole x^{-1}, not the tilt, converts a power law in the sample size into a logarithm, and no choice of θ recovers the power.

5.5.2 The Ewens Sampling Formula and its derivation from the subordinator

The partition structure of $\mathrm{PD}(0, \theta)$ is the Ewens Sampling Formula (Ewens, 1972, 2004). For a sample of n genes with a_j alleles represented exactly j times, $\sum_j j a_j = n$,

$$\mathbb{P}(a_1, \ldots, a_n) = \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{1}{a_j!} \Big(\frac{\theta}{j}\Big)^{a_j}, \qquad \theta_{(n)} = \frac{\Gamma(\theta + n)}{\Gamma(\theta)}, \tag{5.6}$$

and equivalently, for a particular partition of $\{1, \ldots, n\}$ into k blocks of sizes $n_1, \ldots, n_k$,

$$p(n_1, \ldots, n_k \mid \theta) = \frac{\theta^k}{\theta_{(n)}} \prod_{i=1}^{k} (n_i - 1)!. \tag{5.7}$$

PROPOSITION 5.11 (Hoppe urn, Hoppe 1984). *Start with an urn containing a black ball of mass θ. At each of n steps draw a ball with probability proportional to mass. If the black ball is drawn, return it together with a new ball of a new colour and unit mass; otherwise return the drawn ball together with a further ball of the same colour and unit mass. The colour partition of the n draws has law* (5.7).

Proposition 5.11 makes the formula simulable in n steps and exhibits θ in its role as the rate at which new types appear. It is the source of the two facts we

use repeatedly: that K_n is a sum of independent indicators, and that θ enters the likelihood only through K_n.

From the subordinator to the sampling formula. The correspondence of Section 5.5 produces (5.7) explicitly, and the calculation exhibits the exact point at which the exponential tilt enters. Define the Laplace exponent and the tilted moments of ρ,

$$\psi(t) = \int_0^\infty \bigl(1 - e^{-tx}\bigr)\rho(x)\,\mathrm{d}x, \qquad \xi(m,t) = \int_0^\infty x^m e^{-tx}\rho(x)\,\mathrm{d}x.$$

PROPOSITION 5.12 (Poisson Kingman partition formula). *For the normalised subordinator* (5.5)*, the probability of a particular partition of* $\{1,\dots,n\}$ *into blocks of sizes* $n_1,\dots,n_k$ *summing to* n *is*

$$p(n_1,\dots,n_k) = \frac{1}{\Gamma(n)} \int_0^\infty t^{\,n-1} e^{-\psi(t)} \prod_{i=1}^{k} \xi(n_i,t)\,\mathrm{d}t. \tag{5.8}$$

EXAMPLE 5.13 (The gamma subordinator gives the Ewens Sampling Formula). Take $\rho(x) = \theta x^{-1}e^{-x}$. Then $\psi(t) = \theta\int_0^\infty (1-e^{-tx})x^{-1}e^{-x}\,\mathrm{d}x = \theta\log(1+t)$ and $\xi(m,t) = \theta\int_0^\infty x^{m-1}e^{-(1+t)x}\,\mathrm{d}x = \theta\,\Gamma(m)(1+t)^{-m}$. Substituting in (5.8),

$$\begin{aligned} p(n_1,\dots,n_k) &= \frac{\theta^k \prod_i (n_i-1)!}{\Gamma(n)} \int_0^\infty t^{\,n-1}(1+t)^{-\theta-n}\,\mathrm{d}t \\ &= \theta^k \prod_i (n_i-1)!\,\frac{B(n,\theta)}{\Gamma(n)} = \frac{\theta^k}{\theta_{(n)}} \prod_i (n_i-1)!, \end{aligned}$$

which is (5.7).

EXAMPLE 5.14 (Kingman's correspondence at $n = 3$). A sample of three genes can be partitioned in three ways up to labelling: all three alike, written (3); two alike and one different, $(2,1)$, which can happen in three labelled ways; and all different, $(1,1,1)$. Proposition 5.12 gives the probability of each labelled partition from the Lévy density alone.

For the gamma subordinator with $\theta = 2$ the formula of Example 5.13 reads $p(n_1,\dots,n_k) = 2^k\prod_i (n_i-1)!/(2\cdot 3\cdot 4)$, so

$$p(3) = \tfrac{2\cdot 2}{24} = \tfrac{1}{6}, \qquad p(2,1) = \tfrac{4\cdot 1}{24} = \tfrac{1}{6}, \qquad p(1,1,1) = \tfrac{8}{24} = \tfrac{1}{3},$$

and $p(3) + 3p(2,1) + p(1,1,1) = \tfrac{1}{6} + \tfrac{1}{2} + \tfrac{1}{3} = 1$, as it must.

For the stable subordinator, $\rho(x) = \alpha x^{-1-\alpha}/\Gamma(1-\alpha)$, the same computation is a little longer and shows the tilt doing its work. Here $\psi(t) = t^\alpha$ and $\xi(m,t) = \alpha\Gamma(m-\alpha)t^{\alpha-m}/\Gamma(1-\alpha)$, so the integrand in (5.8) is a power of t times e^{-t^α}, and the substitution $u = t^\alpha$ gives $\int_0^\infty t^{k\alpha-1}e^{-t^\alpha}\,\mathrm{d}t = \Gamma(k)/\alpha$. Hence

$$p(n_1,\dots,n_k) = \frac{\alpha^{k-1}\,(k-1)!}{(n-1)!} \prod_{i=1}^{k} (1-\alpha)_{(n_i-1)},$$

which is the two parameter formula (5.10) below at $\theta = 0$. At $\alpha = 1/2$ and $n = 3$,

$$p(3) = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot \tfrac{3}{2} = \tfrac{3}{8}, \qquad p(2,1) = \tfrac{\frac{1}{2} \cdot 1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{8}, \qquad p(1,1,1) = \tfrac{\frac{1}{4} \cdot 2}{2} = \tfrac{1}{4},$$

again summing to one with the factor three on the middle term. The expected number of alleles is $\mathbb{E}K_3 = p(3) + 6p(2,1) + 3p(1,1,1)$, which is $13/6 \approx 2.167$ for the gamma case with $\theta = 2$ and $15/8 = 1.875$ for the stable case. Both values agree with the exact formula (5.12) below, and the integrals in (5.8) were also checked by numerical quadrature.

The stable case puts more mass on the extreme partitions (3) and $(1,1,1)$ than a gamma case with the same $\mathbb{E}K_3$ would: the gamma case with $\theta = 1.092$ also has $\mathbb{E}K_3 = 1.875$, but gives $p(3) = 0.309$ and $p(1,1,1) = 0.184$ against 0.375 and 0.250. That is the small sample face of the fluctuation result of Section 5.6.3: a stable subordinator either has one dominant atom or many small ones, and three genes are enough to see it.

Example 5.13 shows where θ comes from and where the logarithm in $\mathbb{E}K_n \sim \theta \log n$ comes from. The factor θ^k arises because $\xi(m,t)$ carries one factor of θ per block, and the exponential tilt e^{-x} in ρ is what makes $\xi(m,t)$ decay like $(1+t)^{-m}$ rather than like a power of t alone. Replacing the simple pole x^{-1} at the origin by $x^{-1-\alpha}$, with or without the tilt, replaces the logarithm by n^α, since the tilt is harmless at the origin. This is the mechanism behind Corollary 5.22, stated at the level of the Lévy measure rather than at the level of the partition.

5.5.3 The two families to be identified

We now place side by side the two objects that Section 5.6 identifies.

Pitman's two parameter family.

The two parameter family $\mathrm{PD}(\alpha,\theta)$, $0 \le \alpha < 1$, $\theta > -\alpha$, has stick breaking representation $\mathrm{GEM}(\alpha,\theta)$,

$$W_1 = V_1, \qquad W_i = V_i \prod_{j<i}(1 - V_j), \qquad V_i \sim \mathrm{Be}(1-\alpha,\ \theta + i\alpha), \tag{5.9}$$

independently, and exchangeable partition probability function

$$p(n_1,\dots,n_k \mid \alpha,\theta) = \frac{\prod_{i=1}^{k-1}(\theta + i\alpha)}{(\theta+1)_{(n-1)}} \prod_{i=1}^{k} (1-\alpha)_{(n_i-1)}, \tag{5.10}$$

where $x_{(m)} = x(x+1)\cdots(x+m-1)$ is the rising factorial, so that $\theta_{(n)} = \theta\,(\theta+1)_{(n-1)}$. Setting $\alpha = 0$ recovers (5.7), and the form with $(\theta+1)_{(n-1)}$ rather than $\theta_{(n)}$ in the denominator is the one that remains meaningful at $\theta = 0$. Dividing (5.10) for $n+1$ by its value for n gives the predictive rule: given k blocks of sizes $n_1,\dots,n_k$, the next draw joins block i with probability $(n_i - \alpha)/(\theta+n)$ and starts a new block

with probability $(\theta + k\alpha)/(\theta + n)$. The number of distinct blocks satisfies

$$\mathbb{E}K_n \sim \begin{cases} \dfrac{\Gamma(\theta+1)}{\alpha\Gamma(\theta+\alpha)}\, n^{\alpha}, & 0 < \alpha < 1, \\ \theta \log n, & \alpha = 0. \end{cases} \tag{5.11}$$

In fact the mean is available exactly: for $0 < \alpha < 1$,

$$\mathbb{E}K_n = \frac{\Gamma(\theta+1)}{\alpha\,\Gamma(\theta+\alpha)}\,\frac{\Gamma(\theta+\alpha+n)}{\Gamma(\theta+n)} - \frac{\theta}{\alpha}, \tag{5.12}$$

and for $\alpha = 0$, $\mathbb{E}K_n = \sum_{j=0}^{n-1} \theta/(\theta+j)$; both are proved in Section 5.13, and the asymptotics (5.11) follow from $\Gamma(x+\alpha)/\Gamma(x) \sim x^{\alpha}$. The dichotomy in (5.11) is the whole story of what follows. A positive α gives a power law in the sample size; $\alpha = 0$ gives a logarithm, however large θ is made.

Global local shrinkage priors. A global local prior for a many parameter mean vector is

$$\beta_i \mid \lambda_i, \tau \sim N(0, \tau^2\lambda_i^2), \qquad \lambda_i \sim \pi(\lambda), \tag{5.13}$$

and with unit observation variance the shrinkage weight is $\kappa_i = 1/(1+\tau^2\lambda_i^2)$, so that $\mathbb{E}(\beta_i \mid y_i) = (1 - \mathbb{E}(\kappa_i \mid y_i))y_i$ and $\kappa_i = 1$ is total shrinkage while $\kappa_i = 0$ is none. The standard family is the three parameter beta normal class of Armagan, Dunson and Clyde (2011), in which $\kappa \sim \mathrm{Be}(a, b)$ with $a \in (0, 1]$, $b > 0$, the density being $f_\kappa(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$. The exponent a governs the behaviour near $\kappa = 0$, that is how readily large effects escape shrinkage, and b the behaviour near $\kappa = 1$, how hard small effects are pulled to zero. The horseshoe of Carvalho, Polson and Scott (2010) is $a = b = 1/2$, the arcsine law, with poles at both ends. The prior of Strawderman (1971) and Berger (1980) is $(a, b) = (1/2, 1)$, with the pole at zero but none at one. The uniform shrinkage prior, κ uniform on $(0, 1)$, is $a = b = 1$ and has no pole at all.

Polson and Scott (2012) showed that (5.13) is generated by the increments of a subordinator, the object that Kingman's construction normalises. It is therefore unsurprising that the two families should be related. What is perhaps surprising is that the relation is an identity.

5.6 Sparsity level is Pitman's alpha

This section proves that the two families of Section 5.5 are linked by an exact identity of marginal laws. The idea is simple. A shrinkage weight κ near zero means an effect left almost untouched, and under a species sampling prior a stick breaking weight near zero means a rare type; both families have a single parameter, a on one side and α on the other, that controls how much mass the law places near zero. Matching those two parameters, and then the second parameter on each side, gives an exact identity.

prior	$\kappa \sim$	(α, θ)	$\mathbb{E}K_n$
horseshoe	$\mathrm{Be}(1/2, 1/2)$	$(1/2,\ 0)$	$\asymp n^{1/2}$
three parameter beta, general	$\mathrm{Be}(a, b)$	$(1-a,\ a+b-1)$	$\asymp n^{1-a}$
Strawderman Berger	$\mathrm{Be}(1/2, 1)$	$(1/2,\ 1/2)$	$\asymp n^{1/2}$
uniform shrinkage	$\mathrm{Be}(1, 1)$	$(0,\ 1)$	$\asymp \log n$
ridge like, $b > 1$	$\mathrm{Be}(1, b)$	$(0,\ b)$	$\asymp b \log n$
Bayesian lasso, normal gamma	no pole at $\kappa = 0$	$\alpha = 0$	$\asymp \log n$

TABLE 5.1. The dictionary of Theorem 5.15. The final row is an application neither of the theorem nor of Corollary 5.22: it reads these priors through the gamma subordinator, an $\alpha = 0$ object with $\mathbb{E}K_n \sim \theta \log n$. Used directly at $\tau_n = n^{-1/2}$ their effective model size is bounded (see the paragraph after Corollary 5.22), so for them $\log n$ is an upper bound only. The horseshoe plus and Dirichlet Laplace priors are outside the $\mathrm{Be}(a, b)$ family and are not covered by the table.

THEOREM 5.15 (Shrinkage weights are stick breaking weights). *Let $\kappa \sim \mathrm{Be}(a, b)$ with $a \in (0, 1]$ and $b > 0$. Then κ has the law of the first stick breaking weight W_1 of $\mathrm{GEM}(\alpha, \theta)$ in* (5.9) *under*

$$\alpha = 1 - a, \qquad \theta = a + b - 1,$$

and this map is a bijection from $\{a \in (0, 1],\ b > 0\}$ onto $\{0 \le \alpha < 1,\ \theta > -\alpha\}$.

Matching parameters in (5.9) gives $1 - \alpha = a$ and $\theta + \alpha = b$; that the map is onto is checked in Section 5.13.

COROLLARY 5.16. *The horseshoe, $\kappa \sim \mathrm{Be}(1/2, 1/2)$, corresponds to $(\alpha, \theta) = (1/2, 0)$, that is to $\mathrm{PD}(1/2, 0)$, the normalised one half stable subordinator. The Strawderman Berger prior, $\kappa \sim \mathrm{Be}(1/2, 1)$, corresponds to $(\alpha, \theta) = (1/2, 1/2)$. The uniform shrinkage prior, $\kappa \sim \mathrm{Be}(1, 1)$, corresponds to $(\alpha, \theta) = (0, 1)$, that is to the Dirichlet process with $\theta = 1$, whose partition structure is* (5.6) *at $\theta = 1$.*

The arcsine law that gives the horseshoe its name is therefore not an algebraic convenience. It is the signature of $\alpha = 1/2$, and the same law appears in Lévy's arcsine theorem for Brownian motion for the same reason, the one half stable subordinator being the inverse of Brownian local time. Table 5.1 lists the priors of this chapter with their pairs (α, θ) and the order of $\mathbb{E}K_n$.

5.6.1 Consequences: the frequency spectrum and the discovery rate

Theorem 5.15 is an identity for a single weight. The following extends the identification from one weight to the mean intensity of the entire spectrum, and is the result on which the genetic reading of the chapter rests.

Theorem 5.15 identifies κ with the first stick breaking weight W_1, and what the next result needs is a size biased pick. These are the same object, by a classical theorem of Pitman which we record because the identification is the hinge of the argument.

LEMMA 5.17 (Stick breaking is size biased). *The sequence (W_i) of* (5.9) *is the size biased permutation of the ranked frequencies of* $\mathrm{PD}(\alpha,\theta)$. *In particular W_1 is a size biased pick, that is* $\mathbb{P}(W_1 \in A) = \mathbb{E}\sum_i P_i \mathbf{1}(P_i \in A)$.

Lemma 5.17 is Pitman's residual allocation result (Pitman, 1996; Pitman and Yor, 1997) and we do not reprove it. Its consequence here is that the density of κ supplied by Theorem 5.15 is the density of a size biased pick, which is exactly the input required below.

THEOREM 5.18 (Spectrum and shrinkage profile). *Let (P_i) be a random discrete distribution and let ν be the mean intensity of its atoms, so that* $\mathbb{E}\sum_i h(P_i) = \int_0^1 h(x)\nu(x)\,\mathrm{d}x$ *for bounded h vanishing at the origin. Let W be a size biased pick from (P_i), with density f_W. Then*

$$f_W(x) = x\,\nu(x), \qquad \textit{equivalently} \qquad \nu(x) = x^{-1} f_W(x).$$

Consequently, for a global local prior whose shrinkage weight density f_κ is matched by Theorem 5.15, so that $f_\kappa = f_W$ by Lemma 5.17, the induced allele frequency spectrum satisfies

$$\mathbb{E}\,a_j = \binom{n}{j} \int_0^1 x^{j-1}(1-x)^{n-j} f_\kappa(x)\,\mathrm{d}x. \tag{5.14}$$

COROLLARY 5.19 (Fisher's log series). *For $\kappa \sim \mathrm{Be}(1,\theta)$, that is $(\alpha,\theta) = (0,\theta)$,* (5.14) *gives*

$$\mathbb{E}a_j = \frac{\theta}{j} \cdot \frac{n!}{(n-j)!} \cdot \frac{\Gamma(\theta+n-j)}{\Gamma(\theta+n)} \;\longrightarrow\; \frac{\theta}{j}$$

as $n \to \infty$ with j fixed, which is Fisher's log series and the expected frequency spectrum of the Ewens Sampling Formula.

COROLLARY 5.20 (Singletons under the horseshoe). *For the horseshoe, $f_\kappa(x) = \{\pi\sqrt{x(1-x)}\}^{-1}$,* (5.14) *at $j = 1$ gives $\mathbb{E}a_1 = n\,B(1/2, n-1/2)/\pi \sim \sqrt{n/\pi}$.*

EXAMPLE 5.21 (Two spectra with the same number of alleles). Take $n = 20$ genes and compare the Ewens Sampling Formula at $\theta = 2$, that is $\kappa \sim \mathrm{Be}(1,2)$, with the horseshoe, $\mathrm{PD}(1/2,0)$. Their expected numbers of alleles are close, 5.29 and 5.01, but (5.14) shows the alleles are spread differently:

j	1	2	3	4	5	6
Ewens, $\theta = 2$	1.905	0.905	0.571	0.405	0.305	0.238
log series θ/j	2.000	1.000	0.667	0.500	0.400	0.333
horseshoe, $\mathrm{PD}(1/2,0)$	2.572	0.660	0.340	0.219	0.158	0.123

The first entry can be checked by hand: Corollary 5.19 at $j = 1$ gives $\mathbb{E}a_1 = \theta n/(\theta+n-1) = 40/21 = 1.905$. In both cases $\sum_j j\,\mathbb{E}a_j = 20$, as it must, since every gene belongs to exactly one allele. The horseshoe spectrum is more top heavy at both ends: more singletons, and more of the sample concentrated in one or two common alleles, with a thinner middle. With only twenty genes the log series is

still a rough guide to the Ewens means, and the convergence in Corollary 5.19 is visibly slow at larger j, where the factor $n!\,\Gamma(\theta+n-j)/\{(n-j)!\,\Gamma(\theta+n)\}$ is furthest from one.

Corollary 5.20 is a sharp consistency check, not merely a plausibility one. Under $\mathrm{PD}(1/2,0)$, (5.11) gives $\mathbb{E}K_n \sim \Gamma(1)n^{1/2}/\{\frac{1}{2}\Gamma(\frac{1}{2})\} = 1.128\,n^{1/2}$, while Corollary 5.20 gives $\mathbb{E}a_1 \sim \sqrt{n/\pi} = 0.564\,n^{1/2}$. The ratio is exactly $1/2$. That is the value required by the general fact that under $\mathrm{PD}(\alpha,\theta)$ with $\alpha > 0$ the proportion of blocks of size j converges to $\alpha(1-\alpha)_{(j-1)}/j!$, which at $j=1$ is α (Pitman, 2006). Three independently derived quantities, the spectrum formula (5.14), the growth rate (5.11) and the singleton fraction, therefore agree. (On the shrinkage side Theorem 5.23 gives the constant 1 rather than 1.128, the factor $\Gamma(3/2) = 0.886$ of Corollary 5.24.) It is also a statement about rare variant discovery with a direct empirical content. An $\alpha = 1/2$ prior expects the number of singleton variants to grow as the square root of the sample size; an $\alpha = 0$ prior expects it to converge to θ. These are different biological claims about the same data, and modern sequencing studies, in which singletons dominate the discovered spectrum, appear to favour the former.

Discovery rates.

COROLLARY 5.22 (Discovery rates). *Under Theorem 5.15 and* (5.11)*, a global local prior with* $\kappa \sim \mathrm{Be}(a,b)$ *induces an expected number of occupied blocks in a sample of size* n *growing like* n^{1-a} *when* $a<1$ *and like* $(a+b-1)\log n$ *when* $a=1$*. In particular the horseshoe discovers of the order of* $n^{1/2}$ *signals, and every prior with* $a=1$*, that is the uniform shrinkage prior and the ridge like priors* $\mathrm{Be}(1,b)$*, discovers of the order of* $b\log n$*.*

Laplace type priors such as the Bayesian lasso are not covered by Corollary 5.22, because they are not in the $\mathrm{Be}(a,b)$ family: with λ^2 exponential, the density of κ vanishes at the origin faster than any power, so no finite a matches it. The last row of Table 5.1 reads such priors through their species sampling counterpart, the gamma subordinator, which is the boundary $\alpha = 0$ with $\mathbb{E}K_n \sim \theta\log n$. Used directly, with independent local scales at $\tau_n = n^{-1/2}$, the same priors have a bounded effective model size: the expected value of (5.15) tends to $\mathbb{E}\lambda^2$, and with λ^2 exponential of mean one it is 0.981 at $n = 10^2$ and 1.000 at $n = 10^6$, against $\log n = 4.61$ and 13.82. For them $\log n$ is therefore an upper bound, not a rate.

The reading in the last row of Table 5.1 is, to our knowledge, a new explanation of a well documented empirical phenomenon. It is standard that the lasso underselects in the ultra sparse regime while the horseshoe does not, and the usual explanation is in terms of tail thickness and minimax risk. The explanation here is Kingman's: a Lévy density that behaves like x^{-1} at the origin, as that of the gamma subordinator does, is an $\alpha = 0$ object, and an $\alpha = 0$ object cannot discover more than logarithmically many species however many are present. The exponential tilt is not the point, since it only tames the large jumps and equals one at the origin; what matters is the order of the pole of the Lévy density at the origin, which is where the rare types live. The lesson transfers back to genetics unchanged: a prior on effect sizes whose Lévy density has a simple pole at the origin is a prior

which asserts, before seeing any data, that the number of contributing variants is logarithmic in the sample.

5.6.2 *The rate holds for the shrinkage prior itself*

Corollary 5.22 is a statement about the matched species sampling model, and a reader is entitled to ask whether it says anything about the prior (5.13) as actually used, in which the weights are independent across coordinates rather than sampled from a normalised subordinator. It does, and the following makes the statement directly, without passing through the partition.

The natural count on the shrinkage side is the effective model size

$$m_p \;=\; \sum_{i=1}^{p} (1 - \kappa_i), \tag{5.15}$$

the sum of the fractions of the weighted least squares coefficients (5.4) that survive the penalty. It is the shrinkage analogue of K_n, and it is the quantity the horseshoe literature reports as the effective number of parameters.

THEOREM 5.23 (Effective model size). *Let $\kappa_\tau = 1/(1+\tau^2\lambda^2)$ where λ is such that $\kappa_1 \sim \mathrm{Be}(a,b)$ with $a \in (0,1]$, $b > 0$. Then as $\tau \downarrow 0$*

$$\mathbb{E}(1-\kappa_\tau) \;\sim\; \begin{cases} \dfrac{\pi}{B(a,b)\sin(\pi a)}\,\tau^{2a}, & 0 < a < 1, \\ 2b\,\tau^2 \log(1/\tau), & a = 1. \end{cases}$$

Consequently, for $p = n$ coordinates at the canonical global scale $\tau_n = n^{-1/2}$,

$$\mathbb{E}\, m_n \;\sim\; \begin{cases} \dfrac{\pi}{B(a,b)\sin(\pi a)}\, n^{1-a}, & 0 < a < 1, \\ b \log n, & a = 1. \end{cases}$$

COROLLARY 5.24 (The rates agree, and the constants nearly do). *The rates in Theorem 5.23 are exactly those of $\mathbb{E}K_n$ under $\mathrm{PD}(\alpha,\theta)$ with $(\alpha,\theta) = (1-a,\ a+b-1)$, that is under the matching of Theorem 5.15. The constants agree exactly when $a = 1$, both being b, and for $a < 1$ they differ by the factor $\Gamma(2-a)$, which lies in $[\Gamma(1.4616), 1] \approx [0.8856, 1]$ for every $a \in (0,1]$.*

Theorem 5.23 and Corollary 5.24 are, in our view, the results that make the chapter more than a translation. They say that the global local prior in its ordinary form, with independent local scales and the standard global scale $\tau_n = n^{-1/2}$, produces the discovery rate of the matched Poisson Dirichlet process, with the same exponent and a constant agreeing to within twelve per cent uniformly in a. Nothing in the derivation uses the partition structure. The exponent $2a$ in Theorem 5.23 arises from the behaviour of the beta prime density at the origin, and $1-a$ appears in $\mathbb{E}K_n$ because $\alpha = 1-a$; that these produce the same power of n at $\tau_n = n^{-1/2}$ is what has to be checked, and it is what the theorem checks.

(a,b)	(α,θ)	$n=10^3$	$n=10^4$	$n=10^5$	predicted at $n=10^5$
$(1/2,1/2)$ horseshoe	$(1/2,0)$	30.65	99.01	315.23	316.23
$(1,1)$ uniform shrinkage	$(0,1)$	5.92	8.21	10.51	11.51
$(1,3)$	$(0,3)$	15.29	22.14	29.04	34.54

TABLE 5.2. Expected effective model size $\mathbb{E}m_n$ of (5.15) at $\tau_n = n^{-1/2}$, computed by numerical quadrature of the one dimensional integral in the proof of Theorem 5.23, against the leading term of that theorem. The horseshoe row is $n^{1/2}$ to within 0.4 per cent at $n = 10^5$; the two rows with $a = 1$ grow by $b \log 10$ per decade, that is by 2.30 and 6.91, against computed increments of 2.29 and 2.30, and 6.85 and 6.90.

It also explains a convention. The choice $\tau_n = n^{-1/2}$ is standard in the horseshoe literature and is usually justified by risk considerations. Theorem 5.23 gives a second reading: it is the unique power of n at which the effective model size reproduces the species sampling discovery rate for every a simultaneously. At any other power the two sides disagree, and they disagree differently for different a.

Table 5.2 evaluates both sides. The horseshoe attains 315.23 against the predicted $\sqrt{10^5} = 316.23$, and the two priors with $a = 1$ increase by a fixed amount per decade of n, which is the signature of logarithmic growth. The finite n agreement is better for $a < 1$ than for $a = 1$, as it should be, since the logarithmic case carries a relative correction of order $1/\log(1/\tau)$.

EXAMPLE 5.25 (Three members of the beta normal class, side by side). Take the horseshoe $(a,b) = (1/2,1/2)$, the uniform shrinkage prior $(1,1)$ and the Strawderman Berger prior $(1/2,1)$. Theorem 5.15 sends them to GEM$(1/2,0)$, GEM$(0,1)$ and GEM$(1/2,1/2)$ respectively, and Theorem 5.18 gives the frequency spectrum intensities

$$\nu_{\mathrm{HS}}(x) = \frac{1}{\pi}\, x^{-3/2}(1-x)^{-1/2}, \qquad \nu_{\mathrm{U}}(x) = x^{-1}, \qquad \nu_{\mathrm{SB}}(x) = \tfrac{1}{2}\, x^{-3/2},$$

on $0 < x < 1$, the constants being $1/B(a,b)$, namely $1/\pi$, 1 and $1/B(\frac{1}{2},1) = \frac{1}{2}$. The power of x at the origin is $a-2$: it is $x^{-3/2}$ for both priors with $a = 1/2$, which is what makes their discovery counts grow like $n^{1/2}$, and x^{-1} for the uniform prior, which is the borderline case giving a logarithm. The two $a = 1/2$ priors differ only near $x = 1$, that is in how they treat common alleles, and this changes the level of the discovery curve but not its rate.

Table 5.3 evaluates four quantities at $n = 10^2$, 10^3 and 10^4: the exact $\mathbb{E}K_n$ from (5.12), its leading term from (5.11), the expected number of singletons $\mathbb{E}a_1 = n\,B(a,b+n-1)/B(a,b)$ from (5.14), and the expected effective model size $\mathbb{E}m_n$ of (5.15) at $\tau_n = n^{-1/2}$, computed by quadrature. Several entries can be checked by hand. For the uniform prior $\mathbb{E}K_n$ is the harmonic number $\sum_{j=1}^{n} j^{-1}$, which is 5.187 at $n = 100$, and $\mathbb{E}a_1 = n\,B(1,n) = 1$ exactly, for every n: under the Ewens Sampling Formula with $\theta = 1$ one expects a single singleton however large the

prior, (α, θ)	n	$\mathbb{E}K_n$	leading term	$\mathbb{E}a_1$	$\mathbb{E}m_n$
horseshoe, $(1/2, 0)$	10^2	11.27	11.28	5.66	9.09
	10^3	35.68	35.68	17.85	30.65
	10^4	112.84	112.84	56.42	99.01
uniform, $(0, 1)$	10^2	5.19	4.61	1.00	3.69
	10^3	7.49	6.91	1.00	5.92
	10^4	9.79	9.21	1.00	8.21
Strawderman Berger, $(1/2, 1/2)$	10^2	16.75	17.72	8.87	13.92
	10^3	55.06	56.05	28.03	47.74
	10^4	176.25	177.25	88.62	155.10

TABLE 5.3. Discovery counts for three members of the three parameter beta normal class. $\mathbb{E}K_n$ is exact, from (5.12); the leading term is $1.128\, n^{1/2}$, $\log n$ and $\sqrt{\pi}\, n^{1/2} = 1.772\, n^{1/2}$ respectively; $\mathbb{E}a_1$ is the expected number of singletons from (5.14); $\mathbb{E}m_n$ is the expected effective model size of the shrinkage prior at $\tau_n = n^{-1/2}$, whose leading terms by Theorem 5.23 are $n^{1/2}$, $\log n$ and $(\pi/2)n^{1/2}$.

sample. For the horseshoe (5.12) reduces to $\mathbb{E}K_n = 2\Gamma(n + \frac{1}{2})/\{\sqrt{\pi}\,\Gamma(n)\}$, which at $n = 100$ is $1.1284 \times \sqrt{100} \times (1 - 1/800) = 11.27$ to four figures.

Three things are visible in the table. First, the two $a = 1/2$ priors multiply their discoveries by $\sqrt{10} \approx 3.16$ per decade of n, while the uniform prior adds $\log 10 \approx 2.30$ per decade; by $n = 10^4$ the Strawderman Berger prior expects 176 alleles and the uniform prior fewer than ten. Second, the singleton fraction $\mathbb{E}a_1/\mathbb{E}K_n$ is 0.500 for the horseshoe and 0.503 for the Strawderman Berger prior at $n = 10^4$, both close to $\alpha = 1/2$, while for the uniform prior it is $1/9.79 = 0.10$ and falling towards zero; the singleton fraction therefore separates α even when the levels are matched. Third, the effective model size of the shrinkage prior tracks $\mathbb{E}K_n$ with ratio 0.88 for both $a = 1/2$ priors at $n = 10^4$, close to $\Gamma(3/2) = 0.886$, exactly as Corollary 5.24 predicts; for the uniform prior the ratio $8.21/9.79 = 0.84$ is still converging to one at the slow logarithmic rate.

5.6.3 Where the correspondence stops: fluctuations

Theorem 5.23 matches first moments. It is worth asking whether the agreement extends to second moments, and the answer is that it does not. The failure is instructive, because it says exactly what the unproved partition identity would buy.

LEMMA 5.26 (Second moment). *Under the hypotheses of Theorem 5.23 with* $0 < a < 1$,

$$\mathbb{E}(1 - \kappa_\tau)^2 \;\sim\; \frac{B(a, 2-a)}{B(a, b)}\, \tau^{2a} \qquad \text{as } \tau \downarrow 0,$$

so that $\mathrm{Var}(1 - \kappa_\tau) \sim \{B(a, 2-a)/B(a, b)\}\tau^{2a}$, *the square of the mean being of the smaller order* τ^{4a}.

	K_n under PD$(1/2,0)$			m_n, horseshoe at $\tau_n = n^{-1/2}$		
n	mean	s.d.	c.v.	mean	s.d.	c.v.
200	15.91	11.86	0.745	13.20	2.53	0.192
1000	35.36	25.57	0.723	30.56	3.86	0.126
5000	76.66	57.87	0.755	69.51	6.01	0.086

TABLE 5.4. Four hundred replicates at each n. The means agree to within the factor $\Gamma(2-a)$ of Corollary 5.24, as they should. The coefficients of variation do not. That of K_n stays near the Mittag Leffler value 0.7555 at every n, while that of m_n decays: the fitted exponent across the three sample sizes is -0.250 against the predicted $-1/4$. The closed form prediction of Proposition 5.27, which at $a = b = 1/2$ has constant $2^{-1/2}$ and so gives 0.188, 0.126 and 0.084, agrees with the simulated values to within the Monte Carlo error of a coefficient of variation estimated from four hundred replicates, which is about four per cent.

PROPOSITION 5.27 (The two constructions fluctuate differently). *Fix $a \in (0,1)$ and $b > 0$, let $\kappa_1, \dots, \kappa_n$ be independent with the law of κ_{τ_n} at $\tau_n = n^{-1/2}$, put $m_n = \sum_{i=1}^n (1-\kappa_i)$, and let K_n be the number of blocks in a sample of size n from* PD$(\alpha, 0)$ *with $\alpha = 1-a$. Then*

$$\frac{m_n}{\mathbb{E}m_n} \xrightarrow{\mathbb{P}} 1, \qquad \text{with}$$

$$\frac{\{\mathrm{Var} m_n\}^{1/2}}{\mathbb{E}m_n} \sim \frac{\sin(\pi a)}{\pi} \big\{ B(a, 2-a)\, B(a,b) \big\}^{1/2} n^{-(1-a)/2},$$

whereas $K_n / n^\alpha \to M_\alpha$ almost surely, where M_α is Mittag Leffler with $\mathbb{E}M_\alpha^r = r!/\Gamma(1+r\alpha)$. The limit is nondegenerate, so the coefficient of variation of K_n converges to $\{2\Gamma(1+\alpha)^2/\Gamma(1+2\alpha) - 1\}^{1/2}$, equal to 0.7555 at $\alpha = 1/2$, while that of m_n converges to zero.

Proof. By Lemma 5.26 and Theorem 5.23, $\mathrm{Var} m_n = n\mathrm{Var}(1-\kappa_{\tau_n}) \sim \{B(a,2-a)/B(a,b)\} n^{1-a}$ and $\mathbb{E}m_n \sim \{\pi/(B(a,b)\sin\pi a)\} n^{1-a}$, so with $C_2 = B(a,2-a)/B(a,b)$ and $C_1 = \pi/\{B(a,b)\sin(\pi a)\}$ the ratio of standard deviation to mean is $C_2^{1/2} C_1^{-1} n^{-1/2} \tau_n^{-a} = C_2^{1/2} C_1^{-1} n^{-(1-a)/2}$, and $C_2^{1/2}/C_1 = \{B(a,2-a)B(a,b)\}^{1/2} \sin(\pi a)/\pi$. Chebyshev then gives convergence in probability, since the ratio tends to zero for every $a < 1$. The statement for K_n is Pitman's almost sure limit theorem for the two parameter family, and the moments of M_α are classical. □

Table 5.4 confirms both.

The reason is dependence. Under PD$(\alpha, 0)$ the atom frequencies are the normalised jumps of an α stable subordinator, whose total is itself heavy tailed, so a sample either lands in a realisation with many large atoms or in one with few, and that randomness never averages away. The independent weights of (5.13) have no such common factor and obey a law of large numbers.

We regard this as clarifying rather than damaging. Corollary 5.22 is a statement about how many signals a prior expects to find, and Theorem 5.23 establishes it for the prior as used. What Proposition 5.27 shows is that the correspondence is one of means and not of laws, and that a practitioner who wants a prior which is genuinely uncertain about the number of signals, rather than merely uncertain about which ones, must introduce the dependence explicitly, for instance by estimating τ rather than fixing it. That is the practical content of the open problem in the remark below: the partition identity, if true, would settle whether estimating τ is the device which supplies the missing common factor.

REMARK 5.28 (What is and is not proved). Theorems 5.15 and 5.18 concern the random discrete distribution generated by a normalised subordinator, whereas (5.13) places independent weights on a fixed index set. We have not shown that the exchangeable partition induced on coefficient indices by (5.13) coincides with the PD(α, θ) partition, and we do not claim it.

What Theorem 5.23 shows is that the consequence which matters, namely the discovery rate, transfers without that identification. The rate is derived directly from the law of κ under (5.13) and agrees with the species sampling rate in exponent exactly and in constant to within $\Gamma(2-a)$. So Corollary 5.22 may be read as a statement about the shrinkage prior as used, and Theorem 5.15 may be read as the explanation of why that rate takes the form it does rather than as a premise the rate depends on.

The partition level identity remains open and would still be worth having, and Proposition 5.27 says precisely why: the correspondence established here is one of first moments, and the two constructions differ at second order, the species sampling side retaining a nonvanishing coefficient of variation which the independent weights side does not. A partition level identity would settle whether estimating the global scale supplies the missing dependence.

5.7 Inference for the mutation parameter

By Corollary 5.16, θ is the parameter fixing how many discoveries the prior expects. It is therefore worth inferring rather than fixing. This section collects the three facts that make inference about θ easy in principle and hard in practice: the likelihood depends on the data only through the number of alleles, that number is a sum of independent Bernoulli variables, and the awkward ratio of gamma functions in the likelihood can be removed by one latent beta variable per locus. Worked examples on a small sample, on several loci and on a classical ecological data set follow.

PROPOSITION 5.29 (Ewens 1972). *Under* (5.7) *the number of distinct alleles K_n is sufficient for θ, and it is minimal sufficient.*

Proof. Write (5.7) as $h(\pi)g(K_n(\pi), \theta)$ with $h(\pi) = \prod_i (n_i - 1)!$ and $g(k, \theta) = \theta^k / \theta_{(n)}$; Fisher Neyman gives sufficiency. For two partitions of the same n,

$p(\pi \mid \theta)/p(\pi' \mid \theta) = \{h(\pi)/h(\pi')\}\theta^{K_n(\pi)-K_n(\pi')}$ is free of θ if and only if $K_n(\pi) = K_n(\pi')$, so by the Lehmann Scheffé criterion K_n is minimal sufficient. □

In the terms of Edwards (1972), the content of Proposition 5.29 is that the likelihood function for θ, considered as a function on the parameter space, is determined up to a multiplicative constant by the single integer K_n: the entire allelic configuration, which for $n = 1000$ genes may specify a partition of enormous complexity, carries no further information about the mutation parameter. This is an unusually clean instance of the reduction that Edwards takes to be the central object of inference, and it is what makes the model the natural place to test any procedure that claims to find such reductions automatically.

PROPOSITION 5.30 (Feller coupling). *K_n has the law of $\sum_{j=0}^{n-1} B_j$ with B_j independent Bernoulli of success probability $\theta/(\theta+j)$. Hence $\mathbb{E}K_n = \sum_{j=0}^{n-1} \theta/(\theta+j)$ and $\mathrm{Var}K_n = \sum_{j=0}^{n-1} \theta j/(\theta+j)^2$.*

Proof. In the urn of Proposition 5.11, let B_j be the indicator that the $(j+1)$th draw is black, $j = 0, \dots, n-1$. Before that draw the urn holds j coloured balls of unit mass and the black ball of mass θ, so $\mathbb{P}(B_j = 1 \mid \text{past}) = \theta/(\theta+j)$ whatever the past. The B_j are therefore independent with the stated success probabilities, and $K_n = \sum_j B_j$ because each black draw creates exactly one new colour. The mean and variance follow by summing those of the Bernoulli variables. □

The coupling gives exact simulation of K_n in one vectorised operation, the asymptotic $\mathbb{E}K_n \sim \theta \log n$, and the observation that $\mathrm{Var}K_n \sim \theta \log n$ as well, so that the relative precision of any estimate of θ improves only like $(\log n)^{-1/2}$. Section 5.9 shows what that means in practice.

The likelihood is $L(\theta) \propto \theta^{K_n}\Gamma(\theta)/\Gamma(\theta+n)$. The ratio of gamma functions is why the genetics literature usually reports a point estimate solving $\sum_{j=0}^{n-1} \theta/(\theta+j) = K_n$ rather than a posterior. It is removed by a beta integral.

LEMMA 5.31. *For $\theta > 0$ and integer $n \geq 1$, $\Gamma(\theta)/\Gamma(\theta+n) = \Gamma(n)^{-1} \int_0^1 t^{\theta-1}(1-t)^{n-1}\,\mathrm{d}t$.*

PROPOSITION 5.32 (Augmented scheme). *For m independent loci with sample sizes $n_1, \dots, n_m$ and allele counts $k_1, \dots, k_m$, introduce latent $t_1, \dots, t_m \in (0,1)$. Then*

$$t_r \mid \theta \sim \mathrm{Be}(\theta, n_r), \qquad p(\theta \mid t_{1:m}) \propto p(\theta)\, \theta^{\sum_r k_r} \exp\Big(\theta \sum_r \log t_r\Big).$$

If $p(\theta)$ is $\mathrm{Ga}(c_0, d_0)$, with shape c_0 and rate d_0, then $\theta \mid t_{1:m} \sim \mathrm{Ga}(c_0 + \sum_r k_r,\ d_0 - \sum_r \log t_r)$.

For a single locus this is a variant of the scheme of Escobar and West (1995) for the Dirichlet process concentration parameter and we claim no novelty for it. What we claim is that it is not used in population genetics, that it extends unchanged to many loci sharing a common θ, and that the gamma prior is a poor

default because it has light tails, which matters precisely when θ is the parameter controlling discovery. We therefore use the Pochhammer prior of Wang and Polson (2024),

$$p(\theta) \propto \frac{1}{(\theta+1)(\theta+2)}, \tag{5.16}$$

proper, with tail θ^{-2}, and belonging to the family $p(\theta) \propto \theta^m \prod_r \Gamma(\theta+a_r)/\Gamma(\theta+b_r)$, which is closed under multiplication by $\theta^k\Gamma(\theta)/\Gamma(\theta+n)$. Closure means that the posterior after any number of loci is described by the bookkeeping $m \mapsto m + \sum_r k_r$ together with the appending of pairs $(0, n_r)$, the normalising constant being a one dimensional integral.

EXAMPLE 5.33 (Ten genes, four alleles). Suppose $n = 10$ genes are sampled at a locus and fall into $K_{10} = 4$ allelic classes, say with configuration $(5,2,2,1)$. By Proposition 5.29 the configuration is irrelevant and the likelihood is

$$L(\theta) \propto \frac{\theta^4\,\Gamma(\theta)}{\Gamma(\theta+10)} = \frac{\theta^3}{(\theta+1)(\theta+2)\cdots(\theta+9)}.$$

The maximum likelihood estimate solves $\sum_{j=0}^{9} \theta/(\theta+j) = 4$, which a few steps of bisection place at $\hat\theta = 1.956$. Since K_n is a sufficient statistic in a one parameter exponential family with natural parameter $\log\theta$, the Fisher information is $\mathrm{Var}(K_n)/\theta^2$, and Proposition 5.30 gives a standard error $\{\sum_j j/(\hat\theta+j)^2\}^{-1/2}\hat\theta^{1/2} = 1.46$, three quarters of the estimate. The likelihood is correspondingly flat: relative to its maximum it is 0.24 at $\theta = 0.5$, 0.69 at $\theta = 1$ and 0.43 at $\theta = 5$, so values from one half to five are all reasonably supported by ten genes.

Under the Pochhammer prior (5.16) the exact posterior, computed on a fine grid, has mean 2.19, median 1.76 and 95% interval $[0.42, 6.48]$. The right skew is the honest summary of the flat likelihood, and the posterior mean exceeds the maximum likelihood estimate because the heavy θ^{-2} tail of the prior does not cut the long right tail of the likelihood off.

EXAMPLE 5.34 (Pooling three loci). Now suppose three unlinked loci, assumed to share a common θ, give $(n_r, k_r) = (10,4)$, $(20,6)$ and $(15,3)$. The pooled likelihood is the product of three factors $\theta^{k_r}\Gamma(\theta)/\Gamma(\theta+n_r)$, its maximum solves $\sum_r \sum_{j<n_r} \theta/(\theta+j) = 13$ at $\hat\theta = 1.707$, and the standard error falls to 0.66. The Pochhammer posterior has mean 1.76, median 1.65 and 95% interval $[0.76, 3.36]$, whose length is less than half that of the single locus interval although the total number of genes has only risen from 10 to 45. A single locus of 45 genes would have done worse, since by Proposition 5.30 the information in one locus grows like $\log n$, whereas independent loci add their information.

The augmented scheme of Proposition 5.32 reproduces this posterior by simulation. With a $\mathrm{Ga}(1, 0.1)$ prior, so that the conditional for θ is gamma, 6×10^4 Gibbs sweeps alternating $t_r \mid \theta \sim \mathrm{Be}(\theta, n_r)$ and $\theta \mid t \sim \mathrm{Ga}(14,\ 0.1 - \sum_r \log t_r)$ give posterior mean 1.99 and 95% interval $[0.86, 3.78]$, against 2.00 and $[0.86, 3.81]$ from

exact quadrature under the same prior. Each sweep costs three beta draws and one gamma draw.

EXAMPLE 5.35 (Corbet's Malayan butterflies). For a realistic case we take the butterfly collection that led Fisher to the logarithmic series (Fisher, Corbet and Williams, 1943), analysed at length in Chapter 6: $K_n = 620$ species were collected, of which 118 are singletons and 74 doubletons. Species play the role of alleles and individuals of genes. Corbet stopped counting a species after 24 specimens, so the 119 commonest species are censored and n is not reported; the censored likelihood of Chapter 6 estimates it at $\hat{n} = 11068$. The maximum likelihood estimate is $\hat{\theta} = 141.75$ with standard error 7.0, and the Pochhammer posterior given $n = \hat{n}$ has mean 141.5 and 95% interval $[129.2, 154.5]$; with 620 observed classes the prior is irrelevant, in contrast to Example 5.33.

The model is nonetheless wrong in an informative way, and the diagnostic is the one proposed in this chapter. At $\hat{\theta}$ the Ewens Sampling Formula predicts $\mathbb{E}a_1 = \hat{\theta} n/(\hat{\theta} + n - 1) = 140.0$ singletons and $\mathbb{E}a_2 = 69.1$ doubletons at $n = \hat{n}$. The doubletons fit; the singletons are overpredicted by about a fifth. In the language of Corollary 5.20, the observed singleton fraction $118/620 = 0.190$ is smaller than the 0.226 that an $\alpha = 0$ model with this θ requires, which points towards a negative rather than a positive discount, as Chapter 6 finds by fitting the two parameter model directly. For prediction the Ewens model says that a second collection of the same size would add $\sum_{j=n}^{2n-1} \hat{\theta}/(\hat{\theta} + j) = 97.4$ new species, a figure that the singleton misfit suggests is too high.

5.8 The mutation parameter is time scale dependent

There is a point in Hössjer and Ewens (2026) which bears directly on the interpretation of θ, and hence, through Corollary 5.16, on the interpretation of the implied sparsity level.

The Ewens Sampling Formula is the stationary sampling distribution of the infinite alleles diffusion, and its parameter is the compound quantity $\theta = 4N_e u$, mutation rate per gene per generation multiplied by four times an effective population size. Two of the three inputs to that product are themselves contested. Ewens and Hössjer (2026) show that N_e is not one quantity but at least three, the inbreeding, variance and eigenvalue effective sizes, which agree only for the idealised monoecious model and can differ substantially for two sex or subdivided populations; their Table 1 gives cases with $N_m = 1$ where the eigenvalue effective size is 4.56 and the inbreeding effective size is 4.00, a discrepancy of fourteen per cent that propagates directly into θ.

Hössjer and Ewens (2026) then show that the remaining input, the generation, is not well defined in a continuous time model without further work. Their construction defines the per generation time scale by $t' = \int_0^t f(0; s)\, \mathrm{d}s$, where $f(0; s)$ is the density at zero of the age of a parent of an individual born at time s, and they show that the conversion $m_{ij} = \log w_{ij}$ used by Crow and Kimura

(1970) is correct only on that scale, differing otherwise by the factor $f(0)$. Their effective Malthusian parameters $\tilde{m}_{ij}(t)$ depend on the mating scheme and typically differ from the nominal m_{ij}, and under assortative mating with parameter γ their Equation (44) shows the difference is of order γ.

The consequence for the present chapter is worth stating plainly. If θ is measured per generation and the generation is defined by a time change which itself depends on the mating scheme and the age structure, then θ, and by Corollary 5.16 the implied Pitman parameter, is defined only relative to that time change. Two analyses of the same population using different but individually defensible generation conventions will report different θ and therefore different expected discovery counts. The magnitude is not negligible: since $\mathbb{E}K_n \sim \theta \log n$ at $\alpha = 0$, a fourteen per cent discrepancy in N_e produces a fourteen per cent discrepancy in the predicted number of alleles.

We regard this as an argument for reporting θ with a posterior rather than a point, as in Section 5.7, and for reporting the convention alongside it. It is also an argument for the reparametrisation implicit in Theorem 5.15: the pair (α, θ) separates the discovery *rate*, governed by α and invariant under the time changes just described, from the discovery *level*, governed by θ and not invariant. In the polygenic setting the rate is the biologically interesting quantity and the level is a nuisance, which is the reverse of the classical emphasis.

5.9 Numerical illustrations

5.9.1 Discovery rates

This section reports four simulation studies, each designed to test one claim of the preceding sections on a case large enough to be realistic and small enough to run in minutes. Full designs are in Section 5.9.6.

Figure 5.2 simulates the exchangeable partition directly from the Chinese restaurant construction for three members of the family, chosen through Theorem 5.15 to correspond to $\kappa \sim \mathrm{Be}(1/2, 1/2)$, $\kappa \sim \mathrm{Be}(1, 1)$ and $\kappa \sim \mathrm{Be}(1, 3)$, and records $\mathbb{E}K_n$ at sample sizes from 10^2 to 3×10^4 over 60 replicates.

The horseshoe case gives a fitted log log slope of 0.498 against the value $1/2$ predicted by Corollary 5.22. The two cases with $a = 1$ show no power law: their log log slopes are 0.133 and 0.157 and decreasing with n, while their growth is linear in $\log n$ with slopes 1.005 and 2.835 against the predicted $\theta = 1$ and $\theta = 3$. The difference is visible by eye: at $n = 3 \times 10^4$ the horseshoe case has occupied 166 blocks on average and the $\mathrm{Be}(1, 1)$ case 10.6, against exact means of 195.4 and 10.9 from (5.12). The horseshoe shortfall is within Monte Carlo error: by Proposition 5.27 the coefficient of variation of K_n is about 0.76, so the mean of sixty replicates has a standard error near 19.

Three points about how to read the figure. First, the diagnostic is the *shape* of the curve rather than its level, since the level can always be matched by adjusting θ; the two panels are drawn on complementary axes so that exactly one of them

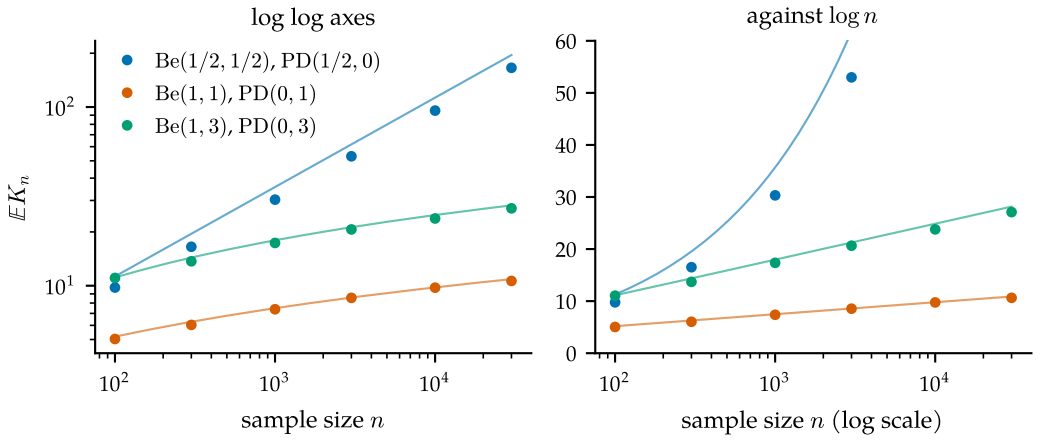


FIGURE 5.2. Expected number of occupied blocks under the species sampling prior matched by Theorem 5.15 to three shrinkage priors. Left, log log axes, where only $a < 1$ produces a straight line; the horseshoe slope of 0.498 agrees with the predicted $n^{1/2}$. Right, plotted against $\log n$, where only $a = 1$ produces a straight line, with slopes 1.005 and 2.835 against the predicted $\theta = 1$ and 3. Sixty replicates of the Chinese restaurant process at each of six sample sizes. Dots are the replicate means and curves the exact means from (5.12).

straightens each curve, and which one it straightens is the test. Second, the apparent log log slope of the $a = 1$ cases is not zero and decreases with n, which is what a logarithm looks like on log log axes over a finite range, and is a reminder that a power law cannot be confirmed from a single decade. Third, the separation is not subtle at genomic sample sizes. Continuing the two curves to $n = 10^6$ with (5.12) gives 1128 for $a = 1/2$ against 14.4 for $\mathrm{Be}(1, 1)$, a factor of nearly eighty, and the factor grows without bound.

5.9.2 *Inflation of the additive genetic variance, and its repair*

We simulate $L = 400$ unlinked biallelic loci with frequencies uniform on $[0.1, 0.9]$, genotypes in Hardy Weinberg proportions, five per cent of loci causal with standard normal effects, a pairwise epistatic contribution among the causal loci, and independent environmental noise. This is deliberately favourable to the additive model: the loci are independent, so $N^{-1}X^\top X$ is I_p up to sampling fluctuations of order $N^{-1/2}$ and linkage disequilibrium plays no part. Because $p = 400$ is at least N at the two smallest sample sizes, the least squares projection itself is not available there (Corollary 5.10), and we use instead its orthogonal design form, the marginal coefficients $\hat\beta = X^\top y/N$ and $\hat\sigma_A^2 = \|\hat\beta\|^2$, which coincide with the projection when $X^\top X = NI$. This is a linear smoother and Proposition 5.5 applies to it, with one difference worth noting: each marginal coefficient absorbs the effects of the other loci as noise, so its noise variance is about $(\sigma_A^2 + \sigma^2)/N$ rather than σ^2/N. The predicted ratio of fitted to true additive variance is therefore $1 + p(\sigma_A^2 + \sigma^2)/(N\sigma_A^2)$. With the simulated averages $\sigma_A^2 \approx 20$ and $\sigma^2 \approx 1.9$ this is

N	200	400	800	1600	3200
unregularised projection, orthogonal design form	3.09	2.09	1.55	1.27	1.14
species sampling prior, $\tau = L^{-1/2}$	0.65	0.81	0.89	0.94	0.98
species sampling prior, oracle τ	0.80	0.89	0.93	0.96	0.98

TABLE 5.5. Estimated σ_A^2 divided by the true σ_A^2, $L = 400$ loci, 40 replicates. The first row is the orthogonal design form of Fisher's projection (5.1) fitted from data, the marginal coefficients $\hat{\beta} = X^\top y/N$, which coincide with the projection when $X^\top X = NI$. Values above one indicate inflation. The second row uses the global scale $\tau = L^{-1/2}$ that Theorem 5.23 identifies as canonical, in units of the coefficient standard error, which is estimated by the median absolute deviation of the coefficients; nothing about the truth is used. The third row uses the oracle scale $\sqrt{q/L}$ with q the true number of causal loci. Monte Carlo standard errors of the first row are 0.053 at $N = 200$ falling to 0.008 at $N = 3200$.

3.18, 2.10, 1.55, 1.27 and 1.14 at the five sample sizes. We compare this estimator with the horseshoe posterior mean, computed by one dimensional quadrature in κ.

At $N = 200$, that is at half the number of loci, the fitted projection, in its orthogonal design form, reports an additive genetic variance three times the truth. The observed ratios 3.09, 2.09, 1.55, 1.27 and 1.14 track the predicted 3.18, 2.10, 1.55, 1.27 and 1.14, the largest discrepancy being under two Monte Carlo standard errors, and the bias is still fourteen per cent at $N = 3200$. The species sampling prior is conservative, understating σ_A^2 at small N and approaching the truth from below, but it never inflates. For heritability estimation, which is the applied form of this question, the direction of the error matters as much as its size: a procedure that overstates additive variance overstates the response to selection, and by Example 5.2 it may report a substantial additive variance in a population sitting exactly at an equilibrium where the true value is zero.

The comparison between the second and third rows of Table 5.5 is the one we regard as most informative, and it is a direct test of Theorem 5.23. The second row is fully data driven: the global scale is set to $L^{-1/2}$, the value the theorem identifies as the unique power of L at which the effective model size reproduces the species sampling discovery rate, and σ is estimated from the median absolute deviation of the fitted coefficients. The third row is given the true number of causal loci. The gap between them is 0.15 at $N = 200$ and closes to 0.01 by $N = 3200$. The theory driven choice therefore recovers almost all of what the oracle knows, and the value of Theorem 5.23 is precisely that it removes the need for the oracle.

It is worth recording why the theory driven scale performs so well here, since the reason is partly an accident of the design. At $\tau = L^{-1/2}$ the prior effective model size (5.15) is $L\,\mathbb{E}(1-\kappa) = L\tau/(1+\tau) = 19.05$, and Corollary 5.22 gives the leading term $L^{1-a} = L^{1/2} = 20$. The simulation has twenty causal loci, because five per cent of four hundred is twenty and $\sqrt{400}$ is also twenty. The prior is therefore expecting almost exactly the right number of signals, which is flattering. What is not an accident is that the prior arrives at that expectation without being told, and that Corollary 5.22 is what tells one what to expect at other values of L.

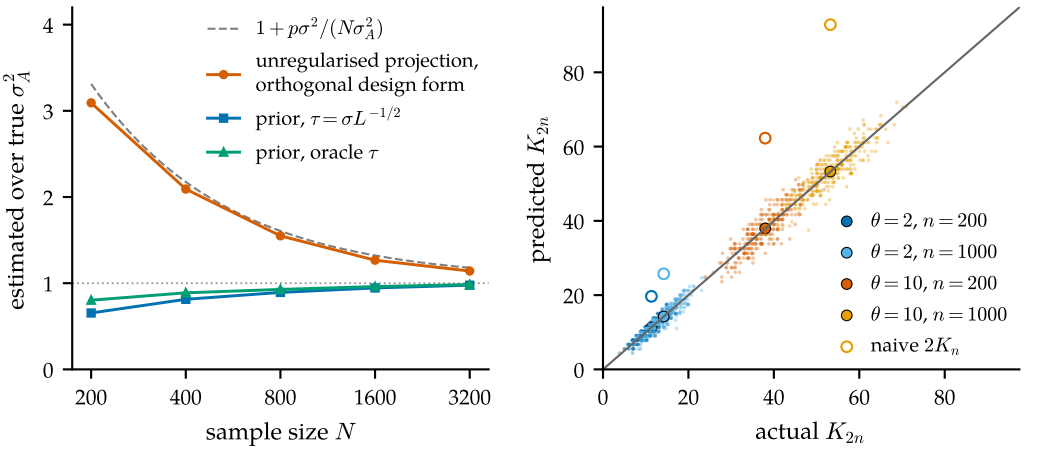


FIGURE 5.3. Left, estimated over true additive genetic variance against sample size with $L = 400$ loci; the unregularised projection, in its orthogonal design form, is inflated by $p\sigma^2/(N\sigma_A^2)$, where here σ^2 is the noise variance of a coefficient times N (the dashed reference curve), while the species sampling prior never inflates. In the legend $\sigma L^{-1/2}$ denotes the canonical global scale $L^{-1/2}$ in units of the estimated coefficient standard error. Right, predicting the number of distinct alleles in $2n$ genes from a sample of n using the Ewens Sampling Formula, 300 replicates in each of four settings, against the naive prediction $2K_n$.

5.9.3 *Rare variant discovery*

The right panel of Figure 5.3 addresses a question of current applied interest: how many new variants appear if the sample is doubled? Estimating θ from K_n by Propositions 5.29 and 5.30 and predicting $\mathbb{E}K_{2n}$ gives, over 300 replicates:

	$\theta = 2,\ n = 200$	$\theta = 2,\ n = 1000$	$\theta = 10,\ n = 200$	$\theta = 10,\ n = 1000$
K_n observed	9.84	12.87	31.14	46.42
predicted K_{2n}	11.25	14.25	37.96	53.28
actual K_{2n}	11.34	14.22	37.97	53.23
mean relative error	9.3%	7.3%	6.0%	4.3%
naive $2K_n$ over actual K_{2n}	1.74	1.81	1.64	1.74

The prediction is essentially unbiased. This is the species sampling prior doing what a sparsity prior is asked to do, namely to say how many discoveries to expect, and doing it from the model rather than from a tuning constant. The relevant comparison is not against a competitor procedure but against the naive answer, which is that doubling the sample doubles the discoveries; the table shows the true increase is between ten and twenty two per cent across the four settings, so the naive prediction $2K_n$ overstates K_{2n} by a factor of between 1.6 and 1.8.

The structure of the calculation is worth making explicit, because it is the same in every application of the frequency spectrum. One observes K_n, inverts

$\mathbb{E}K_n(\theta) = K_n$ using Proposition 5.30 to obtain $\hat{\theta}$, and evaluates $\mathbb{E}K_{2n}(\hat{\theta})$. No new data enters; the extrapolation is carried entirely by the functional form of $\mathbb{E}K_n(\theta)$, which is $\sum_{j=0}^{n-1} \theta/(\theta + j)$ at $\alpha = 0$. The accuracy in the table therefore measures whether that functional form is right, and the errors of four to nine per cent are consistent with the sampling variability of $\hat{\theta}$ implied by Proposition 5.30 rather than with any bias in the extrapolation itself.

Two cautions. The relative error improves with n and with θ, as it should, since both increase K_n and hence the information about θ; a study with few observed alleles will predict badly, and the interval of Section 5.9.5 is the right way to report that. And doubling is a short extrapolation. Predicting K_{10n} from K_n rests much more heavily on the assumed value of α, since by (5.11) the two candidate growth laws diverge, and there Corollary 5.20 rather than K_n alone should be doing the work.

5.9.4 Sensitivity to the assumed sparsity level

Corollary 5.22 is a statement about what a prior will do, not about what is true, so it is worth asking what happens when the assumed α is wrong. The two failure modes are asymmetric and it is useful to say why.

Suppose the truth has q_n contributing variants in dimension n and the analyst uses a prior with parameter a, hence assumed rate n^{1-a}. If $q_n \gg n^{1-a}$, the prior is too conservative: by (5.14) the expected number of occupied blocks is a hard ceiling in expectation, and the excess signals are shrunk toward zero regardless of their magnitude. This is the lasso failure in the ultra sparse regime, and no adjustment of the global scale repairs it, because the global scale moves θ and by (5.11) at $\alpha = 0$ the rate is $\theta \log n$, which is logarithmic for every θ. Multiplying θ by ten buys a factor of ten, not a change of rate.

If instead $q_n \ll n^{1-a}$, the prior is too liberal, and the consequence is overdiscovery at a rate the multiplicity structure does not control. This failure is visible: it shows up as a fitted spectrum with too many singletons relative to the observed one, and by Corollary 5.20 the discrepancy is measurable. The asymmetry is therefore practical as well as formal. Assuming too little sparsity produces an error one can detect from the data; assuming too much produces an error one cannot, because the missing signals never appear in any output. When in doubt the argument favours the smaller a.

5.9.5 Uncertainty in the mutation parameter

Using (5.16) and the exact posterior, at $n = 100$ and $\theta = 5$ the median 95% credible interval over 400 replicates is $[2.53, 8.52]$, a width of 5.99, that is 120% of θ. Its frequentist coverage, computed exactly by summing over the law of K_n given by Proposition 5.30, is 0.943; the empirical coverage in the 400 replicates was 0.915, which is 2.4 Monte Carlo standard errors below that. At $n = 1000$ the width is still 83% of θ, with exact coverage 0.938, and at $\theta = 1$, $n = 100$ it is twice θ. The slow

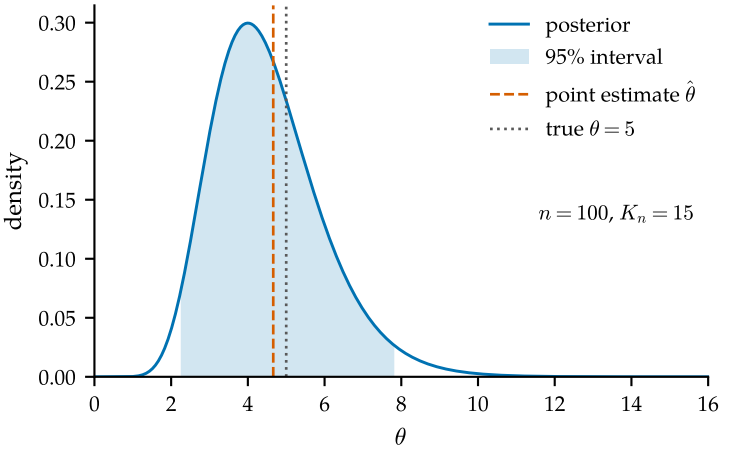


FIGURE 5.4. Exact posterior for θ under the Pochhammer prior (5.16) for a representative sample with $n = 100$ and $K_n = 15$ drawn at $\theta = 5$, against the point estimate $\hat{\theta} = 4.66$ solving $\mathbb{E}K_n(\theta) = K_n$. The posterior mean is 4.53 and the 95% interval $[2.25, 7.83]$. The point estimate is accurate on average and conceals uncertainty of order θ itself.

improvement reflects the $(\log n)^{-1/2}$ rate implied by Proposition 5.30. Figure 5.4 displays a representative posterior against the customary point estimate.

Since by Corollary 5.16 θ fixes the implied sparsity level, uncertainty of this magnitude propagates directly into any downstream discovery statement, and reporting a single value of θ understates it.

The slowness is structural rather than a defect of the estimator. By Proposition 5.30, K_n is a sum of independent indicators with $\mathbb{E}K_n \sim \theta \log n$ and $\mathrm{Var}K_n \sim \theta \log n$, so the Fisher information about θ accumulates like $\log n$ and the standard error of any estimator falls like $(\log n)^{-1/2}$. Going from $n = 100$ to $n = 1000$ multiplies the information at $\theta = 5$ by about 2.1, not by ten (the leading order ratio is $3/2$). This is a feature of the infinite alleles model and not of the Bayesian treatment: the point estimate is exactly as uncertain, and merely does not say so.

There is a practical consequence for the design of studies. If the object of interest is θ itself, or a discovery count derived from it, then increasing the sample size is an inefficient way to buy precision, and pooling independent loci that share a common θ is far better, since by Proposition 5.32 the information from m loci adds. The augmented scheme of Section 5.7 is written for exactly that case.

5.9.6 Simulation design

For reproducibility we record the design of the four studies above. Discovery rates: the two parameter Chinese restaurant process is run to $n = 3 \times 10^4$ with 60 independent replicates per parameter setting, and $\mathbb{E}K_n$ is recorded at six checkpoints; the log log slope is fitted by least squares across those checkpoints. Additive genetic variance: $L = 400$ loci, allele frequencies uniform on $[0.1, 0.9]$, genotypes drawn as $\mathrm{Bin}(2, p_\ell)$ and standardised, twenty causal loci with standard

normal effects, an epistatic term formed from ten products of disjoint pairs of causal loci scaled by 0.3, environmental noise standard normal, 40 replicates at each of five sample sizes. The horseshoe posterior mean of each coefficient is computed by a 400 point midpoint rule in κ on $(0,1)$ in the normal means model $\hat{\beta}_j \sim N(\beta_j, s^2)$, with s estimated by the median absolute deviation of the fitted coefficients, and with the global scale set either to the canonical value $L^{-1/2}$ or to the oracle value $\sqrt{q/L}$ where q is the number of causal loci; the gap between the two rows of Table 5.5 is exactly what knowing q is worth. Rare variant discovery: partitions generated by the Feller coupling of Proposition 5.30, subsampled to size n from a realisation of size $2n$, 300 replicates. Mutation parameter: 400 replicates, posterior computed on a 4000 point geometric grid in θ on $[10^{-2}, 400]$ with the prior (5.16).

Two caveats about what these studies do and do not establish. They are illustrations of the theory, not comparisons against competing methods, and in particular we make no claim about the relative performance of the horseshoe against genomic prediction methods designed for this problem. And the additive genetic variance study uses a nearly orthogonal design, so by Proposition 5.3 it isolates the sampling contribution to the inflation and excludes the linkage disequilibrium contribution of Example 5.4, which is present in real data and is additional.

5.9.7 What the four studies show, taken together

The four studies are not four unrelated demonstrations. Read in order they trace one argument.

Section 5.9.1 establishes that the dictionary of Theorem 5.15 has operational content, in that the partition matched to a prior labelled by (a, b) produces the discovery rate n^{1-a} that Corollary 5.22 predicts, to within the half per cent that the fitted slope 0.498 against 0.5 represents. Without this the dictionary would be a curiosity.

Section 5.9.2 establishes that the problem the dictionary is meant to solve is real, and quantifies it in Fisher's own estimand. The threefold inflation at $N = 200$ is not an artefact of a badly chosen design; the design is nearly orthogonal and the loci are independent. By Proposition 5.3 realistic linkage disequilibrium adds a further, separately signed error to the estimand itself.

Section 5.9.3 shows that when the species sampling prior is used for the thing it is naturally suited to, namely predicting how many further variants a larger sample will reveal, it is accurate to within about nine per cent, against a naive doubling rule which overstates the count by a factor of between 1.6 and 1.8. This is the positive case for the machinery, as against the negative case made by Section 5.9.2.

Section 5.9.5 then closes the loop by showing what is not known. If the prior is indexed by θ and θ is uncertain by an amount comparable to itself, then the implied discovery count inherits that uncertainty, and the customary practice of reporting

a point estimate suppresses it. Combined with Section 5.8, which shows that θ is in addition convention dependent, the practical recommendation is to report the pair (α, θ) with α as the invariant of scientific interest and θ with a posterior attached.

5.10 Genetic implications

Heritability. Corollary 5.6 and Table 5.5 identify an upward bias in additive genetic variance proportional to the number of fitted average effects. Any procedure that estimates σ_A^2 by fitting an additive model across many loci and reporting the explained sum of squares inherits this bias exactly. The correction is not a matter of taste in regularisation; it is forced by the dimension of the projection in Fisher's own definition. Proposition 5.5 gives the general form and shows what any competing estimator must trade.

The frequency spectrum as a prior. Fisher's log series is usually read descriptively. Theorem 5.18 shows it is the shrinkage profile with its size bias removed, so reading it as a prior is not a metaphor but an identity, and Corollary 5.16 sharpens the point: the uniform shrinkage prior corresponds to the Ewens Sampling Formula at $\theta = 1$, and the Strawderman Berger prior is its $\alpha = 1/2$ counterpart $\mathrm{PD}(1/2, 1/2)$. This gives the genetics literature a second use for a quantity it already computes, and gives the statistics literature a generative account of priors it has been choosing on grounds of convenience.

Which prior for which regime. Corollary 5.22 gives a design rule. If the biology says the number of contributing variants grows polynomially in the sample, a prior with $a < 1$ is required and no amount of tuning rescues $a = 1$. If the number is genuinely logarithmic, an $a = 1$ prior is appropriate and an $a < 1$ prior overdiscovers. The choice is a genetic statement, not a computational one, and by Corollary 5.20 it can be checked against the observed singleton count rather than asserted.

Multiple testing. The multiplicity penalty in a global local analysis is usually described as arising automatically from the shared global scale. Theorem 5.18 says what it is: the expected number of discoveries is $\mathbb{E}K_n = \sum_j \mathbb{E}a_j$ with $\mathbb{E}a_j$ given by (5.14), so the penalty is a functional of the frequency spectrum. In a genetic application the multiplicity adjustment can therefore be calibrated against the observed spectrum.

Effective sizes and effective sparsity. Ewens and Hössjer (2026) show that the inbreeding, variance and eigenvalue effective population sizes are distinct quantities which coincide for monoecious Wright Fisher populations and diverge otherwise, and that which to use depends on whether one cares about short or long term loss of heterozygosity. The identity by descent probability π_2 on which the first of these rests is developed in detail by Thompson (2013). The parallel is exact: there is no such thing as *the* sparsity level either. Theorem 5.15 says which two parameter family the alternatives inhabit and Corollary 5.22 says what distinguishes them.

5.11 Limitations

We set out what we take to be the weak points of the argument, in decreasing order of seriousness.

The partition identity is unproved. As Remark 5.28 states, Theorems 5.15 and 5.18 concern the random discrete distribution generated by a normalised subordinator, while the working object in (5.13) is a set of independent weights on a fixed index set. The two agree in the marginal law of a size biased pick and in the mean intensity of the atoms, which is what Theorem 5.18 establishes, but we have not shown that the exchangeable partitions agree. What survives without that identification is the first moment: Theorem 5.23 proves the discovery rate directly for (5.13), and Table 5.3 confirms it numerically. Statements about the whole law of the number of discoveries, by contrast, do not transfer, as Proposition 5.27 shows.

The global scale is doing work. The prior (5.13) has two levels, and the $\mathrm{Be}(a, b)$ description of κ holds at a fixed global scale τ. Allowing τ to be estimated changes the marginal law of κ and therefore, formally, moves the implied (α, θ). Our reading is that estimating τ is how the data speaks about θ while a continues to fix α, which is consistent with the separation of rate from level described in Section 5.8, but we have not proved it.

The additive genetic variance study is idealised. Section 5.9.6 records that the design is nearly orthogonal, which favours the shrinkage estimator; the oracle global scale is reported only for comparison and the headline row does not use it. The orthogonality also suppresses the linkage disequilibrium contribution of Example 5.4, which acts on the estimand rather than the estimator and which no regularisation addresses. The net effect on a real analysis is not something we can predict from these simulations.

Epistasis is treated as noise. Throughout, epistatic contributions are absorbed into η in (5.2) and therefore into σ^2. This is the same convention Fisher adopts and that Ewens (2024) analyses, and it is what makes σ^2_A a projection rather than a decomposition. It means, however, that the inflation $p\sigma^2 / N$ grows with the amount of epistasis, so that the very phenomenon which makes the additive projection an incomplete summary also makes its naive estimate worse.

One parameter, one spectrum. The Ewens Sampling Formula assumes selective neutrality and a single time homogeneous mutation rate. Real spectra are shaped by selection, by demography and by ascertainment. Corollary 5.20 offers a diagnostic, in that the growth rate of the singleton count identifies α, but a mismatch between the observed and predicted spectrum has many possible causes and the prior is only one of them.

5.12 Discussion

The mathematical claims are Theorems 5.15 and 5.18, which place the shrinkage priors of sparse Bayesian statistics inside the two parameter Poisson Dirichlet family and identify the frequency spectrum with the shrinkage density once its size bias is removed. The practical claim is Corollary 5.22, which converts the identification into a statement about how many signals a given prior will ever find. The genetic claim is Corollary 5.6, that Fisher's additive genetic variance, estimated by fitting his own definition, is inflated by an amount proportional to the ratio of parameter count to sample size.

Four problems remain open. The first is the partition level identification of Remark 5.28. The second is whether the $n^{1/2}$ rate for the horseshoe coincides with the sparsity regime in which its posterior is known to contract at the minimax rate; if so, the optimal global scale is a statement about θ. The third is the behaviour of Proposition 5.5 under realistic linkage disequilibrium, where $\operatorname{tr}(H^\top H)$ must be computed from an empirical Gram matrix with strong local correlation, and where the effective number of independent parameters is smaller than p. The fourth is inference for α itself. Corollary 5.20 suggests that the singleton count identifies α, which would make the sparsity level estimable from the frequency spectrum rather than chosen, and would close the circle by letting the genetics determine the prior.

5.13 Proofs

Proof of Proposition 5.1. Write $Q(\beta) = \sum_u \sum_v p_u p_v (m_{uv} - \beta_u - \beta_v)^2$. Each β_u occurs in the terms with first index u and in those with second index u, and by symmetry of m and of P these contribute equally, so $\partial Q / \partial \beta_u = -4 \sum_v p_u p_v (m_{uv} - \beta_u - \beta_v)$. Setting this to zero and dividing by $4 p_u > 0$ gives $m_{u\cdot} - \beta_u - \bar\beta = 0$ with $\bar\beta = \sum_v p_v \beta_v$, that is $\beta_u = m_{u\cdot} - \bar\beta$. Multiplying by p_u and summing gives $\bar\beta = \bar m - \bar\beta$, so $\bar\beta = \bar m / 2$ and $\beta_u = m_{u\cdot} - \bar m / 2$. Then $\bar m = 2\bar\beta = 2 \sum_u p_u \beta_u$ and $\beta_u - \bar\beta = m_{u\cdot} - \bar m = a_u$. The fitted value for genotype (u, v) is $\beta_u + \beta_v$, with mean $2\bar\beta = \bar m$, so the variance removed is $\sum_u \sum_v p_u p_v (\beta_u + \beta_v - \bar m)^2 = \sum_u \sum_v p_u p_v (a_u + a_v)^2 = 2 \sum_u p_u a_u^2$, the cross term vanishing because $\sum_u p_u a_u = \bar m - \bar m = 0$. $\square$

Proof of Proposition 5.9. We assume that the genotype set is finite, that the ε_i are independent of the g_i and of each other and are identically distributed with mean zero and variance σ^2, and that redundant gene content functions have been removed (for instance one allele per locus, since allele counts at a locus sum to two), so that the population Gram matrix $A = \sum_g P_g \, \mathbf{a}_g \mathbf{a}_g^\top = \mathcal{A}^\top W \mathcal{A}$ is nonsingular. Here $\mathbf{a}_g$ is the vector of gene contents of genotype g, so that $\sum_{(g)} \beta_{ij} = \mathbf{a}_g^\top \beta$. Expanding the square,

$$\frac{1}{N} \sum_{i=1}^{N} \left(y_i - \mathbf{a}_{g_i}^\top \beta \right)^2 = \sum_g \hat P_g \left(m_g - \mathbf{a}_g^\top \beta \right)^2 + \frac{2}{N} \sum_{i=1}^{N} \varepsilon_i \left(m_{g_i} - \mathbf{a}_{g_i}^\top \beta \right) + \frac{1}{N} \sum_{i=1}^{N} \varepsilon_i^2,$$

with $\hat{P}_g = N^{-1}\#\{i : g_i = g\}$. The first term is (5.1) at $\hat{P}_N$. The whole expression is a quadratic $\beta^\top A_N \beta - 2\beta^\top b_N + c_N$ whose coefficients are finitely many sample averages; by the strong law $A_N \to A$, $b_N \to b = \sum_g P_g \mathbf{a}_g m_g$ and $c_N \to c = \sum_g P_g m_g^2 + \sigma^2$ almost surely, the cross terms involving ε_i having mean zero. The difference between the criterion and $Q(\beta) + \sigma^2$ is bounded by $\|A_N - A\|\,\|\beta\|^2 + 2\|b_N - b\|\,\|\beta\| + |c_N - c|$, which tends to zero uniformly on compacta. Finally A_N is nonsingular for all large N, and the minimiser $A_N^{-1} b_N$ converges to $A^{-1} b$, which is the minimiser of Q, that is Fisher's vector of average effects. □

Proof of Proposition 5.5. $\mathbb{E}\|Hy\|^2 = \|HX\beta\|^2 + \mathbb{E}\|H\eta\|^2 = \|HX\beta\|^2 + \sigma^2 \mathrm{tr}(H^\top H)$, using $\mathbb{E}\eta = 0$ and $\mathrm{Var}\eta = \sigma^2 I$. Divide by N. □

Proof of Proposition 5.11. Fix a partition of $\{1, \dots, n\}$ into blocks $B_1, \dots, B_k$ of sizes $n_1, \dots, n_k$, listed in order of their least elements. The urn produces this partition exactly when draw j is black whenever j is the least element of its block, and otherwise draws the colour of the block containing j; the names of the colours do not matter. Before draw j the total mass is $\theta + j - 1$. If j is the least element of a block the required probability is $\theta/(\theta + j - 1)$; if j is the rth smallest element of B_i, $r \geq 2$, the colour of B_i then has mass $r - 1$ and the required probability is $(r - 1)/(\theta + j - 1)$. Multiplying over j, the denominators give $\prod_{j=1}^{n} (\theta + j - 1) = \theta_{(n)}$, and the numerators give θ once per block and $1 \cdot 2 \cdots (n_i - 1)$ within block i, that is $\theta^k \prod_i (n_i - 1)!$. This is (5.7). □

Proof that the map of Theorem 5.15 is onto. Let $a \in (0, 1]$, $b > 0$ and set $\alpha = 1 - a \in [0, 1)$, $\theta = a + b - 1$. The GEM constraint $\theta > -\alpha$ reads $a + b - 1 > a - 1$, that is $b > 0$, which holds. Conversely for $\alpha \in [0, 1)$ and $\theta > -\alpha$ set $a = 1 - \alpha \in (0, 1]$ and $b = \theta + \alpha > 0$. The two maps are mutual inverses. □

Proof of Theorem 5.18. By definition a size biased pick satisfies $\mathbb{P}(W \in A) = \mathbb{E}\sum_i P_i \mathbf{1}(P_i \in A)$. Taking $h(x) = x\mathbf{1}(x \in A)$, which is bounded and vanishes at the origin, in the definition of ν gives $\mathbb{P}(W \in A) = \int_A x\nu(x)\,\mathrm{d}x$ for every Borel $A \subset (0, 1)$, so $f_W(x) = x\nu(x)$. For (5.14), condition on (P_i). Given the frequencies, the number of the n independent draws that land on atom i is $\mathrm{Bin}(n, P_i)$, so the number of atoms represented exactly $j \geq 1$ times is $\sum_i \mathbf{1}\{\mathrm{Bin}(n, P_i) = j\}$, with conditional expectation $\sum_i h_j(P_i)$ where $h_j(x) = \binom{n}{j} x^j (1 - x)^{n-j}$. This h_j is bounded and vanishes at the origin because $j \geq 1$, so taking expectations gives $\mathbb{E}a_j = \binom{n}{j} \int_0^1 x^j (1 - x)^{n-j} \nu(x)\,\mathrm{d}x$. Substituting $\nu = x^{-1} f_\kappa$ gives the display. □

Proof of Theorem 5.23. Write $Z = \kappa_1/(1 - \kappa_1)$. If $\kappa_1 \sim \mathrm{Be}(a, b)$ then Z has the beta prime law with density $f(z) = z^{a-1}(1 + z)^{-a-b}/B(a, b)$ on $(0, \infty)$. Since $\kappa_\tau/(1 - \kappa_\tau) = \tau^{-2} Z$ by the definition of κ_τ, we have $1 - \kappa_\tau = \tau^2/(\tau^2 + Z)$ and therefore

$$\mathbb{E}(1 - \kappa_\tau) = \int_0^\infty \frac{\tau^2}{\tau^2 + z} f(z)\,\mathrm{d}z.$$

Substitute $z = \tau^2 w$, so that $\mathrm{d}z = \tau^2\,\mathrm{d}w$ and

$$\mathbb{E}(1-\kappa_\tau) = \tau^2 \int_0^\infty \frac{f(\tau^2 w)}{1+w}\,\mathrm{d}w = \frac{\tau^{2a}}{B(a,b)} \int_0^\infty \frac{w^{a-1}(1+\tau^2 w)^{-a-b}}{1+w}\,\mathrm{d}w.$$

For $0 < a < 1$ the integrand is dominated by $w^{a-1}/(1+w)$, which is integrable at both ends, so dominated convergence gives $\int_0^\infty w^{a-1}(1+w)^{-1}\,\mathrm{d}w = \pi/\sin(\pi a)$ in the limit, which is the first display. For $a = 1$ the limiting integrand $(1+w)^{-1}$ is not integrable at infinity, and the cutoff is exhibited by the further substitution $w = v/\tau^2$, under which

$$\int_0^\infty \frac{(1+\tau^2 w)^{-1-b}}{1+w}\,\mathrm{d}w = \int_0^\infty \frac{(1+v)^{-1-b}}{\tau^2+v}\,\mathrm{d}v = 2\log(1/\tau) + O(1).$$

To see the last step, split the range at $v = 1$. On $(0,1)$ write $(1+v)^{-1-b} = 1 + \{(1+v)^{-1-b} - 1\}$: the first part integrates to $\log\{(1+\tau^2)/\tau^2\} = 2\log(1/\tau) + o(1)$, and the second has absolute value at most $(1+b)v/(\tau^2+v) \le 1+b$, so contributes $O(1)$. On $(1,\infty)$ the integrand is at most $(1+v)^{-1-b}$, which is integrable, so that part is $O(1)$ too. With $B(1,b) = 1/b$ this is the second display. Setting $p = n$ and $\tau_n = n^{-1/2}$ gives $\tau_n^{2a} = n^{-a}$, hence $\mathbb{E}m_n = n\,\mathbb{E}(1-\kappa_{\tau_n}) \sim C\,n^{1-a}$ for $a < 1$, and for $a = 1$, $\mathbb{E}m_n = n \cdot 2b\,n^{-1} \cdot \frac{1}{2}\log n = b\log n$. □

Proof of Lemma 5.26. As in the previous proof, $1-\kappa_\tau = \tau^2/(\tau^2+Z)$ with Z beta prime, so

$$\mathbb{E}(1-\kappa_\tau)^2 = \int_0^\infty \Big(\frac{\tau^2}{\tau^2+z}\Big)^2 f(z)\,\mathrm{d}z = \frac{\tau^{2a}}{B(a,b)} \int_0^\infty \frac{w^{a-1}(1+\tau^2 w)^{-a-b}}{(1+w)^2}\,\mathrm{d}w$$

after $z = \tau^2 w$. For $0 < a < 1$ the integrand is dominated by $w^{a-1}(1+w)^{-2}$, which is integrable, and dominated convergence gives $\int_0^\infty w^{a-1}(1+w)^{-2}\,\mathrm{d}w = B(a,2-a)$. □

Proof of Corollary 5.24. Under Theorem 5.15 the matched parameters are $\alpha = 1-a$ and $\theta = a+b-1$, so $n^\alpha = n^{1-a}$, which is the exponent in Theorem 5.23, and at $a = 1$ the species sampling rate is $\theta\log n = b\log n$, which is the second display. For the constants, (5.11) gives $\Gamma(\theta+1)/\{\alpha\Gamma(\theta+\alpha)\} = \Gamma(a+b)/\{(1-a)\Gamma(b)\}$, while Theorem 5.23 gives $\pi/\{B(a,b)\sin(\pi a)\}$. Using $\Gamma(a)\Gamma(1-a) = \pi/\sin(\pi a)$ and $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, the latter equals $\Gamma(1-a)\Gamma(a+b)/\Gamma(b)$. The ratio of the two constants is therefore $(1-a)\Gamma(1-a) = \Gamma(2-a)$, which on $a \in (0,1]$ attains its minimum $\Gamma(1.4616) \approx 0.8856$ and its maximum $\Gamma(1) = 1$ at $a = 1$. □

Proof of Corollary 5.19. With $f_\kappa(x) = \theta(1-x)^{\theta-1}$, (5.14) gives $\mathbb{E}a_j = \binom{n}{j}\theta B(j, n-j+\theta) = \frac{n!}{j!(n-j)!}\theta\frac{\Gamma(j)\Gamma(n-j+\theta)}{\Gamma(n+\theta)} = \frac{\theta}{j}\frac{n!}{(n-j)!}\frac{\Gamma(\theta+n-j)}{\Gamma(\theta+n)}$. As $n \to \infty$ with j fixed, $n!/(n-j)! \sim n^j$ and $\Gamma(\theta+n-j)/\Gamma(\theta+n) \sim n^{-j}$, giving θ/j. □

Proof of Corollary 5.20. With $f_\kappa(x) = \{\pi\sqrt{x(1-x)}\}^{-1}$ and $j = 1$, (5.14) gives $\mathbb{E}a_1 = n\pi^{-1}\int_0^1 x^{-1/2}(1-x)^{n-3/2}\,\mathrm{d}x = n\pi^{-1}B(1/2, n-1/2)$, and $B(1/2, n-1/2) = \Gamma(1/2)\Gamma(n-1/2)/\Gamma(n) \sim \sqrt{\pi}\,n^{-1/2}$. □

Proof of Proposition 5.3. The additive projection of m on $z = (z_1, \dots, z_L)$ has coefficient vector $R^{-1}c$ and explained variance $c^\top R^{-1}c$. Fitting locus ℓ alone gives coefficient c_ℓ and explained variance c_ℓ^2. Equality of $c^\top R^{-1}c$ and $c^\top c$ for all c forces $R^{-1} = I$. □

Proof of Proposition 5.12. Let $\mathcal{N} = \sum_i \delta_{J_i}$ be the Poisson process, so that $T = T(\mathcal{N}) = \sum_i J_i$. Given the frequencies $P_i = J_i/T$, the n draws are independent, and the probability that they realise a given partition with blocks of sizes $n_1, \dots, n_k$ is $\sum^{\neq} \prod_{l=1}^k P_{i_l}^{n_l}$, the sum running over ordered k tuples of distinct atoms $(i_1, \dots, i_k)$. Hence

$$p(n_1, \dots, n_k) = \mathbb{E}\Big[T^{-n} \sum_{l=1}^{k}{}^{\neq} \prod_{l=1}^k J_{i_l}^{n_l}\Big].$$

The awkward factor T^{-n} is removed by the gamma integral $T^{-n} = \Gamma(n)^{-1}\int_0^\infty t^{n-1}e^{-tT}\,\mathrm{d}t$, and since every term is nonnegative Tonelli's theorem allows the expectation to be taken inside the t integral. It remains to show that, for each $t > 0$,

$$\mathbb{E}\Big[e^{-tT} \sum_{l=1}^{k}{}^{\neq} \prod_{l=1}^k J_{i_l}^{n_l}\Big] = e^{-\psi(t)} \prod_{l=1}^k \xi(n_l, t).$$

This is the multivariate Mecke equation (Last and Penrose, 2017, Theorem 4.4): for a Poisson process with intensity ρ and nonnegative F, $\mathbb{E}\sum^{\neq} F(J_{i_1}, \dots, J_{i_k}; \mathcal{N}) = \int\cdots\int \mathbb{E}F(x_1, \dots, x_k; \mathcal{N} + \delta_{x_1} + \cdots + \delta_{x_k}) \prod_l \rho(x_l)\,\mathrm{d}x_l$. Apply it with $F(x_1, \dots, x_k; \mathcal{N}) = \prod_l x_l^{n_l} e^{-tT(\mathcal{N})}$. Adding the k points x_l adds $\sum_l x_l$ to T, so the integrand factorises as $\mathbb{E}e^{-tT(\mathcal{N})} \prod_l x_l^{n_l} e^{-tx_l}\rho(x_l)$; the integrals over the x_l give $\prod_l \xi(n_l, t)$, and $\mathbb{E}e^{-tT} = e^{-\psi(t)}$ is Campbell's formula for the Laplace functional of a Poisson process. Substituting into the t integral gives (5.8). □

Proof of Corollary 5.22 and of (5.12). We first prove the exact formula (5.12). Dividing (5.10) for the partition with an added singleton block by (5.10) itself, the product in the numerator gains the factor $\theta + k\alpha$, the denominator gains $\theta + n$, and the new block contributes $(1-\alpha)_{(0)} = 1$. So, given $K_n = k$, draw $n+1$ starts a new block with probability $(\theta + k\alpha)/(\theta + n)$, and taking expectations, $u_n = \mathbb{E}K_n$ satisfies

$$u_{n+1} = u_n + \frac{\theta + \alpha u_n}{\theta + n}, \qquad n \geq 1, \qquad u_1 = 1.$$

For $\alpha = 0$ this gives $u_n = 1 + \sum_{j=1}^{n-1} \theta/(\theta + j) = \sum_{j=0}^{n-1} \theta/(\theta + j)$, in agreement with Proposition 5.30. For $\alpha > 0$ put $v_n = u_n + \theta/\alpha$. Then $v_{n+1} = v_n(\theta + \alpha + n)/(\theta + n)$

and $v_1 = (\theta+\alpha)/\alpha$, so

$$v_n = \frac{\theta+\alpha}{\alpha}\prod_{j=1}^{n-1}\frac{\theta+\alpha+j}{\theta+j} = \frac{\theta+\alpha}{\alpha}\cdot\frac{\Gamma(\theta+\alpha+n)\,\Gamma(\theta+1)}{\Gamma(\theta+\alpha+1)\,\Gamma(\theta+n)}$$
$$= \frac{\Gamma(\theta+1)}{\alpha\,\Gamma(\theta+\alpha)}\cdot\frac{\Gamma(\theta+\alpha+n)}{\Gamma(\theta+n)},$$

which is (5.12). Since $\Gamma(x+\alpha)/\Gamma(x) \sim x^\alpha$ as $x\to\infty$, $u_n \sim \Gamma(\theta+1)n^\alpha/\{\alpha\Gamma(\theta+\alpha)\}$, while for $\alpha = 0$ the sum is $\theta\log n + O(1)$; this proves (5.11). The corollary follows on substituting the matched parameters $(\alpha,\theta) = (1-a, a+b-1)$ of Theorem 5.15: the rate is n^{1-a} for $a<1$ and $b\log n$ for $a=1$. □

Proof of Lemma 5.31. $\int_0^1 t^{\theta-1}(1-t)^{n-1}\,\mathrm{d}t = B(\theta,n) = \Gamma(\theta)\Gamma(n)/\Gamma(\theta+n)$. □

Proof of Proposition 5.32. By (5.7) and Lemma 5.31 the contribution of locus r is $\theta^{k_r}\Gamma(n_r)^{-1}\int_0^1 t_r^{\theta-1}(1-t_r)^{n_r-1}\,\mathrm{d}t_r$. Introducing t_r as latent, the joint density of $(\theta, t_{1:m})$ given the data is proportional to $p(\theta)\theta^{\sum_r k_r}\prod_r t_r^{\theta-1}(1-t_r)^{n_r-1}$. Reading off the conditionals gives $t_r \mid \theta \sim \mathrm{Be}(\theta, n_r)$ and $p(\theta \mid t_{1:m}) \propto p(\theta)\theta^{\sum_r k_r}e^{\theta\sum_r \log t_r}$. With $p(\theta)\propto\theta^{c_0-1}e^{-d_0\theta}$ this is $\mathrm{Ga}(c_0+\sum_r k_r,\ d_0-\sum_r\log t_r)$, noting $\log t_r<0$ so that the rate is positive. The Gibbs sampler alternating the two conditionals therefore has the posterior of $(\theta, t_{1:m})$ as its stationary law, and its θ marginal is $p(\theta\mid k_{1:m})$ because integrating out $t_{1:m}$ returns the likelihood by Lemma 5.31. □

5.14 Notation

symbol	meaning
n	number of genes in a sample
N	number of individuals in a regression sample
$p,\ L$	number of fitted average effects, number of loci
K_n	number of distinct alleles in a sample of n genes
a_j	number of alleles represented exactly j times
θ	mutation parameter, equivalently Pitman's second parameter
α	Pitman's first parameter, the discovery exponent
$\theta_{(n)}$	Pochhammer symbol $\Gamma(\theta+n)/\Gamma(\theta)$
$\beta_u,\ a_u$	average effect and average excess of allele A_u
σ_A^2	additive genetic variance
κ	shrinkage weight $1/(1+\tau^2\lambda^2)$
a, b	parameters of the beta law of κ
$\rho,\ \nu$	Lévy density of the subordinator, mean intensity of the atoms
$\psi,\ \xi(m,t)$	Laplace exponent and tilted moments of ρ
H	linear smoother, $\hat y = Hy$
R	correlation matrix of standardised coded loci

The symbol κ is reserved for the shrinkage weight throughout. The tilted moments of Proposition 5.12 are written $\tilde{\xi}(m,t)$ so that the two cannot be confused.

5.15 Exercises

1. *Additive fitness.* In Proposition 5.1 suppose $m_{uv} = \mu + \delta_u + \delta_v$ for constants δ_u. Show that σ_A^2 equals the whole genotypic variance $\sum_{u,v} p_u p_v (m_{uv} - \bar{m})^2$.
   Hint. Show $a_u = \delta_u - \bar{\delta}$ with $\bar{\delta} = \sum_u p_u \delta_u$, so the fit is exact.

2. *The largest additive variance under overdominance.* In Example 5.2, find the allele frequency maximising $\sigma_A^2 = 2p_1p_2s^2(1-2p_1)^2$ and the maximum.
   Hint. With $x = p_1p_2$ the expression is $2s^2x(1-4x)$, maximised at $x = 1/8$, that is $p_1 = (1-2^{-1/2})/2 \approx 0.146$ or its complement, with maximum $s^2/8$.

3. *An unbiased ridge.* In Example 5.7, show that $\mathbb{E}\hat{\sigma}_A^2$ is decreasing in λ and find the λ at which it equals $\sigma_A^2 = 7/6$. Why is this value of no practical use?
   Hint. Each term $\{d_j/(d_j+\lambda)\}^2$ decreases; numerically $\lambda \approx 1.03$. It depends on the unknown β and σ^2.

4. *Saturation.* Prove Corollary 5.10, and show that if the rows of X are independent with a continuous distribution then X has rank $\min(p, N)$ almost surely.
   Hint. For $p \geq N$ the column space is $\mathbb{R}^N$, so $H = I_N$. For the rank, a fixed proper subspace has Lebesgue measure zero.

5. *The horseshoe from the arcsine law.* Show that if $\kappa = 1/(1+\lambda^2)$ has the $\mathrm{Be}(1/2, 1/2)$ law then λ is half Cauchy, and that if $\kappa \sim \mathrm{Be}(1/2, 1)$ then λ^2 has density $\frac{1}{2}(1+u)^{-3/2}$ on $u > 0$.
   Hint. $\lambda^2 = (1-\kappa)/\kappa$ has the beta prime law $u^{b-1}(1+u)^{-a-b}/B(a,b)$ with the roles of a and b exchanged.

6. *An exact effective model size.* For the horseshoe show that $\mathbb{E}(1-\kappa_\tau) = \tau/(1+\tau)$ for every $\tau > 0$, and deduce that $\mathbb{E}m_n = n^{1/2}/(1+n^{-1/2})$ at $\tau_n = n^{-1/2}$. Compare with Table 5.2.
   Hint. Put $z = u^2$ in the integral of the proof of Theorem 5.23 and use $\int_0^\infty \{(c^2+u^2)(1+u^2)\}^{-1}\,\mathrm{d}u = \pi/\{2c(1+c)\}$. At $n = 10^5$ this gives 315.23.

7. *Kingman's correspondence at $n = 4$.* Using the formula of Example 5.14 at $\alpha = 1/2$, compute the probabilities of the five partition types of four genes and check that they sum to one when weighted by the number of labelled partitions of each type.
   Hint. The types (4), $(3,1)$, $(2,2)$, $(2,1,1)$, $(1,1,1,1)$ have 1, 4, 3, 6, 1 labelled versions and probabilities $5/16$, $1/16$, $1/48$, $1/24$, $1/8$.

8. *Singletons and the discount.* From (5.14) show that under $\mathrm{PD}(\alpha,\theta)$, $\mathbb{E}a_1 = n\,(\theta+\alpha)_{(n-1)}/(\theta+1)_{(n-1)}$, and deduce from (5.12) that $\mathbb{E}a_1/\mathbb{E}K_n \to \alpha$ when $\alpha > 0$.

Hint. $\mathbb{E}a_1 = n\,B(a, b+n-1)/B(a,b)$ with $(a,b) = (1-\alpha, \theta+\alpha)$; both numerator and denominator grow like n^{α}.

9. *The law of K_4.* Use Proposition 5.30 to find the exact law of K_4 when $\theta = 1$, and compare with the unsigned Stirling numbers of the first kind.
Hint. $\mathbb{P}(K_4 = k)$ is $6/24$, $11/24$, $6/24$, $1/24$ for $k = 1, 2, 3, 4$, that is $|s(4,k)|/4!$.

10. *How slowly information about θ accumulates.* Show that the Fisher information about θ in a sample of n genes is $I_n(\theta) = \theta^{-1} \sum_{j=0}^{n-1} j/(\theta + j)^2 \sim \theta^{-1} \log n$. At $\theta = 5$, how large must n be to halve the standard error obtained at $n = 100$?
Hint. The standard errors are 1.55 at $n = 100$, 0.87 at $n = 10^4$ and 0.67 at $n = 10^6$; halving needs about 5.7×10^4 genes, nearly six hundred times as many, whereas four independent loci of 100 genes halve it exactly.

11. *A Gibbs sampler.* Implement the sampler of Proposition 5.32 for the three loci of Example 5.34 with a $\mathrm{Ga}(1, 0.1)$ prior, and compare its output with the quadrature values quoted there.
Hint. Ten thousand sweeps already give the mean to about one per cent; the chain mixes quickly, the lag one autocorrelation of θ being about 0.5.

12. *Equicorrelated loci.* In Proposition 5.3 let $c = (1, \ldots, 1)^{\top}$ and let R have all off diagonal entries equal to $r > -1/(L-1)$. Show that $c^{\top} R^{-1} c = L/\{1 + (L-1)r\}$, so that the sum of one locus values overstates the whole genome value by the factor $1 + (L-1)r$. Evaluate it for $L = 100$ and $r = 0.1$.
Hint. c is an eigenvector of R with eigenvalue $1 + (L-1)r$; the factor is 10.9.

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PART II

The Sampling Theory of Neutral Alleles

CHAPTER 6

The Ewens Sampling Formula by de Finetti's Theorem

We derive the Ewens sampling formula from de Finetti's theorem in the elementary form proved by Heath and Sudderth, where the mixing measure appears as a weak limit of the laws of the empirical distribution function F_n. Helly selection and the Helly and Bray lemma supply the representation, and the only inequality needed is that drawing k items with and without replacement from n differs by $k(k-1)/(2n)$ in total variation. If the one step forecast depends on the past only through F_n and is affine in it, exchangeability alone forces the weights $\theta/(\theta+n)$ and $n/(\theta+n)$: the directing measure is a Dirichlet process, ties obey the sampling formula, and θ is identified by an exact finite sample variance identity. The number of jumps of F_n is sufficient for θ, with information only $\log n/\theta$, which we identify with the expected Kingman coalescent tree length divided by 2θ. Two extensions move the directing measure but leave the representation intact: a Rao weighted paintbox for ascertainment, and a two parameter predictive rule giving Pitman's formula. Corbet's Malayan butterflies favour the second once the censoring of the published table is modelled, with posterior probability 0.995 on a finite pool of about 1800 species (ninety percent interval 1100 to 3200), which keeps the genealogy and drops the infinitude of alleles.

6.1 Introduction

The Ewens sampling formula (Ewens, 1972) gives the distribution of the allelic partition of a sample of size n under the infinitely many alleles model with mutation parameter θ. Writing A_j for the number of alleles represented exactly j times in the sample, so that $\sum_{j=1}^{n} jA_j = n$, the formula reads

$$\mathbb{P}(A_1 = a_1, \ldots, A_n = a_n) = \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{\theta^{a_j}}{j^{a_j} a_j!}, \qquad \theta_{(n)} = \theta(\theta+1)\cdots(\theta+n-1). \tag{6.1}$$

Kingman (1978) recognised the formula as the law of an exchangeable random partition of $\mathbb{N}$ and gave the paintbox representation, the partition analogue of de Finetti's theorem. We obtain (6.1) instead from de Finetti's theorem in the elementary form of Heath and Sudderth (1976), where the mixing measure is produced as a weak limit of the laws of the empirical distribution function. Heath and Sudderth treat two valued variables; the argument below carries their device over to a general Borel type space. That route suits the problem for two reasons.

The Helly and Bray step delivers the de Finetti measure as the law of $\widetilde{F} = \lim_n F_n$, so every parameter is a functional of $\widetilde{F}$; and F_n is exactly the statistic through which the Ewens predictive rule sees the past. The mutation parameter is therefore read off a moment of $\widetilde{F}$, with no appeal to genealogy, the coalescent, or Kingman's characterisation.

Section 6.2 is model free: the sampling lemma, the representation, and the finite sequence bound. Section 6.3 imposes the predictive hypothesis and follows it to the harmonic weights, the covariance kernel of $\widetilde{F}$, an exact finite n variance identity, the Dirichlet process, the sampling formula, and the sufficiency of the jump count. Section 6.4 gives a simulation study and a first analysis of Corbet's Malayan butterflies, whose misfit is specific enough to point at its own explanation. Sections 6.5 and 6.6 pursue the two candidates, ascertainment and a finite species pool. Section 6.7 sets the whole argument against the usual genealogical derivation of the sampling formula. Section 6.8 turns the fits into posteriors, by a Beta augmentation for θ and quadrature for the pair, and answers Corbet's question with a predictive distribution. Section 6.9 draws the threads together, and Section 6.10 collects exercises. Small worked examples, with every number checkable by hand, are placed next to the results they illustrate.

Throughout, the sample space of types is the unit interval, which entails no loss of generality for a Borel type space.

6.2 Exchangeability and the de Finetti representation

Nothing in this section uses the Ewens hypothesis, which is why it is isolated: it survives the extensions of Sections 6.5 and 6.6 unchanged.

6.2.1 Setup and hypotheses

Let $(X_1, X_2, \dots)$ be an infinite sequence of random variables with values in $S = [0,1]$, defined on a common probability space, and let

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}, \qquad x \in [0,1],$$

denote the empirical distribution function of the first n observations. The associated empirical measure is $P_n = n^{-1} \sum_{i \le n} \delta_{X_i}$. Let $\mathcal{F}$ be the set of distribution functions on $[0,1]$, equipped with the topology of weak convergence, that is, pointwise convergence at continuity points of the limit.

ASSUMPTION 6.1 (Exchangeability). For every n and every permutation σ of $\{1, \dots, n\}$ the vectors $(X_1, \dots, X_n)$ and $(X_{\sigma(1)}, \dots, X_{\sigma(n)})$ have the same law.

ASSUMPTION 6.2 (Predictive sufficiency of the empirical distribution function). There is a diffuse (that is, atomless) distribution function H on $[0,1]$ with $X_1 \sim H$, and constants $c_n \in (0,1)$, such that for every $n \ge 1$

$$\mathbb{P}\big(X_{n+1} \le x \mid X_1, \dots, X_n\big) = c_n H(x) + (1 - c_n) F_n(x) \qquad \text{almost surely, for all } x. \tag{6.2}$$

Assumption 6.2 says that the past enters the forecast only through the empirical distribution function, and does so affinely: the predictive law is a compromise between the prior guess H and what has been seen. It is the natural nonparametric expression of the idea that only the sample, not its order, matters. It does not presuppose any value of θ, nor the form of c_n beyond it being free of the data. Section 6.3.1 shows that exchangeability then pins down c_n completely.

Because H is diffuse and $X_1 \sim H$, the event $\{X_i = X_j\}$ for $i \neq j$ has positive probability only through the atom that F_n places at an earlier observation. Define the random partition Π_n of $\{1, \dots, n\}$ by the equivalence relation $i \sim j$ if and only if $X_i = X_j$. Let K_n be the number of blocks of Π_n, equivalently the number of jumps of F_n, and let $A_j^{(n)}$ be the number of blocks of size j, so $\sum_j j A_j^{(n)} = n$ and $\sum_j A_j^{(n)} = K_n$. The object of the chapter is the law of Π_n. In genetic language X_i is the allele carried by the ith sampled gene, labelled by a point of $[0,1]$ that carries no meaning beyond its identity, Π_n groups the sampled genes by allele, K_n is the number of distinct alleles in the sample, and $A_j^{(n)}$ is the number of alleles seen exactly j times.

6.2.2 *Sampling without replacement*

The first step is the combinatorial heart of the Heath and Sudderth argument. It says that for an exchangeable sequence the first k coordinates look like a draw without replacement from the empirical measure of the first n, and that a draw without replacement from n items is close in total variation to a draw with replacement. We keep the exact constant, which is worth doing because it is the only quantitative ingredient in the proof and it yields the finite sequence version of de Finetti's theorem as a corollary.

LEMMA 6.3 (Without replacement versus with replacement). *Let $x_1, \dots, x_n \in S$ be fixed, let $\bar{P} = n^{-1} \sum_{i \le n} \delta_{x_i}$, and let $g_1, \dots, g_k : S \to \mathbb{R}$ satisfy $\|g_i\|_\infty \le 1$ with $k \le n$. Write $D_{n,k}$ for the set of k tuples $(j_1, \dots, j_k)$ of distinct indices in $\{1, \dots, n\}$ and set*

$$p_{n,k} \;=\; 1 - \prod_{i=0}^{k-1} \Big(1 - \frac{i}{n}\Big) \;\le\; \frac{1}{n}\binom{k}{2}.$$

Then

$$\left| \frac{(n-k)!}{n!} \sum_{(j_1,\dots,j_k) \in D_{n,k}} \prod_{i=1}^{k} g_i(x_{j_i}) \;-\; \prod_{i=1}^{k} \int_S g_i \, d\bar{P} \right| \;\le\; 2\, p_{n,k} \;\le\; \frac{k(k-1)}{n},$$

and the constant 2 improves to 1 when all $g_i \ge 0$.

Proof. Let ν be the uniform distribution on all n^k tuples $(j_1, \dots, j_k)$, which corresponds to sampling with replacement, and let ν_0 be the uniform distribution

on $D_{n,k}$, which corresponds to sampling without replacement and assigns mass $(n-k)!/n!$ to each of its $n!/(n-k)!$ points. Expanding the product,

$$\prod_{i=1}^{k}\int_S g_i\,d\bar{P} = \frac{1}{n^k}\sum_{(j_1,\dots,j_k)}\prod_{i=1}^{k} g_i(x_{j_i}) = \int \Psi\,d\nu, \qquad \Psi(j_1,\dots,j_k)=\prod_{i=1}^{k} g_i(x_{j_i}),$$

while the left hand sum in the statement is $\int \Psi\,d\nu_0$. Under ν the conditional law of the tuple given that all coordinates are distinct is exactly ν_0, and the probability of a repeat is $p_{n,k}$ by the birthday computation, the stated upper bound following from $\prod(1-i/n)\geq 1-\sum i/n$. Hence $\nu=(1-p_{n,k})\nu_0+p_{n,k}\nu_1$ for some probability measure ν_1, so

$$\int \Psi\,d\nu - \int \Psi\,d\nu_0 = p_{n,k}\Big(\int \Psi\,d\nu_1 - \int \Psi\,d\nu_0\Big),$$

whose modulus is at most $2p_{n,k}$ since $\|\Psi\|_\infty\leq 1$, and at most $p_{n,k}$ when Ψ takes values in $[0,1]$. □

The proof shows more than the statement: $p_{n,k}$ is exactly the total variation distance between ν and ν_0 as laws on index tuples, because ν_0 puts no mass on tuples with a repeat and ν puts mass $p_{n,k}$ there. The law of the values $(x_{j_1},\dots,x_{j_k})$ is the image of ν or ν_0 under a fixed map, and total variation can only shrink under a map, so the distance between the value laws is at most $p_{n,k}$, with equality when the x_i are distinct. The next example shows both cases.

EXAMPLE 6.4 (Ten balls). Take $n=10$ and first let the x_i be ten distinct labels. Then

$$p_{10,2}=\tfrac{1}{10}, \qquad p_{10,3}=1-\tfrac{9}{10}\cdot\tfrac{8}{10}=\tfrac{7}{25}=0.28, \qquad p_{10,4}=1-\tfrac{9\cdot 8\cdot 7}{1000}=\tfrac{62}{125}=0.496,$$

against the cruder bounds $k(k-1)/(2n)=0.1$, 0.3 and 0.6; with distinct labels these are the exact distances between drawing with and without replacement. Now let the urn hold six balls of colour 0 and four of colour 1. For $k=2$, with replacement the four ordered colour pairs $00,01,10,11$ have probabilities $0.36,0.24,0.24,0.16$, and without replacement $\frac{30}{90},\frac{24}{90},\frac{24}{90},\frac{12}{90}$. Half the sum of the absolute differences is

$$\tfrac{1}{2}\big(\tfrac{2}{75}+\tfrac{2}{75}+\tfrac{2}{75}+\tfrac{2}{75}\big)=\tfrac{4}{75}\approx 0.053,$$

about half the bound $p_{10,2}=0.1$. Exact enumeration gives 0.080 at $k=3$ and 0.118 at $k=4$, against 0.28 and 0.496. These are also the distances between the hypergeometric and binomial laws of the number of balls of colour 1, since in both schemes the colour sequence is exchangeable and the count is sufficient for it. Coarse types make sampling with and without replacement much closer than the bound says, because a repeated index often lands on a colour that would have been drawn anyway. For large n the bound is nearly exact: at $n=1000$, $p_{n,10}=0.0441$ against $k(k-1)/(2n)=0.045$, while $p_{n,30}=0.356$ against 0.435.

LEMMA 6.5 (Exchangeable reduction). *Let (X_i) satisfy Assumption 6.1, let $k \le n$ and let $g_1, \dots, g_k$ be bounded by one in supremum norm. Then*

$$\left| \mathbb{E} \prod_{i=1}^{k} g_i(X_i) \;-\; \mathbb{E} \prod_{i=1}^{k} \int_S g_i \, dP_n \right| \;\le\; 2p_{n,k} \;\le\; \frac{k(k-1)}{n}.$$

Proof. Let $\mathcal{E}_n = \sigma(P_n, X_{n+1}, X_{n+2}, \dots)$ be the sigma field generated by the unordered first n values together with the remainder of the sequence. Exchangeability of $(X_1, \dots, X_n)$ implies that conditionally on $\mathcal{E}_n$ the vector $(X_1, \dots, X_n)$ is a uniformly random rearrangement of the atoms of P_n, so that $(X_1, \dots, X_k)$ is a uniform draw without replacement from those n atoms. Therefore

$$\mathbb{E}\left[\prod_{i=1}^{k} g_i(X_i) \;\middle|\; \mathcal{E}_n \right] = \frac{(n-k)!}{n!} \sum_{(j_1,\dots,j_k) \in D_{n,k}} \prod_{i=1}^{k} g_i(X_{j_i}).$$

Apply Lemma 6.3 pathwise to the atoms of P_n and take expectations. □

COROLLARY 6.6 (Finite exchangeable sequences). *Let $(X_1, \dots, X_N)$ be exchangeable and let μ_N be the law of F_N. Then, for $k \le N$,*

$$d_{\mathrm{TV}}\Big(\mathcal{L}(X_1, \dots, X_k),\; \int_{\mathcal{F}} F^{\otimes k} \, \mu_N(dF) \Big) \;\le\; p_{N,k} \;\le\; \frac{k(k-1)}{2N}.$$

Proof. Total variation distance equals one half the supremum of $|\int \phi \, dP - \int \phi \, dQ|$ over measurable ϕ with $\|\phi\|_\infty \le 1$. Conditionally on $\mathcal{E}_N$ the vector $(X_1, \dots, X_k)$ is a draw without replacement and $F_N^{\otimes k}$ is the corresponding draw with replacement, and the proof of Lemma 6.3 bounds their total variation distance by $p_{N,k}$ pathwise. Average over $\mathcal{E}_N$ and use convexity of total variation. □

Corollary 6.6 is the bound of Diaconis and Freedman (1980), obtained here as a by product. It quantifies the failure of de Finetti's theorem for finite sequences and shows that the representation is accurate as soon as $k = o(\sqrt{N})$, which by the birthday problem is sharp.

6.2.3 *Helly selection and the Helly and Bray limit*

Lemma 6.5 says that, to within $k(k-1)/n$, the first k observations behave like an independent sample from the empirical distribution function F_n of a long stretch of the sequence. If F_n had a limit, the first k observations would be an independent sample from that limit. The theorem makes this precise: compactness of $\mathcal{F}$ supplies a limit in law, and the error term disappears as $n \to \infty$.

THEOREM 6.7 (de Finetti, after Heath and Sudderth). *Let (X_i) be exchangeable with values in $[0,1]$. Then there exists a random distribution function $\widetilde{F}$, with law μ on $\mathcal{F}$, such that for every k and all bounded continuous $g_1, \dots, g_k$*

$$\mathbb{E} \prod_{i=1}^{k} g_i(X_i) = \int_{\mathcal{F}} \prod_{i=1}^{k} \left(\int_S g_i \, dF \right) \mu(dF). \tag{6.3}$$

The measure μ is unique, and $F_n \to \widetilde{F}$ uniformly with probability one, so that μ is the law of the limiting empirical distribution function. Conditionally on $\widetilde{F}$ the variables $X_1, X_2, \ldots$ are independent and identically distributed with distribution function $\widetilde{F}$.

Proof. Let μ_n be the law of F_n, a Borel probability measure on $\mathcal{F}$. By Helly's selection theorem every sequence in $\mathcal{F}$ has a subsequence converging weakly to an element of $\mathcal{F}$; since the type space is the compact interval $[0,1]$ no mass can escape and the limit is again a distribution function. Thus $\mathcal{F}$ is compact and metrisable, for instance under the Lévy metric, and the family $\{\mu_n\}$ is automatically tight. By Prohorov's theorem there is a subsequence $n_m \uparrow \infty$ and a probability measure μ on $\mathcal{F}$ with $\mu_{n_m} \Rightarrow \mu$.

Fix k and bounded continuous $g_1, \ldots, g_k$ with $\|g_i\|_\infty \le 1$. The map

$$\Phi : \mathcal{F} \to \mathbb{R}, \qquad \Phi(F) = \prod_{i=1}^{k} \int_S g_i \, dF,$$

is bounded by one, and it is continuous: this is precisely the Helly and Bray lemma, which states that $F_m \to F$ weakly implies $\int g \, dF_m \to \int g \, dF$ for every bounded continuous g. Consequently

$$\int_{\mathcal{F}} \Phi \, d\mu_{n_m} \longrightarrow \int_{\mathcal{F}} \Phi \, d\mu.$$

On the other hand $\int_{\mathcal{F}} \Phi \, d\mu_n = \mathbb{E} \prod_{i \le k} \int g_i \, dP_n$, which by Lemma 6.5 differs from the fixed number $\mathbb{E} \prod_{i \le k} g_i(X_i)$ by at most $k(k-1)/n$. Letting $m \to \infty$ along the subsequence, the error term vanishes and (6.3) follows.

For uniqueness, note that products of the form Φ with g_i continuous separate points of $\mathcal{F}$ and form an algebra containing the constants, so by the Stone and Weierstrass theorem their linear span is dense in the continuous functions on the compact space $\mathcal{F}$. Hence the numbers $\int \Phi \, d\mu$ determine μ. Since every subsequential limit satisfies (6.3), all subsequential limits coincide and the full sequence converges, $\mu_n \Rightarrow \mu$.

Finally, (6.3) says that conditionally on the random distribution function $\widetilde{F}$ drawn from μ the sequence is independent and identically distributed with distribution function $\widetilde{F}$. The strong law applied conditionally gives $F_n(x) \to \widetilde{F}(x)$ almost surely at every continuity point, and the convergence is uniform by the Glivenko and Cantelli theorem applied conditionally on $\widetilde{F}$; so μ is the law of the almost sure limit of the empirical distribution functions.

One point in the last paragraph deserves care, since $\widetilde{F}$ was produced only as a random element with law μ and not yet as a function on the original probability space. The limit can be built there directly. For fixed x, the conditional rearrangement argument in the proof of Lemma 6.5 gives $\mathbb{E}[\mathbf{1}\{X_1 \le x\} \mid \mathcal{E}_n] = F_n(x)$, and the sigma fields $\mathcal{E}_n$ decrease in n. Hence $(F_n(x), \mathcal{E}_n)$ is a reverse martingale and converges almost surely; call the limit $\widetilde{F}(x)$, defined simultaneously for all rational x off a single null set and extended by right continuity. The moment

identities above, applied with g_i indicators approximated by continuous functions, show that the law of this $\widetilde{F}$ has the same mixed moments as μ, hence equals μ by the uniqueness argument, and the conditional statements then hold on the original space. □

Theorem 6.7 is the version of de Finetti's theorem we use. Its value here is not only the representation but the concrete description of the mixing measure: μ is the law of $\lim_n F_n$. Any parameter of an exchangeable model is therefore a functional of the limiting empirical distribution function, and the next section computes that functional for the Ewens case.

6.3 The Ewens model

Assumption 6.2 now enters, and everything follows from it: the weights are forced, θ becomes a moment functional of the Helly and Bray limit, the de Finetti measure is a Dirichlet process, ties obey the sampling formula, and the jump count of F_n is sufficient.

6.3.1 Identification of the parameter

The first observation is that exchangeability alone determines the weights c_n up to a single free constant.

PROPOSITION 6.8 (The weights are forced). *Under Assumptions 6.1 and 6.2, set*

$$\theta = \frac{c_1}{1 - c_1} \in (0, \infty).$$

Then for every $n \geq 1$

$$c_n = \frac{\theta}{\theta + n}, \qquad 1 - c_n = \frac{n}{\theta + n},$$

so that (6.2) *becomes the predictive rule*

$$\mathbb{P}\big(X_{n+1} \in \cdot \mid X_1, \ldots, X_n\big) = \frac{\theta}{\theta + n} H(\cdot) + \frac{1}{\theta + n} \sum_{i=1}^{n} \delta_{X_i}(\cdot). \tag{6.4}$$

Proof. Take $n = 1$ in (6.2). Since H is diffuse and $X_1 \sim H$, the predictive law places an atom of size $1 - c_1$ at X_1 and the diffuse part contributes no atom, so

$$q := \mathbb{P}(X_1 = X_2) = 1 - c_1 = \frac{1}{1 + \theta}.$$

By exchangeability $\mathbb{P}(X_i = X_j) = q$ for every pair $i \neq j$.

Fix $n \geq 1$ and let $N_n = \#\{i \leq n : X_i = X_n\}$ be the size of the block of Π_n containing the index n. Since H is diffuse, evaluating (6.2) at the atom X_n gives

$$\mathbb{P}\big(X_{n+1} = X_n \mid X_1, \ldots, X_n\big) = (1 - c_n)\, P_n(\{X_n\}) = (1 - c_n) \frac{N_n}{n}.$$

Taking expectations and using exchangeability once more on the left hand side,

$$q = \mathbb{P}(X_{n+1} = X_n) = (1 - c_n)\frac{\mathbb{E}N_n}{n}.$$

It remains to compute $\mathbb{E}N_n$. Writing $N_n = 1 + \sum_{i=1}^{n-1} \mathbf{1}\{X_i = X_n\}$ and using $\mathbb{P}(X_i = X_n) = q$ for $i < n$,

$$\mathbb{E}N_n = 1 + (n-1)q = 1 + \frac{n-1}{1+\theta} = \frac{\theta + n}{1+\theta}.$$

Therefore $q = (1 - c_n)(\theta + n)/\{n(1+\theta)\}$, and since $q = 1/(1+\theta)$ we obtain $1 - c_n = n/(\theta + n)$, as claimed. Substituting into (6.2) and writing nF_n as the sum of point masses gives (6.4). □

Rule (6.4) is the urn scheme of Blackwell and MacQueen (1973), equivalently the urn of Hoppe (1984) with a black ball of weight θ. It has been derived here, not assumed. Note that the proof used no induction and no distributional calculation beyond the pairwise tie probability: exchangeability supplies the identity $\mathbb{P}(X_i = X_n) = \mathbb{P}(X_1 = X_2)$, and that is enough to force the harmonic form of c_n.

In urn language, (6.4) reads as follows. An urn holds one black ball of weight θ and, after n draws, n coloured balls of weight one. A ball is drawn with probability proportional to its weight. If it is black, it is returned together with a ball of a brand new colour; if it is coloured, it is returned together with another ball of the same colour. The colours are the alleles. The same scheme is often told as a Chinese restaurant: customer $n+1$ sits at a new table with probability $\theta/(\theta+n)$ and joins a table already holding r customers with probability $r/(\theta+n)$.

EXAMPLE 6.9 (Five draws from Hoppe's urn). Label the alleles in order of first appearance and suppose the first five draws give the colour sequence $1, 1, 2, 1, 3$. Reading (6.4) step by step, the five factors are

$$\underbrace{\frac{\theta}{\theta}}_{\text{new}} \cdot \underbrace{\frac{1}{\theta+1}}_{\text{joins 1}} \cdot \underbrace{\frac{\theta}{\theta+2}}_{\text{new}} \cdot \underbrace{\frac{2}{\theta+3}}_{\text{joins 1}} \cdot \underbrace{\frac{\theta}{\theta+4}}_{\text{new}} = \frac{2\,\theta^3}{\theta(\theta+1)(\theta+2)(\theta+3)(\theta+4)} = \frac{2\,\theta^3}{\theta_{(5)}}.$$

At $\theta = 1$ this is $2/120 = 1/60$, and at $\theta = 2$ it is $16/720 = 1/45$. Notice what the product depends on: the denominators are fixed, three factors θ come from the three new colours, and colour 1, which ends with three members, contributes $1 \cdot 2 = 2!$. Any other order of arrival producing the same set partition $\{\{1,2,4\},\{3\},\{5\}\}$ gives the same numbers in a different order, which is the content of Proposition 6.14 below. There are $5!/(3!\,2!) = 10$ set partitions of $\{1,\dots,5\}$ with block sizes $3,1,1$, so the probability that the sample contains one allele three times and two alleles once each is $10 \cdot 2\theta^3/\theta_{(5)}$, which is $1/6$ at $\theta = 1$ and $2/9$ at $\theta = 2$.

The parameter θ is now identified from the de Finetti measure produced in Theorem 6.7. We compute the entire second order structure of $\widetilde{F}$, not merely its variance at a point.

PROPOSITION 6.10 (Covariance kernel of the de Finetti limit). *Let $\widetilde{F}$ be the almost sure limit of F_n and $\widetilde{P}$ the associated random measure. Then $\mathbb{E}\widetilde{F}(x) = H(x)$ and, for all $x \le y$,*

$$\operatorname{Cov}\big(\widetilde{F}(x), \widetilde{F}(y)\big) = \frac{H(x)\big(1 - H(y)\big)}{1+\theta} = \frac{H(x \wedge y) - H(x)H(y)}{1+\theta}. \tag{6.5}$$

More generally, for every $g \in L^2(H)$,

$$\operatorname{Var}\left(\int_S g\, d\widetilde{P}\right) = \frac{\operatorname{Var}_H(g)}{1+\theta}, \qquad \text{so} \qquad \theta = \frac{\operatorname{Var}_H(g)}{\operatorname{Var}\big(\int g\, d\widetilde{P}\big)} - 1 \tag{6.6}$$

for any g that is not H almost surely constant.

Proof. By Theorem 6.7 the variables are conditionally independent and identically distributed given $\widetilde{P}$, so $\mathbb{E}\int g\, d\widetilde{P} = \mathbb{E}g(X_1) = \int g\, dH$ and $\mathbb{E}(\int g\, d\widetilde{P})^2 = \mathbb{E}[g(X_1)g(X_2)]$. Conditioning on X_1 and using (6.2) with $n = 1$ and $c_1 = \theta/(1+\theta)$,

$$\begin{aligned}\mathbb{E}[g(X_1)g(X_2)] &= \mathbb{E}\Big[g(X_1)\Big\{c_1 \int g\, dH + (1-c_1)g(X_1)\Big\}\Big] \\ &= c_1\Big(\int g\, dH\Big)^2 + (1-c_1)\int g^2\, dH.\end{aligned}$$

Subtracting $(\int g\, dH)^2$ gives $(1-c_1)\operatorname{Var}_H(g) = \operatorname{Var}_H(g)/(1+\theta)$, which is (6.6). Taking $g = \mathbf{1}\{\cdot \le x\}$ and polarising, or repeating the display with $g_1 = \mathbf{1}\{\cdot \le x\}$ and $g_2 = \mathbf{1}\{\cdot \le y\}$ and using $\mathbf{1}\{X_1 \le x\}\mathbf{1}\{X_1 \le y\} = \mathbf{1}\{X_1 \le x\}$ for $x \le y$, gives

$$\mathbb{E}\big[\widetilde{F}(x)\widetilde{F}(y)\big] = \mathbb{P}(X_1 \le x, X_2 \le y) = H(x)\frac{\theta H(y) + 1}{1+\theta},$$

and subtracting $H(x)H(y)$ yields (6.5). Indicators are not continuous, but $\widetilde{F}(x)$ is the bounded almost sure limit of $F_n(x)$, so the moment identities pass to the limit by bounded convergence. □

Formula (6.5) says that $\sqrt{1+\theta}\,(\widetilde{F} - H)$ has exactly the covariance kernel of a Brownian bridge time changed by H. The prior spread of the de Finetti measure around its centre is the bridge, scaled by $(1+\theta)^{-1/2}$: large θ means little variability in the limiting empirical distribution function and a sample close to H, small θ means $\widetilde{F}$ concentrates on a few types.

The next identity is exact for every n, not only in the limit, and is what we use in the simulations of Section 6.4.

PROPOSITION 6.11 (Exact finite n variance and a moment estimator). *Under Assumptions 6.1 and 6.2, for every $n \ge 1$, every $g \in L^2(H)$ and every x,*

$$\operatorname{Var}\left(\int_S g\, dP_n\right) = \operatorname{Var}_H(g)\,\frac{\theta + n}{n(1+\theta)}, \qquad \operatorname{Var}\big(F_n(x)\big) = H(x)\big(1 - H(x)\big)\frac{\theta + n}{n(1+\theta)}. \tag{6.7}$$

Consequently, if v_n denotes the left hand side of (6.7)*, then*

$$\theta = \frac{n\big(\mathrm{Var}_H(g) - v_n\big)}{n v_n - \mathrm{Var}_H(g)}, \tag{6.8}$$

an exact inversion with no asymptotic approximation. The limit $n \to \infty$ recovers (6.6)*, and the case $\theta \to \infty$ recovers the independent sampling variance* $\mathrm{Var}_H(g)/n$.

Proof. Write $T_n = \int g\, dP_n = n^{-1}\sum_{i\le n} g(X_i)$. By exchangeability all the variances and all the covariances in the expansion are equal, so

$$\mathrm{Var}(T_n) = \frac{1}{n}\,\mathrm{Var}_H(g) + \frac{n-1}{n}\,\mathrm{Cov}\big(g(X_1), g(X_2)\big).$$

The proof of Proposition 6.10 gives $\mathrm{Cov}(g(X_1), g(X_2)) = \mathrm{Var}_H(g)/(1+\theta)$. Hence

$$\begin{aligned}\mathrm{Var}(T_n) &= \mathrm{Var}_H(g)\left\{\frac{1}{n} + \frac{n-1}{n(1+\theta)}\right\} \\ &= \mathrm{Var}_H(g)\frac{(1+\theta)+(n-1)}{n(1+\theta)} = \mathrm{Var}_H(g)\frac{\theta+n}{n(1+\theta)}.\end{aligned}$$

Solving (6.7) for θ gives (6.8). □

Proposition 6.11 is the identification statement promised by the heading of this subsection. The Helly and Bray step of Theorem 6.7 produced the de Finetti measure as the law of $\lim_n F_n$; Propositions 6.10 and 6.11 show that the mutation parameter is a ratio of two variances, the variance that g would have under independent sampling from H divided by the variance actually exhibited by the sample mean of g, suitably corrected for n. The correction matters: Table 6.3 shows that dropping it, that is using the limiting form (6.6) at finite n, produces a visible downward bias when θ is large relative to n.

6.3.2 *The de Finetti measure is a Dirichlet process*

PROPOSITION 6.12. *Under Assumptions 6.1 and 6.2, the random measure $\widetilde{P}$ associated with $\widetilde{F}$ is a Dirichlet process with base measure θH. That is, for every finite measurable partition $A_1, \dots, A_m$ of $[0,1]$,*

$$\big(\widetilde{P}(A_1), \dots, \widetilde{P}(A_m)\big) \sim \mathrm{Dir}\big(\theta H(A_1), \dots, \theta H(A_m)\big).$$

Proof. Fix such a partition and set $Y_i = j$ when $X_i \in A_j$. Integrating (6.4) over A_j,

$$\mathbb{P}\big(Y_{n+1} = j \mid X_1, \dots, X_n\big) = \frac{\theta H(A_j) + \#\{i \le n : Y_i = j\}}{\theta + n},$$

and the right hand side depends on the past only through the counts, so (Y_i) is itself a Markov chain of Pólya urn type with initial ball weights $\theta H(A_1), \dots, \theta H(A_m)$ and reinforcement one. For the Pólya urn the vector of limiting frequencies exists almost

surely and is $\mathrm{Dir}(\theta H(A_1), \dots, \theta H(A_m))$ distributed, and given those frequencies the Y_i are independent and identically distributed multinomial draws. By Theorem 6.7 the limiting frequency of category j is $\lim_n P_n(A_j) = \widetilde{P}(A_j)$ almost surely, which proves the claim. Since the partition was arbitrary, $\widetilde{P}$ has the defining finite dimensional distributions of the Dirichlet process $\mathcal{D}(\theta H)$ of Ferguson (1973). □

Proposition 6.10 is consistent with this: for a Dirichlet process, $\widetilde{P}([0, x])$ is Beta with parameters $\theta H(x)$ and $\theta(1 - H(x))$, whose variance is $H(x)(1 - H(x))/(1 + \theta)$. The identification of θ from the empirical distribution function is thus the statement that the concentration parameter of a Dirichlet process is determined by the variance of any one of its linear functionals.

EXAMPLE 6.13 (The median allele class at $\theta = 2$). Let H be uniform and $\theta = 2$, and ask what fraction of the population carries an allele labelled below one half. By Proposition 6.12 this fraction, $\widetilde{P}([0, 1/2])$, is $\mathrm{Beta}(1, 1)$, that is uniform on $[0, 1]$: every split of the population between the two halves of the label space is equally likely, with variance $1/12 \approx 0.0833$, which is $H(1/2)(1 - H(1/2))/(1 + \theta) = 0.25/3$ as Proposition 6.10 requires. A sample of $n = 50$ genes sees this fraction with additional binomial noise, and Proposition 6.11 gives

$$\operatorname{Var} F_{50}(1/2) = 0.25 \cdot \frac{52}{50 \cdot 3} = 0.0867,$$

the entry in Table 6.3. Inverting with the asymptotic formula (6.6) would return $0.25/0.0867 - 1 = 1.88$ rather than 2; the exact inversion (6.8) returns $50(0.25 - 0.0867)/(50 \cdot 0.0867 - 0.25) = 2.00$. By contrast at $\theta = 0.5$ the fraction is $\mathrm{Beta}(1/4, 1/4)$, a U shaped law piling up near 0 and 1: with few alleles in the population, one of them is likely to dominate whichever half it lands in.

6.3.3 *The sampling formula*

We now compute the law of the partition Π_n generated by ties. Let $B_1, \dots, B_k$ denote the blocks listed in order of first appearance and let $n_1, \dots, n_k$ be their sizes, $\sum_j n_j = n$.

PROPOSITION 6.14 (Exchangeable partition probability function). *Under Assumptions 6.1 and 6.2,*

$$\mathbb{P}\big(\Pi_n = \{B_1, \dots, B_k\}\big) = \frac{\theta^k \prod_{j=1}^{k} (n_j - 1)!}{\theta_{(n)}}, \qquad \theta_{(n)} = \theta(\theta + 1) \cdots (\theta + n - 1), \tag{6.9}$$

which depends on the partition only through the multiset of block sizes.

Proof. Read the sequence $X_1, X_2, \dots, X_n$ one index at a time and use (6.4). At step i the denominator is $\theta + i - 1$, whatever happens. If index i opens a new block, the diffuse component is used and the numerator contributes θ, since H is diffuse and the new value is almost surely distinct from all previous ones. If index i

joins an existing block that currently holds r indices, the numerator contributes r. Multiplying over $i = 1, \dots, n$, the denominators give $\theta_{(n)}$, the new block events give θ^k, and a block that ends with n_j members contributes the factors $1, 2, \dots, n_j - 1$ as its second through last members arrive, that is $(n_j - 1)!$. This yields (6.9). The right hand side is symmetric in $n_1, \dots, n_k$, so the partition is exchangeable and the formula is a genuine exchangeable partition probability function in the sense of Pitman (1995). □

Proposition 6.14 gives the probability of one particular grouping of labelled genes. A geneticist rarely cares which genes are grouped together, only how many alleles appear once, twice and so on; and since every grouping with the same block sizes has the same probability, the law of these counts is the partition probability times the number of groupings with the given sizes.

THEOREM 6.15 (Ewens sampling formula). *Under Assumptions 6.1 and 6.2, with* $\theta = c_1/(1 - c_1)$ *identified by* (6.8), *the counts* $A_j = A_j^{(n)}$ *of blocks of size* j *satisfy, for every vector of nonnegative integers* $a_1, \dots, a_n$ *with* $\sum_j j a_j = n$,

$$\mathbb{P}\big(A_1 = a_1, \dots, A_n = a_n\big) = \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{\theta^{a_j}}{j^{a_j}\, a_j!}. \tag{6.10}$$

Proof. The number of set partitions of $\{1, \dots, n\}$ having exactly a_j blocks of size j for each j is the multinomial count

$$\frac{n!}{\prod_{j=1}^{n} (j!)^{a_j}\, a_j!},$$

the factor $a_j!$ removing the ordering among blocks of equal size. Each such partition has $k = \sum_j a_j$ blocks and, by Proposition 6.14, the same probability

$$\frac{\theta^k \prod_{j=1}^{n} \big((j-1)!\big)^{a_j}}{\theta_{(n)}}.$$

Multiplying and using $(j-1)!/j! = 1/j$ together with $\theta^k = \prod_j \theta^{a_j}$,

$$\mathbb{P}(A_1 = a_1, \dots, A_n = a_n) = \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{\theta^{a_j}\, ((j-1)!)^{a_j}}{(j!)^{a_j} a_j!} = \frac{n!}{\theta_{(n)}} \prod_{j=1}^{n} \frac{\theta^{a_j}}{j^{a_j} a_j!},$$

which is (6.10). □

That the right hand side of (6.10) sums to one over the admissible vectors is Cauchy's formula for the number of permutations of n objects with a given cycle type (Arratia, Barbour and Tavaré, 2003) together with the identity $\sum_{\sigma \in S_n} \theta^{\#\text{cycles}(\sigma)} = \theta_{(n)}$; in the present derivation normalisation is automatic, since (6.10) was obtained by summing a probability distribution over partitions.

The factor $n!/\prod_j j^{a_j}a_j!$ in (6.10) is Cauchy's count of the permutations of n objects with a_j cycles of length j, so the formula can be written as

$$\mathbb{P}(A_1 = a_1, \dots, A_n = a_n) = \frac{c(a)\,\theta^k}{\theta_{(n)}}, \qquad c(a) = \frac{n!}{\prod_j j^{a_j}a_j!}, \quad k = \sum_j a_j. \tag{6.11}$$

At $\theta = 1$, where $\theta_{(n)} = n!$, this is the law of the cycle type of a uniformly random permutation, a fact worth keeping in mind when checking small cases.

EXAMPLE 6.16 (Every allelic partition of four and five genes). Table 6.1 lists all allelic partitions for $n = 4$ and $n = 5$ with their probabilities from (6.11), using $\theta_{(4)} = \theta(\theta+1)(\theta+2)(\theta+3)$, which is 24 at $\theta = 1$ and 120 at $\theta = 2$, and $\theta_{(5)} = 120$ and 720. For instance at $n = 4$ and $\theta = 2$ the shape $2+1+1$, one allele seen twice and two seen once, has $c(a) = 4!/(2\cdot 2!) = 6$ and probability $6\cdot 2^3/120 = 2/5$. Each column sums to one, as it must. The columns also give the moments of K_n by direct summation. At $n = 4$ and $\theta = 2$,

$$\mathbb{E}K_4 = 1\cdot\tfrac{1}{10} + 2\big(\tfrac{4}{15} + \tfrac{1}{10}\big) + 3\cdot\tfrac{2}{5} + 4\cdot\tfrac{2}{15} = \tfrac{77}{30} \approx 2.567, \qquad \mathbb{E}K_4^2 = \tfrac{73}{10},$$

so $\operatorname{Var} K_4 = 73/10 - (77/30)^2 = 641/900 \approx 0.712$. Proposition 6.19 below gives the same two numbers as $\sum_{i=0}^{3} 2/(2+i) = 1 + \frac{2}{3} + \frac{1}{2} + \frac{2}{5} = \frac{77}{30}$ and $\sum_{i=0}^{3} 2i/(2+i)^2 = 0 + \frac{2}{9} + \frac{1}{4} + \frac{6}{25} = \frac{641}{900}$. At $n = 5$ the same calculation gives $\mathbb{E}K_5 = 137/60 \approx 2.283$ and $\operatorname{Var} K_5 = 2951/3600 \approx 0.820$ for $\theta = 1$, and $\mathbb{E}K_5 = 29/10$ and $\operatorname{Var} K_5 = 841/900 \approx 0.934$ for $\theta = 2$. The expected number of alleles seen exactly j times, computed from the table, is $\mathbb{E}A_j = 1/j$ for every $j \le n$ when $\theta = 1$: the mean number of j cycles of a random permutation. At $\theta = 2$ and $n = 5$ it is $(\frac{5}{3}, \frac{2}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{15})$ for $j = 1, \dots, 5$, in agreement with Proposition 6.17.

The table makes the role of θ concrete. At $\theta = 1/2$ the most likely outcome for five genes is a single allele, with probability 0.41; at $\theta = 2$ the single allele has probability $1/15$ and the most likely shapes are $3+1+1$ and $2+1+1+1$. Small θ means a population dominated by a few old alleles, large θ a population in which mutation keeps supplying new ones.

PROPOSITION 6.17 (Moments of the abundance counts). *Under* (6.10), *for* $1 \le j \le n$,

$$\mathbb{E}A_j = \frac{\theta}{j}\cdot\frac{n!}{(n-j)!}\cdot\frac{\Gamma(\theta+n-j)}{\Gamma(\theta+n)}, \qquad \mathbb{E}A_j \;\approx\; \frac{\theta x^j}{j}, \quad x = \frac{n}{n+\theta}, \tag{6.12}$$

the approximation being Fisher's logarithmic series with $\alpha = \theta$. *It is an approximation and not a limit, since* x *depends on* n; *at the values fitted to the butterflies below,* $n = 11068$ *and* $\theta = 141.75$, *the two sides of the second display differ by at most* 0.013 *over* $j \le 24$.

Proof. Write $A_j = \sum_B \mathbf{1}\{B \text{ is a block of } \Pi_n\}$, the sum running over the $\binom{n}{j}$ subsets B of $\{1, \dots, n\}$ with $|B| = j$. By exchangeability each term has the same expectation, so $\mathbb{E}A_j = \binom{n}{j}\mathbb{P}(\{1, \dots, j\} \text{ is a block})$. That event is the union over all partitions of

TABLE 6.1. All allelic partitions of $n = 4$ and $n = 5$ genes. The column $c(a)$ counts permutations of the given cycle type; the probability is $c(a)\theta^k/\theta_{(n)}$.

Shape	k	$c(a)$	$\theta = 1/2$	$\theta = 1$	$\theta = 2$
$n = 4$					
4	1	6	$16/35$	$1/4$	$1/10$
$3+1$	2	8	$32/105$	$1/3$	$4/15$
$2+2$	2	3	$4/35$	$1/8$	$1/10$
$2+1+1$	3	6	$4/35$	$1/4$	$2/5$
$1+1+1+1$	4	1	$1/105$	$1/24$	$2/15$
$n = 5$					
5	1	24	$128/315$	$1/5$	$1/15$
$4+1$	2	30	$16/63$	$1/4$	$1/6$
$3+2$	2	20	$32/189$	$1/6$	$1/9$
$3+1+1$	3	20	$16/189$	$1/6$	$2/9$
$2+2+1$	3	15	$4/63$	$1/8$	$1/6$
$2+1+1+1$	4	10	$4/189$	$1/12$	$2/9$
$1+1+1+1+1$	5	1	$1/945$	$1/120$	$2/45$

$\{j+1,\dots,n\}$, with blocks of sizes $m_1,\dots,m_r$ say, of the partition $\{\{1,\dots,j\},\dots\}$, and by (6.9) each has probability

$$\frac{\theta^{1+r}(j-1)!\prod_{i=1}^r(m_i-1)!}{\theta_{(n)}} = \frac{\theta\,(j-1)!\,\theta_{(n-j)}}{\theta_{(n)}} \times \frac{\theta^r\prod_{i=1}^r(m_i-1)!}{\theta_{(n-j)}}.$$

The second factor is (6.9) for a sample of size $n-j$, which sums to one over all partitions of $n-j$ indices. Hence $\mathbb{E}A_j = \binom{n}{j}(j-1)!\,\theta\,\theta_{(n-j)}/\theta_{(n)}$, and since $\theta_{(n-j)}/\theta_{(n)} = \Gamma(\theta+n-j)/\Gamma(\theta+n)$ and $\binom{n}{j}(j-1)! = n!/\{j(n-j)!\}$ this is the first display (with $\theta_{(0)} = 1$ when $j = n$). For the approximation, write

$$\frac{n!}{(n-j)!}\cdot\frac{\Gamma(\theta+n-j)}{\Gamma(\theta+n)} = \prod_{i=0}^{j-1}\frac{n-i}{\theta+n-1-i},$$

a product of j factors. Cross multiplying shows that the ith factor equals $x = n/(n+\theta)$ times $1+(n-i\theta)/\{n(\theta+n-1-i)\}$, a relative error of order $(1+i)/n$ when $\theta \le n$; the product is therefore $x^j\{1+O(j^2/n)\}$, which is close to x^j for $j \ll \sqrt{n}$. □

6.3.4 *Likelihood, sufficiency and information*

The factorisation of (6.10) into a term free of θ and the term $\theta^k/\theta_{(n)}$ has an immediate consequence.

COROLLARY 6.18 (Sufficiency). *K_n, the number of jumps of the empirical distribution function, is sufficient for θ; it is also minimal sufficient and complete, the model being a one parameter exponential family in K_n with natural parameter* $\log\theta$.

Proof. By (6.9) the probability of any particular partition is $h(n_1,\dots,n_k)\exp\{k\log\theta-\log\theta_{(n)}\}$ with $h=\prod_j(n_j-1)!$ free of θ, so the factorisation theorem gives sufficiency of K_n. This is a one parameter exponential family with natural parameter $\eta=\log\theta$ ranging over the open set $\mathbb{R}$ and canonical statistic K_n, which for $n\geq 2$ takes the values $1,\dots,n$ with positive probability (for $n=1$ there is nothing to estimate), so completeness and minimality are the standard facts for full rank exponential families. □

Thus all the information about θ carried by F_n resides in the number of its jumps, not in their locations or their heights. This sharpens the identification of Section 6.3.1: the variance functional (6.8) identifies θ from the population moments, but it is not efficient at fixed n, and it requires replication to be estimable, whereas K_n is computable from a single sample.

PROPOSITION 6.19 (Law of K_n, maximum likelihood and information). *Let ψ denote the digamma function. Under* (6.10),

$$\mathbb{E}K_n=\sum_{i=0}^{n-1}\frac{\theta}{\theta+i}=\theta\big\{\psi(\theta+n)-\psi(\theta)\big\}=\theta\log n-\theta\psi(\theta)+O(1/n), \tag{6.13}$$

$$\operatorname{Var}K_n=\mathbb{E}K_n-\theta^2\big\{\psi'(\theta)-\psi'(\theta+n)\big\}=\theta\log n+O(1), \tag{6.14}$$

and $(K_n-\theta\log n)/\sqrt{\theta\log n}\Rightarrow N(0,1)$. The maximum likelihood estimator solves $K_n=\mathbb{E}_{\hat\theta}K_n$, the expected Fisher information is

$$I_n(\theta)=\sum_{i=0}^{n-1}\frac{i}{\theta(\theta+i)^2}=\frac{\log n}{\theta}+O(1),$$

and $\sqrt{\log n}\,(\hat\theta_n-\theta)\Rightarrow N(0,\theta)$. The simple estimator $K_n/\log n$ has the same asymptotic variance but a bias

$$\mathbb{E}\left[\frac{K_n}{\log n}\right]-\theta=-\frac{\theta\psi(\theta)}{\log n}+O\Big(\frac{1}{n\log n}\Big)$$

which is of order $1/\log n$ against a standard deviation of order $(\log n)^{-1/2}$: of smaller order, yet larger than the standard deviation at realistic n when θ is large.

Proof. By (6.4) the indicator ξ_{i+1} that index $i+1$ opens a new block is Bernoulli with success probability $\theta/(\theta+i)$ conditionally on the past; since that probability is deterministic, the ξ_i are independent. Summing means and variances gives the two displays, and the Lindeberg condition holds because the summands are bounded

and the variance diverges. The log likelihood is $\ell(\theta) = K_n \log\theta - \log\theta_{(n)}$ up to a constant, so $\ell'(\theta) = K_n/\theta - \sum_{i<n}(\theta+i)^{-1}$, whose root is the stated equation, and

$$I_n(\theta) = \frac{\mathbb{E}K_n}{\theta^2} - \sum_{i=0}^{n-1}\frac{1}{(\theta+i)^2} = \sum_{i=0}^{n-1}\left\{\frac{1}{\theta(\theta+i)} - \frac{1}{(\theta+i)^2}\right\} = \sum_{i=0}^{n-1}\frac{i}{\theta(\theta+i)^2}.$$

The bias statement is (6.13) divided by $\log n$. □

Two consequences deserve emphasis. First, the information grows like $\log n$, not like n: the Ewens model is a hard estimation problem, and doubling the sample buys almost nothing. Second, the popular estimator $K_n/\log n$ is asymptotically efficient yet badly biased at any realistic n, by a factor $\psi(\theta)/\log n$ that is substantial for large θ. Table 6.4 quantifies both.

A third consequence is a convenient identity. Comparing the displays for $\operatorname{Var} K_n$ and $I_n(\theta)$,

$$I_n(\theta) = \frac{\operatorname{Var} K_n}{\theta^2}, \tag{6.15}$$

which is the general rule for an exponential family with natural parameter $\log\theta$: the information about $\log\theta$ is the variance of the canonical statistic, and the change of variable from $\log\theta$ to θ divides it by θ^2. In Example 6.16, $I_4(2) = (641/900)/4 = 641/3600 \approx 0.178$, so four genes carry a standard error for θ of about 2.4 when $\theta = 2$: a sample of four can tell $\theta = 0.5$ from $\theta = 10$ but very little else.

The independence of the new allele indicators also gives the whole law of K_n.

COROLLARY 6.20 (Law of the number of alleles). *Let $s(n,k)$ be the signless Stirling numbers of the first kind, defined by $x(x+1)\cdots(x+n-1) = \sum_{k=1}^{n} s(n,k)x^k$. Under* (6.10),

$$\mathbb{E}z^{K_n} = \frac{(z\theta)_{(n)}}{\theta_{(n)}}, \qquad \mathbb{P}(K_n = k) = \frac{s(n,k)\,\theta^k}{\theta_{(n)}}, \quad k = 1,\dots,n. \tag{6.16}$$

Proof. $K_n = \sum_{i=0}^{n-1}\xi_{i+1}$ with independent $\xi_{i+1} \sim \text{Bernoulli}(\theta/(\theta+i))$, so $\mathbb{E}z^{K_n} = \prod_{i=0}^{n-1}(z\theta+i)/(\theta+i)$, which is the first formula. Expanding the numerator by the definition of $s(n,k)$ with $x = z\theta$ and reading off the coefficient of z^k gives the second. □

For $n = 5$ the Stirling numbers are $s(5,k) = 24, 50, 35, 10, 1$, which sum to $5! = 120$. At $\theta = 1$ the law of K_5 is therefore $(24,50,35,10,1)/120$, and at $\theta = 2$, where $\theta_{(5)} = 720$, it is $(48,200,280,160,32)/720$, with mean $2088/720 = 2.9$ as in Example 6.16. Summing the rows of Table 6.1 with the same k gives the same numbers, because $s(n,k)$ is the number of permutations of n objects with k cycles.

6.4 Empirical illustration

All computations below use the accompanying script, which simulates the predictive rule (6.4) directly, without assuming (6.10), so that Table 6.2 is a genuine check of Theorem 6.15 rather than a consistency check of the code.

6.4.1 *The sampling formula itself*

Table 6.2 compares the exact probabilities (6.10) with the empirical frequencies of 200,000 realisations of the urn (6.4) at $n = 8$ and $\theta = 2$. All 22 partition shapes were observed; the twelve most probable are shown, the shape $2+2+2+1+1$ being omitted although it ties with the last three rows at 0.0370. The largest discrepancies, 0.0014 each, are at the shapes $3+2+2+1$ and $6+1+1$, against a Monte Carlo standard error of 0.0006; with the ten omitted shapes pooled into one cell, the tabulated frequencies give a chi squared goodness of fit statistic of about 23 on 12 degrees of freedom (between 21 and 26 allowing for rounding), a p value near 0.03, so the agreement is within what sampling error allows, though toward the upper end for a single run.

TABLE 6.2. Partition shapes at $n = 8$, $\theta = 2$. Simulation of the predictive rule (6.4) against the Ewens sampling formula (6.10).

Shape	Monte Carlo	Ewens formula
$4+2+1+1$	0.1115	0.1111
$3+2+1+1+1$	0.0985	0.0988
$5+2+1$	0.0896	0.0889
$3+2+2+1$	0.0755	0.0741
$4+3+1$	0.0729	0.0741
$6+1+1$	0.0727	0.0741
$7+1$	0.0634	0.0635
$5+1+1+1$	0.0592	0.0593
$3+3+1+1$	0.0486	0.0494
$2+2+1+1+1+1$	0.0378	0.0370
$4+1+1+1+1$	0.0375	0.0370
$6+2$	0.0368	0.0370

6.4.2 *The de Finetti limit and its variance*

Figure 6.1 shows six realisations of F_{500} for $\theta \in \{1, 5, 25\}$ with H uniform on $[0,1]$. These are draws from the measure μ constructed by the Helly and Bray argument. Small θ gives coarse staircases far from the diagonal; large θ gives many small jumps hugging H.

Figure 6.2 plots the Monte Carlo variance of $F_{500}(x)$ against x together with the exact curve $H(x)(1-H(x))(\theta+n)/\{n(1+\theta)\}$ of Proposition 6.11. The agreement is exact to simulation error at every x and every θ, confirming the diagonal of the covariance kernel (6.5).

Table 6.3 inverts the variance. The column headed $\hat{\theta}$ uses the exact inversion (6.8); the column headed asymptotic uses the limiting form (6.6), which ignores the factor $(\theta+n)/n$. At $\theta = 10$ and $n = 50$ the exact inversion returns 9.81 while the asymptotic form returns 8.04, a twenty percent error that disappears only slowly.

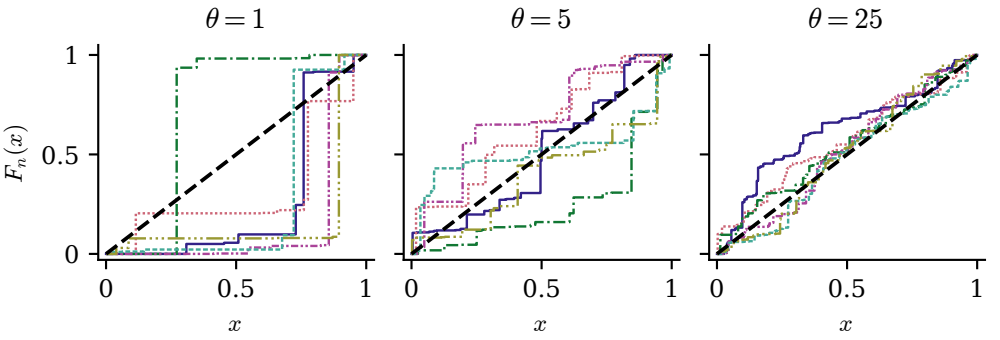


FIGURE 6.1. Realisations of the empirical distribution function F_{500} for $\theta \in \{1, 5, 25\}$ under the Ewens predictive rule with H uniform. The dashed line is H. The de Finetti measure μ is the law of the limit of these staircases.

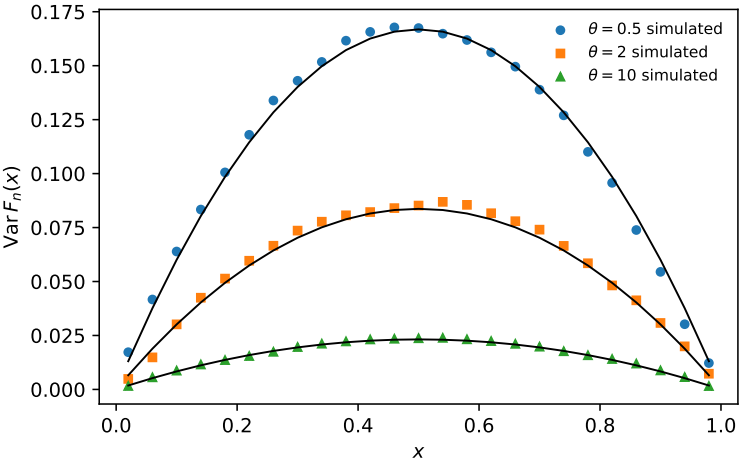


FIGURE 6.2. Variance of $F_n(x)$ at $n = 500$ from 1500 replicates (markers) against the exact formula (6.7) (solid lines).

TABLE 6.3. Identifying θ from $\operatorname{Var} F_n(1/2)$, from 4000 replicates with H uniform. Exact variance from (6.7), exact inversion from (6.8), asymptotic inversion from (6.6).

θ	n	$\operatorname{Var} F_n(1/2)$	exact variance	$\hat\theta$ exact	$\hat\theta$ asymptotic
0.5	50	0.16953	0.16833	0.489	0.475
0.5	200	0.16878	0.16708	0.485	0.481
0.5	5000	0.16564	0.16668	0.509	0.509
2.0	50	0.08607	0.08667	2.022	1.905
2.0	200	0.08257	0.08417	2.059	2.028
2.0	5000	0.08248	0.08337	2.032	2.031
10.0	50	0.02767	0.02727	9.809	8.036
10.0	200	0.02418	0.02386	9.849	9.340
10.0	5000	0.02268	0.02277	10.045	10.023

6.4.3 *The number of types*

Table 6.4 and Figure 6.3 illustrate Proposition 6.19. The predicted bias of $K_n/\log n$, namely $-\theta\psi(\theta)/\log n$, is confirmed to within Monte Carlo error: at $n = 10^4$ the predicted values of $\mathbb{E}[K_n/\log n]$ are 0.607, 1.908 and 7.556 for $\theta = 0.5, 2, 10$, against simulated 0.607, 1.905 and 7.572. The maximum likelihood estimator is centred at the truth for $\theta \geq 2$ and for $n = 10^4$, but at $\theta = 0.5$ it is 15 and 6 percent too high at $n = 100$ and $n = 1000$, and its Monte Carlo standard deviation tracks $I_n(\theta)^{-1/2}$ closely. Note how little the standard error improves between $n = 10^3$ and $n = 10^4$: this is the $\log n$ information at work.

TABLE 6.4. The number of types, 4000 replicates, maximum likelihood over the first 800. The last column is the inverse root expected Fisher information.

θ	n	mean K_n	$\mathbb{E}K_n$	mean $K_n/\log n$	mean $\hat\theta$	sd $\hat\theta$	$I_n^{-1/2}$
0.5	100	3.286	3.284	0.714	0.574	0.338	0.349
0.5	1000	4.524	4.436	0.655	0.530	0.286	0.279
0.5	10000	5.591	5.587	0.607	0.505	0.243	0.240
2.0	100	8.393	8.395	1.822	2.037	0.870	0.827
2.0	1000	12.987	12.973	1.880	2.035	0.642	0.620
2.0	10000	17.550	17.575	1.905	2.012	0.511	0.516
10.0	100	24.459	24.442	5.311	10.198	2.789	2.596
10.0	1000	46.390	46.655	6.716	9.892	1.645	1.661
10.0	10000	69.743	69.595	7.572	9.992	1.252	1.301

6.4.4 *Corbet's Malayan butterflies*

We close with the data that prompted Fisher's logarithmic series, the abundances of Malayan butterflies collected by Corbet and analysed in Fisher, Corbet and

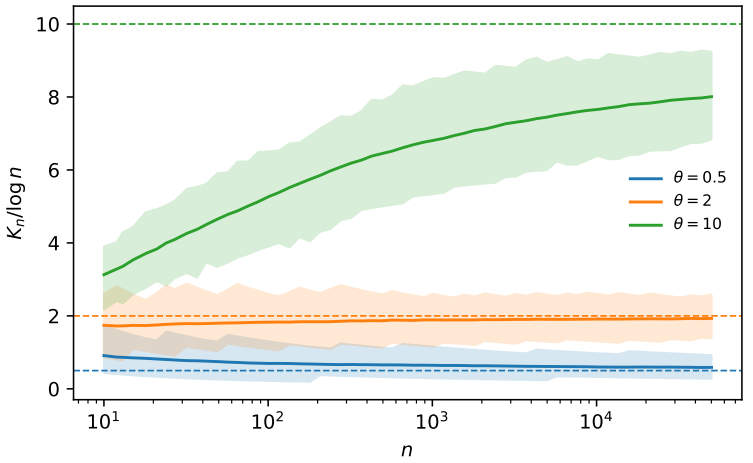


FIGURE 6.3. $K_n / \log n$ against n on a logarithmic scale, mean and central ninety percent band over 200 paths. Dashed lines mark the true θ. Convergence is at rate $1/\log n$ and is slow for large θ.

Williams (1943). The collection held 620 species, but the published table gives the abundance classes $j = 1, \dots, 24$ only, because Corbet took every individual seen of a species only until 24 specimens had been taken (Fisher, Corbet and Williams, 1943, p. 43), so a count above 24 is only a lower bound for the number seen and the classes above 24 are censored by the collecting, not merely grouped. They comprise 501 species and 3306 individuals; the other 119 species are known only to have more than 24 individuals, and the total sample size n is not reported. The data are therefore censored, and the likelihood is the probability under (6.1) of the observed $a_1, \dots, a_{24}$ together with 119 further blocks of size at least 25 whose sizes sum to $n - 3306$. Summing (6.1) over those blocks gives

$$L(\theta, n) \;\propto\; \frac{\theta^{620}\,\Gamma(\theta)}{\Gamma(\theta + n)}\; n!\,\big[u^{\,n-3306}\big]\Big(\sum_{j \ge 25} \frac{u^j}{j}\Big)^{119},$$

where $[u^m]$ extracts a coefficient. For fixed n the likelihood depends on θ only through $K_n = 620$, so Proposition 6.19 still gives $\hat\theta$ given n, and n is then profiled out, the coefficient being computed by exponential tilting and the fast Fourier transform. The result is

$$\hat\theta = 141.75, \qquad \hat n = 11068, \qquad \text{standard error } 7.0,$$

the standard error coming from the curvature of the profile likelihood of θ. Fisher's own form of the model, in which the A_j are independent Poisson variables with means $\theta x^j / j$ and n is left random, gives $\hat\theta = 141.64$ on the same data. By Proposition 6.17 θ is also Fisher's α, since the logarithmic series approximation to $\mathbb{E}A_j$ agrees with the exact Ewens mean to within 0.013 at every $j \le 24$ for these values of n and θ.

EXAMPLE 6.21 (Estimating θ for the butterflies). The estimate $\hat\theta$ above is the estimator proposed by Ewens (1972): solve $K_n = \mathbb{E}_\theta K_n$, which is at once the method of moments equation for the sufficient statistic and, by Proposition 6.19, the likelihood equation. At the profiled sample size $n = 11068$, with $K_n = 620$, the score is $s(\theta) = 620/\theta - \{\psi(\theta + 11068) - \psi(\theta)\}$ and the observed information is $J(\theta) = 620/\theta^2 - \{\psi'(\theta) - \psi'(\theta + 11068)\}$. Newton's method $\theta \leftarrow \theta + s(\theta)/J(\theta)$, started at the naive value $K_n / \log n = 620/9.312 = 66.58$, gives

$$66.58 \;\to\; 100.11 \;\to\; 128.50 \;\to\; 140.36 \;\to\; 141.73 \;\to\; 141.747 \;\to\; 141.747,$$

converging in six steps. The starting value is off by a factor of more than two, which is the bias $-\theta\psi(\theta)/\log n$ of Proposition 6.19 at work: $\psi(142) \approx 4.95$, so $K_n / \log n$ estimates roughly $\theta\{1 - 4.95/9.31\} \approx 0.47\,\theta$.

At $\hat\theta$ and $n = 11068$, $\operatorname{Var} K_n = 479.5$ and so, by (6.15), $I_n(\hat\theta) = 479.5/141.7^2 = 0.0239$ and the standard error given n is $0.0239^{-1/2} = 6.47$. Not knowing n widens it to the 7.0 of the profile likelihood. Two ninety five percent intervals agree closely: the Wald interval $141.75 \pm 1.96 \times 7.0 = (128.0, 155.5)$, and the likelihood ratio interval, the set where the profile log likelihood is within 1.92 of its maximum, $(128.6, 156.0)$. The same profile puts n between 9890 and 12436. The naive asymptotic standard error $\sqrt{\theta/\log n} = 3.90$ is well below the correct value, because $\log n$ is a poor proxy for $\sum_i i/(\theta + i)^2$ when θ is a substantial fraction of n.

A different moment estimator uses the sample homozygosity, the proportion of pairs of individuals of the same species,

$$\hat F = \sum_j a_j \frac{j(j-1)}{n(n-1)}, \qquad \mathbb{E}\hat F = \frac{1}{1+\theta}, \qquad \hat\theta_F = \frac{1}{\hat F} - 1,$$

the expectation following from $\mathbb{P}(X_i = X_\ell) = 1/(1+\theta)$ for every pair. In 2000 simulated samples of $n = 11068$ from the urn at $\theta = 142$, with n known, the maximum likelihood estimator had mean 141.9 and standard deviation 6.6, matching $I_n^{-1/2} = 6.47$, while $\hat\theta_F$ had mean 143.6 and standard deviation 16.4. The homozygosity estimator, which ignores the sufficient statistic, has about six times the variance of $\hat\theta$. Worse, as n grows $\hat F$ converges not to $1/(1+\theta)$ but to the random population homozygosity $\sum_\ell p_\ell^2 = \mathbb{P}(X_1 = X_2 \mid \widetilde P)$, so $\hat\theta_F$ is not even consistent: no number of genes drawn from one population removes the variability of that population's own frequencies. On the butterfly table $\hat F$ cannot be computed at all: the homozygosity is dominated by the common species, and the counts of the 119 species with more than 24 individuals were not recorded. The censoring removes $\hat F$ but leaves $K_n = 620$ intact, a further reason to base inference on K_n.

Figure 6.4 and Table 6.5 show that the fit is fair but misses in a specific direction. The model predicts 140.0 singleton species and 69.1 doubletons; the data show 118 and 74. The rarest class is over predicted by about a fifth, some two Poisson standard deviations, while the moderately abundant classes $j = 5, \dots, 12$ are under predicted, 151 species against 131.0 expected. The expected counts over $j \le 24$

sum to 495.4 against the observed 501, leaving 124.6 expected above 24 against the observed 119, and the Pearson discrepancy over the 24 tabulated classes is 30.4. The misfit is far milder than a fit that ignores the censoring suggests: treating the 24 classes as the whole sample, with $n = 3306$ and $K_n = 501$, gives $\hat\theta = 163.99$, 156.3 predicted singletons and a discrepancy of 100.5. What remains is the pattern that ascertainment would produce. A model that fits the tie structure of a simple random sample cannot simultaneously fit a collection in which the rarest species are systematically less likely to be caught, and the observed sample may be a weighted, rather than a simple, draw from the Kingman paintbox. Section 6.5 formalises that idea, and Section 6.6 enlarges the predictive rule; Section 6.6.3 returns to these data and reports which of the two mechanisms the likelihood actually prefers.

TABLE 6.5. Corbet's Malayan butterflies. Observed number of species with j individuals, the exact Ewens mean (6.12) at $\hat\theta = 141.75$ and $\hat n = 11068$, and Fisher's logarithmic series approximation. A further 119 species have more than 24 individuals.

j	observed	Ewens	log series	j	observed	Ewens	log series
1	118	139.97	139.95	13	6	9.25	9.24
2	74	69.10	69.09	14	12	8.48	8.47
3	44	45.49	45.48	15	6	7.82	7.81
4	24	33.69	33.68	16	9	7.24	7.23
5	29	26.61	26.60	17	9	6.73	6.72
6	22	21.90	21.89	18	6	6.27	6.26
7	20	18.53	18.52	19	10	5.87	5.86
8	19	16.01	16.00	20	10	5.50	5.49
9	20	14.06	14.05	21	11	5.18	5.17
10	15	12.49	12.48	22	5	4.88	4.87
11	12	11.21	11.20	23	3	4.61	4.60
12	14	10.15	10.14	24	3	4.36	4.35

6.5 Ascertainment and the weighted paintbox

Fisher (1934) named the problem: a sample is rarely a simple random sample, because the probability that an item enters the record may depend on the very quantity being measured. Rao (1965) gave the general device, the weighted distribution. We apply it to Kingman's paintbox and ask what survives.

6.5.1 The construction

Let $P = (p_1 \ge p_2 \ge \cdots)$ be the ranked frequencies of the paintbox, so that $\widetilde P = \sum_{\ell \ge 1} p_\ell \delta_{\xi_\ell}$ with ξ_ℓ independent draws from the diffuse H; under Proposition 6.12 the law of P is Poisson Dirichlet with parameter θ. Let $w : (0,1] \to [0,\infty)$ be measurable with $0 < \sum_\ell w(p_\ell) p_\ell < \infty$ almost surely.

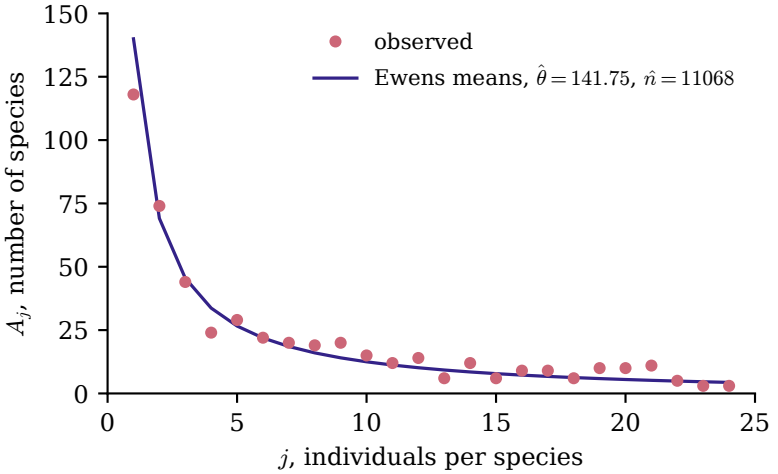


FIGURE 6.4. Corbet's Malayan butterflies: observed abundance counts against the fitted Ewens means. The rarest class is over predicted and the moderately abundant classes are under predicted, the pattern ascertainment would produce.

DEFINITION 6.22 (Rao weighted paintbox). The ascertained sample is a sequence $(X_i^w)_{i\geq 1}$ which, given P and (ξ_ℓ), is independent and identically distributed with common distribution

$$\widetilde{P}^w \;=\; \frac{\sum_{\ell\geq 1} w(p_\ell)\, p_\ell\, \delta_{\xi_\ell}}{\sum_{\ell\geq 1} w(p_\ell)\, p_\ell}. \tag{6.17}$$

The interpretation is that a species of relative abundance p contributes to the record in proportion to $w(p)p$ rather than to p: the collector's effort per individual depends on how common that individual's species is. Taking w increasing represents a collector who overlooks rarities; w decreasing represents a specialist hunting them.

PROPOSITION 6.23 (The de Finetti architecture is invariant). *For every admissible w the sequence (X_i^w) is exchangeable, and Theorem 6.7 applies to it verbatim: the empirical distribution functions F_n^w converge almost surely and uniformly to a random distribution function, whose law is the de Finetti measure, and that limit is the distribution function of $\widetilde{P}^w$. Only the mixing measure changes; Sections 6.2.2 and 6.2.3 are untouched.*

Proof. Conditionally on $(P, (\xi_\ell))$ the sequence is independent and identically distributed, hence unconditionally exchangeable. Theorem 6.7 was proved for an arbitrary exchangeable sequence on $[0,1]$, so it applies, and the uniqueness clause identifies the de Finetti measure with the law of the conditional distribution, which is that of $\widetilde{P}^w$. □

This is the point of interest for the present chapter. Helly selection and the Helly and Bray lemma never used the Ewens hypothesis; they used only exchangeability. Weighting can move the de Finetti measure off the Dirichlet process and so break

the sampling formula, but it does not touch the representation that produced the sampling formula. What replaces (6.9) is the weighted paintbox partition function.

PROPOSITION 6.24 (Weighted exchangeable partition probability function). *The partition of $\{1,\dots,n\}$ induced by ties in (X_i^w) is exchangeable with*

$$p^w(n_1,\dots,n_k) = \mathbb{E}\left[\Big(\sum_\ell w(p_\ell)p_\ell\Big)^{-n} \sum_{\ell_1,\dots,\ell_k \text{ distinct}} \prod_{j=1}^{k} \big(w(p_{\ell_j})p_{\ell_j}\big)^{n_j}\right], \tag{6.18}$$

which reduces to (6.9) *when w is constant.*

Proof. Condition on P and (ξ_ℓ), use conditional independence and the fact that H is diffuse, so that two observations are tied exactly when they land on the same ξ_ℓ, and sum over the assignment of blocks to distinct atoms. Constant w cancels from numerator and denominator, leaving the Poisson Dirichlet paintbox, which by Theorem 6.15 is the Ewens sampling formula. □

6.5.2 *Effort that ignores abundance is invisible*

Before fitting anything it is worth ruling out the simplest explanation of a misfit, namely that the collector's effort varied over the collection period.

PROPOSITION 6.25 (Uniform thinning). *Let Π_N follow the Ewens sampling formula with parameter θ and suppose each of the N indices is retained independently with probability $\pi \in (0,1]$, the retention being independent of the types. Let n be the number retained. Then, conditionally on n, the partition of the retained indices follows the Ewens sampling formula with the same θ and sample size n. The same holds if π varies across indices, provided it does not depend on the types.*

Proof. The retained index set R is independent of $(X_1,\dots,X_N)$, so conditionally on $|R| = n$ the restricted sequence $(X_i)_{i\in R}$ is exchangeable with the same finite dimensional laws as $(X_1,\dots,X_n)$. Its tie partition is therefore Π_n, and Theorem 6.15 applies. This is Kingman's consistency, or deletion, property: the Ewens family is a partition structure. □

Proposition 6.25 says that a tired collector, one whose effort declines over time but who is equally likely to miss a common and a rare specimen, produces data indistinguishable from a smaller simple random sample. Any explanation of the butterfly misfit must therefore involve dependence on abundance, which is exactly what (6.17) supplies. A second negative result explains when the weighting must break Assumption 6.2.

PROPOSITION 6.26 (Only a Dirichlet directing measure admits the affine predictive). *Among species sampling sequences, the affine predictive rule* (6.2) *with coefficients free of the data holds if and only if the directing measure is a Dirichlet process. Consequently, if w is not P almost surely constant and $\widetilde{P}^w$ is not a Dirichlet process, then (X_i^w) violates Assumption 6.2.*

This is Johnson's sufficientness postulate in the form given by Zabell (1982), whose argument requires the diffuse base measure assumed here; the forward direction is Proposition 6.8 above. The stated consequence needs one further step, that a nonconstant weighting actually moves $\widetilde{P}^{w}$ off the Dirichlet process, which we do not prove in general. For the power weights used below it is visible in the simulations: at $\theta = 142$ and $\beta = 0.5$ the weighted paintbox produces 308 species with at most 24 individuals at $n = 11068$, whereas the Ewens law with the same count (θ near 85.5) gives about 85 singletons and 42 doubletons against the 89.5 and 43.8 of Table 6.6. It tells us where to look for an enlargement: if we want tractability back, we must relax the predictive rule itself rather than tinker with the weights, and Section 6.6 does so.

6.5.3 What power weighting can and cannot do

Take $w(p) = p^{\beta}$, so that the recorded frequencies are $q_\ell \propto p_\ell^{1+\beta}$; $\beta = 0$ is simple random sampling and $\beta > 0$ favours the abundant species. Table 6.6 and Figure 6.5 report the abundance spectrum at $\theta = 142$ and $n = 11068$, the values fitted to the butterflies, averaged over 60 realisations of the paintbox.

TABLE 6.6. Power weighted paintbox, $\theta = 142$, $n = 11068$, $w(p) = p^{\beta}$, 60 replicates. Increasing β suppresses the singletons and the richness in nearly the same proportion. The column K_n counts the species represented by at most 24 individuals, the quantity tabulated for the butterflies, so at $\beta = 0$ it is 497 and not $\mathbb{E}K_n = 620$.

β	A_1	A_2	A_5	A_{10}	A_{20}	K_n
0.00	139.3	70.9	26.2	12.9	5.8	496.9
0.15	121.1	60.8	23.8	10.5	4.6	427.5
0.30	108.1	52.6	20.1	9.1	4.0	368.7
0.50	89.5	43.8	16.7	7.1	3.0	308.0
observed	118	74	29	15	10	501

The message is negative and useful. Raising β does pull the singleton count down from 139 to the observed 118 near $\beta = 0.2$, but it drags the number of species seen at most 24 times down with it, from 497 to 308 against an observed 501, and it leaves the moderately abundant classes too thin: A_{20} falls from six to three where the data show ten. A single tilt of the one parameter paintbox trades rare species against total richness along a one dimensional curve, and the butterflies do not lie on it. The data want more species and a flatter abundance profile at the same time, which is a statement about the paintbox itself, not about how it is sampled.

6.6 The two parameter family and a reanalysis

A weighted paintbox has no closed form partition function and, by Proposition 6.26, no affine predictive rule. We take the other route: rather than reweight a

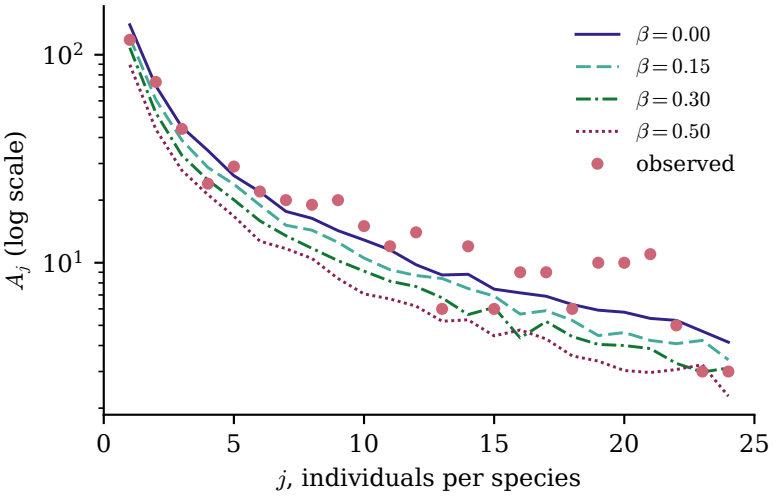


FIGURE 6.5. Abundance spectra under the power weighted paintbox at $\theta = 142$ and $n = 11068$, on a logarithmic vertical scale, against the observed butterfly counts.

fixed paintbox, enlarge the predictive rule itself, in the smallest way that keeps the empirical distribution function as the summary of the past. One extra parameter buys back tractability.

6.6.1 *The predictive rule*

Assumption 6.2 allowed the predictive law to load only two measures, the base H and the empirical distribution function F_n, with coefficients free of the data. There is a third measure naturally available from F_n, namely the empirical distribution function of its distinct jump locations,

$$G_n(x) = \frac{1}{K_n} \sum_{\text{distinct values } v \le x} 1,$$

which puts equal mass on each of the K_n observed types regardless of how often each was seen. Loading G_n as well, and allowing the coefficients to depend on K_n, gives the following relaxation.

ASSUMPTION 6.27 (Two parameter predictive rule). With H diffuse and $X_1 \sim H$, for every $n \ge 1$,

$$\mathbb{P}\big(X_{n+1} \le x \mid X_1, \dots, X_n\big) = \frac{\theta + \alpha K_n}{\theta + n} H(x) \; + \; \frac{n}{\theta + n} F_n(x) \; - \; \frac{\alpha K_n}{\theta + n} G_n(x), \quad (6.19)$$

where either $0 \le \alpha < 1$ and $\theta > -\alpha$, or $\alpha < 0$ and $\theta = m|\alpha|$ for an integer $m \ge 1$.

Equivalently, in urn form, index $n+1$ opens a new block with probability $(\theta + \alpha K_n)/(\theta + n)$ and joins an existing block of current size n_j with probability $(n_j - \alpha)/(\theta + n)$. Setting $\alpha = 0$ recovers (6.4). The coefficients still depend on

the data only through the empirical distribution function, now through its jump count as well as its heights, which is precisely the relaxation of the sufficientness postulate studied by Zabell (1997). We do not reprove the characterisation; we verify that (6.19) is exchangeable by exhibiting its symmetric partition function.

THEOREM 6.28 (Pitman's sampling formula). *Under Assumption 6.27, the tie partition of* $\{1,\dots,n\}$ *satisfies*

$$\mathbb{P}\big(\Pi_n = \{B_1,\dots,B_k\}\big) = \frac{\prod_{i=1}^{k-1}(\theta+i\alpha)}{(\theta+1)_{n-1}} \prod_{j=1}^{k} (1-\alpha)_{n_j-1}, \tag{6.20}$$

where $(x)_m = x(x+1)\cdots(x+m-1)$ *denotes the rising factorial, and consequently*

$$\mathbb{P}(A_1=a_1,\dots,A_n=a_n) = n!\,\frac{\prod_{i=1}^{k-1}(\theta+i\alpha)}{(\theta+1)_{n-1}} \prod_{j=1}^{n} \frac{1}{a_j!}\left(\frac{(1-\alpha)_{j-1}}{j!}\right)^{a_j}, \qquad k=\sum_j a_j. \tag{6.21}$$

Both reduce to (6.9) *and* (6.10) *at* $\alpha = 0$.

Proof. Identical in structure to Proposition 6.14 and Theorem 6.15. Read the indices in order. The denominator at step $i+1$ is $\theta+i$, giving $\prod_{i=1}^{n-1}(\theta+i) = (\theta+1)_{n-1}$. The ith new block, for $i = 1,\dots,k-1$, is opened when i blocks are already present and contributes $\theta+i\alpha$; the first block contributes nothing since index one opens it with probability one. A block that ends with n_j members contributes $(1-\alpha)(2-\alpha)\cdots(n_j-1-\alpha) = (1-\alpha)_{n_j-1}$ as its second through last members arrive. This gives (6.20), which is symmetric in the block sizes, so the partition is exchangeable. Multiplying by the number $n!/\prod_j (j!)^{a_j} a_j!$ of set partitions with the given shape gives (6.21). At $\alpha = 0$, $\prod_{i=1}^{k-1}\theta = \theta^{k-1}$, $(\theta+1)_{n-1} = \theta_{(n)}/\theta$ and $(1)_{n_j-1} = (n_j-1)!$. □

EXAMPLE 6.29 (Four genes under the two parameter rule). Take $n = 4$, so the denominator in (6.20) is $(\theta+1)_3 = (\theta+1)(\theta+2)(\theta+3)$. Table 6.7 gives the five shape probabilities for three parameter choices. For the single block at $\alpha = 1/2$, $\theta = 1$, the product over new blocks is empty and the block contributes $(1/2)_3 = \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2} = 15/8$, so the probability is $(15/8)/24 = 5/64$. Positive α moves mass towards many alleles: $\mathbb{E}K_4 = 187/64 \approx 2.92$ against $25/12 \approx 2.08$ for the Ewens rule with the same θ. The choice $\alpha = -1$, $\theta = 2$ is the finite pool of Remark 6.32 with $m = 2$ types, the frequency of the first type being uniform on $[0,1]$: the factor $\theta + 2\alpha = 0$ kills every partition with three or more blocks, and the remaining three probabilities $2/5, 2/5, 1/5$ can be checked directly, since with $P \sim \mathrm{Beta}(1,1)$ a given sequence with c genes of the first type has probability $\int_0^1 p^c(1-p)^{4-c}\,dp = c!\,(4-c)!/5!$; for instance the shape 4 has probability $2\cdot 4!/5! = 2/5$.

PROPOSITION 6.30 (Exact abundance means). *Under* (6.21), *for* $1 \le j \le n$,

$$\mathbb{E}A_j = \binom{n}{j} \frac{(1-\alpha)_{j-1}\,(\theta+\alpha)_{n-j}}{(\theta+1)_{n-1}}, \tag{6.22}$$

TABLE 6.7. The two parameter partition law at $n = 4$. The column N is the number of set partitions with the given shape.

Shape	N	$\alpha = 1/2,\ \theta = 1$	$\alpha = 0,\ \theta = 1$	$\alpha = -1,\ \theta = 2$
4	1	$5/64$	$1/4$	$2/5$
$3+1$	4	$3/16$	$1/3$	$2/5$
$2+2$	3	$3/64$	$1/8$	$1/5$
$2+1+1$	6	$3/8$	$1/4$	0
$1+1+1+1$	1	$5/16$	$1/24$	0
$\mathbb{E}K_4$		$187/64$	$25/12$	$8/5$

which reduces to (6.12) *at* $\alpha = 0$.

Proof. By exchangeability $\mathbb{E}A_j = \binom{n}{j}\mathbb{P}(B)$ where B is the event that $\{1, \dots, j\}$ is a block of Π_n. Summing (6.20) over all partitions of the remaining $n - j$ indices into blocks of sizes $m_1, \dots, m_r$, the summand is

$$\frac{\prod_{i=1}^{r}(\theta + i\alpha)}{(\theta+1)_{n-1}}(1-\alpha)_{j-1}\prod_{i=1}^{r}(1-\alpha)_{m_i-1}$$

$$= \frac{(\theta+\alpha)(1-\alpha)_{j-1}(\theta+\alpha+1)_{n-j-1}}{(\theta+1)_{n-1}} \times p_{\alpha,\theta+\alpha}(m_1, \dots, m_r),$$

because $\prod_{i=1}^{r}(\theta + i\alpha) = (\theta+\alpha)\prod_{i=1}^{r-1}(\theta+\alpha+i\alpha)$. The second factor is the partition function (6.20) for the two parameter family with parameters $(\alpha, \theta+\alpha)$ on $n-j$ indices, and it sums to one over all such partitions. Since $(\theta+\alpha)(\theta+\alpha+1)_{n-j-1} = (\theta+\alpha)_{n-j}$, this gives (6.22) for $j < n$; for $j = n$ the event B is the single block partition, whose probability $(1-\alpha)_{n-1}/(\theta+1)_{n-1}$ gives (6.22) with $(\theta+\alpha)_0 = 1$. At $\alpha = 0$ the constant becomes $\theta(j-1)!\,\Gamma(\theta+n-j)/\Gamma(\theta+n)$, which is (6.12). □

6.6.2 *What the empirical distribution function identifies*

PROPOSITION 6.31 (Second order structure and what it identifies). *Under Assumption 6.27, write* $\rho = (1-\alpha)/(1+\theta)$. *Then* $\mathbb{P}(X_1 = X_2) = \rho$ *and, for every* $g \in L^2(H)$ *and every* n,

$$\operatorname{Var}\left(\int g\, d\widetilde{P}\right) = \rho\, \operatorname{Var}_H(g), \qquad \operatorname{Var}\left(\int g\, dP_n\right) = \operatorname{Var}_H(g)\left\{\frac{1}{n} + \frac{n-1}{n}\rho\right\}, \tag{6.23}$$

and $\operatorname{Cov}(\widetilde{F}(x), \widetilde{F}(y)) = \rho\{H(x\wedge y) - H(x)H(y)\}$. *Hence the covariance kernel of the Helly and Bray limit identifies only the single number* ρ, *and not the pair* (α, θ). *The two are separated by the growth of the number of jumps of* F_n:

$$\frac{K_n}{\log n} \to \theta \ \ (\alpha = 0), \qquad \frac{K_n}{n^\alpha} \to S_\alpha \ \ (0 < \alpha < 1), \qquad K_n \uparrow m \ \ (\alpha < 0),$$

almost surely, where S_α *is a strictly positive random variable.*

Proof. The predictive rule at $n = 1$ gives $\mathbb{P}(X_2 = X_1) = (1-\alpha)/(\theta+1) = \rho$, since the single existing block has size one. The proofs of Propositions 6.10 and 6.11 used only the value of this tie probability, through $\mathrm{Cov}(g(X_1), g(X_2)) = \rho \,\mathrm{Var}_H(g)$, so they carry over with $1/(1+\theta)$ replaced by ρ. Any (α, θ) on the line $1-\alpha = \rho(1+\theta)$ therefore gives the same de Finetti covariance kernel. The three limit laws for K_n are classical: for $\alpha = 0$ it follows from the independent indicators in the proof of Proposition 6.19 and the strong law for independent summands, for $\alpha > 0$ it is the law of large numbers of Pitman and Yor (1997), and for $\alpha < 0$ the number of types is bounded above by m and reaches it almost surely. □

Proposition 6.31 sharpens the theme of the chapter. The limiting empirical distribution function identifies the model through exactly two functionals of a different character: the variance of its heights, which gives ρ, and the growth rate of the number of its jumps, which gives α. In the Ewens case the second functional is degenerate, $\alpha = 0$ is known, and the first suffices; that is why Section 6.3.1 could identify θ from a single moment.

REMARK 6.32 (The negative branch is a finite species pool). When $\alpha < 0$ and $\theta = m|\alpha|$, the directing random measure is $\widetilde{P} = \sum_{i=1}^{m} P_i \delta_{\xi_i}$ with $(P_1, \ldots, P_m) \sim \mathrm{Dir}(|\alpha|, \ldots, |\alpha|)$ symmetric on m types, and (6.21) is the Dirichlet multinomial partition law. The urn of Assumption 6.27 then closes: once $K_n = m$ the probability of a new type is $(\theta + m\alpha)/(\theta + n) = 0$. This branch is not an ascertainment model at all; it is a statement that the pool of species is finite, of size m, with a symmetric Dirichlet abundance profile of concentration $|\alpha|$.

REMARK 6.33 (Weighting and tilting are different operations). On the negative branch the two parameter family is not the weighted paintbox (6.17) of the one parameter family for any choice of θ and w; Proposition 6.34 below proves it. For $0 < \alpha < 1$ we do not know whether some weight reproduces the two parameter law exactly: a growth rate argument does not exclude it, since a weight with $w(p)\, p$ of order $(\log 1/p)^{-1/\alpha}$ turns the geometric decay of the ranked Poisson Dirichlet frequencies into the power decay of the two parameter family, and we record the question in Section 6.9. For $0 < \alpha < 1$ the two parameter family arises instead from a tilt of the stable subordinator that generates the frequencies (Perman, Pitman and Yor, 1992; Pitman and Yor, 1997), which changes the law of P itself rather than reweighting a fixed P. The two mechanisms are genuinely distinct hypotheses about the same data, and Sections 6.5 and 6.6.3 compare them informally on the butterflies.

PROPOSITION 6.34 (The negative branch is not a weighted paintbox). *Let $\alpha < 0$ and $\theta = m|\alpha|$ with m a positive integer. For no $\theta' > 0$ and no admissible weight w is the weighted paintbox* (6.17) *built on Poisson Dirichlet frequencies with parameter θ' equal in law to the two parameter paintbox with parameters (α, θ).*

Proof. Let $S = \{p \in (0,1] : w(p) > 0\}$ and, for $0 < r \le 1$, let $N_r = \#\{\ell : r p_\ell \in S\}$. Since the ξ_ℓ are distinct, $\widetilde{P}^w$ has exactly N_1 atoms, while by Remark 6.32 the two

parameter paintbox has exactly m; so it suffices to show that $N_1 = m$ almost surely is impossible. By the stick breaking construction (Pitman and Yor, 1997), the frequencies are V together with $(1-V)$ times the frequencies P' of an independent Poisson Dirichlet paintbox with parameter θ', where $V \sim \mathrm{Beta}(1,\theta')$ has a positive density on $(0,1)$. Hence $N_r = \mathbf{1}_S(rV) + N'_{r(1-V)'}$, with N' computed from P'. If $N_1 = m$ almost surely then, by Fubini's theorem, N'_{1-v} is almost surely constant for almost every v, so there is a function g with $N_r = g(r)$ almost surely for almost every $r \in (0,1)$; applying the same decomposition to such an N_r gives $g(r) = \mathbf{1}_S(rv) + g(r(1-v))$ for almost every $(r,v) \in (0,1)^2$. In the variables $u = rv$ and $s = r(1-v)$ this reads $g(u+s) = \mathbf{1}_S(u) + g(s)$ for almost every (u,s) with $u+s < 1$. Expanding $g(u+s+t)$ once as $\mathbf{1}_S(u+s) + g(t)$ and once as $\mathbf{1}_S(u) + \mathbf{1}_S(s) + g(t)$ yields

$$\mathbf{1}_S(u+s) = \mathbf{1}_S(u) + \mathbf{1}_S(s) \quad \text{for almost every } (u,s) \text{ with } u+s<1.$$

The left side is at most one, so $S \cap (0, \frac{1}{2})$ is a null set; then the right side vanishes for almost every $u, s \in (0, \frac{1}{2})$, and since $u+s$ has a positive density on $(0,1)$ when u and s are independent and uniform on $(0, \frac{1}{2})$, $S \cap (0,1)$ is a null set. The atom sizes are almost surely smaller than one and their intensity $\theta' x^{-1}(1-x)^{\theta'-1}\,dx$ is absolutely continuous, so $N_1 = 0$ almost surely, which contradicts $m \geq 1$ (and the admissibility condition $\sum_\ell w(p_\ell)p_\ell > 0$). □

6.6.3 *The butterflies revisited*

We now refit Corbet's data, 620 species of which 119 are censored above 24 individuals, under (6.21). The censored likelihood is built as in Section 6.4, with the weight $1/j$ of a block of size j replaced by $(1-\alpha)_{j-1}/j!$ and $\theta^{620}\Gamma(\theta)/\Gamma(\theta+n)$ by $\prod_{i=1}^{619}(\theta+i\alpha)/(\theta+1)_{n-1}$, and n is again profiled out. Unlike the Ewens case, the two parameter likelihood depends on the whole vector $(a_1,\ldots,a_{24})$ and not only on K_n, since the sufficiency of Corollary 6.18 is a property of the one parameter family alone. Maximising it numerically gives the results in Table 6.8; the log likelihoods are those of the censored data, constants included.

TABLE 6.8. Fits to Corbet's Malayan butterflies. The two parameter maximum lies on the negative α branch; restricted to $\alpha \geq 0$ the maximum is attained at the Ewens boundary $\alpha = 0$.

Model	$\hat{\alpha}$	$\hat{\theta}$	$\hat{n}$	log likelihood	Pearson discrepancy
Ewens, $\alpha = 0$	0	141.75	11068	−70.13	30.4
Two parameter, $\alpha \geq 0$	0.000	141.75	11068	−70.13	30.4
Two parameter, unrestricted	−0.1454	209.70	9747	−65.85	20.0

Three things are worth drawing out.

First, the likelihood moves to the negative branch. The improvement in log likelihood is 4.3, a likelihood ratio statistic of 8.57; since $\alpha = 0$ is an interior point

of the range searched, the reference distribution is chi squared on one degree of freedom, whose ninety nine percent point is 6.63, so the evidence is clear though far from overwhelming. Ignoring the censoring, with $n = 3306$ and $K_n = 501$, exaggerates it: that fit gives $\hat{\alpha} = -0.49$ and a statistic of 44.3. By Remark 6.32 the fitted model is a finite pool of $\hat{m} = \hat{\theta}/|\hat{\alpha}| = 1443$ species with a symmetric Dirichlet abundance profile of concentration 0.15. The optimisation treated m as continuous; the model is proper only at integer m, and the likelihood is flat enough at this scale that the rounding is immaterial. The data are not saying that rare butterflies were overlooked; they are saying that the pool of species is finite, of the order of twice the 620 already seen.

Second, the fit to the rare classes improves in exactly the place the diagnostics pointed. The predicted singleton count falls from 140.0 to 117.1 against an observed 118, the doubletons move from 69.1 to 65.7 against 74, and the Pearson discrepancy $\sum_j (A_j - \mathbb{E}A_j)^2/\mathbb{E}A_j$ over the tabulated classes falls from 30.4 to 20.0. That quantity is descriptive only: the cells are dependent, being counts of a single partition, and the expected values do not sum to the observed total, so it should not be referred to a chi squared distribution. Table 6.9 and Figure 6.6 give the detail. The remaining misfit lies largely in the classes $j = 19$ to 24: the data show 31 species in classes 19 to 21 against 19.4 expected under the two parameter fit, and 11 in classes 22 to 24 against 16.2, an irregularity that neither model reproduces.

TABLE 6.9. Corbet's butterflies: observed counts against the exact means (6.12) and (6.22) at the fitted parameters, each at its own $\hat{n}$.

j	observed	Ewens	two parameter	j	observed	Ewens	two parameter
1	118	139.97	117.09	13	6	9.25	10.80
2	74	69.10	65.65	14	12	8.48	9.93
3	44	45.49	45.96	15	6	7.82	9.17
4	24	33.69	35.39	16	9	7.24	8.49
5	29	26.61	28.72	17	9	6.73	7.90
6	22	21.90	24.12	18	6	6.27	7.36
7	20	18.53	20.73	19	10	5.87	6.89
8	19	16.01	18.13	20	10	5.50	6.45
9	20	14.06	16.06	21	11	5.18	6.06
10	15	12.49	14.38	22	5	4.88	5.70
11	12	11.21	12.98	23	3	4.61	5.38
12	14	10.15	11.81	24	3	4.36	5.08

Third, the choice of branch matters for the question Corbet actually asked Fisher, how many new species a second two year expedition would yield. Taking a second sample of the same size $\hat{n}$ as each fit, and computing the expected number of species in the combined sample of $2\hat{n}$ given the 620 already seen, gives

Ewens: 97.4 new species, two parameter: 77.7,

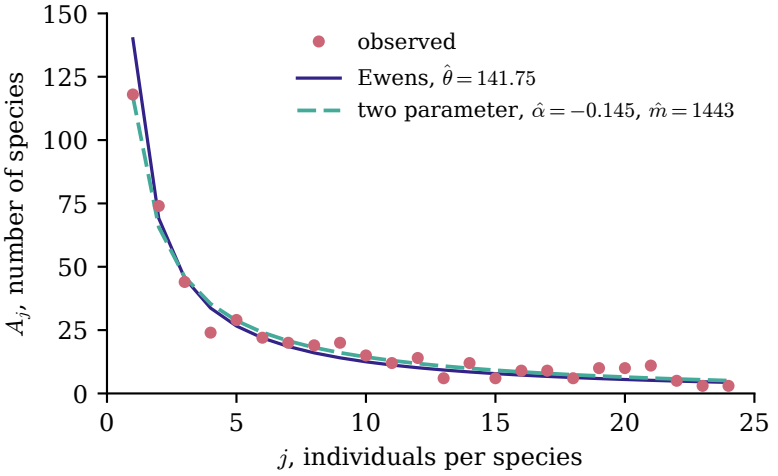


FIGURE 6.6. Corbet's Malayan butterflies with the one and two parameter fits. The negative α fit corrects the over prediction of singletons and lifts the middle of the spectrum.

against the model free alternating sum of Good and Toulmin (1956) evaluated on the tabulated classes, $\sum_{j=1}^{24}(-1)^{j+1}A_j = 75$, which is the figure Fisher reported (the terms with $j > 24$ are unknown, and under either fit their expected contribution is small, about two species under the Ewens fit). The one parameter model is the optimistic one, as it must be: with $\alpha = 0$ the species pool is inexhaustible and K_n grows without bound like $\theta \log n$, whereas the fitted negative branch has seen 620 of a ceiling of about 1443 and grows more slowly. The nonparametric answer lies close to the two parameter one and well below the Ewens one.

REMARK 6.35 (Caveats). Two should be recorded. First, the tabulation used here is the published one, censored at $j = 24$, and the fits use only the fact, reported by Fisher, Corbet and Williams (1943), that 119 further species had more than 24 individuals; the total n is estimated, not observed, and an analysis that treats the 24 classes as the whole sample overstates both the misfit of the Ewens model and the evidence for the negative branch. The fitted θ of 142 should not be compared with the value near 40 that is often quoted for Fisher's α, since the latter belongs to the Rothamsted light trap moths, a different data set with 240 species and about 15,600 individuals. Second, Proposition 6.24 and Theorem 6.28 are not nested, so the comparison in Table 6.8 is between the two parameter family and the Ewens submodel, not between weighting and tilting; the weighted paintbox is ruled out here informally, by Table 6.6, rather than by a likelihood ratio.

6.7 The Kingman coalescent

The derivation above makes no reference to genealogy, which invites an obvious objection: the Ewens sampling formula is usually, and most naturally, obtained from Kingman's coalescent (Kingman, 1982a,b), where it falls out of the

genealogical process directly (Griffiths, 1979; Ewens, 2004). This section sets the two derivations side by side. The relationship is closer than it first appears, and looking at it settles which of the two arguments assumes what.

6.7.1 *The coalescent with killing is the predictive rule*

Consider the standard construction. Trace the ancestry of a sample of n genes backwards in time under the infinitely many alleles model: each pair of ancestral lineages coalesces at rate one, and each lineage mutates at rate $\theta/2$, a mutation conferring an allele never seen before and resolving the type of every sample member below it. Every event, coalescence or mutation, reduces the number of active lineages by one, so the construction terminates after exactly n events.

PROPOSITION 6.36 (The competing exponentials give the predictive weights). *With m active lineages, the probability that the next event backwards in time is a mutation rather than a coalescence is*

$$\frac{m\theta/2}{\binom{m}{2} + m\theta/2} = \frac{\theta}{\theta + m - 1}, \tag{6.24}$$

and the complementary probability of a coalescence is $(m-1)/(\theta+m-1)$. Setting $m = n+1$, these are exactly the weights $c_n = \theta/(\theta+n)$ and $1-c_n$ of Proposition 6.8.

Proof. The total coalescence rate among m lineages is $\binom{m}{2} = m(m-1)/2$ and the total mutation rate is $m\theta/2$; the ratio of competing exponential rates gives (6.24) after cancelling $m/2$. □

EXAMPLE 6.37 (Three genes traced back by hand). Start with three lineages. The first event is a mutation with probability $\theta/(\theta+2)$ and a coalescence with probability $2/(\theta+2)$.

If it is a coalescence, two of the genes now share one ancestral lineage, and two lineages remain. The next event is a coalescence with probability $1/(\theta+1)$, in which case all three genes descend from one lineage and carry one allele whatever happens later; or a mutation with probability $\theta/(\theta+1)$, which separates the two lineages for good and produces the shape $2+1$.

If the first event is a mutation, the gene on that lineage carries an allele no other sample member shares, and two lineages remain; they coalesce next with probability $1/(\theta+1)$, giving shape $2+1$, or mutate with probability $\theta/(\theta+1)$, giving three distinct alleles. Collecting the paths,

$$\begin{aligned}
\mathbb{P}(\text{shape } 3) &= \frac{2}{\theta+2}\cdot\frac{1}{\theta+1} = \frac{2}{(\theta+1)(\theta+2)},\\
\mathbb{P}(\text{shape } 2+1) &= \frac{2}{\theta+2}\cdot\frac{\theta}{\theta+1} + \frac{\theta}{\theta+2}\cdot\frac{1}{\theta+1} = \frac{3\theta}{(\theta+1)(\theta+2)},\\
\mathbb{P}(\text{shape } 1+1+1) &= \frac{\theta}{\theta+2}\cdot\frac{\theta}{\theta+1} = \frac{\theta^2}{(\theta+1)(\theta+2)}.
\end{aligned}$$

These are exactly (6.11) at $n = 3$, with $c(a) = 2, 3, 1$ and $\theta_{(3)} = \theta(\theta+1)(\theta+2)$; they sum to $(\theta^2 + 3\theta + 2)/\{(\theta+1)(\theta+2)\} = 1$. At $\theta = 1$ they are $1/3, 1/2, 1/6$, the cycle type probabilities of a random permutation of three objects. The urn produces the same numbers forwards in time: the factors $1/(\theta+1)$ and $2/(\theta+2)$ are the joining probabilities for the second and third genes.

So Hoppe's urn is the killed coalescent read backwards, one event at a time: adding the $(n+1)$st observation to the urn corresponds to removing one of $n+1$ lineages from the genealogy, a new colour corresponding to a mutation and a repeated colour to a coalescence. That the allelic partition of the killed coalescent is the Ewens sampling formula is classical (Kingman, 1982b; Griffiths, 1979), and we do not reprove it; Table 6.10 instead reports a direct simulation of the genealogy, with the pair chosen uniformly at each coalescence, checked against (6.10).

TABLE 6.10. Allelic partitions from simulating the coalescent with killing, against the Ewens sampling formula (6.10). Each row is 200,000 independent genealogies; the statistic is the chi squared goodness of fit over all observed partition shapes.

n	θ	shapes observed	chi squared	degrees of freedom
6	0.5	11	8.5	10
8	2.0	22	24.7	21
10	5.0	42	57.0	41

6.7.2 *Which way the implication runs*

The genealogical argument and the present one use the same arithmetic in opposite directions, and that is the substantive point. The coalescent derivation posits a population model, extracts the rates, and deduces the predictive weights; θ enters as a ratio of rates, $\theta = 4N\mu$, four times the product of the diploid population size N and the mutation rate μ per gene per generation, and is a modelling input. Proposition 6.8 runs the other way. It assumes nothing about time, population size, reproduction or mutation. It assumes only that the sample is exchangeable and that the one step forecast is affine in the empirical distribution function, and it then forces $c_n = \theta/(\theta+n)$ using nothing beyond the equality of pairwise tie probabilities. The competing exponential structure of the coalescent is therefore not needed to obtain the Ewens formula: it is implied by exchangeability together with predictive sufficiency of F_n.

The parameter changes character in the same way. Genealogically θ is a rate ratio; here, by Propositions 6.10 and 6.11, it is

$$\theta = \frac{1-q}{q} = \frac{\mathrm{Var}_H(g)}{\mathrm{Var}\big(\int g\, d\widetilde{P}\big)} - 1,$$

where $q = \mathbb{P}(X_1 = X_2)$ is the homozygosity, so that a ratio of rates in the genealogy is a ratio of variances of the limiting empirical distribution function.

This cuts both ways, and the negative half is worth stating plainly. Any mechanism whatever that delivers the predictive rule (6.4) delivers the Ewens sampling formula, so the allelic partition of a sample carries no information about the genealogy beyond that rule. The sampling formula cannot be used to confirm the coalescent; it can only fail to contradict it.

6.7.3 Why the information grows like a logarithm

The genealogy does explain something the urn leaves mysterious, namely the very slow information growth of Proposition 6.19. Mutations fall on the coalescent tree as a Poisson process of rate $\theta/2$ per unit length, so the information about θ should be proportional to the expected total branch length

$$\mathbb{E}L_n = 2\sum_{k=1}^{n-1}\frac{1}{k},$$

and indeed

$$I_n(\theta) = \frac{\mathbb{E}L_n}{2\theta} + O(1), \tag{6.25}$$

the two sides differing by a constant that depends on θ alone: numerically the difference settles at 2.162, 1.145 and 0.388 for $\theta = 0.5, 2, 10$, stable to three decimal places between $n = 10^4$ and $n = 10^6$.

Both halves of this statement can be checked directly. While k lineages remain in the coalescent without mutation, the waiting time T_k to the next coalescence is exponential with rate $\binom{k}{2}$, and those k lineages contribute kT_k to the total length, so

$$\mathbb{E}L_n = \sum_{k=2}^{n} k\,\mathbb{E}T_k = \sum_{k=2}^{n}\frac{2}{k-1} = 2\sum_{k=1}^{n-1}\frac{1}{k}.$$

For the difference, $\mathbb{E}L_n/(2\theta) = \theta^{-1}\sum_{i=1}^{n-1} i^{-1}$ and $I_n(\theta) = \sum_{i=1}^{n-1} i/\{\theta(\theta+i)^2\}$, so

$$\frac{\mathbb{E}L_n}{2\theta} - I_n(\theta) = \frac{1}{\theta}\sum_{i=1}^{n-1}\left\{\frac{1}{i} - \frac{i}{(\theta+i)^2}\right\} = \sum_{i=1}^{n-1}\frac{2i+\theta}{i\,(\theta+i)^2}, \tag{6.26}$$

a series of positive terms of order $2/i^2$. It increases to a finite limit $c(\theta)$, which is 2.162, 1.145 and 0.388 at $\theta = 0.5, 2, 10$, and this proves (6.25).

Why compare with $\mathbb{E}L_n/(2\theta)$ at all? Suppose an oracle revealed the tree and the number S_n of mutations on it, as the infinitely many sites model would. Given the tree, S_n is Poisson with mean $\theta L_n/2$, whose Fisher information about θ is $(L_n/2)/\theta$, with expectation $\mathbb{E}L_n/(2\theta)$. So (6.25) says that the allelic partition carries all but a bounded amount of the information that the full tree and its mutations would carry, and (6.26) measures exactly what is lost by seeing only which genes share an allele.

EXAMPLE 6.38 (Four genes). For $n = 4$, $\mathbb{E}T_4 = 1/6$, $\mathbb{E}T_3 = 1/3$ and $\mathbb{E}T_2 = 1$, so $\mathbb{E}L_4 = 4/6 + 3/3 + 2 = 11/3$, which is $2(1 + \frac{1}{2} + \frac{1}{3})$, and $\operatorname{Var} L_4 = \sum_k k^2 \operatorname{Var} T_k = 16/36 + 9/9 + 4 = 49/9$. At $\theta = 2$ the oracle information is $\mathbb{E}L_4/(2\theta) = 11/12 \approx 0.917$, while from Example 6.16 and (6.15) the allelic partition carries $I_4(2) = 641/3600 \approx 0.178$, under a fifth. At $\theta = 1$ the figures are $11/6$ and $95/144$, a ratio of 0.36. With so few genes most mutations fall on lineages that are already distinguished, and are wasted. Table 6.11 and Figure 6.7 show how slowly the ratio recovers as n grows.

TABLE 6.11. Fisher information about θ in the allelic partition, $I_n(\theta)$, against the oracle information $\mathbb{E}L_n/(2\theta)$ from the tree and its mutations.

	$\theta = 2$			$\theta = 10$		
n	I_n	$\mathbb{E}L_n/(2\theta)$	ratio	I_n	$\mathbb{E}L_n/(2\theta)$	ratio
4	0.178	0.917	0.19	0.004	0.183	0.02
10	0.452	1.414	0.32	0.018	0.283	0.06
100	1.464	2.589	0.57	0.148	0.518	0.29
1000	2.599	3.742	0.69	0.362	0.748	0.48
10^4	3.749	4.894	0.77	0.591	0.979	0.60
10^6	6.051	7.196	0.84	1.051	1.439	0.73

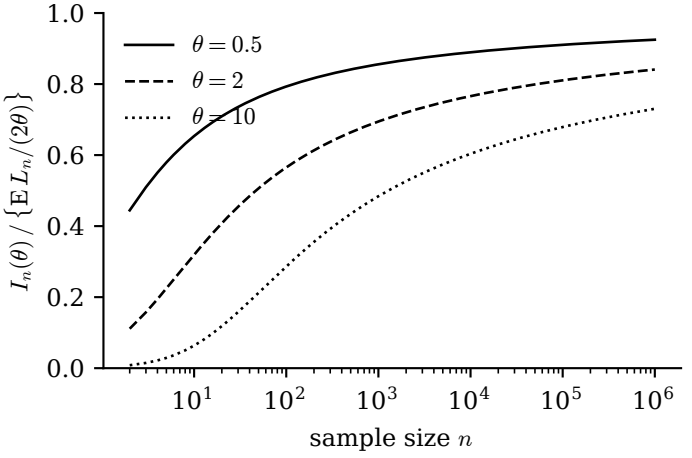


FIGURE 6.7. The fraction of the oracle tree information retained by the allelic partition, $I_n(\theta)/\{\mathbb{E}L_n/(2\theta)\}$, against n on a logarithmic scale. The gap $c(\theta)$ is fixed, so the ratio tends to one, but only at rate $1/\log n$.

The estimation problem is hard because the tree is short. Each new sample member attaches to a genealogy whose total length is already about $2\log n$

and lengthens it by only $2/n$, so doubling the sample adds a fixed quantity of information rather than doubling it.

6.7.4 *Where the two parameter family leaves the coalescent*

In the Kingman construction the event probabilities depend on the number of active lineages and on nothing else. Assumption 6.27 allows them to depend on K_n, the number of alleles already resolved, which no rate specification in the lineage count alone can produce. The Kingman genealogy with infinitely many alleles is therefore exactly the case $\alpha = 0$, and the butterfly estimate $\hat{\alpha} = -0.15$ is a statement that the sample is not the allelic partition of such a model. It does not follow that the genealogy is wrong. The negative branch is itself a coalescent model, with the infinitude of alleles dropped.

PROPOSITION 6.39 (The negative branch is the finite allele model). *Let $\alpha = -a < 0$ and $\theta = ma$ with m a positive integer. Then* (6.20) *equals the partition law of a sample of size n from m types with symmetric Dirichlet frequencies* $\mathrm{Dir}(a, \dots, a)$*, namely*

$$m(m-1)\cdots(m-k+1)\,\frac{\prod_{j=1}^{k}(a)_{n_j}}{(ma)_n}, \tag{6.27}$$

which is the stationary sampling law of the m allele model with parent independent mutation under a Kingman genealogy.

Proof. Substituting $\alpha = -a$ and $\theta = ma$ in (6.20), $\prod_{i=1}^{k-1}(\theta + i\alpha) = a^{k-1}\prod_{i=1}^{k-1}(m-i)$, while $(\theta+1)_{n-1} = (ma)_n/(ma)$ and $(1-\alpha)_{n_j-1} = (1+a)_{n_j-1} = (a)_{n_j}/a$. Multiplying,

$$\frac{a^{k-1}\prod_{i=1}^{k-1}(m-i)\cdot ma\cdot a^{-k}\prod_j (a)_{n_j}}{(ma)_n} = \frac{m(m-1)\cdots(m-k+1)\prod_j (a)_{n_j}}{(ma)_n},$$

which is (6.27). The right hand side is the Dirichlet multinomial partition probability obtained by integrating a multinomial sample against a symmetric $\mathrm{Dir}(a, \dots, a)$ frequency vector and choosing which of the m types the k occupied ones are. □

So the butterflies are not evidence against Kingman's genealogy. They are evidence against the infinitely many alleles assumption laid on top of it: the fit keeps the coalescent and replaces the inexhaustible allele space by a finite one of about 1443 types with per type parameter 0.15. That is a considerably milder conclusion than abandoning the genealogical model, and it is the one the data actually support.

For completeness, the remaining branch $0 < \alpha < 1$ does leave the Kingman genealogy, since no finite type space will produce a power law abundance spectrum. The natural home for that case is the family of coalescents with multiple mergers (Pitman, 1999), in which several lineages may coalesce at once; we do not pursue the correspondence here, and note only that the butterflies give it no support.

6.8 Bayesian inference and prediction

Everything so far has been likelihood based, which understates the case: the whole construction is a statement about a forecaster's predictive rule, and the natural summary of what the butterflies say about θ is a posterior rather than a standard error. This section supplies one, and then answers Corbet's question with a predictive distribution rather than a point.

6.8.1 Data augmentation for the mutation parameter

By Corollary 6.18 the likelihood is $L(\theta) \propto \theta^{K_n}\Gamma(\theta)/\Gamma(\theta+n)$, a function of the data through K_n alone, so the posterior inherits that reduction. The ratio of gamma functions is awkward, but it has an exact integral representation that makes the posterior a two block Gibbs sampler.

PROPOSITION 6.40 (Beta augmentation). *For $\theta > 0$ and $n \geq 1$,*

$$\frac{\Gamma(\theta)}{\Gamma(\theta+n)} = \frac{\theta+n}{\theta\,(n-1)!}\int_0^1 \eta^{\theta}(1-\eta)^{n-1}\,d\eta. \tag{6.28}$$

Consequently, under a Gamma(a,b) *prior on θ with rate b, the joint posterior of (θ,η) given $K_n = k$ satisfies*

$$\begin{aligned} \eta \mid \theta &\sim \text{Beta}(\theta+1, n), \\ \theta \mid \eta &\sim \pi_\eta\,\text{Gamma}(a+k,\, c_\eta) + (1-\pi_\eta)\,\text{Gamma}(a+k-1,\, c_\eta), \end{aligned}$$

where $c_\eta = b - \log\eta$ and $\pi_\eta/(1-\pi_\eta) = (a+k-1)/(n\,c_\eta)$.

Proof. The Beta integral gives $\int_0^1 \eta^{\theta}(1-\eta)^{n-1}d\eta = B(\theta+1,n) = \Gamma(\theta+1)\Gamma(n)/\Gamma(\theta+n+1)$, and $\Gamma(\theta+1) = \theta\Gamma(\theta)$, $\Gamma(\theta+n+1) = (\theta+n)\Gamma(\theta+n)$, which rearranges to (6.28). Multiplying the likelihood by the prior and by the integrand gives joint density proportional to $\theta^{a+k-2}(\theta+n)e^{-c_\eta\theta}(1-\eta)^{n-1}$, and the bracket $\theta^{a+k-1} + n\theta^{a+k-2}$ is an unnormalised mixture of the two gamma densities with the stated odds. □

This is the device of Escobar and West (1995), where it appears as one node of a larger sampler for Dirichlet process mixtures (Antoniak, 1974). Here θ is the only unknown, so the posterior is one dimensional and can be obtained by quadrature; we compute it both ways, which checks the sampler against an exact answer.

For the butterflies n is not observed. Given n the censored likelihood of Section 6.4 is $\theta^{620}\Gamma(\theta)/\Gamma(\theta+n)$ times a factor free of θ, so the proposition applies with $k = 620$ at any fixed n; we run the sampler at $n = \hat{n} = 11068$ and check it against quadrature at the same n, and we also report the posterior with n integrated out under a flat prior, by quadrature over (θ, n).

To see why the sampler mixes well, follow one sweep from $\theta = 142$ with $n = 11068$, $k = 620$ and the prior Gamma$(1, 0.01)$ used below. The augmentation variable has mean $\mathbb{E}(\eta \mid \theta) = 143/11211 = 0.01276$; at that value $c_\eta = 0.01 -$

$\log 0.01276 = 4.372$, the mixing weight is $\pi_\eta = 0.013$, and the conditional law of θ is essentially Gamma$(620, 4.372)$, with mean 141.8 and standard deviation $\sqrt{620}/4.372 = 5.7$. That conditional spread is most of the posterior standard deviation of about 6.5, so successive draws of θ are only weakly correlated (the lag one autocorrelation is 0.22) and a single sweep can move across the bulk of the posterior. Table 6.12 and the left panel of Figure 6.8 show that the sampler and quadrature agree to two decimal places in the mean and to within Monte Carlo error at the quantiles. With $k = 620$ the prior is irrelevant: a Gamma$(1, 0.01)$ prior, with mean 100, moves the posterior mean to 141.67 against a maximum likelihood value of 141.75. Integrating over n lowers the mean slightly and widens the interval a little.

TABLE 6.12. Posterior for θ on Corbet's butterflies under a Gamma$(1, 0.01)$ prior, computed two ways at the profiled n and by quadrature with n integrated out under a flat prior, with the maximum likelihood fit for comparison.

	mean	median	90 percent interval
Quadrature, $n = 11068$	141.67	141.55	(131.23, 152.48)
Beta augmentation, $n = 11068$, 2×10^5 sweeps	141.67	141.54	(131.25, 152.50)
Quadrature, n integrated out	141.16	140.84	(129.75, 152.66)
Maximum likelihood, n profiled	141.75		$141.75 \pm 1.645 \times 7.0$

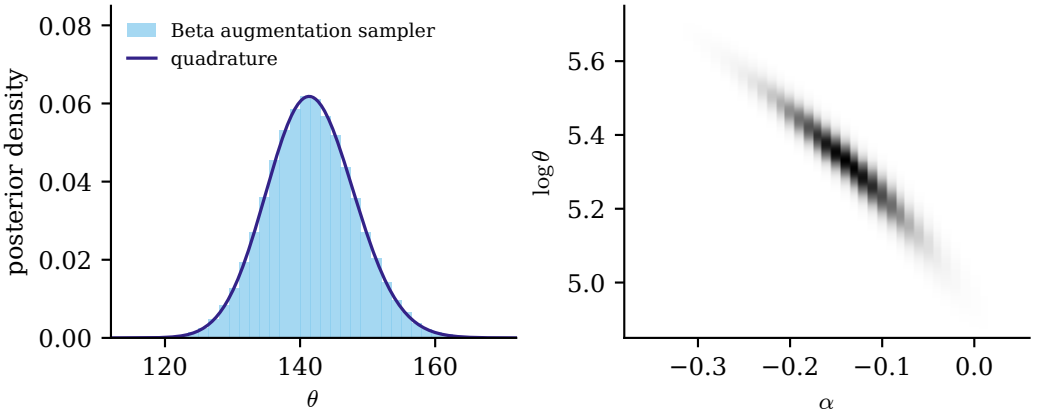


FIGURE 6.8. Left: posterior for θ in the Ewens model at $n = \hat{n} = 11068$, quadrature against the Beta augmentation sampler. Right: joint posterior for $(\alpha, \log\theta)$ in the two parameter model, with n integrated out.

6.8.2 *The two parameter posterior*

For the two parameter model the likelihood (6.21) is no longer a function of K_n alone, and no augmentation of the kind above is available, but the parameter space is two dimensional and quadrature on a grid is exact enough. Under a

uniform prior on α and on $\log\theta$, restricted to the admissible region, the posterior is summarised in the right panel of Figure 6.8 and by

$$\alpha:\ -0.137\ (-0.23,\ -0.05), \qquad \theta:\ 206\ (163,\ 251),$$
$$m = \theta/|\alpha|:\ 1773\ (1098,\ 3160),$$

posterior means with ninety percent intervals, n being integrated out under a flat prior. The posterior probability that $\alpha < 0$ is 0.995. The evidence for a finite species pool is therefore not an artefact of maximising, but the pool size is poorly determined: the interval runs from about 1100 to 3200 species against the 620 already seen, and the posterior of m has a long right tail, since $m = \theta/|\alpha|$ grows without bound as α approaches the Ewens value.

6.8.3 How many new species?

Corbet's question was what a second expedition of the same size would add. Under either model the answer has an exact conditional law. Given (α, θ) and the current state, the indicator that individual $i+1$ starts a new species is Bernoulli with probability $(\theta + \alpha K_i)/(\theta + i)$, so for the Ewens model $\alpha = 0$ the number of new species in a further t individuals is a sum of independent Bernoulli variables,

$$K_{n+t} - K_n \ \sim\ \sum_{i=n}^{n+t-1} \text{Bernoulli}\left(\frac{\theta}{\theta+i}\right), \tag{6.29}$$

free of K_n, while for $\alpha \neq 0$ the chain must be run forward because each new species changes the next probability. Mixing (6.29) over the posterior for θ, and simulating the urn forward under the posterior for (α, θ), gives the posterior predictive distributions in Figure 6.9:

$$\text{Ewens: } 97.2\ (80,\ 116), \qquad \text{two parameter: } 79.2\ (60,\ 98),$$

for a further sample of the same size n as the first, drawn with n from its posterior, against the model free alternating sum of Good and Toulmin (1956), which is 75 on the tabulated classes and is the figure Fisher reported.

The Ewens model assigns posterior probability 0.98 to a count exceeding Fisher's own answer, and its ninety percent interval excludes 75; the two parameter model puts 75 near the middle of its predictive distribution, with probability 0.66 of a count of at least 75. An inexhaustible pool is not merely optimistic here, it is hard to reconcile with the nonparametric extrapolation from the same table, although, as the next example shows, that extrapolation is itself noisy. The discrepancy is a consequence of the logarithmic growth $\mathbb{E}K_n \sim \theta \log n$ that Assumption 6.2 imposes.

EXAMPLE 6.41 (How noisy is the Good and Toulmin figure?). The alternating sum on Corbet's table is

$$118 - 74 + 44 - 24 + 29 - 22 + 20 - 19 + 20 - 15 + 12 - 14$$
$$+ 6 - 12 + 6 - 9 + 9 - 6 + 10 - 10 + 11 - 5 + 3 - 3 = 75.$$

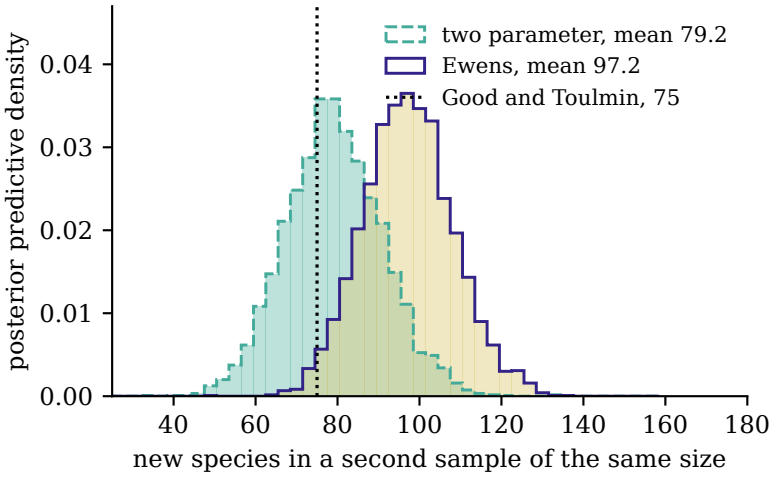


FIGURE 6.9. Posterior predictive distribution of the number of new species in a further sample of the same size as the first, under both models, with the Good and Toulmin value marked.

Under the Ewens model at $\hat{\theta} = 141.75$ and $\hat{n} = 11068$ its expectation, computed from (6.12) over all $j \leq n$, is 97.36, equal to two decimals to the expected number of new species in a second sample of size $\hat{n}$; restricted to $j \leq 24$ it is 95.2. So the alternating sum is, for this model, a nearly unbiased predictor, and the comparison in the text is fair. It is, however, a noisy one. In 2000 samples of $n = 11068$ simulated from the urn at $\theta = 142$, the alternating sum over $j \leq 24$ had mean 95.7 and standard deviation 21.6, and fell at or below 75 in 17.7 percent of samples. Taken alone, then, the Good and Toulmin figure is no real evidence against the Ewens model; the main evidence is the likelihood comparison of Table 6.8, and the predictive comparison should be read as showing that the two models give materially different answers to Corbet's question, not as an independent test.

6.9 Discussion

6.9.1 What each hypothesis contributes

It is worth separating the three ingredients, because they do very different amounts of work. Exchangeability alone yields the representation (6.3) with an arbitrary mixing measure μ on $\mathcal{F}$, and that is all it yields; it is far too weak to produce a sampling formula, and Corollary 6.6 shows that even its finite sample content is a statement about k being small relative to n rather than about the shape of anything. The second ingredient, Assumption 6.2, is where the model enters: the one step forecast is an affine function of the empirical distribution function with coefficients free of the data. Proposition 6.8 shows that this collapses the whole family of weight sequences to the single harmonic family indexed by θ, using nothing but the equality of pairwise tie probabilities. The third ingredient is diffuseness of H,

which is what makes ties informative: without it, two observations could agree by landing on a common atom of the base measure rather than by sharing a type, and the partition would no longer be the object the sampling formula describes.

In Kingman's language, Assumption 6.2 selects the Ewens member of the family of exchangeable partitions, the Gibbs partitions of type zero (Gnedin and Pitman, 2006); in genealogical language, by Proposition 6.36, it selects the Kingman coalescent with killing. Assumption 6.27 enlarges the selection to the two parameter family by letting the coefficients see the number of jumps of F_n and by loading the empirical distribution function of the distinct values alongside F_n itself. Both assumptions are statements about which functionals of the empirical distribution function the forecaster is permitted to use, which is the sense in which this chapter treats the sampling formula as a theorem about F_n rather than about genealogy.

6.9.2 The paintbox, Poisson Dirichlet and homozygosity

By Proposition 6.12 and the stick breaking construction of Sethuraman (1994), $\widetilde{P}$ can be written as $\sum_{\ell \geq 1} p_\ell \delta_{\xi_\ell}$ with ξ_ℓ independent draws from H and (p_ℓ) obtained from independent $\mathrm{Beta}(1, \theta)$ sticks. The ranked rearrangement of (p_ℓ) has the Poisson Dirichlet law with parameter θ (Pitman, 2006). Conditionally on $\widetilde{P}$ the sequence is independent and identically distributed, so the partition by ties is exactly Kingman's paintbox applied to that ranked sequence, and Theorem 6.15 recovers the classical statement that the paintbox of a Poisson Dirichlet frequency vector produces the Ewens sampling formula.

The route taken here reverses the usual order of exposition. The standard development starts from the frequencies, defines the paintbox, and derives the sampling formula by combinatorics; the present one starts from the forecaster's rule, derives the sampling formula by a telescoping product, and arrives at the frequencies only afterwards, as the atoms of the Helly and Bray limit. The two meet at the homozygosity: the random quantity $\sum_\ell p_\ell^2$ equals $\mathbb{P}(X_1 = X_2 \mid \widetilde{P})$ and has expectation $1/(1+\theta)$, which is the pairwise tie probability q of Proposition 6.8 and the single number that drove the entire identification argument. Everything this chapter proves about θ is, in the end, a restatement of how often two randomly chosen individuals share a type.

6.9.3 Two ways to read the parameter off the empirical distribution function

Proposition 6.11 identifies θ from the heights of F_n, exactly and at every n, but it identifies it from a variance, so it requires either independent replication or knowledge of H. Corollary 6.18 says that the jumps alone are sufficient, so a single sample suffices, but it gives up the connection to the de Finetti limit. The two are complementary rather than competing. The variance identity is what makes the de Finetti argument close, since it exhibits θ as a functional of the object that Helly and Bray produced; K_n is what one computes from data. Proposition 6.31 shows that in the two parameter model both are needed at once and neither suffices alone: the

heights deliver only $\rho = (1-\alpha)/(1+\theta)$, and the growth rate of the jump count must supply the second coordinate.

The estimation problem is genuinely hard in a way the algebra conceals. The information about θ grows like $\log n$, so that the standard error in Table 6.4 improves by less than a quarter between $n = 10^3$ and $n = 10^4$, and the widely used estimator $K_n / \log n$ carries a bias $-\theta\psi(\theta)/\log n$ of smaller order than its standard deviation yet larger than it at realistic n when θ is large. At $\theta = 10$ and $n = 10^4$ that bias pulls the estimate down by a quarter of the parameter. Solving the likelihood equation costs nothing and removes it.

6.9.4 Sharpness of the sampling bound

The bound $2p_{n,k}$ of Lemma 6.3 is of the right order. For k growing like $\sqrt{n}$ the birthday problem shows that sampling with and without replacement are not asymptotically equivalent, so no bound of this type can hold beyond that range; and since de Finetti's theorem is a statement about fixed k and large n, the bound is more than sufficient for Theorem 6.7. Corollary 6.6 records the corresponding statement for finite exchangeable sequences, recovering the estimate of Diaconis and Freedman (1980) as a by product rather than as a separate theorem. It is worth noticing that this is the only quantitative inequality in the entire derivation of the sampling formula: everything else is exact algebra.

6.9.5 What the butterflies did and did not settle

The data were brought in to illustrate the theory and ended by discriminating between two extensions of it. The one parameter fit misfits in a specific direction, over predicting the rarest class by about a fifth and under predicting the moderately abundant classes, though once the censoring of the table is modelled the misfit is modest; that pattern is what an ascertainment story predicts, and it is why Section 6.5 was written. But the power weighted paintbox cannot produce it: as Table 6.6 shows, raising β buys a lower singleton count only by lowering the richness in nearly the same proportion, and the data want more species and a flatter profile simultaneously. The two parameter family delivers both, and it does so on the negative branch, which by Remark 6.32 is not an ascertainment model at all but a finite pool, estimated at about 1400 by maximum likelihood and 1800 by the posterior mean, with a symmetric Dirichlet abundance profile. The likelihood ratio statistic of 8.57 is clear evidence but not overwhelming, and Section 6.8 shows the posterior agrees: probability 0.995 on $\alpha < 0$, a poorly determined pool size, and a predictive interval for the second expedition that covers Fisher's answer where the Ewens interval misses it.

Two things this does not settle. The comparison is not a like for like test: the weighted paintbox and the two parameter family are not nested and were compared informally, by simulation against likelihood, rather than by a common criterion. And the tabulation is the published one, censored at $j = 24$, so the total sample size is estimated rather than observed; an analysis that treats the 24 classes

as the whole sample gives $\hat{\theta} = 164$ and a much stronger, but spurious, case against the Ewens model. The $\hat{\theta}$ of 142 reported here should in any case not be compared with the value near 40 often quoted as Fisher's α, which belongs to the Rothamsted light trap moths rather than to these butterflies.

6.9.6 Open questions

Four seem worth recording.

First, Proposition 6.26 is stated with a gap. Zabell's theorem characterises the Dirichlet process among species sampling sequences with an affine predictive; what is missing is the complementary statement that a nonconstant weight function actually moves $\widetilde{P}^w$ off the Dirichlet class. The simulations make this plain for power weights, but a proof covering all admissible w, presumably through the induced law on ranked frequencies, would let Proposition 6.26 be stated as the clean dichotomy it appears to be.

Second, the characterisation direction of Assumption 6.27 is quoted rather than proved. The forcing argument of Proposition 6.8 is short because exchangeability of pairwise ties determines everything once the coefficients are deterministic; when the coefficients are allowed to depend on K_n the same device no longer closes, since the relevant expectation involves the joint law of the block size and the jump count. A self contained proof in the style of Proposition 6.8 would complete the parallel between the two assumptions.

Third, the weighted paintbox of (6.17) and the tilt of Section 6.6 are distinct operations on the same object, and nothing forbids applying both. A weighted two parameter paintbox has three parameters and would let ascertainment and pool size be estimated jointly rather than compared informally; the obstacle is (6.18), which has no closed form, so the comparison would have to be made by simulation based inference. A prior question is whether the two operations ever coincide: Proposition 6.34 shows that no weighted Kingman paintbox has the law of the negative branch, but for $0 < \alpha < 1$ we do not know whether some weight w applied to a Poisson Dirichlet paintbox reproduces the two parameter law exactly.

Fourth, the identification result of Proposition 6.31 is a statement about two functionals of the limiting empirical distribution function, the variance of its heights and the growth rate of its jump count. It is natural to ask how far this goes: for which families of exchangeable partitions is the de Finetti limit identified by a finite list of such functionals, and is there a general principle assigning one functional per parameter? The Gibbs partitions of type α (Gnedin and Pitman, 2006) would be the place to start.

6.10 Exercises

1. Six balls, three of colour 0 and three of colour 1, are in an urn. Compute the exact total variation distance between the laws of the ordered colours of two balls drawn with and without replacement, and compare it with $p_{6,2}$ and with $k(k-1)/(2n)$. *Hint.* The four ordered pairs have probabilities $1/4$ each

with replacement and $1/5, 3/10, 3/10, 1/5$ without; the answer is $1/10$, against $p_{6,2} = 1/6$.

2. Show that for distinct $x_1, \dots, x_n$ the bound of Lemma 6.3 is attained: the total variation distance between the value laws equals $p_{n,k}$. Use the birthday approximation $p_{n,k} \approx 1 - e^{-k(k-1)/(2n)}$ to find the k at which it first exceeds one half when $n = 365$. *Hint.* The value map is one to one on tuples; $k = 23$.

3. Write out the law of the allelic partition for $n = 3$ from (6.10), and verify by expanding that $\mathbb{E}K_3 = 1 + \theta/(\theta+1) + \theta/(\theta+2)$. *Hint.* Example 6.37 lists the three probabilities.

4. Prove that at $\theta = 1$ the sampling formula is the law of the cycle type of a uniform random permutation of $\{1, \dots, n\}$, and deduce that $\mathbb{E}A_j = 1/j$ for $1 \le j \le n$. *Hint.* Use (6.11) and, for the mean, (6.12) with $\theta = 1$.

5. Show that $\mathbb{P}(K_n = 1) = (n-1)!/(\theta+1)_{n-1}$ and $\mathbb{P}(K_n = n) = \theta^{n-1}/(\theta+1)_{n-1}$, and evaluate both at $n = 5$, $\theta = 2$. *Hint.* $s(n,1) = (n-1)!$ and $s(n,n) = 1$ in Corollary 6.20; the values are $1/15$ and $2/45$, as in Table 6.1.

6. Show that the sample homozygosity $\hat{F} = \sum_j a_j j(j-1)/\{n(n-1)\}$ is unbiased for $1/(1+\theta)$ under (6.10), and compute its variance for $n = 3$. *Hint.* $\hat{F}$ is the proportion of the $\binom{n}{2}$ pairs that are tied, and each pair is tied with probability $q = 1/(1+\theta)$. For $n = 3$, $\hat{F}$ takes the values $1, 1/3, 0$ on the three shapes.

7. Prove (6.15), $I_n(\theta) = \operatorname{Var} K_n/\theta^2$, directly from the formulas of Proposition 6.19, and explain why the identity must hold for any family of the form $p_\theta(x) = h(x)\theta^{t(x)}/Z(\theta)$. *Hint.* Differentiate $\log Z$ twice in $\log\theta$.

8. For Corbet's data, with the censored fit $\hat{\theta} = 141.75$ and $\hat{n} = 11068$ of Section 6.4, show that under the Ewens model the expected number of new species in a second sample of the same size is $\sum_{i=n}^{2n-1} \hat{\theta}/(\hat{\theta}+i) \approx 97.4$, and compare with the approximation $\hat{\theta}\log 2$. *Hint.* Use (6.29); $141.75 \log 2 = 98.3$, the difference coming from θ not being small compared with n.

9. In the two parameter model of Assumption 6.27 with $n = 3$, show that the three shapes have probabilities $(1-\alpha)(2-\alpha)$, $3(1-\alpha)(\theta+\alpha)$ and $(\theta+\alpha)(\theta+2\alpha)$, each divided by $(\theta+1)(\theta+2)$, and check that they sum to one. *Hint.* Apply (6.20) and count set partitions of each shape.

10. For the finite pool of Remark 6.32 with m types and concentration a, show that $\mathbb{E}K_n = m\{1 - ((m-1)a)_n/(ma)_n\}$, where $(x)_n$ is the rising factorial. Evaluate it at the butterfly fit $m = 1442.6$, $a = 0.1454$, $n = 9747$. *Hint.* A given type is absent from the sample with probability $\mathbb{E}(1-P_1)^n$, and $1 - P_1 \sim \mathrm{Beta}((m-1)a, a)$; the value is 619.8, close to the observed 620.

11. Show that the total length of the coalescent tree has $\operatorname{Var} L_n = 4\sum_{k=1}^{n-1} k^{-2}$, so that $L_n/\mathbb{E}L_n \to 1$ in probability, and compute the mean and variance of the number of mutations S_n on the tree, given that S_n is Poisson with mean $\theta L_n/2$

conditionally on L_n. *Hint.* $\mathrm{Var}(kT_k) = 4/(k-1)^2$; $\mathbb{E}S_n = \theta \sum_{k=1}^{n-1} k^{-1}$ and $\mathrm{Var}\, S_n = \mathbb{E}S_n + \theta^2 \sum_{k=1}^{n-1} k^{-2}$, Watterson's formulas.

12. Simulate Hoppe's urn at $n = 8$, $\theta = 2$ and reproduce Table 6.2, then simulate the coalescent with killing of Section 6.7 and check that the two empirical laws of the allelic partition agree. *Hint.* For the urn, a new allele arrives at step $i+1$ with probability $\theta/(\theta+i)$; otherwise copy the allele of a uniformly chosen earlier gene.

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CHAPTER 7

Ascertainment and the Two Parameter Poisson Dirichlet Family

Sampling a Poisson Dirichlet subdivision of the unit interval with a nonuniform density is a Rao weighted distribution problem for Kingman's paintbox, and in the one parameter case it has an exact answer: a $\mathrm{Beta}(1,\alpha)$ *weight carries the Ewens sampling formula with parameter* θ *to the one with parameter* θ/α*. We ask what survives in the two parameter family. Three things happen. The reduction survives untouched, because its proof uses nothing about the law of the subdivision. Exactness does not survive, and the obstruction is a rigidity theorem: the residual allocation structure is preserved only by power weights, and among power weights the Ewens family alone is preserved, so no sampling density that charges every interval carries the size biased sticks of* $\mathrm{PD}(\sigma,\theta)$ *with* $\sigma > 0$ *to the sticks of another member of the Ewens Pitman family. What replaces exactness is a first order law of the same shape as the one parameter result, in which ascertainment multiplies the odds* $(1-\sigma)/\sigma$ *by* α *and leaves the concentration* θ *untouched. Going further, the limiting* σ *diversity of the weighted problem is the unweighted one raised to a power,* $S_* = CS^\kappa$ *with* $\kappa = \alpha\sigma_*/\sigma$ *and* C *in closed form, which follows from an identity between the diversity and the stick envelope,* $S = (M/\sigma)^\sigma$*. Finally the coincidence theory of the one parameter case collapses entirely: for* $\sigma > 0$ *the probability that no two intervals receive the same number of points tends to zero under every weight with* $\sigma_* > 0$*. The abundance spectrum degenerates: every* A_j/n^{σ_*} *converges to a deterministic multiple of one random variable, so the shape of the spectrum is free of everything but* σ_**. Two consequences are practical. The singleton fraction* A_1/K_n *estimates* σ_* *with no normalisation at all, and the bias from nonuniform sampling inverts exactly.*

7.1 Introduction

Warren Ewens asked what becomes of the Ewens sampling formula when the n points dropped on a Poisson Dirichlet subdivision of $(0,1)$ are sampled from a nonuniform density. In the one parameter case the answer is unusually clean. Pushing the sample through the probability integral transform absorbs the weighting into new paintbox frequencies, so the problem is a Rao weighted distribution version of Kingman's paintbox. If the intervals are then laid down in size biased rather than ranked order, a $\mathrm{Beta}(1,\alpha)$ density converts $\mathrm{ESF}(\theta)$ into $\mathrm{ESF}(\theta/\alpha)$, exactly and at every sample size. In Sethuraman's stick breaking form the same statement reads: ascertainment carries a Dirichlet process to a

Dirichlet process, fixing the base measure and the atoms and rescaling only the concentration.

The natural next question, and the one this chapter answers, is how much of that survives the two parameter extension of Pitman and Yor. It is a fair question to ask sceptically. A result that holds for a one parameter family and fails for its two parameter enlargement is often an accident of the smaller family, and one wants to know which it is here.

7.1.1 What we find

The answer separates into three parts, and they behave differently.

The reduction is universal. Lemma 7.3 uses nothing about the law of the subdivision: not exchangeability of the atoms, not Poisson Dirichlet structure, not even that the lengths are random. It is also free of the direction in which the intervals are laid, by Lemma 7.4, so a density and a direction enter only through G. The density disappears into new paintbox frequencies and only the tail sums remain. So the entire framework is available for $\mathrm{PD}(\sigma, \theta)$, for Ewens Pitman, and for any other paintbox one cares to write down.

Exactness is exactly the Ewens case, and rigidly so. Theorem 7.11 says that the residual allocation structure, independence of the successive stick factors, is preserved only by power weights $G(r) = r^{\alpha}$, by a Cauchy equation that we derive through quantiles. Theorem 7.9 then says that among power weights the Ewens family alone is preserved. Together these give the following statement: for $\sigma \in (0, 1)$ no sampling density that charges every interval, apart from the uniform, carries the size biased sticks of $\mathrm{PD}(\sigma, \theta)$ to the sticks of a member of the Ewens Pitman family. Whether a density could still produce an Ewens Pitman partition with the induced sticks out of size biased order is settled here only at particular parameter values (Section 7.3). The stability of the one parameter theory is therefore a coincidence between two one parameter families rather than an instance of a general principle, and the two parameter case is where one sees that.

The first order generalises, in the right coordinate. Proposition 7.13 gives

$$\frac{\log K_n}{\log n} \longrightarrow \sigma_* = \frac{\sigma}{\alpha(1-\sigma)+\sigma}, \qquad \text{equivalently} \qquad \frac{1-\sigma}{\sigma} \longmapsto \alpha\,\frac{1-\sigma}{\sigma}.$$

Ascertainment multiplies the odds of the discount parameter by α, as it divides the Ewens parameter by α. The concentration θ does not appear.

Beyond the first order we can say more than we expected. The limiting σ diversity satisfies $S = (M/\sigma)^{\sigma}$, where M is the stick envelope (Lemma 7.16). The weighted diversity is then $S_* = CS^{\kappa}$ almost surely (Proposition 7.17), with C in closed form and free of θ (Proposition 7.18). So the weighting does not produce a new random object: it reproduces the old one raised to a power. The abundance spectrum follows suit, and degenerately: by Proposition 7.24 every A_j/n^{σ_*} converges to a deterministic multiple of the same S_*.

Section 7.10 drops the Beta form altogether. For an arbitrary weight that charges every interval the exact theory is gone, by Theorem 7.11. The asymptotic theory

survives with the regular variation index ρ of G in place of α: the growth rates of K_n fall into four regimes (Theorem 7.33), the singleton fraction estimates σ_* for every weight of positive index whose density is regularly varying at the far endpoint (Corollary 7.34), and ρ itself can be read off recorded positions by Hill's estimator (Corollary 7.35).

Section 7.4 reads all of this genealogically. The coalescent derivation of the one parameter formula cannot produce $\sigma > 0$: its type count is a sum of independent indicators and the two parameter count is overdispersed (Proposition 7.12). The two repairs in the literature are the Chinese restaurant and multiple merger coalescents, and ascertainment biases the index a geneticist would infer from either.

Section 7.7 records the one part of the one parameter theory that does not survive at all. The probability that no two intervals receive the same number of points tends to $e^{-\gamma\vartheta}/\Gamma(1+\vartheta)$ when $\sigma = 0$, where ϑ is the Ewens parameter of the sampled law ($\vartheta = \theta/\alpha$ under a $\mathrm{Beta}(1,\alpha)$ density), and to zero for every $\sigma > 0$ under every weight with $\sigma_* > 0$. Section 7.8 draws the statistical consequence: an estimator of σ from a nonuniformly sampled population is biased in a way that inverts exactly, and we give the correction.

7.1.2 Why "ascertainment"

The word is inherited from human genetics, and it means what it says: a unit is *ascertained* when it comes to the investigator's attention, and the difficulty is that the probability of coming to attention depends on the very quantity under study.

The canonical case is Fisher's. To estimate the segregation ratio for a recessive trait one collects families and counts affected children, but families cannot be sampled at random, because a family with no affected child never reaches a clinic. Families are found *through* their affected children. Under single ascertainment a family with r affected children enters the sample with probability proportional to r; if each affected child independently brings the family to attention with probability π, the probability is $1-(1-\pi)^r$. Either way the sample overrepresents families with many affected children and the naive ratio is biased upward. Fisher (1934) derived the correction, formalising what Weinberg's proband method had attempted, and "ascertainment bias" entered the vocabulary there.

Rao (1965) turned this into a general framework, in a paper titled, literally, "On discrete distributions arising out of methods of ascertainment". Every such scheme has one structure: the population has density f, a unit with value x is ascertained with probability proportional to a weight $w(x)$, and the sample follows the weighted density $w(x)f(x)/\mathbb{E}[w]$. Size biased sampling is the case $w(x) = x$. Patil and Rao (1978) developed the framework for ecology and wildlife, which is the lineage of the tired collector of Section 7.11.

In the present problem the weight is the sampling density itself. An interval of the paintbox is ascertained, that is, receives at least one specimen, with probability governed by the mass the density places on it, so the induced frequencies of

Lemma 7.3 are the Rao weighted version of the paintbox frequencies. This is why the confounding warning of Bayarri and DeGroot applies with full force, and why it can be made exact here: the weight and the underlying law are learnable only through a single combination, θ/α in the one parameter case and $\sigma_* = \sigma/\{\rho(1-\sigma)+\sigma\}$ in the two parameter one.

7.1.3 Related work

The weighted distribution formulation and its identifiability difficulty are set out in Section 7.1.2; the confounding is the theme of Bayarri and DeGroot's work on the subject, and the present chapter is a case where it can be described exactly rather than warned about. The paintbox is Kingman's (Kingman, 1978), the one parameter sampling formula is Ewens's (Ewens, 1972), the stick breaking form we use throughout is Sethuraman's (Sethuraman, 1994), and the two parameter family, its partition probability function and its σ diversity are due to Pitman and Yor (Pitman, 1995; Pitman and Yor, 1997), with Pitman (2006) the standard account. The occupancy asymptotics we lean on, the correspondence between the counting function of the frequencies and the growth of the number of occupied cells, are Karlin's (Karlin, 1967). What appears to be new here is the rigidity of Theorem 7.11, the exponent map of Proposition 7.13 and its odds form, and the diversity identities of Section 7.6.

7.1.4 Notation

Write $\mathrm{PD}(\sigma,\theta)$ for the two parameter Poisson Dirichlet law, $0 \le \sigma < 1$ and $\theta > -\sigma$, whose size biased stick fractions are independent with

$$B_k \sim \mathrm{Beta}(1-\sigma,\ \theta+k\sigma), \qquad W_k = B_k \prod_{i<k}(1-B_i), \qquad \widetilde{R}_k = \prod_{i\le k}(1-B_i). \quad (7.1)$$

At $\sigma = 0$ this is $\mathrm{GEM}(\theta)$ and the induced partition of a uniform sample is $\mathrm{ESF}(\theta)$. The subdivision of $(0,1)$ is laid down in some order, n points are dropped with density f rather than uniformly, F is its distribution function, and

$$G(r) = 1 - F(1-r), \qquad f_\alpha(x) = \alpha(1-x)^{\alpha-1} \iff G(r) = r^\alpha. \quad (7.2)$$

Here $\alpha > 1$ concentrates the sample near $x = 0$ and $\alpha < 1$ near $x = 1$. At $\sigma = 0$, ϑ denotes the Ewens parameter of the sampled law: $\vartheta = \theta$ under uniform sampling and $\vartheta = \theta/\alpha$ under f_α. We take the layout to run from the left throughout, which by Lemma 7.4 costs no generality. Write A_j for the number of intervals hit exactly j times, $K_n = \sum_j A_j$ for the number of intervals hit at all, and $N(x) = \#\{k : p_k > x\}$ for the counting function of the induced frequencies. Rising factorials are written $(x)_m = x(x+1)\cdots(x+m-1)$, with $(x)_0 = 1$, and the odds of the discount parameter, $(1-\sigma)/\sigma$, are written β inside proofs and ω when they are viewed as a coordinate on the parameter space.

7.1.5 *Small samples by hand*

Before any asymptotics it helps to have the exact law of a small sample in front of us, because several statements below (the exact one parameter theorem, its failure at $\sigma > 0$) are statements about these finite dimensional laws. Under uniform sampling the probability that the partition of $\{1,\dots,n\}$ induced by a $\mathrm{PD}(\sigma,\theta)$ paintbox equals a particular set partition with k blocks of sizes $n_1,\dots,n_k$ is Pitman's exchangeable partition probability function (Pitman, 1995),

$$p_{\sigma,\theta}(n_1,\dots,n_k) \;=\; \frac{\prod_{i=1}^{k-1}(\theta+i\sigma)}{(\theta+1)_{n-1}} \prod_{j=1}^{k}(1-\sigma)_{n_j-1}. \tag{7.3}$$

At $\sigma = 0$ this is the Ewens sampling formula written for a single set partition, $\theta^{k-1}\prod_j (n_j-1)!/(\theta+1)_{n-1}$. The probability of a partition *type*, a multiset of block sizes, is (7.3) multiplied by the number of set partitions of that type: for $n = 4$ these multiplicities are $1,4,3,6,1$ for the types 4, 3+1, 2+2, 2+1+1 and 1+1+1+1, fifteen set partitions in all.

EXAMPLE 7.1 (Pitman's formula at $n = 3$ and $n = 4$). Take $(\sigma,\theta) = (\frac12, 1)$ and the set partition $\{1,2\}\{3\}\{4\}$. Here $k = 3$, the numerator is $(\theta+\sigma)(\theta+2\sigma) = \frac32 \cdot 2 = 3$, the denominator is $(2)_3 = 2\cdot 3\cdot 4 = 24$, and the block factors are $(\frac12)_1 = \frac12$ for the pair and 1 for each singleton. So $p = 3/48 = 1/16$, and the type 2+1+1 has probability $6/16 = 3/8$. Table 7.1 lists every type for $n = 3$ and $n = 4$ under five parameter pairs; each column sums to one, which is a useful check on (7.3).

Three things are visible in the table. First, at $n = 2$ the formula gives the chance that two sampled genes are of the same allele, the homozygosity

$$\mathbb{P}(\text{two draws in one block}) \;=\; \frac{1-\sigma}{1+\theta},$$

which is $1/(1+\theta)$ in the Ewens case. One number cannot separate σ from θ; the pairs $(0,1)$ and $(\frac12,0)$ both give $\frac12$. Second, the two pairs are separated at $n = 3$: the probability that all three draws agree is $(1-\sigma)(2-\sigma)/\{(1+\theta)(2+\theta)\}$, which is $\frac13$ for $(0,1)$ and $\frac38$ for $(\frac12,0)$. Third, at a fixed homozygosity a positive σ moves mass to both extremes at once: compared with $\mathrm{ESF}(1)$, the law $\mathrm{PD}(\frac12,0)$ puts more probability on a single block of four (5/16 against 1/4) and more on four distinct blocks (1/8 against 1/24), less on the shapes 3+1 and 2+2, and the same on 2+1+1. A few very common types together with many rare ones is the small sample face of the heavy tail.

7.1.6 *An occupancy lemma*

Most asymptotic statements in this chapter pass from the induced frequencies to the counts through one correspondence, due to Karlin (1967), and we state it once so that later proofs can cite it rather than gesture at it. Throughout, the frequencies are held fixed (we condition on the paintbox) and L denotes a slowly varying function.

TABLE 7.1. Exact probabilities of each partition type at $n = 3$ and $n = 4$ under (7.3). Each column, headed by a pair (σ, θ), gives the probability of the whole type, including its multiplicity.

n	type	$(0,1)$	$(0,2)$	$(\frac{1}{2},0)$	$(\frac{1}{2},1)$	$(\frac{1}{4},2)$
3	3	1/3	1/6	3/8	1/8	7/64
	2+1	1/2	1/2	3/8	3/8	27/64
	1+1+1	1/6	1/3	1/4	1/2	15/32
4	4	1/4	1/10	5/16	5/64	77/1280
	3+1	1/3	4/15	1/4	3/16	63/320
	2+2	1/8	1/10	1/16	3/64	81/1280
	2+1+1	1/4	2/5	1/4	3/8	27/64
	1+1+1+1	1/24	2/15	1/8	5/16	33/128

LEMMA 7.2 (Karlin's occupancy correspondence). *Let (p_k) be a fixed probability sequence with counting function $N(x) = \#\{k : p_k > x\}$, and suppose $N(x) = x^{-\gamma}L(1/x)$ as $x \downarrow 0$ for some $\gamma \in (0,1)$. Then, for an i.i.d. sample of size n from (p_k),*

$$K_n \sim \Gamma(1-\gamma)\, n^{\gamma} L(n) \quad \text{and} \quad A_j \sim \frac{\gamma\,\Gamma(j-\gamma)}{j!}\, n^{\gamma} L(n)$$

almost surely, the second for each fixed $j \geq 1$. If instead $N(x) \sim c\log(1/x)$, then $K_n \sim c\log n$ almost surely. In the boundary case $\gamma = 1$, $N(x) = x^{-1}L(1/x)$, the function $L_1(y) = \int_y^{\infty} L(u)u^{-1}du$ is finite and slowly varying with $L_1(y)/L(y) \to \infty$, and

$$K_n \;\sim\; nL_1(n) \;=\; \sum_k \min(np_k, 1)$$

almost surely; so $K_n/n \to 0$, but more slowly than any negative power of n.

Proof sketch. Replace the fixed sample size by a Poisson number of draws with mean t, so that the cells are filled independently and cell k is occupied with probability $1 - e^{-tp_k}$. Writing the sum over cells as a Stieltjes integral against $-dN$ and integrating by parts,

$$\mathbb{E}[K(t)] \;=\; \sum_k \bigl(1 - e^{-tp_k}\bigr) \;=\; \int_0^{\infty} te^{-tx} N(x)\, dx \;=\; \int_0^{\infty} e^{-u} N(u/t)\, du.$$

The boundary terms vanish because $N(x) = 0$ for $x \geq 1$ and $xN(x) \to 0$ as $x \downarrow 0$, the latter since $xN(x) \leq xN(\varepsilon) + \sum_{k:\, p_k \leq \varepsilon} p_k$ for every $\varepsilon > 0$. Now $N(u/t) = t^{\gamma}u^{-\gamma}L(t/u)$, and $L(t/u)/L(t) \to 1$ for each u; Potter's bounds for slowly varying functions dominate the ratio by a multiple of $u^{-\delta} + u^{\delta}$ for small $\delta > 0$, so dominated convergence gives $\mathbb{E}[K(t)] \sim t^{\gamma}L(t)\int_0^{\infty} u^{-\gamma}e^{-u}du = \Gamma(1-\gamma)t^{\gamma}L(t)$. The same calculation with $(tx)^j e^{-tx}/j!$ in place of $1 - e^{-tx}$ gives the mean of A_j, after one integration by parts: $\mathbb{E}[A_j(t)] = \int_0^{\infty} \pi_j'(u) N(u/t)\, du$ with $\pi_j(u) = u^j e^{-u}/j!$, and $\int_0^{\infty} \pi_j'(u)\, u^{-\gamma} du = \gamma \int_0^{\infty} \pi_j(u) u^{-\gamma-1} du = \gamma\Gamma(j-\gamma)/j!$. In

the Poissonised scheme $K(t)$ is a sum of independent indicators, so $\mathrm{Var}K(t) \le \mathbb{E}K(t)$; Chebyshev's inequality along $t_m = (1+\eta)^m$ and the Borel and Cantelli lemma give $K(t_m)/\mathbb{E}K(t_m) \to 1$ almost surely, and monotonicity of K in t fills the gaps up to a factor $(1+\eta)^\gamma$, which is released as $\eta \downarrow 0$. Finally a Poisson variable with mean t lies in $[t - t^{2/3},\, t + t^{2/3}]$ eventually, and K_n is monotone in n, so the statement transfers to fixed n. The counts A_j are not monotone in n, but each is a difference of the monotone quantities $\#\{k : \mathrm{count}_k \ge j\}$, which are handled the same way. The logarithmic case is identical with $c \log t$ in place of $t^\gamma L(t)$.

At $\gamma = 1$ the constant $\Gamma(1-\gamma)$ is infinite and the mean needs a different treatment. The substitution $y = 1/x$ turns $\int_0^1 N(x)\,dx = \sum_k p_k = 1$ into $\int_1^\infty L(y)y^{-1}dy = 1$, so L_1 is finite, and $\int_0^{1/t} N(x)\,dx = \sum_k \min(p_k, 1/t)$ is $L_1(t)$. For $\lambda > 1$, $L_1(y) - L_1(\lambda y) = \int_y^{\lambda y} L(u)u^{-1}du \sim L(y)\log\lambda$, so $\liminf L_1(y)/L(y) \ge \log\lambda$ for every λ; hence $L_1/L \to \infty$ and $L_1(\lambda y)/L_1(y) \to 1$. Now split $\mathbb{E}[K(t)] = t\int_0^\infty e^{-u}u^{-1}L(t/u)\,du$ at $u = 1$. The part $u > 1$ is $O(L(t))$ by Potter's bounds, which is $o(L_1(t))$. The part $u \le 1$ is $\int_t^\infty e^{-t/y}L(y)y^{-1}dy$, which lies between $e^{-1/\lambda}L_1(\lambda t)$ and $L_1(t)$; since $L_1(\lambda t) \sim L_1(t)$ and λ is arbitrary, it is $L_1(t)(1+o(1))$. So $\mathbb{E}[K(t)] \sim tL_1(t)$, which is regularly varying with index one, and the almost sure statement and the passage to fixed n go through as before. □

Two features of the lemma recur below. The constant $\Gamma(1-\gamma)$ is the only place the sampling enters: everything else is a property of the frequencies. And the profile $\gamma\Gamma(j-\gamma)/j!$ of the counts depends on γ alone, which is the source of every estimator of σ in Section 7.8.

7.2 The reduction, and why it is universal

LEMMA 7.3 (Reduction). *Let the intervals be laid down in any order, with tail sums R_k, and let $X_1, \dots, X_n$ be i.i.d. with density f. Then the induced partition of $\{1, \dots, n\}$ is a Kingman paintbox with frequencies*

$$p_k \;=\; \mathbb{P}(X \in V_k) \;=\; G(R_{k-1}) - G(R_k), \qquad \sum_k p_k = 1.$$

Proof. The kth interval is $(1-R_{k-1},\, 1-R_k]$, so $\mathbb{P}(X \in V_k) = F(1-R_k) - F(1-R_{k-1})$, which is $G(R_{k-1}) - G(R_k)$ by (7.2). The sum telescopes to $G(1) - \lim_k G(R_k) = 1$ since $R_k \downarrow 0$ and G is continuous at the origin. The points remain i.i.d., so the induced partition is exchangeable and is the paintbox with those frequencies. □

Two structural facts follow and hold for every paintbox. Under uniform sampling $G(r) = r$ gives $p_k = V_k$ for every arrangement, so the order the atoms are laid down in is irrelevant, which is why the classical theory never mentions it. Under any other weight the order matters, because p_k depends on where the kth interval sits and not only on how long it is. And the induced frequencies are again a paintbox, so we have not left Kingman's framework but moved inside it, from

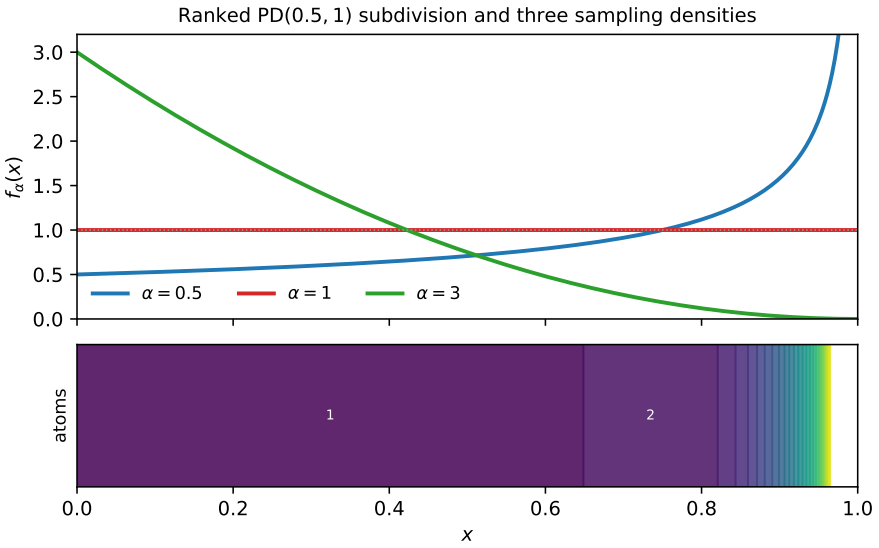


FIGURE 7.1. The construction. Below, a ranked PD$(0.5, 1)$ subdivision of $(0, 1)$, the first twenty five atoms shaded and the largest few labelled. Above, the sampling densities f_α for $\alpha = 0.5$, 1 and 3. Uniform sampling is the dotted line; every other density makes the position of an atom, and not merely its length, matter.

one paintbox to another. Everything that follows is a question about which laws of (R_k) leave the new frequencies tractable.

Figure 7.1 shows the objects. The lower panel is a ranked PD$(0.5, 1)$ subdivision; the upper panel is the sampling density for three values of α. A collector using $\alpha = 3$ takes almost half of the sample from the first fifth of the interval, which under a ranked layout is where the large atoms are, so a few atoms absorb almost all of the mass. A collector using $\alpha = 0.5$ does the reverse and reaches deep into the dust.

7.2.1 *The direction of the layout, and a reflection*

Lemma 7.3 was stated for intervals laid down from the left, so that the kth interval is $(1 - R_{k-1}, 1 - R_k]$ and the weight enters through $G(r) = 1 - F(1 - r)$. Nothing forces that choice, and the alternative is not a new case.

LEMMA 7.4 (Reflection). *Fix a subdivision with tail sums (R_k). Lay the intervals from the left and sample with density f; or lay them from the right, so that the kth interval is $(R_k, R_{k-1}]$, and sample with the reflected density $\check{f}(x) = f(1 - x)$. The induced frequencies coincide, both being $G(R_{k-1}) - G(R_k)$ with $G(r) = 1 - F(1 - r)$.*

Proof. The reflected distribution function is $\check{F}(x) = 1 - F(1 - x) = G(x)$. Laid from the right, the kth frequency is $\check{F}(R_{k-1}) - \check{F}(R_k)$, which is $G(R_{k-1}) - G(R_k)$. □

TABLE 7.2. The one parameter cases, reduced to G. The size biased layout is a residual allocation model in either direction; the ranked layout is not. The image is ESF when G is a power and the layout is size biased; rows four and five are not ESF at the values checked in the text.

layout	density	$G(r)$	image
size biased, from the left	1	r	ESF(θ)
size biased, from the left	$\alpha(1-x)^{\alpha-1}$	r^{α}	ESF(θ/α)
size biased, from the right	$\alpha x^{\alpha-1}$	r^{α}	ESF(θ/α)
size biased, from the left	$\alpha x^{\alpha-1}$	$1-(1-r)^{\alpha}$	not ESF; $K_n \sim \theta \log n$
ranked, either direction	$\alpha(1-x)^{\alpha-1}$	r^{α}	not ESF; $K_n \sim (\theta/\alpha)\log n$

So a density and a direction are not two pieces of data but one. Everything that follows is a statement about G alone, and a table of arrangements against densities collapses to a table of values of G. Table 7.2 records the collapse at $\sigma = 0$, where the exact theory lives.

Rows two and three are one statement by Lemma 7.4. Taking $\alpha = \theta$ in either gives ESF(1): the weight $\theta(1-x)^{\theta-1}$ undoes the stick breaking exactly, since its distribution function is the one that carries each Beta$(\theta, 1)$ residual factor to a uniform. Rows four and five are the two ways exactness can fail, and they fail for different reasons. In row four G is not multiplicative, which is the failure Theorem 7.11 shows to be generic. In row five G is multiplicative but the layout is not a residual allocation model, so Theorem 7.11 does not apply in the first place. The two conditions, $G(r) = r^{\alpha}$ and a size biased layout, are independent, and exactness needs both. The check is at $n = 2$: if $K_n \sim c \log n$ the only candidate law is ESF(c), under which two points share an interval with probability $1/(1+c)$, and at $\theta = 1$, $\alpha = 2$ row four gives 0.445 against $1/2$ and row five gives 0.728 against $2/3$.

The last column of Table 7.2 is asymptotic in its last two rows, and those are the content of the next two statements.

PROPOSITION 7.5 (The index is the contact order at the far endpoint). *Let the intervals be laid from the left and let f be continuous on $(0,1)$ with $f(x) \sim a(1-x)^{\alpha-1}$ as $x \uparrow 1$, for some $a > 0$ and $\alpha > 0$. Then*

$$G(r) = \int_0^r f(1-u)\,du \;\sim\; \frac{a}{\alpha}\, r^{\alpha}, \qquad r \downarrow 0,$$

so the density $G'(r) = f(1-r) \sim a\, r^{\alpha-1}$ is regularly varying at the origin with index $\alpha - 1$, G is regularly varying with index $\rho = \alpha$, and Proposition 7.32 below applies with that ρ. Since $G'(r)/r^{\alpha-1} \to \alpha\,(a/\alpha)$, the density condition of Section 7.10.3 holds with $c = a/\alpha$, and in the size biased layout $S_* = (a/\alpha)^{\sigma_*} C\, S^{\kappa}$ with C the constant of (7.12).

Only the behaviour of the sampling density at the endpoint the layout runs towards matters, and it matters through one number. Two consequences answer the cases that looked intractable.

If f has a positive finite limit a at the far endpoint, then $\alpha = 1$ and $\rho = 1$, so $\sigma_* = \sigma$ and $\kappa = 1$. The weight does not move the exponent at all, in either family; its whole effect is the constant a^σ multiplying the diversity. Row four of Table 7.2 is this case with $a = \alpha$: sampling a forward size biased layout with $\alpha x^{\alpha-1}$ gives no exact law, but $\sigma_* = \sigma$ and $S_* = \alpha^\sigma S$ almost surely, and at $\sigma = 0$ it gives $K_n \sim \theta \log n$ with the concentration untouched. The absence of a clean exact form is not the absence of an effect to report; the effect is a scale factor and nothing more.

If f vanishes at the far endpoint, $\alpha > 1$, the collector is fading and the exponent falls; if f blows up there, $\alpha < 1$, he is persistent and it rises. This is the reading of α given after Proposition 7.13, now attached to the density rather than to G.

PROPOSITION 7.6 (The ranked layout at $\sigma = 0$). *Lay the* PD(θ) *atoms in decreasing order and sample with* f_α. *Then* $K_n / \log n \to \theta/\alpha$ *almost surely, although at the values* $\alpha \neq 1$ *computed below and in the discussion of Table 7.2 the induced partition is not* ESF(θ/α).

Proof. The ranked PD(θ) atoms satisfy $-k^{-1} \log V_{(k)} \to 1/\theta$ almost surely, so $R_k = \sum_{i>k} V_{(i)} = e^{-k(1+o(1))/\theta}$ and, by Lemma 7.3, $R_k^\alpha = e^{-\alpha k(1+o(1))/\theta}$. Differencing, the induced frequencies have counting function $N(x) = (\theta/\alpha) \log(1/x)\,(1 + o(1))$, and Karlin's occupancy correspondence in the logarithmic case converts this to $K_n = (\theta/\alpha) \log n\,(1 + o(1))$. That the partition is not ESF does not follow from Theorem 7.11, which concerns the size biased layout (see the discussion of Table 7.2), but is a computation: for $\theta = \alpha = 2$ the ranked layout puts two points in the same interval with probability about 0.56, against $1/2$ for ESF(1), and Table 7.3 shows the difference at $n = 4$. □

REMARK 7.7. Propositions 7.6 and 7.15 are the same statement in the two families. In both, the arrangement of the atoms decides exactness and is invisible to the leading order, and in both the constant is known in closed form only in the size biased layout. What changes across the families is not the role of the layout but what survives it: at $\sigma = 0$ the leading order retains the whole parameter map $\theta \mapsto \theta/\alpha$, while at $\sigma > 0$ it retains the exponent map (7.9) and loses the law.

7.3 Rigidity: no weight preserves the two parameter family

In the one parameter case the exact theory rests on a single line. In size biased order $G(r) = r^\alpha$ gives induced residual $\widetilde{R}_k^\alpha = \prod_{i \le k}(1 - B_i)^\alpha$, and for $\sigma = 0$ the factors satisfy

$$1 - B_i \sim \text{Beta}(\theta, 1) \stackrel{d}{=} U^{1/\theta} \quad \Longrightarrow \quad (1 - B_i)^\alpha \stackrel{d}{=} U^{\alpha/\theta} \sim \text{Beta}(\theta/\alpha, 1), \tag{7.4}$$

so the induced subdivision is GEM(θ/α) and the catch is exactly ESF(θ/α) at every n. The middle equality in (7.4) is the whole calculation: if $\mathbb{P}(1 - B \le u) = u^\theta$ then $\mathbb{P}\{(1 - B)^\alpha \le v\} = \mathbb{P}(1 - B \le v^{1/\alpha}) = v^{\theta/\alpha}$, which is the distribution function of Beta$(\theta/\alpha, 1)$. Independence of the factors is untouched because each is transformed separately.

TABLE 7.3. Partition types of $n = 4$ points sampled with density $2(1-x)$ from a GEM(2) subdivision, over 10^6 replicates; standard errors are at most 0.0005. The theorem predicts ESF(1) in the size biased layout. The ranked layout uses the same sticks sorted.

type	size biased, simulated	ESF(1)	ESF(2)	ranked, simulated
4	0.2499	0.2500	0.1000	0.3146
3+1	0.3337	0.3333	0.2667	0.3575
2+2	0.1253	0.1250	0.1000	0.1134
2+1+1	0.2498	0.2500	0.4000	0.1912
1+1+1+1	0.0413	0.0417	0.1333	0.0233

EXAMPLE 7.8 (The one parameter theorem by simulation). Take $\theta = 2$ and $\alpha = 2$, so the theorem predicts that a sample drawn with density $f_2(x) = 2(1-x)$ from a size biased GEM(2) subdivision has exactly the law ESF(1), although under uniform sampling the same subdivision gives ESF(2). The two laws are far apart at $n = 4$: by Table 7.1, ESF(1) puts probability $1/4$ on a single block and $1/24$ on four singletons, while ESF(2) puts $1/10$ and $2/15$.

To check this we simulated 10^6 weighted paintboxes. Each replicate draws eighty sticks $B_k \sim \text{Beta}(1,2)$, lays them from the left, and drops four points with $1 - X = U^{1/2}$, so that $\mathbb{P}(1 - X \le r) = r^2 = G(r)$. The omitted mass after eighty sticks has mean $(2/3)^{80}$, below 10^{-14}, so the truncation is invisible. Table 7.3 gives the frequencies of the five types. In the size biased layout every one agrees with ESF(1) to within two standard errors, and none is anywhere near ESF(2). Sorting the same sticks into decreasing order before laying them down, and changing nothing else, gives the last column, which is neither law: the ranked layout puts the largest atoms where the fading collector works hardest, so a single block becomes more likely (0.315) and four singletons less likely (0.023) than under ESF(1). That column is Proposition 7.6 at work: the same G, the same leading order $(\theta/\alpha)\log n$, and no exact law.

An exact check is also available. The moment recursion of Example 7.10 below, run with $\sigma = 0$, returns the probabilities $1/4$, $1/12$, $1/24$, $1/24$, $1/24$ for the five set partition shapes to six decimal places, which are the ESF(1) values of (7.3).

The step that fails is visible in (7.4). It works because $\text{Beta}(\theta, 1)$ is the law of a power of a uniform, and powers of a uniform are closed under further powers. For $\sigma > 0$ the residual factor is $1 - B_k \sim \text{Beta}(\theta + k\sigma,\, 1 - \sigma)$, whose second parameter is not 1, so it is not a power of a uniform and $(1 - B_k)^\alpha$ leaves the Beta family altogether.

THEOREM 7.9. *Within the Ewens Pitman family,* $\text{PD}(\sigma, \theta)$ *is carried to another member of the family by the weight* f_α *in the size biased layout if and only if* $\sigma = 0$ *or* $\alpha = 1$.

Proof. Sufficiency at $\sigma = 0$ is (7.4), and at $\alpha = 1$ the map is the identity. For necessity, suppose the image is $\text{PD}(\sigma', \theta')$. Its residual factors are independent with $1 - B'_k \sim \text{Beta}(\theta' + k\sigma',\, 1 - \sigma')$, while by Lemma 7.3 the image factors are

$(1-B_k)^\alpha$ with $1-B_k \sim \mathrm{Beta}(\theta+k\sigma,\,1-\sigma)$. Matching laws, the αth power of a $\mathrm{Beta}(a,b)$ variable must be $\mathrm{Beta}(a',b')$ for every k, with $a=\theta+k\sigma$, $b=1-\sigma$. By the change of variables $y=x^\alpha$, the law of $Y=X^\alpha$ for $X\sim\mathrm{Beta}(a,b)$ has density proportional to $y^{a/\alpha-1}(1-y^{1/\alpha})^{b-1}$ on $(0,1)$. Suppose this equals a constant multiple of $y^{a'-1}(1-y)^{b'-1}$. Then

$$\bigl(1-y^{1/\alpha}\bigr)^{b-1} \;=\; c\,y^{a'-a/\alpha}(1-y)^{b'-1}, \qquad 0<y<1.$$

As $y\downarrow 0$ the left side tends to 1, so $a'=a/\alpha$ and $c=1$. As $y\uparrow 1$, $1-y^{1/\alpha}\sim(1-y)/\alpha$, so the left side behaves like $\alpha^{1-b}(1-y)^{b-1}$ and $b'=b$. Dividing, the function $y\mapsto\{(1-y^{1/\alpha})/(1-y)\}^{b-1}$ is identically one. But $(1-y^{1/\alpha})/(1-y)$ tends to 1 at the origin and to $1/\alpha$ at the other end, so for $\alpha\neq 1$ it is not constant, and the identity forces $b-1=0$, that is $\sigma=0$. □

What the image *is*, explicitly, is the residual allocation model with independent factors of density proportional to

$$y^{(\theta+k\sigma)/\alpha-1}\bigl(1-y^{1/\alpha}\bigr)^{-\sigma}, \qquad 0<y<1, \tag{7.5}$$

obtained by pushing $\mathrm{Beta}(\theta+k\sigma,1-\sigma)$ through $y\mapsto y^\alpha$. It is a perfectly explicit law; it is simply not Ewens Pitman.

How far from Ewens Pitman is it? Theorem 7.9 is a statement about the infinite sequence of factors, and one may wonder whether the failure is visible in a sample one could actually collect. It is, already at $n=4$, and the calculation is exact.

EXAMPLE 7.10 (The failure at $n=4$, computed exactly). For any residual allocation model with independent factors Y_k and frequencies $p_k=R_{k-1}(1-Y_k)$, $R_k=R_{k-1}Y_k$, the power sums $m_j=\mathbb{E}\sum_k p_k^j$ satisfy

$$m_j \;=\; \sum_{k\ge 1}\Bigl\{\prod_{i<k}\mathbb{E}\bigl[Y_i^j\bigr]\Bigr\}\,\mathbb{E}\bigl[(1-Y_k)^j\bigr],$$

and the second moment $Q=\mathbb{E}\{(\sum_k p_k^2)^2\}$ follows from the backward recursion $T_k=(1-Y_k)^2+Y_k^2T_{k+1}$ for the scaled tail $T_k=\sum_{i\ge k}(p_i/R_{k-1})^2$:

$$\mathbb{E}T_k^2 \;=\; \mathbb{E}(1-Y_k)^4+2\,\mathbb{E}\bigl[(1-Y_k)^2Y_k^2\bigr]\mathbb{E}T_{k+1}+\mathbb{E}\bigl[Y_k^4\bigr]\mathbb{E}T_{k+1}^2,$$

with $Q=\mathbb{E}T_1^2$. For the image (7.5), $Y_k=X_k^\alpha$ with $X_k\sim\mathrm{Beta}(\theta+k\sigma,1-\sigma)$, every moment needed is a ratio of Beta functions, $\mathbb{E}[Y_k^m]=B(\theta+k\sigma+\alpha m,\,1-\sigma)/B(\theta+k\sigma,\,1-\sigma)$, and $\mathbb{E}[(1-Y)^jY^i]$ follows by the binomial theorem. The set partition probabilities at $n=4$ are then

$$p(4)=m_4, \qquad p(3,1)=m_3-m_4, \qquad p(2,2)=Q-m_4,$$
$$p(2,1,1)=m_2-2m_3-Q+2m_4,$$

and $p(1,1,1,1)$ from normalisation. (The expression for $p(2,1,1)$ comes from adding a fourth point to the partition $\{1,2\}\{3\}$: it joins the pair, joins the singleton, or starts a block.) Run with $\alpha=1$ the recursion reproduces (7.3) exactly, as it must.

TABLE 7.4. Partition types at $n = 4$ for the weighted image of PD$(\frac{1}{2}, 1)$, computed exactly, against the unique Ewens Pitman law with the same m_2 and m_3.

	$\alpha = 2$, fit $(0.3642, 0.5299)$			$\alpha = 4$, fit $(0.2530, 0.2919)$		
type	image	fitted	ratio	image	fitted	ratio
4	0.20131	0.20068	1.0031	0.36899	0.36776	1.0033
3+1	0.26975	0.27227	0.9907	0.28693	0.29182	0.9832
2+2	0.08113	0.07937	1.0221	0.09716	0.09358	1.0382
2+1+1	0.31437	0.31411	1.0008	0.20013	0.19994	1.0010
1+1+1+1	0.13344	0.13357	0.9990	0.04679	0.04688	0.9980

Now take $(\sigma, \theta, \alpha) = (\frac{1}{2}, 1, 2)$. The recursion, truncated at 2×10^5 factors, gives $m_2 = 0.41563$ and $m_3 = 0.26875$. Any Ewens Pitman law is pinned down by these two numbers, since $m_2 = (1 - \sigma')/(1 + \theta')$ and $m_3 = (1 - \sigma')(2 - \sigma')/\{(1 + \theta')(2 + \theta')\}$; solving gives $(\sigma', \theta') = (0.3642, 0.5299)$. If the image belonged to the family it would have to be PD$(0.3642, 0.5299)$, and Table 7.4 compares the two at $n = 4$. The type 2+2 is off by about two percent, and at $\alpha = 4$ by almost four. The other types are matched more closely, which is partly forced: the fit uses m_2 and m_3, and these fix the laws of the first two and first three points exactly, so the fourth point is the only room left for a discrepancy. A Monte Carlo run of 4×10^5 weighted paintboxes at $\alpha = 2$ gives 0.2000, 0.2707, 0.0803, 0.3152, 0.1339 for the five types, in agreement with the exact column to within about two standard errors.

Two further points are worth noticing. The fitted $\sigma' = 0.364$ is not the asymptotic $\sigma_* = 1/3$ of Section 7.5: matching small samples and matching growth rates are different questions once the family is left. And the discrepancies are small. A collector with four specimens could not detect them, which is why the rigidity theorem matters as a statement of principle rather than as a test: the obstruction is real but it is the asymptotic theory of the following sections, not exact small sample formulae, that carries the information.

It is natural to ask whether some other, necessarily larger, family of weights does better. The answer is no, and the obstruction sits one level deeper than the Beta algebra of (7.4): the residual allocation structure itself already pins the weight down.

THEOREM 7.11. *Let G be continuous and strictly increasing with $G(0) = 0$, $G(1) = 1$ (equivalently, the sampling density charges every interval), and lay the atoms in size biased order. Then the induced subdivision, read in the order in which its intervals are laid, is again a residual allocation model with independent factors if and only if $G(r) = r^\alpha$ for some $\alpha > 0$. Consequently, for $\sigma \in (0, 1)$ no such sampling density and direction makes the induced subdivision, in its layout order, the stick breaking sequence of another member of the Ewens Pitman family, apart from the uniform density; by Lemma 7.4 the enlargement by reflection is vacuous.*

Proof. By Lemma 7.3 the induced residual is $R'_k = G(\widetilde{R}_k)$, so the induced residual

factors are

$$\frac{R'_k}{R'_{k-1}} = \frac{G\big(\widetilde{R}_{k-1}(1-B_k)\big)}{G(\widetilde{R}_{k-1})} =: H\big(\widetilde{R}_{k-1},\, 1-B_k\big).$$

For the induced sequence to be a residual allocation model this factor must be independent of R'_{k-1}, hence of $\widetilde{R}_{k-1}$, which is a strictly increasing function of it. Write $U = 1 - B_k$, independent of $\widetilde{R}_{k-1}$ and supported on $(0,1)$ with a density. For each fixed r the map $s \mapsto H(r,s) = G(rs)/G(r)$ is strictly increasing, so the q quantile of $H(r,U)$ is $H(r,s_q)$ with s_q the q quantile of U. Independence forces every quantile of $H(r,U)$ to be free of r, so $r \mapsto H(r,s_q)$ is constant for each $q \in (0,1)$. The quantiles s_q are dense in $(0,1)$ because U has a density there, and H is continuous, so $H(r,s)$ is constant in r for every s. Letting $r \to 1$ identifies the constant as $G(s)$, giving $G(rs) = G(r)G(s)$ on $(0,1)^2$. Since G is strictly increasing with $G(0) = 0$, it is positive on $(0,1]$, so $g(u) = -\log G(e^{-u})$ is well defined, continuous and increasing on $[0,\infty)$ with $g(0) = 0$, and the multiplicative equation becomes Cauchy's equation $g(u+v) = g(u) + g(v)$. Its continuous solutions are linear, $g(u) = \alpha u$, and monotonicity forces $\alpha > 0$; returning to G, $G(r) = r^\alpha$.

For the second claim, the stick breaking sequence of every Ewens Pitman law is a residual allocation model with independent factors. So if the induced sequence in layout order is such a sequence, G is a power by the first part, and Theorem 7.9 forces $\sigma = 0$ or $\alpha = 1$. □

The theorem is narrower than the question that prompted it, in two ways, and we state both. First, the quantile step uses strict monotonicity of G, so a density that vanishes on an interval is not covered. Second, membership of the family is read in the strong sense that the induced frequencies, in the order in which they are laid, are the sticks of a member of the family. The weaker statement, that the induced partition has an Ewens Pitman law although its layout order is not a size biased order, asks only that the ranked induced frequencies be Poisson Dirichlet, and the residual allocation argument does not reach it. For Beta weights Example 7.10 settles the weaker statement at the points it computes: at $(\sigma,\theta) = (\frac{1}{2},1)$ with $\alpha = 2$ or 4 the partition law at $n = 4$ is not that of any Ewens Pitman law, because the only one with the same m_2 and m_3 misses the type 2+2 by about two and four percent. For other parameter values and for a general weight the weaker statement is not proved here.

The rigidity has two independent halves, and they fail in different places. Independence of the residual factors is preserved only by powers; that half holds for every paintbox and has nothing to do with Ewens or Pitman Yor. Among powers, the Ewens family alone is preserved; that half is the Beta algebra of (7.4). The one parameter theory passes both. The two parameter family passes the first and fails the second, and no enlargement of the weight class can help, because the first half has already used up the freedom.

7.4 A genealogical reading

The one parameter sampling formula has a derivation from the coalescent that is more transparent than any of its algebraic ones, and Warren Ewens asks whether it has an analogue for $\sigma > 0$. The answer says something about why the two parameter family is a different kind of object, not merely a richer one.

7.4.1 The defining event construction, and why it forces $\sigma = 0$

Trace backwards up Kingman's coalescent until a defining event occurs. With j branches present, the event in $(t, t+dt)$ is a mutation with probability $j\theta\,dt/2$ or a coalescence with probability $j(j-1)\,dt/2$; in both cases one stops tracing that lineage, so $j-1$ branches remain. The defining event is a mutation with probability

$$p_j \;=\; \frac{j\theta/2}{j\theta/2 + j(j-1)/2} \;=\; \frac{\theta}{\theta + j - 1}, \tag{7.6}$$

and iterating from $j = n$ down to $j = 1$ produces $\mathrm{ESF}(\theta)$. The mutation probability (7.6) is a function of the number of branches alone.

That last sentence is the whole obstruction.

PROPOSITION 7.12. *Consider any genealogical rule of the above form: with j branches present the defining event is a mutation with probability $p_j \in [0,1]$ and a binary coalescence otherwise, with p_j depending only on j. Then the number of types satisfies*

$$K_n = \sum_{j=1}^{n} \xi_j, \qquad \xi_j \sim \mathrm{Bernoulli}(p_j) \text{ independent}, \qquad \text{so} \qquad \mathrm{Var}[K_n] \le \mathbb{E}[K_n].$$

No such rule produces $\mathrm{PD}(\sigma,\theta)$ with $\sigma > 0$, for which $\mathrm{Var}[K_n] \asymp n^{2\sigma} \gg \mathbb{E}[K_n] \asymp n^{\sigma}$.

Proof. Each defining event reduces the branch count by one and creates a new type precisely when it is a mutation, so K_n counts the mutations among the n defining events. Given the branch count the type of each event is decided by a coin that depends on nothing else, so the indicators are independent and $\mathrm{Var}[K_n] = \sum_j p_j(1-p_j) \le \sum_j p_j = \mathbb{E}[K_n]$. Under $\mathrm{PD}(\sigma,\theta)$ with $\sigma > 0$ and $\theta \neq 0$ the mean is exact,

$$\mathbb{E}[K_n] \;=\; \frac{\theta}{\sigma}\Big\{\frac{(\theta+\sigma)_n}{(\theta)_n} - 1\Big\} \;\sim\; \frac{\Gamma(\theta+1)}{\sigma\,\Gamma(\theta+\sigma)}\, n^{\sigma} \;=\; \mathbb{E}[S]\, n^{\sigma},$$

the last equality being (7.10) at $p = 1$. (At $n = 3$, $\sigma = \frac{1}{2}$, $\theta = 1$ the formula gives 2.375, which is $1\cdot\frac{1}{8} + 2\cdot\frac{3}{8} + 3\cdot\frac{1}{2}$ from Table 7.1.) Since $K_n n^{-\sigma} \to S$ almost surely, $(K_n - \mathbb{E}K_n)n^{-\sigma} \to S - \mathbb{E}S$ almost surely, and Fatou's lemma gives $\liminf_n n^{-2\sigma}\mathrm{Var}[K_n] \ge \mathrm{Var}[S] > 0$. So $\mathrm{Var}[K_n]$ is of order at least $n^{2\sigma}$, which is of larger order than $\mathbb{E}[K_n]$, and the inequality $\mathrm{Var}[K_n] \le \mathbb{E}[K_n]$ fails for all large n. (For $\theta \in (-\sigma, 0)$ the same formula holds, and at $\theta = 0$ it holds in the limiting form $\mathbb{E}[K_n] = (\sigma)_n/\{\sigma\,(n-1)!\}$, with the same asymptotics.) As a check, (7.6) gives $\mathbb{E}[K_n] = \sum_{j=1}^{n} \theta/(\theta+j-1)$, the $\mathrm{ESF}(\theta)$ mean, and $\mathrm{Var}[K_n] =$

$\mathbb{E}[K_n] - \sum_j \theta^2/(\theta + j - 1)^2$, so the variance is smaller than the mean by a bounded amount, which is what the Feller coupling gives. □

The two parameter family is overdispersed in its type count, and a branch counting rule with independent defining events cannot be. This is the genealogical shadow of Proposition 7.24: for $\sigma > 0$ the whole abundance spectrum is one random variable times a deterministic profile, and that random variable is the diversity. A construction in which the randomness is a sequence of independent coin flips has nowhere to put it.

7.4.2 The two ways out

The literature resolves this in two ways, and they correspond to relaxing the two hypotheses of Proposition 7.12.

Let the rate depend on how many types have appeared. Keep one event per step but replace (7.6) by

$$p_{n,k} \;=\; \frac{\theta + k\sigma}{\theta + n}, \tag{7.7}$$

the probability that the $(n+1)$th draw is a new type when k have been seen. This is the (σ, θ) Chinese restaurant, it produces $\mathrm{PD}(\sigma, \theta)$ exactly, and it is the literal analogue of Warren's rule: (7.6) is (7.7) at $\sigma = 0$ read in the other direction. What it is not is a rule about branches. The dependence on k means the construction must remember how many types it has already made, which a coalescent traced upwards does not. The randomness that Proposition 7.12 had nowhere to put is exactly this feedback: a run of new types makes further new types more likely, and that is what generates the nondegenerate S.

Let several branches merge at once. Keep the mutation rate proportional to the number of branches and replace Kingman's binary coalescent by a Λ coalescent with multiple collisions (Pitman, 1999; Sagitov, 1999). With Λ the $\mathrm{Beta}(2 - a, a)$ distribution one obtains the genealogy of an a stable branching process (Birkner et al., 2005), embedded in continuous stable random trees (Berestycki, Berestycki and Schweinsberg, 2007), and the total branch length of order n^{2-a} makes the number of observed types grow like a power of n rather than like $\log n$. Möhle (2010) treats the two parameter Poisson Dirichlet coalescent directly. Matching the growth exponents requires $2 - a = \sigma$. The resulting partition is not $\mathrm{PD}(\sigma, \theta)$ on the nose, and the exact correspondence between Λ coalescents and Ewens Pitman laws is delicate. But the mechanism is the one a geneticist would want: a single individual leaving a large number of offspring, which produces both multiple mergers and a heavy tailed abundance distribution.

7.4.3 What ascertainment does to the genealogy

The results of this chapter now say something a geneticist can use, because the exponent is what the genealogical models are calibrated on.

A sample drawn with effort profile f_α from a two parameter population has $\log K_n / \log n \to \sigma_*$ by Proposition 7.13, and the whole abundance spectrum follows

the profile $q_j(\sigma_*)$ by Proposition 7.24. An analyst who fits a multiple merger model to such a sample, whether by the growth of the type count or by the shape of the frequency spectrum, therefore infers σ_* and not σ. He concludes that offspring numbers are less or more heavy tailed than they are, according to whether his effort faded or persisted. By Corollary 7.22 the inference is repaired exactly, $\sigma = \alpha\sigma_*/(\alpha\sigma_* + 1 - \sigma_*)$, and by Corollary 7.35 the required index can be read off the recorded sampling positions. A biologist inferring reproductive skew from an unevenly sampled population is estimating a compound of skew and sampling effort, and the two are separable only with the positions.

At $\sigma = 0$ the corresponding statement is the classical one. The catch is exactly $\mathrm{ESF}(\theta/\alpha)$, so the defining event rule (7.6) still describes the sample with θ replaced by θ/α: ascertainment rescales the mutation rate and the coalescent is untouched. That is the cleanest way to say what the one parameter theorem means genealogically, and by Theorem 7.11 it has no counterpart for $\sigma > 0$: there is no reparameterisation of the genealogy that absorbs the weight.

7.5 The first order law

Exactness goes and a first order law of the same shape takes its place.

PROPOSITION 7.13. *Lay the* $\mathrm{PD}(\sigma, \theta)$ *atoms in size biased order,* $0 < \sigma < 1$, *and sample with* f_α. *Then almost surely*

$$\widetilde{R}_k\, k^{(1-\sigma)/\sigma} \longrightarrow M \in (0, \infty), \qquad \textit{and} \qquad \frac{\log K_n}{\log n} \longrightarrow \sigma_* = \frac{\sigma}{\alpha(1-\sigma) + \sigma}. \tag{7.8}$$

Proof. By Lemma 7.3 the induced residual is $\widetilde{R}_k^{\alpha}$ exactly, so it suffices to treat $\alpha = 1$ and raise to the power at the end. The factors in (7.1) are independent, and $1 - B_i \sim \mathrm{Beta}(\theta + i\sigma,\, 1 - \sigma)$. For $X \sim \mathrm{Beta}(a, b)$ one has $\mathbb{E}[\log X] = \psi(a) - \psi(a + b)$ and $\mathrm{Var}[\log X] = \psi'(a) - \psi'(a + b)$, with ψ the digamma function. Since $\psi(x + b) - \psi(x) = b/x + O(x^{-2})$ and $\psi'(x) - \psi'(x + b) = O(x^{-2})$ as $x \to \infty$, taking $x = \theta + i\sigma$ and $b = 1 - \sigma$ gives

$$\mathbb{E}\big[-\log(1 - B_i)\big] = \frac{1 - \sigma}{i\sigma} + O(i^{-2}), \qquad \mathrm{Var}\big[\log(1 - B_i)\big] = O(i^{-2}).$$

The centred variables $\log(1 - B_i) - \mathbb{E}\log(1 - B_i)$ are independent with summable variances, so by Kolmogorov's convergence criterion their partial sums converge almost surely. The partial sums of the means are $\beta(\log k + \gamma) + c + o(1)$ for a finite constant c, where $\beta = (1 - \sigma)/\sigma$ and γ is Euler's constant, because the $O(i^{-2})$ corrections are summable. Adding, $\log \widetilde{R}_k + \beta \log k$ converges almost surely to a finite limit, which is the first claim with M the exponential of that limit, so $0 < M < \infty$.

Raising to the power α gives $R'_k k^{\alpha\beta} \to M^{\alpha}$. The individual induced frequencies $p'_k = \widetilde{R}_{k-1}^{\alpha}\{1 - (1 - B_k)^{\alpha}\}$ are not of exact order $k^{-\alpha\beta-1}$, because B_k fluctuates on its own scale, and here some care is needed: the sandwich argument in the proofs

of Lemma 7.16 and Proposition 7.18, which uses only the first claim of the present proposition, shows that $x^{\sigma_*} N_*(x)$ converges almost surely to a positive finite limit, where $1/\sigma_* = \alpha\beta + 1$. That relation is $(1-\sigma_*)/\sigma_* = \alpha\beta$, which rearranges to the stated σ_*. Lemma 7.2 then gives $K_n \sim \text{const} \cdot n^{\sigma_*}$ almost surely, and in particular $\log K_n / \log n \to \sigma_*$. □

Three readings of the formula follow.

The natural coordinate is the odds. Writing $\omega = (1-\sigma)/\sigma$, the conclusion is

$$\omega \;\longmapsto\; \alpha\,\omega. \tag{7.9}$$

Ascertainment multiplies the odds of the discount parameter by α, exactly as it divides the Ewens parameter by α. In both cases the effect is a scaling in the coordinate that linearises the growth of K_n, and (7.9) is the cleaner statement of the two. It is also the reason the curves in the left panel of Figure 7.2 look unrelated while those in the right panel are parallel lines.

The concentration does not move. The parameter θ enters Proposition 7.13 only through $O(i^{-2})$ terms, so it does not appear in σ_* at all. At first order ascertainment acts on the discount parameter and leaves the concentration alone. This is the reverse of the one parameter picture, where θ is the only parameter and takes the whole effect.

The limits are consistent. As $\sigma \downarrow 0$, $\sigma_* \to 0$ and the growth returns to the logarithmic scale where the one parameter theory lives. At $\alpha = 1$, $\sigma_* = \sigma$. A fading weight, $\alpha > 1$, gives $\sigma_* < \sigma$ and a persistent one gives $\sigma_* > \sigma$, so an uneven sample makes a two parameter community look less or more diverse in exactly the sense the exponent measures.

EXAMPLE 7.14 (The odds law with numbers). Take a community with $\sigma = \frac{1}{2}$, so that the odds are $\omega = 1$, and a collector whose effort fades linearly along the transect, $f_2(x) = 2(1-x)$, so $\alpha = 2$. The odds double to 2, and $\sigma_* = 1/(1+2) = 1/3$: types accumulate like $n^{1/3}$ instead of $n^{1/2}$. A community with $\sigma = 0.6$, odds $2/3$, sampled with $\alpha = 1.5$ has odds 1 after ascertainment and so $\sigma_* = \frac{1}{2}$, the same exponent as an unweighted survey of the first community. Neither survey can tell which of these two situations it is in; that is the confounding of Section 7.1.2 in its simplest arithmetic form.

The constants can be filled in too. With $\theta = 1$ the results of Section 7.6 give $\kappa = 4/3$, $C = 0.8624$ and $\mathbb{E}[S_*] = 2.654$, against $\mathbb{E}[S] = 2.257$ for the uniform survey. So the prediction for the weighted survey is $\mathbb{E}[K_n] \approx 2.654\, n^{1/3}$, to which Proposition 7.27 adds a correction of $-\frac{1}{2}$, while a uniform survey would give $\mathbb{E}[K_n] \approx 2.257\, n^{1/2}$. At $n = 10^6$ these are about 265 and 2257 types. Table 7.5 compares the weighted prediction with simulation over 1500 independent stick realisations, each with 2×10^5 sticks (ample, by the criterion (7.19) of Section 7.9.3) and Rao Blackwellised counts $\mathbb{E}[K_n \mid P] = \sum_k \{1 - (1-p_k)^n\}$. The ratio approaches one from below and is within one percent of it from $n = 10^5$ on; the successive local slopes of $\log \mathbb{E}[K_n]$ against $\log n$ are 0.375, 0.350, 0.341, 0.337, 0.335, settling on

TABLE 7.5. The weighted type count at $\sigma = \frac{1}{2}$, $\theta = 1$, $\alpha = 2$: simulated mean with its standard error over 1500 replicates, against $\mathbb{E}[S_*]\, n^{1/3} - \frac{1}{2}$.

n	simulated $\mathbb{E}[K_n]$	s.e.	predicted	ratio
10^2	10.50	0.13	11.82	0.888
10^3	24.86	0.33	26.04	0.955
10^4	55.66	0.77	56.68	0.982
10^5	122.04	1.71	122.68	0.995
10^6	265.22	3.74	264.89	1.001
10^7	573.40	8.11	571.27	1.004

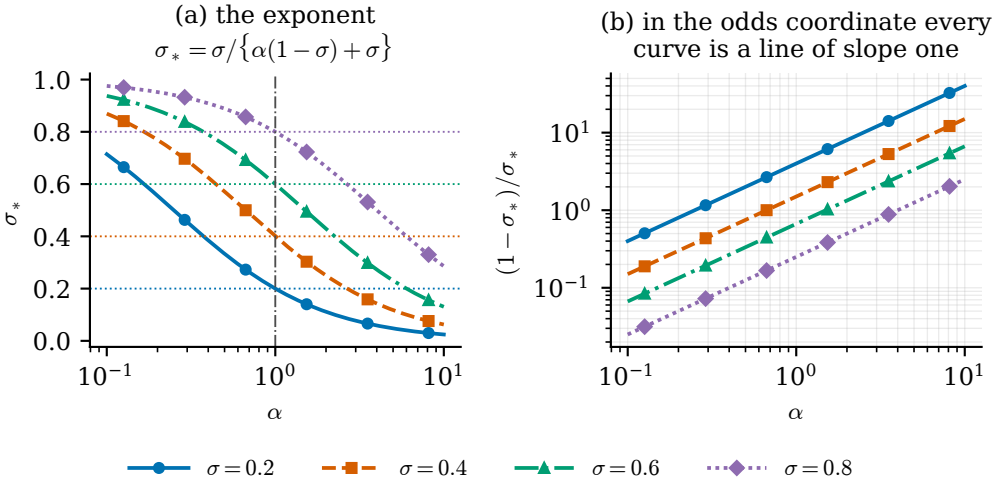


FIGURE 7.2. The exponent map. Left, σ_* against α for four values of σ, with the unweighted values dotted and $\alpha = 1$ marked. Right, the same information in the odds coordinate: every curve becomes a line of slope one through (7.9), which is why two different (σ, α) pairs with the same $\alpha(1-\sigma)/\sigma$ give the same exponent.

1/3 from above. The slow approach of the slopes is the reason, given in Section 7.8, not to estimate σ_* from the growth of K_n.

Figure 7.3 shows the exponent directly, by fitting the slope of $\log \mathbb{E}[N(x)]$ on $\log(1/x)$ over the range where the truncated stick is still exact. The legend fits are close for $\sigma = 0.5$ with $\alpha = 1$ and 2 and rough for the other two curves (0.542 against 0.667 and 0.208 against 0.176). Table 7.6 collects a wider set, including the control $\alpha = 1$ where the answer must be σ, and two rows sharing a σ_* through (7.9).

7.5.1 The layout does not change the exponent

In the one parameter theory the arrangement of the atoms is decisive for the constant term and irrelevant for the leading order. The same is true here, and for a

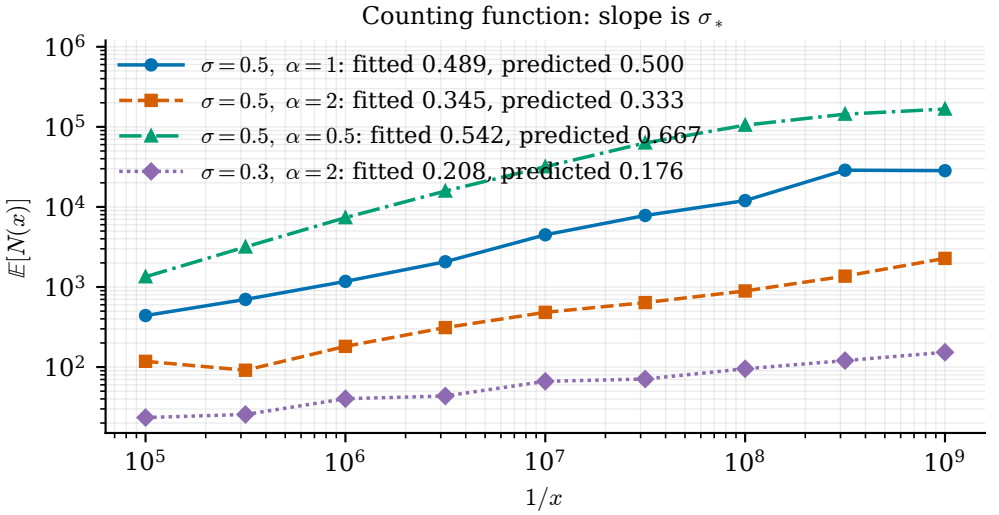


FIGURE 7.3. Counting functions of the induced frequencies over four decades in x, on log log axes. The slope is σ_*. Fitted and predicted values are in the legend; the pair $(\sigma, \alpha) = (0.5, 1)$ and $(0.3, 2)$ shows how far the exponent can move.

reason that is visible in one line.

PROPOSITION 7.15. *Lay the* PD(σ, θ) *atoms in decreasing rather than size biased order and sample with* f_α. *Then* $\log K_n / \log n \to \sigma_*$, *the same exponent as in Proposition 7.13.*

Proof. The ranked atoms of PD(σ, θ) satisfy $V_{(k)} k^{1/\sigma} \to T$ a.s. for a positive random T, so the ranked tail sums satisfy

$$R_k = \sum_{i>k} V_{(i)} \asymp \int_k^\infty T u^{-1/\sigma} du = \frac{\sigma T}{1-\sigma} k^{-(1-\sigma)/\sigma},$$

which is the same power of k as the size biased residual of Proposition 7.13. Since Lemma 7.3 makes the induced frequencies a function of the tail sums alone, and since the decreasing layout case of the proof of Proposition 7.32, with $G(r) = r^\alpha$ and $\rho = \alpha$, uses only that power and the decay $V_{(k)} / R_{k-1} \sim \beta / k$, not the constant, the conclusion is unchanged. □

Two remarks. First, this is not a triviality even though the proof is short: the two layouts have genuinely different tail sum *processes*, and what the argument uses is only that they share a power. Second, the constant is a different matter. In the size biased layout the residual is a product of independent factors, so M is the limit of a normalised random walk. In the decreasing layout the tail sums are neither independent nor Markov in any convenient sense, and the analogue of Lemma 7.16 would need the joint law of the ranked atoms. This is exactly the split found in the one parameter case, where the leading order is layout free and

TABLE 7.6. The exponent σ_*, predicted and fitted. Fitted values are the slope of $\log \mathbb{E}[N(x)]$ on $\log(1/x)$ over a window above the truncation floor of a 2×10^6 stick approximation, over 40 replicates. The first and last rows have $\alpha(1-\sigma)/\sigma = 1$ and so share an exponent.

σ	θ	α	predicted σ_*	fitted
0.5	1	1	0.5000	0.4996
0.5	1	2	0.3333	0.3331
0.5	1	0.5	0.6667	0.6669
0.3	1	2	0.1765	0.1763
0.7	0	3	0.4375	0.4360
0.6	1	1.5	0.5000	0.5005

the constant is known in closed form only in the size biased layout. Everything in Section 7.6 should therefore be read as a size biased statement.

7.6 The diversity is a power of the unweighted diversity

Under PD(σ, θ) the ratio K_n/n^σ converges almost surely to the σ diversity $S = S_{\sigma,\theta}$, a Mittag Leffler variable with

$$\mathbb{E}[S^p] \;=\; \frac{\Gamma(\theta+1)\,\Gamma(\theta/\sigma+p+1)}{\Gamma(\theta/\sigma+1)\,\Gamma(\theta+p\sigma+1)}, \qquad p > -1-\theta/\sigma. \tag{7.10}$$

Proposition 7.13 gives the exponent of the weighted problem but says nothing about the limit of $K_n n^{-\sigma_*}$. In fact the two limits are the same random variable in disguise, and the route to that goes through an identity worth recording on its own.

LEMMA 7.16. *With $M = \lim_k \widetilde{R}_k k^{(1-\sigma)/\sigma}$ as in Proposition 7.13,*

$$S \;=\; \Big(\frac{M}{\sigma}\Big)^{\sigma} \qquad \textit{almost surely.}$$

Proof. Represent the stick fractions through independent Gamma variables, $B_k = \Gamma_k/(\Gamma_k+\Gamma'_k)$ with $\Gamma_k \sim \Gamma(1-\sigma,1)$ and $\Gamma'_k \sim \Gamma(\theta+k\sigma,1)$, all independent. Then

$$p_k \;=\; \widetilde{R}_{k-1}B_k \;=\; \widetilde{R}_{k-1}\,\frac{\Gamma_k}{\Gamma_k+\Gamma'_k} \;=\; \frac{M}{\sigma}\,\Gamma_k\,k^{-1/\sigma}\,\varepsilon_k, \qquad \varepsilon_k := \frac{\widetilde{R}_{k-1}k^{\beta}}{M}\cdot\frac{k\sigma}{\Gamma_k+\Gamma'_k}.$$

The first factor of ε_k tends to 1 almost surely by (7.8). For the second, Γ'_k has mean $\theta+k\sigma$ and a Gamma tail, so $\mathbb{P}(|\Gamma'_k/(k\sigma)-1|>\eta) \le 2e^{-c(\eta)k}$ is summable and $\Gamma'_k/(k\sigma) \to 1$ almost surely; since $\Gamma_k = O(\log k)$ almost surely by the same summability, $k\sigma/(\Gamma_k+\Gamma'_k) \to 1$ as well. Hence for every $\eta>0$ there is an almost surely finite k_0 with $1-\eta \le \varepsilon_k \le 1+\eta$ for $k \ge k_0$.

Fix η and write $c_\pm = \sigma x/\{(1\mp\eta)M\}$. For $k \ge k_0$,

$$\mathbf{1}\{\Gamma_k > c_+k^{1/\sigma}\} \;\le\; \mathbf{1}\{p_k > x\} \;\le\; \mathbf{1}\{\Gamma_k > c_-k^{1/\sigma}\},$$

so $N(x)$ is sandwiched, up to the k_0 early terms, between two sums $N_\pm(x) = \sum_{k\ge k_0} \mathbf{1}\{\Gamma_k > c_\pm k^{1/\sigma}\}$ of independent indicators. Their means are

$$\Lambda_\pm(x) = \sum_{k\ge k_0} \mathbb{P}\big(\mathcal{G} > c_\pm k^{1/\sigma}\big) \;\sim\; \int_0^\infty \mathbb{P}\big(\mathcal{G} > c_\pm u^{1/\sigma}\big)\,du \;=\; c_\pm^{-\sigma}\,\mathbb{E}[\mathcal{G}^\sigma] \quad (x\downarrow 0),$$

the last step by the substitution $t = c_\pm u^{1/\sigma}$ and the identity $\sigma\int_0^\infty t^{\sigma-1}\mathbb{P}(\mathcal{G} > t)\,dt = \mathbb{E}[\mathcal{G}^\sigma]$. Since $\mathrm{Var} N_\pm \le \Lambda_\pm$ and $\Lambda_\pm(x)\to\infty$, Kolmogorov's criterion along $x_j = (1+\delta)^{-j}$ gives $N_\pm(x_j)/\Lambda_\pm(x_j)\to 1$ almost surely, and monotonicity of N in x extends this to all $x\downarrow 0$ up to a factor $(1+\delta)^{\pm\sigma}$, which is released as $\delta\downarrow 0$. Therefore

$$(1-\eta)^\sigma \;\le\; \liminf_{x\downarrow 0} \frac{x^\sigma N(x)}{(M/\sigma)^\sigma \mathbb{E}[\mathcal{G}^\sigma]} \;\le\; \limsup_{x\downarrow 0} \frac{x^\sigma N(x)}{(M/\sigma)^\sigma \mathbb{E}[\mathcal{G}^\sigma]} \;\le\; (1+\eta)^\sigma,$$

and η was arbitrary. For a $\Gamma(1-\sigma,1)$ variable $\mathbb{E}[\mathcal{G}^p] = \Gamma(1-\sigma+p)/\Gamma(1-\sigma)$, so $\mathbb{E}[\mathcal{G}^\sigma] = 1/\Gamma(1-\sigma)$ and $x^\sigma N(x)\to (M/\sigma)^\sigma/\Gamma(1-\sigma)$ almost surely. Karlin's occupancy correspondence, $K_n n^{-\sigma}\to\Gamma(1-\sigma)\lim x^\sigma N(x)$, multiplies by $\Gamma(1-\sigma)$ and the two factors cancel. □

That the Γ factors cancel is the reason the rest of this section is clean. The σ diversity is a deterministic power of the stick envelope with no constant left over, so $M = \sigma S^{1/\sigma}$ can be substituted wherever the envelope appears. Since everything downstream rests on the identity, we checked it directly: estimating M from the tail of the stick and S from the counting function on the same realisations, the ratio $S/(M/\sigma)^\sigma$ has mean 0.9995, 1.0020, 0.9970 and 0.9991 at $(\sigma,\theta) = (0.5,1)$, $(0.3,1)$, $(0.6,0)$ and $(0.5,2)$, over 80 replicates with 4×10^5 sticks.

PROPOSITION 7.17. *Write S_* for the almost sure limit of $K_n n^{-\sigma_*}$ under the weighted paintbox. Then*

$$S_* \;=\; C\,S^\kappa \quad \textit{almost surely}, \qquad \kappa \;=\; \frac{\alpha\sigma_*}{\sigma} \;=\; \frac{\alpha}{\alpha(1-\sigma)+\sigma}, \tag{7.11}$$

for a constant $C = C(\sigma,\alpha)$.

Proof. Both problems inherit the same envelope M: the unweighted frequencies satisfy $p_k \asymp (M/\sigma)k^{-1/\sigma}$ and the weighted ones $p'_k \asymp (\alpha M^\alpha/\sigma)k^{-1/\sigma_*}$. Counting as in Lemma 7.16 gives $x^\sigma N(x)\to Z$ and $x^{\sigma_*}N_*(x)\to Z_*$ almost surely, with Z and Z_* each a deterministic function of the single random variable M. Eliminating M and applying Karlin's correspondence to both gives (7.11), with the exponent $\alpha\sigma_*/\sigma = \kappa$ read off the two envelopes. □

PROPOSITION 7.18. *In* (7.11),

$$C \;=\; \alpha^{\sigma_*}\,\sigma^{(\alpha-1)\sigma_*}\,\frac{\Gamma(1-\sigma_*)\,\Gamma(1-\sigma+\sigma_*)}{\Gamma(1-\sigma)}. \tag{7.12}$$

In particular C does not depend on θ, and $C = 1$ at $\alpha = 1$.

Proof. The weighted frequencies are $p'_k = \widetilde{R}^{\alpha}_{k-1}\{1-(1-B_k)^{\alpha}\}$. Since $B_k \to 0$ almost surely and $\{1-(1-B)^{\alpha}\}/(\alpha B) \to 1$ as $B \to 0$, the argument of Lemma 7.16 applies verbatim with $\widetilde{R}^{\alpha}_{k-1}$ in place of $\widetilde{R}_{k-1}$ and one further factor, tending to one, absorbed into ε_k: for $k \geq k_0$,

$$p'_k \;=\; \frac{\alpha M^{\alpha}}{\sigma}\,\Gamma_k\, k^{-1/\sigma_*}\,\varepsilon'_k, \qquad 1-\eta \leq \varepsilon'_k \leq 1+\eta,$$

with the same Gamma variables Γ_k, because they come from the same sticks, and with $\alpha\beta + 1 = 1/\sigma_*$. The sandwich then gives, with σ_* in place of σ in the substitution,

$$x^{\sigma_*} N_*(x) \;\longrightarrow\; \Big(\frac{\alpha M^{\alpha}}{\sigma}\Big)^{\sigma_*} \mathbb{E}[\mathcal{G}^{\sigma_*}] \;=\; \Big(\frac{\alpha M^{\alpha}}{\sigma}\Big)^{\sigma_*} \frac{\Gamma(1-\sigma+\sigma_*)}{\Gamma(1-\sigma)} \quad \text{almost surely,}$$

and $S_* = \Gamma(1-\sigma_*)$ times that. Substituting $M = \sigma S^{1/\sigma}$ from Lemma 7.16 and using $\alpha\sigma_*/\sigma = \kappa$ gives (7.12). □

COROLLARY 7.19. $\mathbb{E}[S_*^p] = C^p\,\mathbb{E}[S^{\kappa p}]$ *for* $p > -(1+\theta/\sigma)/\kappa$, *with the right side given by* (7.10) *at* κp. *Comparing that expression, as a function of* p, *with* (7.10) *shows it is of Mittag Leffler form only when* $\kappa = 1$, *that is only when* $\alpha = 1$: *the gamma factors in* p *are spaced by* $1/\kappa$ *on one side and by* 1 *on the other.*

Two comments. First, that θ is absent from (7.12) is the second appearance of the phenomenon noted after Proposition 7.13. It does not mean θ is irrelevant to the weighted problem: it enters through the law of S itself, by (7.10). What it means is that the *map* from the unweighted diversity to the weighted one is free of θ, so ascertainment acts on the diversity by a transformation fixed by σ and α alone. Second, the Mittag Leffler family is not closed under powers, so Corollary 7.19 is a second sense in which ascertainment leaves the standard families, and it matches the first: exactness fails at the level of the law, and closure fails at the level of the diversity.

Table 7.7 checks (7.12) against simulation. What is tested is the ratio Z_*/Z^{κ}, estimated on common stick realisations so the comparison is pathwise; the last column is the relative standard deviation of that ratio across replicates, which measures how nearly (7.11) holds pathwise rather than in mean.

The first three rows agree to within Monte Carlo error and the $\alpha = 1$ row is exact, as it must be. Doubling θ moves the simulated constant by less than half a percent, well within the sampling error at this x.

REMARK 7.20. The proofs of Lemma 7.16 and Proposition 7.18 use two facts about the Gamma representation of the sticks: that the large shape Gamma variable concentrates exponentially fast, and that the resulting count is a sum of independent indicators to which Kolmogorov's criterion applies. Neither step is delicate, and the sandwich makes the nonuniformity of the approximations harmless, since it costs only a factor $(1\pm\eta)^{\sigma}$ that is released at the end. The one place where care is needed is the passage from a geometric grid $x_j = (1+\delta)^{-j}$, with $\delta \downarrow 0$ at the end, to all x, and there the monotonicity of N does the work.

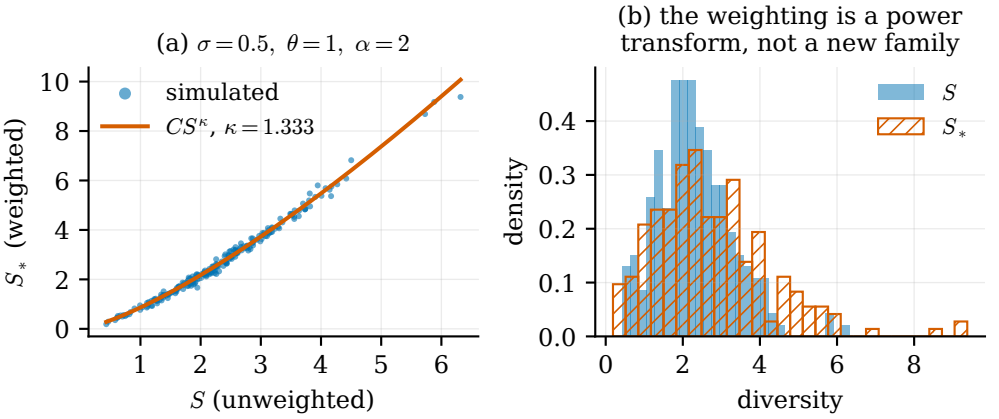


FIGURE 7.4. The diversity map at $\sigma = 0.5$, $\theta = 1$, $\alpha = 2$, so $\kappa = 4/3$. Left, the weighted diversity against the unweighted one on common stick realisations, with the predicted curve CS^κ of (7.11) and (7.12) overlaid; the identity is pathwise, not merely in distribution. Right, the two densities: the weighting widens the diversity and changes its shape, because a power of a Mittag Leffler variable is not Mittag Leffler.

7.7 The coincidence theory collapses

The other half of the one parameter theory asks for the probability that no two intervals receive the same number of points. Under ESF(ϑ) that probability tends to $e^{-\gamma\vartheta}/\Gamma(1+\vartheta)$, and the whole analysis turns on the small frequencies forming a point process of intensity $\vartheta u^{-1}du$, which is scale invariant. None of it survives $\sigma > 0$, and the reason is the same intensity.

PROPOSITION 7.21. *For a* PD(σ,θ) *paintbox with* $\sigma > 0$*, and for every weight* f *of positive index in the sense of Proposition 7.32, in particular every* f_α*, the probability that all occupied intervals receive distinct counts tends to zero.*

Under f_α the small frequencies have intensity proportional to $x^{-1-\sigma_*}dx$, and under any weight of positive index in the sense of Proposition 7.32 their counting function is regularly varying with index $-\sigma_*$, again strictly negative. It is never $\vartheta u^{-1}du$. So $A_j/K_n \to q_j(\sigma_*) > 0$ for each fixed j (Section 7.10.3), and since $K_n \to \infty$, in particular $A_1 \to \infty$: there are eventually many singletons and ties are certain. A power intensity is also not scale invariant, so the invariance result of the one parameter theory, that any weight continuous and positive at the right endpoint leaves the constant untouched, has nothing to act on either.

Figure 7.5 shows both halves. On the left, $\mathbb{E}[A_1]$ is bounded at $\sigma = 0$ and grows like n^σ beyond it; the slopes on log log axes are the exponents. On the right, the probability that all abundances are distinct approaches its limit $e^{-\gamma}$ for the one parameter case and falls to zero for every $\sigma > 0$, faster the larger σ is.

That is a negative result but a clarifying one. The two questions one is naturally

TABLE 7.7. The constant of (7.12) tested through the ratio $Z_*/Z^\kappa = C\,\Gamma(1-\sigma)^\kappa/\Gamma(1-\sigma_*)$, predicted and simulated, using 2×10^6 sticks at level $x=10^{-7}$ over 60 replicates with Z and Z_* computed on the same realisation. The $\alpha=1$ row is exact; the last row tests the θ independence of (7.12).

σ	θ	α	σ_*	predicted	simulated	rel. s.d.
0.5	1	2	0.3333	1.3661	1.3588	0.065
0.4	1	2	0.2500	1.1623	1.1559	0.101
0.6	1	1.5	0.5000	1.4058	1.4034	0.019
0.5	1	1	0.5000	1.0000	1.0000	0.000
0.5	2	2	0.3333	1.3661	1.3650	0.050

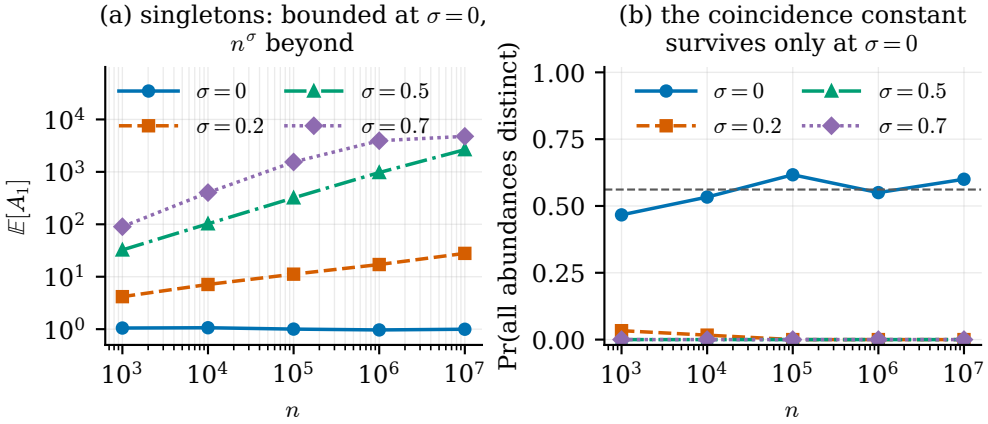


FIGURE 7.5. Why the coincidence constant is a one parameter phenomenon. Left, the expected number of singletons: bounded at $\sigma=0$, growing like n^σ for $\sigma>0$. Right, the probability that no two intervals receive the same number of points; the dashed line is $e^{-\gamma}=0.5615$, the ESF(1) limit. Uniform sampling, $\alpha=1$, throughout.

led to ask, the first order profile and the coincidence law, are not two aspects of a single theory. The first generalises with a change of coordinate; the second is specific to $\sigma=0$ and to the scale invariance of the logarithmic intensity.

7.8 Estimating the discount parameter from a biased sample

The statistical content of Proposition 7.13 is a bias with an exact inverse. A practitioner who estimates the discount parameter by the growth of the number of types, and whose sample is drawn with effort profile f_α rather than uniformly, does not estimate σ. He estimates σ_*, and by (7.9) the map is invertible.

COROLLARY 7.22. *If α is known by design and $\hat\sigma_*$ is any consistent estimator of the*

exponent from the weighted sample, then

$$\hat{\sigma} \;=\; \frac{\alpha\hat{\sigma}_*}{\alpha\hat{\sigma}_* + 1 - \hat{\sigma}_*} \tag{7.13}$$

is consistent for σ. Equivalently, in the odds coordinate the correction is a division by α.

Proof. Invert $\sigma_* = \sigma/\{\alpha(1-\sigma)+\sigma\}$, or equivalently divide $(1-\sigma_*)/\sigma_*$ by α and solve. □

Three estimators of σ_* are now available, and they are not equivalent. The exponent estimator $\log K_n / \log n$ is the obvious one and the worst: it converges only logarithmically, and at $n = 10^6$ with $\log n = 13.8$ an $O(1)$ error in $\log K_n$ is already an error of order 0.07 in the exponent. The two spectrum estimators, $1 - 2A_2/A_1$ from Corollary 7.26 and A_1/K_n from Corollary 7.25, are ratios of counts at the same n and converge at the rate the counts do.

Figure 7.6 compares them at $\sigma = 0.5$, $n = 10^6$. For $\alpha \geq 1$, the fading effort of a tiring collector and the common case in practice, the two spectrum estimators are close to exact. At $\alpha = 1, 2, 4$ they return $0.500, 0.341, 0.199$ and $0.500, 0.344, 0.201$ against targets $0.500, 0.333, 0.200$; $\log K_n / \log n$ returns $0.547, 0.397, 0.263$. For $\alpha < 1$ the picture appears to reverse: at $\alpha = 0.25$, where $\sigma_* = 0.8$, the spectrum estimators read 0.61 and 0.67 while the exponent estimator reads 0.798. That comparison is an artefact and should not be believed, for the reason set out in Section 7.9.3: a persistent weight makes the truncated stick a poor approximation, and the omitted mass depresses the spectrum estimators specifically. Raising the stick count at fixed n moves A_1/K_n steadily back towards σ_*. What the left panel shows for $\alpha < 1$ is the limit of the simulation, not the limit of the estimator.

The right panel applies the correction. It is exact for all three in the limit, being the inverse group element of Proposition 7.23, so what the panel shows is the accuracy of the underlying estimator and nothing else. The spectrum estimators recover $\sigma = 0.5$ to within about 0.01 for $\alpha \geq 1$ and lose accuracy as α falls; the exponent estimator drifts across the whole range, from 0.50 to 0.59, its errors partly cancelling in the correction. In practice one would use A_1/K_n, check it against $1 - 2A_2/A_1$, and treat disagreement between them as a warning that the sample is too small for the stable regime to have set in.

7.8.1 *The effort profiles form a group, and the odds coordinate linearises it*

Write T_α for the map on the discount parameter induced by the weight f_α, so $T_\alpha(\sigma) = \sigma/\{\alpha(1-\sigma)+\sigma\}$.

PROPOSITION 7.23. *$T_\alpha \circ T_{\alpha'} = T_{\alpha\alpha'}$ for all $\alpha, \alpha' > 0$, with $T_1 = \mathrm{id}$ and $T_\alpha^{-1} = T_{1/\alpha}$. The family $\{T_\alpha\}$ is therefore a one parameter group isomorphic to the multiplicative positive reals, and* (7.9) *is the coordinate in which the action is multiplication.*

Proof. Composition of weights is composition of the maps G, and $r^{\alpha'} \mapsto (r^{\alpha'})^\alpha = r^{\alpha\alpha'}$. In the odds coordinate $\omega = (1-\sigma)/\sigma$ the action is $\omega \mapsto \alpha\omega$ by (7.9), and multiplication composes. □

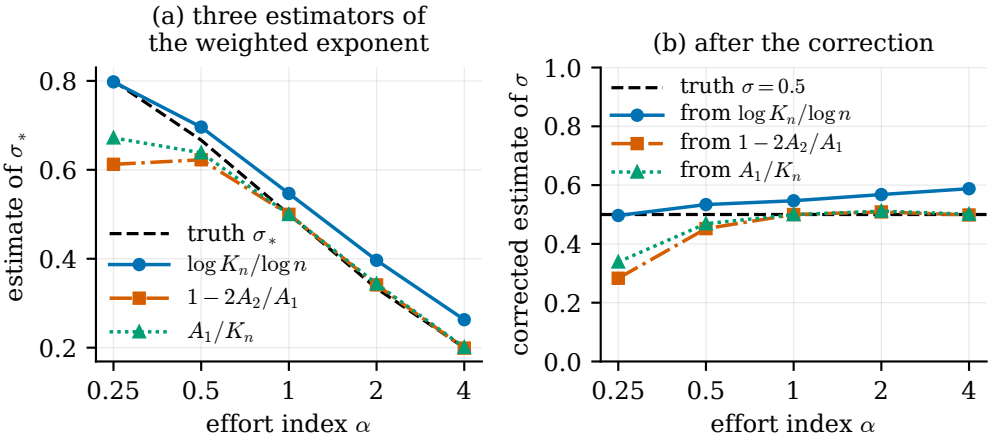


FIGURE 7.6. Estimating $\sigma = 0.5$ from a sample drawn with effort profile f_α, at $n = 10^6$ over 24 replicates. Left, three estimators of the weighted exponent against its true value. Right, the same three after the correction (7.13), whose target is the dashed line. The spectrum estimators are near exact for $\alpha \geq 1$ and degrade as σ_* approaches one; the exponent estimator is mediocre everywhere.

The group property is a consistency check on the whole framework, and it says something about design. Applying two effort profiles in succession, a fading pass and then a persistent one, is the same as applying a single profile with the product index; there is no path dependence and no interaction. It also means the correction (7.13) is exactly the inverse group element, which is why it is exact and not merely first order.

The exponent inherits the group structure: writing $\kappa(\sigma, \alpha)$ for the power in (7.11),

$$\kappa(\sigma, \alpha\alpha') \;=\; \kappa(\sigma, \alpha')\, \kappa\big(T_{\alpha'}(\sigma), \alpha\big) \tag{7.14}$$

identically, which we verified at forty five parameter triples to machine precision.

The constant does not, and the failure is instructive rather than embarrassing. One might expect the cocycle relation

$$C(\sigma, \alpha\alpha') \;\stackrel{?}{=}\; C\big(T_{\alpha'}(\sigma), \alpha\big)\, C(\sigma, \alpha')^{\kappa(T_{\alpha'}(\sigma), \alpha)}, \tag{7.15}$$

obtained by applying (7.11) twice. It fails, by the amounts computed below. The reason is Theorem 7.11. The second application of (7.11) is only legitimate if the intermediate object is a $\mathrm{PD}(\sigma', \theta')$ system, and it is not: after the first weighting the subdivision is the residual allocation model (7.5), which lies outside the Ewens Pitman family. The exponent survives the detour because it depends only on the power of k in the tail sums, which (7.5) shares with a genuine two parameter law; the constant does not, because it depends on the law of the stick fractions, which (7.5) does not share.

TABLE 7.8. Moments of the weighted σ diversity at $\sigma = 0.5$, $\theta = 1$, from $\mathbb{E}[S_*^p] = C^p \mathbb{E}[S^{\kappa p}]$. The row $\alpha = 1$ is the Mittag Leffler law itself.

α	σ_*	κ	C	$\mathbb{E}[S_*]$	$\mathrm{sd}[S_*]$	c.v.
0.25	0.800	0.400	1.1623	1.572	0.283	0.180
0.5	0.667	0.667	1.1129	1.875	0.544	0.290
1	0.500	1.000	1.0000	2.257	0.952	0.422
2	0.333	1.333	0.8624	2.654	1.465	0.552
4	0.200	1.600	0.7423	2.962	1.944	0.656

So the size of the gap in (7.15) is a quantitative measure of how far the image is from the family it left, and it is a measure computable in closed form from (7.12), with no simulation. It is zero when $\alpha = 1$ or $\alpha' = 1$, and, at $\sigma = 0.65$, largest when both weights fade: about four percent at $(\sigma, \alpha, \alpha') = (0.65, 2, 3)$, seven percent at $(0.65, 3, 3)$ and eleven percent at $(0.65, 4, 4)$. It is negative when one weight fades and the other persists, and small when both persist, 0.12 percent at $(0.65, 0.5, 0.5)$. That is a second and sharper reading of Theorem 7.11: the family is not merely unpreserved, it is unpreserved by an amount one can compute.

7.8.2 Moments of the weighted diversity

Corollary 7.19 makes every moment explicit, and the numbers show that the weighting does not merely shift the diversity but changes its shape. Table 7.8 gives the mean, standard deviation and coefficient of variation of S_* at $\sigma = 0.5$, $\theta = 1$, computed from (7.10) and (7.12) with no simulation.

The coefficient of variation moves by a factor of nearly four across the range. A fading effort profile, $\alpha > 1$, raises the diversity to a power greater than one and so stretches its upper tail: the type count becomes both larger in mean, on the n^{σ_*} scale, and much more variable between replicate surveys. A persistent profile does the reverse and produces an artificially stable diversity. Either way an analyst who fits a Mittag Leffler law to replicate surveys, as the two parameter model instructs, will find it does not fit, and the reason is not the model but the sampling.

7.9 The joint law of the counts

Everything so far concerns K_n and the diversity. The abundance spectrum $(A_1, A_2, \dots)$ is a finer object, and in the one parameter case it is the one that carries the interesting structure: there the counts converge to *independent* Poisson variables with means ϑ/j, a genuinely infinite dimensional limit. Two points of precision are owed here, both of which Warren Ewens insisted on. At finite n the A_j under ESF(ϑ) are *not* independent: the exact statement is the conditioning relation, that $(A_1, \dots, A_n)$ has the law of independent $Z_j \sim \mathrm{Poisson}(\vartheta/j)$ conditioned on $\sum_j jZ_j = n$ (Arratia, Barbour and Tavaré, 2003), and independence emerges only in the limit, for $(A_1, \dots, A_b)$ with b fixed as $n \to \infty$. Likewise the coincidence

constant $e^{-\gamma\vartheta}/\Gamma(1+\vartheta)$ is the limiting value of the probability that all abundances are distinct and not its value at any finite n. For $\sigma > 0$ the answer is the opposite, and simpler.

PROPOSITION 7.24. *Under the weighted paintbox with $\sigma > 0$, for each fixed j,*

$$\frac{A_j}{n^{\sigma_*}} \;\longrightarrow\; q_j(\sigma_*)\, S_*, \qquad q_j(\sigma) \;=\; \frac{\sigma\,\Gamma(j-\sigma)}{\Gamma(1-\sigma)\,j!}, \tag{7.16}$$

jointly in j and almost surely, with S_* the limit of Proposition 7.17.

Proof. Poissonise and condition on the frequencies. With $\pi_j(u) = u^j e^{-u}/j!$ we have $\mathbb{E}[A_j \mid P] = \sum_k \pi_j(np_k)$, and the induced frequencies have counting function $N(x) \sim Z_* x^{-\sigma_*}$, so the number of them in $(x, x+dx)$ is $Z_*\sigma_* x^{-\sigma_*-1}dx$. Hence

$$\begin{aligned}\sum_k \pi_j(np_k) \;&\approx\; \int_0^\infty \pi_j(nx)\, Z_*\sigma_* x^{-\sigma_*-1}dx \\ &=\; Z_*\sigma_* n^{\sigma_*}\frac{1}{j!}\int_0^\infty u^{\,j-\sigma_*-1}e^{-u}du \;=\; Z_*\,\frac{\sigma_*\Gamma(j-\sigma_*)}{j!}\,n^{\sigma_*}.\end{aligned}$$

Since $S_* = \Gamma(1-\sigma_*)Z_*$, this is $q_j(\sigma_*)S_* n^{\sigma_*}$. The display is the heuristic; the rigorous version is Lemma 7.2 applied conditionally on the frequencies, with $\gamma = \sigma_*$ and the constant slowly varying function $L \equiv Z_*$, which is legitimate because $x^{\sigma_*}N_*(x) \to Z_*$ almost surely by the proof of Proposition 7.18. The lemma gives $A_j/n^{\sigma_*} \to \sigma_*\Gamma(j-\sigma_*)Z_*/j!$ almost surely for each j, and since countably many almost sure statements hold simultaneously, the convergence is joint in j with the same S_* throughout. □

The limit is degenerate given S_*: every coordinate is a deterministic multiple of one random variable. Two checks confirm the normalisation. At $j = 1$ the constant is $q_1 = \sigma_*$, so $A_1/n^{\sigma_*} \to \sigma_* S_*$; and $\sum_{j\ge1} q_j(\sigma) = 1$ by the identity $\sum_{j\ge1}\Gamma(j-\sigma)/j! = \Gamma(1-\sigma)/\sigma$, which recovers $K_n/n^{\sigma_*} \to S_*$. The identity is the binomial series at its boundary. For $|x| \le 1$,

$$(1-x)^\sigma \;=\; \sum_{j\ge0}\frac{(-\sigma)_j}{j!}\,x^j \;=\; 1+\sum_{j\ge1}\frac{\Gamma(j-\sigma)}{\Gamma(-\sigma)\,j!}\,x^j,$$

the series converging at $x = 1$ because its terms are of order $j^{-1-\sigma}$. Setting $x = 1$ gives $\sum_{j\ge1}\Gamma(j-\sigma)/j! = -\Gamma(-\sigma) = \Gamma(1-\sigma)/\sigma$. The terms $q_j(\sigma) = \sigma\Gamma(j-\sigma)/\{\Gamma(1-\sigma)j!\}$ are therefore a probability distribution on $j \ge 1$, Sibuya's distribution, and Proposition 7.24 says that the abundance of a randomly chosen observed type is asymptotically Sibuya distributed with parameter σ_*. Its first terms are $q_1 = \sigma_*$, $q_2 = \sigma_*(1-\sigma_*)/2$ and $q_3 = \sigma_*(1-\sigma_*)(2-\sigma_*)/6$.

Together the proposition and its normalisation give a prediction with no constant in it at all.

COROLLARY 7.25 (The singleton fraction). *$A_1/K_n \to \sigma_*$ almost surely. In a two parameter community the fraction of observed types seen exactly once estimates the discount parameter directly; under a biased sample it estimates σ_*, which Corollary 7.22 inverts.*

Proof. Divide (7.16) at $j = 1$, where $q_1(\sigma_*) = \sigma_*$, by $K_n/n^{\sigma_*} \to S_*$; the random factor S_* cancels. □

Notice how little this asks. No normalisation, no knowledge of n, no estimate of S_* or θ: two numbers a survey already reports, the count of types and the count of types seen once. At $n = 10^8$ simulation gives $A_1/K_n = 0.3326$ against $\sigma_* = 1/3$ and 0.1752 against 0.1765. At $\sigma_* = 1/2$ the same simulation gives 0.4724, which is not a property of the estimator but of the simulation: raising the number of sticks from 3×10^5 to 1.5×10^6 moves it to 0.4885 and then 0.4953. Section 7.9.3 below says how many sticks are needed, and why the answer is sometimes impossible.

7.9.1 A dichotomy, and an estimator

The contrast with $\sigma = 0$ is complete. There the spectrum converges to an infinite family of independent Poisson variables, and no scaling makes it degenerate; the interesting questions are about the joint law, which is where the coincidence probability of the one parameter theory lives. Here the spectrum converges, after a single scaling, to a deterministic profile multiplied by one random number. All the randomness in the abundance spectrum of a two parameter community is the randomness of its diversity.

That degeneracy is a practical gift, because it makes the shape of the spectrum a deterministic function of σ_* alone, free of S_*, of θ and of n.

COROLLARY 7.26. *For each $j \geq 2$,*

$$\frac{A_j}{A_1} \longrightarrow \frac{\Gamma(j-\sigma_*)}{j!\,\Gamma(1-\sigma_*)}, \qquad \textit{in particular} \qquad \frac{A_2}{A_1} \longrightarrow \frac{1-\sigma_*}{2}. \tag{7.17}$$

Hence $\hat\sigma_* = 1 - 2A_2/A_1$ is consistent for σ_*, and $\hat\sigma = \alpha\hat\sigma_*/(\alpha\hat\sigma_* + 1 - \hat\sigma_*)$ by Corollary 7.22 is consistent for σ.

7.9.2 The bias of the singleton fraction

How large must a survey be before A_1/K_n can be trusted? The answer is: not very, and the calculation is a Mellin transform.

PROPOSITION 7.27. *Suppose the induced frequencies follow the envelope $p_k = c\,k^{-1/\sigma_*}$ exactly, and write $A = nc$. Then, as $A \to \infty$,*

$$\mathbb{E}[A_1 \mid P] = \sigma_*\Gamma(1-\sigma_*)A^{\sigma_*} + O(A^{-\delta}),$$
$$\mathbb{E}[K_n \mid P] = \Gamma(1-\sigma_*)A^{\sigma_*} - \tfrac{1}{2} + O(A^{-\delta}),$$

for some $\delta > 0$, and consequently

$$\frac{\mathbb{E}[A_1 \mid P]}{\mathbb{E}[K_n \mid P]} = \sigma_*\Big(1 + \frac{1}{2\,\mathbb{E}[K_n \mid P]}\Big) + o\Big(\frac{1}{\mathbb{E}[K_n \mid P]}\Big). \tag{7.18}$$

Proof. For f with Mellin transform $\tilde{f}(s) = \int_0^\infty f(u)u^{-s-1}du$, substituting $v = Ak^{-1/\sigma_*}$ gives $\int_0^\infty f(Ak^{-1/\sigma_*})A^{-s-1}dA = k^{-s/\sigma_*}\tilde{f}(s)$, so $\sum_{k\geq 1} f(Ak^{-1/\sigma_*})$ has Mellin transform $\zeta(s/\sigma_*)\tilde{f}(s)$ and equals $(2\pi i)^{-1}\int \zeta(s/\sigma_*)\tilde{f}(s)A^s ds$. Shifting the contour left collects residues. For A_1 take $f(u) = ue^{-u}$, whose transform is $\Gamma(1-s)$; the only pole crossed is that of ζ at $s = \sigma_*$, with residue $\sigma_*\Gamma(1-\sigma_*)A^{\sigma_*}$. For K_n take $f(u) = 1 - e^{-u}$, whose transform is $\Gamma(1-s)/s$ on $0 < \Re s < 1$; here the contour crosses the same pole, giving $\Gamma(1-\sigma_*)A^{\sigma_*}$, and additionally the pole of $1/s$ at the origin, whose residue is $\zeta(0)\Gamma(1) = -\frac{1}{2}$. Dividing and expanding gives (7.18). □

Two things follow. The deterministic bias is $O(1/K_n)$, whereas the sampling standard deviation of A_1/K_n is $O(K_n^{-1/2})$: at any sample size where the estimator has appreciable noise the bias is invisible beside it, and there is no bias correction worth making. And the coefficient is $\sigma_*/2$, so the bias is upward, not downward, which is worth saying because every deficit of A_1/K_n observed up to here has been downward and every one of them was truncation.

Simulating the deterministic envelope directly, with 4×10^7 frequencies and no random sticks, isolates (7.18) from the difficulty of Section 7.9.3. At $\sigma_* = 0.3$ the quantity $(A_1/K_n - \sigma_*)K_n$ equals 0.1500 at $n = 10^5$, 10^6 and 10^7, against the predicted $\sigma_*/2 = 0.15$, to four decimal places at every sample size. At $\sigma_* = 0.5$ it gives 0.249 and 0.242 against 0.25 at the first two sample sizes. At $\sigma_* = 0.7$ it fails, and fails worse as n grows, for the reason (7.19) predicts: the envelope $k^{-1/0.7}$ has so heavy a tail that 4×10^7 frequencies do not cover the range a sample of 10^7 explores.

The doubleton ratio $1 - 2A_2/A_1$ is a better estimator than the exponent estimator of Section 7.8 for the reason given there: $\log K_n / \log n$ converges to σ_* only logarithmically, whereas (7.17) is a ratio of two counts at the same n and converges at the rate the counts themselves do. It needs neither n nor any normalisation, only the number of singletons and the number of doubletons, both of which a survey reports. At $\sigma_* \to 0$ it returns $A_2/A_1 \to 1/2$, which is the one parameter value $\mathbb{E}[A_j] \propto 1/j$, so the estimator degrades gracefully to the boundary of the family.

Table 7.9 tests (7.16) and (7.17) by Rao Blackwellised simulation. The predicted and observed ratios A_3/A_1 and A_4/A_1 agree closely wherever the truncated stick resolves the relevant range, and the estimator $1 - 2A_2/A_1$ recovers σ_*. The last two rows are the same configuration at two truncation levels, and they show where the residual error comes from. That configuration has $\alpha = 1/2$, so by the criterion (7.19) of Section 7.9.3 below an honest simulation would need $K \gg n^2$ sticks: 3×10^5 sticks at $n = 10^8$ are short by some ten orders of magnitude and the estimator reads -0.09. Raising the stick count to 2×10^6 and lowering n to 10^7 brings it to 0.578 against a target of 0.667, still far inside the regime (7.19) forbids. The deficit is ours and not the estimator's.

REMARK 7.28. Proposition 7.24 also settles, negatively, a question one might ask by analogy with the one parameter theory. There the interesting phenomenon is

TABLE 7.9. The spectrum profile (7.16) and the estimator (7.17). Counts are Rao Blackwellised, $\mathbb{E}[A_j \mid P] = \binom{n}{j} \sum_k p_k^j (1-p_k)^{n-j}$, over 40 replicates at $n = 10^8$ with 3×10^5 sticks unless marked. The final row repeats the fifth at $n = 10^7$ with 2×10^6 sticks.

σ	θ	α	σ_*	$1 - 2A_2/A_1$	A_3/A_1 pred/obs	A_4/A_1 pred/obs
0.5	1	1	0.5000	0.451	0.125/0.138	0.078/0.086
0.5	1	2	0.3333	0.331	0.185/0.186	0.124/0.125
0.3	1	2	0.1765	0.183	0.250/0.248	0.177/0.177
0.6	0	1.5	0.5000	0.473	0.125/0.132	0.078/0.082
0.5	2	0.5	0.6667	−0.090	0.074/0.344	0.043/0.236
0.5	2	0.5	0.6667	0.578	0.074/0.094	·

that the layout of the atoms changes the joint law of the counts while leaving the profile $\mathbb{E}[A_j]$ alone. Here there is no room for such an effect: by Proposition 7.15 the exponent is layout free, and by (7.16) the whole spectrum is determined by the exponent and by S_*, so a change of layout can move only the law of S_* and not the shape. The delicate part of the one parameter theory has no two parameter counterpart, for the same reason the coincidence constant does not: the stable regime is too rigid to support it.

7.9.3 How many sticks a simulation needs

Every numerical statement in this chapter rests on a truncated stick, and the truncation is not innocuous. The criterion explains the residual errors above and is a trap for anyone repeating this work.

Simulating K sticks omits the mass R_K^α beyond them. Those omitted atoms are the smallest, so almost all of them would have been singletons; omitting them therefore depresses A_1 more than K_n and biases the spectrum estimators *downwards*. For the omission to be harmless the missing mass must be small against the resolution $1/n$ of a sample of size n, that is $R_K^\alpha \ll 1/n$. Using $R_K \asymp MK^{-(1-\sigma)/\sigma}$ from Proposition 7.13,

$$K \gg \left(M^\alpha n\right)^{\sigma/\{\alpha(1-\sigma)\}}. \tag{7.19}$$

The exponent is the whole story. At $\sigma = 1/2$ and $\alpha = 1$ it is 1, so K must exceed n: our runs at $n = 10^8$ used 1.5×10^6 sticks and were accordingly short by about two orders of magnitude, at a cost of about one percent in A_1/K_n, which is exactly the shortfall reported after Corollary 7.25. At $\sigma = 1/2$ and $\alpha = 1/4$ the exponent is 4, so $K \gg n^4$, which at $n = 10^5$ is 10^{20} sticks and is not going to happen. The diagnostic is direct: at $n = 10^5$, $\sigma_* = 0.8$, increasing K through 5×10^5, 2×10^6 and 8×10^6 reduces the omitted mass from 5.0% to 3.3% to 2.3% and moves A_1/K_n from 0.717 to 0.748 to 0.763, climbing towards the true 0.8 at every step.

So a persistent weight is expensive to simulate and a strongly persistent one is effectively unsimulable by this route, while a fading weight, $\alpha > 1$, makes the exponent in (7.19) smaller than its uniform value, and for $\sigma \le 1/2$ smaller than

one, so the simulation is easy. That asymmetry is a fact about the method and not about the theory, and none of the analytic results above depends on it. It does mean that the numerical support in this chapter is strong for $\alpha \geq 1$ and weak for $\alpha \ll 1$, and we have marked the places where that matters.

7.9.4 *Estimation in practice: a simulated survey and two classic data sets*

The estimators of this section are simple enough to apply by hand, and this subsection does so, first on a simulated survey where the truth is known and then on two published frequency of frequency tables.

EXAMPLE 7.29 (Estimating σ from a fading survey). Let the community be $\mathrm{PD}(0.6, 5)$ and let the collector's effort fade linearly, $\alpha = 2$. The odds law gives $\omega = 2/3 \mapsto 4/3$, so $\sigma_* = 3/7 = 0.4286$. We drew one stick realisation with 10^6 sticks (by (7.19) the requirement is $K \gg (M^2 n)^{3/4}$, about 10^5 here), formed the induced frequencies $p'_k = G(\widetilde{R}_{k-1}) - G(\widetilde{R}_k)$ with $G(r) = r^2$, and drew a multinomial sample of $n = 10^6$ specimens. The survey reports

$$K_n = 1087, \qquad A_1 = 454, \qquad A_2 = 130.$$

The three estimates of the weighted exponent are

$$\frac{A_1}{K_n} = 0.4177, \qquad 1 - \frac{2A_2}{A_1} = 0.4273, \qquad \frac{\log K_n}{\log n} = 0.5060.$$

An analyst who ignored the sampling design would report one of the two spectrum estimates as σ and be wrong by about 0.18. Applying (7.13) with the known $\alpha = 2$,

$$\hat{\sigma} = \frac{2(0.4177)}{2(0.4177) + 0.5823} = 0.589, \qquad \hat{\sigma} = \frac{2(0.4273)}{2(0.4273) + 0.5727} = 0.599,$$

from the two spectrum estimators, while the exponent estimator corrects to 0.672. Repeating the whole experiment 100 times, the singleton fraction has mean 0.4294 and standard deviation 0.0135 and corrects to 0.601 ± 0.013; the doubleton estimator corrects to 0.599 ± 0.048; the exponent estimator corrects to 0.702 ± 0.018, consistently too high. The singleton fraction is the estimator of choice: it is unbiased to within Monte Carlo error and has about a quarter of the spread of the doubleton one, because it uses all K_n types rather than the two smallest abundance classes. Its spread is close to the binomial value $\{\sigma_*(1-\sigma_*)/K_n\}^{1/2} = 0.015$, as if each observed type were independently a singleton with probability σ_*. Finally, the $O(1/K_n)$ bias of Proposition 7.27, here $\sigma_*/(2K_n) \approx 0.0002$, is far below that spread: the mean of $A_1/K_n - \sigma_*\{1 + 1/(2K_n)\}$ over the replicates is 0.0007, within its Monte Carlo error of 0.0014.

The variance of the corrected estimator follows from that of $\hat{\sigma}_*$ by the delta method. Writing $h(s) = \alpha s/(\alpha s + 1 - s)$, one has $h'(s) = \alpha/(\alpha s + 1 - s)^2$, which at $\alpha = 2$ and $s = 3/7$ equals $2/(10/7)^2 = 0.98$. So the correction carries the standard

deviation through almost unchanged here; for a strongly fading profile the factor $h'(\sigma_*) = \sigma(1-\sigma)/\{\sigma_*(1-\sigma_*)\}$ (as one checks by substituting $\sigma_* = T_\alpha(\sigma)$) can be large, and the price of a biased design is then paid in precision as well as in bias.

EXAMPLE 7.30 (Shakespeare's vocabulary). Efron and Thisted (1976) tabulate the word frequencies of the Shakespeare canon: $n = 884{,}647$ word tokens of $K_n = 31{,}534$ distinct word types, of which $A_1 = 14{,}376$ occur once and $A_2 = 4{,}343$ twice; the next eight classes are 2292, 1463, 1043, 837, 638, 519, 430 and 364. Word frequencies are a standard instance of a power law abundance distribution, so the two parameter model is the natural first try, and the estimators give

$$\frac{A_1}{K_n} = 0.456, \qquad 1 - \frac{2A_2}{A_1} = 0.396, \qquad \frac{\log K_n}{\log n} = 0.757.$$

The exponent estimator is far from the other two, as Section 7.8 warned. The two spectrum estimators disagree by 0.06, and with counts in the thousands that is not sampling error: the binomial standard deviation of the singleton fraction is 0.003. A least squares fit of the profile $A_j/A_1 = \Gamma(j-\sigma)/\{j!\,\Gamma(1-\sigma)\}$ of (7.17) to the classes $j = 2, \ldots, 10$ on the log scale gives 0.411. Table 7.10 shows the observed ratios against the profiles at the two spectrum estimates. The profile at $\sigma = 0.396$ fits the low classes well, which is no surprise since it was fitted to A_2/A_1, and drifts above the data from $j = 7$ on, by about a tenth at $j = 10$; the profile at 0.456 lies below the data throughout.

The diagnosis suggested at the end of Section 7.8 applies: the disagreement says that the exchangeable two parameter paintbox describes these data only approximately, which is not surprising for a text, where words are not drawn independently. It is worth noticing what the disagreement does *not* indicate. By Corollary 7.34, every ascertainment scheme of positive index in the sense of Proposition 7.32 moves both spectrum estimators to the same σ_*, so no such scheme can open a gap between them. The comparison of A_1/K_n with $1 - 2A_2/A_1$ is a check on the model that the sampling design cannot spoil, and here it fails.

EXAMPLE 7.31 (Corbet's butterflies, once more). Corbet's Malayan butterflies (Fisher, Corbet and Williams, 1943), analysed in Chapter 6, have $A_1 = 118$ and $A_2 = 74$ among $K_n = 620$ species, 119 of them censored above 24 specimens so that n is estimated, at $\hat{n} = 11068$. The doubleton estimator gives $1 - 2(74)/118 = -0.254$, outside $[0, 1)$ altogether, and the reason is that $A_2/A_1 = 0.627$ exceeds one half. For every $\sigma \in (0, 1)$ and every weight of positive index in the sense of Proposition 7.32 the limit $(1-\sigma_*)/2$ of A_2/A_1 is below one half, and at $\sigma = 0$ under f_α the limiting Poisson means $\vartheta/2$ and ϑ are in ratio exactly one half, so in the limit no member of the family, under none of these ascertainments, produces more doubletons than half the singletons. The counts are small, and the ratio has a standard error of about 0.09 under a Poisson approximation, so on its own this is suggestive rather than decisive: the exact Ewens means at the fitted $\theta = 141.75$ and $n = \hat{n}$ give a ratio of 0.494, about 1.4 standard errors below the data. But it points

TABLE 7.10. Shakespeare word frequencies (Efron and Thisted, 1976): observed A_j/A_1 against the two parameter profile $\Gamma(j-\sigma)/\{j!\,\Gamma(1-\sigma)\}$ at the two spectrum estimates.

j	A_j	observed A_j/A_1	profile, $\sigma = 0.396$	profile, $\sigma = 0.456$
1	14376	1.000	1.000	1.000
2	4343	0.302	0.302	0.272
3	2292	0.159	0.162	0.140
4	1463	0.102	0.105	0.089
5	1043	0.073	0.076	0.063
6	837	0.058	0.058	0.048
7	638	0.044	0.047	0.038
8	519	0.036	0.038	0.031
9	430	0.030	0.032	0.026
10	364	0.025	0.028	0.022

the same way as the likelihood analysis of Chapter 6, which moves to a negative discount parameter, a finite pool of species. A singleton deficit of this shape is a statement about the community and not, as one might first suspect, about the collector.

7.10 An arbitrary weight

Everything so far has used the weight f_α, whose distribution function is the power $G(r) = r^\alpha$. Warren Ewens observes that this family behaves like a conjugate prior, and that the neatness of the exact results came from there. The observation is exactly right, and Theorem 7.11 is its precise form: the powers are the multiplicative weights, $G(rs) = G(r)G(s)$, and multiplicativity is what keeps the residual allocation structure intact. For any other weight with G strictly increasing the induced subdivision is not a residual allocation model at all, so there is no reparameterisation to look for and everything must be asymptotic. This section asks how much is. The answer is that the weight enters the asymptotic theory through a single number, and that the practically useful results survive without change.

7.10.1 Only the index at the endpoint matters

PROPOSITION 7.32. *Suppose that near the origin G has a density $g = G'$ that is regularly varying with index $\rho - 1$, for some $\rho \in (0, \infty)$; then G is regularly varying with index ρ, that is $G(r) = \ell(r)r^\rho$ with ℓ slowly varying. In either layout,*

$$\frac{\log K_n}{\log n} \longrightarrow \sigma_* = \frac{\sigma}{\rho(1-\sigma)+\sigma} \qquad \textit{almost surely,}$$

and more precisely $K_n = n^{\sigma_*}L(n)$ for a slowly varying L depending on ℓ.

Proof. Write $\beta = (1-\sigma)/\sigma$, so that $\rho\beta + 1 = 1/\sigma_*$. By Karamata's theorem $G(r) = \int_0^r g \sim rg(r)/\rho$, which is the regular variation of G. The induced frequencies are

increments of G over relative distances that shrink to zero, and the density is what controls them. By the uniform convergence theorem $g(ur)/g(r) \to u^{\rho-1}$ uniformly in $u \in [\frac{1}{2}, 1]$, so

$$G(r) - G\big(r(1-\delta)\big) \;=\; r\int_{1-\delta}^{1} g(ur)\,du \;=\; \delta\, r g(r)\,(1+o(1)) \;=\; \rho\,\delta\, G(r)\,(1+o(1)) \tag{7.20}$$

as $r \downarrow 0$ and $\delta \downarrow 0$ jointly.

In the size biased layout $p'_k = G(\widetilde{R}_{k-1}) - G(\widetilde{R}_{k-1}(1-B_k))$, with $\widetilde{R}_{k-1} \to 0$ and $B_k \to 0$ almost surely. By (7.20), the Gamma representation $B_k = \Gamma_k/(\Gamma_k + \Gamma'_k)$ of Lemma 7.16 with $(\Gamma_k + \Gamma'_k)/(k\sigma) \to 1$, the limit $\widetilde{R}_{k-1} k^{\beta} \to M$ of Proposition 7.13, and $\ell(\widetilde{R}_{k-1})/\ell(k^{-\beta}) \to 1$ (the uniform convergence theorem for ℓ, since $\widetilde{R}_{k-1}k^{\beta}$ stays in a compact subset of $(0,\infty)$),

$$p'_k \;=\; \rho\, G(\widetilde{R}_{k-1})\, B_k\,(1+o(1)) \;=\; \frac{\rho M^{\rho}}{\sigma}\,\Gamma_k\, k^{-1/\sigma_*}\,\ell(k^{-\beta})\,\varepsilon'_k, \qquad \varepsilon'_k \to 1.$$

This is the expression in the proof of Proposition 7.18 with one more slowly varying factor. A pure power $k^{-\rho\beta-1}$ is not available pathwise, because Γ_k fluctuates on its own scale, so the counting goes through the Gamma sandwich of Lemma 7.16. Put $\phi(k) = k^{1/\sigma_*}/\ell(k^{-\beta})$, regularly varying with index $1/\sigma_*$, and let ψ be an asymptotic inverse of ϕ, regularly varying with index σ_*. With $c_{\pm} = \sigma/\{(1 \mp \eta)\rho M^{\rho}\}$, $N(x)$ lies, up to finitely many terms, between the sums of independent indicators $\mathbf{1}\{\Gamma_k > c_{\pm} x\,\phi(k)\}$, whose means are

$$\begin{aligned}\Lambda_{\pm}(x) \;&=\; \sum_k \mathbb{P}\big(\mathcal{G} > c_{\pm} x\,\phi(k)\big) \;\sim\; \int_0^{\infty} \mathbb{P}\big(\mathcal{G} > c_{\pm} x\,\phi(u)\big)\,du \\ &\sim\; c_{\pm}^{-\sigma_*}\,\mathbb{E}[\mathcal{G}^{\sigma_*}]\,\psi(1/x) \qquad (x \downarrow 0),\end{aligned}$$

the last step by the substitution $t = c_{\pm} x \phi(u)$, Potter's bounds for ψ and dominated convergence. Running the Borel and Cantelli step of Lemma 7.16 along $x_j = (1+\eta)^{-j}$ and letting $\eta \downarrow 0$,

$$N(x) \;\sim\; \Big(\frac{\rho M^{\rho}}{\sigma}\Big)^{\sigma_*}\mathbb{E}[\mathcal{G}^{\sigma_*}]\,\psi(1/x) \qquad \text{almost surely.}$$

At $\ell \equiv 1$, $\psi(y) = y^{\sigma_*}$ and this is the limit in the proof of Proposition 7.18 with $\rho = \alpha$.

In the decreasing layout $p'_k = G(R_{k-1}) - G(R_{k-1} - V_{(k)})$, and by the proof of Proposition 7.15 $V_{(k)} k^{1/\sigma} \to T$ and $R_{k-1} k^{\beta} \to T/\beta$, so $\delta_k = V_{(k)}/R_{k-1} \sim \beta/k$. Now (7.20) gives $p'_k \sim \rho\beta (T/\beta)^{\rho}\, k^{-1/\sigma_*}\ell(k^{-\beta})$ with no random factor at all, and $N(x) \sim \psi\big(\rho\beta(T/\beta)^{\rho}/x\big)$.

In both layouts, therefore, $N(x) = x^{-\sigma_*}\ell_{**}(1/x)$ with ℓ_{**} slowly varying and $\sigma_* \in (0,1)$, and Lemma 7.2 gives $K_n \sim \Gamma(1-\sigma_*)\, n^{\sigma_*}\ell_{**}(n)$ almost surely. □

The density hypothesis is not a technicality. Regular variation of G controls $G(\lambda r)/G(r)$ for fixed λ, but the induced frequencies are increments of G over relative distances of order $1/k$, and regular variation says nothing about those. A continuous G with $G(r)/r^2 \to 1$ can rise only on very short intervals next to the points r_j of a grid with $r_{j+1}/r_j = 1 - \varepsilon_j$, $\varepsilon_j \to 0$ slowly, and be flat in between, so that its density vanishes on most of every cell. When each rise is narrower than one stick at its scale, the whole mass $G(r_j) - G(r_{j+1})$ of a cell falls into a single induced frequency, and $N(x)$ grows only like a power of $\log(1/x)$. With $\varepsilon_j = \{2 + \log(1/r_j)\}^{-1/2}$, $\sigma = \frac{1}{2}$ and 10^7 sticks, $N(10^{-14})$ is 36 for this weight against 15361 for $G(r) = r^2$ on the same sticks, and the local slope of $\log \mathbb{E}[K_n]$ against $\log n$ falls from 0.18 to 0.06 between $n = 10^4$ and $n = 10^{12}$, while for the power it stays near $\sigma_* = 1/3$.

By Proposition 7.5 the index is not an abstract attribute of G but the contact order of f at the endpoint the layout runs towards, which is what a practitioner can inspect. Two densities with the same index give the same exponent however different they are elsewhere, and a density that is not regularly varying at the endpoint is outside the proposition, where, by the example above, the exponent can be lost. The Beta case is recovered at $\ell \equiv 1$, $\rho = \alpha$, and every statement about α in Sections 7.5 to 7.8 that concerns the exponent is a statement about ρ: the odds coordinate still linearises, $\omega \mapsto \rho\,\omega$, and composing weights multiplies indices.

The slowly varying part is not invisible, only subleading. It does not touch σ_*, but it enters K_n through $L(n)$ and it biases finite sample exponent fits. Taking $G(r) = r^2/(1 - \frac{1}{2}\log r)$ at $\sigma = \frac{1}{2}$, so $\rho = 2$ and $\sigma_* = 1/3$, a fit of $\log \mathbb{E}[N(x)]$ on $\log(1/x)$ over four decades returns 0.318 rather than 0.333. The slowly varying factor tilts the line by five percent over that window. Anyone fitting exponents to real data, where ℓ is never known to be constant, should know this.

7.10.2 Four regimes

At $\sigma = 0$ the tail sums decay geometrically, $\widetilde{R}_k \approx e^{-k/\theta}$, and a weight acts through G along that sequence. A G regularly varying with index $\rho > 0$ gives $\log n$ growth, a G vanishing exponentially at the origin gives $\log\log n$, and a G vanishing identically near it gives a bounded count. A slowly varying G is a fourth class, and not a single behaviour. For $G(r) = (1 - \frac{1}{2}\log r)^{-1}$, $G(\widetilde{R}_k) \approx (1 + k/2\theta)^{-1}$, the induced frequencies are of order k^{-2}, and $\mathbb{E}[K_n] \sim \Gamma(\frac{1}{2})\Gamma(\frac{3}{2})(2\theta n)^{1/2}$: power growth, with no σ in sight. (At $\theta = 1$ the constant is 2.221, and simulation gives $\mathbb{E}[K_n]/\sqrt{n} = 2.215$ at $n = 10^6$ and 2.222 at $n = 10^8$.) A weight with $G(r) \sim \{\log(1/r)\}^{-a}$ gives $n^{1/(1+a)}$ in the same way, so at $\sigma = 0$ the exponent of a slowly varying weight is set by the slowly varying function itself. The two parameter theory has the same four classes. What $\sigma > 0$ changes is the exponents: every regularly varying weight with the regular density of part (ii) below gives power growth n^{σ_*} in place of $\log n$, and every slowly varying weight with the regular density of part (i) below gives the single exponent $\sigma_* = 1$, whatever its slowly varying function.

THEOREM 7.33. *Lay the* $\mathrm{PD}(\sigma,\theta)$ *atoms in either layout,* $0<\sigma<1$*, and sample with a weight whose G is continuous and increasing. Then, according to the behaviour of G at the origin:*

(i) *G slowly varying,* $\rho=0$*, with a density g near the origin such that* $rg(r)$ *is slowly varying:* $\sigma_*=1$ *and* $K_n=n/L(n)$ *for a slowly varying L. The fraction* K_n/n *of observations that are new types tends to zero, but only at the slowly varying rate* $1/L(n)$*.*

(ii) *G regularly varying with index* $\rho\in(0,\infty)$*, with a density near the origin regularly varying with index* $\rho-1$*:* $K_n=n^{\sigma_*}L(n)$ *with* $\sigma_*=\sigma/\{\rho(1-\sigma)+\sigma\}$*.*

(iii) *G rapidly varying,* $G(r)=\exp\{-r^{-\tau}(1+o(1))\}$ *for some* $\tau>0$*:*

$$K_n=(\log n)^{\sigma/\{\tau(1-\sigma)\}}\,L(\log n)$$

for a slowly varying L.

(iv) $G\equiv 0$ *on* $(0,r_0)$ *for some* $r_0>0$*:* K_n *is bounded almost surely.*

Proof. Part (ii) is Proposition 7.32. Part (i) is the case $\rho=0$, which that proposition excludes, because Lemma 7.2 with $\gamma=\sigma_*=1$ needs its boundary form. Put $\lambda(r)=rg(r)$, slowly varying by hypothesis. The uniform convergence argument behind (7.20) now gives $G(r)-G(r(1-\delta))=\delta\lambda(r)(1+o(1))$, so in the size biased layout $p'_k=\Gamma_k\,\lambda(k^{-\beta})(\sigma k)^{-1}\varepsilon'_k$ with $\varepsilon'_k\to 1$, and in the decreasing layout the same holds with β in place of Γ_k/σ. The sandwich of that proof, with $\phi(k)=k/\lambda(k^{-\beta})$ regularly varying with index one, gives $N(x)=x^{-1}L_0(1/x)$ with L_0 slowly varying, and the boundary case of Lemma 7.2 gives $K_n\sim nL_1(n)$ with $L_1(y)=\int_y^\infty L_0(u)u^{-1}du$ slowly varying and tending to zero; that is (i) with $L=1/L_1$. The slowly varying function of K_n is not that of the counting function, since $L_1/L_0\to\infty$. For (iii), $R'_k=G(R_k)\approx\exp\{-c\,k^{\tau\beta}\}$ for a random $c>0$, so $-\log p'_k=k^{\tau\beta}\times$ (slowly varying) and $N(x)=\{\log(1/x)\}^{1/(\tau\beta)}\times$ (slowly varying in $\log(1/x)$); the occupancy correspondence with a counting function of this form gives $K_n=(\log n)^{1/(\tau\beta)}L(\log n)$, and $1/(\tau\beta)=\sigma/\{\tau(1-\sigma)\}$. For (iv), $R_k\downarrow 0$, so $R_k<r_0$ for all k beyond some a.s. finite K and $p'_k=0$ thereafter. □

Regime (i) is easiest to see on the first weight of Table 7.11. For $G(r)=(1-\frac12\log r)^{-1}$ at $\sigma=\frac12$, $\lambda(r)=\frac12(1-\frac12\log r)^{-2}$, $N(x)\sim 2x^{-1}\{\log(1/x)\}^{-2}$, $L_1(n)\sim 2/\log n$, and so $K_n\sim 2n/\log n$. In the form $K_n\sim\sum_k\min(np'_k,1)$ the lemma says what is going on: a fraction of the sample of order $1/\log n$ lands in types of frequency below $1/n$, and each of those is seen about once. In simulation, with 3×10^7 sticks and the remaining mass added, the ratio of $\mathbb{E}[K_n]$ to $\sum_k\min(np'_k,1)$ is 0.94 to 0.95 at $n=10^3$ and 0.97 at $n=10^8$, while $n/\mathbb{E}[K_n]$ is 7.1 to 7.7 at $n=10^8$ against $\frac12\log n=9.2$: the approach to the limit is logarithmically slow.

Regime (iii) has no one parameter analogue in this form. A weight that vanishes faster than every power at the endpoint destroys the power law growth entirely

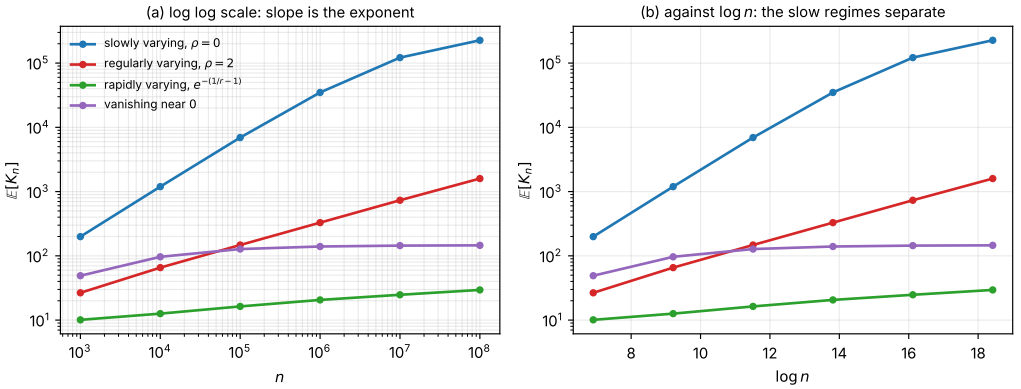


FIGURE 7.7. The four regimes of Theorem 7.33 at $\sigma = \frac{1}{2}$, $\theta = 1$, over five decades of n. Left, on log log axes: the slowly varying and regularly varying weights have limiting slopes $\sigma_* = 1$ and $\sigma_* = 1/3$. The second is straight. The first bends, its local slope falling from 0.78 to 0.27 over this range, because the 3×10^5 simulated sticks leave out about a seventh of the probability mass, the rarest types, and with them about forty percent of $\mathbb{E}[K_n]$ at $n = 10^3$ and ninety eight percent at $n = 10^8$; without that loss its slope would be 0.88 to 0.95. Right, the same counts against $\log n$ on a logarithmic vertical scale, which separates the two slow regimes: the rapidly varying weight keeps climbing, the truncated one has stopped.

and replaces it by a power of a logarithm, with the two parameter structure surviving only in the exponent $\sigma/\{\tau(1-\sigma)\}$ of that logarithm. A community with a genuinely heavy tailed abundance distribution, sampled by a collector whose effort dies off fast enough at the far end, will look logarithmic, which is to say it will look like a one parameter community. Ascertainment can disguise the family, not merely the parameter.

Figure 7.7 shows all four at once, and Table 7.11 gives representative weights. Simulation confirms regime (iii), the least obvious of the four. With $G(r) = \exp\{-(r^{-1}-1)\}$ at $\sigma = \frac{1}{2}$ the prediction is $K_n \asymp \log n$. Over eight decades, at $n = 10^3, 10^5, \ldots, 10^{11}$, the simulated $\mathbb{E}[K_n]$ at $\theta = 1$, over 3×10^5 replicates, is 15.8, 29.4, 43.2, 57.0, 70.8, increasing by about 13.8 per two decades: linear in $\log n$ to within a percent.

7.10.3 What survives, and what needs the power

The results of Sections 7.6 to 7.9 fare differently under a general weight, and the split follows Theorem 7.11 exactly: what depends on the induced frequencies only through the tail of their counting function survives, and what depends on the induced subdivision being a residual allocation model does not.

The spectrum degeneracy survives in full. Proposition 7.24 used only that the counting function is regularly varying with index $-\sigma_*$, which is Proposition 7.32. So the profile $A_j/K_n \to q_j(\sigma_*)$ of (7.16) holds for every weight in regime (ii), Beta or not, and with it the singleton fraction.

TABLE 7.11. Representative weights and their regimes, with the growth of the number of observed types at $\sigma = \frac{1}{2}$.

$G(r)$	regime	K_n at $\sigma = \frac{1}{2}$
$\{1 - \frac{1}{2}\log r\}^{-1}$	slowly varying, $\rho = 0$	$n/L(n)$
r^ρ, $\rho > 0$	regularly varying	$n^{1/(1+\rho)}$
$r^\rho/(1 - \frac{1}{2}\log r)$	regularly varying, $\ell \not\to c$	$n^{1/(1+\rho)}L(n)$
$\exp\{-(r^{-\tau} - 1)\}$	rapidly varying	$(\log n)^{1/\tau}$
$(r - r_0)^+/(1 - r_0)$	vanishing near 0	bounded

COROLLARY 7.34. *For every weight of index $\rho \in (0, \infty)$ in the sense of Proposition 7.32, $A_1/K_n \to \sigma_*$ almost surely, and*

$$\hat{\sigma} \;=\; \frac{\rho\hat{\sigma}_*}{\rho\hat{\sigma}_* + 1 - \hat{\sigma}_*}, \qquad \hat{\sigma}_* = A_1/K_n,$$

is consistent for σ.

Figure 7.8 is the claim in picture form. Three weights of index $\rho = 2$ but quite different shapes, a pure power, a power times a slowly varying factor, and a power times a polynomial, all drive A_1/K_n to the same $\sigma_* = 1/3$. At $n = 10^8$ they read 0.336, 0.312 and 0.332, the middle one slowest because its slowly varying factor does not settle. So correcting a survey drawn with an unknown effort profile does not need the profile. It needs a single number, the index of G at the far end of the transect, and nothing else about f matters even in principle. Two collectors with wildly different habits but the same ρ produce statistically indistinguishable catches at this order.

The index can be read off the positions, which breaks the confounding. In an observational survey ρ is unknown. But recorded positions X_i are an i.i.d. sample from f, and ρ is by definition the lower tail index of $Y = 1 - X$, since $\mathbb{P}(Y \le y) = G(y) = \ell(y)y^\rho$. Estimating a regular variation index from extreme order statistics is Hill's problem, with Hill's answer (Hill, 1975).

COROLLARY 7.35. *Let $Y_{(1)} \le \cdots \le Y_{(n)}$ be the ordered values of $1 - X_i$, and for $m = m_n \to \infty$ with $m/n \to 0$ let*

$$\hat{\rho} \;=\; \Big\{ \frac{1}{m} \sum_{i=1}^{m} \log \frac{Y_{(m+1)}}{Y_{(i)}} \Big\}^{-1}.$$

Then $\hat{\rho} \to \rho$ in probability, and Corollary 7.34 with $\hat{\rho}$ in place of ρ is consistent for σ from the recorded catch alone.

The two ingredients use disjoint information: the abundances give σ_* and know nothing about where the specimens were found, the positions give ρ and know nothing about which species they belonged to. Numerically, with 10^6 positions and the top 10^5 order statistics, Hill returns 2.003 for $G(r) = r^2$ and 0.49 for

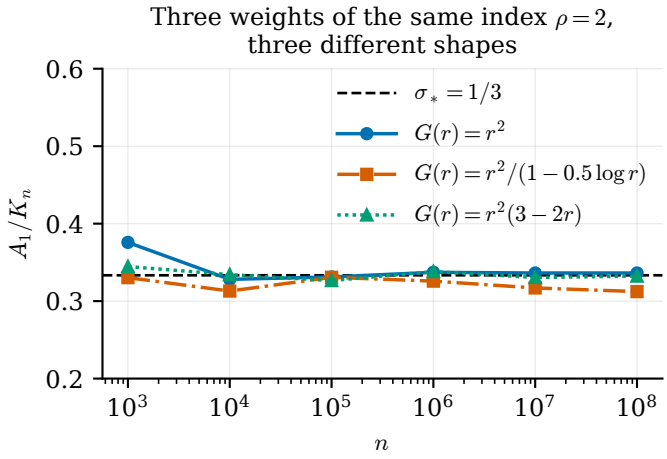


FIGURE 7.8. Only the index matters. Three weights with $\rho = 2$ and different shapes, at $\sigma = \frac{1}{2}$, $\theta = 1$: the singleton fraction converges to $\sigma_* = 1/3$ for all three, most slowly for the weight whose slowly varying factor does not settle.

$G(r) = 2r^{1/2} - r$. For $G(r) = r^2/(1 - \frac{1}{2}\log r)$ it returns 2.20, 2.24, 2.30 as m runs through 10^3, 10^4, 10^5: the slowly varying factor biases Hill upward, the well known behaviour of the estimator under a nonconstant ℓ. The same factor biased the exponent fit of Section 7.10.1 towards a larger effective ρ as well; both estimators see the local index $\rho + d\log\ell/d\log r$, so feeding the Hill $\hat{\rho}$ into the correction partly compensates for the bias in $\hat{\sigma}_*$.

The compensation can be explained heuristically, and the explanation gives a rule for choosing m. Write $\rho(r) = d\log G(r)/d\log r = \rho + d\log\ell/d\log r$ for the local index of G at scale r. Both estimators see a local index rather than ρ itself, and the question is at which scale.

REMARK 7.36 (Scale matching, a heuristic). The singleton fraction is governed by the induced frequencies of order $1/n$, which sit at tail sum scale $r_A(n) \asymp n^{-(1-\sigma)/\{\rho(1-\sigma)+\sigma\}}$, so to first order in the slowly varying factor $(1 - \hat{\sigma}_*)/\hat{\sigma}_* \approx \rho(r_A)\,\omega$. Hill's estimator with m order statistics is governed by G at scale $r_H = G^{-1}(m/n)$, so $\hat{\rho} \approx \rho(r_H)$. The two scales coincide when $m \asymp n\,G(r_A) \asymp n^{\sigma_*} \asymp K_n$. Hence with $m_n \asymp K_n$ the composed estimator $\hat{\omega} = \{(1 - \hat{\sigma}_*)/\hat{\sigma}_*\}/\hat{\rho}$ is free of ℓ to first order, and the corrected $\hat{\sigma}$ is less sensitive to the slowly varying factor than either ingredient. This is a heuristic, not a proved result: the approximations are first order in the variation of ℓ with no error term, and the support for the rule is the arithmetic of the scales together with Table 7.12.

The rule is memorable: take as many extreme positions as there are observed types. It has the right order of magnitude rather than a sharp optimum, because the local index varies slowly, so any m within a constant factor of K_n sees nearly

TABLE 7.12. The corrected estimate of $\sigma = \frac{1}{2}$ under $G(r) = r^2/(1 - \frac{1}{2}\log r)$ at $n = 10^6$, as a function of the number m of extreme positions used in Hill's estimator; mean and standard deviation over 12 independent position samples, the abundance side held fixed at $A_1/K_n = 0.320$ with $K_n = 151$. Correcting with the true $\rho = 2$ gives 0.485.

m	30	50	100	150	300	1000	3000	10^4	10^5
$\hat{\sigma}$	0.507	0.502	0.503	0.508	0.511	0.508	0.510	0.514	0.520
s.d.	0.038	0.036	0.023	0.019	0.016	0.007	0.006	0.004	0.001

the same $\rho(r)$.

Table 7.12 tests this at $G(r) = r^2/(1 - \frac{1}{2}\log r)$, $\sigma = \frac{1}{2}$, $n = 10^6$, where $K_n \approx 150$ and the singleton fraction reads 0.320 against $\sigma_* = 1/3$. Correcting with the *true* $\rho = 2$ gives 0.485, the downward bias of the abundance side passing straight through. Correcting with Hill's $\hat{\rho}$ instead gives 0.502 to 0.511 for every m from 50 to 1000, within about a hundredth of the truth and on the other side of it, because the upward bias of Hill offsets the downward bias of the singleton fraction. The compensation is broad rather than sharp, as the remark predicts, and it degrades only when m is far above K_n, where Hill is looking at a scale the abundances never reach.

The diversity identity survives only up to the slowly varying factor. Proposition 7.17 used $R_k' = \widetilde{R}_k^{\alpha}$ exactly, and for a general G one has $R_k' = \ell(\widetilde{R}_k)\widetilde{R}_k^{\rho}$ instead. The exponent $\kappa = \rho\sigma_*/\sigma$ survives; the constant does not, and $K_n n^{-\sigma_*}$ need not converge at all if ℓ oscillates. If G has a density g near the origin with $g(r)/r^{\rho-1} \to \rho c$, $c \in (0, \infty)$, then everything goes through with α replaced by ρ and an extra factor c^{σ_*} in (7.12). The clean diversity theory needs the weight to be asymptotically a power, not merely regularly varying.

7.11 A worked example: heavy tailed abundance surveys

The one parameter theory is the right model for a community whose species abundance curve is logarithmic, and the two parameter family is the right model when it is not. Power law abundance is the rule rather than the exception in microbial surveys, in word frequency data and in network degree data. It is exactly what $\sigma > 0$ produces: the number of types in a sample of size n grows like n^{σ} rather than like $\log n$. The results above say what a nonuniform sampling effort does to such a survey, and the answer is arithmetically severe.

Suppose the truth is $\sigma = 0.5$, so that types accumulate like $\sqrt{n}$, and the sampling effort follows f_{α}. Table 7.13 reads off the consequences from (7.9), (7.11) and (7.12). The column σ_* is what an analyst would estimate from the growth of the type count. The type count ratio is the factor by which that count at $n = 10^6$ differs from what uniform sampling would have produced, namely $n^{\sigma_* - \sigma}$ times the constant of (7.12), for a community whose diversity is $S = 1$; in general the factor $S^{\kappa-1}$

TABLE 7.13. Consequences of an effort profile f_α for a community with true discount parameter $\sigma = 0.5$. The type count ratio is at $n = 10^6$ and combines the change of exponent with the constant (7.12).

α	effort	σ_*	κ	type count ratio at $n = 10^6$
0.25	strongly persistent	0.800	0.400	73.3
0.5	persistent	0.667	0.667	11.1
1	uniform	0.500	1.000	1.000
2	fading	0.333	1.333	0.086
4	strongly fading	0.200	1.600	0.012

multiplies it, and the ratio of mean counts at $\theta = 1$ is 44.0, 8.31, 1, 0.118 and 0.021. And κ is the power to which the whole diversity distribution is raised.

Three things follow. First, the effect is a change of *exponent*, so it compounds with sample size: a fading collector with $\alpha = 2$ recovers about nine percent of the types a uniform sample would have found at $n = 10^6$, and under three percent at $n = 10^9$ (both figures are for $S = 1$; the ratios of mean counts at $\theta = 1$ are twelve percent, as in Example 7.14, and under four percent). No amount of sampling repairs it, because the deficit grows. This is qualitatively worse than the one parameter case, where an effort profile changes the constant multiplying $\log n$ and the relative deficit is fixed.

Second, the direction is not obvious in advance. A persistent collector, one who works harder at the far end of the transect where the rare types live, inflates the apparent diversity, and at $\alpha = 0.25$ he reports seventy times too many types when $S = 1$; at $\theta = 1$ the ratio of mean counts is forty four. Ascertainment is not simply loss.

Third, and most useful in practice, the correction (7.13) is exact and needs only α. If the effort profile is part of the design, as in a stratified or adaptive survey, then α is known and the bias is removable in full. If it is not, then σ and α enter the type count only through σ_* and are not separately identifiable from it. The confounding that Bayarri and DeGroot describe for weighted distributions in general appears here in its sharpest form, as a single ratio.

7.12 What remains

Three problems are left, the last of them a matter of refinement.

The random envelope in the bias calculation. Proposition 7.27 gives the bias of the singleton fraction under a deterministic envelope, and finds it negligible. The sandwich argument of Lemma 7.16 controls the leading order of the random case but not the $O(1/K_n)$ term, where the fluctuation of the Gamma factors about their mean contributes at the same order as the $-\frac{1}{2}$ from the zeta function. Carrying the Mellin calculation through with the Gamma noise included, and doing the same for $1 - 2A_2/A_1$, is the natural next step. Neither question is the simulation

difficulty of Section 7.9.3: a real survey has no truncated stick, so (7.19) constrains our numerics and not the estimator.

Two parameter weights. Theorem 7.11 closes the question, for densities that charge every interval and in the stick breaking sense, for weights acting as $G(R_k)$ on the tail sums. It says nothing about ascertainment schemes that are not of this form, for instance those depending on the atom index as well as its position. Whether some such scheme preserves Ewens Pitman is open.

Second order scale matching. Remark 7.36 cancels the slowly varying factor to first order by matching the scales at which the two estimators read the local index. The residual, which Table 7.12 shows to be about a hundredth here, comes from the variation of $\rho(r)$ across the two windows and from the fact that the abundance side averages over a range of scales while Hill averages over another. Identifying the constant in $m \asymp K_n$ that minimises it, and the second order term itself, would turn a rule of thumb into a procedure.

7.13 Summary

In one sentence: the reduction is universal in the law of the subdivision and in the direction it is laid down; exactness is the Ewens case and no other weight that charges every interval rescues it; the first order rescaling generalises with the odds $(1-\sigma)/\sigma$ in place of θ; the limiting diversity is the unweighted one raised to the power κ, by a map free of θ; and the coincidence theory does not generalise at all.

7.14 Exercises

1. Verify the $n = 4$ column for $(\sigma, \theta) = (\frac{1}{4}, 2)$ in Table 7.1 directly from (7.3), and check that the column sums to one. *Hint.* The denominator is $(3)_3 = 60$; the multiplicities are $1, 4, 3, 6, 1$.

2. Show that under $\mathrm{PD}(\sigma, \theta)$ the probability that n uniformly sampled points are all distinct is $\prod_{i=1}^{n-1}(\theta + i\sigma)/(\theta+1)_{n-1}$, and that it tends to zero for every $\sigma < 1$. *Hint.* Take all $n_j = 1$ in (7.3); for the limit note that the ith factor $(\theta + i\sigma)/(\theta + i)$ tends to $\sigma < 1$.

3. Let $1 - B \sim \mathrm{Beta}(\theta, 1)$. Show that $(1-B)^\alpha \sim \mathrm{Beta}(\theta/\alpha, 1)$, and deduce that the weighted image of a size biased $\mathrm{GEM}(\theta)$ subdivision under $f_\alpha(x) = \alpha(1-x)^{\alpha-1}$ is $\mathrm{GEM}(\theta/\alpha)$. *Hint.* Compute $\mathbb{P}\{(1-B)^\alpha \le v\}$ and use Lemma 7.3 with $G(r) = r^\alpha$.

4. Under $\mathrm{ESF}(\theta)$ the homozygosity is $1/(1+\theta)$. Show that a collector who samples a size biased $\mathrm{GEM}(\theta)$ subdivision with f_α observes homozygosity $\alpha/(\alpha+\theta)$, and find the α that doubles it when $\theta = 3$. *Hint.* Use Exercise 3; the answer is $\alpha = 3$.

5. Prove the composition law $T_\alpha \circ T_{\alpha'} = T_{\alpha\alpha'}$ of Proposition 7.23 directly from the formula $T_\alpha(\sigma) = \sigma/\{\alpha(1-\sigma) + \sigma\}$, without the odds coordinate. *Hint.* Compute $1/T_\alpha(\sigma) - 1$.

TABLE 7.14. What the ascertainment calculus does to the two families.

	one parameter, $\sigma = 0$	two parameter, $\sigma > 0$
the reduction	holds	holds, unchanged
genealogical construction	defining events, θ/α	feedback or multiple mergers
layout matters for exactness	yes	yes
layout matters for leading order	no (Prop 7.6)	no (Prop 7.15)
direction of layout	absorbed into G (Lemma 7.4)	
family preserved	yes, $\theta \mapsto \theta/\alpha$	no, by no weight charging every interval (Thm 7.11)
image of the weighting	GEM(θ/α)	explicit RAM, (7.5)
growth of K_n	$(\theta/\alpha)\log n$	$n^{\sigma_*+o(1)}$
parameter rescaled	$\theta \mapsto \theta/\alpha$	$(1-\sigma)/\sigma \mapsto \alpha(1-\sigma)/\sigma$
role of the other parameter	·	θ absent from σ_* and from C
limit law of the diversity	degenerate	$C\,S^\kappa$, C by (7.12)
limit of (A_j)	independent Poisson(ϑ/j)	$q_j(\sigma_*)S_*$, degenerate
coincidence probability	$\to e^{-\gamma\vartheta}/\Gamma(1+\vartheta)$	$\to 0$, any weight with $\sigma_* > 0$
correction for a biased sample	$\theta = \alpha\vartheta$	(7.13)
estimator of the parameter	$K_n/\log n$	A_1/K_n, or $1 - 2A_2/A_1$
arbitrary weight, index ρ	$K_n \sim (\theta/\rho)\log n$	$\alpha \mapsto \rho$ throughout; four regimes

6. Show that $\kappa = \alpha\sigma_*/\sigma$ satisfies the composition rule (7.14). *Hint.* Write $\kappa(\sigma,\alpha) = \alpha/\{\alpha(1-\sigma)+\sigma\}$ and use $1 - T_{\alpha'}(\sigma) = \alpha'(1-\sigma)/\{\alpha'(1-\sigma)+\sigma\}$.

7. Show that the Sibuya probabilities $q_j(\sigma)$ of (7.16) satisfy $q_{j+1}/q_j = (j-\sigma)/(j+1)$, and deduce that $A_2/A_1 \to (1-\sigma_*)/2$ and $A_3/A_2 \to (2-\sigma_*)/3$. Propose a third estimator of σ_* from A_3/A_2 and compute it for the Shakespeare data of Example 7.30. *Hint.* $\Gamma(j+1-\sigma) = (j-\sigma)\Gamma(j-\sigma)$; the estimate is $2 - 3(2292)/4343 = 0.417$.

8. A survey reports $K_n = 640$ types of which $A_1 = 192$ are singletons, collected with a design known to have $\alpha = 3$. Estimate σ_* and σ, and give an approximate standard error for each. *Hint.* $\hat\sigma_* = 0.30$ and $\hat\sigma = 0.5625$; use $\{\hat\sigma_*(1-\hat\sigma_*)/K_n\}^{1/2}$ and the delta method factor $\sigma(1-\sigma)/\{\sigma_*(1-\sigma_*)\}$.

9. Let $f(x) = 6x(1-x)$ and lay the intervals from the left. Find the index ρ of G at the origin, and hence σ_* when $\sigma = \frac{1}{2}$. Repeat for $f(x) = 2x$. *Hint.* Proposition 7.5: $f(x) \sim 6(1-x)$ at $x = 1$ gives $\rho = 2$ and $\sigma_* = 1/3$; $f(x) = 2x$ tends to 2 at $x = 1$, so $\rho = 1$ and $\sigma_* = \frac{1}{2}$.

10. The defining event construction of Proposition 7.12 gives $K_3 = 1 + \xi_2 + \xi_3$ with independent $\xi_j \sim \text{Bernoulli}(p_j)$. Show that no choice of $p_2, p_3 \in [0,1]$ reproduces the law of K_3 under $\text{PD}(\frac{1}{2}, 1)$, so the obstruction is visible already at $n = 3$, before any variance comparison can detect it. *Hint.* From Table 7.1, $\mathbb{P}(K_3 = 1) = 1/8$ and $\mathbb{P}(K_3 = 3) = 1/2$; these force $p_2 + p_3 = 11/8$ and $p_2 p_3 = 1/2$, and the quadratic with these roots has negative discriminant.

11. (Computation.) Implement the moment recursion of Example 7.10 and reproduce the $\alpha = 2$ column of Table 7.4. Then vary α over $(0, 5)$ and plot the ratio for the type 2+2; confirm that it equals one at $\alpha = 1$. *Hint.* Run the recursion backwards from a truncation index of 10^5; all moments are Beta function ratios, best computed through log gamma functions.

12. (Computation.) Simulate a ranked $\text{PD}(\frac{1}{2}, 1)$ subdivision, sample it with f_2, and compare A_1/K_n at $n = 10^6$ with the value $1/3$ given by Proposition 7.15 and Corollary 7.25. *Hint.* Generate size biased sticks, sort them, and use (7.19) to choose the number of sticks; Rao Blackwellise the counts as in Table 7.9.

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CHAPTER 8

Pólya Inverse Gamma Posteriors, Fisher's Species Model, and Good's Predictive Rule for the Ewens Sampling Problem

McCullagh (2016) observes that Fisher et al. (1943) almost discovered the Ewens sampling formula and that Good (1953) and Good and Toulmin (1956) asked the right predictive questions but closed neither. This chapter gives an inferential account that connects the two through Pólya inverse gamma (PIG) augmentation (He et al., 2019). Three points organise it. First, the Ewens richness likelihood is a regular one parameter natural exponential family in $\log\theta$ *with cumulant* $\log\{\Gamma(\theta+n)/\Gamma(\theta)\}$ *and sufficient statistic* K*, the number of species; the maximum likelihood equation is pure moment matching,* $K = \mathbb{E}[K \mid \theta]$*, and a Bernstein and von Mises argument shows that the Bayes point estimator agrees with the maximum likelihood estimate to relative order* $O(1/K)$ *and the credible interval with the Wald interval to leading order. In this precise sense the Bayes estimator of* θ *sheds little light. Second, Fisher's two parameters map exactly onto the two PIG representations: the richness* θ *needs only the gamma ratio (Beta) block* $\Gamma(\theta)/\Gamma(\theta+n)$*, while the shape* ν *that governs departure from the log series or Ewens limit is a reciprocal gamma* $1/\Gamma(\nu)$ *problem requiring the full PIG(0) engine. The gamma subordinator, the prototypical generalized gamma convolution, is the structure common to Fisher's model, the Ewens partition, and PIG. Third, Good's nonparametric estimator, once given a parametric skeleton, collapses exactly to the Chinese restaurant rule* $\mathbb{E}[q_r \mid \textit{data}] = r/(n+\theta)$*, so the PIG posterior yields a fully Bayesian Good Turing smoothing with honest bands. Worked examples on small cases, on Corbet's Malayan butterflies and on Williams' Rothamsted Macrolepidoptera reproduce Williams' fit (*$\hat{\theta} = 40.146$*) and the temporal homogeneity check (58 expected new species against 62 observed), check the PIG(0) simulator against* $e^{-\gamma s}/\Gamma(1+s)$ *to within* 0.07%*, and show the Beta augmentation sampler delivering about seven times the effective samples per second of random walk Metropolis, and about half that when the random walk proposal is optimally scaled.*

8.1 Introduction

A sample of n specimens from a large population falls into K distinct species. Writing a_j for the number of species represented exactly j times, the abundance vector $a = (a_1, \ldots, a_n)$ satisfies $\sum_j a_j = K$ and $\sum_j j a_j = n$. Fisher et al. (1943) modelled the frequency of frequencies and, with Corbet's Malayan butterflies and Williams' Rothamsted Macrolepidoptera, introduced a richness index that we now

recognise as the parameter θ of the Ewens sampling formula (Ewens, 1972). In population genetics the same $\theta = 4M\mu$ is the scaled mutation rate, the species are allele types, and a_j counts the alleles carried by exactly j of the n sampled genes. McCullagh (2016) shows that Fisher's two parameter Poisson process model contains the Ewens distribution as the conditional law of a given n, so that Fisher "had it in his grasp", and that Good (1953) and Good and Toulmin (1956), who pioneered sequential prediction of new species, missed the Chinese restaurant description for want of a parametric skeleton.

The history repays a moment's attention because it explains what is and is not at stake. Fisher was handed a table of frequencies of frequencies, the counts $a_1, a_2, \ldots$ of species seen once, twice, and so on, and asked how many species the population holds. His answer, the index of diversity, is the modern θ, and his log series curve fits the Malayan and Rothamsted data strikingly well. What Fisher did not write down is the exchangeable partition process whose one step ahead rule, the probability $\theta/(n+\theta)$ that the next specimen is new, generates the whole distribution. Good and Toulmin, working a decade later on the same prediction problem, built nonparametric estimators of exactly that quantity but declined to name a model, so their formulas never simplified. Ewens, arriving from population genetics, wrote the distribution in one line. The three approaches are a single object seen from estimation, prediction, and combinatorics, and the modern task is less to choose among them than to say precisely how they fit together and where a Bayesian treatment adds something the others lack.

We revisit the problem from the standpoint of Pólya inverse gamma augmentation (He et al., 2019), a data augmentation calculus for densities in which a parameter is nested inside gamma functions. The Ewens richness likelihood is $L(\theta) \propto \theta^K \Gamma(\theta)/\Gamma(\theta+n)$, a gamma ratio in θ; Fisher's second parameter enters through a reciprocal gamma $1/\Gamma(\nu)$; and Good's smoothing is a ratio of expected multiplicities. All three are PIG objects, and organising them this way makes precise both what Bayesian inference contributes and what it does not. The organising claim is a division of labour. The richness θ is estimated almost as well by maximum likelihood as by any Bayesian rule, because the model reduces to a one parameter exponential family whose only statistic is the species count K, and K accumulates information slowly. Fisher's shape ν and every predictive quantity, by contrast, live where the posterior genuinely matters, and it is there that the PIG machinery earns its place.

What the chapter does. Section 8.2 shows that the Ewens richness likelihood is a regular natural exponential family (NEF) in $\varphi = \log\theta$ with sufficient statistic K, so that maximum likelihood is moment matching, and works the case $n = 4$ by hand. Section 8.3 sets out the two PIG representations, the Beta block for the gamma ratio and the PIG(0) block for a reciprocal gamma, and runs the resulting Gibbs sampler on a small sample line by line. Section 8.4 places Fisher's model, Kingman's Ewens partition, and the PIG calculus inside one gamma subordinator. Section 8.5 proves that Good's estimator collapses to $r/(n+\theta)$ and applies a Bayesian Good Turing

rule to Corbet's butterflies. Section 8.6 states the Bernstein and von Mises theorem that makes the Bayes estimator of θ redundant, with a table across K. Section 8.7 studies the two regimes, small K and K near n, where the posterior is strongly skewed, and heavy tailed when K is near n, and recommends a weakly informative prior in the PIG family. Section 8.8 untangles the two variances of the species count that appear in Fisher's and Ewens' treatments and gives an exact Gibbs sampler under a heavy tailed prior. Section 8.9 collects the simulation checks and the Williams analysis, Section 8.10 discusses, and the proofs, sampler details and exercises close the chapter.

8.2 The Ewens richness model as a natural exponential family

This section shows that all the information about richness in a sample is carried by one number, the species count K, and that the model is the simplest kind of exponential family. Everything later in the chapter, the samplers, the Bernstein and von Mises statement, and the heavy tails, rests on this structure.

The Ewens sampling formula assigns the abundance vector the probability

$$p(a \mid \theta) = \frac{n!}{(\theta)_{(n)}} \prod_{j=1}^{n} \frac{1}{a_j!} \Big(\frac{\theta}{j}\Big)^{a_j}, \qquad (\theta)_{(n)} = \theta(\theta+1)\cdots(\theta+n-1) = \frac{\Gamma(\theta+n)}{\Gamma(\theta)}. \tag{8.1}$$

Isolating θ gives $L(\theta) \propto \theta^K \Gamma(\theta)/\Gamma(\theta+n)$, so $K = \sum_j a_j$ is minimal sufficient: given K the abundance profile has a distribution free of θ, and all information about richness is the single count K. This is the Fisher Neyman factorisation at work and, as we now record, the whole model is a one parameter exponential family.

EXAMPLE 8.1 (The Ewens formula for four genes). Take $n = 4$. There are five abundance vectors, one for each way of writing 4 as a sum of positive integers, and (8.1) gives the probabilities in Table 8.1. At $\theta = 1$ the rising factorial is $(1)_{(4)} = 24$ and the formula reduces to the cycle type law of a uniformly random permutation of four objects: there are 1, 6, 3, 8 and 6 permutations with the five cycle types, and the probabilities are these counts divided by 24. At $\theta = 2$ the rising factorial is $2 \cdot 3 \cdot 4 \cdot 5 = 120$ and the mass shifts toward many species. In both columns the probabilities add to one, and the mean number of species, $\sum_a K(a)\, p(a \mid \theta)$, equals $\sum_{i=0}^{3} \theta/(\theta+i)$: $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = 25/12 \approx 2.083$ at $\theta = 1$ and $1 + \frac{2}{3} + \frac{2}{4} + \frac{2}{5} = 2.567$ at $\theta = 2$. Reading the table down the K column shows sufficiency at work: the two vectors with $K = 2$, namely $(0,2,0,0)$ and $(1,0,1,0)$, have probabilities in the fixed ratio $3 : 8$ at every θ, because θ enters both only through $\theta^2/(\theta)_{(4)}$.

The intuition for the next result is that a likelihood of the form $\theta^K \times$ (function of θ) is an exponential family in $\log\theta$, so its cumulant function delivers the mean and variance of K by differentiation.

TABLE 8.1. The Ewens sampling formula for $n = 4$. The numerator column is $n!\prod_j (\theta/j)^{a_j}/a_j!$ evaluated at the stated θ; dividing by $(\theta)_{(4)}$ gives the probability.

$a = (a_1, a_2, a_3, a_4)$	K	general numerator	$\theta = 1$		$\theta = 2$	
			numer.	prob.	numer.	prob.
$(4,0,0,0)$	4	θ^4	1	0.0417	16	0.1333
$(2,1,0,0)$	3	$6\theta^3$	6	0.2500	48	0.4000
$(0,2,0,0)$	2	$3\theta^2$	3	0.1250	12	0.1000
$(1,0,1,0)$	2	$8\theta^2$	8	0.3333	32	0.2667
$(0,0,0,1)$	1	6θ	6	0.2500	12	0.1000
total		$(\theta)_{(4)}$	24	1	120	1

PROPOSITION 8.2 (NEF structure). *With natural parameter $\varphi = \log\theta$ and statistic K,* (8.1) *is a regular NEF, $p(a \mid \varphi) = h(a)\exp\{\varphi K - A(\varphi)\}$, with cumulant*

$$A(\varphi) = \log\frac{\Gamma(e^\varphi + n)}{\Gamma(e^\varphi)} = \log(\theta)_{(n)}.$$

Its derivatives are $A'(\varphi) = \theta[\psi(\theta+n) - \psi(\theta)] = \mathbb{E}[K \mid \theta]$ and $A''(\varphi) = \mathrm{Var}(K \mid \theta) = \sum_{i=0}^{n-1}\frac{\theta i}{(\theta+i)^2} > 0$ for $n \geq 2$, where $\psi = \Gamma'/\Gamma$. Hence $\log L$ is strictly concave in φ, the maximum likelihood estimate (MLE) is unique when $1 < K < n$, and it solves the moment equation

$$K = \theta[\psi(\theta+n) - \psi(\theta)] = \mathbb{E}[K \mid \theta], \tag{8.2}$$

and the Fisher information in θ is $I(\theta) = \mathrm{Var}(K \mid \theta)/\theta^2 = \theta^{-1}\sum_{i=0}^{n-1} i/(\theta+i)^2$.

The Feller coupling writes $K = \sum_{i=0}^{n-1} B_i$ with independent $B_i \sim \mathrm{Bernoulli}(\theta/(\theta+i))$, from which the moments in Proposition 8.2 follow at once; the proof is in Section 8.11. Equation (8.2) is the exponential family identity "observed = expected": the MLE sets the number of species equal to its mean. Since $\mathbb{E}[K \mid \theta]$ increases continuously from 1 (as $\theta \to 0$) to n (as $\theta \to \infty$), the equation has a finite positive root exactly when $1 < K < n$; at the boundaries $K = 1$ and $K = n$ the MLE is 0 or $+\infty$, a fact that returns with force in Section 8.7.

The coupling is worth reading directly, because it is the mechanism behind every result in the chapter. Imagine the specimens arriving one at a time. The arrival that finds i specimens already seated either founds a new species with probability $\theta/(\theta+i)$ or joins an old one; the indicator B_i of founding is independent across i because the Chinese restaurant rule depends only on the count i, not on the configuration. The number of species is then a sum of independent coin flips whose success probabilities decay like θ/i, so K behaves like $\theta\sum_{i<n} 1/i \approx \theta\log n$: information about θ enters one nearly fair coin at the start and progressively rarer coins thereafter. Sufficiency of K now has a plain reading. Two samples with the same species count carry identical evidence about richness however differently their specimens are distributed among species, because given the sequence of

founding events the way later specimens join old species does not involve θ. The moment equation (8.2) inherits the same simplicity: since the log likelihood is concave in $\varphi = \log\theta$ with curvature equal to $\mathrm{Var}(K)$, there is a single hump, no multimodality to worry a sampler, and the MLE is found by matching one number.

EXAMPLE 8.3 (Moments by coin flips). For $n = 4$ and $\theta = 1$ the coins have success probabilities $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$, so $\mathbb{E}[K] = 25/12$ and $\mathrm{Var}(K) = 0 + \frac{1}{4} + \frac{2}{9} + \frac{3}{16} = 0.6597$, which agrees with the variance computed from the $\theta = 1$ column of Table 8.1. For a sample of $n = 50$ with $K = 10$ species, solving (8.2) numerically gives $\hat\theta = 3.480$; at that value $\mathrm{Var}(K \mid \hat\theta) = 6.20$, so $I(\hat\theta) = 6.20/3.480^2 = 0.512$ and the standard error is $I(\hat\theta)^{-1/2} = 1.40$. Ten species in fifty specimens therefore pin θ down only to within a factor of about two either way. Note how much smaller $\mathrm{Var}(K) = 6.20$ is than $\mathbb{E}[K] = 10$: the coins with large i are rarely heads, and a sum of rare Bernoulli variables has variance below its mean.

Conjugate prior. The NEF carries a Diaconis and Ylvisaker conjugate family (Diaconis and Ylvisaker, 1979). For $n_0 > 1$, $\tau > 0$ and $\tau - 1 < \kappa < \tau n_0 - 1$,

$$\pi(\theta \mid \kappa, \tau, n_0) \;\propto\; \theta^{\kappa} \Big[\frac{\Gamma(\theta)}{\Gamma(\theta + n_0)}\Big]^{\tau}, \tag{8.3}$$

which is proper (Section 8.11) and reads as τ prior Ewens samples of size n_0 displaying κ species in total, measured against the base measure $\mathrm{d}\theta$. Given independent samples $(n_j, K_j)_{j=1}^{J}$ the posterior is $\pi(\theta \mid \text{data}) \propto \theta^{\kappa + K_\bullet}[\Gamma(\theta)/\Gamma(\theta + n_0)]^{\tau} \prod_j \Gamma(\theta)/\Gamma(\theta + n_j)$ with $K_\bullet = \sum_j K_j$, closed in the family when the n_j are all equal to n_0. In practice we default to a diffuse gamma prior, which is not conjugate but is handled *exactly* by the augmentation of the next section. Throughout the chapter the "diffuse prior" is $\mathrm{Ga}(10^{-2}, 10^{-2})$ unless stated otherwise.

The information grows logarithmically. The Feller representation gives the Korwar and Hollander (1973) law of large numbers, $K/\log n \to \theta$ almost surely, and a central limit theorem $K \approx \mathcal{N}(\theta \log n, \theta \log n)$. Since $\mathbb{E}[K] = \theta[\psi(\theta + n) - \psi(\theta)] = \theta \log n - \theta\psi(\theta) + o(1)$, the effective number of observations for θ is K itself, which grows only like $\log n$: even Williams' $n = 15{,}609$ specimens supply just $K = 240$ effective data points, and $K/\log n = 24.9$ still sits well below $\theta = 40$ because convergence is logarithmically slow. This is the quantitative reason the richness parameter is hard to pin down no matter how many specimens are collected, and the backdrop for the Bernstein and von Mises statement of Section 8.6. In genetic terms, sequencing ten times as many genes from a neutral population adds only about $\theta \log 10 \approx 2.3\,\theta$ new allele types, and so only that many new effective observations on the mutation rate.

8.3 Pólya inverse gamma augmentation: two representations

This section supplies the computational tool. The obstruction to conjugacy is that θ sits inside a gamma ratio, and He et al. (2019) give two integral representations

that remove such obstructions; the Ewens problem uses the first, Fisher's second parameter the second.

The idea of data augmentation is to trade a hard density for a tractable joint density with an extra latent variable, so that each conditional is a standard draw. A density is awkward when its normaliser or a factor cannot be written in closed exponential family form; the trick is to represent that factor as an expectation over a latent variable chosen so that, given the latent, the target becomes conjugate. The gamma ratio $\Gamma(\theta)/\Gamma(\theta+n)$ and the reciprocal $1/\Gamma(\nu)$ are the two shapes in which a gamma function can trap a parameter, and PIG supplies a latent for each. What makes the calculus attractive here is that the latents are elementary, a Beta variable in the first case and an infinitely divisible positive variable in the second, so the resulting samplers avoid any evaluation of Γ or its logarithm inside the inner loop and mix close to independently.

R1: the gamma ratio (Beta block). The ratio in (8.1) has the Beta representation, row three of Table 2 in He et al. (2019),

$$\frac{\Gamma(\theta)}{\Gamma(\theta+n)} = \frac{1}{\Gamma(n)} \int_0^1 \eta^{\theta-1}(1-\eta)^{n-1}\,\mathrm{d}\eta, \tag{8.4}$$

which is nothing more than the Beta integral $B(\theta,n) = \Gamma(\theta)\Gamma(n)/\Gamma(\theta+n)$ divided by $\Gamma(n)$. The factor $\eta^{\theta-1} = e^{(\theta-1)\log\eta}$ is exponential in θ, which is the whole point. Under a $\mathrm{Ga}(a_0,b_0)$ prior the augmented posterior $p(\theta,\eta \mid K) \propto \theta^{K+a_0-1}e^{-b_0\theta}\eta^{\theta-1}(1-\eta)^{n-1}$ has the original posterior as its θ marginal and gives the two block Gibbs sampler

$$\theta \mid \eta \sim \mathrm{Ga}\big(K+a_0,\ b_0-\log\eta\big), \qquad \eta \mid \theta \sim \mathrm{Be}(\theta,n), \tag{8.5}$$

both draws standard, no gamma function evaluation required. This is a variant of the Escobar and West (1995) augmentation for the Dirichlet process concentration; each factor $\Gamma(\theta)/\Gamma(\theta+n_j)$ contributes one latent η_j, so the multisample posterior of (8.3) needs one Beta draw per sample.

EXAMPLE 8.4 (Checking the Beta identity). At $\theta = 2.5$, $n = 10$ the left side of (8.4) is $\Gamma(2.5)/\Gamma(12.5) = 9.7143\times10^{-9}$, and numerical integration of the right side returns the same value to a relative error below 10^{-9}. At the Williams scale, $\theta = 40.146$ and $n = 50$, both sides equal 1.0967×10^{-90}; at $\theta = 0.5$, $n = 5$ both equal $(0.5\cdot1.5\cdot2.5\cdot3.5\cdot4.5)^{-1} = 0.033862$. The latent η has a direct meaning: since $\mathbb{E}[-\log\eta \mid \theta] = \psi(\theta+n)-\psi(\theta)$, the gamma rate $b_0-\log\eta$ in (8.5) is a one draw estimate of $\mathbb{E}[K \mid \theta]/\theta$, and the Gibbs step for θ is a randomised version of the moment equation (8.2).

EXAMPLE 8.5 (The Beta augmentation sampler, line by line). Take $n = 50$, $K = 10$ and the diffuse prior $a_0 = b_0 = 0.01$, and start the chain at $\theta^{(0)} = 1$. Each iteration draws $\eta \sim \mathrm{Be}(\theta,50)$, forms the rate $0.01-\log\eta$, and draws $\theta \sim \mathrm{Ga}(10.01,\mathrm{rate})$. Table 8.2 lists the first six iterations from one run. The conditional mean $10.01/\mathrm{rate}$ moves from 1.7 to about 3.9 within three steps, the chain forgets its poor starting

TABLE 8.2. First six iterations of the Beta augmentation sampler (8.5) for $n = 50$, $K = 10$, prior $\mathrm{Ga}(0.01, 0.01)$, started at $\theta^{(0)} = 1$.

t	$\eta^{(t)} \sim \mathrm{Be}(\theta^{(t-1)}, 50)$	rate $0.01 - \log \eta^{(t)}$	$\mathbb{E}[\theta \mid \eta^{(t)}] = 10.01/\text{rate}$	$\theta^{(t)}$
1	0.0029	5.857	1.709	2.510
2	0.0331	3.419	2.928	2.630
3	0.0607	2.813	3.559	2.805
4	0.0781	2.560	3.910	4.635
5	0.0591	2.838	3.527	5.111
6	0.0796	2.541	3.939	5.377

value at once, and the draws scatter around the posterior centre. Over 99,000 further iterations the chain gives posterior mean 3.596, standard deviation 1.468 and central 95% interval $[1.43, 7.11]$, against 3.602, 1.469 and $[1.43, 7.10]$ from exact quadrature of $\theta^{K+a_0-1} e^{-b_0\theta} \Gamma(\theta)/\Gamma(\theta + n)$. The integrated autocorrelation time is 2.1, so each draw is worth about half an independent one. The posterior mean sits above the MLE 3.480 and the interval is lopsided about it, $[\hat\theta - 2.05, \hat\theta + 3.62]$, a first sign of the right skew studied in Section 8.7.

REMARK 8.6 (Drawing η when θ is small). When K is small the chain visits values of θ near zero, where a $\mathrm{Be}(\theta, n)$ draw can underflow to $\eta = 0$ in floating point and the rate $-\log \eta$ becomes infinite. The sampler needs only $\log \eta$, which is computed stably as $\log \eta = \log G_1 - \log(G_1 + G_2)$ with $G_2 \sim \mathrm{Ga}(n, 1)$ and $\log G_1 = \log G' + (\log U)/\theta$, $G' \sim \mathrm{Ga}(\theta + 1, 1)$, $U \sim \mathrm{U}(0, 1)$, using the identity $G' U^{1/\theta} \sim \mathrm{Ga}(\theta, 1)$. All chains in this chapter are run this way.

R2: the reciprocal gamma (PIG(0) block). When a parameter appears as $1/\Gamma(\alpha)$ rather than in a fixed n ratio, the relevant object is the Pólya inverse gamma law of He et al. (2019, Definition 2.1): $X_0 \sim \mathrm{PIG}(0)$ if and only if

$$\mathbb{E}\left[e^{-s^2 X_0}\right] = \frac{e^{-\gamma s}}{\Gamma(1+s)}, \qquad s > 0, \tag{8.6}$$

with γ the Euler Mascheroni constant. Rearranged, $1/\Gamma(\nu) = \nu\, e^{\gamma\nu}\, \mathbb{E}[e^{-\nu^2 X_0}]$, so a reciprocal gamma factor becomes an average of Gaussian kernels in ν, and given X_0 the parameter ν enters the augmented density through $e^{\gamma\nu - \nu^2 X_0}$. The next proposition, which is Theorem 1 of He et al. (2019) with a short self contained proof in Section 8.11, makes X_0 simulable.

PROPOSITION 8.7 (Inverse gamma series for PIG(0)). *Let* $Y_1, Y_2, \ldots$ *be independent* $\mathrm{InvGa}(3/2, 1)$ *variables, that is* $1/Y_k \sim \mathrm{Ga}(3/2, 1)$. *Then* $X_0 = \sum_{k \ge 1} Y_k/(4k^2)$ *converges almost surely and satisfies* (8.6). *In particular* $\mathbb{E}[X_0] = \sum_k 1/(2k^2) = \zeta(2)/2 = \pi^2/12$, *while* $\mathrm{Var}(X_0) = \infty$.

The intuition is that each inverse gamma term contributes one Weierstrass factor $(1 + s/k)e^{-s/k}$ of $1/\Gamma(1+s)$, and the infinite product of these factors is

exactly $e^{-\gamma s}/\Gamma(1+s)$. Representation R2 is what one needs for Fisher's shape ν and for the vector concentration $\alpha = (\alpha_1, \ldots, \alpha_K)$ of the finite Dirichlet multinomial (He et al., 2019, Section 4.3), where individual factors $1/\Gamma(\alpha_k)$ appear. The infinite variance is worth noticing: X_0 is itself heavy tailed, so its sample mean settles slowly, as Section 8.9 shows.

REMARK 8.8. The scalar Ewens richness problem needs only R1. That the harder representation R2 is unnecessary here is the computational shadow of the inferential fact, made precise in Section 8.6, that the Bayes estimator of θ adds little.

8.4 Fisher's two parameters and the gamma subordinator picture

This section explains why the Ewens partition, Fisher's log series, and the PIG calculus are one object, and it identifies which of Fisher's two parameters calls for which PIG representation.

Fisher's model posits species intensities λ distributed as a Poisson process with a gamma mean measure G on $(0,\infty)$, and Poisson(λ) specimen counts. Concretely $G(\mathrm{d}\lambda) = c\,\lambda^{\nu-1}e^{-\lambda/\beta}\,\mathrm{d}\lambda$ with shape ν, scale β and a constant c that we absorb into θ. The joint intensity and count process is $Z \sim \mathrm{PP}(\mu)$ with $\mu(\mathrm{d}\lambda, r) = e^{-\lambda}\lambda^r/r!\; G(\mathrm{d}\lambda)$, and the frequency of frequencies $m_r = \#\{\lambda : (\lambda, r) \in Z\}$ is Poisson with

$$\mu_r = \mathbb{E}[m_r] = \theta\,\frac{\Gamma(r+\nu)}{r!\,\Gamma(\nu)}\,x^r(1-x)^\nu, \qquad r \geq 1, \qquad x = \frac{\beta}{1+\beta}, \tag{8.7}$$

a negative binomial series (McCullagh, 2016). As $\nu \to 0$, with θ held fixed in this parametrisation, $\Gamma(r+\nu)/\Gamma(\nu) \sim \nu\,(r-1)!$; absorbing the factor ν into θ gives $\mu_r \to \theta x^r/r$, Fisher's log series, with total specimen count $N = \sum_{r\geq 1} r\,m_r$ and species count $K = \sum_{r\geq 1} m_r$ having means $\mathbb{E}[N] = \theta x/(1-x)$ and $\mathbb{E}[K] = -\theta\log(1-x)$. Eliminating x between the last two gives Fisher's famous relation $\mathbb{E}[K] = \theta\log(1+\mathbb{E}[N]/\theta)$ (Fisher et al., 1943). Fisher concluded from the butterfly data that $\nu \approx 0$.

The common structure. The gamma subordinator $\{G_t\}_{t\geq 0}$ has Lévy measure $\rho(\mathrm{d}\lambda) = \lambda^{-1}e^{-\lambda}\,\mathrm{d}\lambda$, the prototypical generalized gamma convolution (GGC) with a degenerate Thorin measure (Bondesson, 1992). Normalising by the total mass $G_\theta \sim \mathrm{Ga}(\theta, 1)$ yields the Dirichlet process (Ferguson, 1973), whose sample partition is Ewens(θ) (Kingman, 1978). The log series multiplicities $\mu_r \propto x^r/r$ are the discretised Lévy density of that subordinator, and $m_r \sim \mathrm{Po}(\mu_r)$ is exactly the Arratia et al. (1992) Poisson process approximation. The GGC calculus that PIG uses to build augmentations (He et al., 2019, Remark 4) is therefore the same calculus that generates the Fisher and Ewens model, so the two are not analogies but one object.

It is worth tracing the chain slowly, since it is what licenses treating estimation, prediction, and combinatorics as one problem. A subordinator is a nondecreasing

process built by summing a Poisson rain of positive jumps, and the gamma subordinator is the case where a jump of size near λ arrives at rate $\lambda^{-1}e^{-\lambda}$. Small jumps are abundant and large jumps rare, and the weight λ^{-1} near the origin is precisely what makes infinitely many tiny species possible in a finite population. Read the jumps of G_θ as the unnormalised abundances of the species and divide by the total mass: the resulting random probability vector is a draw from the Dirichlet process, and a sample of n specimens from it partitions into species according to the Ewens formula (8.1). The mass parameter θ of the subordinator is the richness parameter of the partition, one quantity wearing two hats. Now count, rather than normalise. If each species of intensity λ contributes Poisson(λ) specimens, the number of species represented exactly r times is itself Poisson, with mean obtained by integrating the jump rate against the Poisson weight. Carrying out the integral for the scaled jump rate $\theta\lambda^{-1}e^{-\lambda/\beta}$,

$$\int_0^\infty \frac{e^{-\lambda}\lambda^r}{r!}\,\theta\lambda^{-1}e^{-\lambda/\beta}\,\mathrm{d}\lambda = \frac{\theta\,(r-1)!}{r!}\Big(\frac{\beta}{1+\beta}\Big)^r = \frac{\theta x^r}{r},$$

returns Fisher's log series. So Fisher's frequency of frequencies curve, Kingman's partition, and Ferguson's process are three exposures of the same gamma photograph, and the Thorin measure that indexes the PIG family is the dial that was set to a point mass all along. The practical consequence is the parameter map below: moving the Thorin dial off its point mass, which is what a nonzero ν does, is exactly the reciprocal gamma problem R2 solves.

The parameter map. Fisher's two parameters split cleanly across the two PIG representations, and across the three worked examples of He et al. (2019):

Fisher parameter	PIG representation	He, Polson and Xu example
richness or mass θ	R1 gamma ratio $\Gamma(\theta)/\Gamma(\theta+n)$	Dirichlet concentration (Section 4.3)
shape ν, departure from log series	R2 reciprocal gamma $1/\Gamma(\nu)$	gamma shape, negative binomial dispersion (Sections 4.1 and 4.2)

Inference for θ is the near trivial NEF problem of Section 8.2; inference for ν is a genuine gamma shape problem that only R2 renders a clean Gibbs step.

Two parameter Pitman and Yor. The natural discrete parameter analogue of Fisher's richness and shape pair is the two parameter Poisson Dirichlet process (θ, σ), $0 \le \sigma < 1$, $\theta > -\sigma$ (Pitman, 2006), with sequential predictive rule

$$\begin{aligned} \Pr(\text{new species}) &= \frac{\theta + \sigma K}{n+\theta}, \\ \Pr(\text{join species seen } n_j \text{ times}) &= \frac{n_j - \sigma}{n+\theta}. \end{aligned} \tag{8.8}$$

Setting $\sigma = 0$ recovers the Ewens rule $\theta/(n+\theta)$ of Section 8.5. The sampling distribution of the block sizes carries factors $\Gamma(n_j-\sigma)/\Gamma(1-\sigma)$ and $\prod_{i=1}^{K-1}(\theta+i\sigma)$,

again products of gamma ratios, but they are rising factorials in the numerator, which the Beta block R1 does not remove, so Chapter 6 computes the two parameter posterior by quadrature; σ tunes the power law tail of the species abundance curve that the one parameter Ewens model cannot fit. We record this only to place Fisher's model in the wider family and do not fit it to the butterfly data, for which McCullagh finds σ superfluous.

8.5 Good's predictive and the Ewens reconciliation

This section turns from estimation to prediction. It proves that Good's nonparametric estimator is the Chinese restaurant rule in disguise, gives the predictive law of new species in closed form, and works both on small cases and on Corbet's butterflies.

Good (1953) estimated the population frequency q_r of a species seen r times by the nonparametric ratio

$$\mathbb{E}[q_r \mid \text{data}] = \frac{r+1}{n+1}\,\frac{\mu_{r+1,\,n+1}}{\mu_{r,\,n}}, \qquad \mu_{r,n} = \mathbb{E}[a_r(n)], \tag{8.9}$$

the Good Turing estimator, deliberately avoiding parametric assumptions. In practice $\mu_{r,n}$ is replaced by the observed a_r, and the ratio $r^* = (r+1)a_{r+1}/a_r$ is called the adjusted count. Under the Ewens model the expected multiplicities are available in closed form, and (8.9) telescopes.

Good's reasoning is a small marvel of empirical Bayes and deserves stating in words. He wanted the population frequency of a species observed r times but had no model for how frequencies are distributed, so he estimated it indirectly. A species seen r times in a sample of n would, in a sample of $n+1$, tend to be seen $r+1$ times, and the expected number of such species is $\mu_{r+1,n+1}$; dividing by the number seen r times and rescaling by $(r+1)/(n+1)$ converts a ratio of observable counts into an estimate of an unobservable frequency. The construction is entirely nonparametric, which was the point, and it is why the answer arrived as a ratio of two expectations he had no way to evaluate. The moment he grants himself a model for the $\mu_{r,n}$, the ratio can be computed, and under the Ewens model it collapses.

PROPOSITION 8.9 (Good's rule is the Ewens predictive). *Under* (8.1), $\mathbb{E}[a_r(n)] = \dfrac{\theta}{r}\,\dfrac{n!}{(n-r)!}\,\dfrac{\Gamma(\theta+n-r)}{\Gamma(\theta+n)}$ *for* $1 \le r \le n$, *and consequently*

$$\mathbb{E}[q_r \mid \textit{data}] = \frac{r}{n+\theta}, \qquad r \ge 1,$$

with unseen mass $q_0 = \theta/(n+\theta)$, *the probability the next specimen is a new species.*

The proof (Section 8.11) is a few lines of gamma algebra; McCullagh's remark that Good "could easily have arrived at" $r/(n+\theta)$ is thus an identity, not an approximation. Good's insistence on nonparametric smoothing left him with (8.9)

in a form that never simplified, whereas the parametric skeleton collapses it to the Chinese restaurant rule. The resulting formula is almost embarrassingly simple: the estimated frequency of a species seen r times is $r/(n+\theta)$, the observed proportion r/n shrunk toward zero by the factor $n/(n+\theta)$, with the mass $\theta/(n+\theta)$ freed up and assigned to species not yet seen. This is the Bayesian smoothing that Good approximated from the data alone, now with θ as the single knob controlling how much probability is held in reserve for the unseen.

EXAMPLE 8.10 (Good's rule by hand). Take $\theta = 1$ and $n = 4$. Proposition 8.9 gives $\mu_{1,4} = \frac{1}{1} \cdot 4 \cdot \Gamma(4)/\Gamma(5) = 1$, which matches Table 8.1: the expected number of singletons is $4(1/24) + 2(6/24) + 1(8/24) = 1$. In a sample of five, $\mu_{2,5} = \frac{1}{2} \cdot 20 \cdot \Gamma(4)/\Gamma(6) = \frac{1}{2}$. Good's ratio is $\frac{2}{5} \cdot \frac{1/2}{1} = \frac{1}{5} = 1/(n+\theta)$, as claimed. In the same way $\mu_{2,4} = \frac{1}{2}$ and $\mu_{3,5} = \frac{1}{3}$ give $\frac{3}{5} \cdot \frac{1/3}{1/2} = \frac{2}{5} = 2/(n+\theta)$, and the unseen mass is $1/5$. The three estimates $\frac{1}{5}$, $\frac{2}{5}$ and the mass $\frac{1}{5}$ for new species are exactly the Chinese restaurant probabilities for the fifth customer.

Bayesian Good Turing. Proposition 8.9 depends on θ, which the PIG posterior supplies. The Bayesian Good Turing estimate is $\mathbb{E}[q_r \mid K] = \mathbb{E}[\, r/(n+\theta) \mid K\,]$ with a posterior credible band obtained by pushing the draws (8.5) through $r/(n+\theta)$. For large n the relative uncertainty in $q_r = r/(n+\theta)$ is about $\mathrm{sd}(\theta)/n$, which is negligible, whereas the relative uncertainty in $q_0 = \theta/(n+\theta)$ is about $\mathrm{sd}(\theta)/\theta$, which is not; the posterior matters for the unseen mass, not for the seen frequencies (Figure 8.8(b)). The Good and Toulmin (1956) extension follows the same route and, through the Feller coupling, is available in closed form.

PROPOSITION 8.11 (Predictive law of new species). *Let U_m be the number of new species among m further specimens. Then, given θ, $U_m \overset{d}{=} \sum_{i=n}^{n+m-1} B_i$ with independent $B_i \sim \mathrm{Bernoulli}(\theta/(\theta+i))$, independent of the first n specimens, so*

$$\mathbb{E}[U_m \mid \theta] = \theta[\psi(\theta+n+m) - \psi(\theta+n)],$$

$$\mathrm{Var}[U_m \mid \theta] = \sum_{i=n}^{n+m-1} \frac{\theta\, i}{(\theta+i)^2}.$$

The unconditional predictive variance adds parameter uncertainty by the law of total variance, $\mathrm{Var}(U_m \mid K) = \mathbb{E}[\mathrm{Var}(U_m \mid \theta) \mid K] + \mathrm{Var}(\mathbb{E}[U_m \mid \theta] \mid K)$*, both terms computed from the PIG draws* (8.5).

This gives the honest predictive interval of Figures 8.9 and 8.10, the object Good and Toulmin sought. The sampling term grows like $\theta \log(1+m/n)$ while the parameter term grows like its square, $\mathrm{sd}(\theta)^2 \log^2(1+m/n)$, so the sampling term dominates for small enlargements and the parameter term for large ones, and the accumulation band widens without ever closing. For a doubling, $m = n$ with $n \gg \theta$, the mean is close to $\theta \log 2$: about 27 new species for Williams' 1933 sample.

EXAMPLE 8.12 (Good Turing and log series smoothing for Corbet's butterflies). Corbet's Malayan butterfly collection, analysed in Fisher et al. (1943) and revisited

by Efron and Hastie (2016, Chapter 6), gives for $r = 1, \ldots, 15$ the frequencies of frequencies

$$a_r = 118,\ 74,\ 44,\ 24,\ 29,\ 22,\ 20,\ 19,\ 20,\ 15,\ 12,\ 14,\ 6,\ 12,\ 6,$$

which account for 435 species and 2029 specimens; we use only these fifteen cells. The raw adjusted counts $r^* = (r+1)a_{r+1}/a_r$ begin sensibly, $r^* = 1.25,\ 1.78,\ 2.18$ for $r = 1, 2, 3$, and then become erratic: 6.04 at $r = 4$, 14.0 at $r = 11$, 5.57 at $r = 12$ and 28.0 at $r = 13$, because a single small a_r in the denominator can double or halve the estimate (Figure 8.1(b)). This is the well known instability of unsmoothed Good Turing.

Under Fisher's model the a_r are independent Poisson counts with means $\theta x^r/r$, where x is the probability parameter of (8.7) with $\nu = 0$, and Good's ratio has expectation $(r+1)\mu_{r+1}/\mu_r = rx$. Smoothing therefore means estimating x. With a prior flat in $\log\theta$ and in $x \in (0.85, 1]$, the posterior from the fifteen Poisson cells has θ mean 135.3 with 95% interval $[121.6, 150.7]$ and x mean 0.993 with interval $[0.979, 1.000]$; the fitted multiplicities 134.3, 66.7, 44.1, 32.9, 26.1, ... match the data with a Pearson statistic of 13.5 on 13 degrees of freedom (Figure 8.1(a)). The smoothed adjusted counts are rx, so $r^* = 0.993,\ 1.986,\ 2.979, \ldots$ with 95% intervals $[0.979r, 1.000r]$, about 2% of r wide. The truncated table identifies θ well and x only weakly, as it must: x^r barely varies over $r \le 15$ when x is close to one, which is the case $n \gg \theta$ in Ewens' notation, since $x = n/(n+\theta)$ is precisely the shrinkage factor of Proposition 8.9.

The same posterior predicts the number of new species U in a second collection of equal effort. Under the log series the expected number is $\sum_r (-1)^{r+1}\theta x^r/r = \theta\log(1+x)$ (Exercise 6), with posterior mean 93.3 and interval $[84.1, 103.4]$; adding the Poisson noise of the new collection gives a 95% predictive interval $[73, 115]$. Good and Toulmin's nonparametric estimate is the alternating sum $\sum_r (-1)^{r+1}a_r = 75$ with standard error $(\sum_r a_r)^{1/2} = 20.9$. The two answers are compatible, and the difference has a clear source: the data hold 118 singletons against the log series' 134, a deficit of 1.4 Poisson standard deviations, and a_1 is the largest cell in the alternating sum. The parametric answer is four times more precise about the mean because it borrows strength across all fifteen cells; the price is reliance on the log series shape, which the Pearson statistic does not reject.

8.6 The Bayes estimate of θ sheds little; the posterior sheds light on prediction and uncertainty

This section makes precise the claim that, for the richness parameter itself, a Bayesian point estimate and interval add almost nothing to maximum likelihood once K is moderate. The NEF structure delivers a clean Bernstein and von Mises statement. Write $\varphi = \log\theta$, $\hat\varphi$ for the MLE, and $J_n = A''(\hat\varphi) = \mathrm{Var}(K \mid \hat\theta)$ for the observed information in φ, which by Proposition 8.2 is of the same order as $\mathbb{E}[K] = \theta[\psi(\theta+n) - \psi(\theta)]$, growing like $\theta\log n$.

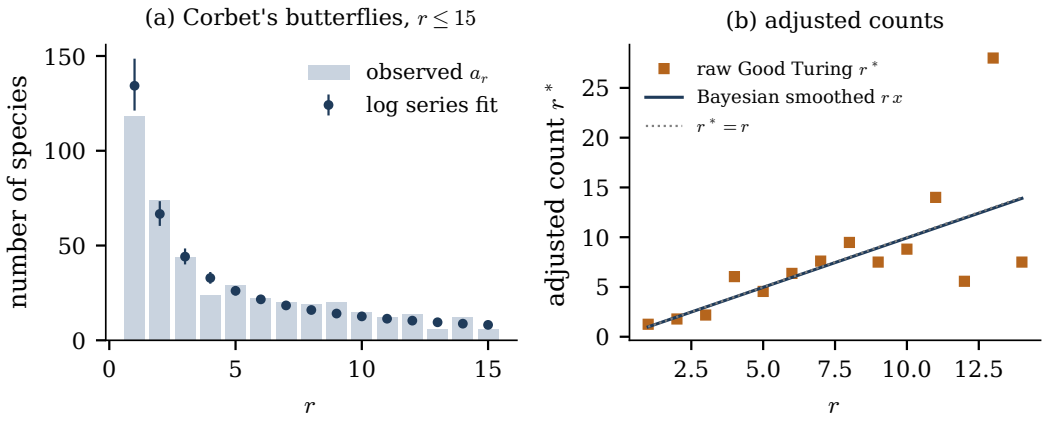


FIGURE 8.1. Corbet's Malayan butterflies, $r \le 15$. (a) Observed frequencies of frequencies (bars) and posterior mean log series fit $\theta x^r / r$ with 95% posterior intervals. (b) Raw Good Turing adjusted counts $r^* = (r+1)a_{r+1}/a_r$ (squares) against the Bayesian smoothed rx (line); the dotted line is $r^* = r$.

Bernstein and von Mises is the theorem that, with enough information, the posterior forgets its prior and lines up with the sampling distribution of the MLE. What makes it bite cleanly here is that a one parameter exponential family has a single Fisher information J_n, and that J_n equals the variance of the sufficient statistic K. The posterior for φ is therefore a Gaussian whose width is fixed by how variable the species count is, and the prior contributes a correction of relative order $1/J_n$. The one subtlety, and the reason the phrase "sheds little" needs the qualifier "estimator", is that J_n grows only logarithmically in the sample size. Doubling the number of specimens does not halve the posterior variance of θ; it barely moves it. The governing quantity is K, and K is a slow accumulator.

PROPOSITION 8.13 (Asymptotic redundancy). *For any prior with a continuous positive density at $\hat\theta$, the posterior of φ is asymptotically $\mathcal{N}(\hat\varphi, J_n^{-1})$ as $J_n \to \infty$. The posterior mean satisfies $\mathbb{E}[\theta \mid K] = \hat\theta + O(\hat\theta J_n^{-1})$, the central 95% credible interval equals the Wald interval $\hat\theta \pm 1.96\, I(\hat\theta)^{-1/2}$ up to $o(\hat\theta J_n^{-1/2})$, and two priors differing by a smooth reweighting give posteriors within total variation $O(J_n^{-1/2})$ and posterior means within $O(\hat\theta J_n^{-1})$.*

Thus the effective sample size is K, not n: with $K = 27$ the posterior already tracks the likelihood, and with Williams' $K = 240$ the prior is a rounding error (Section 8.9). This is the precise content of "sheds little." The statement is specific to θ. It does not transfer to the shape ν, whose information accumulates differently, nor to prediction of unseen mass q_0, where the posterior over θ is exactly what supplies the uncertainty. Bayesian machinery is redundant for estimating richness and indispensable for the predictive questions that motivated Good. The distinction is not that estimation is frequentist and prediction Bayesian, but that the two questions load on different features of the model: a point estimate of

TABLE 8.3. Maximum likelihood against the diffuse prior posterior for $n = 1000$ as the species count K varies, by exact quadrature. $J_n = \mathrm{Var}(K \mid \hat{\theta})$ is the information in $\log\theta$; "log Wald" is $\hat{\theta}\exp(\pm 1.96 J_n^{-1/2})$.

K	$\hat{\theta}$	J_n	SE	Wald 95%	log Wald 95%	post. mean	credible 95%	skew
3	0.282	1.9	0.204	$[-0.118, 0.682]$	$[0.068, 1.164]$	0.289	$[0.034, 0.823]$	1.53
5	0.589	3.7	0.306	$[-0.011, 1.189]$	$[0.213, 1.632]$	0.599	$[0.157, 1.355]$	1.11
10	1.444	8.0	0.512	$[0.440, 2.447]$	$[0.721, 2.893]$	1.458	$[0.639, 2.647]$	0.77
20	3.430	16.0	0.857	$[1.751, 5.109]$	$[2.102, 5.596]$	3.447	$[1.990, 5.349]$	0.55
40	8.210	31.3	1.467	$[5.335, 11.084]$	$[5.785, 11.652]$	8.224	$[5.630, 11.375]$	0.40
80	20.295	59.6	2.629	$[15.143, 25.448]$	$[15.745, 26.161]$	20.285	$[15.514, 25.800]$	0.30
160	53.541	108.7	5.136	$[43.475, 63.608]$	$[44.364, 64.616]$	53.392	$[43.928, 63.980]$	0.23

θ asks for the centre of a well behaved likelihood, which any sensible method locates, whereas a predictive interval for the unseen must propagate the residual uncertainty in θ, which only a posterior carries honestly.

EXAMPLE 8.14 (Bernstein and von Mises across K). Fix $n = 1000$ and let the observed species count range from $K = 3$ to $K = 160$. Table 8.3 compares the MLE and Wald interval with the posterior mean and 95% credible interval under the diffuse prior, all computed by exact quadrature. Three patterns are visible. First, the posterior mean is within 0.04 standard errors of the MLE at every K, even $K = 3$: the point estimates are interchangeable. Second, the intervals are not: at $K = 3$ and $K = 5$ the Wald interval runs below zero, an impossible value, while the credible interval is asymmetric about the MLE, extending 2.7 standard errors above it but only 1.2 below at $K = 3$. Third, the disagreement fades steadily as K grows, from a gross mismatch at $K = 3$ to endpoint differences under 10% of a standard error at $K = 160$, at the rate $J_n^{-1/2}$ that Proposition 8.13 predicts. The Wald interval computed on the natural scale, $\hat{\theta}\exp(\pm 1.96 J_n^{-1/2})$, respects positivity and captures the direction of the asymmetry, though it overshoots on the right; at small K neither Gaussian interval reproduces the credible interval, which is why Section 8.7 recommends reporting the posterior itself, summarised on the $\log\theta$ scale.

The sharper statement is therefore the one in the section title: the Bayes estimate of θ sheds little, while the posterior sheds real light on prediction and uncertainty quantification, the Chinese restaurant rule, the unseen mass, and the Good and Toulmin curve of Section 8.5, each delivered with a calibrated band rather than a point.

8.7 Skewed and heavy tailed concentration posteriors

The Gaussian picture of Section 8.6 is an assurance about large K, and this section examines where it fails. Two regimes that occur constantly in practice violate it, and in both the posterior for the concentration is strongly right skewed: small K, where few species are seen, and K near n, the hyperdiverse case of a sample that is

TABLE 8.4. Posterior of the concentration under a diffuse $\mathrm{Ga}(10^{-2}, 10^{-2})$ prior, by exact quadrature. The mode is that of the posterior density of θ. Skewness rises and the MLE detaches from the posterior centre as K approaches its boundaries.

regime	MLE	mean	median	mode	sd	skew	95% CI
$n = 200$, $K = 19$ (moderate)	4.97	5.02	4.88	4.60	1.37	0.64	$[2.77, 8.08]$
$n = 200$, $K = 4$ (small K)	0.58	0.60	0.52	0.37	0.36	1.34	$[0.12, 1.50]$
$n = 30$, $K = 28$ ($K \approx n$)	198.1	135.9	119.6	92.0	71.7	1.67	$[46.7, 319.2]$

almost all singletons, where under weak priors it is also genuinely heavy tailed. Both sit at the boundary of the range $1 < K < n$ where the moment equation (8.2) has a well separated interior root, and the tail is controlled by the same gamma ratio that drives everything else in the chapter. The intuition for the next result is simply that $\Gamma(\theta)/\Gamma(\theta + n)$ is a product of n factors $1/(\theta + i)$, each of which decays like $1/\theta$, so the likelihood has a polynomial right tail of degree $K - n$.

PROPOSITION 8.15 (Tail of the concentration posterior). *Under the prior $\pi(\theta) \propto \theta^{a_0 - 1} e^{-b_0 \theta}$ with $a_0 > 0$, $b_0 \geq 0$, the posterior $p(\theta \mid K) \propto \theta^{K + a_0 - 1} e^{-b_0 \theta}\, \Gamma(\theta)/\Gamma(\theta + n)$ has right tail $p(\theta \mid K) \sim c\, \theta^{K + a_0 - 1 - n} e^{-b_0 \theta}$ as $\theta \to \infty$. If $b_0 > 0$ the tail is light and every posterior moment is finite. If $b_0 = 0$ the tail is Pareto of index $n - K - a_0$: the posterior is proper if and only if $K + a_0 < n$, and $\mathbb{E}[\theta^p \mid K] < \infty$ if and only if $p < n - K - a_0$. Under the flat prior $(a_0, b_0) = (1, 0)$ the posterior is proper if and only if $K \leq n - 2$, has finite mean if and only if $K \leq n - 3$, and finite variance if and only if $K \leq n - 4$.*

The proof is a power law integral test (Section 8.11). The reading is that each specimen beyond the species count, each "repeat", buys one power of decay in the right tail; a sample with only one or two repeats cannot rule out arbitrarily large richness. Even a proper but diffuse $\mathrm{Ga}(10^{-2}, 10^{-2})$ prior leaves the shape strongly regime dependent (Table 8.4, Figure 8.2(a)). At $n = 200$, $K = 19$ the posterior of θ is mildly skewed, skewness 0.64, and the Bernstein and von Mises Gaussian is accurate. At $K = 4$ the skewness is 1.34, and at $n = 30$, $K = 28$, the hyperdiverse case, it is 1.67: there the likelihood maximiser $\hat{\theta} = 198$ sits far out in the right tail while the posterior median is 120 and the mean 136, so the point estimate is a poor summary of the posterior it is supposed to anchor. Reporting inference on the natural parameter $\varphi = \log \theta$, where the family is regular and the tails are exponential, restores approximate symmetry and is the practical remedy; on the raw θ scale a symmetric interval about the mean can exclude the mode.

Under a genuinely flat prior the effect is sharper (Figure 8.2(b)). With $n = 20$ the posterior mean is finite for $K \leq n - 3$, but at $K = n - 2 = 18$ the truncated mean integral grows without bound as the cutoff is raised, taking values $111, 403, 748, 1101$ at $\theta \leq e^6, e^8, e^{10}, e^{12}$, an increase of about 175 for every unit of $\log c$, which is the signature of a θ^{-2} tail: the posterior is proper yet has no finite

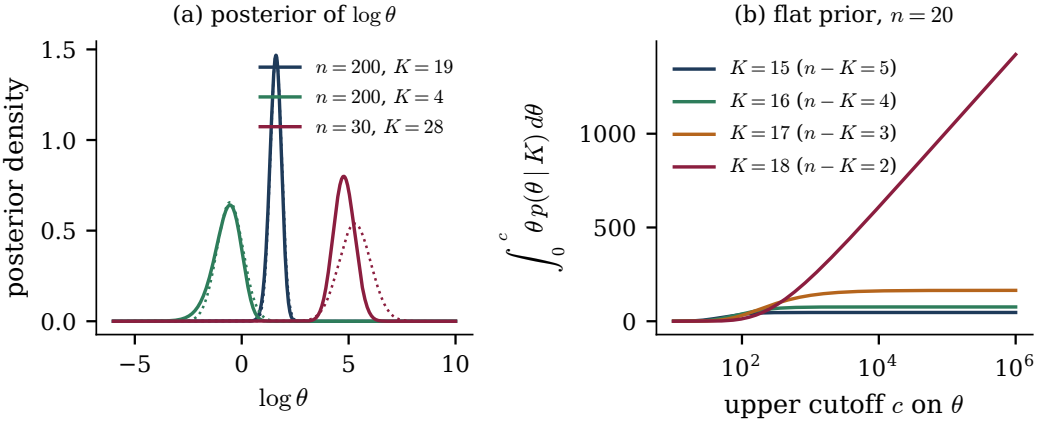


FIGURE 8.2. (a) Posterior of $\log\theta$ under a diffuse $\mathrm{Ga}(10^{-2}, 10^{-2})$ prior for three regimes; dotted curves are the Bernstein and von Mises Gaussians $\mathcal{N}(\log\hat{\theta}, J_n^{-1})$, accurate for moderate K but not for small K or $K \approx n$. (b) Under a flat prior with $n = 20$, the posterior mean integral truncated at an increasing cutoff c on θ: finite and stable for $K \le n - 4$, converging slowly at $K = n - 3$, and divergent at $K = n - 2$, as Proposition 8.15 predicts.

mean. At $K = n - 1$ it fails to integrate at all. A sample in which all but one or two specimens are distinct species carries almost no information from above about how much mass to hold in reserve for the unseen, and a flat prior does not supply the missing constraint.

A weakly informative PIG prior. The remedy is a prior in the PIG family itself, specifically the $\tau = 1$ case of the Diaconis and Ylvisaker prior (8.3) with $\kappa = K_0 - 1$: a single pseudosample

$$\pi(\theta) \propto \theta^{K_0 - 1} \frac{\Gamma(\theta)}{\Gamma(\theta + n_0)}, \tag{8.10}$$

read as "in a pilot of n_0 specimens I saw K_0 species." Precisely, (8.10) is the posterior that the scale invariant prior $\mathrm{d}\theta/\theta$ would give after such a pilot sample. We take $(n_0, K_0) = (10, 3)$, a prior with median $\theta \approx 0.93$ carrying the weight of one small sample; it is proper because $0 < K_0 - 1 < n_0 - 1$. The posterior $p(\theta \mid K) \propto \theta^{K + K_0 - 1}[\Gamma(\theta)/\Gamma(\theta + n)][\Gamma(\theta)/\Gamma(\theta + n_0)]$ has tail $\theta^{K + K_0 - 1 - n - n_0}$, so its mean and variance are finite for every $K \le n$; the pseudosample supplies exactly the upper tail information the data lack. It is sampled by the same augmentation with one extra latent, $\eta \sim \mathrm{Be}(\theta, n)$, $\eta_0 \sim \mathrm{Be}(\theta, n_0)$, and $\theta \sim \mathrm{Ga}(K + K_0, -\log\eta - \log\eta_0)$. Being worth $K_0 = 3$ pseudospecies, it is negligible whenever the data are informative: for Williams it gives posterior mean 38.96 (SD 2.79) against the maximum likelihood value 40.15, a shift of under half a posterior standard deviation. But it regularises the pathological cases. In the $n = 30$, $K = 28$ example the flat prior posterior mean is infinite and the diffuse $\mathrm{Ga}(10^{-2}, 10^{-2})$ gives 136 (an artefact of that prior's hidden exponential scale $1/b_0$), while the PIG prior returns a proper 40.7 with a sensible interval $[17.3, 86.2]$. A diffuse gamma is

not in fact noninformative for a parameter that may be large, since its rate fixes a scale; the PIG pseudosample regularises without imposing one.

These boundary regimes are what the posterior is for. The Beta augmentation sampler (8.5) stays valid throughout the proper range, since the conditional $\theta \mid \eta \sim \mathrm{Ga}(K + a_0,\, b_0 - \log \eta)$ is a proper gamma draw with positive rate $b_0 - \log \eta > 0$ whatever the marginal tail, so the chain explores a skewed or infinite mean posterior without breaking and simply returns wide asymmetric intervals; when $K \geq n - 1$ under a flat prior the chain instead drifts off to infinity, which is itself the signal that a proper prior is needed. Where Section 8.6 found the Bayes estimator of θ redundant, here it is the maximum likelihood point that misleads and the full posterior, with its skew and its honest tail, that is indispensable. The heavy tail is moreover native to the gamma ratio that the Beta block represents: its polynomial decay in θ drives Proposition 8.15, and the augmentation meets it without special treatment.

A real hyperdiverse sample. Estimating how large a vocabulary an author commands is the unseen species problem in disguise, the reading Efron and Thisted (1976) gave the Shakespeare canon, and n gram frequency smoothing remains the main modern home of the Good Turing estimator through the Katz backoff of language models. Taking a Shakespeare corpus of about 2×10^5 word tokens, the one parameter concentration behaves exactly as Proposition 8.15 anticipates, and the choice of unit decides the regime. For word unigrams a sample of $n = 500$ tokens contains $K = 240$ types, $K/n = 0.48$, and the likelihood is tight and nearly symmetric about the MLE 181: richness is well identified and the Bayes estimator sheds little, as in Section 8.6. For word trigrams the same corpus is hyperdiverse. The first 50, 100, or 200 trigrams are all distinct, $K = n$, and the likelihood $\theta^K \Gamma(\theta)/\Gamma(\theta + n)$ tends to a nonzero constant as $\theta \to \infty$, since $\theta^n \Gamma(\theta)/\Gamma(\theta + n) \to 1$: the maximum likelihood estimate is $+\infty$ and a flat prior posterior is improper (Figure 8.3(b)). Here the weakly informative PIG prior does its work. Under the pseudosample prior $(n_0, K_0) = (10, 3)$ the unigram posterior is barely moved (mean 170, skewness 0.28, against MLE 181), while the trigram posteriors, improper under a flat prior when $K = n$, are proper: $n = 100$ trigrams with $K = 100$ give posterior mean 757 with interval $[331, 1649]$, and $n = 400$, $K = 396$ give mean 7700 with interval $[4100, 14100]$ (Figure 8.3(a)). These posteriors are proper and finite but strongly prior influenced and enormous, the honest verdict that the data do not bound the concentration from above once almost every unit is a singleton.

Two lessons follow. First, the pathology is common, not exotic: any sparse categorical sample, n grams, rare genetic variants, long tail catalogues, lands in it, and reporting an MLE or a symmetric interval for θ there would mislead badly where the honest posterior is heavy tailed or improper. In population genetic samples the same happens when nearly every sampled haplotype is unique, as in highly variable regions sequenced at modest depth. Second, near total distinctness is the fingerprint of a power law, which the one parameter Ewens model cannot represent; the second Pitman and Yor parameter σ of (8.8) absorbs it, trading an

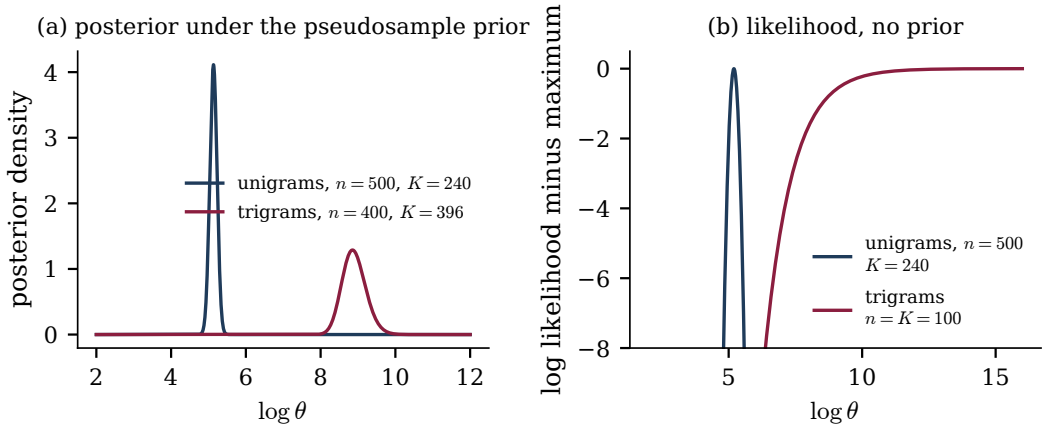


FIGURE 8.3. Real hyperdiverse data: Shakespeare word units under the one parameter Ewens model. (a) Posterior of $\log \theta$ under the weakly informative PIG prior (8.10) with $(n_0, K_0) = (10, 3)$; unigrams ($n = 500$, $K = 240$) give a tight symmetric posterior, trigrams ($n = 400$, $K = 396$) a proper but broad, prior regularised one near $\theta \approx 7700$. (b) Log likelihood relative to its maximum, prior free; unigrams peak at $\hat{\theta} \approx 181$, whereas trigrams with $K = n = 100$ rise monotonically to a plateau, so θ has no finite maximiser and any flat prior posterior is improper.

exploding θ for a finite pair (θ, σ) with σ near one. The heavy tailed θ posterior is therefore also a specification diagnostic, and the PIG augmentation that samples it cleanly is what makes the diagnostic cheap to read.

8.8 Fisher's two variances: a worked example

This section works one example in detail and settles a point that confuses many readers of the classical literature: Fisher's unconditional model and Ewens' conditional model attach two different variances to the species count, and only one of them is the right yardstick for the uncertainty in θ. We write k for the observed number of species, as in the Dirichlet process literature. The concentration parameter governs partition diversity wherever exchangeable clustering arises. In population genetics, $\theta = 4M\mu$ is the scaled mutation rate driving allele diversity (Ewens, 1972). In Bayesian nonparametrics, θ is the Dirichlet process concentration: $\mathbb{E}[K_n \mid \theta] \sim \theta \log n$, so misspecifying θ directly miscounts mixture components. In community ecology, θ is Fisher's diversity index, one of the oldest biodiversity statistics still in routine use (Fisher et al., 1943). In every case the parameter is weakly identified from a single partition and the likelihood is far from Gaussian.

The two variances. In Fisher's model of Section 8.4 with $\nu = 0$, the species count K is a sum of independent Poisson multiplicities and is therefore itself Poisson, with variance equal to its mean, $\mathbb{E}[K] = \theta \log(1 + \mathbb{E}[N]/\theta)$. This is the relation of Fisher et al. (1943), and solving $K = \theta \log(1 + N/\theta)$ for θ is Fisher's estimator. If one treats K as Poisson with $N = n$ fixed and mean $m(\theta) = \theta \log(1 + n/\theta)$, the

TABLE 8.5. The two variances at the conditional MLE. $\mathbb{E}[K]$ and $\mathrm{Var}(K \mid n)$ from Proposition 8.2; SE (Ewens) $= V_E^{1/2}$ and SE (Poisson) $= V_P^{1/2}$ from (8.11); the last column is Fisher's unconditional estimate from $K = \theta \log(1 + n/\theta)$.

sample	$\hat\theta$	$\mathbb{E}[K]$	$\mathrm{Var}(K \mid n)$	ratio	SE (Ewens)	SE (Poisson)	Fisher's $\tilde\theta$
$n = 200$, $k = 25$	7.333	25.0	17.40	0.696	1.758	2.082	7.542
Williams 1933, $n = 3540$, $k = 178$	39.355	178.0	138.57	0.779	3.343	3.784	39.496
Williams, $n = 15{,}609$, $k = 240$	40.146	240.0	199.45	0.831	2.843	3.115	40.247

delta method gives the Poisson variance

$$V_P(\theta) = \frac{m(\theta)}{m'(\theta)^2}, \qquad m'(\theta) = \log\Big(1 + \frac{n}{\theta}\Big) - \frac{n}{n+\theta}. \tag{8.11}$$

In the Ewens model, by contrast, the sample size n is fixed and the variance of K is

$$\mathrm{Var}(K \mid \theta, n) = \sum_{j=0}^{n-1} \frac{\theta\, j}{(\theta + j)^2} = \theta[\psi(\theta + n) - \psi(\theta)] - \theta^2[\psi'(\theta) - \psi'(\theta + n)], \tag{8.12}$$

as in Proposition 8.2, and the variance of the conditional MLE is $V_E(\theta) = \theta^2 / \mathrm{Var}(K \mid \theta, n) = 1/I(\theta)$. Both have the same form, (variance of K) divided by (slope of $\mathbb{E}[K]$ in θ) squared; they differ in which variance of K is used. For $1 \ll \theta \ll n$ the expansions $\psi(\theta + n) - \psi(\theta) = \log(n/\theta) + O(1/\theta)$ and $\theta\psi'(\theta) = 1 + O(1/\theta)$ give

$$\frac{\mathrm{Var}(K \mid \theta, n)}{\mathbb{E}[K \mid \theta, n]} \approx 1 - \frac{1}{\log(n/\theta)}, \qquad V_E(\theta) \approx \frac{\theta}{\log(n/\theta) - 1},$$

whereas $V_P(\theta) \approx \theta \log(n/\theta)/\{\log(n/\theta) - 1\}^2$. To leading order both are $\theta/\log(n/\theta)$, but the conditional variance is smaller by the factor $1 - 1/\log(n/\theta)$, which converges to one only logarithmically.

Which variance is right? Conditioning on n removes the part of the variation in K that is due to the random total N, and in Fisher's model N carries almost no information about θ: N is negative binomial with shape θ and probability x, and once x is profiled out the information about θ left in N is about $1/(2\theta^2)$ (Exercise 11), against $I(\theta) \approx \{\log(n/\theta) - 1\}/\theta$ in K given N. For Williams that is 0.0003 against 0.124, a quarter of one percent. So N is effectively ancillary for θ, the conditional variance V_E is the relevant one, and the Poisson variance overstates the standard error.

Table 8.5 puts numbers on this. For the small sample the Poisson standard error is 18% too large; for Williams it is 13% too large in 1933 and 10% too large for the full data. The approximation $V_E \approx \theta/\{\log(n/\theta) - 1\}$ is excellent, giving the standard errors 1.783, 3.354 and 2.844 against the exact 1.758, 3.343 and 2.843, while the leading order $\theta/\log(n/\theta)$ gives 1.489, 2.958 and 2.595, too small. Fisher's unconditional estimate for the full Williams data, 40.247, matches the value 40.248

quoted by McCullagh (2016), and it differs from the conditional MLE by 0.1, about 4% of a standard error. Whichever estimator is used, the posterior of the next paragraphs is built on the exact conditional likelihood and so inherits the correct variance automatically; no choice between the two is needed.

Beta augmentation. The likelihood of $K = k$ is $L(\theta \mid k) \propto \theta^k/(\theta)_{(n)}$, and the rising factorial in the denominator blocks conjugacy. Since $\Gamma(\theta)/\Gamma(\theta+n) = B(\theta,n)/\Gamma(n)$, the representation (8.4) introduces $\eta \in (0,1)$ as a latent and yields $p(\theta,\eta \mid k) \propto \pi(\theta)\,\theta^k\,\eta^{\theta-1}(1-\eta)^{n-1}$. With $\pi(\theta) = \mathrm{Ga}(a,b)$ the conditionals are those of (8.5) with $(K, a_0, b_0) = (k, a, b)$, a variant of the Escobar and West sampler (Escobar and West, 1995).

A heavy tailed prior. For small k the gamma prior in (8.5) leaks into the posterior through its rate. A natural heavy tailed alternative puts a half Cauchy prior on $\sqrt{\theta}$, the horseshoe choice for a scale parameter. If $\sqrt{\theta} \sim \mathrm{C}^+(0,1)$ then θ has the beta prime density $\pi(\theta) = \pi^{-1}\theta^{-1/2}(1+\theta)^{-1}$ and $\varphi = \log\theta$ has density $\{2\pi\cosh(\varphi/2)\}^{-1}$, a hyperbolic secant law. More generally the beta prime prior

$$\pi(\theta) \propto \theta^{\chi/2-1}(1+\theta)^{-\chi}, \qquad \text{equivalently} \qquad p(\varphi) \propto \cosh^{-\chi}(\varphi/2), \tag{8.13}$$

gives φ a symmetric law with exponential tails of rate $\chi/2$. The Pólya Gamma identity of Polson et al. (2013), $\cosh^{-\chi}(\omega/2) = \mathbb{E}[e^{-\omega^2\Lambda/2}]$ with $\Lambda \sim \mathrm{PG}(\chi,0)$, writes this prior as a scale mixture of normals on $\log\theta$, but a Gaussian factor in $\log\theta$ does not combine conjugately with η^θ. The gamma integral $(1+\theta)^{-\chi} = \Gamma(\chi)^{-1}\int_0^\infty \xi^{\chi-1}e^{-\xi(1+\theta)}\,\mathrm{d}\xi$ does, and it gives a sampler in which every step is standard.

PROPOSITION 8.16 (Three block sampler). *Under the prior* (8.13) *the joint density*

$$p(\theta,\eta,\xi \mid k) \propto \theta^{k+\chi/2-1}\,\eta^{\theta-1}(1-\eta)^{n-1}\,\xi^{\chi-1}e^{-\xi(1+\theta)}$$

on $(0,\infty)\times(0,1)\times(0,\infty)$ *has* θ *marginal equal to the posterior* $p(\theta \mid k) \propto \pi(\theta)\theta^k\Gamma(\theta)/\Gamma(\theta+n)$, *and its full conditionals are*

$$\begin{aligned} \eta \mid \theta &\sim \mathrm{Be}(\theta, n), \\ \xi \mid \theta &\sim \mathrm{Ga}(\chi,\, 1+\theta), \\ \theta \mid \eta, \xi &\sim \mathrm{Ga}\big(k+\tfrac{\chi}{2},\, \xi - \log\eta\big). \end{aligned}$$

Proof. Integrating out η with (8.4) produces $\Gamma(n)\Gamma(\theta)/\Gamma(\theta+n)$, and integrating out ξ with the gamma integral produces $\Gamma(\chi)(1+\theta)^{-\chi}$, so the θ marginal is proportional to $\theta^{k+\chi/2-1}(1+\theta)^{-\chi}\Gamma(\theta)/\Gamma(\theta+n)$, which is the stated posterior. Each conditional is read off the joint density: in η it is a $\mathrm{Be}(\theta,n)$ kernel; in ξ it is $\xi^{\chi-1}e^{-\xi(1+\theta)}$; in θ it is $\theta^{k+\chi/2-1}e^{-\theta(\xi-\log\eta)}$, a gamma kernel with positive rate since $\xi > 0$ and $\log\eta < 0$. □

The half Cauchy case $\chi = 1$ has θ shape $k+\frac{1}{2}$, which is why the half integer appears. The tail of the posterior is $\theta^{k-n-\chi/2-1}$, so the prior adds $1+\chi/2$ powers

TABLE 8.6. Fisher and Bayes for $n = 200$, $k = 25$. Spreads agree; shapes do not. Posterior summaries by exact quadrature. "Right/left" is $(q_{0.975} - \text{median})/(\text{median} - q_{0.025})$.

	Fisher (asymptotic)	Ga(0.01, 0.01) prior	half Cauchy on $\sqrt{\theta}$
point estimate	$\hat{\theta}_{\text{MLE}} = 7.33$	median 7.22, mean 7.39	median 7.09, mean 7.26
spread	SE = 1.76	SD = 1.78	SD = 1.75
95% interval	[3.89, 10.78]	[4.40, 11.33]	[4.31, 11.15]
skewness	0 (by construction)	0.58	0.58
right/left	1.00	1.45	1.46

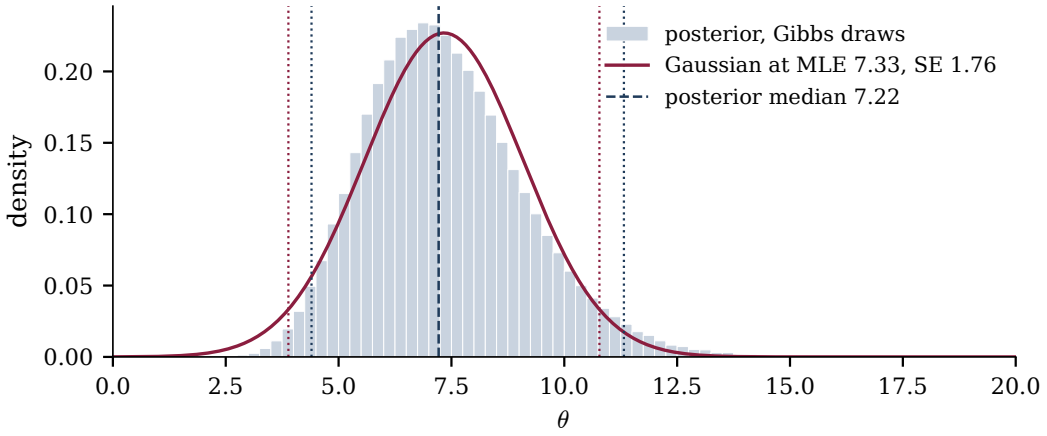


FIGURE 8.4. Posterior of θ for $n = 200$, $k = 25$ from the Beta augmentation sampler (histogram) overlaid with Fisher's information based Gaussian at the MLE (curve). Dotted lines mark the two 95% intervals: the posterior interval $[4.40, 11.32]$ and the Wald interval $[3.89, 10.78]$. The posterior's right tail is about 45% longer than its left.

of decay to the likelihood's $k - n$; with $\chi = 1$ the posterior mean is finite whenever $k \leq n - 1$, two repeats better than the flat prior of Proposition 8.15.

Numerical example: $n = 200$, $k = 25$. Table 8.6 and Figure 8.4 compare Fisher's asymptotic answer with the posteriors under the diffuse gamma prior and under the half Cauchy prior on $\sqrt{\theta}$. The posterior summaries are by exact quadrature; the Beta augmentation sampler (200,000 draws) reproduces them to Monte Carlo error, with median 7.22 and interval $[4.40, 11.32]$, and the three block sampler of Proposition 8.16 gives median 7.10 and interval $[4.32, 11.14]$ for the half Cauchy prior with integrated autocorrelation time 1.9.

The summaries agree on the centre to within 0.3 and on the spread to within 0.03, yet disagree materially on coverage: Fisher's lower endpoint 3.89 lies below the posterior 2.5% quantile 4.40, and Fisher's upper endpoint 10.78 lies below the posterior 97.5% quantile 11.33. The entire Fisher interval is shifted left relative to the posterior because the Gaussian forces symmetry on a genuinely skewed object. The two priors differ by about 0.13 in the centre, less than a tenth of a standard

deviation, as Proposition 8.13 predicts for $k = 25$.

Improvement over Fisher. Fisher's interval treats the curvature at $\hat{\theta}$ as if it characterised the whole likelihood. In the Ewens problem it does not. The score is $k/\theta - \sum_{i<n} 1/(\theta+i) \approx k/\theta - \log(n/\theta)$, so to the left of $\hat{\theta}$ the log likelihood plunges like $(k-1)\log\theta$ as $\theta \to 0$, while to the right it declines only slowly, roughly linearly in θ while $\theta \ll n$ and eventually like $(k-n)\log\theta$. The right tail is therefore longer and the posterior is right skewed whenever $k \ll n$. Three concrete improvements follow.

(i) Honest coverage. The Bayesian interval is asymmetric by construction and reproduces the true tail behaviour (Figure 8.4); Fisher's symmetric interval puts too much probability on the left and too little on the right.

(ii) Robustness in the data poor regime. When k is small the gamma prior's rate and shape both matter. At $n = 200$, $k = 3$ the diffuse gamma prior gives posterior mean 0.38 and interval $[0.045, 1.11]$, the half Cauchy prior 0.43 and $[0.067, 1.16]$; the heavy tailed prior keeps the lower endpoint from being dragged toward zero by the gamma shape $a_0 = 0.01$. At the other boundary, $n = 30$, $k = 28$, the half Cauchy prior gives a posterior median of 175 and interval $[54, 958]$ with a finite mean but an infinite third moment, honestly reporting how little the data say; the three block sampler then mixes slowly (integrated autocorrelation time about 30 to 35), a computational echo of the heavy tail. The weakly informative PIG prior of Section 8.7 is the better choice there.

(iii) One variance instead of two. The posterior is built from the exact conditional likelihood, so its spread is automatically governed by $\mathrm{Var}(K \mid \theta, n)$ and not by the Poisson variance (8.11); Table 8.6 shows the posterior SD 1.78 agreeing with the Ewens SE 1.76 and not with the Poisson SE 2.08. The construction is closed form Gibbs throughout.

8.9 Numerical study

This section collects the simulation checks behind the claims of the chapter and completes the analysis of Williams' Rothamsted Macrolepidoptera. Unless stated otherwise the prior is the diffuse $\mathrm{Ga}(10^{-2}, 10^{-2})$ and every chain uses the stable Beta draw of Remark 8.6.

Good's rule (Table 8.7, Figure 8.8(a)). Monte Carlo expected multiplicities from the Chinese restaurant process match the closed form of Proposition 8.9, and Good's estimator (8.9) evaluated with the exact $\mu_{r,n}$ reproduces $r/(n+\theta)$ to rounding error, as an identity must.

Posteriors reproduce the likelihood (Figure 8.5). With $\theta = 5$ and $n = 1000$ the expected species count is $\mathbb{E}[K \mid \theta] = 27.03$, and we condition on the typical value $K = 27$. The diffuse prior posterior mean is then 5.01 (SD 1.08) against MLE 4.99 (SE 1.08). For Williams ($n = 15{,}609$, $K = 240$) the conditional MLE is $\hat{\theta} = 40.146$ (SE 2.843),

TABLE 8.7. Good Turing reconciliation, $\theta = 3$, $n = 40$. Monte Carlo $\mathbb{E}[a_r]$ (6×10^4 simulated partitions) against the closed form, and Good's estimator with exact $\mu_{r,n}$ against $r/(n+\theta)$.

r	$\mathbb{E}[a_r]$ Monte Carlo	$\mathbb{E}[a_r]$ formula	Good's (8.9)	$r/(n+\theta)$
1	2.858	2.857	0.02326	0.02326
2	1.361	1.359	0.04651	0.04651
3	0.863	0.861	0.06977	0.06977
4	0.616	0.612	0.09302	0.09302
5	0.460	0.464	0.11628	0.11628
6	0.368	0.366	0.13953	0.13953

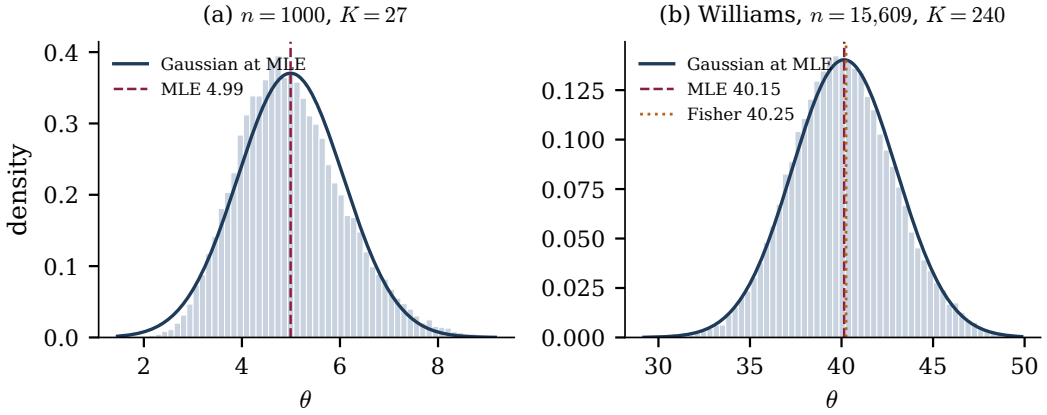


FIGURE 8.5. Posterior of θ (histogram of 40,000 Beta augmentation draws) against the Gaussian centred at the MLE with the asymptotic standard error (Proposition 8.13). (a) $n = 1000$, $K = 27$, the expected species count when $\theta = 5$. (b) Williams' Macrolepidoptera, $n = 15{,}609$, $K = 240$. In both the Bayes point estimate and interval reproduce the likelihood.

matching McCullagh (2016) to three decimals; the posterior mean is 40.09 (SD 2.84), with Fisher's unconditional 40.248 marked for comparison.

Computation and prior sensitivity (Table 8.8, Figure 8.6). The Beta augmentation sampler (8.5) is nearly i.i.d. (integrated autocorrelation time 1.4) and delivers about seven times the effective samples per second of a random walk Metropolis sampler on $\log\theta$ with proposal standard deviation 0.07, whose chain has autocorrelation time 8.9. That proposal is narrower than optimal: at the best scale, standard deviation about 0.17 with acceptance rate 0.44, the autocorrelation time is about 4.4, which roughly halves the advantage. The timings are for plain Python loops and only their ratio is meaningful. Across 150 simulated data sets with $\theta = 6$, $n = 800$ the posterior mean exceeds the MLE by 0.019 on average and never by more than 0.02, under two percent of a standard error. A coverage experiment with the same design and 400 data sets (mean $K = 30.1$) gives 92.8% empirical coverage of the nominal 95% credible interval, 91.3% for the Wald interval and 93.3% for the log

TABLE 8.8. Sampler efficiency for the Williams posterior ($n = 15{,}609$, $K = 240$), 16,000 retained draws. The exact posterior mean and SD are 40.088 and 2.838. IAT is the integrated autocorrelation time, ESS the effective sample size; ESS per second is for plain Python loops on one core.

sampler	mean	SD	IAT	ESS	ESS/sec	acc.
PIG Beta augmentation (8.5)	40.127	2.860	1.40	11,410	147,600	n/a
random walk Metropolis	40.056	2.871	8.92	1,794	20,571	0.71

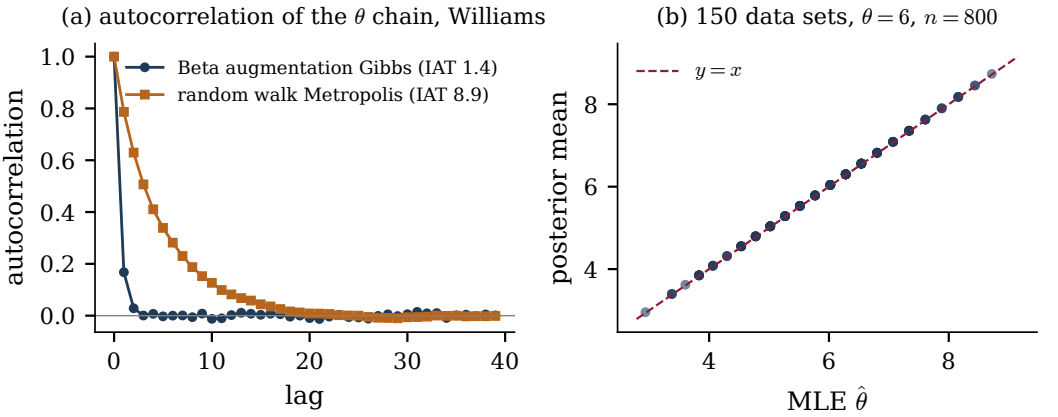


FIGURE 8.6. (a) Autocorrelation of the θ chain for the Williams posterior: the PIG Beta augmentation sampler decorrelates in one or two steps, random walk Metropolis in about ten. (b) Posterior mean against MLE over 150 data sets ($\theta = 6$, $n = 800$); the estimators are interchangeable.

scale Wald interval; with 400 replications the Monte Carlo standard error is about 1.3 points, and the shortfall reflects the moderate information at $K \approx 30$ and the discreteness of K.

Prior sensitivity. Proposition 8.13 concerns priors flat near $\hat{\theta}$, and among these the Williams posterior is stable: the diffuse $\mathrm{Ga}(10^{-3}, 10^{-3})$ gives posterior mean 40.16, the default $\mathrm{Ga}(10^{-2}, 10^{-2})$ gives 40.09, and a $\mathrm{Ga}(2, 0.05)$ prior with mean 40 gives 40.16, all within 0.03 posterior standard deviations of each other. The claim is not that no prior can matter at finite K. An aggressive $\mathrm{Ga}(1, 1)$, effectively $\mathrm{Exp}(1)$ favouring small θ, pulls the mean to 33.69, a 16% shift, because its rate 1 competes with the modest data rate $\mathbb{E}[-\log \eta] = \psi(\theta + n) - \psi(\theta) \approx \log\{(n + \theta)/\theta\} \approx 6$ in the gamma step of (8.5). With effective sample size $K = 240$ the likelihood dominates any diffuse prior but not an informative one, exactly as the logarithmic information growth predicts.

The PIG(0) engine and Fisher's ν (Table 8.9, Figure 8.7). Simulating X_0 by the inverse gamma series of Proposition 8.7, truncated at 1500 terms with the mean of the remainder, $\sum_{k>1500} 1/(2k^2) = 3.3 \times 10^{-4}$, added as a constant, and comparing $\widehat{\mathbb{E}}[e^{-s^2 X_0}]$ over 2×10^5 draws to the exact reciprocal gamma transform $e^{-\gamma s}/\Gamma(1 + s)$

TABLE 8.9. PIG(0) simulator (2×10^5 draws, 1500 term truncation plus remainder mean) against the exact reciprocal gamma transform (8.6).

s	$\widehat{\mathbb{E}}[e^{-s^2 X_0}]$	$e^{-\gamma s}/\Gamma(1+s)$	rel. error
0.25	0.95532	0.95501	3.3×10^{-4}
0.50	0.84603	0.84550	6.2×10^{-4}
1.00	0.56177	0.56146	5.5×10^{-4}
1.50	0.31651	0.31648	0.9×10^{-4}
2.00	0.15756	0.15762	3.9×10^{-4}

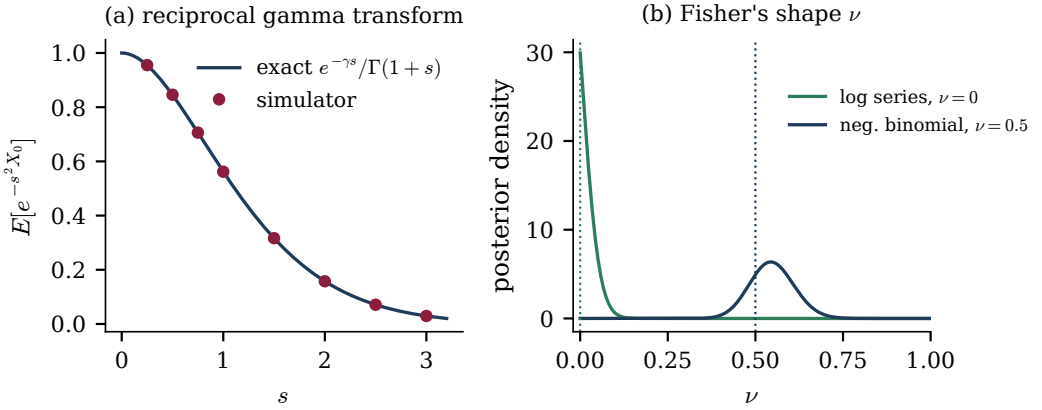


FIGURE 8.7. (a) The PIG(0) simulator (points) reproduces the exact reciprocal gamma transform $e^{-\gamma s}/\Gamma(1+s)$ (curve). (b) Posterior of Fisher's shape ν from 400 simulated species: log series data concentrate at $\nu = 0$ (the Ewens limit), and $\nu = 0.5$ data are recovered.

gives agreement within 0.07% over $s \in [0.25, 2]$. The sample mean is 0.801 against $\pi^2/12 = 0.822$; the gap is not an error but a consequence of $\mathrm{Var}(X_0) = \infty$, which makes the sample mean converge slowly and erratically, and it is why the Laplace transform, a bounded function of X_0, is the right check. For Fisher's ν we simulated 400 species abundances from the zero truncated negative binomial of (8.7) with $x = 0.98$ and computed the joint posterior of (ν, x) under flat priors on $\nu \in (0, 1.2]$ and $x \in [0.9, 1)$ by quadrature, which is the target that the R2 Gibbs sampler draws from. Log series data (true $\nu = 0$, 5568 specimens) yield posterior mean 0.027 with 95% interval $[0, 0.075]$, recovering Fisher's empirical conclusion, while $\nu = 0.5$ data (11,886 specimens) are recovered at 0.55 with interval $[0.43, 0.67]$.

Good Turing for Williams (Figure 8.8(b)). Pushing the posterior draws through $r/(n+\theta)$ gives Bayesian Good Turing frequencies. For the seen species the posterior is degenerate for practical purposes: q_1 has posterior mean 6.390×10^{-5} with 95% interval $[6.388, 6.392] \times 10^{-5}$, a relative width of 0.07%, slightly below the raw proportion $1/n = 6.407 \times 10^{-5}$ because of the shrinkage factor $n/(n+\theta) = 0.9974$. The unseen mass is where the posterior matters: $q_0 = \theta/(n+\theta)$ has posterior

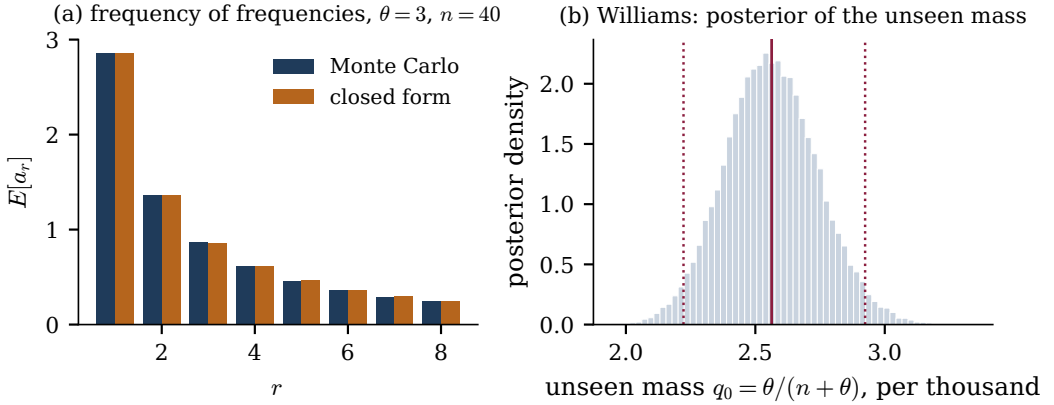


FIGURE 8.8. (a) Expected frequency of frequencies for $\theta = 3$, $n = 40$: Monte Carlo against the closed form of Proposition 8.9. (b) Posterior of the unseen mass $q_0 = \theta/(n+\theta)$ for Williams, in units of one per thousand; the solid line is the posterior mean and the dotted lines the 95% interval. The corresponding bands for the seen frequencies $q_r = r/(n+\theta)$, $r \geq 1$, have relative width below 0.1% and are not shown.

mean 2.56 per thousand with interval $[2.22, 2.92]$ per thousand, a relative width of 27%.

Williams temporal homogeneity (Figure 8.9). Refitting the year 1933 alone ($n = 3540$, $K = 178$) gives $\hat{\theta} = 39.355$, exactly McCullagh (2016), with posterior mean 39.3 (SD 3.3). For the 12,069 specimens of 1934 to 1936, Proposition 8.11 gives the expected number of new species, $\mathbb{E}[U_m \mid \theta]$, which has posterior mean 58.0 with 95% interval $[48.8, 68.0]$, reproducing McCullagh's plug in value 58.06. The full predictive distribution of the count itself, computed exactly by mixing the Poisson binomial law of U_m over the posterior of θ, has mean 58.0 and standard deviation 9.0, of which the sampling term contributes variance 57.6 and the parameter term 24.0; its 95% interval is $[41, 77]$ and the probability of 62 or more new species is 0.34. The observed 62 is well inside both intervals: no evidence against temporal homogeneity, now with a calibrated interval rather than a point.

The full Good and Toulmin (1956) accumulation curve, obtained by propagating the same posterior through the predictive law for every m, is shown in Figure 8.10. The 95% predictive band is $[9, 27]$ at $m = 2000$, $[26, 53]$ at $m = 6000$ and $[41, 76]$ at $m = 12{,}000$; the observed count lies inside the band, and the band widens without ever narrowing, the honest reflection of the logarithmic information in K: extrapolation is possible but its uncertainty never collapses. This curve, with its interval, is the deliverable Good and Toulmin were reaching for and the Bayesian object that the richness point estimate cannot provide.

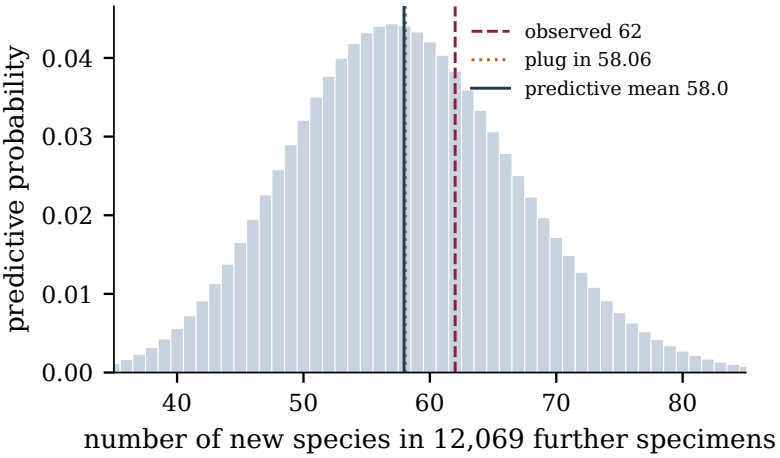


FIGURE 8.9. Posterior predictive distribution of the number of new species among 12,069 further specimens under the Ewens process fitted to 1933, including both sampling and parameter uncertainty. The observed 62 and McCullagh's plug in 58.06 both fall in the bulk.

8.10 Discussion

The Fisher and Ewens richness problem is the cleanest instance of a division of labour that recurs whenever a model reduces to a low dimensional sufficient statistic in a regular exponential family. For estimating θ the Bayes rule is a reparametrised maximum likelihood estimate and the prior is a rounding error, a fact Proposition 8.13 makes quantitative through the effective sample size K. The genuinely Bayesian content lies elsewhere, in the predictive law the model induces: the Chinese restaurant rule $r/(n+\theta)$, its unseen mass specialisation $\theta/(n+\theta)$, and the Good and Toulmin accumulation curve, each carried with an honest posterior band. This is the de Finetti reading of Good and Toulmin (1956): had they grafted a parametric skeleton onto their sequential prediction, the exchangeable partition process would have appeared, and Proposition 8.9 shows their own estimator was already that process in disguise.

Pólya inverse gamma augmentation is the computational thread that ties the three classical viewpoints together. Its two representations answer to Fisher's two parameters: the gamma ratio for richness, where inference is easy and the Bayesian gain is slight, and the reciprocal gamma for the shape ν, where inference is genuinely hard and the PIG engine earns its place. Beneath both sits the gamma subordinator, the prototypical generalized gamma convolution, whose normalisation is Ferguson's Dirichlet process and whose sample partition is Kingman's Ewens formula, and whose discretised Lévy density is Fisher's log series. Read this way, Fisher's estimation, Good's smoothing, and PIG augmentation are three views of one gamma process structure, and the division of labour between what the Bayes estimator can and cannot add follows from where in that structure information accumulates: logarithmically for θ, and not at all from

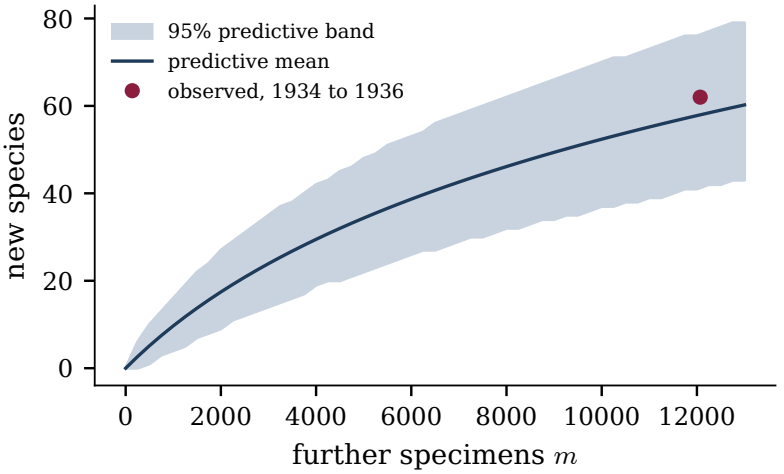


FIGURE 8.10. Posterior predictive species accumulation curve (Good and Toulmin) from the 1933 fit: predictive mean and 95% predictive band for the number of new species among m further specimens. The observed 62 new species at $m = 12{,}069$ falls within the band.

above once the sample is saturated with singletons.

That last qualification is the second half of the chapter and it tempers the first. "Sheds little" is a statement about the interior of the range $1 < K < n$. At the boundaries the posterior for the concentration is strongly skewed, and heavy tailed when K is near n, and the roles reverse: for a hyperdiverse sample it is the maximum likelihood point that misleads, running off to infinity when every unit is distinct, while the full posterior with its honest tail is indispensable. A flat prior is not merely vague there but improper, and a diffuse gamma is not innocent either, since its rate quietly fixes a scale. Our recommendation is the weakly informative PIG prior (8.10), a single pseudosample in the same gamma ratio family, which guarantees a proper posterior with finite mean and variance for every $K \leq n$, costs one extra Beta draw, and is dominated by any informative sample. It also serves as a specification diagnostic: an exploding concentration under this prior is the fingerprint of a power law that the one parameter model cannot represent, and it points directly to the second parameter that can.

Several limitations bound the claims and mark the natural next steps. The one parameter Ewens model is misspecified for power law data, and the honest response is the two parameter Pitman and Yor process of (8.8), whose discount σ absorbs the heavy tail; we placed it in the family but did not fit it, so the Shakespeare trigram posteriors should be read as diagnostics rather than estimates. Our recovery of Fisher's shape ν used simulated frequencies because the Williams analysis here uses only the pair (n, K) for each period, so a full two parameter fit to primary frequency data through the reciprocal gamma augmentation R2 remains open; the butterfly example of Section 8.5 shows that even fifteen cells of a frequency table identify the log series richness well, though not its second

parameter x. The Bernstein and von Mises statement is asymptotic in K, and at the moderate values typical of ecological samples the Gaussian approximation is only approximate, as the 93% interval coverage at $K \approx 30$ records. None of these caveats disturbs the reconciliations, which are exact identities, but each marks where the modelling, as opposed to the mathematics, must be taken further.

The structure generalises beyond the scalar problem. The reciprocal gamma representation R2 handles the vector concentration of the finite Dirichlet multinomial and the shape of a gamma mixing measure, so the same augmentation supports species models with covariates and hierarchical concentrations shared across sites or years, all within one gamma and Beta sampler. More broadly, placing Fisher, Good, and Ewens inside the generalized gamma convolution class connects a classical problem to the wider PIG program and to the Thorin measure calculus that indexes it. On the applied side, the Good Turing rule remains the workhorse of n gram smoothing in language models, and a Bayesian version that reports the unseen mass with a calibrated interval, rather than a point, may be useful wherever that mass carries decision weight, from estimating undetected biodiversity to sizing a vocabulary to bounding the burden of rare genetic variants.

8.11 Proofs

Proof of Proposition 8.2. From (8.1), $p(a \mid \theta) = h(a)\,\theta^K/(\theta)_{(n)}$ with $h(a) = n!\prod_j (j^{a_j} a_j!)^{-1}$. Writing $\varphi = \log\theta$ gives $p = h(a)\exp\{\varphi K - A(\varphi)\}$ with $A(\varphi) = \log(\theta)_{(n)} = \log\Gamma(\theta+n) - \log\Gamma(\theta)$. The natural parameter space is all of $\mathbb{R}$, which is open, so the family is regular. Then $\mathrm{d}A/\mathrm{d}\varphi = (\mathrm{d}A/\mathrm{d}\theta)\theta = \theta[\psi(\theta+n) - \psi(\theta)]$. For the Feller coupling, run the Chinese restaurant process and let B_i indicate that the specimen arriving when i are present founds a new species; it does so with probability $\theta/(\theta+i)$ whatever the earlier configuration, so the B_i are independent and $K = \sum_{i=0}^{n-1} B_i$. Hence $\mathbb{E}[K] = \sum_i \theta/(\theta+i)$, which equals $\theta[\psi(\theta+n) - \psi(\theta)]$ by the recurrence $\psi(x+1) = \psi(x) + 1/x$, and this is $A'(\varphi)$ as the exponential family identity requires. Likewise $\mathrm{Var}(K) = \sum_i \frac{\theta}{\theta+i}\cdot\frac{i}{\theta+i} = \sum_i \theta i/(\theta+i)^2$; differentiating $A'(\varphi) = \sum_i e^{\varphi}/(e^{\varphi}+i)$ once more gives the same sum, so $A''(\varphi) = \mathrm{Var}(K)$, positive as soon as $n \geq 2$. Convexity of A gives strict concavity of the log likelihood $\varphi K - A(\varphi)$ and uniqueness of the root of (8.2), which exists when $1 < K < n$ because A' increases from 1 to n. Finally

$$I(\theta) = A''(\varphi)\Big(\frac{\mathrm{d}\varphi}{\mathrm{d}\theta}\Big)^2 = \frac{\mathrm{Var}(K)}{\theta^2},$$

and expanding, $I(\theta) = \theta^{-1}\sum_i i/(\theta+i)^2$. □

Propriety of (8.3). As $\theta \to 0^+$, $\Gamma(\theta)/\Gamma(\theta+n_0) \sim 1/(\theta\,\Gamma(n_0))$, so the integrand behaves like $\theta^{\kappa-\tau}$, integrable near zero if and only if $\kappa - \tau > -1$. As $\theta \to \infty$, $\Gamma(\theta)/\Gamma(\theta+n_0) \sim \theta^{-n_0}$, so the integrand behaves like $\theta^{\kappa-\tau n_0}$, integrable at infinity if and only if $\kappa - \tau n_0 < -1$. The integrand is continuous and positive in between. Both conditions hold for some κ precisely when $\tau - 1 < \tau n_0 - 1$, that is $n_0 > 1$. □

Proof of Proposition 8.7. For $Y \sim \mathrm{InvGa}(3/2, 1)$ and $t > 0$, substituting $y = 1/u$,

$$\mathbb{E}[e^{-tY}] = \frac{1}{\Gamma(3/2)} \int_0^\infty u^{1/2} e^{-u-t/u}\, \mathrm{d}u = (1 + 2\sqrt{t})\, e^{-2\sqrt{t}},$$

by the standard integral $\int_0^\infty u^{1/2} e^{-u-t/u}\, \mathrm{d}u = \frac{\sqrt{\pi}}{2}(1 + 2\sqrt{t})e^{-2\sqrt{t}}$. With $t = s^2/(4k^2)$ this is $(1 + s/k)e^{-s/k}$. The terms $Y_k/(4k^2)$ are independent and nonnegative with $\sum_k \mathbb{E}[Y_k]/(4k^2) = \sum_k 1/(2k^2) < \infty$, since $\mathbb{E}[Y] = 1/(3/2 - 1) = 2$; so the series converges almost surely and in mean, and by monotone convergence

$$\mathbb{E}[e^{-s^2 X_0}] = \prod_{k \geq 1} \Big(1 + \frac{s}{k}\Big) e^{-s/k} = \frac{e^{-\gamma s}}{\Gamma(1 + s)},$$

the last step being the Weierstrass product $1/\Gamma(1 + s) = e^{\gamma s} \prod_k (1 + s/k)e^{-s/k}$. The mean is $\sum_k 2/(4k^2) = \zeta(2)/2 = \pi^2/12$. Finally $\mathbb{E}[Y^2] = \infty$ because the $\mathrm{InvGa}(3/2, 1)$ density decays like $y^{-5/2}$, and since $X_0 \geq Y_1/4$ we have $\mathbb{E}[X_0^2] = \infty$. As a cross check on the mean, $\log \mathbb{E}[e^{-s^2 X_0}] = -\gamma s - \log \Gamma(1 + s) = -\sum_{k \geq 2} (-1)^k \zeta(k) s^k/k$, so $\mathbb{E}[e^{-s^2 X_0}] = 1 - \frac{1}{2}\zeta(2)s^2 + O(s^3)$; the presence of an s^3 term, odd in s, is another sign that X_0 has no second moment. □

Proof of Proposition 8.9. By exchangeability, label the specimens $1, \ldots, n$; then

$$\mathbb{E}[a_r(n)] = \binom{n}{r} \Pr(\text{a fixed set } S \text{ of } r \text{ specimens is exactly one block}),$$

and since the law is exchangeable we may take $S = \{1, \ldots, r\}$, the first r arrivals. Under the Chinese restaurant process the first arrival opens a block, and the arrival that finds i specimens all in that block joins it with probability $i/(\theta + i)$, so the first r form one block with probability $\prod_{i=1}^{r-1} i/(\theta + i) = (r - 1)!\, \Gamma(\theta + 1)/\Gamma(\theta + r)$. Each of the remaining arrivals, finding $r + j$ specimens present ($j = 0, \ldots, n - r - 1$), avoids that block of size r with probability $(\theta + j)/(\theta + r + j)$, whatever happens elsewhere. The product of these is $\frac{\Gamma(\theta+n-r)}{\Gamma(\theta)} \cdot \frac{\Gamma(\theta+r)}{\Gamma(\theta+n)}$, and multiplying the two factors gives $\theta(r - 1)!\, \Gamma(\theta + n - r)/\Gamma(\theta + n)$ for the probability that S is a block. Multiplying by $\binom{n}{r}$,

$$\mathbb{E}[a_r(n)] = \frac{\theta}{r} \frac{n!}{(n - r)!} \frac{\Gamma(\theta + n - r)}{\Gamma(\theta + n)},$$

which at $r = n$ reduces to $\theta\Gamma(n)\Gamma(\theta)/\Gamma(\theta + n)$, the probability of a single block. Substituting $\mu_{r,n}$ and $\mu_{r+1,n+1}$ into (8.9), the factors $\Gamma(\theta + n - r)$ cancel and

$$\frac{\mu_{r+1,n+1}}{\mu_{r,n}} = \frac{r}{r + 1}\, (n + 1)\, \frac{\Gamma(\theta + n)}{\Gamma(\theta + n + 1)} = \frac{r}{r + 1}\, \frac{n + 1}{\theta + n},$$

so $\mathbb{E}[q_r \mid \text{data}] = \frac{r+1}{n+1} \cdot \frac{r}{r+1} \cdot \frac{n+1}{\theta+n} = \frac{r}{n+\theta}$. The unseen mass $q_0 = \theta/(n + \theta)$ is the Chinese restaurant probability of opening a new block, and the masses add up correctly: $\sum_r a_r\, r/(n + \theta) + \theta/(n + \theta) = (n + \theta)/(n + \theta) = 1$. □

Proof of Proposition 8.11. Continue the Chinese restaurant process past n. The arrival that finds $i \geq n$ specimens present founds a new species with probability $\theta/(\theta+i)$, independently of the configuration and hence of the first n specimens, and every species founded after time n is new relative to the sample. So $U_m = \sum_{i=n}^{n+m-1} B_i$ with independent Bernoulli B_i, and the mean and variance follow as in the proof of Proposition 8.2, using $\sum_{i=n}^{n+m-1} 1/(\theta+i) = \psi(\theta+n+m) - \psi(\theta+n)$. Given the data, U_m depends on the sample only through θ, whose posterior depends only on K; the law of total variance then gives the stated decomposition. □

Proof of Proposition 8.13. This is Bernstein and von Mises for a regular one parameter NEF (van der Vaart, 1998, Chapter 10), with $J_n = A''(\hat{\varphi}) \to \infty$ playing the role of the sample size: the log likelihood $\varphi K - A(\varphi)$ is concave with curvature A'', and $A'''(\varphi)/A''(\varphi)^{3/2} \to 0$ as $J_n \to \infty$ because the cumulants of the sum of independent Bernoulli variables K are all of order $\mathbb{E}[K]$. Hence the posterior of φ converges in total variation to $\mathcal{N}(\hat{\varphi}, J_n^{-1})$. A Laplace expansion of $\mathbb{E}[e^{\varphi} \mid K]$ about $\hat{\varphi}$ gives $\mathbb{E}[\theta \mid K] = \hat{\theta}\{1 + O(J_n^{-1})\}$, and the delta method maps the Gaussian interval for φ to the Wald interval for θ with error $o(\hat{\theta} J_n^{-1/2})$. Reweighting the prior by a smooth positive g shifts the posterior mean of φ by $J_n^{-1}(\log g)'(\hat{\varphi}) + o(J_n^{-1})$, hence the posterior mean of θ by

$$\hat{\theta}\, J_n^{-1}(\log g)'(\hat{\varphi}) + o(\hat{\theta} J_n^{-1}),$$

and changes the posterior by $O(J_n^{-1/2})$ in total variation. Since J_n is of the order of $\mathbb{E}[K]$, the effective sample size is K. □

Proof of Proposition 8.15. Since $\Gamma(\theta)/\Gamma(\theta+n) = \prod_{i=0}^{n-1}(\theta+i)^{-1}$ and each factor is asymptotic to θ^{-1}, the ratio is $\sim \theta^{-n}$ as $\theta \to \infty$, so the posterior integrand satisfies

$$\theta^{K+a_0-1} e^{-b_0\theta} \frac{\Gamma(\theta)}{\Gamma(\theta+n)} \sim \theta^{K+a_0-1-n} e^{-b_0\theta}.$$

If $b_0 > 0$ the exponential dominates and every moment is finite. If $b_0 = 0$ the tail is θ^{K+a_0-1-n}; $\int^{\infty} \theta^{K+a_0-1-n}\, \mathrm{d}\theta < \infty$ if and only if $K + a_0 - 1 - n < -1$, that is $K + a_0 < n$, and $\int^{\infty} \theta^{p} \theta^{K+a_0-1-n}\, \mathrm{d}\theta < \infty$ if and only if $p < n - K - a_0$. Integrability at $\theta \to 0^+$ holds throughout since $\Gamma(\theta)/\Gamma(\theta+n) \sim 1/(\theta\Gamma(n))$ makes the integrand $O(\theta^{K+a_0-2})$ with $K + a_0 - 2 > -1$ for $K \geq 1$. Substituting $(a_0, b_0) = (1, 0)$ gives propriety for $K < n-1$, finite mean for $K < n-2$, and finite variance ($p = 2$) for $K < n-3$, that is $K \leq n-2,\ n-3,\ n-4$ respectively. □

8.12 Sampler details

Richness θ. Prior $\mathrm{Ga}(a_0, b_0)$; iterate (8.5), drawing $\log \eta$ as in Remark 8.6, and replacing the single Beta by one draw per sample for multiyear data, in which case the gamma rate is $b_0 - \sum_j \log \eta_j$. For the conjugate prior (8.3) with integer τ, augment with τ additional $\mathrm{Be}(\theta, n_0)$ latents; for the PIG prior (8.10) this is the

single extra latent η_0. For the beta prime prior (8.13) use the three block sampler of Proposition 8.16.

Shape ν. The zero truncated negative binomial likelihood $p(r \mid \nu, x) = \frac{\Gamma(r+\nu)}{r!\,\Gamma(\nu)}(1-x)^{\nu}x^{r}/\{1-(1-x)^{\nu}\}$, $r \geq 1$, places ν inside $1/\Gamma(\nu)$, which the PIG(0) augmentation of He et al. (2019, Section 4.1) turns into a Gaussian factor in ν through (8.6); the factors $\Gamma(r+\nu)$ and the truncation term need their own latents, as described there. For the moderate sample sizes in this chapter the two dimensional posterior of (ν, x) can also be computed directly by quadrature, which is what Figure 8.7(b) shows. Near $\nu = 0$ the truncation term should be evaluated as $1-(1-x)^{\nu} = -\mathrm{expm1}(\nu \log(1-x))$ to avoid cancellation. The scripts that produce every number and figure in this chapter accompany the book.

8.13 Exercises

1. Write down the Ewens probabilities of the three abundance vectors for $n = 3$ and check that they sum to one for every θ. Verify that $\mathbb{E}[K] = 1 + \theta/(\theta+1) + \theta/(\theta+2)$.
   Hint. The probabilities are $2/\{(\theta+1)(\theta+2)\}$, $3\theta/\{(\theta+1)(\theta+2)\}$ and $\theta^2/\{(\theta+1)(\theta+2)\}$.

2. Show that $A''(\varphi) = \theta\,\frac{\mathrm{d}}{\mathrm{d}\theta}\mathbb{E}[K \mid \theta]$ and deduce that the Fisher information can be written $I(\theta) = \{\mathrm{d}\mathbb{E}[K]/\mathrm{d}\theta\}^2/\operatorname{Var}(K)$, the form used in Section 8.8.
   Hint. Use $\mathrm{d}/\mathrm{d}\varphi = \theta\,\mathrm{d}/\mathrm{d}\theta$ and $A'' = \operatorname{Var}(K)$.

3. Prove the Beta identity (8.4) and derive the conditionals (8.5). Show that the θ marginal of the augmented density is the posterior.
   Hint. $B(\theta, n) = \Gamma(\theta)\Gamma(n)/\Gamma(\theta+n)$; write $\eta^{\theta-1} = e^{(\theta-1)\log\eta}$.

4. Show that if $K = n$ the likelihood $\theta^n\Gamma(\theta)/\Gamma(\theta+n)$ is increasing in θ with limit one as $\theta \to \infty$, and that if $K = 1$ the likelihood $\theta\,\Gamma(\theta)/\Gamma(\theta+n)$ is decreasing in θ with limit $1/(n-1)!$ as $\theta \to 0$. Conclude that the MLE is $+\infty$ and 0 respectively, and interpret both in terms of the moment equation (8.2).
   Hint. Write the likelihood as $\prod_{i=1}^{n-1}\theta/(\theta+i)$ when $K = n$ and as $\prod_{i=1}^{n-1}1/(\theta+i)$ when $K = 1$.

5. For $\theta = 2$ and $n = 3$ compute $\mu_{1,3}$ and $\mu_{2,4}$ from Proposition 8.9 and check that Good's ratio (8.9) equals $1/(n+\theta) = 0.2$.
   Hint. $\mu_{1,3} = 1.5$ and $\mu_{2,4} = 0.6$.

6. In Fisher's log series model with parameters (θ, x), show that the expected number of species absent from a first collection and present in a second collection of equal effort is $\theta\log(1+x)$, and that this equals $\mathbb{E}[\sum_{r\geq 1}(-1)^{r+1}a_r]$, the expectation of the Good and Toulmin alternating sum.
   Hint. Integrate $(e^{-\lambda} - e^{-2\lambda})$ against $\theta\lambda^{-1}e^{-\lambda/\beta}$ and use $x = \beta/(1+\beta)$; then expand $\log(1+x)$.

7. Show that if $\sqrt{\theta}$ is half Cauchy then θ has the beta prime density $\pi^{-1}\theta^{-1/2}(1+\theta)^{-1}$ and $\log\theta$ has density $\{2\pi\cosh(\varphi/2)\}^{-1}$. Prove Proposition 8.16 for general χ and find the tail exponent of the posterior.
Hint. Change variables twice; $(1+\theta)^{-\chi}$ is a gamma integral in ξ.

8. Show that the PIG prior (8.10) is the posterior from the improper prior $\mathrm{d}\theta/\theta$ after a pilot sample with n_0 specimens and K_0 species, and find the condition on (n_0, K_0) for it to be proper. For which K does the resulting posterior have a finite pth moment?
Hint. Proper if and only if $1 < K_0 < n_0$; the posterior tail is $\theta^{K+K_0-1-n-n_0}$.

9. Take $n = 100$ and $K = 12$. Solve the moment equation for $\hat\theta$, compute the standard error from Proposition 8.2, and run the sampler (8.5) for 10^4 iterations under the diffuse prior. Compare the posterior and Wald intervals, and repeat on the $\log\theta$ scale.
Hint. Use a bracketing root finder on $\varphi = \log\theta$; the posterior interval should be visibly asymmetric about $\hat\theta$.

10. Using Proposition 8.7, show that $\mathbb{E}[X_0] = \pi^2/12$ and $\mathrm{Var}(X_0) = \infty$. Simulate X_0 with a truncated series and observe how slowly the sample mean settles compared with the sample average of e^{-X_0}.
Hint. $\mathbb{E}[Y] = 2$ and $\mathbb{E}[Y^2] = \infty$ for $Y \sim \mathrm{InvGa}(3/2, 1)$; compare with $e^{-\gamma}/\Gamma(2) = 0.5615$.

11. In Fisher's model N is negative binomial with shape θ and success parameter x, $p(N) = \frac{\Gamma(N+\theta)}{N!\Gamma(\theta)} x^N (1-x)^\theta$. Show that the information about θ in N after profiling out x is $\psi'(\theta) - \mathbb{E}[\psi'(N+\theta)] - x/\theta$, and that for large $\mathbb{E}[N]$ this is approximately $1/(2\theta^2)$. Compare with $I(\theta)$ from Proposition 8.2 for Williams' data.
Hint. The information matrix has $I_{\theta x} = 1/(1-x)$ and $I_{xx} = \theta/\{x(1-x)^2\}$; use $\psi'(\theta) = 1/\theta + 1/(2\theta^2) + O(\theta^{-3})$.

12. For Williams' 1933 sample, $\hat\theta = 39.355$ and $n = 3540$. Use Proposition 8.11 to find the expected number of new species if the sample is doubled, and compare with $\hat\theta \log 2$.
Hint. The exact value is 27.06, and $\hat\theta \log 2 = 27.28$.

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CHAPTER 9

Rational Conjugacy: Pochhammer Priors and Gamma Ratios

Pochhammer priors, introduced by Wang and Polson (2024), give conjugate Bayesian inference for concentration parameters that enter a likelihood through ratios of Gamma functions with integer shifts. This chapter develops the general theory of that family around a single principle, rational conjugacy: the Pochhammer family is not closed under Bayesian updating, but the proper rational densities on $\mathbb{R}^+$ are, and on that class every normalizing constant, moment and predictive probability is a finite sum of residues.

Five results follow, each generalizing or correcting a statement of Wang and Polson (2024). A general residue formula admits poles of arbitrary multiplicity anywhere in the cut plane, as required for Pitman Yor, where poles repeat, and for Erlang's loss system, where they are complex. The Mellin transform of a Pochhammer density is obtained in closed form on its maximal strip, delivering fractional and negative moments, $\mathbb{E}\log\alpha$, the exact tail index $b-m-d-1$ (with a second, probabilistic proof in Corollary 9.10), and a corrected reading of the Kullback Leibler risk bound: every m attains the superefficient rate $N^{-1}\log N$, and $m=0$ merely minimizes the constant. A completely monotone representation then identifies $m=d=0$ on structural grounds: the half horseshoe is a scale mixture of exponentials whose mixing law is a size debiased sum of independent exponentials, so a latent scale reduces the prior to a tilt exactly as an inverse Gamma scale reduces the half Cauchy to a Gaussian. The consequence is practical. Wherever a Gamma prior on a concentration parameter admits an exact draw, the half horseshoe does too, at the cost of b extra exponential variates; for Negative Binomial regression this yields a Gibbs sampler in which every block is a standard draw, no Metropolis step appears, and no residue is ever evaluated. An exponentially tilted family completes the picture, its normalizer a finite sum of exponential integrals, and the tilt removes the degree restriction $b \geq m+d+2$ altogether.

We then locate the boundary of the method: $\Gamma(\alpha+u)/\Gamma(\alpha+v)$ is rational if and only if $u-v\in\mathbb{Z}$, so Damsleth's (1975) conjugate class for the two parameter Gamma lies outside it, and we give three repairs, two of them exact, of which Poisson augmentation is the sharpest, together with a fourth that is not ours: the reciprocal Gamma mixtures of He, Polson and Xu reach the transcendental factor directly, and we set out the division of labour between augmentation and residues. A power law theorem shows the Negative Binomial, Beta, Pochhammer hierarchy generates $p(n)\sim Kn^{-(1+a)}(\log n)^{-\{2+\mathbf{1}(m\geq 1)\}}$, so a Zipf exponent may be elicited as a prior hyperparameter. Applications include Pólya urns, the Ewens sampling formula, Pitman Yor concentration, Yule processes and Erlang's formula, and a Negative Binomial regression study of two count data sets in which a $\mathrm{Ga}(2,1)$ prior on the dispersion holds the posterior far below the near Poisson values the likelihood supports and doubles the width of the predictive bands, while the half horseshoe does not. Finally

we quantify a limitation: the residue expansion is exact but ill conditioned, provably losing $n \log_{10} 2$ *digits in a solvable case, and we give three remedies.*

Because this is a book about genetics, the running example is the concentration parameter θ *of the Ewens sampling formula, the scaled mutation rate of the infinitely many alleles model, together with its Pitman Yor extension and the Hoppe urn that generates both. A dedicated section works the residue calculus by hand for a sample of four genes carrying two alleles, obtaining the normalizing constant, the posterior mean and the probability that the next gene carries a new allele as exact combinations of* $\log 2$ *and* $\log 3$*; shows how many digits double precision loses as the sample grows; gives an exact two line sampler for the half horseshoe; and analyses Corbet's Malayan butterflies, where a light tailed* $\mathrm{Ga}(2, 1)$ *prior of the kind used in the Dirichlet process literature moves the posterior for* θ *far below the maximum likelihood value while the half horseshoe leaves it in place.*

9.1 Introduction

Let $[x]^n = x(x+1)\cdots(x+n-1) = \Gamma(x+n)/\Gamma(x)$ be the rising factorial. A large part of applied probability depends on a concentration parameter $\alpha > 0$ only through such symbols. The Dirichlet Multinomial likelihood is $\prod_k [\alpha_k]^{n_k} / [A]^N$; the Negative Binomial is $\prod_i [\alpha]^{n_i} \pi^{K\alpha}$; the Ewens sampling formula (Ewens, 1972) is $\alpha^k / [\alpha]^n$; the Pitman Yor law of the block count is $\prod_{i<k} (\theta + i\sigma) / [\theta+1]^{n-1}$; the Pólya urn probability of k white in n draws is $[a]^k [\beta]^{n-k} / [a+\beta]^n$. In each case the parameter is hard to learn: the log likelihood is generally not concave, and the standard responses are empirical Bayes point estimates (Minka, 2000), auxiliary variables (Escobar and West, 1995), or Metropolis steps.

For the readers of this book the most important instance is the Ewens sampling formula. In the infinitely many alleles model a sample of n genes falls into K_n allelic classes, and the probability of the observed configuration depends on the scaled mutation rate $\theta = 4N_e\mu$ only through $\theta^{K_n} / [\theta]^n$ (Chapter 6). The same symbol governs the Hoppe urn (Hoppe, 1984), in which the ith gene sampled is a new allele with probability $\theta/(\theta + i - 1)$; the Dirichlet process concentration in species sampling and clustering; and, with one extra parameter, the Pitman Yor model. Two facts make θ an awkward parameter. The information about it grows only like $\log n$ (Section 9.6.2), so a sample of thousands of genes pins it down less well than a sample of a dozen pins down a normal mean; and its plausible range spans several orders of magnitude, from $\theta < 1$ for a nearly monomorphic locus to θ in the hundreds for a hypervariable one. A prior for θ must therefore be heavy tailed, and a posterior for it must be computed with care. This chapter supplies both.

Wang and Polson (2024), hereafter WP, introduced the Pochhammer family

$$p(\alpha \mid m, a, b, c) \;\propto\; \frac{[\alpha]^m}{[c\alpha + a]^b}, \qquad \alpha \geq 0, \tag{9.1}$$

with m, b nonnegative integers, $b \geq m+2$, $a, c > 0$, and its size biased extension $\mathrm{PPH}(m, a, b, c, d) \propto \alpha^d [\alpha]^m / [c\alpha + a]^b$, of which (9.1) is the case $d = 0$.[1] WP establish conjugacy for the Dirichlet Multinomial concentration parameter, closed form normalizers by residues when all poles are simple, and that the choice $m = 0$, $b = 2$ behaves like a horseshoe on the half line. Everything below takes that paper as its starting point.

Five questions are left open by that treatment. Which likelihoods does the family serve, and is it the right family? What happens at repeated and complex poles? How far does the calculus reach towards the Gamma shape parameter, and towards Damsleth's (1975) conjugate class in particular? What count distribution does the hierarchy induce? And is the closed form usable in practice? The organizing idea fits in a sentence.

> *Rational conjugacy.* Proper rational densities on $\mathbb{R}^+$ are conjugate to rational likelihoods, because rational functions form a field; partial fractions then reduce every integral to elementary residues.

Pochhammer symbols are tractable only because integer shifts make them polynomials, a remark that sounds trivial and is not: Proposition 9.18 turns it into an exact criterion for membership, and thereby into an exact description of where the method stops.

What is new

Table 9.1 maps each result of Wang and Polson (2024) to its generalization. Three statements in that paper are also sharpened or corrected: the moment formula extends from integer to complex order (Theorem 9.6); the heavy tail proposition is replaced by the exact regular variation index (Corollary 9.7); and the claim that $m = 0$ is required for Kullback Leibler superefficiency is corrected, since every m attains the rate and $m = 0$ only minimizes the constant (Corollary 9.8). The last correction removes the stated justification for the default prior, and Theorem 9.9 supplies a better one in its place.

9.2 Rational Conjugacy

9.2.1 The class

DEFINITION 9.1 (Pochhammer rational likelihood). A likelihood L belongs to $\mathcal{R}$ if

$$L(\alpha) \;=\; \kappa \, \frac{\prod_{j=1}^{J} [u_j \alpha + v_j]^{p_j}}{\prod_{l=1}^{J'} [w_l \alpha + x_l]^{q_l}}, \qquad u_j \geq 0,\; v_j \geq 0,\; w_l > 0,\; x_l > 0, \tag{9.2}$$

with κ free of α and p_j, q_l nonnegative integers. Let $\mathcal{P}$ denote the set of probability densities on $\mathbb{R}^+$ that are rational functions with no poles in $[0, \infty)$.

[1] Definition 2 of Wang and Polson (2024) states the case as $d = 1$; the intended convention is $d = 0$, which we adopt throughout.

Wang and Polson (2024)	This chapter
Residues, simple poles only; one double root example; general recipe deferred	Theorem 9.3: arbitrary multiplicity, arbitrary complex location, improper case included
Integer moments by size biasing, $b \geq m + d + 2$	Theorem 9.6: Mellin transform on the maximal strip; fractional, negative and log moments
Their Proposition 1: tail is heavier than exponential	Corollary 9.7: regularly varying with exact index $b - m - d - 1$
Their Proposition 2: prior mass $\asymp \varepsilon^{1/2}$, "$m = 0$ needed"	Corollary 9.8: rate $N^{-1} \log N$ for all m, constant $\{1 + \mathbf{1}(m \geq 1)\}/2$
Conjugacy for DM, NB, NM, GDM, Yule Simon, Ewens	Definition 9.1 and Proposition 9.2: characterization and minimality of the rational class
NB with coupled Beta prior, their Figure 3 sparsity patterns	Theorem 9.20: exact power law tail $Kn^{-(1+a)}(\log n)^{-\{2+\mathbf{1}(m\geq 1)\}}$
"Default horseshoe type prior" with $m = 0$, $b = 2$	Theorem 9.9: $m = d = 0$ is exactly the completely monotone case; scale mixture of exponentials
Metropolis within Gibbs for heterogeneous α	Algorithms 1 and 2: tilted family, and a sampler with every block exact and no residues
Residue conditioning not discussed	Proposition 9.13: exponential ill conditioning, proved in a solvable case, and three remedies
"Further calibration required beyond DM"	Sections 9.4 to 9.7: Gamma shapes, urns, Ewens and Pitman Yor, Yule, credibility, Erlang

TABLE 9.1. Correspondence between the earlier results and their generalizations here.

Every $L \in \mathcal{R}$ is rational with all poles in the open left half plane, and $\mathcal{R}$ is closed under products, so independent experiments compound. The following is elementary but is the reason for everything that follows, and it says that $\mathcal{P}$ is not merely convenient but forced.

PROPOSITION 9.2 (Conjugacy and minimality). *(i) If $p \in \mathcal{P}$ and $L \in \mathcal{R}$ then $p(\cdot \mid data) \propto Lp$ lies in $\mathcal{P}$, provided* $\deg(denominator) \geq \deg(numerator) + 2$ *after cancellation. (ii)* $\mathrm{PH}(m, a, b, c) \subset \mathrm{PPH}(m, a, b, c, d) \subset \mathcal{P}$. *(iii) The Pochhammer family is itself* not *closed. Updating* $\mathrm{PPH}(m, a, b, c, d)$ *by* (9.2) *multiplies the numerator by $\prod_j [u_j\alpha + v_j]^{p_j}$ and adjoins poles at $-(x_l + i - 1)/w_l$, $i = 1, \ldots, q_l$, so the posterior is again Pochhammer if and only if every $u_j\alpha + v_j$ and $w_l\alpha + x_l$ is an affine reparametrization of α and of $c\alpha + a$ respectively. Otherwise the pole set strictly grows. Since w_l and x_l range over all of $(0, \infty)^2$ and multiplicities compound, the closure of the Pochhammer family under $\mathcal{R}$ updating contains every element of $\mathcal{P}$ with real poles and real nonpositive numerator roots whose numerator and denominator retain the initial prior factors, and no smaller family of densities is closed.*

Model	$L(\alpha)$ up to constants	Section
Dirichlet Multinomial	$\prod_k [\alpha]^{n_k} / [K\alpha]^N$	Wang and Polson (2024)
Heterogeneous DM, Gibbs conditional	$[\alpha_k]^{n_k} / [\alpha_k + A_{-k}]^N$	9.3.2
Negative Binomial, π integrated	$\prod_k [\alpha]^{n_k} [c\alpha + a]^b / [(c + K)\alpha + a]^{N+b}$	9.5
Negative Multinomial	$\alpha / [\alpha + N]^{K+1}$	Wang and Polson (2024)
Beta Binomial, Pólya urn	$\prod_i [\alpha]^{y_i} [\beta]^{n_i - y_i} / [\alpha + \beta]^{n_i}$	9.6.1
Ewens, Dirichlet process	$\alpha^k / [\alpha]^n$	9.6.2
Pitman Yor, σ fixed	$\prod_{i=1}^{k-1} (\theta + i\sigma) / [\theta + 1]^{n-1}$	9.6.3
Yule Simon	$\rho^K / \prod_k [\rho + 1]^{n_k}$	9.6.4
Beta Negative Binomial, Waring	$[\rho]^n [\alpha]^r / [\alpha + \rho]^{n+r}$	9.5.4
Erlang loss, offered load	$B(c, A)^y \{1 - B(c, A)\}^{M-y}$	9.6.6

TABLE 9.2. Rational likelihoods treated in this chapter. All are rational in the parameter shown; all but the last are of the Pochhammer form (9.2) and so lie in $\mathcal{R}$, while the Erlang likelihood lies in the wider class of rational likelihoods with complex poles.

Part (iii) is the practical message: one leaves the Pochhammer family immediately but never leaves $\mathcal{P}$, and an element of $\mathcal{P}$ is completely described by its pole locations $\{r_i\}$, multiplicities $\{\mu_i\}$ and coefficients $\{\gamma_{ij}\}$, all available in closed form. The larger class $\mathcal{P}$, rather than its real pole part, is genuinely needed: the Erlang likelihood of Section 9.6.6 has complex poles, which no Pochhammer factor can produce. Table 9.2 lists members of $\mathcal{R}$.

9.2.2 *The general residue formula*

Repeated poles are not exceptional. They arise whenever a Pochhammer prior with $a/c \in \mathbb{Z}$ meets a likelihood with integer shifts, which is the generic situation for the Dirichlet process and Pitman Yor. Complex poles arise in Section 9.6.6.

THEOREM 9.3 (Normalizing constant). *Let $R = P/Q$ be rational with poles at $-r_1, \dots, -r_J$ of multiplicities $\mu_1, \dots, \mu_J$, where $r_i \in \mathbb{C} \setminus (-\infty, 0]$, and write the Laurent coefficients*

$$\gamma_{ij} \;=\; \frac{1}{(\mu_i - j)!}\,\frac{d^{\mu_i - j}}{d\alpha^{\mu_i - j}}\Big\{(\alpha + r_i)^{\mu_i} R(\alpha)\Big\}\Big|_{\alpha = -r_i},$$

$$R(\alpha) = \Pi(\alpha) + \sum_i \sum_{j=1}^{\mu_i} \frac{\gamma_{ij}}{(\alpha + r_i)^j},$$

with Π the polynomial part. If $\deg Q \geq \deg P + 2$ then $\Pi \equiv 0$, $\sum_i \gamma_{i1} = 0$, and

$$\int_0^\infty R(\alpha)\,\mathrm{d}\alpha \;=\; -\sum_{i=1}^{J} \gamma_{i1} \log r_i + \sum_{i=1}^{J} \sum_{j=2}^{\mu_i} \frac{\gamma_{ij}}{(j-1) r_i^{j-1}}, \tag{9.3}$$

with the principal logarithm. If R is real the right side is real, a conjugate pair $r, \bar{r}$ contributing $-2\,\mathrm{Re}\{\gamma \log r\}$. If $\deg Q < \deg P + 2$ the integral diverges, but (9.3) *remains valid after tilting, by Theorem 9.16.*

The identity $\sum_i \gamma_{i1} = 0$ is the statement that R decays faster than α^{-1}, and it is what cancels the divergent logarithms. It is also a free numerical diagnostic, and the first of a whole ladder.

For the Pochhammer kernel itself the poles are always simple, since $[c\alpha + a]^b = \prod_{i=1}^{b} (c\alpha + a + i - 1)$ has the distinct roots $-r_i$, $r_i = (a + i - 1)/c$. We fix once and for all the convention, used in every result below, that when $b \geq m + d + 1$

$$\frac{\alpha^d [\alpha]^m}{[c\alpha + a]^b} \;=\; \sum_{i=1}^{b} \frac{\gamma_i}{c\alpha + a + i - 1} \;=\; \sum_{i=1}^{b} \frac{\gamma_i / c}{\alpha + r_i}, \qquad \gamma_i = \frac{(-1)^{i-1} (-r_i)^d [-r_i]^m}{(i-1)!\,(b-i)!}, \tag{9.4}$$

so that γ_i is the coefficient of the linear factor $c\alpha + a + i - 1$ and the residue in the sense of Theorem 9.3 is $\gamma_{i1} = \gamma_i / c$. (The formula for γ_i follows by multiplying through by $c\alpha + a + i - 1$ and setting $\alpha = -r_i$, since $c(-r_i) + a + l - 1 = l - i$ and $\prod_{l \neq i} (l - i) = (-1)^{i-1} (i-1)!\,(b-i)!$.) Where $b \leq m + d$ the expansion acquires a polynomial part Π, which we add explicitly.

LEMMA 9.4 (Vanishing moments of the residue measure). *Let $R = \alpha^d [\alpha]^m / [c\alpha + a]^b$ with $b \geq m + d + 1$, poles $-r_i$ and coefficients γ_i as in* (9.4). *Then*

$$\sum_{i=1}^{b} \gamma_i r_i^k = 0 \quad \textit{for } 0 \leq k \leq b - m - d - 2,$$

$$\sum_{i=1}^{b} \gamma_i r_i^{-k} = 0 \quad \textit{for } 1 \leq k \leq d + \mathbf{1}(m \geq 1).$$

The first family comes from expanding R at infinity, where $R = O(\alpha^{m+d-b})$; the second from expanding at the origin, where $R = O(\alpha^{d+\mathbf{1}(m\geq 1)})$, the indicator appearing because $[\alpha]^m$ contributes exactly one factor of α when $m \geq 1$. The lemma is proved in the Proofs section at the end of the chapter and is used twice below: to determine the exact strip of the Mellin transform, and as a graded accuracy check on implementations, since $|\sum_i \gamma_i r_i^k| / \max_i |\gamma_i r_i^k|$ reports the achieved precision directly.

EXAMPLE 9.5 (Double poles in the Dirichlet process posterior). Combine the Ewens likelihood $\alpha^k/[\alpha]^n$ with the half horseshoe prior $\mathrm{PH}(0,1,2,1) \propto \{(\alpha+1)(\alpha+2)\}^{-1}$. After cancelling one factor of α,

$$p(\alpha \mid n,k) \;\propto\; \frac{\alpha^{k-1}}{(\alpha+1)^2(\alpha+2)^2(\alpha+3)\cdots(\alpha+n-1)},$$

of numerator degree $k-1$ and denominator degree $n+1$, so properness holds for every $k \leq n$. The poles at -1 and -2 are double and (9.3) is required. For $n = 20$, $k = 5$ residues and quadrature agree at $1.5311352489324 \times 10^{-19}$ to fourteen digits.

9.2.3 *Mellin transform, fractional and negative moments*

Each simple pole has an elementary Mellin transform, and the partial fraction expansion diagonalizes the density into simple poles. Lemma 9.4 then determines exactly how far the resulting identity continues.

THEOREM 9.6 (Mellin transform). *Let $R(\alpha) = \alpha^d[\alpha]^m/[c\alpha+a]^b$ with $b \geq m+d+1$, poles $-r_i$ and coefficients γ_i as in* (9.4)*. On the maximal strip*

$$-\{d+\mathbf{1}(m\geq 1)\} \;<\; \mathrm{Re}(s) \;<\; b-m-d,$$

$$\mathcal{M}(s) \;=\; \int_0^\infty \alpha^{s-1}R(\alpha)\,\mathrm{d}\alpha \;=\; \frac{\pi}{c\sin(\pi s)}\sum_{i=1}^{b}\gamma_i\, r_i^{s-1}. \tag{9.5}$$

The apparent poles of $\pi/\sin(\pi s)$ inside the strip are removable, cancelled term by term by Lemma 9.4. In particular $s = 1$ gives the normalizing constant $C = \mathcal{M}(1) = -c^{-1}\sum_i \gamma_i \log r_i$, and

$$\mathbb{E}(\alpha^{s-1}) = \frac{\mathcal{M}(s)}{\mathcal{M}(1)}, \qquad \mathbb{E}(\log\alpha) = \frac{\mathcal{M}'(1)}{\mathcal{M}(1)} = \frac{\sum_i \gamma_i \log^2 r_i}{2\sum_i \gamma_i \log r_i}.$$

The theorem needs $b \geq m+d+2$ for the strip to contain $s = 1$, which is exactly the condition for R to be integrable. As a check on the sign of the last formula, take $R = \{(\alpha+1)(\alpha+2)(\alpha+3)\}^{-1}$, so $c = 1$, $(\gamma_1,\gamma_2,\gamma_3) = (\tfrac12, -1, \tfrac12)$ and $r_i = i$. Then $C = \log 2 - \tfrac12\log 3 = 0.143841$ and $\mathbb{E}(\log\alpha) = \{\tfrac12\log^2 3 - \log^2 2\}/\{2(\tfrac12\log 3 - \log 2)\} = -0.427630$, in agreement with quadrature, so the geometric mean of this law is $e^{-0.427630} = 0.652$.

Three consequences. First, the earlier moment formula $\mathbb{E}(\alpha^k) = C_{(m,a,b,c,d+k)}/C_{(m,a,b,c,d)}$ requires k to be a nonnegative integer, because it is

obtained by size biasing inside the family; (9.5) gives $\mathbb{E}(\alpha^{1/2})$, $\mathbb{E}(\alpha^{-1/2})$ and $\mathbb{E}(\log\alpha)$ directly, the last being the quantity required for entropies, Kullback Leibler divergences and the score of the Gamma shape. Second, $\mathbb{E}(\alpha^{-\nu})$ is finite exactly for $\nu < 1 + d + \mathbf{1}(m \geq 1)$. The half horseshoe $\mathrm{PH}(0,1,2,1)$, whose density is positive at the origin, therefore has $\mathbb{E}(\alpha^{-\nu}) < \infty$ for $\nu < 1$ but $\mathbb{E}(\alpha^{-1}) = \infty$, and this, rather than any statement about integer moments, is the correct description of its behaviour at the origin. Third, the pole structure of $\mathcal{M}$ reads off both tails, and the rightmost pole gives the following.

COROLLARY 9.7 (Exact tail index). *Let $\alpha \sim \mathrm{PPH}(m,a,b,c,d)$. Then*

$$\mathbb{P}(\alpha > x) \;\sim\; \frac{x^{-(b-m-d-1)}}{C\,c^{b}\,(b-m-d-1)}, \qquad x \to \infty,$$

so the law is regularly varying with index $b-m-d-1$ and has exactly $b-m-d-2$ finite moments. The half horseshoe $m=0$, $b=2$, $d=0$ has index one and no moments. Proposition 1 of Wang and Polson (2024), that $e^{tx}\mathbb{P}(\alpha > x) \to \infty$ for every $t > 0$, is the weaker statement that regular variation excludes exponential tails.

COROLLARY 9.8 (Shrinkage rate, corrected). *For $\mathrm{PH}(m,a,b,c)$ let $\nu = 1 + \mathbf{1}(m \geq 1)$. Then $\int_0^\delta p \asymp \delta^\nu$ as $\delta \downarrow 0$. The Cesàro risk bound of Polson and Scott (2011), $R_N \leq \varepsilon - N^{-1}\log\int_0^{\sqrt{\varepsilon}} p$, therefore reads*

$$R_N \;\leq\; \varepsilon + \frac{\nu}{2N}\log\frac{1}{\varepsilon} + O(N^{-1}),$$

minimized at $\varepsilon = \nu/(2N)$ to give $R_N = \frac{\nu}{2}N^{-1}\log N\{1+o(1)\}$. Hence every *$m$ attains the Kullback Leibler superefficient rate $N^{-1}\log N$; what $m=0$ buys is the smaller constant, $1/2$ against 1 for every $m \geq 1$. For the specific half horseshoe $\mathrm{PH}(0,1,2,1)$, $\int_0^{\sqrt{\varepsilon}} p = (\log 2)^{-1}\log\{1+\sqrt{\varepsilon}/(2+\sqrt{\varepsilon})\} \sim \sqrt{\varepsilon}/(2\log 2)$.*

The exponent is ν and not $m+1$ because $[\alpha]^m = \alpha(\alpha+1)\cdots(\alpha+m-1)$ vanishes only to first order at the origin, whatever $m \geq 1$; numerically, $\log_{10}\{\int_0^{10^{-4}} p / \int_0^{10^{-5}} p\}$ equals 1.0000 for $m=0$ and 2.0000 for $m = 1,2,3$ (with $a=c=1$, $b=m+2$). This corrects the reading of Proposition 2 in Wang and Polson (2024). The case for $m=0$ is real but rests on the constant and on the sparsity behaviour of Section 9.5, not on the rate.

9.2.4 Pochhammer priors are scale mixtures of exponentials

The horseshoe owes its practicality to a representation: a half Cauchy scale is an inverse Gamma mixture, so conditionally on the scale everything is Gaussian and Gibbs sampling is routine. The Pochhammer family has an exact counterpart, and it identifies $m = d = 0$ as the distinguished member on structural rather than asymptotic grounds.

THEOREM 9.9 (Conditional exponentiality). *Let $R(\alpha) = \alpha^d[\alpha]^m/[c\alpha+a]^b$ with $b \geq m+d+2$, poles $-r_i$, $r_i = (a+i-1)/c$, and coefficients γ_i as in* (9.4)*, so that the residue at $-r_i$ is γ_i/c. Then R is completely monotone on $\mathbb{R}^+$ if and only if $m = d = 0$. In that case*

$$R(\alpha) \;=\; \int_0^\infty e^{-\alpha\tau} g(\tau)\,\mathrm{d}\tau, \qquad g(\tau) = \frac{1}{c}\sum_{i=1}^{b} \gamma_i e^{-r_i\tau} \;\geq\; 0, \tag{9.6}$$

where $g = f_T/(c^b \prod_i r_i)$ and f_T is the density of $T = E_1 + \cdots + E_b$ with E_i independent and $E_i \sim \mathrm{Exp}(r_i)$. Equivalently,

$$\alpha \mid \tau \;\sim\; \mathrm{Exp}(\tau), \qquad p(\tau) \;=\; C^{-1}\,\frac{g(\tau)}{\tau}, \tag{9.7}$$

and the normalizing constant is Frullani's integral, $C = \int_0^\infty \tau^{-1} g(\tau)\,\mathrm{d}\tau = -c^{-1}\sum_i \gamma_i \log r_i$, the same expression as in Theorem 9.3.

Proof. If $m = d = 0$ then R is a product of the completely monotone functions $(c\alpha + a + i - 1)^{-1}$, and products of completely monotone functions are completely monotone. If $m \geq 1$ or $d \geq 1$ then $R(0) = 0$ while $R > 0$ on $(0,\infty)$, so R is not nonincreasing and cannot be completely monotone. Given $m = d = 0$, (9.6) is the term by term Laplace inversion of the partial fraction expansion, and $g \geq 0$ because $\prod_i r_i/(\alpha + r_i)$ is the Laplace transform of the hypoexponential law of T. Bernstein's theorem identifies the mixing measure uniquely. For (9.7), write $e^{-\alpha\tau} = \tau^{-1}\cdot\tau e^{-\alpha\tau}$ and read $\tau e^{-\alpha\tau}$ as the $\mathrm{Exp}(\tau)$ density. Integrability of $g(\tau)/\tau$ at the origin is Lemma 9.4, and Frullani's identity $\int_0^\infty \tau^{-1}\sum_i \gamma_i e^{-r_i\tau}\,\mathrm{d}\tau = -\sum_i \gamma_i \log r_i$ is valid precisely because $\sum_i \gamma_i = 0$. □

Lemma 9.4 thus acquires a probabilistic reading. Its identities $\sum_i \gamma_i r_i^k = 0$ for $k \leq b-2$ say exactly that the hypoexponential density g vanishes to order $b-1$ at the origin, which is what makes the size debiased law $g(\tau)/\tau$ proper. The vanishing moments of the residue measure, the finiteness of the normalizing constant and the smoothness of a sum of b exponentials at zero are one fact viewed three ways.

COROLLARY 9.10 (Shape and tail, second proof). *For $m = d = 0$ the density $\mathrm{PH}(0,a,b,c)$ is strictly decreasing and log convex on $\mathbb{R}^+$, with its maximum at the origin. Moreover $\mathbb{P}(\alpha > x) = \mathbb{E}\{e^{-\tau x}\}$, and since $p(\tau) \propto \tau^{b-2}$ as $\tau \downarrow 0$, Karamata's Tauberian theorem gives $\mathbb{P}(\alpha > x) \asymp x^{-(b-1)}$, recovering Corollary 9.7 at $m = d = 0$ without any residue computation.*

COROLLARY 9.11 (Infinite divisibility). *For $m = d = 0$ the law $\mathrm{PH}(0,a,b,c)$ is infinitely divisible and belongs to the Goldie, Steutel and Bondesson class $\mathcal{B}(\mathbb{R}^+)$, the smallest class of infinitely divisible laws on the half line closed under convolution and weak limits that contains all mixtures of exponentials.*

This is immediate from Theorem 9.9 and the Goldie and Steutel theorem, that every law on $(0,\infty)$ with completely monotone density is infinitely divisible

(Goldie, 1967; Steutel, 1967). For $m \geq 1$ or $d \geq 1$ the argument does not apply, the density not being completely monotone.

Whether $\mathrm{PH}(0, a, b, c)$ lies in the smaller Thorin class of generalized gamma convolutions, which would follow if the mixing density $g(\tau)/\tau$ were hyperbolically completely monotone in the sense of Bondesson (1992), we leave open. The question is not idle: membership would supply a Wiener gamma integral representation and hence a second route to simulation.

Theorem 9.9 answers a question Corollary 9.8 could not. The Kullback Leibler rate does not distinguish m, and the constant $\nu/2$, which is $1/2$ for $m = 0$ against 1 for $m \geq 1$, is a weak argument. Complete monotonicity is not weak: only $m = d = 0$ makes the prior conditionally exponential, and conditional exponentiality is what turns the prior into a tilt. The practical consequence is Proposition 9.12.

PROPOSITION 9.12 (Plug in replacement for Gamma priors). *Under* (9.7) *the pair* (α, τ) *has joint density proportional to* $e^{-\alpha\tau} g(\tau)$, *so*

$$\alpha \mid \tau \sim \mathrm{Exp}(\tau), \qquad \tau \mid \alpha \sim \mathrm{Exp}(r_1 + \alpha) * \cdots * \mathrm{Exp}(r_b + \alpha),$$

the second because tilting by $e^{-\alpha\tau}$ *shifts every rate of the convolution by* α. *Consequently, if* $L \in \mathcal{R}$ *is a likelihood for which a* $\mathrm{Ga}(\nu, \lambda)$ *prior on* α *admits an exact conditional draw, possibly after augmentation, then* $\mathrm{PH}(0, a, b, c)$ *admits one too: adjoin* τ, *replace the* $\mathrm{Ga}(\nu, \lambda)$ *prior by* $\mathrm{Exp}(\tau)$, *that is* $\nu = 1$ *and* $\lambda = \tau$, *and draw* τ *as a sum of* b *independent exponentials.*

The τ step needs no special functions, no inversion and no residues, and its cost is b exponential variates. The marginal law of τ is the size debiased hypoexponential $g(\tau)/\tau$; its conditional law given α is a plain hypoexponential. The contrast with the half Cauchy is instructive: there the latent scale enters a Gaussian precision, here it enters an exponential rate, and in both cases the heavy tail is bought by letting the latent scale approach zero.

The scope is wide. Escobar and West (1995) obtain a Gamma mixture draw for the Dirichlet process concentration after a Beta auxiliary variable; Zhou and Carin (2015) obtain a Gamma draw for the Negative Binomial dispersion after the Chinese restaurant table augmentation. In both, and in every descendant, a light tailed Gamma prior may now be replaced by a heavy tailed half horseshoe at the cost of one extra univariate draw, with no Metropolis step and no change to the rest of the sampler. Section 9.5 gives the details for Negative Binomial regression, where the conditioning problem of Proposition 9.13 disappears entirely: the residue expansion is used only for the two pole prior, never for the N pole posterior.

9.2.5 Conditioning of the residue expansion

Closed form is not the same as computable. The coefficients γ_{ij} alternate in sign and are enormous relative to their sum, which is precisely the content of Lemma 9.4. A sum $\sum_i x_i$ evaluated in floating point with unit roundoff u carries an error of

order $u \max_i |x_i|$, so the natural condition number of the residue formula is

$$\kappa \;=\; \frac{\max_{i,j} |\gamma_{ij}|}{|\int_0^\infty R|},$$

and about $\log_{10} \kappa$ decimal digits are lost. (The logarithms $\log r_i$ are of moderate size and do not change the order of magnitude.) We write κ_n when the kernel comes from a sample of size n.

PROPOSITION 9.13 (Precision requirement). *Let $R(\alpha) = \alpha^{k-1} / \prod_{i=1}^{n-1} (\alpha + i)$ be the Ewens kernel underlying Example 9.5, the two poles contributed by the prior being omitted since they do not affect the growth rate. This is* $\mathrm{PPH}(0, 1, n-1, 1, k-1)$, *and by* (9.4) *its residues are $\gamma_j = (-1)^{j-1}(-j)^{k-1}/\{(j-1)!\,(n-1-j)!\}$, $j = 1, \dots, n-1$.*

(i) For $k = 1$ the growth is exactly solvable: $\max_j |\gamma_j| = \binom{n-2}{\lfloor (n-2)/2 \rfloor} / (n-2)!$ and $\int_0^\infty R \sim \{(n-1)! \log n\}^{-1}$, whence

$$\kappa_n \;=\; 2^{n-2}(n-1)\sqrt{\frac{2}{\pi n}}\, \log n \,\{1 + O(1/\log n)\},$$
$$\log_{10} \kappa_n \;=\; 0.301\, n + O(\log n).$$

(ii) For $k = \rho n$ with $\rho \in (0,1)$ fixed, set $\Phi(x) = \rho \log x - x \log x - (1-x)\log(1-x) + 1$ and let $x^* \in (0,1)$ be the unique root of $\rho/x = \log\{x/(1-x)\}$. Then the largest residue satisfies

$$\log \max_j |\gamma_j| \;=\; (\rho - 1)\, n \log n \;+\; \Phi(x^*)\, n \;+\; O(\log n),$$

the maximum being attained at $j \sim x^* n$. For $\rho = 1/8$, $x^* = 0.5560$ and $\Phi(x^*) = 1.6135$, and the formula matches the exact maximum to within 0.03% at $n = 1600$.

In either regime double precision is exhausted by n of a few tens, and D correct digits require $\Theta(n) + D$ digits of working precision.

Both parts are proved in the Proofs section at the end of the chapter. Part (ii) describes the numerator only. Since the observed growth of the full ratio at $\rho = 1/8$ is $\log_{10} \kappa_n = 0.502\, n - 2.30$ over $10 \le n \le 160$ (the Ewens kernel with $k = \lceil n/8 \rceil$ under $\mathrm{PH}(0, 1, 2, 1)$), with $R^2 > 0.999$ and no visible curvature, the $(\rho - 1) n \log n$ term in (ii) must cancel against an identical term in $\log \int_0^\infty R$, leaving growth exactly exponential in n rather than superexponential. We have not proved that cancellation and flag it as the open half of the problem: what is needed is a Laplace analysis of $\int_0^\infty \alpha^{k-1} \Gamma(\alpha+1) \Gamma(\alpha+n)^{-1}\, \mathrm{d}\alpha$ in the regime where the mode sits at $\alpha \asymp \rho n / \log n$, so that both arguments of the Gamma ratio are large. The constant 0.502 is in any case not universal, depending as it does on ρ.

Three remedies. (i) Exact arithmetic: for rational a, c the coefficients are rational and (9.3) is exact in $O(N^2)$ rational operations, at the cost of coefficient swell. (ii) Adaptive precision. The largest coefficient is available in logarithms before

any cancellation occurs, and a crude estimate of the integral (from a Laplace approximation, say) is good enough to fix the order of magnitude of κ_n; one then works with $\lceil \log_{10} \kappa_n \rceil + D$ digits. For the Ewens kernel with k/n near $1/8$ the simpler rule of $\lceil N/2 \rceil + D$ digits suffices for n below about 400 and is what we use in Section 9.8, but it is not universal: with $n = 3306$ and $k = 501$, so k/n near 0.15, the half horseshoe kernel needs about $0.54N$ digits, while Example 9.29, with k/n near 0.056, needs about $0.42N$. (iii) The Stirling dual of Section 9.4.3, whose terms are positive. Crucially, the ill conditioning afflicts only the normalizer and moments, not sampling: the unnormalized kernel is evaluated stably in logarithms, so Metropolis and slice steps are unaffected. Figure 9.1 displays the growth.

EXAMPLE 9.14 (Digits lost in double precision). Take the Ewens likelihood with $k = 3$ alleles and the prior $\mathrm{PH}(0, \tfrac{1}{2}, 2, 1) \propto \{(\theta + \tfrac{1}{2})(\theta + \tfrac{3}{2})\}^{-1}$, chosen so that every pole of

$$R(\theta) \;=\; \frac{\theta^2}{(\theta + \tfrac{1}{2})(\theta + \tfrac{3}{2}) \prod_{i=1}^{n-1} (\theta + i)}$$

is simple and the residues are given by a one line formula. We evaluate $C = -\sum_i \gamma_i \log r_i$ twice, once in IEEE double precision (unit roundoff 1.1×10^{-16}) and once with 120 significant digits, and confirm the latter by adaptive quadrature. Table 9.3 records the outcome. The number of correct digits in double precision is close to $16 - \log_{10} \kappa_n$ throughout, and each additional five genes costs about one and a half digits. At $n = 40$ the double precision answer, 8.38×10^{-49}, is wrong in the first significant figure (the true value is 7.572×10^{-49}), although every individual residue was computed to full accuracy and the identity $\sum_i \gamma_i = 0$ holds to 4×10^{-50}, which is comparable to the error in the answer. The lesson is that the diagnostic of Lemma 9.4 must be read relative to $\max_i |\gamma_i|$, as the implementation notes at the end of the chapter prescribe, and not relative to one.

n	C (120 digits)	$\log_{10} \kappa_n$	relative error in double	correct digits
10	$1.38817920005 \times 10^{-7}$	4.30	8.4×10^{-13}	12.1
15	$3.79758380833 \times 10^{-13}$	6.00	2.7×10^{-10}	9.6
20	$2.11300775768 \times 10^{-19}$	7.65	2.2×10^{-8}	7.7
25	$3.46954782558 \times 10^{-26}$	9.25	1.9×10^{-6}	5.7
30	$2.12766470822 \times 10^{-33}$	10.85	6.6×10^{-5}	4.2
35	$5.72071973778 \times 10^{-41}$	12.42	1.9×10^{-3}	2.7
40	$7.57232314072 \times 10^{-49}$	14.00	1.1×10^{-1}	1.0

TABLE 9.3. The residue formula for the Ewens posterior with $k = 3$ under $\mathrm{PH}(0, \tfrac{1}{2}, 2, 1)$, evaluated in double precision. Correct digits are $-\log_{10}$ of the relative error.

9.3 Extending the Family

9.3.1 Exponential tilting

Several likelihoods contribute a factor z^α, $z \in (0,1]$: the Negative Binomial contributes π^α, Gamma sampling contributes $(\prod_i x_i)^\alpha$, and log linear links contribute $e^{-\lambda\alpha}$. Such a factor is not rational, but it is exactly integrable against a simple pole, and it strictly enlarges what the family can do.

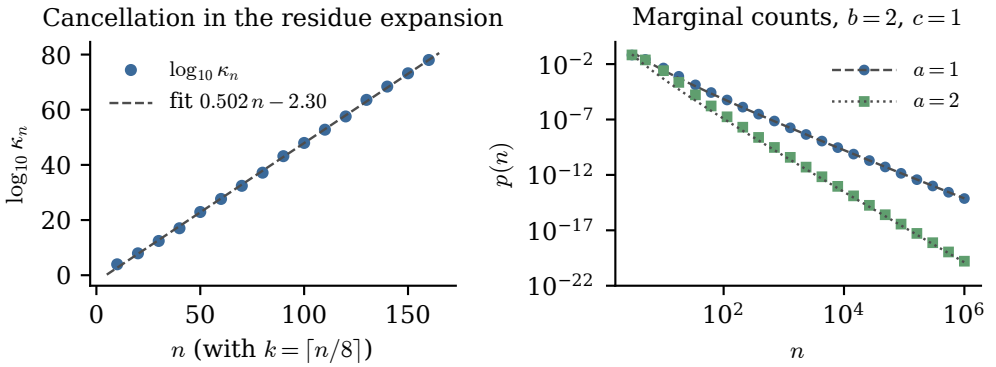


FIGURE 9.1. Left: $\log_{10}\kappa_n$ for the Dirichlet process posterior, the Ewens kernel with $k = \lceil n/8 \rceil$ under PH$(0,1,2,1)$, with its least squares line of slope 0.502 (dashed). Right: marginal counts of the Negative Binomial, Beta, Pochhammer hierarchy, $b = 2$, $c = 1$ (points, quadrature) against the asymptote of Theorem 9.20 (dashed for $a = 1$, dotted for $a = 2$).

DEFINITION 9.15. $\alpha \sim \mathrm{TPH}(m,a,b,c,d,z)$ if $p(\alpha) \propto z^\alpha \alpha^d [\alpha]^m / [c\alpha + a]^b$ on $\alpha \geq 0$, $0 < z \leq 1$. More generally, for a polynomial P that is nonnegative on $[0,\infty)$ we write $\alpha \sim \mathrm{TPH}(P;a,b,c,z)$ if $p(\alpha) \propto z^\alpha P(\alpha)/[c\alpha + a]^b$, so that $\mathrm{TPH}(m,a,b,c,d,z) = \mathrm{TPH}(\alpha^d[\alpha]^m;a,b,c,z)$.

The general form is needed because updating by a Negative Binomial sample multiplies the numerator by $\prod_i [\alpha]^{n_i}$, which is a polynomial of degree $N = \sum_i n_i$ but is not $[\alpha]^N$.

THEOREM 9.16 (Tilted normalizer). *Let $\lambda = -\log z > 0$ and let P be a polynomial. Write the partial fraction expansion*

$$\frac{P(\alpha)}{[c\alpha + a]^b} \;=\; \Pi(\alpha) + \sum_{i=1}^{b} \frac{\gamma_i}{c\alpha + a + i - 1}, \qquad \Pi(\alpha) = \sum_k \pi_k \alpha^k,$$

with the convention of (9.4) *and poles $-r_i$, $r_i = (a+i-1)/c$. Then*

$$\int_0^\infty z^\alpha \frac{P(\alpha)}{[c\alpha + a]^b}\,\mathrm{d}\alpha \;=\; \sum_k \frac{k!\,\pi_k}{\lambda^{k+1}} \;+\; \frac{1}{c}\sum_{i=1}^{b} \gamma_i e^{\lambda r_i} E_1(\lambda r_i), \tag{9.8}$$

where $E_1(x) = \int_x^\infty t^{-1}e^{-t}\,\mathrm{d}t$. *No degree condition is required:* (9.8) *holds for every* $b \geq 0$, *and all moments exist, obtained by replacing* P *with* $\alpha^k P$. *As* $\lambda \downarrow 0$, *when* $\deg P \leq b-2$ *so that* $\Pi \equiv 0$ *and* $\sum_i \gamma_i = 0$, (9.8) *converges to the untilted normalizer* $-c^{-1}\sum_i \gamma_i \log r_i$ *of Theorems 9.3 and 9.6 via* $e^x E_1(x) = -\log x - \gamma_{\mathrm{E}} + O(x \log x)$.

The intuition is that the tilt $e^{-\lambda\alpha}$ turns each simple pole into an exponential integral, exactly as Laplace transforming $1/(\alpha + r)$ does, and turns each power α^k of the polynomial part into a Gamma integral; nothing else can appear.

The removal of the degree condition matters. In the untilted family, conjugate updating by $\prod_i [\alpha]^{n_i}$ requires $b \geq m + N + 2$, so a prior hyperparameter must grow with the total count, which is both awkward to elicit and, by Proposition 9.13, computationally punishing. With a tilt present, however small, b may be held fixed at the default 2. The scaled function $e^x E_1(x)$ is in every standard library and is stable for all $x > 0$; (9.8) inherits the conditioning of the untilted case and nothing worse.

COROLLARY 9.17 (Tilted distribution function). *For* $\alpha \sim \mathrm{TPH}(P; a, b, c, z)$, *with the notation of Theorem 9.16 and* Z *the right side of* (9.8),

$$\mathbb{P}(\alpha \leq A) \;=\; \frac{1}{Z}\Big[\sum_k \frac{\pi_k\,\gamma_{\mathrm{inc}}(k+1, \lambda A)}{\lambda^{k+1}} + \frac{1}{c}\sum_i \gamma_i e^{\lambda r_i}\big\{E_1(\lambda r_i) - E_1(\lambda(r_i + A))\big\}\Big],$$

where $\gamma_{\mathrm{inc}}(s, x) = \int_0^x t^{s-1}e^{-t}\,\mathrm{d}t$ *is the lower incomplete Gamma function. This is a monotone function of* A *computable at the same cost as the normalizer, so exact draws follow by bisection on the uniform scale. When* $\Pi \equiv 0$ *the first sum is absent.*

9.3.2 *Global local Pochhammer priors*

The half horseshoe is purely local: each α_k gets an independent $\mathrm{PH}(0, a, 2, c)$. The horseshoe of Carvalho et al. (2010) owes much of its performance to the global scale. We therefore set

$$\alpha_k \mid \tau \;\sim\; \mathrm{PH}(0, a, 2, c/\tau) \text{ independently}, \qquad \tau \sim p(\tau),$$

so α_k/τ has the standard shape. Since c enters only through $r_i = (a + i - 1)\tau/c$, the conditional of τ given α is rational whenever $p(\tau)$ is, and each α_k conditional remains in $\mathcal{P}$. In the Dirichlet Multinomial the global scale plays the role of the total mass $A = \sum_k \alpha_k$, which is exactly the quantity the heterogeneous Gibbs conditional $p(\alpha_k \mid n_k, N, A_{-k}) \propto [\alpha_k]^{n_k}[\alpha_k]^m / \{[\alpha_k + A_{-k}]^N [c\alpha_k + a]^b\}$ couples through. Placing a prior on τ therefore regularizes the coupling directly rather than through the data alone.

9.3.3 *Product form*

Definition 9.1 already permits several denominators, and Theorem 9.3 applies verbatim to $p(\alpha) \propto \alpha^d \prod_j [\alpha + v_j]^{p_j} / \prod_l [c_l\alpha + a_l]^{q_l}$. This matters because, by Proposition 9.2(iii), a posterior after several rounds of updating is of product

form even when the prior was not. Sequential updating is therefore closed but not stationary in the hyperparameters, and the pole set is the right bookkeeping device.

9.4 Gamma Shapes and Damsleth's Class

9.4.1 The class and the obstruction

Let $x_1, \dots, x_n$ be independent $\mathrm{Ga}(\alpha, \beta)$. The likelihood is $(\beta^\alpha/\Gamma(\alpha))^n (\prod_i x_i)^{\alpha-1} e^{-\beta \sum_i x_i}$, with sufficient statistic $(n, \sum_i x_i, \prod_i x_i)$. Damsleth (1975) studies the natural conjugate class

$$p(\alpha, \beta \mid \nu, \rho, \eta) \;\propto\; \Big(\frac{\beta^\alpha}{\Gamma(\alpha)}\Big)^\nu \rho^\alpha e^{-\eta\beta} \beta^{-1}, \tag{9.9}$$

which updates by $(\nu, \rho, \eta) \mapsto (\nu + n, \rho \prod_i x_i, \eta + \sum_i x_i)$ and has $\beta \mid \alpha \sim \mathrm{Ga}(\nu\alpha, \eta)$, so only the marginal

$$p(\alpha) \;\propto\; \frac{\Gamma(\nu\alpha)}{\Gamma(\alpha)^\nu} \cdot \frac{\rho^\alpha}{\eta^{\nu\alpha}} \tag{9.10}$$

is problematic. It is the reference point for the modern literature, including the quasiconjugate transfer to generalized Pareto tails and the Gamma approximation of Miller (2019).

PROPOSITION 9.18 (The rational boundary). *$\Gamma(\alpha + u)/\Gamma(\alpha + v)$ is a rational function of α if and only if $u - v \in \mathbb{Z}$.*

Proof. If $u - v = k \geq 0$ the ratio is $[\alpha + v]^k$, a polynomial, and for $k < 0$ a reciprocal polynomial. Conversely the ratio has poles at $\alpha = -u - j$, $j \geq 0$, and zeros at $\alpha = -v - j$. These sets are disjoint when $u - v \notin \mathbb{Z}$, so the ratio has infinitely many uncancelled poles, whereas a rational function has finitely many. □

Hence a likelihood lies in $\mathcal{R}$ if and only if its Gamma functions appear in ratios whose arguments differ by integers *and* share the same affine function of α. The Damsleth marginal (9.10) fails on both counts: $\Gamma(\nu\alpha)$ and $\Gamma(\alpha)^\nu$ involve different affine maps, and $\Gamma(\alpha)^{-\nu}$ is not a rational function of α, so it admits no finite partial fraction expansion. The tilt $\rho^\alpha \eta^{-\nu\alpha}$, by contrast, is harmless: it is exactly the z^α of Theorem 9.16. The sole obstruction is $\Gamma(\nu\alpha)/\Gamma(\alpha)^\nu$.

9.4.2 Exact repairs

Lattice shapes. If $\alpha \in \{\alpha_0 + j : j \geq 0\}$, as in Erlang and phase type models, then $\Gamma(\alpha)/\Gamma(\alpha_0) = [\alpha_0]^j$ and the likelihood is Pochhammer rational in j. A Pochhammer prior on j gives exact conjugate inference for the number of phases, a quantity of direct interest in queueing and reliability.

Poisson augmentation. The sharpest repair replaces continuous observations by counts. If $\lambda_i \sim \mathrm{Ga}(\alpha, \beta)$ and $n_i \mid \lambda_i \sim \mathrm{Poi}(\lambda_i)$ then $n_i \sim \mathrm{NB}(\alpha, \pi)$, $\pi = \beta/(1+\beta)$, and the Gamma functions appear as $\Gamma(\alpha + n_i)/\Gamma(\alpha)$, with integer shift and common argument.

PROPOSITION 9.19 (Exact conjugacy for the Gamma shape). *In the Gamma Poisson model with $\pi \mid \alpha \sim \mathrm{Be}(c\alpha + a, b)$ and $\alpha \sim \mathrm{PH}(m, a, b, c)$,*

$$p(\alpha \mid \boldsymbol{n}) \;\propto\; \frac{\prod_k [\alpha]^{n_k} [\alpha]^m}{[(c+K)\alpha + a]^{N+b}} \;\in\; \mathcal{P},$$

so all normalizers, moments and predictives follow from Theorem 9.3. If instead $\pi \sim \mathrm{Be}(a_0, b_0)$ independently of α, then conditionally on π the posterior of α is $\mathrm{TPH}(P_{\boldsymbol{n}}; a, b, c, \pi^K)$ with $P_{\boldsymbol{n}}(\alpha) = [\alpha]^m \prod_k [\alpha]^{n_k}$, exactly, and $\pi \mid \alpha, \boldsymbol{n} \sim \mathrm{Be}(K\alpha + a_0, N + b_0)$, giving a two block Gibbs sampler with both blocks drawn exactly.

The interpretation is clean: the Gamma shape is learnable in closed form exactly when the information about it arrives through counts. That is not restrictive in the applications motivating Damsleth's class, namely overdispersion, credibility and degradation, where the shape is identified through discrete events.

Tilted approximation. When continuous Gamma data must be used, Miller (2019) shows the full conditional of α is well approximated by a Gamma. A $\mathrm{TPH}(0, a, b, c, d, e^{-\lambda})$ retains a heavy body and a pole free tail while matching the Gamma in the tilt; its log derivative is $d/\alpha - \lambda - c\sum_{i=1}^{b}(c\alpha + a + i - 1)^{-1}$, so (d, λ) can be matched to the first two log derivatives of (9.10) at the mode by solving two equations. This yields a proposal or variational family with exact normalizer, not an exact posterior, and we flag it as such.

Reciprocal Gamma augmentation. The three repairs above all work by altering the model until the likelihood becomes rational. A fourth leaves the model alone and augments instead. He et al. (2019, 2021) attack $1/\Gamma(\alpha)$ head on: building on the classical normal variance mean mixture representations of the reciprocal Gamma function, they construct the Pólya Inverse Gamma class, and its exponential counterpart, so that conditionally on a latent Pólya Inverse Gamma variable the shape parameter has a tractable full conditional. Their scheme requires no integer shift and no common affine argument, and so reaches exactly the transcendental factor $\Gamma(\nu\alpha)/\Gamma(\alpha)^\nu$ that Proposition 9.18 places outside $\mathcal{R}$. It applies to Damsleth's class directly.

The relation to the present chapter is complementary rather than competitive, and the division of labour is clean.

- Residues apply only when the Gamma functions appear in ratios with integer shifts of a common affine argument, but within that range they return exact normalizers, exact moments and exact marginal likelihoods with no latent variables at all.

- Pólya Inverse Gamma augmentation carries no restriction on the arguments, and so covers the cases residues cannot, but it returns a Markov chain rather than a closed form, and marginal likelihoods must then be estimated.

Where both apply, as for the Dirichlet Multinomial and Negative Binomial models treated by He et al. (2019) and here, the two are naturally used together: augmentation for estimation, residues for the marginal likelihood and for Bayes factors. Section 9.5.3 is an instance of the same pattern with a different augmentation.

9.4.3 The Chinese restaurant table dual

There is a complementary expansion. The unsigned Stirling numbers of the first kind give

$$[\alpha]^n = \sum_{\ell=1}^{n} |s(n,\ell)|\, \alpha^\ell, \tag{9.11}$$

which Zhou and Carin (2015) exploit as the Chinese restaurant table augmentation: with $\ell \mid n, \alpha \sim \mathrm{CRT}(n,\alpha)$, $p(\ell \mid n,\alpha) = |s(n,\ell)|\alpha^\ell/[\alpha]^n$, so $p(\ell, n \mid \alpha) \propto |s(n,\ell)|\alpha^\ell \pi^\alpha$ and a Gamma prior becomes conjugate.

These two devices are duals. Partial fractions expand the *denominator* into simple poles; (9.11) expands the *numerator* into monomials. Together with the reciprocal Gamma mixtures of He et al. (2019, 2021), which dissolve the Gamma function itself into a normal variance mean mixture, they make three distinct ways of rendering a Gamma ratio tractable. The tradeoffs of the first two are opposite:

- Stirling: all terms positive, perfectly conditioned, but $O(N^2)$ enormous integers and, being an augmentation, a Markov chain rather than a closed form.
- Residues: closed form, no auxiliary variables, exact moments and marginal likelihoods, but ill conditioned by Proposition 9.13.

The hybrid is obvious and we recommend it: residues for marginal likelihoods and Bayes factors, where one accurate number is wanted; CRT augmentation inside long chains, where stability over millions of iterations matters. Equating the two expansions of $\int_0^\infty [\alpha]^n \{[c\alpha + a]^b\}^{-1}\, \mathrm{d}\alpha$ gives, for $b \geq n+2$,

$$\sum_{\ell=1}^{n} |s(n,\ell)|\, \mathcal{M}_{0,a,b,c}(\ell+1) \;=\; -\frac{1}{c} \sum_{i=1}^{b} \gamma_i^{(n,a,b,c)} \log \frac{a+i-1}{c},$$

a signed sum of residues written as a positive combination of Mellin values.

9.5 The Negative Binomial Family

9.5.1 The induced count distribution is a power law

Wang and Polson (2024) couple π to α via $\pi \mid \alpha \sim \mathrm{Be}(c\alpha + a, b)$ for joint conjugacy, and display the resulting sparsity patterns numerically. The construction in fact determines the marginal law of the counts exactly, and it is a Zipf law.

THEOREM 9.20 (Zipf tails). *Let $\alpha \sim \mathrm{PH}(m, a, b, c)$ with normalized density p_α, $\pi \mid \alpha \sim \mathrm{Be}(c\alpha + a, b)$ and $n \mid \alpha, \pi \sim \mathrm{NB}(\alpha, \pi)$. Then as $n \to \infty$*

$$\begin{aligned} p(n) \;&\sim\; K\, n^{-(1+a)} (\log n)^{-(1+\iota)}, \\ K = \frac{\Gamma(a)\,\Gamma(1+\iota)\,\varpi}{B(a,b)\,c^{1+\iota}}, &\qquad \iota = 1 + \mathbf{1}(m \geq 1), \end{aligned} \tag{9.12}$$

where $\varpi = \lim_{\alpha \downarrow 0} \alpha^{-\mathbf{1}(m \geq 1)} p_\alpha(\alpha)$. For the default $m = 0$, $\iota = 1$ and the tail is $K n^{-(1+a)} (\log n)^{-2}$ with $K = \Gamma(a) p_\alpha(0) / \{B(a,b) c^2\}$; for every $m \geq 1$, $\iota = 2$ and the correction is $(\log n)^{-3}$. The indicator, rather than m itself, appears because $[\alpha]^m$ vanishes only to first order at the origin.

The proof, given at the end of the chapter, is Watson's lemma: $p(n \mid \alpha) \sim \mathcal{K}(\alpha) n^{-c\alpha - a - 1}$ with $\mathcal{K}(\alpha) \to \alpha \Gamma(a) / B(a,b)$ as $\alpha \downarrow 0$, so the integral over α concentrates on a neighbourhood of the origin of width $1/(c \log n)$, and the order of vanishing of $\mathcal{K} p_\alpha$ at the origin, namely ι, produces the power of the logarithm.

Three consequences. The tail exponent is the prior hyperparameter a alone; b and c affect only the constant, and m only the logarithmic correction. At the default $a = 1, b = 2, c = 1, m = 0$ one gets $p(n) \asymp n^{-2} (\log n)^{-2}$, so $\mathbb{E}(n) < \infty$ and $\mathbb{E}(n^2) = \infty$, the standard regime for word frequencies, degrees and claim severities, obtained without any explicit power law modelling. And the mechanism is transparent: the zero of $\Gamma(\alpha)^{-1}$ at the origin together with nondecaying prior mass near zero converts geometric decay into polynomial decay. Excess zeros and Zipf tails are two faces of the same prior mass at $\alpha = 0$. This is the count model form of the quasisparsity that motivated Datta and Dunson (2016), and it is the analytic content of Figure 3 of Wang and Polson (2024). The right panel of Figure 9.1 verifies (9.12) to $n = 10^6$; convergence to K is $O(1/\log n)$, as the argument predicts.

9.5.2 *Negative Binomial regression: Pólya Gamma meets Pochhammer*

Let $n_i \sim \mathrm{NB}(\alpha, \pi_i)$ with $\psi_i = \log\{(1 - \pi_i)/\pi_i\} = \boldsymbol{x}_i^\top \boldsymbol{\beta}$, so that

$$p(n_i \mid \alpha, \boldsymbol{\beta}) = \frac{[\alpha]^{n_i}}{n_i!} \, \frac{(e^{\psi_i})^{n_i}}{(1 + e^{\psi_i})^{n_i + \alpha}}.$$

Polson et al. (2013) give $(e^\psi)^n (1 + e^\psi)^{-(n+\alpha)} = 2^{-(n+\alpha)} e^{\kappa\psi} \mathbb{E}_\omega\{e^{-\omega\psi^2/2}\}$ with $\kappa = (n - \alpha)/2$ and $\omega \sim \mathrm{PG}(n + \alpha, 0)$, so $\boldsymbol{\beta} \mid \boldsymbol{\omega}$ is Gaussian. The obstruction has always been α, which appears both in $[\alpha]^{n_i}$ and in the Pólya Gamma *index*. He et al. (2019) resolve it by a Pólya Inverse Gamma draw for α; we resolve it differently, and without leaving the prior family. The index dependence is not rational and cannot be handled by conditioning on $\boldsymbol{\omega}$. It can be handled by not conditioning on $\boldsymbol{\omega}$ at all.

PROPOSITION 9.21 (Exact dispersion block). *Marginally in $\boldsymbol{\omega}$,*

$$p(\alpha \mid \boldsymbol{n}, \boldsymbol{\beta}) \;\propto\; \prod_{i=1}^{K} [\alpha]^{n_i} \pi_i^\alpha \, p(\alpha) \;=\; \prod_i [\alpha]^{n_i} z^\alpha p(\alpha), \qquad z = \prod_{i=1}^{K} \frac{1}{1 + e^{\psi_i}},$$

so with $p = \mathrm{PH}(m, a, b, c)$ *one has* $\alpha \mid \boldsymbol{n}, \boldsymbol{\beta} \sim \mathrm{TPH}(P_{\boldsymbol{n}}; a, b, c, z)$, $P_{\boldsymbol{n}}(\alpha) = [\alpha]^m \prod_i [\alpha]^{n_i}$, *exactly, for* any $b \geq 0$, *with* $\lambda = \sum_i \log(1 + e^{\psi_i})$. *Drawing* α *from this law before redrawing* $\boldsymbol{\omega}$ *gives a partially collapsed Gibbs sampler with the correct stationary distribution.*

Algorithm 1 records the sampler. It is blocked, free of Metropolis steps, and the ordering matters: $\boldsymbol{\omega}$ must be redrawn after α and before $\boldsymbol{\beta}$, in the manner of van Dyk and Park (2008); reversing the order targets the wrong distribution. Theorem 9.16 is what makes the α block exact, and its removal of the degree condition is what keeps b fixed at 2 rather than growing with N. For large N the residue draw may be replaced by a CRT draw, per Section 9.4.3, without changing the target.

Zero inflation needs no mixture component: when many n_i vanish the tilted posterior retains its concentration at the origin, which is Remark 4 of Wang and Polson (2024) in the presence of covariates.

ALGORITHM 1 Negative Binomial regression with Pochhammer dispersion

Input counts $\boldsymbol{n}$, design X, prior $\mathrm{PH}(m, a, b, c)$ on α, $\boldsymbol{\beta} \sim N(0, B)$
for $t = 1, \ldots, T$ do
  $\psi_i \leftarrow \boldsymbol{x}_i^\top \boldsymbol{\beta}$; $\lambda \leftarrow \sum_i \log(1 + e^{\psi_i})$
  $\alpha \sim \mathrm{TPH}(P_{\boldsymbol{n}}; a, b, c, e^{-\lambda})$ ▷ exact, by Corollary 9.17; $\boldsymbol{\omega}$ marginalized
  $\omega_i \sim \mathrm{PG}(n_i + \alpha, \psi_i)$, $i = 1, \ldots, K$
  $\boldsymbol{\beta} \sim N(\Sigma X^\top \boldsymbol{\kappa}, \Sigma)$, $\Sigma = (X^\top \Omega X + B^{-1})^{-1}$, $\kappa_i = (n_i - \alpha)/2$

ALGORITHM 2 Negative Binomial regression, half horseshoe dispersion, all blocks exact

Input counts $\boldsymbol{n}$, design X, prior $\mathrm{PH}(0, a, b, c)$ with poles $r_i = (a + i - 1)/c$, $\boldsymbol{\beta} \sim N(0, B)$
for $t = 1, \ldots, T$ do
  $\ell_i \sim \mathrm{CRT}(n_i, \alpha)$, $L \leftarrow \sum_i \ell_i$ ▷ $\ell_i = \sum_{j=1}^{n_i} \mathrm{Bern}\{\alpha/(\alpha + j - 1)\}$
  $\tau \leftarrow \sum_{i=1}^{b} E_i$ with $E_i \sim \mathrm{Exp}(r_i + \alpha)$ independent ▷ $b = 2$ at the default prior
  $\psi_i \leftarrow \boldsymbol{x}_i^\top \boldsymbol{\beta}$, $\lambda \leftarrow \sum_i \log(1 + e^{\psi_i})$
  $\alpha \sim \mathrm{Ga}(L + 1, \lambda + \tau)$
  $\omega_i \sim \mathrm{PG}(n_i + \alpha, \psi_i)$; $\boldsymbol{\beta} \sim N(\Sigma X^\top \boldsymbol{\kappa}, \Sigma)$, $\Sigma = (X^\top \Omega X + B^{-1})^{-1}$, $\kappa_i = (n_i - \alpha)/2$

9.5.3 *An exact sampler with no ill conditioning*

Algorithm 1 draws α from a tilted law whose numerator $P_{\boldsymbol{n}}$ has degree $m + N$. Its pole set does not grow, but dividing $P_{\boldsymbol{n}}$ by $[c\alpha + a]^b$ to obtain the polynomial part is a long signed recursion of the same kind as the residue sums of Proposition 9.13, and it too loses digits: for $K = 40$ counts with $N = 59$ under $\mathrm{PH}(0, \frac{1}{2}, 2, 1)$ and $\lambda = 40 \log 2$, the normalizer (9.8) is 53.8513331487, but computed with sixteen significant digits it comes out negative, and about thirty digits are needed for

twelve correct ones. Extended precision is therefore required once N is large. Theorem 9.9 removes the difficulty at the default prior. Augmenting by τ turns the prior into a tilt, and augmenting by the Chinese restaurant tables turns $\prod_i [\alpha]^{n_i}$ into a monomial, so the two devices of Section 9.4.3 combine into a sampler in which every block is an exact standard draw.

Write $\lambda = \sum_i \log(1 + e^{\psi_i})$, so that $\prod_i \pi_i^{\alpha} = e^{-\lambda\alpha}$. Conditionally on the tables ℓ_i and on τ,

$$p(\alpha \mid \boldsymbol{\ell}, \tau, \boldsymbol{\beta}) \;\propto\; \alpha^{L} e^{-(\lambda+\tau)\alpha}, \qquad L = \sum_i \ell_i,$$

which is $\mathrm{Ga}(L+1, \lambda+\tau)$. Conditionally on α, the latent scale has the closed form distribution function of Proposition 9.12. Proposition 9.21 supplies the remaining block, and Algorithm 2 is the result.

Every step is an exponential, Gamma, Bernoulli, Gaussian or Pólya Gamma draw. No residue is computed anywhere, so Proposition 9.13 is irrelevant however large N becomes, and the sampler is exact in the sense that each block is drawn from its full conditional without approximation. The price is two latent variables in place of one closed form draw, together with the usual loss of the marginal likelihood: Algorithm 1 returns $p(\boldsymbol{n})$ as a byproduct and Algorithm 2 does not. Use Algorithm 2 for estimation and Algorithm 1, at extended precision, for model comparison.

As a check we ran Algorithm 2 on $K = 40$ counts drawn from $\mathrm{NB}(1.7, \frac{1}{2})$ under $\mathrm{PH}(0, 1, 2, 1)$, with no covariates so that the exact posterior is available by quadrature. Five chains of 1.2×10^5 draws give posterior mean 1.6026 and 95% interval $(1.127, 2.152)$, against the exact 1.6031 and $(1.126, 2.154)$; the discrepancy is within two Monte Carlo standard errors of 3×10^{-4}.

9.5.4 Waring and Beta Negative Binomial

For integer r,

$$p(n \mid r, \alpha, \rho) = \frac{[r]^n}{n!} \, \frac{B(\alpha + r, \rho + n)}{B(\alpha, \rho)} = \frac{[r]^n}{n!} \cdot \frac{[\rho]^n [\alpha]^r}{[\alpha + \rho]^{n+r}},$$

Pochhammer rational in α, in ρ and in r separately. Setting $r = 1$ gives Waring, and the further choice $\rho = 1$ makes $n + 1$ Yule Simon with parameter α. Pochhammer priors on α and ρ, and a lattice Pochhammer prior on r, give a fully conjugate three parameter analysis by Gibbs. Since the tail index of the Beta Negative Binomial is α, this is a second and more direct route to a learnable power law, complementary to Theorem 9.20: there the exponent is a hyperparameter, here it is a parameter with a conjugate posterior.

9.6 Examples in Applied Probability

This section runs through the likelihoods of Table 9.2 one at a time, to show what the general theory delivers in each and where its limits lie. The Ewens and Pitman

Yor cases, which matter most for genetics, are introduced here and worked through numerically in Section 9.7.

9.6.1 Pólya urns and reinforcement

An urn holds a white and β black; a drawn ball is returned with one of the same colour. After n draws with k white, $\mathbb{P} = \binom{n}{k}[a]^k[\beta]^{n-k}/[a+\beta]^n$, the Beta Binomial law, rational in a for fixed β and conversely. With experiments (n_i, k_i),

$$p(a \mid \text{data}) \;\propto\; \frac{[a]^m \prod_i [a]^{k_i}}{[c_0 a + a_0]^{b_0} \prod_i [a+\beta]^{n_i}},$$

with poles at $-(a_0 + j - 1)/c_0$ and at $-(\beta + j)$ of multiplicity $\#\{i : n_i > j\}$, so Theorem 9.3 is needed as soon as two experiments overlap. The quantity of interest is usually the limiting white fraction $W_\infty \sim \mathrm{Be}(a, \beta)$, and $\mathbb{E}\{a/(a+\beta) \mid \text{data}\}$ is again a ratio of normalizers. This delivers uncertainty quantification for the strength of reinforcement, which the exchangeable urn literature typically fixes by fiat.

9.6.2 Ewens, cycles and the Feller coupling

Under the Ewens sampling formula (Ewens, 1972) only $\theta^k/[\theta]^n$ depends on θ, so the number of alleles k is sufficient and the likelihood is Pochhammer rational. Under the Feller coupling, $k = \sum_{i=1}^n \xi_i$ with independent $\xi_i \sim \mathrm{Bern}\{\theta/(\theta + i - 1)\}$, whence the Fisher information is

$$I_n(\theta) = \sum_{i=1}^{n} \frac{i-1}{\theta(\theta+i-1)^2} \;\sim\; \frac{\log n}{\theta},$$

so $\hat\theta_n$ is asymptotically $N(\theta, \theta/\log n)$: the effective sample size is $\log n$. Uncertainty quantification therefore matters more here than almost anywhere, and a point estimate is close to useless. It also means the likelihood is too flat to overcome a light tailed prior, so the heavy tail of Corollary 9.7 is a necessity rather than a luxury.

Two further payoffs. The one step predictive probability of a new block is $\mathbb{P}(\xi_{n+1} = 1 \mid k) = \mathbb{E}\{\theta/(\theta+n) \mid k\}$, a ratio of two normalizers in $\mathcal{P}$, since multiplying the posterior kernel by $\theta/(\theta+n)$ raises the numerator degree by one and adjoins a pole at $-n$. Sequential prediction is thus exact and simulation free, which matters in streaming settings. And the case $\theta = 1$ is the uniform random permutation, for which k is the number of cycles with $\mathbb{E}k = \log n + \gamma_{\mathrm{E}} + o(1)$, so the same formulas give exact posterior inference for the reinforcement of a Chinese restaurant process from a single observation of the block count.

9.6.3 Dirichlet process and Pitman Yor concentration

For a Dirichlet process mixture, $p(K_n = k \mid \theta) = |s(n,k)|\theta^k/[\theta]^n$, the Ewens likelihood again. Escobar and West (1995) introduce a Beta auxiliary variable so

that a Gamma prior is conditionally conjugate, at the cost of a latent variable and a two component Gamma mixture. The Pochhammer prior removes the auxiliary variable and returns the exact marginal posterior and its moments; Example 9.5 is the calculation.

The Stirling gamma distribution of Zito et al. (2026) has a similar functional form and was built for exactly this problem. The two lines of work are independent, as Wang and Polson (2024) record, and Zito et al. (2026) appeared first, in June 2023. The underlying device is older than either: the partial fraction and residue calculus applied here to Gamma ratios is the one used for the integrals in Polson et al. (2013), and Section 2.2 of Wang and Polson (2024) identifies that lineage explicitly. On the Dirichlet process itself the Stirling gamma is the better instrument, being designed for that problem, having hyperparameters with a direct partition interpretation, and coming with an efficient rejection sampler and a careful treatment of elicitation through the induced law of the number of clusters.

What the Pochhammer view adds is scope rather than a better answer to the same question. Stirling gamma conjugacy for the partition requires one hyperparameter to be tied to the sample size n, whereas Proposition 9.2 gives closure for the whole rational class $\mathcal{R}$ with no such tie, so a single theorem covers the Dirichlet Multinomial, Negative Binomial, urn, Waring and Erlang likelihoods as well. Three further items are new here: repeated and complex poles in Theorem 9.3, which the Pitman Yor case immediately below forces; the Mellin transform and its corollaries; and the exponential mixture of Theorem 9.9, which yields the sampler of Section 9.5.3.

For $\mathrm{PY}(\sigma, \theta)$ with $0 \leq \sigma < 1$, $\theta > -\sigma$, the exchangeable partition probability function is (Pitman, 2006)

$$p(\text{partition}) = \frac{\prod_{i=1}^{k-1}(\theta + i\sigma)}{[\theta+1]^{n-1}} \prod_{j=1}^{k} [1-\sigma]^{n_j - 1},$$

so for fixed σ the likelihood in θ depends on the data only through (n, k) and is rational of numerator degree $k-1$ and denominator degree $n-1$, with poles at $-1, \ldots, -(n-1)$. Combined with a prior $\mathrm{PH}(0, a, b, c)$, $b \geq 2$, the posterior has numerator degree $k-1$ and denominator degree $n-1+b \geq k+1$, so it is proper for every $k \leq n$ and the prior is exactly conjugate. We restrict to $\theta \geq 0$, the part of the parameter space $\theta > -\sigma$ on which the Pochhammer prior lives. A prior pole $-(a+i-1)/c$ coincides with a likelihood pole when $(a+i-1)/c \in \{1, \ldots, n-1\}$, for instance for every i when $c = 1$ and a is a positive integer, and the posterior then has double poles; otherwise all poles are simple. The numerator roots $-i\sigma$, $1 \leq i \leq k-1$, lie on the negative axis among the poles, and when $i\sigma$ is an integer a root cancels a pole outright; at $\sigma = \frac{1}{2}$, for example, every second root does so. This interplay between σ and the pole set is one analytic expression of the known weak identifiability of (σ, θ) from the block count. Conditionally on θ, the likelihood in σ is a polynomial but lies outside $\mathcal{R}$, because $[1-\sigma]^{n_j-1}$ violates the sign condition $u_j \geq 0$ of Definition 9.1, so we recommend an exact residue draw

for θ and a griddy or Metropolis step for σ. The exact marginal likelihood in θ also permits model choice among nonparametric priors by Bayes factor rather than by predictive check.

9.6.4 *Yule processes, preferential attachment and Zipf*

A linear birth process of rate λ per individual with new types arriving at rate μ per existing type gives, at stationarity, the Yule Simon law (Simon, 1955), $p(n \mid \rho) = \rho B(n, \rho + 1)$, with $\rho = \mu/\lambda$, so the likelihood over K observations is $\rho^K / \prod_k [\rho + 1]^{n_k}$, Pochhammer rational. Here ρ is both the scientific quantity in preferential attachment models of citation, degree and firm size, and the power law exponent, $p(n \mid \rho) \sim \rho\, \Gamma(\rho + 1)\, n^{-(1+\rho)}$. A Pochhammer prior therefore yields the exact posterior of a power law exponent from raw counts, in contrast to the Hill estimator and to discrete maximum likelihood, neither of which propagates exponent uncertainty into predictions.

PROPOSITION 9.22 (Boundary Zipf law). *Under $\rho \sim \mathrm{PH}(0, a, b, c)$ with density p_ρ, the Yule Simon prior predictive satisfies $p(n) \sim p_\rho(0)\, n^{-1} (\log n)^{-2}$ as $n \to \infty$. This is a proper distribution, since $\sum_n n^{-1} (\log n)^{-2} < \infty$, but it has no moments of any positive order.*

The exponent one is the boundary of summability, reached only because of the logarithmic correction. This is the count model analogue of the horseshoe sitting exactly at the boundary of integrability, and it is the sense in which the half horseshoe is the least informative proper prior for a Zipf exponent.

9.6.5 *Insurance claim counts and credibility*

The Gamma Poisson mixture is the classical credibility model: policyholder i has latent frequency $\lambda_i \sim \mathrm{Ga}(\alpha, \beta)$ and count $n_i \sim \mathrm{Poi}(\lambda_i t_i)$. Bühlmann credibility replaces the posterior mean of λ_i by a linear estimate with α plugged in from moments or maximum likelihood. Proposition 9.19 makes the exact Bayesian treatment available, so the credibility factor $Z_i = t_i/(t_i + \beta)$ becomes a random variable with a computable distribution rather than a constant. The heavy prior tail of Corollary 9.7 means the data may push α to large values, that is towards Poisson behaviour, without prior resistance, and the excess zeros normal in motor and health portfolios are handled by the mass at the origin rather than by a zero inflation mixture. Covariates enter through Algorithm 1.

9.6.6 *Erlang's loss formula and complex poles*

For c trunks with offered load A and no waiting room, the stationary blocking probability is $B(c, A) = (A^c/c!) / \sum_{j=0}^{c} A^j/j!$, rational in A. Observing y blocked calls among M offered gives the Binomial likelihood $B(c, A)^y \{1 - B(c, A)\}^{M-y}$. Since $1 - B(c, A) = \sum_{j<c} (A^j/j!) / \sum_{j \le c} A^j/j!$, this is rational with numerator degree $cy + (c-1)(M-y) \le cM$ and denominator degree cM, so any proper rational prior yields a proper rational posterior. The poles are the zeros of the truncated

exponential $\sum_{j\le c} A^j/j!$. Its coefficients are positive, so none lies on $[0,\infty)$, which is all Theorem 9.3 requires; for even c none is real and for odd c exactly one is, so complex poles are the rule, and by Szegő's theorem the rescaled zeros A/c approach the Szegő curve $|ze^{1-z}| = 1$, $|z| \le 1$, as c grows. For $c = 4$ the zeros are $-1.7294 \pm 0.8890i$ and $-0.2706 \pm 2.5048i$, all in the left half plane, but from $c = 5$ onwards some lie in the right half plane (the largest real part is 0.2398 at $c = 5$ and 3.3749 at $c = 10$), so the poles of a rational likelihood need not lie where those of a Pochhammer factor do. Theorem 9.3 covers this, each conjugate pair contributing $-2\,\mathrm{Re}\{\gamma \log r\}$, and delivers the exact posterior of the offered load and the exact predictive blocking probability. We are not aware of a previous closed form Bayesian treatment of Erlang's formula. The example also makes the general point that rational conjugacy is not about Gamma ratios: any rational likelihood will do, and Gamma ratios are simply the commonest source of them.

9.6.7 Contingency tables and species sampling

Two examples from Wang and Polson (2024) extend directly. For multiway tables with $K = 4^p$ cells, the heterogeneous half horseshoe places posterior mass near zero for the overwhelming majority of empty cells, and Cramér association statistics inherit valid credible intervals. For species sampling, the discovery probability after n observations is $\theta/(\theta + n)$ under the Dirichlet process, so Section 9.6.3 gives an exact posterior for the probability of a new species, the quantity at issue in the Good and Turing literature.

9.7 The Ewens Concentration Parameter: Worked Examples

This section carries the general results through, with actual numbers, on the problem that motivates the book: inference for the scaled mutation rate θ from a sample of genes classified by allele. We begin with the urn that generates the data, then do one posterior completely by hand, vary the model to Pitman Yor, give an exact sampler, and finish with a classical data set large enough to show both the value of the heavy tail and the cost of the residue formula.

9.7.1 The Hoppe urn

Hoppe (1984) gave the Ewens sampling formula a sequential description. An urn holds one black ball of weight θ and coloured balls of weight one. At each step a ball is drawn with probability proportional to its weight. If it is black, it is returned together with one ball of a colour never seen before, a new allele; otherwise it is returned together with one more ball of its own colour, a copy of an existing allele. The ith gene is therefore new with probability $\theta/(\theta + i - 1)$, independently of the past, which is the Feller coupling of Section 9.6.2. Multiplying the conditional probabilities along any sequence that ends with k alleles of sizes $n_1, \dots, n_k$ gives

$$\frac{\theta^k \prod_{j=1}^{k} (n_j - 1)!}{\theta(\theta+1)\cdots(\theta+n-1)} \;=\; \frac{\theta^k}{[\theta]^n} \prod_{j=1}^{k} (n_j - 1)!,$$

since each allele contributes one factor θ when it is founded and the factors $1, 2, \dots, n_j - 1$ as it grows. Summing over sequences gives the Ewens sampling formula $n!\,[\theta]^{-n} \prod_j (\theta/j)^{a_j}/a_j!$ for the allele frequency spectrum $(a_1, \dots, a_n)$, and summing further gives $\mathbb{P}(K_n = k \mid \theta) = |s(n,k)|\,\theta^k/[\theta]^n$. In every form θ enters only through $\theta^k/[\theta]^n$, so K_n is sufficient, and after cancelling one factor θ the likelihood is $\theta^{k-1}/[\theta+1]^{n-1}$, the kernel $\mathrm{PPH}(0, 1, n-1, 1, k-1)$ of Proposition 9.13. The Hoppe urn is a Pólya urn of the kind in Section 9.6.1 with an immigrant colour, and θ is its reinforcement parameter.

9.7.2 Residues by hand: four genes, two alleles

The smallest case that shows every feature of Theorem 9.3, double poles included, is a sample of $n = 4$ genes carrying $K_4 = 2$ alleles, analysed under the half horseshoe $\mathrm{PH}(0, 1, 2, 1)$, whose density is $\{(\theta+1)(\theta+2)\log 2\}^{-1}$.

EXAMPLE 9.23 (Four genes, two alleles). The posterior kernel is

$$R(\theta) \;=\; \frac{\theta^2}{\theta(\theta+1)(\theta+2)(\theta+3)} \cdot \frac{1}{(\theta+1)(\theta+2)} \;=\; \frac{\theta}{(\theta+1)^2(\theta+2)^2(\theta+3)},$$

with double poles at -1, -2 and a simple pole at -3. For the double pole at -1 put $h(\theta) = (\theta+1)^2 R(\theta) = \theta/\{(\theta+2)^2(\theta+3)\}$. Then $\gamma_{12} = h(-1) = -\frac{1}{2}$, and since $h'/h = 1/\theta - 2/(\theta+2) - 1/(\theta+3)$ equals $-\frac{7}{2}$ at $\theta = -1$, $\gamma_{11} = h'(-1) = \frac{7}{4}$. The same computation at -2 gives $\gamma_{22} = -2$, $\gamma_{21} = -1$, and at -3, $\gamma_{31} = -3/\{(-2)^2(-1)^2\} = -\frac{3}{4}$. Hence

$$R(\theta) = \frac{7/4}{\theta+1} - \frac{1/2}{(\theta+1)^2} - \frac{1}{\theta+2} - \frac{2}{(\theta+2)^2} - \frac{3/4}{\theta+3},$$

and the check $\sum_i \gamma_{i1} = \frac{7}{4} - 1 - \frac{3}{4} = 0$ passes. Formula (9.3) gives the normalizing constant

$$\begin{aligned} C \;&=\; -\Big(\tfrac{7}{4}\log 1 - \log 2 - \tfrac{3}{4}\log 3\Big) - \frac{1/2}{1} - \frac{2}{2} \\ &=\; \log 2 + \tfrac{3}{4}\log 3 - \tfrac{3}{2} \;=\; 0.0171063971. \end{aligned}$$

Multiplying the kernel by θ and repeating the computation gives the coefficients $(-\frac{9}{4}, \frac{1}{2})$ at -1, $(0, 4)$ at -2 and $\frac{9}{4}$ at -3, so $\int_0^\infty \theta R = \frac{5}{2} - \frac{9}{4}\log 3 = 0.0281223505$ and the posterior mean is

$$\mathbb{E}(\theta \mid K_4 = 2) \;=\; \frac{\frac{5}{2} - \frac{9}{4}\log 3}{\log 2 + \frac{3}{4}\log 3 - \frac{3}{2}} \;=\; 1.64397.$$

Similarly $\int_0^\infty \theta^2 R = \frac{27}{4}\log 3 - 4\log 2 - \frac{9}{2}$, so $\mathbb{E}(\theta^2 \mid K_4 = 2) = 8.36203$ and the posterior standard deviation is 2.37895. The posterior has no third moment, since $\theta^3 R(\theta) \sim \theta^{-1}$. Integrating the partial fractions from 0 to A gives the distribution function in closed form, and bisection on it gives the posterior median 1.0153 and the equal tailed 95% interval $(0.1144,\ 6.9330)$. Every one of these numbers agrees with adaptive quadrature of R to the ten digits we carried.

The interval is wide, as it must be: four genes carry almost no information about θ, and Figure 9.2(a) shows that the posterior differs from the prior mainly by removing mass near zero, where two alleles in four genes would be improbable. The quantity a geneticist is most likely to want is predictive.

EXAMPLE 9.24 (Probability that the next gene is a new allele). By the Hoppe urn the fifth gene is a new allele with probability $\theta/(\theta+4)$, so the posterior predictive probability is $\mathbb{E}\{\theta/(\theta+4) \mid K_4 = 2\}$, the ratio of $\int_0^\infty \theta R(\theta)(\theta+4)^{-1}\,\mathrm{d}\theta$ to C. The new kernel $\theta^2/\{(\theta+1)^2(\theta+2)^2(\theta+3)(\theta+4)\}$ has the coefficients $(-\frac{29}{36}, \frac{1}{6})$ at -1, $(-1, 2)$ at -2, $\frac{9}{4}$ at -3 and $-\frac{4}{9}$ at -4; the simple residues again sum to zero, and

$$\mathbb{P}(\text{gene 5 is new} \mid K_4 = 2) \;=\; \frac{\frac{7}{6} + \frac{17}{9}\log 2 - \frac{9}{4}\log 3}{\log 2 + \frac{3}{4}\log 3 - \frac{3}{2}} \;=\; 0.23775.$$

For comparison, the maximum likelihood estimate solves $\mathbb{E}_\theta K_4 = \sum_{i=0}^{3} \theta/(\theta+i) = 2$, giving $\hat\theta = 0.8794$ and the plug in probability $\hat\theta/(\hat\theta+4) = 0.1802$. The Bayesian answer is about a third larger. The reason is not Jensen's inequality, which for the concave map $\theta \mapsto \theta/(\theta+4)$ pushes the other way, but the shape of the posterior: it is strongly right skewed, with median 1.02 and mean 1.64 against $\hat\theta = 0.88$, and the plug in value ignores exactly the uncertainty that Section 9.6.2 says dominates this problem.

EXAMPLE 9.25 (Prior predictive law of the allele count). Before any data are seen, the half horseshoe induces a law on K_4 through $\mathbb{P}(K_4 = k) = |s(4,k)| \int_0^\infty \theta^k [\theta]^{-4} p(\theta)\,\mathrm{d}\theta$, with $|s(4,k)| = 6, 11, 6, 1$ for $k = 1, \dots, 4$. Each integral is a residue sum of the kind just computed. The result, against the uniform random permutation $\theta = 1$, is

	$K_4 = 1$	$K_4 = 2$	$K_4 = 3$	$K_4 = 4$
$\theta \sim \mathrm{PH}(0,1,2,1)$	0.2787	0.2715	0.2434	0.2064
$\theta = 1$	0.2500	0.4583	0.2500	0.0417

The half horseshoe spreads its predictive mass almost evenly over the four possible allele counts, from a monomorphic sample to one in which every gene is distinct. This is the sense in which it is uninformative about diversity, and it is a more useful description for elicitation than any statement about the density of θ itself. The marginal likelihood of the observed $K_4 = 2$, namely $0.2715 = 11C/\log 2$, is the quantity a Bayes factor between priors would use.

9.7.3 The Pitman Yor variant

The Pitman Yor model adds a discount $\sigma \in [0, 1)$ under which the probability of a new type after n draws with k types is $(\theta + k\sigma)/(\theta + n)$, so that types are founded at a rate that grows with their number; $\sigma = 0$ is Ewens. Section 9.6.3 showed that for fixed σ the likelihood in θ is again rational.

EXAMPLE 9.26 (Four genes, two types, $\sigma = \frac{1}{2}$). With $n = 4$, $k = 2$ the likelihood is $(\theta + \frac{1}{2})/\{(\theta+1)(\theta+2)(\theta+3)\}$, and under $\mathrm{PH}(0,1,2,1)$ the posterior kernel is $(\theta + \frac{1}{2})/\{(\theta+1)^2(\theta+2)^2(\theta+3)\}$, the double pole integrand in the first line of Table 9.5. Its partial fractions are

$$\frac{9/8}{\theta+1} - \frac{1/4}{(\theta+1)^2} - \frac{1/2}{\theta+2} - \frac{3/2}{(\theta+2)^2} - \frac{5/8}{\theta+3},$$

so $C = \frac{1}{2}\log 2 + \frac{5}{8}\log 3 - 1 = 0.0332062707$. The posterior mean of θ is 1.1045 and its median 0.5893, both well below their Ewens values 1.6440 and 1.0153, while the predictive probability of a new type at the fifth draw, $\mathbb{E}\{(\theta+1)/(\theta+4) \mid \text{data}\}$, rises to 0.37867 from 0.23775. Part of the novelty that Ewens must attribute to θ is carried here by σ, so θ is smaller, but the discount adds its own contribution $k\sigma/(\theta+n)$ to the predictive. Figure 9.2(a) shows the two posteriors.

9.7.4 Exact draws from the half horseshoe

Theorem 9.9 and Proposition 9.12 promise that wherever a Gamma prior admits an exact draw, the half horseshoe does too. Here is the promise kept in full for the Ewens likelihood.

EXAMPLE 9.27 (Inverse distribution function and the latent scale). For $\mathrm{PH}(0,1,2,1)$ the distribution function is

$$F(A) = \frac{1}{\log 2}\int_0^A \Big(\frac{1}{\theta+1} - \frac{1}{\theta+2}\Big)\,\mathrm{d}\theta = \frac{1}{\log 2}\log\frac{2(A+1)}{A+2},$$

and solving $F(A) = U$ gives the exact two line sampler

$$U \sim \mathrm{U}(0,1), \qquad A = \frac{2(2^U - 1)}{2 - 2^U}.$$

The median, at $U = \frac{1}{2}$, is exactly $\sqrt{2}$; $U = 0.1$ and $U = 0.9$ give 0.1546 and 12.93, a ratio of more than eighty between the deciles. In the notation of Theorem 9.9, $c = 1$, $(\gamma_1, \gamma_2) = (1, -1)$, $(r_1, r_2) = (1, 2)$, so $g(\tau) = e^{-\tau} - e^{-2\tau}$, which is half the density of $\mathrm{Exp}(1) + \mathrm{Exp}(2)$, and the latent scale has density $p(\tau) = (e^{-\tau} - e^{-2\tau})/(\tau \log 2)$, which tends to $1/\log 2$ at the origin and decays like $e^{-\tau}/\tau$. The two step scheme $\tau \mid \alpha \sim \mathrm{Exp}(1+\alpha) + \mathrm{Exp}(2+\alpha)$, $\alpha \mid \tau \sim \mathrm{Exp}(\tau)$ reproduces the law: with 2×10^4 parallel chains run for 300 steps, the fraction of draws below one is 0.4145, against the exact $F(1) = \log(4/3)/\log 2 = 0.4150$.

EXAMPLE 9.28 (Escobar and West with a half horseshoe). Escobar and West (1995) write $\theta^k/[\theta]^n = \theta^{k-1}(\theta+n)B(\theta+1,n)/\Gamma(n)$ and introduce $\eta \mid \theta \sim \mathrm{Be}(\theta+1, n)$, after which a $\mathrm{Ga}(\nu,\lambda)$ prior gives a two component Gamma mixture for θ. Proposition 9.12 replaces the Gamma prior by $\mathrm{Exp}(\tau)$, that is $\nu = 1$ and $\lambda = \tau$, and adds the τ step. One sweep is

(i) $\tau \sim \mathrm{Exp}(1+\theta) + \mathrm{Exp}(2+\theta)$;

(ii) $\eta \sim \mathrm{Be}(\theta+1, n)$;

(iii) with $\lambda' = \tau - \log\eta$, draw $\theta \sim \mathrm{Ga}(k+1, \lambda')$ with probability w and $\theta \sim \mathrm{Ga}(k, \lambda')$ otherwise, where $w/(1-w) = k/(n\lambda')$.

Every step is a standard draw, and no residue is computed. For $n = 4$, $k = 2$, 4000 parallel chains of 400 sweeps, of which the last 300 are kept, give posterior mean 1.6444, median 1.0147 and predictive $\mathbb{E}\{\theta/(\theta+4)\} = 0.23769$, against the exact 1.64397, 1.0153 and 0.23775 of Examples 9.23 and 9.24. The agreement is within Monte Carlo error.

9.7.5 Corbet's Malayan butterflies

Fisher's log series and the Ewens sampling formula describe the same partition structure, and the most famous species abundance data are the Malayan butterflies collected by Corbet and analysed by Fisher et al. (1943). Chapter 6 studies these data in detail and finds that the Ewens fit is fair but misses in a specific direction; here we use them only as a realistic test of the prior and of the arithmetic. Corbet took every individual seen of a species only until 24 specimens had been taken (Fisher et al., 1943, p. 43), so a count above 24 is only a lower bound and the table gives the abundance classes $j = 1, \ldots, 24$, holding 501 species and 3306 individuals, and reports only that 119 further species were taken more than 24 times: $K = 620$, and the total n is unknown. The censored likelihood of Chapter 6 estimates it at $\hat{n} = 11068$. Given n, that likelihood depends on θ only through K, as $\theta^{620}/[\theta]^n$, so we condition on $n = \hat{n}$. This is an approximation, since it ignores the uncertainty in n; Chapter 6 shows that integrating n out instead lowers the posterior mean by about half a unit and widens the posterior by less than a tenth. Treating the 24 tabulated classes as the whole sample, as earlier analyses did, would give $n = 3306$, $K = 501$ and $\hat{\theta} = 163.99$, an estimate that ignores the 119 commonest species.

EXAMPLE 9.29 (Butterflies: prior sensitivity and the cost of residues). At $n = \hat{n} = 11068$ and $K = 620$ the likelihood $\theta^{620}/[\theta]^{11068}$ has its maximum at $\hat{\theta} = 141.75$, where the Fisher information $\sum_{i \le n}(i-1)/\{\theta(\theta+i-1)^2\}$ equals 0.0239, a standard error of 6.47. Three priors give the posteriors in Table 9.4 and Figure 9.2(b). The half horseshoe leaves the likelihood essentially untouched. The diffuse $\mathrm{Ga}(0.1, 0.1)$ moves the median down by about two thirds of a standard error, the effect of its rate 0.1. The $\mathrm{Ga}(2, 1)$ prior, a Gamma prior of the kind Escobar and West (1995) use for Dirichlet process mixtures, moves it to 111, almost five standard errors below $\hat{\theta}$, with a 95% interval $(101.4, 120.5)$ that excludes the maximum likelihood value. A normal approximation explains the size of the shift: the log prior has slope $1/\theta - 1 \approx -1$, and dividing by the information 0.0239 predicts a shift of about -42; the actual shift of -31 is smaller because the information grows as θ falls. For the population genetics reading, in which $\theta = 4N_e\mu$, the $\mathrm{Ga}(2, 1)$ prior would understate diversity by more than a fifth.

The predictive probability that the next individual belongs to a new species is 0.01262 under the half horseshoe, close to the plug in $\hat{\theta}/(\hat{\theta}+n) = 0.01265$, and

0.00990 under $\mathrm{Ga}(2,1)$. Good's estimate $a_1/n = 118/11068 = 0.0107$ is lower than the Ewens value; that discrepancy reflects the misfit documented in Chapter 6, not the prior.

The arithmetic is instructive too. The half horseshoe posterior has 11067 distinct poles, two of them double (denominator degree 11069), and the residue formula needs about 4650 digits merely to cancel: at 4300 digits it returns a negative normalizer, while at 5500 digits it gives $C = 3.83864526979 \times 10^{-38952}$, posterior mean 141.500581202 and predictive 0.0126229439769, agreeing with careful quadrature of the logarithmic kernel to twelve or more digits. The computation took about five minutes of processor time in arbitrary precision arithmetic, against a fraction of a second for quadrature in logarithms and a few seconds for the sampler of Example 9.28, which gives mean 141.50. The practical rule for samples of this size is therefore the one stated in Section 9.5.3: sample or integrate numerically for estimation, and reserve the residue formula for the exact marginal likelihood when one is needed.

Prior on θ	mean	median	2.5%	97.5%	$\mathbb{P}(\text{new species})$
$\mathrm{PH}(0,1,2,1)$	141.50	141.39	129.15	154.49	0.01262
$\mathrm{Ga}(0.1,0.1)$	137.76	137.65	125.79	150.35	0.01229
$\mathrm{Ga}(2,1)$	*110.71*	*110.63*	*101.35*	*120.52*	*0.00990*
maximum likelihood		141.75			0.01265

TABLE 9.4. Posterior for the Ewens concentration of Corbet's butterflies, all $K = 620$ species, conditional on the estimated sample size $\hat{n} = 11068$ of the censored likelihood (Chapter 6), under three priors. Italics mark the $\mathrm{Ga}(2,1)$ row, whose interval excludes the maximum likelihood value. The last column is the posterior predictive probability that individual $n+1$ is of a new species.

9.8 Numerical Verification and a Simulation Study

9.8.1 Verification of the closed forms

The residue computations of this section use working precision $\lceil N/2 \rceil + 15$ digits, the rule of remedy (ii) after Proposition 9.13, and are validated against adaptive quadrature of the unnormalized kernel. Table 9.5 reports the checks; agreement is to every digit shown.

The strip of Theorem 9.6 is sharp in the numerics as well as in Lemma 9.4. For $\mathrm{PPH}(2, 0.7, 6, 1, 0)$ the left edge is -1, since $[\alpha]^2$ vanishes only to first order at the origin; evaluating (9.5) at $s = -1.7$ returns -0.0214, a negative value for a positive integrand, correctly signalling that the identity has been used outside its domain. For $\mathrm{PPH}(1,1,5,2,1)$ the left edge is -2 and $s = -1.7$ is legitimate, as the table confirms. This is a useful practical warning: the formula does not announce its own failure by blowing up.

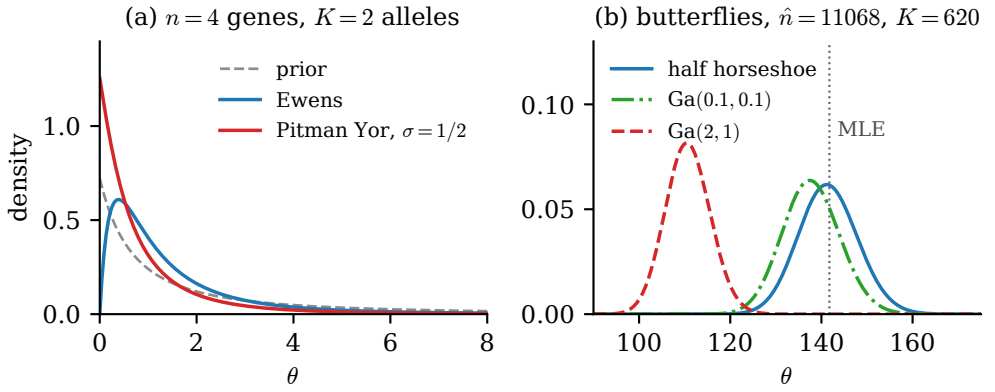


FIGURE 9.2. (a) Four genes carrying two alleles: the half horseshoe prior and the exact posteriors of Examples 9.23 and 9.26. (b) Corbet's butterflies, $K = 620$ species at $n = \hat{n} = 11068$: posteriors for θ under three priors, with the maximum likelihood value dotted (Example 9.29).

Result and setting	Closed form	Quadrature
Thm. 9.3, double poles, $R = (\alpha+\frac{1}{2})\{(\alpha+1)^2(\alpha+2)^2(\alpha+3)\}^{-1}$	0.033206270697541	0.033206270697541
Thm. 9.3, complex poles, $R = \{(\alpha+1)((\alpha+2)^2+9)\}^{-1}$	0.095487677098166	0.095487677098166
Thm. 9.3, Ewens, $n = 20$, $k = 5$, PH$(0,1,2,1)$	$1.5311352489 \times 10^{-19}$	$1.5311352489 \times 10^{-19}$
Thm. 9.6, $s = 0.4$, PH$(0, 1.1, 5, 2)$	0.00891856126471	0.00891856126471
Thm. 9.6, $s = 2.3$, PH$(0, 1.1, 5, 2)$	0.000877909609669	0.000877909609669
Thm. 9.6, $s = -0.5$, PPH$(2, 0.7, 6, 1, 0)$	0.00442197021726	0.00442197021726
Thm. 9.6, $s = -1.7$, PPH$(1, 1, 5, 2, 1)$	0.0166781629184	0.0166781629184
Thm. 9.16, $z = 0.30$, PH$(1, 1, 4, 3)$	0.00140806861934	0.00140806861934
Thm. 9.16, improper part, $R = \alpha^2/(\alpha+1.3)$, $z = 0.45$	0.92288110392933	0.92288110392933

TABLE 9.5. Closed forms against adaptive quadrature at thirty digits of working precision.

9.8.2 Does the heavy tail matter? A simulation study

Corollary 9.7 says the half horseshoe has no moments; Section 9.6.2 says the Ewens likelihood carries only $\log n$ worth of information. Together they predict that a light tailed prior will fail to reach a large true concentration and that its credible intervals will undercover. We test this. For $\theta \in \{0.5, 2, 10\}$ and $n \in \{100, 1000\}$ we draw the block count k by the Feller coupling and compute the exact posterior of θ under three priors: the half horseshoe $\mathrm{PH}(0, 1, 2, 1)$, a $\mathrm{Ga}(2, 1)$ prior of the kind used by Escobar and West (1995), and the diffuse $\mathrm{Ga}(0.1, 0.1)$. Table 9.6 reports root mean squared error of the posterior mean, coverage of the equal tailed 95% interval, and mean interval width over 200 replicates.

		$\mathrm{PH}(0,1,2,1)$			$\mathrm{Ga}(2,1)$			$\mathrm{Ga}(0.1,0.1)$		
θ	n	RMSE	cov	width	RMSE	cov	width	RMSE	cov	width
0.5	100	0.359	0.97	1.44	0.455	0.94	1.62	0.362	0.91	1.36
0.5	1000	0.274	0.93	1.11	0.335	0.94	1.22	0.278	0.93	1.07
2.0	100	0.817	0.94	3.14	*0.641*	0.97	2.80	0.849	0.93	3.20
2.0	1000	0.538	0.96	2.40	*0.474*	0.98	2.25	0.553	0.96	2.43
10.0	100	2.585	0.94	10.15	3.196	*0.61*	6.26	2.434	0.94	9.77
10.0	1000	1.548	0.94	6.41	2.054	*0.78*	5.14	1.520	0.94	6.32

TABLE 9.6. Posterior for the Dirichlet process concentration from the block count, 200 replicates. Italics mark the two effects discussed in the text.

Three readings, and we report the unfavourable one too. When the truth sits at the prior mean, $\theta = 2$, the correctly centred $\mathrm{Ga}(2, 1)$ wins on RMSE by twelve to twenty two per cent, as it should; there is no free lunch, and a heavy tail costs efficiency at the centre. When the truth is large, $\theta = 10$, the $\mathrm{Ga}(2, 1)$ posterior cannot follow, and its coverage collapses to 0.61 at $n = 100$ and 0.78 at $n = 1000$, with intervals that are both too narrow and misplaced. The half horseshoe holds nominal coverage throughout with uniformly lower RMSE than $\mathrm{Ga}(2, 1)$ in that regime. Note also that the diffuse $\mathrm{Ga}(0.1, 0.1)$ is competitive with the half horseshoe everywhere and slightly better at $\theta = 10$: the practical gain over a well chosen diffuse Gamma is small, and the case for the Pochhammer prior rests on conjugacy and closed forms rather than on a large risk improvement. What the comparison does establish is that the failure mode of a light tailed default is real and is exactly the one Corollary 9.7 predicts.

9.8.3 Conditioning and tails

Figure 9.1 reports the two remaining experiments. Left, $\log_{10} \kappa_n$ over $n \in \{10, \dots, 160\}$ with fit $0.502n - 2.30$; at double precision the formula is exhausted by $n \approx 36$, a property of the representation and not of the posterior. Right, the marginal $p(n)$ against (9.12) for $a \in \{1, 2\}$; the ratio $p(n)n^{1+a}(\log n)^2 / K$ is 1.045 at $n = 10^3$ and 0.977 at $n = 10^6$ for $a = 1$, consistent with the predicted $O(1/\log n)$ correction.

9.9 Application: Two Regimes of Overdispersion

Corollary 9.7 says the half horseshoe has no moments, and the simulation study of Section 9.8 (Table 9.6) showed that a light tailed Gamma prior fails when the true concentration is large; Example 9.29 showed the same failure on the butterflies. This section tests the claim on a second kind of real data, counts with covariates. Two data sets from Venables and Ripley (2002) carry the same model and differ only in regime.

- `quine`: days of absence from school for $K = 146$ children in New South Wales, with covariates for ethnicity, sex, age band and learner status. Mean 16.5, variance 262, maximum likelihood dispersion $\hat{\alpha} = 1.27$. Strong overdispersion.
- `Insurance`: third party motor claims in 64 policy cells, with covariates for district, engine capacity and driver age, and the log number of holders as offset. Mean 49.2, variance 4985. Close to Poisson: the profile likelihood of α increases all the way to its Poisson limit, so the maximum likelihood dispersion is $\hat{\alpha} = \infty$, and α is bounded below by the data but not above.

The model is Negative Binomial regression as in Section 9.5, $n_i \sim \mathrm{NB}(\alpha, \pi_i)$ with $\psi_i = o_i + \boldsymbol{x}_i^\top \boldsymbol{\beta}$, and $\boldsymbol{\beta} \sim N(0, 100I)$. We compare three priors on α: the half horseshoe $\mathrm{PH}(0, 1, 2, 1)$, the diffuse $\mathrm{Ga}(0.01, 0.01)$ and the light tailed $\mathrm{Ga}(2, 1)$. Posterior summaries for `quine` use 10^5 retained draws of Algorithm 2. Algorithm 2 does not mix on `Insurance` (Section 9.9.3), so there the summaries use 10^5 draws, four chains of 2.5×10^4, from a blocked Metropolis Hastings sampler for the same model: a random walk step on $\log \alpha$ in which $\boldsymbol{\beta}$ is redrawn from its Laplace approximation given the proposed α and the pair is accepted or rejected together, alternated with an independence refresh of $\boldsymbol{\beta}$.

9.9.1 Where the prior does not matter

On `quine` the three priors are indistinguishable. Panel (a) of Figure 9.3 overlays the three posteriors of α; the posterior medians are 1.21, 1.21 and 1.22 with essentially identical intervals, and predictive coverage is nominal for all three. This is the expected outcome and we report it as such. With $\sum_i n_i = 2403$ the likelihood is sharply informative about α near its mode, and no reasonable prior can move it. In this regime the contribution of the present chapter is exactness and conjugacy, not accuracy: Algorithm 2 runs with every block a standard draw and mixes about as well as a tuned random walk Metropolis step for α in the same sampler, at about the same cost per iteration: in four runs of 2×10^4 draws each, the effective sample size for α is 520 to 890 for Algorithm 2 against 700 to 810 for random walk Metropolis. What it saves is the tuning, not the time.

9.9.2 Where it does

On `Insurance` the picture separates, and in the direction Corollary 9.7 predicts. Panel (b) plots the three posteriors on a logarithmic scale. The $\mathrm{Ga}(2, 1)$ posterior

sits at a median of 13.1 with a 95% interval $(8.2, 20.0)$, in a region the likelihood has all but ruled out: at $\alpha = 20$ the profile log likelihood lies 17.7 units below its Poisson limit. Its light tail simply cannot reach the region the data prefer, and because a small α means a large variance function $\mu + \mu^2/\alpha$, the error propagates straight into prediction: the 90% posterior predictive bands, shown in panel (c), have mean width 55.3 against 24.8 for the half horseshoe, with empirical coverage 1.000 rather than the nominal 0.90. The intervals are not merely wrong, they are more than twice as wide as they should be.

The half horseshoe reaches the near Poisson region the likelihood supports, with median 353 and interval $(66, 7.4 \times 10^3)$. That interval is very wide, and deliberately so. Near Poisson data bound α from below but carry almost no information about how large it is, so an honest posterior must be diffuse to the right, and the heavy tail of Corollary 9.7 is what permits it. The episode also shows what Corollary 9.7 means in practice. The prior has no moments, and the posterior mean of α under the half horseshoe, about 2.1×10^3 by quadrature, is finite only because the $N(0, 100)$ prior on the intercept of ψ penalizes very large α; it is six times the median, it is carried by the far right tail, and the average of the 10^5 draws, 1664, still falls short of it. Under the mean parameterization of Section 9.9.3, where the likelihood in α tends to a positive constant, the posterior has no mean at all. Posterior medians, not means, are the right summary for this family, and we use them throughout Table 9.7.

		posterior for α			90% predictive	
Data	Prior on α	median	2.5%	97.5%	coverage	width
`quine`	PH$(0, 1, 2, 1)$	1.21	0.94	1.54	0.932	49.8
$\hat{\alpha} = 1.27$	Ga$(0.01, 0.01)$	1.21	0.94	1.54	0.938	49.8
	Ga$(2, 1)$	1.22	0.95	1.56	0.932	49.4
`Insurance`	PH$(0, 1, 2, 1)$	353.3	65.5	7433.5	0.953	24.8
$\hat{\alpha} = \infty$	Ga$(0.01, 0.01)$	159.8	55.2	449.2	0.953	27.0
	Ga$(2, 1)$	*13.1*	*8.2*	*20.0*	*1.000*	*55.3*

TABLE 9.7. Negative Binomial regression under three priors on the dispersion. On `Insurance` the likelihood has no finite maximum ($\hat{\alpha} = \infty$, the Poisson limit). Italics mark the failure: the Ga$(2, 1)$ interval lies where the profile log likelihood is at least 17.7 units below its supremum, and its predictive bands are more than twice too wide and overcover.

9.9.3 A limitation of Algorithm 2

The second data set also exposes a genuine weakness of our own sampler, which we report rather than tune away. On `Insurance`, Algorithm 2 yields an effective sample size between 4 and 10 from 2×10^4 draws of α, in four runs started at $\alpha = 20$, 50, 200 and 1000. Panel (d) shows why: the chain drifts rather than mixes.

The cause is not the Chinese restaurant table block and not the exponential scale, since replacing the α draw by a slice sampler on the same conditional changes

nothing, giving effective sample sizes between 4 and 16 in the same four runs. The cause is the parameterization. Algorithm 2 is exact precisely because it works with (α, ψ), in which the mean is αe^{ψ_i}; when α is large the data identify the product far better than either factor, and α and the intercept lie along a ridge. Any update of α that conditions on $\boldsymbol{\beta}$ must crawl along it.

One remedy is to work with the mean, $\mu_i = e^{\eta_i}$ with $\eta_i = o_i + \boldsymbol{x}_i^\top \boldsymbol{\beta}$, and $\pi_i = \alpha/(\alpha + \mu_i)$, which breaks the ridge and raises the effective sample size to between 13256 and 17526 from 2×10^4 draws, a gain of three orders of magnitude at about the same cost per iteration. As run, with $\boldsymbol{\beta} \sim N(0, 100I)$ placed on the coefficients of η_i, this is not a reparameterization of the model stated above but a change of prior: in the coordinates of η the intercept prior moves from $N(\log \alpha, 100)$ to $N(0, 100)$. The change matters only where the likelihood is flat, in the right tail of α: under the half horseshoe the posterior median is 395 and the 97.5% point 1.13×10^4, against 353 and 7.4×10^3 for the stated model. Table 9.7 therefore keeps the stated model and breaks the ridge differently, by moving α and $\boldsymbol{\beta}$ together in the blocked sampler described above, which gives effective sample sizes of 8903, 22253 and 14893 for $\log \alpha$ from 10^5 draws under the three priors. The cost of either remedy is the conjugacy. The blocked sampler replaces the exact draw of α by a joint Metropolis Hastings step, and in the mean parameterization, with μ fixed, the factor π_i^α is no longer of the form z^α, so the tilted Pochhammer draw of Theorem 9.16 is unavailable and α must be slice sampled from its conditional, which is stable in logarithms even though its normalizer is not. The honest summary is that conjugacy and good geometry pull in opposite directions here, that Algorithm 2 is the right choice in the moderate dispersion regime that motivates most count models, and that a sampler which moves α and $\boldsymbol{\beta}$ together should be preferred when α is large.

9.10 Discussion

The Pochhammer family is best understood not as a device for Dirichlet Multinomial models but as the conjugate family for the rational likelihoods of applied probability. Posed that way, the tools are partial fractions and the Mellin transform, and the reach is delimited by one criterion, Proposition 9.18. Damsleth's class sits just outside it, and we have given the exact repairs available inside.

For population genetics the conclusions are concrete. The scaled mutation rate θ of the infinitely many alleles model, and its Pitman Yor generalization for fixed discount, admit exact conjugate inference from the allele count alone; the posterior, its moments and the probability that the next gene sampled carries a new allele are finite combinations of logarithms, as Examples 9.23 to 9.26 show by hand. Because the information about θ grows only like $\log n$, the prior matters at every sample size, and Example 9.29 shows that a light tailed Gamma prior of the kind used in the clustering literature can move the answer by almost five standard errors in a sample of 620 species and about eleven thousand individuals. The half horseshoe does not, and Example 9.28 shows that it costs nothing in sampling convenience.

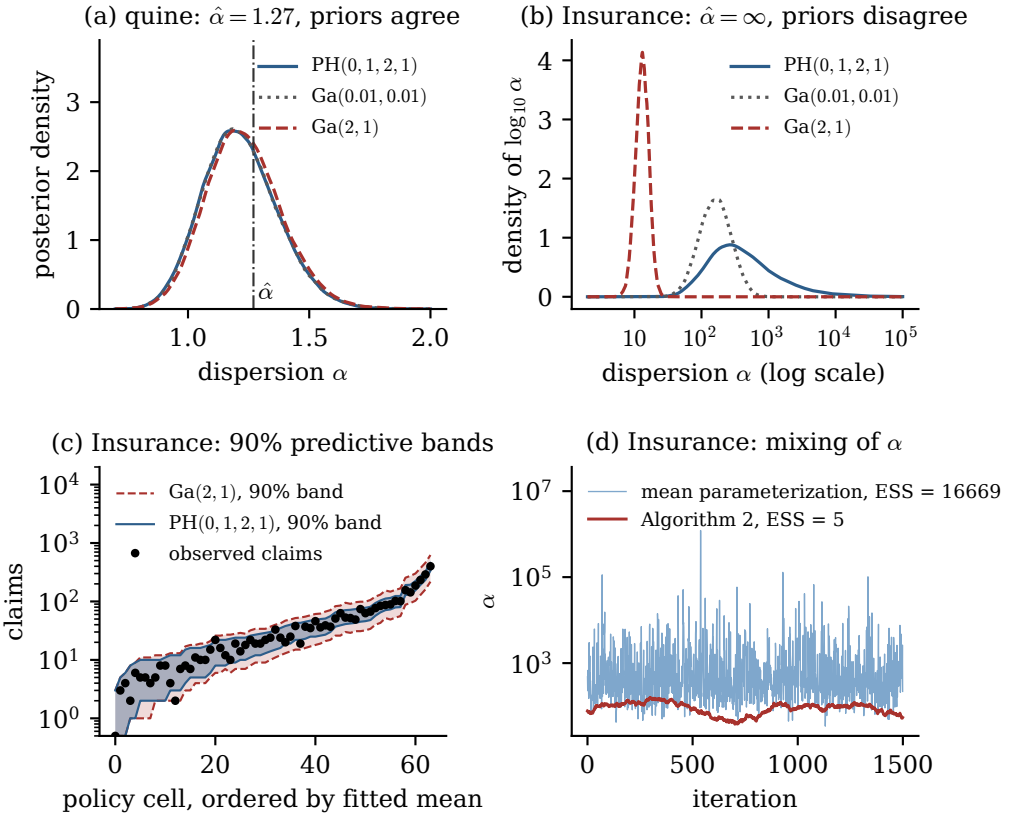


FIGURE 9.3. Negative Binomial regression on two data sets. (a) On `quine` the three priors on the dispersion give the same posterior; the dash dotted line is the maximum likelihood value $\hat{\alpha} = 1.27$. (b) On `Insurance` they do not: the $\text{Ga}(2, 1)$ posterior sits far below the near Poisson values the likelihood supports, while the half horseshoe reaches them with an honestly diffuse right tail. The likelihood has no finite maximum here, so no $\hat{\alpha}$ is marked in (b). (c) The consequence for prediction: the $\text{Ga}(2, 1)$ bands are more than twice as wide. (d) Mixing of the dispersion block on `Insurance`, first 1500 retained draws from a start at $\alpha = 20$, with the effective sample size over all 2×10^4 draws in the legend; the (α, ψ) ridge that makes Algorithm 2 conjugate is what stalls it. Panels (b) and (c) show the model and sampler of Table 9.7; the mean parameterization trace in (d) comes from the run of Section 9.9.3, whose prior on the intercept differs.

Four directions remain. Proposition 9.13(i) solves one case of the conditioning problem exactly and (ii) leaves the general one open; the natural conjecture is $\log \kappa_n \sim n\,\vartheta(k/n)$ for a rate function ϑ determined by the pole configuration, obtainable from the same Laplace analysis that yields the leading cancellation. A stable barycentric form of the partial fraction expansion may also exist, by analogy with barycentric rational interpolation. The Mellin representation (9.5) invites a Mellin Barnes treatment, giving asymptotic expansions of posterior moments in N by shifting contours across poles, and connects the family to generalized Gamma convolutions and Thorin measures; Theorem 9.9 makes that connection concrete, since the mixing law in (9.7) is the size debiased law of a sum of independent exponentials, and whether it lies in the Thorin class is a well posed question we leave open. The regression extension in which each concentration parameter α_k depends on covariates through a log linear link, listed as future work in Wang and Polson (2024), is now available through Theorem 9.16, since a log linear link contributes exactly a factor z^{α}. Finally, Theorem 9.20 and Proposition 9.22 suggest using the hierarchy as a generative model for heavy tailed counts in its own right, with the Zipf exponent elicited as a hyperparameter and the logarithmic correction supplying a testable refinement of the pure power law.

9.11 Exercises

1. Prove the coefficient formula (9.4), and deduce directly that $\sum_i \gamma_i = 0$ for $\mathrm{PH}(0, a, b, c)$ with $b \geq 2$.
   Hint. With $m = d = 0$, $\gamma_i = (-1)^{i-1}\binom{b-1}{i-1}/(b-1)!$; use $\sum_j (-1)^j \binom{b-1}{j} = 0$.

2. For the half horseshoe $\mathrm{PH}(0, 1, 2, 1)$ show that $\mathbb{E}(\alpha^{1/2}) = \pi(\sqrt{2}-1)/\log 2 = 1.8774$ and $\mathbb{E}(\alpha^{-1/2}) = \pi(1-2^{-1/2})/\log 2 = 1.3275$, and that $\mathbb{E}(\alpha^{-1}) = \mathbb{E}(\alpha) = \infty$.
   Hint. Theorem 9.6 at $s = \frac{3}{2}$ and $s = \frac{1}{2}$, or, for $\mathbb{E}(\alpha^{-1/2})$ alone, directly from $\int_0^\infty \alpha^{s-1}(\alpha + r)^{-1}\,\mathrm{d}\alpha = \pi r^{s-1}/\sin(\pi s)$, valid for $0 < s < 1$.

3. Repeat Example 9.23 for four genes carrying three alleles. Show that $C = \frac{5}{2} - \frac{9}{4}\log 3$, that the posterior mean is 5.0865, and that the posterior has no variance.
   Hint. The kernel is $\theta^2/\{(\theta+1)^2(\theta+2)^2(\theta+3)\}$, whose integral was already computed in Example 9.23; for the mean use the integral of $\theta^2 R$ there.

4. Using the Feller coupling, derive the Fisher information $I_n(\theta)$ of Section 9.6.2, and show that $I_n(\theta)\theta/\log n \to 1$.
   Hint. A Bernoulli variable with success probability $p(\theta)$ carries information $p'^2/\{p(1-p)\}$; compare the sum with $\int_1^n \mathrm{d}x/(\theta + x)$.

5. Show that the Hoppe urn assigns probability $\theta^k \prod_j (n_j - 1)!/[\theta]^n$ to every sequence with allele sizes $n_1, \ldots, n_k$, and deduce that K_n is sufficient for θ given

the whole sequence.
Hint. Factorization theorem; the remaining factor does not involve θ.

6. Verify the identity $\theta^{k}/[\theta]^{n} = \theta^{k-1}(\theta+n)B(\theta+1,n)/\Gamma(n)$ and derive the mixture weight $w/(1-w) = k/(n\lambda')$ in Example 9.28.
Hint. Write $B(\theta+1,n) = \int_0^1 \eta^{\theta}(1-\eta)^{n-1}\,\mathrm{d}\eta$, then integrate $\theta^{k-1}(\theta+n)e^{-\lambda'\theta}$ term by term.

7. Use Legendre's duplication formula to show that $\Gamma(2\alpha)/\Gamma(\alpha)^2 = 2^{2\alpha-1}\Gamma(\alpha+\tfrac{1}{2})/\{\sqrt{\pi}\,\Gamma(\alpha)\}$, and explain via Proposition 9.18 why no exponential tilt makes the Damsleth marginal with $\nu = 2$ rational.
Hint. The shifts $\tfrac{1}{2}$ and 0 differ by a non integer; $2^{2\alpha}$ is a tilt and cannot cancel infinitely many poles.

8. Compute $\int_0^{\infty} e^{-\lambda\alpha}\{(\alpha+1)(\alpha+2)\}^{-1}\,\mathrm{d}\alpha$ by Theorem 9.16 and check that it tends to $\log 2$ as $\lambda \downarrow 0$. At $\lambda = 0.01$ it equals 0.65603.
Hint. The answer is $e^{\lambda}E_1(\lambda) - e^{2\lambda}E_1(2\lambda)$.

9. Show that for the Pitman Yor model the posterior predictive probability of a new type, $\mathbb{E}\{(\theta+k\sigma)/(\theta+n) \mid \text{data}\}$, is a ratio of two integrals of rational functions, and reproduce the value 0.37867 of Example 9.26.
Hint. Multiplying the kernel by $(\theta+k\sigma)/(\theta+n)$ adds one numerator factor and one simple pole.

10. (Computation.) For the kernel of Proposition 9.13 with $k = 1$, find the smallest n at which κ_n exceeds 10^{16}, so that double precision returns no correct digits. Then reproduce Table 9.3 and extend it to $n = 60$ with 40 digit arithmetic.
Hint. Use the formula of Proposition 9.13(i) as a first guess; the answer is $n = 51$.

11. (Computation.) Run the sampler of Example 9.28 on Corbet's butterflies, with $n = 11068$ and $k = 620$ as in Example 9.29, and on a sample of your own simulated from the Hoppe urn with $\theta = 10$ and $n = 200$. Compare the posterior median and 95% interval with those from quadrature of $\theta^{k}[\theta]^{-n}p(\theta)$ in logarithms.
Hint. Evaluate $\log[\theta]^{n}$ as $\log\Gamma(\theta+n) - \log\Gamma(\theta)$ to avoid overflow.

12. Show that the posterior of Example 9.23 is unimodal, with mode at the unique positive root of $4\theta^3 + 15\theta^2 + 9\theta - 6 = 0$, namely $\theta = 0.3887$. Compare the mode, median, mean and maximum likelihood estimate, and explain their order.
Hint. Clear denominators in $1/\theta - 2/(\theta+1) - 2/(\theta+2) - 1/(\theta+3) = 0$ and apply Descartes' rule of signs.

Proofs

Proof of Theorem 9.3. Properness gives $R(\alpha) = O(\alpha^{-2})$ at infinity, so the integral converges absolutely; no pole lies on $[0,\infty)$, so the integrand is continuous there.

The stated coefficients are the Laurent coefficients at each pole. Multiplying the expansion by α and letting $\alpha \to \infty$ gives $0 = \lim \alpha R(\alpha) = \sum_i \gamma_{i1}$. For $j \geq 2$ and $r \notin (-\infty, 0]$, $\int_0^\infty (\alpha + r)^{-j} \, \mathrm{d}\alpha = \{(j-1) r^{j-1}\}^{-1}$. For $j = 1$ the individual integrals diverge, so integrate the sum on $[0, T]$: $\sum_i \gamma_{i1} \{\log(T + r_i) - \log r_i\}$. Writing $\log(T + r_i) = \log T + \log(1 + r_i/T)$, the first part contributes $\log T \sum_i \gamma_{i1} = 0$ and the second vanishes as $T \to \infty$. The branch is principal because the path from r_i to $T + r_i$ stays in $\mathbb{C} \setminus (-\infty, 0]$. Nonreal poles occur in conjugate pairs with conjugate coefficients, so the total is real. □

Proof of Lemma 9.4. By (9.4), $cR = \sum_i \gamma_i / (\alpha + r_i)$, and the constant factor c changes none of the orders used below. Expanding each simple pole at infinity, $\gamma_i / (\alpha + r_i) = \sum_{k \geq 0} \gamma_i (-r_i)^k \alpha^{-k-1}$, so the coefficient of α^{-k-1} in cR is $(-1)^k \sum_i \gamma_i r_i^k$. Since $R = O(\alpha^{m+d-b})$, these vanish for $k + 1 < b - m - d$, that is $k \leq b - m - d - 2$. Expanding at the origin, $\gamma_i / (\alpha + r_i) = \sum_{k \geq 0} (-1)^k \gamma_i r_i^{-k-1} \alpha^k$, so the coefficient of α^k is $(-1)^k \sum_i \gamma_i r_i^{-k-1}$. Since $[\alpha]^m = \alpha \prod_{s=1}^{m-1} (\alpha + s)$ for $m \geq 1$ and equals 1 for $m = 0$, we have $R = O(\alpha^{d + \mathbf{1}(m \geq 1)})$ at the origin, so these vanish for $k \leq d + \mathbf{1}(m \geq 1) - 1$, that is $\sum_i \gamma_i r_i^{-k} = 0$ for $1 \leq k \leq d + \mathbf{1}(m \geq 1)$. □

Proof of Theorem 9.6. For $0 < \mathrm{Re}(s) < 1$ and $r > 0$, substituting $\alpha = ru$ gives $\int_0^\infty \alpha^{s-1} (\alpha + r)^{-1} \, \mathrm{d}\alpha = r^{s-1} B(s, 1-s) = \pi r^{s-1} / \sin(\pi s)$. Summing termwise, valid there, gives (9.5) on that strip. The left side is analytic on $-\{d + \mathbf{1}(m \geq 1)\} < \mathrm{Re}(s) < b - m - d$ by the growth of R at both endpoints. The right side is meromorphic with possible poles at the integers; at $s = k + 1$ for $0 \leq k \leq b - m - d - 2$ the pole is cancelled by $\sum_i \gamma_i r_i^k = 0$, and at $s = -k + 1$ for $1 \leq k \leq d + \mathbf{1}(m \geq 1)$ by $\sum_i \gamma_i r_i^{-k} = 0$, both by Lemma 9.4; these are exactly the integers in the strip. Analytic continuation gives the identity throughout. At $s = 1$, using $r_i^{s-1} = 1 + (s-1) \log r_i + \frac{1}{2}(s-1)^2 \log^2 r_i + O((s-1)^3)$ and $\pi / \sin(\pi s) = -(s-1)^{-1} + O(s-1)$ together with $\sum_i \gamma_i = 0$ gives $\mathcal{M}(1) = -c^{-1} \sum_i \gamma_i \log r_i$ and the stated $\mathcal{M}'(1)$. □

Proof of Corollary 9.7. $R(\alpha) \sim c^{-b} \alpha^{m+d-b}$ at infinity, so $\int_x^\infty R \sim c^{-b} x^{m+d-b+1} / (b - m - d - 1)$; dividing by C gives the claim. Equivalently, $\mathcal{M}$ has a simple pole at the right edge $s = b - m - d$ of the strip, and the Tauberian correspondence for Mellin transforms converts it into regular variation of index $b - m - d - 1$. □

Proof of Theorem 9.16. The polynomial part contributes

$$\int_0^\infty e^{-\lambda \alpha} \alpha^k \, \mathrm{d}\alpha = k! / \lambda^{k+1}.$$

For a simple pole, write $c\alpha + a + i - 1 = c(\alpha + r_i)$ with $r_i = (a + i - 1)/c$, and substitute $t = \lambda(\alpha + r_i)$, so that t runs from λr_i to ∞ and $e^{-\lambda \alpha} = e^{\lambda r_i} e^{-t}$:

$$\int_0^\infty \frac{e^{-\lambda \alpha}}{c\alpha + a + i - 1} \, \mathrm{d}\alpha = \frac{1}{c} \int_0^\infty \frac{e^{-\lambda \alpha}}{\alpha + r_i} \, \mathrm{d}\alpha = \frac{1}{c} e^{\lambda r_i} \int_{\lambda r_i}^\infty \frac{e^{-t}}{t} \, \mathrm{d}t = \frac{1}{c} e^{\lambda r_i} E_1(\lambda r_i).$$

Each of the finitely many terms is absolutely integrable against $e^{-\lambda\alpha}$, so the integral of the sum is the sum of the integrals, and no degree condition is needed because $e^{-\lambda\alpha}$ dominates any polynomial. For the limit, $\Pi \equiv 0$ and $e^{\lambda r_i} E_1(\lambda r_i) = -\log\lambda - \log r_i - \gamma_{\mathrm{E}} + O(\lambda\log\lambda)$, so the right side of (9.8) is $-c^{-1}\sum_i \gamma_i \log r_i - c^{-1}(\log\lambda + \gamma_{\mathrm{E}})\sum_i \gamma_i + o(1)$, and $\sum_i \gamma_i = 0$ kills the divergent term. □

Proof of Proposition 9.13(ii). Write $j = xn$ and $k = \rho n$, so that $\log|\gamma_j| = (k-1)\log j - \log\Gamma(j) - \log\Gamma(n-j)$. Stirling's formula gives

$$\log|\gamma_{xn}| = \rho n\log(xn) - \{xn\log(xn) - xn\} \\ - \{(1-x)n\log((1-x)n) - (1-x)n\} + O(\log n).$$

The terms carrying $\log n$ collect to $n\log n\{\rho - x - (1-x)\} = (\rho-1)n\log n$, which is free of x, and the remainder is $n\Phi(x)$. Hence the maximizer is that of Φ, and $\Phi'(x) = \rho/x - \log x + \log(1-x) = 0$ is the stated equation. On $(0,1)$ the left side ρ/x decreases strictly from $+\infty$ to ρ and the right side $\log\{x/(1-x)\}$ increases strictly from $-\infty$ to $+\infty$, so the root is unique and is a maximum. □

Proof of Theorem 9.20. Write $u = (1+c)\alpha + a$. Integrating out π, $p(n \mid \alpha) = [\alpha]^n B(u, b+n)/\{n!\, B(c\alpha + a, b)\}$. As $n \to \infty$, $[\alpha]^n/n! \sim n^{\alpha-1}/\Gamma(\alpha)$ and $B(u, b+n) \sim \Gamma(u) n^{-u}$, uniformly on compact subsets of $[0,\infty)$. Hence $p(n \mid \alpha) \sim \mathcal{K}(\alpha) n^{-c\alpha-a-1}$ with $\mathcal{K}(\alpha) = \Gamma\{(1+c)\alpha + a\}/\{\Gamma(\alpha)B(c\alpha+a, b)\}$. Since $\Gamma(\alpha)^{-1} = \alpha + O(\alpha^2)$ at the origin, $\mathcal{K}(\alpha) = \alpha\Gamma(a)/B(a,b) + O(\alpha^2)$, and $p_\alpha(\alpha) = \varpi\alpha^{\mathbf{1}(m\geq 1)} + O(\alpha^{1+\mathbf{1}(m\geq 1)})$ by the same first order vanishing of $[\alpha]^m$. With $L = \log n$,

$$p(n) \sim n^{-a-1}\int_0^\infty e^{-cL\alpha}\mathcal{K}(\alpha)p_\alpha(\alpha)\,\mathrm{d}\alpha,$$

and Watson's lemma applies. With $\iota = 1 + \mathbf{1}(m \geq 1)$ we have $\mathcal{K}p_\alpha = \varpi\Gamma(a)B(a,b)^{-1}\alpha^\iota + O(\alpha^{\iota+1})$ and $\int_0^\infty \alpha^\iota e^{-cL\alpha}\,\mathrm{d}\alpha = \Gamma(1+\iota)/(cL)^{1+\iota}$, which gives (9.12) with relative error $O(1/L)$. □

Proof of Proposition 9.13(i). With $k = 1$, $R(\alpha) = \prod_{i=1}^{n-1}(\alpha+i)^{-1}$ and $\gamma_j = \prod_{i\neq j}(i-j)^{-1} = (-1)^{j-1}/\{(j-1)!(n-1-j)!\}$, so $|\gamma_j| = \binom{n-2}{j-1}/(n-2)!$. The central binomial coefficient is the largest, so the maximum is attained at $j - 1 = \lfloor (n-2)/2 \rfloor$ and equals $\binom{n-2}{\lfloor (n-2)/2\rfloor}/(n-2)! \sim 2^{n-2}\{2/(\pi n)\}^{1/2}/(n-2)!$ by Stirling. For the normalizer, $R(\alpha) = \Gamma(\alpha+1)/\Gamma(\alpha+n)$. Substitute $\alpha = u/\log n$; since $\Gamma(n)/\Gamma(n+u/\log n) = e^{-u}\{1 + O(1/n)\}$ uniformly for u in compacts, and $\Gamma(1+u/\log n) \to 1$, dominated convergence together with $\Gamma(1+x) = 1 + O(x)$ gives

$$(n-1)!\int_0^\infty R \;=\; \frac{1}{\log n}\int_0^\infty \Gamma\Big(1 + \frac{u}{\log n}\Big)\frac{\Gamma(n)}{\Gamma(n+u/\log n)}\,\mathrm{d}u \;=\; \frac{1 + O(1/\log n)}{\log n}.$$

Dividing, $\kappa_n = \binom{n-2}{\lfloor (n-2)/2\rfloor}(n-1)\log n\{1 + O(1/\log n)\}$, which is the stated formula. Numerically, $\log_{10}\kappa_n$ equals 3.171, 6.465, 12.733, 25.001 and 49.298 at

$n = 10, 20, 40, 80, 160$, against 3.127, 6.425, 12.698, 24.970 and 49.269 from the asymptotic formula; the gap, between 0.03 and 0.05 in the logarithm, is the $O(1/\log n)$ factor. □

Proof of Proposition 9.22. $\rho B(n, \rho+1) = \rho\Gamma(n)\Gamma(\rho+1)/\Gamma(n+\rho+1) \sim \rho\Gamma(\rho+1)n^{-(1+\rho)}$. Integrating against p_ρ and writing $n^{-\rho} = e^{-\rho \log n}$, Watson's lemma with $\rho\Gamma(\rho+1)p_\rho(\rho) = p_\rho(0)\rho + O(\rho^2)$ gives $p(n) \sim p_\rho(0) n^{-1}(\log n)^{-2}$. Summability follows from the integral test. □

Implementation notes

The evaluation is short. Given poles r_i with multiplicities μ_i and numerator P, form $h_i(\alpha) = P(\alpha)\prod_{l \neq i}(\alpha + r_l)^{-\mu_l}$, analytic at $-r_i$, and set $\gamma_{ij} = h_i^{(\mu_i - j)}(-r_i)/(\mu_i - j)!$; then apply (9.3), or (9.8) if a tilt is present. Set working precision to $\lceil N/2 \rceil + 15$ digits for kernels like those of Section 9.8; a kernel with k/n near 0.15 needs about $0.54N$. Before using the result, check the graded diagnostics of Lemma 9.4: the largest k for which $|\sum_i \gamma_i r_i^k| / \max_i |\gamma_i r_i^k|$ is still at the precision floor reports the achieved accuracy, and a failure at $k = 0$ means the answer is noise.

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PART III

Inference Without a Likelihood

CHAPTER 10

Sufficiency Without a Likelihood in Population Genetics

Approximate Bayesian Computation was invented in population genetics because coalescent likelihoods are intractable, and it has carried two acknowledged defects ever since: the analyst must choose summary statistics, so that the object computed is $\pi(\theta \mid S(y))$ rather than $\pi(\theta \mid y)$, and the analyst must choose a tolerance, which trades bias against acceptance rate. This chapter replaces the whole apparatus with a learned transport map. By the noise outsourcing lemma the posterior is *a deterministic function of the data and an independent uniform, and that function is the population minimiser of an expected check loss, so it can be fitted by regression on simulated parameter and data pairs. There is no summary statistic and no tolerance. The central result, Theorem 10.12, shows that if a permutation invariant architecture with a d dimensional bottleneck attains the Bayes risk of the check loss, then the bottleneck is a sufficient statistic. Sufficiency becomes something that can be tested by fitting, and the infinite alleles model, where Ewens proved that K_n is sufficient, becomes the natural validation problem. The method rediscovers that theorem: a one dimensional bottleneck does as well as an eight dimensional one and its learned scalar has Spearman correlation* 0.998 *with K_n, while a plausible but nonsufficient summary is* 56% *worse. On the infinite sites model we prove that Watterson's S is not sufficient, by an explicit calculation with the coalescent, and the learned posterior using the whole site frequency spectrum has* 90% *intervals about a quarter narrower in raw width than the exact posterior given S. Small worked examples, each checkable by a short program or by hand, accompany every result: rejection ABC on ten genes, the check loss for a single sampled gene, the law of S for five sequences, and the failure of sufficiency for three.*

10.1 Introduction

Population genetics has an unusual computational structure. Its forward models are easy to simulate and its likelihoods are essentially impossible to evaluate. One can generate a genealogy, drop mutations on it, allow recombination and selection, and read off the sample that would have been observed, in a few lines of code. To do inference in the classical way one needs instead the probability of the observed configuration as a function of the parameters, and that requires summing over every genealogy that could have produced it (Kingman, 1982).

This is why Approximate Bayesian Computation was invented here rather than anywhere else (Tavaré et al., 1997; Pritchard et al., 1999), and it is why the method has remained unsatisfactory in a way its users readily acknowledge. Rejection ABC

proposes $\theta^{(m)}$ from the prior, simulates $y^{(m)}$, and accepts when $\|S(y^{(m)}) - S(y)\| \leq \epsilon$ for a chosen summary S and a chosen tolerance ϵ. Both choices are consequential and the effect of the summary cannot be checked from the output.

This chapter removes both. Section 10.4 sets out the construction: the posterior is a deterministic transport map from an independent uniform to the parameter, and that map is the population minimiser of an expected check loss, so it is fitted by regression on simulated pairs. We call this Generative Bayesian Computation. It consumes the raw data, it has no tolerance, and it is amortised, so that the marginal cost of analysing a further dataset is a single evaluation of a fitted function.

Section 10.6 contains what we regard as the main idea. Removing the summary statistic is worth doing, but on its own it is a technical improvement. The more interesting question is the one the summary statistic was standing in for: what does a genetic sample actually tell you about the parameters, and how much of it can be discarded without loss? Theorem 10.12 answers this operationally. If one constrains the fitted map to factor through a d dimensional statistic and the constrained fit attains the same risk as the unconstrained one, then that statistic is sufficient. Sufficiency becomes something one can test by fitting rather than prove by algebra.

Sections 10.8 and 10.9 carry this out on two coalescent models. The first is the infinite alleles model of Ewens (1972), studied in Chapter 6, where the answer is known: K_n is sufficient for θ. It is one of the very few models in population genetics where a generative method can be validated against an exact posterior, and it is therefore the right proving ground. The second is the infinite sites model, where we prove (Proposition 10.25) that the number of segregating sites is *not* sufficient, and quantify what is lost.

The reader who wants to see the argument in miniature before the general theory can turn straight to Examples 10.6, 10.9 and 10.26. The first shows rejection ABC on ten genes against the exact posterior, the second fits a posterior quantile function by regression in a model where it is known in closed form, and the third proves by a two line calculation that the number of segregating sites in a sample of three sequences does not carry all the information about θ.

10.1.1 What the chapter adds

The observation that ABC conditions on the wrong thing is not new. The difficulty is acknowledged throughout the literature, and a substantial body of work addresses summary selection by projection, by regression adjustment, by entropy criteria and by information theoretic scores; Fearnhead and Prangle (2012) is a representative and influential example. It is worth saying precisely what is added here.

1. *A criterion about the posterior rather than about the summary, Theorem 10.12.* A bottleneck that attains the Bayes risk of the randomised check loss is Bayes sufficient, and, under the further conditions of that theorem, classically sufficient. Existing summary selection methods score a candidate summary;

this scores the resulting posterior, and it does so without knowing the right answer in advance.

2. *A quantitative version, Theorem 10.16.* The excess risk at a given bottleneck is squeezed between $\frac{\delta}{2}$ and $\frac{\delta}{2}$ times the expected squared Wasserstein distance between the full posterior and the posterior obtained from that bottleneck. Approximate sufficiency therefore has a rate, and the risk curve can be read on a posterior scale rather than in arbitrary loss units.

3. *A converse and a diagnostic, Proposition 10.21 and Corollary 10.22.* Sufficiency implies attainment as well, so within any nested class of statistics the curve $d \mapsto \mathcal{R}_d^\star$ is flat exactly from the smallest dimension at which the class contains a sufficient statistic onward. Remark 10.15 explains why the class must be restricted for the notion of dimension to mean anything.

4. *A validation problem that is not circular, Section 10.8.* Simulation based inference is typically demonstrated where the truth is unavailable, so that the demonstration reduces to agreement with another approximation. Ewens (1972) gives the sufficient statistic for the infinite alleles model and the posterior is then available by one dimensional quadrature, so a claim to have recovered sufficiency can be checked.

5. *A negative result with a proof, Proposition 10.25.* Watterson's S is not sufficient under the infinite sites model. The proof is explicit: given a single segregating site, the probability that it is carried by $n - 1$ of the n sequences varies with θ. The generative posterior then measures the cost.

6. *A reading of the whole construction as generalised weighted least squares, Section 10.5.* The criterion keeps the form of Fisher's, a loss minimised over functions with the data entering through a pooled statistic, and changes two things: the loss identifies a law rather than a mean, and the pooled statistic is fitted rather than specified.

10.1.2 Empirical findings

1. *The method rediscovers Ewens (1972).* On the infinite alleles model the held out check loss is 0.1164, 0.1178 and 0.1175 at bottleneck dimensions 1, 4 and 8, so widening the channel buys nothing, and the learned scalar has Spearman correlation 0.998 with K_n. A plausible but nonsufficient summary, the largest allele frequency, scores 0.1828, which is 56% worse than $d = 8$.

2. *The learned posterior matches the exact one in the centre.* Median relative errors in the $5, 25, 50, 75, 95$ per cent quantiles are 10.4%, 4.9%, 2.5%, 3.6% and 8.1%, with no likelihood evaluated at any point.

3. *Watterson's S discards roughly a tenth of the interval, a quarter in raw width.* On the infinite sites model the median width of the 90% interval, relative to the true θ, is 0.98 using the whole spectrum against 1.29 using the exact posterior given S

alone. The check loss also keeps falling from $d = 1$ to $d = 4$, by a gap whose 95% interval over 50 random seeds excludes zero, and the learned scalar at $d = 1$ already beats S.

4. *ABC pays for acceptance in bias.* At $\epsilon = 0$ the acceptance rate is about 1.05%, so 2×10^5 simulations yield about 2,100 draws for one dataset. At $\epsilon = 20$ acceptance rises to 66% while the posterior median falls from 6.0 to 3.0 and the relative width rises from 1.27 to 3.6.

5. *The generative fit is cheaper at a single dataset and amortises thereafter.* Training used 3×10^4 simulations in total. The crossover against rejection ABC occurs below one dataset, and at 10^4 loci the simulation ratio against ABC with fresh simulations for each locus is about 6×10^4.

6. *Calibration is imperfect, measurable, and repairable.* The fitted 90% interval covers in 83% of held out replicates against 90% for the exact posterior, because the extreme quantiles are pulled inward. Training harder and tail weighting the quantile levels leave this unchanged; making the quantile level enter through an explicit location scale term in $\Phi^{-1}(u)$ repairs it in the fits that converge, 12 of 20 seeds, and the failures are plain from their held out loss.

10.1.3 Relation to the companion literature

Three recent papers of Ewens and of Ewens and Hössjer supply both the motivation and, in one case, the validation problem.

Ewens (2024) argues that the direction of population genetics theory is now retrospective, driven by the volume of genomic data, and it is precisely retrospective inference that runs into intractable likelihoods: the prospective theory needs no likelihood for a sample, while asking what history produced an observed configuration needs one summed over every history that could have produced it. The same paper supplies, indirectly, the reason the validation strategy of this chapter is available at all. Ewens (1972) proved that the number of distinct alleles is sufficient for the mutation parameter in the infinite alleles model, and that theorem, together with the exact posterior of Section 10.8.1, gives the point in this subject at which a simulation based method can be checked against the truth rather than against another approximation.

Hössjer and Ewens (2026) construct a self contained continuous time theory of allele frequency change, showing that the accepted treatment silently borrows its generations concept from a discrete approximation, so that what is presented as a continuous time result is a restatement of a discrete time one. Our objection to ABC has the same shape. A method should not borrow its target from an arbitrary reduction of the data, and Proposition 10.1 says that ABC does exactly that: what is presented as $\pi(\theta \mid y)$ is $\pi(\theta \mid S(y))$, and the substitution leaves no trace in the output. The two remedies are parallel as well, each consisting of building the object from first principles rather than importing it from a neighbouring construction.

Ewens and Hössjer (2026) show that the inbreeding, variance and eigenvalue effective population sizes are three distinct functionals of a transition kernel which agree for the idealised model and disagree otherwise. Read as statistics rather than as genetics they are Kish's effective sample size, asymptotic variance efficiency, and the reciprocal spectral gap. That correspondence bears on the efficiency comparison of Section 10.10, since any honest accounting compares effective rather than nominal draws, and we return to it in Section 10.12.

10.2 The deficiencies of Approximate Bayesian Computation

This section states the four defects of rejection ABC as propositions, so that the size of each can be seen rather than asserted, and then works them out exactly on a sample of ten genes. Let $\pi(\theta)$ be a prior on $\Theta \subseteq \mathbb{R}$, let $p(y \mid \theta)$ be a model on a standard Borel space $\mathcal{Y}$ from which we can simulate but which we cannot evaluate, let P be the marginal law of the data, and let $S : \mathcal{Y} \to \mathbb{R}^{d_S}$ be a measurable summary. Rejection ABC with tolerance ϵ accepts a simulated pair when $\|S(Y) - S(y)\| \le \epsilon$, and the accepted values of θ are then an exact sample from

$$\pi_\epsilon(\theta \mid y) \;\propto\; \pi(\theta)\, \mathbb{P}\big(\|S(Y) - S(y)\| \le \epsilon \,\big|\, \theta\big). \tag{10.1}$$

At $\epsilon = 0$ and for a discrete summary this is to be read as $\pi(\theta)\mathbb{P}(S(Y) = S(y) \mid \theta)$.

D1. The target is the wrong conditional. Even at $\epsilon = 0$, (10.1) is a posterior given $S(y)$, not given y.

PROPOSITION 10.1 (Irreducible bias). *At $\epsilon = 0$ the target* (10.1) *equals $\pi(\theta \mid S(y))$. It equals $\pi(\theta \mid y)$ for P almost every y, for every prior with full support, if and only if S is sufficient for θ.*

The first statement is Bayes' theorem applied to the observation $S(Y) = S(y)$. The second is the definition of Bayes sufficiency together with the theorem of Kolmogorov, Halmos and Savage discussed in Section 10.6. The practical content is that when S is not sufficient the discrepancy is an irreducible bias: it does not shrink with the number of simulations, and no diagnostic computable from ABC output reveals it, because the output is a correct sample from the wrong target. Every convergence diagnostic an ABC user runs will pass.

D2. Tolerance trades bias for acceptance. The next proposition makes the bias of a positive tolerance explicit. Intuitively, a ball of radius ϵ around s averages the sampling density over the ball, and the first correction to an average over a symmetric ball is the Laplacian.

PROPOSITION 10.2 (Tolerance bias). *Let $s = S(y)$, let $f(\cdot \mid \theta)$ denote the density of $S(Y)$ given θ, and suppose that f is four times continuously differentiable in its first argument, with derivatives up to order four bounded uniformly in θ on a neighbourhood of s, and that the prior predictive density $m(s) = \int \pi(\theta) f(s \mid \theta)\, \mathrm{d}\theta$ is positive. Then as*

$\epsilon \downarrow 0$,

$$\mathbb{P}\big(\|S(Y) - s\| \le \epsilon \mid \theta\big) = V_{d_S} \epsilon^{d_S} \Big\{ f(s \mid \theta) + \tfrac{\epsilon^2}{2(d_S+2)}\, \Delta f(s \mid \theta) + O(\epsilon^4) \Big\},$$

where V_{d_S} is the volume of the unit ball in $\mathbb{R}^{d_S}$ and Δ is the Laplacian in s. Consequently

$$\pi_\epsilon(\theta \mid y) = \pi(\theta \mid s) \Big[1 + \tfrac{\epsilon^2}{2(d_S+2)} \Big\{ \frac{\Delta f(s \mid \theta)}{f(s \mid \theta)} - \frac{\Delta m(s)}{m(s)} \Big\} \Big] + O(\epsilon^4). \tag{10.2}$$

The proof is in Section 10.16. The bracket in (10.2) has posterior mean zero, so the tolerance does not move probability in or out, it moves it around: the ABC posterior gains mass where the relative curvature $\Delta f / f$ of the sampling density exceeds its posterior average and loses it elsewhere. The bias is therefore not uniform across θ; it is largest where the sampling density of the summary changes fastest with s, which is where the summary discriminates best. Section 10.10 shows the trade quantitatively.

D3. Acceptance rates degrade geometrically in d_S.

PROPOSITION 10.3 (Curse of summary dimension). *Under the conditions of Proposition 10.2 the unconditional acceptance probability is*

$$\alpha_\epsilon = \int \pi(\theta) \mathbb{P}\big(\|S(Y) - s\| \le \epsilon \mid \theta\big)\, \mathrm{d}\theta = V_{d_S} \epsilon^{d_S}\, m(s) \{1 + O(\epsilon^2)\}.$$

If a summary of dimension d_S with predictive density m_{d_S} is enlarged to one of dimension $d_S + 1$ with predictive density m_{d_S+1}, then holding the acceptance probability fixed requires the tolerance to change from ϵ to $\epsilon' = \{V_{d_S} m_{d_S}(s)\, \epsilon^{d_S} / V_{d_S+1} m_{d_S+1}(s')\}^{1/(d_S+1)}$ to leading order, so that ϵ'/ϵ grows like $\epsilon^{-1/(d_S+1)}$ as $\epsilon \downarrow 0$.

The display follows by integrating the expansion of Proposition 10.2 against the prior, and the statement about ϵ' by equating the two leading terms. Propositions 10.1 and 10.3 together are the central practical dilemma of ABC and the reason it has not been solved by better summaries. Enriching S is the only route to removing the D1 bias, and it is precisely what destroys the acceptance rate. The two defects cannot be attacked simultaneously within the rejection framework.

D4. Nothing is amortised. The value of amortisation is now well established outside this setting: Liu and Wong (2026) show that a single trained generative model can serve arbitrary conditional inferences without retraining, which is the same economy in a different guise.

PROPOSITION 10.4 (Cost accounting). *To obtain n_{eff} posterior draws for each of R datasets, rejection ABC with fresh simulations for each dataset requires on average $R\, n_{\mathrm{eff}} / \alpha_\epsilon$ model simulations. The generative alternative of Section 10.4 requires M simulations in total, independent of R and of n_{eff}.*

The first count is the mean of a negative binomial waiting time, since each proposal is accepted independently with probability α_ϵ. At the values measured in Section 10.10, $\alpha_0 \approx 1.05 \times 10^{-2}$, $n_{\text{eff}} = 2 \times 10^3$ and $M = 3 \times 10^4$, the crossover is at $R = M\alpha_0 / n_{\text{eff}} \approx 0.16$: the generative method is already cheaper for a single dataset, and its advantage grows linearly in the number of datasets thereafter when rejection ABC uses fresh simulations for each dataset. A table of simulated pairs shared across datasets avoids that linear growth, since its size is set by the dataset with the smallest acceptance probability.

REMARK 10.5. Regression adjustment (Beaumont et al., 2002) mitigates D2 and synthetic likelihood mitigates D3, but neither addresses D1, which is the defect that matters, because it is the one that cannot be detected from the output.

10.2.1 The four defects on ten genes

Everything in the propositions above can be computed exactly for a small sample under the infinite alleles model, because the number of possible configurations is small enough to list. The example below does so; it is the first of several in this chapter that a reader can check with a short program or, in places, by hand.

EXAMPLE 10.6 (Rejection ABC on ten genes). Take $n = 10$ genes under the infinite alleles model with the Pochhammer prior $p(\theta) = \{\log 2\,(\theta+1)(\theta+2)\}^{-1}$ on $(0, \infty)$, the heavy tailed default of Wang and Polson (2024), to which the chapter returns in Section 10.8. Its distribution function is $F(\theta) = \log_2\{2(\theta+1)/(\theta+2)\}$, so a prior draw is $\theta = 2(w-1)/(2-w)$ with $w = 2^V$, V uniform, and the prior median is $\sqrt{2} = 1.414$. Suppose the observed sample contains four alleles, carried by 5, 2, 2 and 1 genes. By the Ewens sampling formula (Chapter 6) its probability is

$$\frac{10!}{\theta_{(10)}} \cdot \frac{\theta^4}{1 \cdot 1! \;\cdot\; 2^2\, 2! \;\cdot\; 5 \cdot 1!} = \frac{90{,}720\,\theta^4}{\theta(\theta+1)\cdots(\theta+9)},$$

and the 42 partitions of 10 can be listed, so the law of any summary is a finite sum.

Table 10.1 gives the result of rejection ABC with three summaries. With the sufficient summary K_{10} at $\epsilon = 0$ the ABC target is the exact posterior, here with median 1.763 and 90% interval $(0.534,\ 5.264)$; summing over the 42 partitions confirms to machine precision that the posterior given the whole configuration is the same. One in eight proposals is accepted. Widening the tolerance to $\epsilon = 1$, so that samples with three, four or five alleles are accepted, triples the acceptance rate and lowers the median by 7%; at $\epsilon = 3$ almost nine proposals in ten are accepted and the interval has stretched from $(0.53,\ 5.3)$ to $(0.06,\ 8.2)$, drifting back toward the prior. This is D2: acceptance is bought with bias, and the bias is a smooth function of ϵ with no visible signal in the output.

The last two rows are D1. The largest allele count and the number of singletons are both natural summaries for a population geneticist, and at $\epsilon = 0$ each yields a perfectly valid posterior, but not the posterior of the data: given only the number of singletons the median falls to 1.02 and the interval to $(0.19,\ 4.0)$, a shift of more than 40% in the centre. Nothing in the accepted sample reveals this.

summary	ϵ	acceptance (exact)	acceptance (10^5 draws)	5%	median	95%
K_{10} (sufficient)	0	0.120	0.121	0.534	1.763	5.264
K_{10}	1	0.364	0.365	0.393	1.639	5.797
K_{10}	2	0.615	0.618	0.213	1.399	6.805
K_{10}	3	0.872	0.873	0.063	1.091	8.245
largest allele count	0	0.115	·	0.304	1.561	6.249
number of singletons	0	0.200	·	0.185	1.019	3.992
whole configuration (exact)	·	·	·	0.534	1.763	5.264

TABLE 10.1. Rejection ABC for $n = 10$ genes with observed allele counts $(5, 2, 2, 1)$, under the Pochhammer prior. The ABC target (10.1) is computed exactly by summing over the 42 partitions of 10 and integrating numerically in θ; quantiles are of the ABC target. Only the first row is the posterior of the data.

The Monte Carlo column was produced with 10^5 prior draws and the Feller coupling of Section 10.14; at $\epsilon = 0$ it accepted 12,067 draws with sample median 1.760, against the exact 1.763.

10.3 The other lineage: Monte Carlo likelihood

Before proposing a replacement it is only fair to record that ABC is not the sole response to an intractable coalescent likelihood, and that the alternative response does not suffer the defect we have just spent a section describing.

The Monte Carlo likelihood tradition begins with Geyer and Thompson (1992) and Geyer (1994), who show that a likelihood ratio $L(\theta)/L(\theta_0)$ can be estimated by importance sampling from the model at a reference parameter θ_0, even when neither likelihood is computable, and that maximising the resulting estimate is consistent under conditions they make precise. Geyer and Thompson (1995) then apply annealing and simulated tempering to exactly the setting of this chapter, ancestral inference, on a pedigree of two thousand individuals where a Gibbs sampler would not mix in any usable time. Thompson (1994, 2000) develops the approach systematically for genetic data on pedigrees, and the coalescent versions are the importance sampling of Griffiths and Tavaré (1994), the likelihood surfaces of Kuhner et al. (1995), and the improved proposals of Stephens and Donnelly (2000). The common device, treating the unobserved genealogy as missing data and averaging over it, is the data augmentation idea of Tanner and Wong (1987).

These methods target the full likelihood. They introduce the latent genealogy, sample it, and average, so the data are never reduced and Proposition 10.1 does not apply to them. On the criterion that matters most here they are therefore better placed than ABC, and it would be misleading to present the choice as one between ABC and the present proposal.

The difficulty is elsewhere, and it is a difficulty of variance rather than of bias. The estimated likelihood ratio is an importance sampling average whose weights degrade as θ moves away from the reference θ_0, so the usable range around θ_0 shrinks as the data grow, and one is driven to iterate, retune, or run at many

reference values. The latent space is the space of genealogies, which is enormous and awkwardly connected, so the chain must be helped along by tempering, which is what Geyer and Thompson (1995) supply and why their paper is about mixing as much as about genetics. And nothing is amortised: a second dataset requires the whole computation again, so Proposition 10.4 applies here as it does to ABC.

The generative construction of the next section makes a different trade. It requires only forward simulation, with no importance weights, no reference parameter and no chain on the latent space, and it is amortised. What it gives up is the exactness in the Monte Carlo limit that the likelihood based methods possess: their error vanishes as the sampling effort grows with the model fixed, whereas ours vanishes only as both the simulation budget and the approximating capacity grow. A reader who values that exactness above amortisation should prefer the older lineage, and for a single dataset with a well chosen reference parameter that preference is defensible.

10.4 Generative posteriors

This section shows that a posterior distribution on the line is a function, the conditional quantile function, and that this function can be learned by regression from simulated pairs without ever evaluating a likelihood. A small genetic example in which every quantity is known in closed form follows the two theorems.

10.4.1 The posterior is a transport map, not merely representable by one

Bayes' theorem is an identity among densities. It says what the posterior density is proportional to, and every classical computational method, Markov chain Monte Carlo included, is a scheme for exploring a density known up to a constant. When the likelihood cannot be evaluated, every such scheme fails at the first step, because the density is not merely awkward to normalise but unavailable pointwise. The following theorem removes the density from the problem altogether.

THEOREM 10.7 (Noise outsourcing). *Let (θ, Y) be a random element of $\Theta \times \mathcal{Y}$ with $\mathcal{Y}$ standard Borel. There exist a measurable $H : (0,1) \times \mathcal{Y} \to \Theta$ and $U \sim \mathrm{Unif}(0,1)$ independent of Y such that $\theta = H(U, Y)$ almost surely, on an extension of the probability space if necessary. When $\Theta \subseteq \mathbb{R}$ one may take $H(u, y) = F^{-1}_{\theta \mid Y = y}(u)$, the posterior quantile function, in which case the pair $(H(U, Y), Y)$ has the same joint law as (θ, Y) for any uniform U independent of Y.*

A proof is in Section 10.16. The statement is the transfer theorem of Kallenberg (2002, Theorem 6.10), and the quantile form is his randomisation lemma (Lemma 3.22). Its relevance here is that the posterior is not merely representable by a transport map, it *is* one, and a function can be approximated from examples of its inputs and outputs even when no formula for it exists. The examples are the simulated pairs. What was an integration problem becomes a regression problem, and regression is a problem for which we have machinery that requires evaluating nothing.

The price is that the map must be learned from a finite training set, so the resulting posterior is exact only in the limit of infinite simulation and infinite capacity. That is a different kind of error from the one ABC makes. ABC's error is a bias with a fixed target which no amount of computation removes when S is insufficient. The error here is an estimation error which shrinks with the simulation budget and which can be measured by the calibration diagnostic of Section 10.8.5. We regard the exchange of an unmeasurable bias for a measurable variance as the main practical argument for the method.

One further contrast matters for genetics. Nothing here estimates a density. Simulation based methods that fit $p(y \mid \theta)$ or $p(\theta, y)$ by normalising flows must contend with the dimension of y, which for a genomic configuration is large. The map H has a one dimensional output whatever the dimension of y, and the loss is a scalar regression loss. The dimension of y enters only through the capacity needed in the encoder φ of Section 10.6, and the exchangeability of the configuration, which the architecture (10.6) exploits, cuts that further.

10.4.2 Fitting the map by quantile regression

The tool that turns the transport map into a regression target is the check loss of Koenker and Bassett (1978). Let

$$\rho_u(z) = z\{u - \mathbf{1}(z < 0)\}, \qquad 0 < u < 1,$$

so that positive residuals are charged at rate u and negative ones at rate $1 - u$. For a single u the minimiser of $\mathbb{E}\rho_u(\theta - c)$ over constants c is the u quantile of θ; the next theorem averages over u as well, so that one regression recovers every quantile at once.

THEOREM 10.8 (The check loss identifies the posterior). *Let $\mathbb{E}|\theta| < \infty$ and suppose that for P almost every y the conditional distribution function F_y of θ given $Y = y$ is continuous and strictly increasing on the interior of its support. Then*

$$H^\star \;=\; \arg \min_{H \text{ measurable}} \; \mathbb{E}_{(\theta,Y)}\, \mathbb{E}_{U\sim\mathrm{Unif}(0,1)}\, \rho_U\big(\theta - H(U, Y)\big) \tag{10.3}$$

exists, is unique up to sets of measure zero in (u, y), and satisfies $H^\star(u, y) = F_y^{-1}(u)$.

The proof is in Section 10.16. Two features of (10.3) deserve emphasis. The expectation is over the *joint* law of (θ, Y), so it is estimated by simulating $\theta^{(m)} \sim \pi$ and $y^{(m)} \sim p(\cdot \mid \theta^{(m)})$; the likelihood is never evaluated. And the argument of H is y itself, so no reduction of the data is imposed. A useful identity, used repeatedly below, measures how far a constant c is from being optimal at level u: if F is the distribution function of θ and $q_u = F^{-1}(u)$, then

$$\mathbb{E}\big[\rho_u(\theta - c)\big] - \mathbb{E}\big[\rho_u(\theta - q_u)\big] \;=\; \int_{q_u}^{c} \big(F(t) - u\big)\, \mathrm{d}t, \tag{10.4}$$

valid for every real c (Exercise 1). The right side is nonnegative, since $F(t) - u$ has the sign of $t - q_u$, and it vanishes only at $c = q_u$ when F is strictly increasing there.

Algorithm (Generative Bayesian Computation).

1. Simulate $\{(\theta^{(m)}, y^{(m)})\}_{m=1}^{M}$ from the joint law.
2. Fit H_ϕ by stochastic minimisation of the empirical version of (10.3), drawing a fresh $u \sim \mathrm{Unif}(0,1)$ with each minibatch.
3. For observed y_{obs}, return $\{H_\phi(u_1, y_{\mathrm{obs}}), \dots\}$ for u_i uniform. These are posterior draws; $H_\phi(q, y_{\mathrm{obs}})$ is the qth posterior quantile directly.

Step 1 is the entire model input. Steps 2 and 3 involve no tolerance, no acceptance and no summary. Step 2 is done once.

EXAMPLE 10.9 (One sampled gene). Let θ be the frequency of allele A in a large population, with a uniform prior on $(0,1)$, and let $Y = 1$ if a single gene drawn at random carries A, so that $Y \mid \theta \sim \mathrm{Bernoulli}(\theta)$. The posterior is $\mathrm{Be}(2,1)$, with density $2t$, when $Y = 1$ and $\mathrm{Be}(1,2)$, with density $2(1-t)$, when $Y = 0$. Both distribution functions invert in closed form, $F_1(t) = t^2$ and $F_0(t) = 1 - (1-t)^2$, so the map of Theorems 10.7 and 10.8 is

$$H^\star(u,1) = \sqrt{u}, \qquad H^\star(u,0) = 1 - \sqrt{1-u}.$$

A posterior draw given $Y = 1$ is the square root of a uniform, which is the noise outsourcing of Theorem 10.7 with nothing hidden. The posterior median after seeing A is $1/\sqrt{2} = 0.707$.

The Bayes risk. For $Y = 1$ and level u, direct integration gives $\mathbb{E}[\rho_u(\theta - \sqrt{u}) \mid Y = 1] = \frac{2}{3}(u - u^{3/2})$, and integrating over u gives $\frac{2}{3}(\frac{1}{2} - \frac{2}{5}) = \frac{1}{15}$. By the symmetry $\theta \mapsto 1 - \theta$ the same holds for $Y = 0$, so $\mathcal{R}^\star = 1/15$. A map that ignores the data must use the prior quantile function $H(u) = u$, for which $\mathbb{E}\rho_u(\theta - u) = u(1-u)/2$ when θ is uniform, with average $1/12$. The excess risk of ignoring the sampled gene is therefore $1/12 - 1/15 = 1/60$. This is a first instance of Theorem 10.12 read in reverse: the constant statistic is not sufficient, and the check loss says so.

Learning the map from simulations. Draw M pairs $(\theta^{(m)}, Y^{(m)})$ and minimise the empirical check loss separately for each u and each value of Y. For a fixed u the minimiser of $\sum_m \rho_u(\theta^{(m)} - c)$ over constants c is an empirical u quantile (Koenker and Bassett, 1978), so the fitted map is the empirical quantile function of the simulated θ among pairs with the given Y. Table 10.2 shows the fit at five levels. At $M = 200$ the errors are a few hundredths; at $M = 20{,}000$ they are below 0.01 everywhere, and no likelihood was evaluated at any point.

10.5 Relation to weighted least squares

Before the sufficiency theory we locate the criterion (10.3) relative to weighted least squares. The comparison identifies which classical fact Theorem 10.12 generalises.

	$Y=1$				$Y=0$	
level u	exact $\sqrt{u}$	$M=200$	$M=2{,}000$	$M=20{,}000$	exact $1-\sqrt{1-u}$	$M=2{,}000$
0.05	0.224	0.225	0.244	0.230	0.025	0.025
0.25	0.500	0.516	0.489	0.505	0.134	0.129
0.50	0.707	0.678	0.704	0.714	0.293	0.290
0.75	0.866	0.878	0.864	0.868	0.500	0.494
0.95	0.975	0.968	0.975	0.975	0.776	0.785

TABLE 10.2. Example 10.9: the posterior quantile function of an allele frequency after one sampled gene, fitted by minimising the empirical check loss on M simulated pairs, against the closed form. Simulations are successive independent runs from one random seed (script `checkloss_bernoulli.py`).

10.5.1 Least squares returns one number, the check loss returns a law

Fisher's definition of the average effect, as restated by Ewens (2024, Equation 10), is the weighted least squares criterion

$$Q(\beta) \;=\; \sum_{g} P_g \Big(m_g - \sum_{(g)} \beta_{ij} \Big)^2, \tag{10.5}$$

where g runs over genotypes with frequencies P_g and mean values m_g, and the inner sum adds the effects β_{ij} of the alleles carried by g. Its minimiser is the $L^2(P)$ orthogonal projection of m onto the additive subspace. The general principle behind (10.5) is that minimising expected squared error over a class of functions returns the conditional mean: $\arg\min_f \mathbb{E}(Z - f(X))^2 = \mathbb{E}[Z \mid X]$. It is a first moment criterion and it returns a first moment.

Our criterion is the same idea with two changes. Replace the squared loss by the check loss ρ_u, and randomise over u. At $u = 1/2$, $\rho_{1/2}$ is half the absolute error and the minimiser is the conditional median, so a single u returns one number just as (10.5) does. Averaging over $u \sim \mathrm{Unif}(0,1)$ is what changes the character of the problem: by Theorem 10.8 the minimiser is the whole conditional quantile function, hence by Theorem 10.7 the whole posterior. The relation is therefore

$$\text{weighted least squares : conditional mean} \quad :: \quad \text{randomised check loss : conditional law.}$$

Both are regression problems, solved by fitting rather than by integration. The second is inferential rather than merely predictive because the randomisation over u recovers every quantile, and a distribution on the line is determined by its quantiles on a dense set. This is the precise sense in which the method is Bayesian and not simply a neural network fitted to predict a parameter.

10.5.2 The normal equations are the original bottleneck

The architecture (10.6) of the next section constrains the fitted map to factor through a pooled statistic $T_\varphi(y) = \sum_i \varphi(y_i)$. That constraint has a familiar special case.

Consider the linear model $y = X\beta + \varepsilon$ with $\beta \in \mathbb{R}^p$, $\varepsilon \sim N(0, \sigma^2 W^{-1})$, σ^2 and the diagonal weight matrix W known, and prior $\beta \sim N(0, \Sigma_0)$ with Σ_0 nonsingular. The weighted least squares normal equations are $X^\top W X\hat{\beta} = X^\top W y$, and the data enter only through $X^\top W y$, a p dimensional vector which is sufficient for β. Written in the pooled form,

$$X^\top W y \;=\; \sum_{i=1}^{N} w_i x_i y_i \;=\; \sum_{i=1}^{N} \varphi(x_i, y_i), \qquad \varphi(x_i, y_i) = w_i x_i y_i,$$

which is (10.6) with φ linear in y_i and $d = p$. So the deep sets bottleneck is the normal equations with the map φ learned instead of assumed. The posterior is $\beta \mid y \sim N(\mu(y), V)$ with $V = (X^\top W X/\sigma^2 + \Sigma_0^{-1})^{-1}$ and $\mu(y) = V X^\top W y/\sigma^2$.

Because Theorem 10.8 concerns a scalar parameter, the correspondence has to be stated one target at a time, and the answer then depends on how many targets share the bottleneck.

PROPOSITION 10.10 (Recovery of the classical case). *In the model above, with X of full column rank p:*

(i) For a single scalar target $\psi = c^\top \beta$, $c \neq 0$, the one dimensional linear statistic $T(y) = c^\top V X^\top W y/\sigma^2$ attains the Bayes risk of (10.3) *with $g(u, t) = t + (c^\top V c)^{1/2}\Phi^{-1}(u)$, which is affine in $\Phi^{-1}(u)$.*

(ii) If the p coordinates $\beta_1, \ldots, \beta_p$ are fitted jointly through one shared bottleneck, with loss the sum of the p check losses, then $T(y) = X^\top W y$ with $d = p$ attains the Bayes risk, and no continuous statistic of dimension $d < p$ does when g is continuous.

Proof. (i) The posterior of ψ is $N(c^\top \mu(y), c^\top V c)$, whose quantile function is $c^\top \mu(y) + (c^\top V c)^{1/2}\Phi^{-1}(u) = g(u, T(y))$; Theorem 10.8 gives attainment. (ii) The jth marginal posterior has quantile function $\mu_j(y) + V_{jj}^{1/2}\Phi^{-1}(u)$, so the vector $\mu(y) = V X^\top W y/\sigma^2$, which is an invertible linear image of $X^\top W y$, is what the bottleneck must carry; attainment at $d = p$ follows as in (i). Conversely, suppose a continuous statistic T attains the joint Bayes risk with a continuous g. By the uniqueness in Theorem 10.8 applied to each coordinate, $g_j(u, T(y)) = \mu_j(y) + V_{jj}^{1/2}\Phi^{-1}(u)$ for almost every (u, y), and so for every (u, y), since both sides are continuous and the marginal law of the data has a positive density on $\mathbb{R}^N$. Hence every $\mu_j(y)$ is a function of $T(y)$, so $\mu(y) = h(T(y))$ for some h. Since X has rank p and W is positive definite, $y \mapsto \mu(y)$ maps some p dimensional affine subspace A of the data space bijectively onto $\mathbb{R}^p$, and $\mu = h \circ T$ forces T to be injective on A. A continuous injective map from an open subset of $\mathbb{R}^p$ into $\mathbb{R}^d$ with $d < p$ does not exist, by invariance of domain, so $d \geq p$. □

The proposition fixes the correspondence and contains a warning. For one scalar parameter of interest a single number suffices even when the model has p coefficients, because the quantity of interest is a single linear functional; the dimension that the bottleneck test measures is the dimension needed for the targets

one actually fits. And the word *continuous* in (ii) is not decoration: Remark 10.15 shows that without it the notion of dimension collapses.

10.5.3 What is genuinely new

Three things do not carry over from the least squares picture, and they are the reason this chapter exists.

First, (10.5) requires the model to be written down. Its normal equations involve $X^\top W X$, which for the whole genome problem Ewens (2024) shows cannot be evaluated, and which for a coalescent problem cannot even be defined, since there is no design matrix. The criterion (10.3) requires only that one can simulate, which for coalescent models is easy.

Second, φ is learned. In the linear case the sufficient statistic is linear and known; in the coalescent case it is neither, and the whole difficulty of ABC is that the analyst has been asked to supply it by hand. Corollary 10.22 replaces that choice with a fitted curve. The idea of learning a low dimensional encoding and conducting inference through it is not new, and the closest antecedent we know is the encoding generative modelling of Liu et al. (2024), who learn a bidirectional map between a high dimensional covariate space and a low dimensional latent space and then model the dependence of treatment and response on subsets of the latent coordinates. The difference is what the encoding is asked to preserve. There it is the confounding structure needed for a causal estimand; here it is the whole conditional law of the parameter, and Theorem 10.12 supplies a certificate that the encoding has preserved it.

Third, the answer is a distribution and not a point. Weighted least squares delivers $\hat{\beta}$ and, with additional distributional assumptions, a standard error. The map $H^\star$ delivers the posterior, including its asymmetry and its tails, and it delivers it for every dataset at the cost of a single evaluation.

The summary is therefore this. We keep the form of Fisher's criterion, a loss minimised over a class of functions with the data entering through a pooled statistic. We change the loss so that the criterion identifies a law rather than a mean, and we let the pooled statistic be estimated rather than specified. Everything else, including the sufficiency theory of the next section, follows from those two changes.

10.6 Sufficiency by fitting

This section turns the regression of Section 10.4 into a test of sufficiency. The idea is simple: if the fitted map is forced to see the data only through a few numbers and it still does as well as an unconstrained map, those few numbers must carry everything the data say about θ.

The data in a coalescent problem is an unordered configuration: a set of allele block sizes, or a site frequency spectrum. We therefore build H so as to respect that

exchangeability,

$$H_{\phi,d}(u,y) \;=\; g\Big(u,\, \sum_i \varphi(y_i)\Big), \qquad \varphi : \mathcal{Y}_1 \to \mathbb{R}^d, \tag{10.6}$$

where $y = \{y_i\}$ is the configuration, $\mathcal{Y}_1$ is the space of its elements, φ and g are learned, and the pooled dimension d is a bottleneck we choose. Write $T_\varphi(y) = \sum_i \varphi(y_i) \in \mathbb{R}^d$ and let

$$\mathcal{R}(H) = \mathbb{E}_{(\theta,Y)}\mathbb{E}_U\, \rho_U(\theta - H(U,Y)), \qquad \mathcal{R}^\star = \inf_H \mathcal{R}(H),$$

the Bayes risk, attained by $H^\star$ of Theorem 10.8. Throughout, π denotes the prior used to simulate and P the marginal law of Y.

DEFINITION 10.11. $T : \mathcal{Y} \to \mathbb{R}^d$ is *Bayes sufficient* for θ under the prior π if the conditional law of θ given $Y = y$ equals its conditional law given $T(Y) = T(y)$ for P almost every y.

The next theorem is the central result of the chapter. Its content is that the check loss, having a unique minimiser, leaves no room for a constrained map to tie with the unconstrained one unless the constraint discards nothing.

THEOREM 10.12 (A bottleneck that attains the Bayes risk is sufficient). *Assume the conditions of Theorem 10.8. Suppose $H(u,y) = g(u,T(y))$ for some measurable g and T, and suppose $\mathcal{R}(H) = \mathcal{R}^\star$. Then T is Bayes sufficient for θ. If moreover $\mathcal{Y}$ is countable, as it is for allelic partitions and site frequency spectra, then T is sufficient in the Fisher Neyman sense for the family $\{p(\cdot \mid \theta) : \theta \in \Theta_0\}$ for a set Θ_0 of prior probability one; and if in addition π has full support and $\theta \mapsto p(y \mid \theta)$ is continuous for every y, then T is sufficient for the whole family.*

Sketch; full proof in Section 10.16. By Theorem 10.8 the minimiser is unique up to null sets, so $\mathcal{R}(H) = \mathcal{R}^\star$ forces $g(u,T(y)) = F_y^{-1}(u)$ for almost every (u,y). The left side is a function of $T(y)$ alone, hence so is the conditional quantile function, hence so is the conditional law, which is Bayes sufficiency. The passage to classical sufficiency is the theorem of Kolmogorov in the form of Halmos and Savage (1949); see also Schervish (1995, Theorem 2.14). □

In genetic terms, the theorem says that a network forced to summarise a sample of genes by d numbers, and losing nothing by it, has found a sufficient statistic, whether or not anyone can write down its formula.

COROLLARY 10.13 (The operational test). *Fit* (10.6) *at bottleneck dimensions d and $d' > d$ within a fixed class of encoders. If the held out risks agree to within Monte Carlo error and the class at d' contains a sufficient statistic, the d dimensional learned statistic is sufficient to that accuracy. If they do not, no statistic of dimension d in the class is sufficient, and the excess risk at dimension d measures the information such a summary must discard.*

Corollary 10.13 is what makes the framework useful beyond computation. It converts the question whether a summary statistic is good enough from a question requiring a proof into a question requiring two fits. The excess risk is a number on the scale of the loss, so it can be reported, and Theorem 10.16 below converts it to a posterior scale.

EXAMPLE 10.14 (One sampled gene, continued). In Example 10.9 the data take two values, so a bottleneck of dimension one can carry Y itself and attains $\mathcal{R}^\star = 1/15$, while the bottleneck of dimension zero, a constant statistic, has risk $1/12$. The excess $1/60$ is positive, so by Theorem 10.12 read in contrapositive the constant is not sufficient, as it obviously is not. With n sampled genes and a uniform prior, Y the number carrying A, a one dimensional bottleneck still attains the Bayes risk, because the count is sufficient, and the posterior quantile function given $Y = n$ is $u^{1/(n+1)}$ (Exercise 3).

10.6.1 What dimension means

The test compares bottlenecks of different dimension, and a reader trained in measure theory will object at once that dimension, for a measurable statistic, means nothing. The objection is correct and has to be met before the curve $d \mapsto \mathcal{R}^\star_d$ can be interpreted.

REMARK 10.15 (Dimension requires regularity). Every standard Borel space is Borel isomorphic to a subset of $\mathbb{R}$, so any statistic, including the identity $T(y) = y$, can be encoded measurably in a single real number. Over the class of all measurable statistics a one dimensional sufficient statistic therefore always exists. The encoding can even be chosen additive: for a site frequency spectrum $\xi = (\xi_1, \dots, \xi_{n-1})$ the pooled statistic $\sum_i \xi_i \sqrt{p_i}$, with p_i the ith prime, is injective on vectors of nonnegative integers, because the square roots of distinct primes are linearly independent over the rationals (Exercise 9). It is a one dimensional statistic of the form (10.6), and it is sufficient.

The dimension that the bottleneck test measures is therefore relative to a class $\mathcal{T}_d$ of admissible statistics: continuous maps when the data live in a continuum, as in Proposition 10.10, and for discrete data the statistics representable by networks of the given capacity with smooth activations applied to bounded features. A network of modest width cannot represent the prime encoding, which requires resolving irrational combinations of large integers, so the curve it traces is an *effective* dimension. That is what a scientist wants, a summary that is a smooth function of interpretable features, but it should be called what it is.

With this understood, fix nested classes $\mathcal{T}_1 \subseteq \mathcal{T}_2 \subseteq \cdots$, for instance by padding a d dimensional encoder with a zero coordinate, and define

$$\mathcal{R}^\star_d = \inf\{\mathcal{R}(H) : H(u, y) = g(u, T(y)),\ T \in \mathcal{T}_d,\ g \text{ measurable}\}.$$

10.6.2 A remark on the likelihood principle

Theorem 10.12 sits oddly against the tradition that takes the likelihood function as the primary object of inference, and the tension is worth stating rather than leaving for a reader to notice.

Edwards (1972) argues that the likelihood function, considered as a function of the parameter for the observed data, carries everything the data has to say, and that sufficiency is simply the statement that a reduction of the data leaves that function unchanged up to a factor. On that reading, sufficiency is a property of the likelihood and cannot be discussed without it. The Fisher Neyman criterion used in Proposition 10.24 is a factorisation of the likelihood; the Lehmann Scheffé criterion for minimality is a statement about ratios of likelihoods at two data points.

The construction here never forms that function. It has no access to $p(y \mid \theta)$ at any θ, and Section 10.9 applies it to a model where the function is not computable in practice at realistic sample sizes. Yet Theorem 10.12 recovers sufficiency, and Corollary 10.22 recovers minimal dimension. The resolution is that the reduction Edwards regards as fundamental is a property of the posterior as much as of the likelihood: Bayes sufficiency says the conditional law of θ given y depends on y only through $T(y)$, and for dominated families Kolmogorov's theorem makes the two notions the same when Bayes sufficiency is required under every prior, whereas for the single simulation prior the passage to classical sufficiency needs the further conditions of Theorem 10.12. What the generative construction exploits is that the posterior can be estimated by regression while the likelihood cannot be evaluated at all. That the likelihood function itself has delicate large sample behaviour in high or infinite dimensional parameter spaces, which Wong and Shen (1995) characterise, is a further reason not to make its evaluation a prerequisite for inference.

Two qualifications follow. First, the equivalence runs through a prior, so what is recovered is sufficiency relative to the simulation prior, as Remark 10.23 records; a reader committed to a likelihood only account will regard that as a weaker conclusion than the classical one, and it is. Second, nothing here bears on the likelihood principle as a prescription. We do not claim that inference should ignore the likelihood function, only that its central consequence for data reduction survives when the function is unavailable, which is the situation that population genetics has been in since Tavaré et al. (1997).

10.6.3 A quantitative version: excess risk bounds a posterior distance

Theorem 10.12 is an exact statement: attaining the Bayes risk implies sufficiency. In practice one never attains anything exactly, and the object one computes is a small positive excess risk. Corollary 10.13 then asks the reader to accept that a small excess means approximate sufficiency, which so far has been an appeal to continuity rather than an argument. The following supplies the argument. The intuition is that the check loss near its minimum is a quadratic whose curvature is

the posterior density, so an excess in loss is a squared displacement of quantiles weighted by density.

THEOREM 10.16 (Excess risk is a posterior distance). *Let $H(u,y) = g(u,T(y))$ with $u \mapsto g(u,T(y))$ nondecreasing, and write $\pi_T(\cdot \mid y)$ for the law with quantile function $u \mapsto g(u,T(y))$. Suppose that for P almost every y the posterior $\pi(\cdot \mid y)$ has a density f_y, and that there are constants $0 \le \delta \le \bar{\delta} < \infty$ such that for almost every (u,y) the inequality $\delta \le f_y \le \bar{\delta}$ holds on the closed interval with endpoints $F_y^{-1}(u)$ and $g(u,T(y))$. Then*

$$\frac{\delta}{2}\,\mathbb{E}_Y\, W_2^2\big(\pi(\cdot \mid Y),\, \pi_T(\cdot \mid Y)\big) \;\le\; \mathcal{R}(H) - \mathcal{R}^\star \;\le\; \frac{\bar{\delta}}{2}\,\mathbb{E}_Y\, W_2^2\big(\pi(\cdot \mid Y),\, \pi_T(\cdot \mid Y)\big),$$

where W_2 is the 2 Wasserstein distance on the line.

The proof is in Section 10.16 and rests on the identity (10.4): its right side is squeezed between $\frac{\delta}{2}(c-q_u)^2$ and $\frac{\bar{\delta}}{2}(c-q_u)^2$, and integrating over u turns $\int_0^1 (g(u,T(y)) - F_y^{-1}(u))^2\,\mathrm{d}u$ into W_2^2, which is the standard quantile representation of the Wasserstein distance on the line. The theorem says that the check loss is, up to constants, a squared transport distance between the posterior one has and the posterior one wants.

EXAMPLE 10.17 (The bounds for one sampled gene). In Example 10.9 take the constant map $H(u) = u$, so that π_T is the uniform prior. Given $Y = 1$ the posterior quantile function is $\sqrt{u}$, and

$$W_2^2\big(\mathrm{Be}(2,1), \mathrm{Unif}(0,1)\big) = \int_0^1 (\sqrt{u} - u)^2\,\mathrm{d}u = \tfrac{1}{2} - \tfrac{4}{5} + \tfrac{1}{3} = \tfrac{1}{30},$$

and the same for $Y = 0$ by symmetry. The posterior densities $2t$ and $2(1-t)$ are bounded by $\bar{\delta} = 2$, so the upper bound is $\frac{\bar{\delta}}{2} \cdot \frac{1}{30} = \frac{1}{30}$, against the exact excess $\frac{1}{60}$ computed earlier: the bound holds with a factor of two to spare. The lower bound is vacuous here, because both densities vanish at an endpoint and so $\delta = 0$. The example shows both what the theorem gives, a posterior scale for the excess risk, and what it needs, a density bounded away from zero, which fails in exactly the tails where calibration is hardest.

Three consequences follow.

COROLLARY 10.18 (Approximate sufficiency). *Under the conditions of Theorem 10.16 with $\delta > 0$, if $\mathcal{R}(H) - \mathcal{R}^\star \le \varepsilon$ then $\mathbb{E}_Y W_2\big(\pi(\cdot \mid Y), \pi_T(\cdot \mid Y)\big) \le \sqrt{2\varepsilon/\delta}$. In particular a small excess certifies that the posterior obtained from T is close to the full posterior in a metric that controls the expectations of all Lipschitz functions.*

The bound follows from the left inequality of the theorem and Jensen's inequality, $\mathbb{E}_Y W_2 \le (\mathbb{E}_Y W_2^2)^{1/2}$.

COROLLARY 10.19 (The risk curve is a distance curve). *The curve $d \mapsto \mathcal{R}_d^\star - \mathcal{R}^\star$ of Corollary 10.22 is, up to the factors $\delta/2$ and $\bar{\delta}/2$, the curve of squared Wasserstein*

distances between the full posterior and the best posterior obtainable from a d dimensional summary. Its elbow is therefore not merely where the fit stops improving; it is where the achievable posterior stops moving.

The third consequence is that the numbers reported in Sections 10.8 and 10.9 can be read, roughly, on a posterior scale. The bound of Corollary 10.18 needs $\mathcal{R}^\star$, which is unknown, and a lower bound δ on posterior densities, which does not exist in the tails. As a heuristic we replace $\mathcal{R}^\star$ by the held out risk at the widest bottleneck and take δ as the reciprocal of the prior range in $\log\theta$, so $\delta = 1/\log(25/0.2) = 0.207$ for the infinite alleles study and $\delta = 1/\log(30/0.5) = 0.244$ for the infinite sites study:

comparison	excess risk	$\sqrt{2\varepsilon/\delta}$ in $\log\theta$	factor in θ
largest allele frequency against $d = 8$ (Table 10.3)	0.0653	0.794	2.21
S only against $d = 4$ (Table 10.4)	0.0080	0.256	1.29

The numbers are indicative rather than certified, since neither substitution is exact, and the true displacements are probably smaller because the posterior densities are far from flat in their centres. Their value is that they are on a scale a reader can interpret and that they order the two cases correctly: the badly chosen summary of Section 10.8 can move the posterior by a factor of about two in θ, while Watterson's S, a well motivated summary supported by Proposition 10.27, is confined to a much smaller displacement, consistent with the narrowing of intervals measured directly in Section 10.9.

REMARK 10.20. Theorem 10.16 also explains why a negative result in the bottleneck test is safer than a positive one. A positive excess risk lower bounds a genuine posterior discrepancy through the right hand inequality, since $\mathbb{E}_Y W_2^2 \geq 2(\mathcal{R}(H) - \mathcal{R}^\star)/\bar{\delta}$, and therefore cannot be an artefact of the encoding. A near zero excess risk bounds the discrepancy from above only through δ, which may be tiny, and in any case the fitted H may be suboptimal for reasons of optimisation, which is why a flat curve must be corroborated by holding d fixed and varying only the capacity.

10.6.4 Overparameterisation of the bottleneck

Corollary 10.13 instructs the analyst to widen d until the risk stops falling, and a reader may reasonably worry that this invites overfitting: a bottleneck wider than necessary is an overparameterised model, and the classical account says that such models degrade. They do not degrade here, and it is worth saying why, since the reason is the same one that governs double descent in regression (Belkin et al., 2019; Polson and Sokolov, 2025).

Theorem 10.12 concerns the population risk, and the population risk $\mathcal{R}_d^\star$ is nonincreasing in d by construction, because the classes are nested. Overparameterising the bottleneck therefore cannot hurt the estimand. What

it can do is raise the estimation error at a fixed simulation budget, and that is an empirical question rather than a structural one. Section 10.8.4 answers it for the infinite alleles model: the differences between $d = 1$, 4 and 8 are of the same size as the variation across network widths at fixed $d = 1$, so the cost of an unnecessarily wide bottleneck is lost in the run to run noise.

The contrast with the regression case is instructive. There the pathology at the interpolation threshold arises because the minimum norm rule supplies an implicit and badly conditioned prior. Here there is no interpolation threshold, because the training set is generated on demand and can be enlarged at will, and no implicit prior, because the prior used to simulate is the one being conditioned on. The practical rule that follows is to err on the side of a wider bottleneck and let Corollary 10.22 locate the elbow, which is the opposite of the advice one would give when selecting summary statistics for ABC, where Proposition 10.3 makes every additional dimension expensive.

10.6.5 The converse, and minimal sufficiency

Theorem 10.12 runs one way: attaining the risk implies sufficiency. The converse holds without further conditions and is worth stating, since without it a failure to attain the risk could not be interpreted.

PROPOSITION 10.21 (Converse). *If $T : \mathcal{Y} \to \mathbb{R}^d$ is Bayes sufficient for θ then there exists a measurable $g : (0,1) \times \mathbb{R}^d \to \Theta$ with $\mathcal{R}(g(\cdot, T(\cdot))) = \mathcal{R}^\star$.*

Proof. Sufficiency gives that the law of θ given $Y = y$ equals its law given $T(Y) = T(y)$ for almost every y. Define $g(u,t) = F^{-1}_{\theta \mid T(Y)=t}(u)$, which is jointly measurable because a regular conditional distribution of θ given $T(Y)$ exists and the generalised inverse of a measurable family of distribution functions is measurable in (u,t). Then $g(u, T(y)) = F_y^{-1}(u) = H^\star(u,y)$ for almost every (u,y), and $H^\star$ attains $\mathcal{R}^\star$ by Theorem 10.8. □

Proposition 10.21 and Theorem 10.12 together say that the smallest d at which the Bayes risk is attainable within the classes $\mathcal{T}_d$ is exactly the smallest d for which $\mathcal{T}_d$ contains a sufficient statistic. Call it $d^\star$.

COROLLARY 10.22 (Reading off minimal sufficiency). *$\mathcal{R}^\star_d$ is nonincreasing in d, and if the infimum defining $\mathcal{R}^\star_d$ is attained for each d, then $\mathcal{R}^\star_d = \mathcal{R}^\star$ if and only if $d \geq d^\star$. The curve $d \mapsto \mathcal{R}^\star_d$ is therefore a complete description of how much a summary of each dimension in the class can retain, and its elbow locates $d^\star$.*

Corollary 10.22 is what Tables 10.3 and 10.4 estimate. In the infinite alleles model the curve is flat from $d = 1$, so $d^\star = 1$ and by Proposition 10.24 the minimal sufficient statistic is K_n. In the infinite sites model the curve is still falling at $d = 4$, so $d^\star > 1$ for the class the network represents, which is consistent with Proposition 10.25.

One caution. $d^\star$ as defined is the minimal dimension of a sufficient statistic in the class, not the dimension of the minimal sufficient statistic in the Lehmann

Scheffé sense. The two agree in the smooth cases of interest, for instance in a full rank exponential family with k natural parameters and continuous data, where both equal k, and they can differ otherwise, as Remark 10.15 shows.

REMARK 10.23 (Caveats). Three caveats apply. The theorem is about the population risk; in practice we compare held out empirical risks and the comparison inherits Monte Carlo error. Failure to attain the risk at dimension d may reflect optimisation failure rather than insufficiency, and a flat curve may reflect an optimiser that failed to exploit the extra capacity rather than sufficiency, so either result should be checked by increasing capacity in φ and g while holding d fixed. And the notion is Bayes sufficiency relative to the simulation prior; a prior with restricted support can make an insufficient statistic look sufficient.

10.7 A Gaussian example where everything is explicit

Before the coalescent it is worth running the machinery on a continuous problem where every object can be written down, so that the reader can see what the theorems are asserting.

Let $\theta \sim N(0, \sigma_0^2)$ and $y = (y_1, \dots, y_n)$ with $y_i \mid \theta \sim N(\theta, 1)$ independent. Then $\bar{y}$ is sufficient, and

$$\theta \mid y \sim N\Big(\frac{n\sigma_0^2}{1+n\sigma_0^2}\,\bar{y},\ \frac{\sigma_0^2}{1+n\sigma_0^2}\Big), \qquad H^\star(u, y) = \frac{n\sigma_0^2}{1+n\sigma_0^2}\,\bar{y} + \sqrt{\frac{\sigma_0^2}{1+n\sigma_0^2}}\,\Phi^{-1}(u).$$

Three things are visible in this formula. The map is affine in $\Phi^{-1}(u)$, which is the noise outsourcing of Theorem 10.7 made concrete. The map depends on y only through $\bar{y}$, which is the sufficiency of Theorem 10.12 made concrete: a bottleneck of dimension one attains the Bayes risk here and a bottleneck of dimension zero does not, since a constant map cannot track $\bar{y}$. And the posterior spread does not depend on y at all, which is special to the Gaussian case and is the reason this example is easy rather than typical.

Now consider what ABC would do. With the summary $S = \bar{y}$ and tolerance ϵ, the acceptance event $|\bar{y}_{\text{sim}} - \bar{y}| \le \epsilon$ has probability proportional to the density at $\bar{y}$ of $\bar{y}_{\text{sim}} + V$, where V is uniform on $(-\epsilon, \epsilon)$ and independent. ABC is therefore exact Bayes for a summary observed with additional uniform noise of variance $\epsilon^2/3$ (Exercise 4). Approximating the uniform by a normal of the same variance, the ABC posterior variance is

$$\Big\{\sigma_0^{-2} + \big(n^{-1} + \epsilon^2/3\big)^{-1}\Big\}^{-1},$$

so the tolerance acts exactly like a measurement error on the summary. At $n = 100$ and $\sigma_0 = 1$ the exact posterior variance is $1/101 = 0.00990$; at $\epsilon = 0.1$, about one posterior standard deviation, the formula gives 0.01316 and numerical integration of the exact ABC target gives 0.01316 as well, an inflation of 33%. At $\epsilon = 0.2$ the inflation is 130%, as the formula also gives.

With the summary $S = y_1$, a reasonable looking choice that discards $n-1$ observations, ABC at $\epsilon = 0$ returns $N(\sigma_0^2 y_1/(1+\sigma_0^2),\ \sigma_0^2/(1+\sigma_0^2))$, whose variance exceeds the correct posterior variance by a factor of $(1+n\sigma_0^2)/(1+\sigma_0^2)$. At $n = 100$ and $\sigma_0 = 1$ that factor is 50.5. The ABC output is a perfectly valid sample from a posterior; it is simply not the posterior of the data one collected, and nothing in the output says so.

This is the whole argument of the chapter in a case where it can be checked by hand. The rest of the chapter does the same thing where it cannot.

10.8 Worked example 1: the infinite alleles model

This section applies the sufficiency test to the one coalescent model where the answer is known in advance, so that the test itself is on trial.

10.8.1 Exact sufficiency and the exact posterior

Under the infinite alleles model with scaled mutation rate θ, a sample of n genes induces a random partition Π of $\{1,\dots,n\}$ into allelic classes. For a partition with k blocks of sizes $n_1,\dots,n_k$ the Ewens sampling formula (Chapter 6) gives

$$p(\Pi \mid \theta) \;=\; \frac{\theta^k}{\theta_{(n)}} \prod_{i=1}^{k} (n_i - 1)!, \qquad \theta_{(n)} = \frac{\Gamma(\theta+n)}{\Gamma(\theta)} = \theta(\theta+1)\cdots(\theta+n-1). \tag{10.7}$$

PROPOSITION 10.24 (Ewens, 1972). *K_n, the number of distinct alleles, is sufficient for θ, and it is minimal sufficient.*

Proof. Factorise (10.7) as $h(\Pi)\, g(K_n(\Pi), \theta)$ with $h(\Pi) = \prod_i (n_i - 1)!$ and $g(k,\theta) = \theta^k/\theta_{(n)}$; the Fisher Neyman criterion gives sufficiency. For minimality, note that for two partitions Π, Π' of the same n, $p(\Pi \mid \theta)/p(\Pi' \mid \theta) = \{h(\Pi)/h(\Pi')\}\,\theta^{K_n(\Pi)-K_n(\Pi')}$, which is free of θ if and only if $K_n(\Pi) = K_n(\Pi')$. By the Lehmann Scheffé criterion the partition of the sample space induced by K_n is minimal sufficient. □

The genetic reading is the Feller coupling. Genes arrive one at a time and the ith founds a new allele with probability $\theta/(\theta+i-1)$, independently of how earlier genes were distributed among alleles; K_n counts the foundings, and only the number of foundings, not how later genes were distributed among the founded alleles, depends on θ.

By Proposition 10.24 the likelihood is $L(\theta) \propto \theta^{K_n}\Gamma(\theta)/\Gamma(\theta+n)$, the law of K_n itself being $\mathbb{P}(K_n = k \mid \theta) = |s(n,k)|\,\theta^k/\theta_{(n)}$ with $|s(n,k)|$ the unsigned Stirling numbers of the first kind. With the heavy tailed Pochhammer prior $p(\theta) \propto \{(\theta+1)(\theta+2)\}^{-1}$ of Wang and Polson (2024), the posterior is available to high precision by one dimensional quadrature, and Chapter 9 studies it in detail. This is our gold standard. Example 10.6 has already shown it at $n = 10$; the rest of the section uses $n = 100$.

10.8.2 Architecture and training

We record the implementation so that the numbers below can be reproduced and so that the capacity caveat of Remark 10.23 can be assessed.

The map (10.6) uses φ and g each a three layer perceptron with 64 hidden units and SiLU activations, $\varphi : \mathbb{R}^2 \to \mathbb{R}^d$ acting on per block features and $g : \mathbb{R}^{1+d} \to \mathbb{R}$ acting on the concatenation of u with the pooled statistic. Pooling is a plain sum over blocks with zero padding masked out, so that a constant φ reproduces K_n exactly and the network is not prevented from finding the right answer by its architecture. The target is $\log \theta$ rather than θ, which stabilises the check loss across the prior range. Training uses Adam at learning rate 3×10^{-3} with cosine decay, minibatches of 256, and a fresh $u \sim \mathrm{Unif}(0,1)$ per example per step, so that no quantile level is ever revisited. Held out risk is averaged over six independent draws of u across the test set.

Three design points matter. First, the same architecture and the same budget are used at every d, so the comparison in Table 10.3 isolates the bottleneck; Section 10.8.4 then varies capacity at fixed d to confirm that the budget is not the binding constraint. Second, the network never sees K_n, S, or any other summary. It sees the raw configuration, and any statistic it uses it must construct, which is what makes the recovery of K_n a finding rather than an input.

Third, and less obviously, the per block features are chosen to make the right answer representable without making it easy. Supplying $(\log n_i,\ n_i/n)$ rather than n_i alone matters because block sizes under the Ewens sampling formula are heavy tailed: if C_j denotes the number of alleles carried by exactly j genes, then $\mathbb{E}C_j \to \theta/j$ as $n \to \infty$, so a linear feature is dominated by the few large blocks while the information about θ sits in the many small ones. Neither feature is a constant, so φ must learn to ignore its inputs in order to produce K_n; the architecture permits that, since a bias term suffices, but does not hand it over. Had we included a constant feature the $d = 1$ result would have been vacuous.

The masking deserves a word as well. Configurations are zero padded to a fixed width and the pooled sum excludes the padding, so a sample with few alleles and a sample with many are treated identically up to the number of terms in the sum. Without the mask the padding would itself encode K_n, and the sufficiency test would again be vacuous.

10.8.3 Does the method find the theorem?

We simulate $M = 12{,}000$ training pairs with $n = 100$ genes, drawing θ from the Pochhammer prior truncated to $[0.2, 25]$ and generating the exact partition by the Feller coupling. The map (10.6) takes as input the raw unordered set of block sizes, with per block features $(\log n_i,\ n_i/n)$, and no summary is supplied. We fit at $d \in \{1, 4, 8\}$, and additionally fit a deliberately misspecified model that sees only the largest allele frequency, which is the kind of plausible summary an ABC analyst might choose.

	$d = 1$	$d = 4$	$d = 8$	largest allele frequency only
held out check loss	0.11644	0.11779	0.11749	0.18277
difference from $d = 8$	-0.0011	$+0.0003$	·	$+0.0653$

TABLE 10.3. Infinite alleles model, $n = 100$. Widening the bottleneck from one to eight dimensions does not lower the risk, so by Corollary 10.13 the learned scalar is sufficient to within Monte Carlo error. A plausible but nonsufficient summary is 56% worse than $d = 8$.

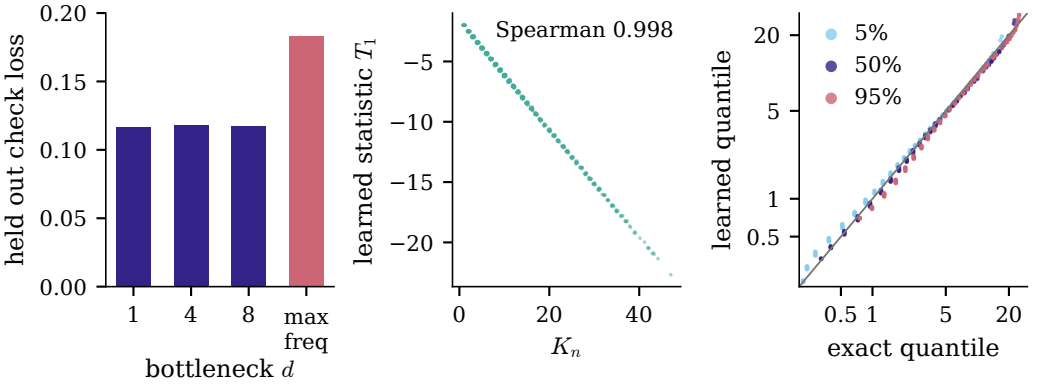


FIGURE 10.1. Infinite alleles model, $n = 100$. Left: held out check loss at bottleneck dimensions 1, 4 and 8 and for the largest allele frequency alone; widening the bottleneck buys nothing, while the nonsufficient summary is badly penalised. Centre: the learned one dimensional statistic against K_n, Spearman 0.998. Right: learned posterior quantiles at levels 5%, 50% and 95% against the exact posterior quantiles, computed with no likelihood evaluation.

Two conclusions follow. First, by Corollary 10.13 the one dimensional bottleneck is sufficient, recovered without any reference to Proposition 10.24; indeed the $d = 1$ fit has the lowest loss of the three, which shows how little the wider channels add. Second, the learned scalar has Spearman correlation 0.998 with K_n in absolute value over held out samples, so the network has not merely found *a* sufficient statistic, it has found Ewens's. Figure 10.1 shows the relation: the learned statistic is a decreasing, almost linear function of K_n.

Against the exact posterior of Section 10.8.1, the median relative errors in the learned $5, 25, 50, 75, 95$ per cent quantiles are 10.4%, 4.9%, 2.5%, 3.6% and 8.1%. The learned 90% interval covers the truth in 82.9% of held out replicates against 89.8% for the exact posterior, so at this training budget the method is overconfident in the tails. We report this because it is the honest limitation: the centre of the posterior is recovered accurately and the extreme quantiles are the hardest part of the fit, which is what one would expect from a check loss with few observations at extreme u. Section 10.8.5 diagnoses the cause and Section 10.8.6 repairs it in the fits that converge.

10.8.4 Capacity control

Remark 10.23 concedes that a flat risk curve could reflect an optimiser which failed to exploit the extra capacity rather than genuine sufficiency. The concession is testable and we test it. In a separate set of runs of the same design, on the same training data with an independent held out sample of 5,000 pairs, each configuration fitted from 20 random seeds, holding $d = 1$ and widening φ and g gave

$d = 1$, hidden width	64	128	256
held out check loss, mean of 20 seeds	0.11688	0.11695	0.11746
standard deviation over seeds	0.00047	0.00037	0.00155

Quadrupling the width does not lower the risk at all; the widest network is if anything slightly worse and less stable, and no difference in loss between widths has a 95% Welch interval that excludes zero. That is the signature of a budget which is not binding, not of capacity starvation. The same runs at $d = 4$ and $d = 8$ give mean losses 0.11689 and 0.11671, so the mean gaps among $d = 1$, 4 and 8 are at most 0.0002, again with intervals that include zero, while single fits at $d = 1$ range from 0.11632 to 0.11813. Run to run variation at a fixed bottleneck is larger than the effect of the bottleneck.

This is what a genuine flat region looks like, and it is the control that Corollary 10.13 requires before a positive result may be believed. It also calibrates how finely the risk curve can be read: at this training budget and with a single run per configuration, differences below roughly 0.002 should not be interpreted. The infinite sites comparison of Section 10.9 therefore uses three random seeds per configuration, and a separate study uses 50, so that its smaller differences can be judged against their own spread.

10.8.5 Calibration

Because the map is fitted, it can be miscalibrated, and because the joint law is available by simulation, miscalibration can be measured exactly. The diagnostic is the one of Cook et al. (2006): if H_ϕ were the true posterior quantile function then for a simulated pair (θ, y) the rank of θ under the fitted map, the u with $H_\phi(u, y) = \theta$, would be uniform, so the proportion of held out replicates with θ below the fitted qth quantile should equal q.

In the main run the fitted 90% central interval covers in 82.9% of held out replicates against 89.8% for the exact posterior. The centre is accurate, with a median relative error of 2.5% in the posterior median, and the error grows toward the tails, reaching 10.4% at the fifth percentile and 8.1% at the ninety fifth. This is the expected failure mode for a check loss: extreme u are visited rarely during training and the corresponding quantiles are estimated from correspondingly less information.

The obvious remedies do not work, and we report the fact because it locates the difficulty correctly. In the second set of runs, with its independent held out sample and 20 seeds per configuration, tripling the number of training steps, and separately drawing u from a $\mathrm{Be}(1/2, 1/2)$ proposal with importance reweighting so that extreme quantiles are visited far more often, both left the coverage where it was (means over seeds):

	coverage of nominal 90%	held out check loss
uniform u, 700 steps	0.836	0.11688
uniform u, 2100 steps	0.838	0.11621
tail weighted u, 2100 steps	0.837	0.11629
exact posterior	0.893	·

The mechanism is not undertraining but shrinkage of the extreme quantiles toward the centre. In the main run the signed median relative errors are $+10.4\%$ at the fifth percentile and -7.4% at the ninety fifth, so the errors point inward from both sides and the fitted 90% interval is 0.86 of the correct width. That is a bias in the fitted quantile function, of the kind familiar from any smooth regression of a quantity that varies sharply with the covariate, and it will not be removed by visiting the tails more often.

It also exposes a real limitation of Theorem 10.16. The excess risk here is small, so the theorem certifies that the fitted posterior is close to the exact one in W_2, and it is. But W_2 integrates the quantile discrepancy uniformly in u, and a fourteen per cent deficit in interval width contributes very little to it relative to the overall posterior spread. Wasserstein closeness therefore controls the bulk of the posterior and is nearly blind to tail calibration, as Example 10.17 anticipated. A small excess risk is not a substitute for calibrating directly.

10.8.6 A parametrisation that fixes it

The diagnosis suggests the repair. If the difficulty is that extreme quantiles are regressed toward the centre, the remedy is to arrange for them to be extrapolated rather than fitted. Replace (10.6) by

$$H(u, y) \;=\; \mu\big(T_\varphi(y)\big) \;+\; \exp\big\{\varsigma(T_\varphi(y))\big\}\,\Phi^{-1}(u) \;+\; \eta\, c\big(\Phi^{-1}(u),\, T_\varphi(y)\big), \quad (10.8)$$

with μ, ς, φ and c learned and η a small fixed constant, taken here as 0.3. The quantile level enters through $\Phi^{-1}(u)$ inside an explicit location scale term, so the map is exactly linear in $z = \Phi^{-1}(u)$ up to the correction c. It is increasing in u, hence a legitimate quantile function, whenever $\eta\,|\partial c/\partial z| < \exp\{\varsigma(T_\varphi(y))\}$ pointwise, a condition we encourage by taking η small. In the fits reported below the five fitted quantile levels stayed in order on every held out sample, but the condition did not hold everywhere: in 8 of the 12 converged fits $\partial H/\partial z$ turns negative somewhere in $|z| \le 3.5$, so taking η small does not by itself guarantee a monotone quantile function, and an application should constrain c or check

monotonicity on a grid as we did. Nothing else changes: the loss is still (10.3), the bottleneck is still T_φ, and Theorems 10.12 and 10.16 apply unaltered, since they constrain only how the data enter. When $c \equiv 0$ the family (10.8) contains the exact answer of the Gaussian example of Section 10.7, which is the reason to expect it to extrapolate well.

On the same held out sample as the second set of runs, again with 20 seeds, the fit is bimodal. In 8 of the 20 seeds the bottleneck never finds K_n (Spearman correlation 0.94 to 0.98 rather than 0.998), and the held out check loss, 0.128 to 0.160, is far worse than any plain fit; these failures are obvious from the loss alone, which is how one would screen them in practice. In the other 12 the repair is complete. The table gives means over seeds, for the plain parametrisation at 2100 steps and for the 12 converged location scale fits.

	coverage	5%	50%	95%	width ratio
plain, u as input (10.6)	0.838	$+9.4\%$	$+0.2\%$	-8.4%	0.86
location scale (10.8)	0.899	-1.8%	$+0.6\%$	$+0.7\%$	1.02
exact posterior	0.893	·	·	·	1.00

Coverage moves from 0.838 to 0.899 against the exact posterior's 0.893 on the same held out sample. The signed errors at the fifth and ninety fifth percentiles fall from $+9.4\%$ and -8.4% to -1.8% and $+0.7\%$, and, decisively, they no longer point inward from both sides, so the systematic narrowing is gone and the width ratio is 1.02. The held out check loss also improves, from 0.11621 to 0.11546, and every converged fit, between 0.11527 and 0.11568, beats every plain fit of the second set of runs, which is the best we report on this problem; the parametrisation is not buying calibration at the expense of accuracy.

We draw two lessons. The narrow one is that (10.8) should be the default architecture for this class of problem, fitted from several seeds and screened by held out loss, and that the cost of the plain parametrisation is paid entirely in the tails where it is least visible. The broader one concerns what the theory does and does not certify. Theorem 10.16 was satisfied by both rows of the table, since both have small excess risk, and it was the direct calibration check rather than the certificate that distinguished them. A generative posterior must be calibrated before it is believed, in a way that an exact posterior need not be, and the diagnostic is cheap because it uses simulations one has already generated.

10.9 Worked example 2: the infinite sites model

This section applies the same test where the classical summary is not sufficient. We first prove that it is not, by an explicit calculation, then compute its exact law, and finally measure what discarding the rest of the data costs.

10.9.1 Watterson's S is not sufficient

Consider n sequences under the standard coalescent (Chapter 11) with the infinite sites mutation model at scaled rate θ, and suppose the ancestral state at each site is known. While there are k lineages the waiting time to the next coalescence is $T_k \sim \text{Exp}\{k(k-1)/2\}$, independently for $k = n, \dots, 2$ and independently of the tree topology. Let ℓ_i be the total length of branches subtending exactly i leaves, $i = 1, \dots, n-1$, let $L = \sum_i \ell_i = \sum_{k=2}^{n} k T_k$ be the total tree length, let $\xi = (\xi_1, \dots, \xi_{n-1})$ be the site frequency spectrum, so that ξ_i is the number of sites at which the derived allele is carried by exactly i sequences, and let $S = \sum_i \xi_i$ be the number of segregating sites. Write $a_n = \sum_{j=1}^{n-1} j^{-1}$ and $b_n = \sum_{j=1}^{n-1} j^{-2}$.

Before the proposition, the intuition. Given the tree, mutations fall on it as a Poisson process, so the spectrum given S and the tree is multinomial with probabilities ℓ/L, free of θ. But the tree is not observed, and the number of mutations S is informative about the tree length L in a way that depends on θ: for the same S, a small θ suggests a long tree and a large θ a short one. Long trees and short trees have different shapes, so the conditional law of the spectrum given S depends on θ through this back door. The proposition makes the back door explicit at $S = 1$.

PROPOSITION 10.25. *Conditionally on the coalescent tree, $\xi_i \sim \text{Poisson}(\theta\ell_i/2)$ independently, so that $S \sim \text{Poisson}(\theta L/2)$ and $\xi \mid S, \text{tree} \sim \text{Multinomial}(S;\ \ell/L)$, free of θ. Unconditionally, let e_{n-1} be the spectrum with a single site carried by $n-1$ sequences, and*

$$\phi_n(\theta) = \mathbb{P}(\xi = e_{n-1} \mid S = 1, \theta) = \frac{\mathbb{E}\left[\ell_{n-1} e^{-\theta L/2}\right]}{\mathbb{E}\left[L\, e^{-\theta L/2}\right]}.$$

Then for $n \geq 3$, $\phi_n(0+) = 1/\{(n-1)a_n\}$ and

$$\phi_n'(0+) = -\frac{a_n - b_n}{(n-1)\, a_n^2} < 0.$$

In particular ϕ_n is not constant in θ, and S is not sufficient for θ.

Proof. Given the tree, $\mathbb{P}(\xi = e_{n-1} \mid \text{tree}) = (\theta\ell_{n-1}/2)e^{-\theta L/2}$ and $\mathbb{P}(S = 1 \mid \text{tree}) = (\theta L/2)e^{-\theta L/2}$; averaging over the tree and dividing gives the formula for ϕ_n. Branches subtending $n-1$ leaves exist only while two lineages remain, and one of the two does so exactly when the root split is $\{1, n-1\}$. Under the coalescent the size of one side of the root split is uniform on $\{1, \dots, n-1\}$ (Kingman, 1982), so $\ell_{n-1} = N T_2$ with N an indicator independent of the T_k and $\mathbb{E}N = 2/(n-1)$. With $\mathbb{E}T_k = 2/\{k(k-1)\}$, $\mathbb{E}T_2^2 = 2$ and the independence of the T_k,

$$\mathbb{E}\ell_{n-1} = \frac{2}{n-1}, \qquad \mathbb{E}[\ell_{n-1} L] = \frac{2}{n-1}\left(2\mathbb{E}T_2^2 + \sum_{k=3}^{n} k\, \mathbb{E}T_k\right) = \frac{2}{n-1}\,(2 + 2a_n),$$

and $\mathbb{E}L = 2a_n$, $\mathrm{Var}L = \sum_k k^2 \mathrm{Var}T_k = 4b_n$, so $\mathbb{E}L^2 = 4b_n + 4a_n^2$. The functions $\lambda \mapsto \mathbb{E}[\ell_{n-1}e^{-\lambda L}]$ and $\lambda \mapsto \mathbb{E}[Le^{-\lambda L}]$ are differentiable at $\lambda = 0$ from the right because L has all moments. Hence, with $\lambda = \theta/2$, $\phi_n(0+) = \mathbb{E}\ell_{n-1}/\mathbb{E}L = 1/\{(n-1)a_n\}$ and

$$\begin{aligned}\frac{\mathrm{d}\phi_n}{\mathrm{d}\lambda}\Big|_{0+} &= \frac{\mathbb{E}\ell_{n-1}}{\mathbb{E}L}\Big\{\frac{\mathbb{E}L^2}{\mathbb{E}L} - \frac{\mathbb{E}[\ell_{n-1}L]}{\mathbb{E}\ell_{n-1}}\Big\} \\ &= \frac{1}{(n-1)a_n}\Big\{2a_n + \frac{2b_n}{a_n} - 2 - 2a_n\Big\} = -\frac{2(a_n - b_n)}{(n-1)a_n^2}.\end{aligned}$$

Dividing by two gives $\phi_n'(0+)$, which is negative because $1/j^2 < 1/j$ for $j \geq 2$, so $a_n > b_n$ when $n \geq 3$. A function with nonzero right derivative at 0 takes different values at two points of $(0, \infty)$. Sufficiency of S would require the conditional law of ξ given $S = 1$, an event of positive probability, to be the same for all θ, which fails. □

The genetic content of Proposition 10.25 is that the *shape* of the spectrum carries information about the total tree length, which in turn carries information about θ. A lone variant carried by all but one sequence sits on a root branch, and root branches are long precisely in long trees; as θ grows, the trees compatible with a single mutation get shorter, and the chance that the mutation sits on a root branch falls. This is why Watterson's estimator is known to be inefficient relative to the full data likelihood (Felsenstein, 1992; Fu and Li, 1993). What the generative framework adds is that the loss can now be measured rather than argued.

EXAMPLE 10.26 (Three sequences). For $n = 3$ everything is explicit. The spectrum is (ξ_1, ξ_2), with $\ell_1 = 3T_3 + T_2$ and $\ell_2 = T_2$, where $T_3 \sim \mathrm{Exp}(3)$ and $T_2 \sim \mathrm{Exp}(1)$. Integrating the Poisson probabilities against these exponentials gives rational functions of θ; for example

$$\mathbb{P}(\xi = (0,1) \mid \theta) = \frac{\theta}{(\theta+1)^2(\theta+2)}, \qquad \mathbb{P}(\xi = (1,0) \mid \theta) = \frac{\theta(3\theta+4)}{(\theta+1)^2(\theta+2)^2},$$

whose sum is $\mathbb{P}(S = 1 \mid \theta) = 2\theta(2\theta+3)/\{(\theta+1)^2(\theta+2)^2\}$, in agreement with Proposition 10.27 below. The first of these takes one line: $\mathbb{E}[(\lambda T_2)e^{-3\lambda T_3 - 2\lambda T_2}] = \lambda(1+\lambda)^{-1}(1+2\lambda)^{-2}$ with $\lambda = \theta/2$. Hence

$$\phi_3(\theta) = \mathbb{P}(\xi = (0,1) \mid S = 1, \theta) = \frac{\theta+2}{2(2\theta+3)},$$

which decreases from $1/3$ at $\theta = 0$ to $1/4$ as $\theta \to \infty$, with slope $-1/18$ at the origin, as Proposition 10.25 predicts since $a_3 = 3/2$ and $b_3 = 5/4$. A simulation of 4×10^5 coalescent trees gives 0.322 at $\theta = 0.2$ and 0.279 at $\theta = 3$, against the exact 0.324 and 0.278.

The failure is visible in likelihood ratios. Two data sets with the same $S = 1$, a singleton and a doubleton, have likelihood ratio

$$\frac{\mathbb{P}(\xi = (0,1) \mid \theta)}{\mathbb{P}(\xi = (1,0) \mid \theta)} = \frac{\theta+2}{3\theta+4},$$

which is 0.43 at $\theta = 1$ and 0.375 at $\theta = 4$; a sufficient S would make it constant. With $S = 2$ the three possible spectra give

$$\frac{L(1)}{L(4)} = \frac{4625}{3136} = 1.47, \qquad \frac{625}{352} = 1.78, \qquad \frac{125}{64} = 1.95$$

for $(2,0)$, $(1,1)$ and $(0,2)$ respectively. Two doubletons, both mutations on the internal branch, favour small θ most strongly, because they point to a long internal branch and hence a long tree. Under the log uniform prior on $[0.5, 30]$ used in Section 10.9.4, the posterior medians given the full spectrum are 1.83, 1.66 and 1.58, against 1.73 given $S = 2$ alone. The differences are modest at $n = 3$, where the spectrum has only two cells, and grow with n as the shape of the spectrum becomes more informative about the tree.

10.9.2 What the second coordinate is

Proposition 10.25 says a second number is needed; it does not say which. The fitted map answers that question empirically, and the answer is recognisable.

Write $\pi_{\mathrm{T}} = \binom{n}{2}^{-1} \sum_{i=1}^{n-1} i(n-i)\xi_i$ for the mean pairwise difference and recall Tajima's D (Tajima, 1989), which contrasts π_{T} with S/a_n, standardised by an estimate of the standard deviation of the difference. On held out data the two learned coordinates (T_1, T_2) of the $d = 2$ fit (not the waiting times T_k above) divide the labour cleanly. The coordinate T_2 carries the overall mutation count, with absolute Spearman correlations 0.988 with S and 0.944 with π_{T}. The coordinate T_1 is almost uncorrelated with both, at 0.04 and 0.12; it is not a measure of how many mutations there are. Figure 10.2 shows the split: colour, which is $\log \theta$, changes along T_2 and hardly at all along T_1.

What then is T_1? Regressing each learned coordinate on a cubic in the separately fitted $d = 1$ scalar, which is itself a near monotone function of S, and correlating the residual with D gives absolute correlations 0.73 for T_1 and 0.44 for T_2. The coordinate that carries no count information is, to a substantial degree, a spectrum shape contrast of Tajima's kind, which is what Proposition 10.25 predicts, since the shape of the spectrum is what carries information about the total tree length.

We read this cautiously. The correlation with D is substantial but not close to one, and under strict neutrality D has mean near zero by construction, so there is limited shape variation for the network to exploit. The clean test of the idea is the one described in Section 10.11: run the same procedure under selection, where D has genuine signal, and see whether the learned second coordinate tracks it. That experiment is not in this chapter.

10.9.3 The exact law of the number of segregating sites

To quantify what S discards we need the exact posterior given S, which is available through a classical representation due to Watterson (1975).

PROPOSITION 10.27. *Under the standard coalescent with infinite sites mutation at scaled rate θ,*

$$S \overset{d}{=} \sum_{j=1}^{n-1} G_j, \qquad G_j \text{ independent}, \quad \mathbb{P}(G_j = k) = \Big(\frac{\theta}{\theta+j}\Big)^k \frac{j}{\theta+j},\ k \geq 0.$$

Consequently $\mathbb{E}S = \theta a_n$, which is Watterson's estimating equation, and $\mathrm{Var}S = \theta a_n + \theta^2 b_n$.

Proof. While the ancestral process has k lineages, mutations arrive at total rate $k\theta/2$ and coalescences at rate $k(k-1)/2$, as independent Poisson processes. By the competing exponentials property, each successive event is a mutation with probability $(k\theta/2)/\{k\theta/2 + k(k-1)/2\} = \theta/(\theta+k-1)$, independently of everything before, until the first coalescence ends the phase. The number of mutations occurring while there are k lineages is thus geometric on $\{0,1,\dots\}$ with success probability $(k-1)/(\theta+k-1)$, and the phases are independent by the strong Markov property. Writing $j = k-1$ and summing over $k = 2,\dots,n$ gives the display. A geometric variable with success probability $j/(\theta+j)$ has mean θ/j and variance $\theta/j + \theta^2/j^2$, and the moments of S follow by summation. □

The probability generating function is the product $\mathbb{E}z^S = \prod_{j=1}^{n-1} j/(j+\theta-\theta z)$, so the law of S is a convolution of $n-1$ geometric mass functions and $\pi(\theta \mid S)$ can be computed to machine precision on a grid or by adaptive quadrature. This is what we use as the comparison in Table 10.5 and Figure 10.2. The proposition also explains why S is a natural summary: it has a clean law, and Watterson's estimator S/a_n falls out of it. Proposition 10.25 says that naturalness is not sufficiency.

EXAMPLE 10.28 (The law of S for five sequences). For $n = 5$, $S = G_1 + G_2 + G_3 + G_4$ with success probabilities $j/(\theta+j)$, $a_5 = 25/12$ and $b_5 = 205/144$. Two values are immediate from the generating function:

$$\mathbb{P}(S = 0 \mid \theta) = \prod_{j=1}^{4} \frac{j}{\theta+j}, \qquad \mathbb{P}(S = 1 \mid \theta) = \mathbb{P}(S = 0 \mid \theta) \sum_{j=1}^{4} \frac{\theta}{\theta+j},$$

so at $\theta = 1$ these are $24/120 = 1/5$ and $\frac{1}{5} \cdot \frac{77}{60} = 77/300 = 0.2567$. The full mass functions at $\theta = 1$ and $\theta = 3$, obtained by convolving the four geometric laws, are

s	0	1	2	3	4	5	6	7	8	≥ 9
$\theta = 1$	0.2000	0.2567	0.2111	0.1423	0.0861	0.0487	0.0264	0.0140	0.0072	0.0075
$\theta = 3$	0.0286	0.0651	0.0935	0.1085	0.1111	0.1049	0.0937	0.0804	0.0670	0.2472

with means $25/12 = 2.083$ and $25/4 = 6.25$ and variances 3.507 and 19.06, exactly as the proposition gives. A simulation of 4×10^5 coalescent trees with Poisson mutations reproduces every entry to within 0.002. The variance at $\theta = 3$ is three times the mean, not equal to it as a Poisson count would be: the extra term $\theta^2 b_n$ is

	S only	$d = 1$	$d = 2$	$d = 4$
held out check loss, mean of three seeds	0.10550	0.09834	0.09831	0.09748
range over seeds	0.10540 to 0.10565	0.09823 to 0.09850	0.09730 to 0.09889	0.09712 to 0.09776

TABLE 10.4. Infinite sites model, $n = 30$. Unlike Table 10.3, the risk falls as the bottleneck widens from $d = 1$ to $d = 4$, and a rerun with 50 seeds per configuration puts the gap between them clear of zero, so by Corollary 10.13 no one dimensional statistic of the kind the network represents is sufficient. The learned scalar at $d = 1$ also beats Watterson's S by a wide margin: even among one dimensional summaries, S is not the best.

the randomness of the tree, and it is the reason a single locus is so uninformative. Under the log uniform prior on $[0.5, 30]$ the posterior given $S = 3$ has median 1.62 and 90% interval $(0.60, 5.91)$, a tenfold range, against Watterson's estimate $3/a_5 = 1.44$.

10.9.4 Numerical results

We simulate $n = 30$ sequences, $M = 30{,}000$ training replicates, θ log uniform on $[0.5, 30]$, using exact coalescent simulation with branch wise Poisson mutation. The input to (10.6) is the raw spectrum as the set $\{(i, \xi_i)\}$. We also fit a baseline that sees only S, and we compute the exact posterior $\pi(\theta \mid S)$ from Proposition 10.27. Because the differences to be resolved here are smaller than in Section 10.8, every configuration is fitted from three random seeds.

Two things in Table 10.4 are worth separating. The gap of 0.0072 between the S only column and $d = 1$ says that S is not the best scalar summary. The gap of 0.0009 between $d = 1$ and $d = 4$ says that no scalar summary in the class suffices, a wider statement than Proposition 10.25, which concerns S alone, and one detected here by fitting; it is small, and three seeds alone cannot settle it, so we refitted every configuration from 50 seeds on the same simulated data. The mean gap between $d = 1$ and $d = 4$ is then 0.00088, with 95% Welch interval $(0.00068, 0.00107)$, and $d = 2$ sits between, 0.00040 below $d = 1$ and 0.00047 above $d = 4$, each with an interval that excludes zero. Single fits still overlap: a $d = 1$ fit beats a $d = 4$ fit in about one pair in ten, which is why one run per configuration cannot resolve a difference of this size. The seeds vary only the training randomness, not the simulated data, so the interval measures optimisation noise at this training budget, and the reading rests on the fitted class, not on a proof. The learned scalar at $d = 1$ has Spearman correlation 0.991 with S in absolute value, so it is a reweighted count of segregating sites rather than something exotic.

In posterior terms the loss is substantial. Over 500 held out datasets the median width of the 90% interval, divided by the true θ, is 0.980 using the whole spectrum against 1.294 using the exact posterior given S alone, a reduction of 24%. On the same datasets the generative interval covers the truth in 85.8% of cases against 92.0% for the exact posterior given S, so part of the narrowing is the tail shrinkage diagnosed in Section 10.8.5, which in Section 10.8 cost about 14% of width. The remainder, roughly ten per cent of the interval, is information in the shape of the

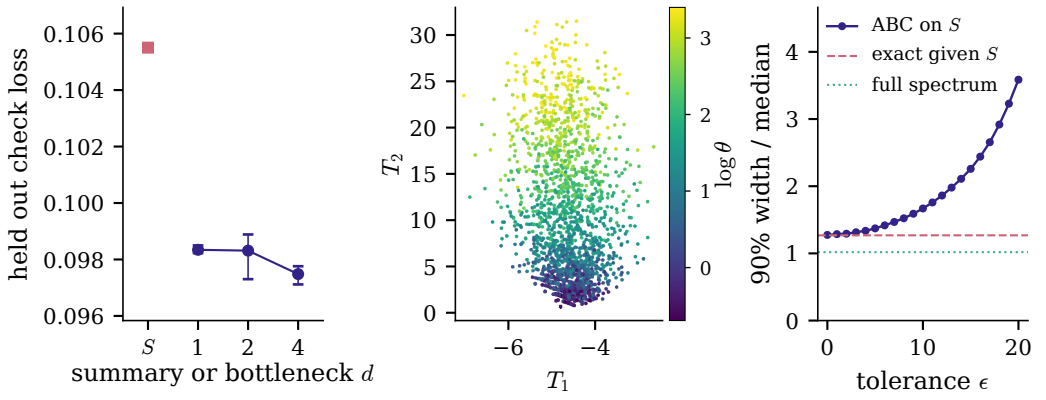


FIGURE 10.2. Infinite sites model, $n = 30$. Left: held out check loss for S alone and for bottlenecks $d = 1, 2, 4$, mean and range over three seeds; unlike the infinite alleles case, widening the bottleneck continues to help, so no scalar summary in the class suffices, a wider statement than Proposition 10.25, which concerns S alone. Centre: the learned two dimensional bottleneck on held out data, coloured by $\log \theta$; T_2 carries the mutation count and T_1 the shape of the spectrum. Right: for the observed dataset of Section 10.10, the ABC posterior width, 90% interval divided by median, as a function of tolerance, against the exact posterior given S (width 1.27) and the generative posterior using the whole spectrum (width 1.02).

spectrum that S throws away, and the location scale parametrisation of Section 10.8.6 would be used in any application to separate the two cleanly. Discarding the shape of the spectrum in favour of its total is not a harmless simplification.

10.9.5 The two examples read together

The two worked examples are chosen to be a matched pair, and the contrast between Tables 10.3 and 10.4 is the empirical core of the chapter.

In both cases the same kind of architecture, the same loss and the same protocol are applied, and the only thing varied is the bottleneck dimension. In the infinite alleles model the risk curve is flat from $d = 1$: the values 0.1164, 0.1178 and 0.1175 at $d = 1, 4, 8$ differ by less than the run to run variation, so by Corollary 10.22 we read $d^\star = 1$. In the infinite sites model the curve is still falling at $d = 4$, from 0.0983 to 0.0975, by a gap whose interval over 50 seeds excludes zero, so $d^\star > 1$.

That contrast is the whole diagnostic, and it is worth being clear about what makes it credible. If the procedure reported $d^\star = 1$ everywhere, it would be uninformative. If it reported $d^\star > 1$ everywhere, it would be detecting optimisation failure rather than insufficiency. It reports $d^\star = 1$ where Ewens (1972) proves that it should and $d^\star > 1$ where Proposition 10.25 proves that S fails, with no tuning between the two runs. The scalar it finds in the first case is K_n, to Spearman 0.998.

The second half of the argument is that the excess risk is not merely detectable but material. A 56% excess for the largest allele frequency summary, a choice an ABC analyst might plausibly have made, corresponds to discarding most of the

information. Watterson's S is a far better summary, chosen for good reasons and supported by Proposition 10.27, and it still costs a substantial part of the posterior interval. The lesson is not that summaries are useless but that their cost is invisible unless something like Corollary 10.22 is computed, and that a well motivated summary can still be expensive.

10.10 Efficiency

This section compares rejection ABC and the generative fit on one dataset, then on many. We use the infinite sites problem of Section 10.9.4 with an observed dataset simulated at $\theta = 8$, and give ABC the benefit of the best simple scalar, S itself.

EXAMPLE 10.29 (One simulated locus). The observed sample, a clearly labelled simulation rather than real data, has $n = 30$ sequences and $S = 23$ segregating sites, with spectrum

$$\xi_1 = 4, \quad \xi_2 = 1, \quad \xi_3 = 3, \quad \xi_7 = 7, \quad \xi_{17} = 1, \quad \xi_{20} = 1, \quad \xi_{23} = 6,$$

and all other $\xi_i = 0$. Watterson's estimate is $23/a_{30} = 23/3.962 = 5.81$, the mean pairwise difference is $\pi_{\mathrm{T}} = 6.73$, and Tajima's $D = 0.56$. Under the log uniform prior on $[0.5, 30]$ the exact posterior given $S = 23$, computed from Proposition 10.27 by adaptive quadrature, has median 6.00, mean 6.37 and 90% interval $(3.19,\ 10.81)$, so its relative width, the interval length divided by the median, is 1.269; the maximum likelihood estimate from S alone is 6.11.

The spectrum has a striking shape. Thirteen of the 23 variants sit at frequencies $7/30$ and $23/30$, which are complementary: they are the mutations on the two branches below the root of a genealogy whose first split separates seven sequences from twenty three. So many mutations on the root branches say that those branches are long, hence that the tree is long, hence that 23 mutations reflect a smaller θ than a typical tree would suggest. This is Proposition 10.25 at work on a single dataset, and the generative posterior using the whole spectrum duly moves down, to a median of 4.76, with relative width 1.02. Here the move is away from the true value 8, which is a reminder that one dataset proves nothing either way: what matters is calibration over many datasets, which the previous section measured.

Table 10.5 shows rejection ABC on this dataset, with 2×10^5 prior draws and acceptance when $|S_{\mathrm{sim}} - 23| \leq \epsilon$. Since the ABC target (10.1) depends only on the law of S, it can also be computed exactly, and the table gives both. At $\epsilon = 0$ rejection ABC samples exactly from $\pi(\theta \mid S = 23)$, so its median, 5.96 from 2,094 draws, agrees with the exact 6.00 to within Monte Carlo error, whose standard deviation is about 0.06.

The bias of tolerance in Table 10.5 is the finite tolerance counterpart of the bias of Proposition 10.2: the Monte Carlo and exact rows agree, so none of the drift is sampling error. Up to $\epsilon = 5$ the median moves by less than three per cent while the acceptance rate rises elevenfold; beyond that the drift accelerates, and at $\epsilon = 20$ the median has halved.

tolerance ϵ	0	2	5	10	20
rejection ABC, 2×10^5 *simulations*					
acceptance rate	1.05%	5.39%	11.9%	24.1%	65.6%
accepted draws	2094	10773	23815	48232	131153
posterior median	5.96	5.94	5.82	5.36	3.02
relative posterior width	1.275	1.293	1.372	1.669	3.586
exact ABC target (10.1)					
acceptance probability	1.06%	5.33%	11.9%	24.1%	65.6%
posterior median	6.00	5.97	5.84	5.37	3.00
relative posterior width	1.269	1.290	1.374	1.668	3.614

TABLE 10.5. Rejection ABC for the simulated locus of Example 10.29, true $\theta = 8$, observed $S = 23$, prior log uniform on $[0.5, 30]$. Relative width is the 90% interval length divided by the median. At $\epsilon = 0$ the target is the exact $\pi(\theta \mid S = 23)$, with median 6.00 and relative width 1.269; the Monte Carlo values agree with it to within sampling error. Tolerance buys acceptance and pays in bias: at $\epsilon = 20$ the posterior has collapsed towards the prior, whose median is 3.87.

The arithmetic of D4 is the decisive point. ABC at $\epsilon = 0$ required 2×10^5 simulations to produce about 2,100 draws for a single dataset. The generative fit used 3×10^4 simulations in total and thereafter returns posterior draws for any dataset at the cost of a single evaluation. For a study with 10^4 independent loci and rejection ABC with fresh simulations for each locus, the comparison is approximately 1.9×10^9 simulations against 3×10^4, a factor of about 6×10^4, and the generative posterior conditions on the whole spectrum while the ABC posterior conditions on S.

10.10.1 The genome scale comparison

Proposition 10.4 becomes vivid when the number of datasets is realistic. A modern study analyses each of many loci, or each of many windows along a genome, under a common model. Writing R for that count and taking the measured $\alpha_0 = 1.05 \times 10^{-2}$ and a target of $n_{\text{eff}} = 2 \times 10^3$ draws per locus, for rejection ABC with fresh simulations for each locus:

number of loci R	1	10^2	10^4	10^6
rejection ABC simulations	1.9×10^5	1.9×10^7	1.9×10^9	1.9×10^{11}
generative simulations	3×10^4	3×10^4	3×10^4	3×10^4
ratio	6.3	6.3×10^2	6.3×10^4	6.3×10^6

Two qualifications. The generative row assumes a common sample size across loci, and the caveat of Section 10.11 about varying n applies; training across a range of n raises M by a modest factor, not by a factor of R. And the comparison is against rejection ABC; sequential methods reduce the ABC row, though they do

not amortise it, since each new dataset restarts the sequence. A table of simulated pairs shared across loci avoids the linear growth of the ABC row, since its size is set by the locus with the smallest acceptance probability.

The more important point is not the ratio but what is being purchased. The ABC column buys R posteriors conditioned on S; the generative column buys R posteriors conditioned on the whole configuration, which by Section 10.9 are materially narrower. The method is simultaneously cheaper and conditioning on more.

10.11 Extensions and limitations

This section records where the method should be taken next and where it is weakest.

10.11.1 Where the programme goes next

The two examples were chosen because in one the answer is known and in the other it can be computed. Neither is where the method is needed. Three targets follow immediately.

The ancestral recombination graph. With recombination the data is a collection of correlated genealogies and the likelihood is intractable in a stronger sense than under the infinite sites model, since even the one dimensional summaries lack closed form laws. The architecture of Section 10.8.2 extends by treating the data as a set of site patterns with position features, and the bottleneck test of Corollary 10.13 applies unchanged. The interesting output is not the posterior for the recombination rate, which several methods already estimate, but the dimension at which the held out risk stops falling, which is an estimate of how many numbers a recombining sample is worth.

Selection. Under directional or balancing selection the genealogy is no longer the neutral coalescent and the frequency spectrum is distorted in ways that Tajima's D and its relatives are designed to detect. Those statistics are hand built two dimensional summaries. The framework here fits the analogous object rather than positing it, and the comparison between the learned two dimensional bottleneck and the pair (S, π_{T}) is a direct test of how much the classical construction leaves on the table.

Multiple merger coalescents. For Λ coalescents the sufficiency question is genuinely open and the simulation is straightforward, which is the combination the method wants. Here the bottleneck dimension is itself the scientific quantity: it says how many independent features of the spectrum distinguish a multiple merger history from a neutral one.

10.11.2 Limitations

The test is a comparison of fits, not a proof. Corollary 10.13 concerns population risks; we compare held out empirical risks. A negative result, meaning that the risk keeps falling as d grows, is the safer of the two, since it exhibits an actual improvement. A positive result, meaning that widening the bottleneck buys nothing, is weaker: it could reflect an optimiser that failed to exploit the extra capacity. The remedy is the one adopted in Section 10.8.2, holding the architecture and budget fixed and varying only d, and then checking that increasing capacity at fixed d does not help either. Section 10.8.4 carries out that check. And, as Remark 10.15 explains, the dimension measured is relative to what the network can represent.

Sufficiency is relative to the simulation prior. Theorem 10.12 delivers Bayes sufficiency with respect to the prior used to generate training data, and the passage to classical sufficiency requires that prior to have full support. A prior concentrated on a narrow range of θ can make an insufficient statistic appear sufficient, because within that range there is little for the discarded information to distinguish. Example 10.26 shows the mechanism: the likelihood ratio between two spectra with the same S changes only from 0.45 to 0.33 as θ runs from 0.5 to infinity, so a prior on a short interval would barely see it. The prior range must therefore be reported, and it should be wider than the range of scientific interest.

Amortisation assumes a fixed sample size and design. The trained map takes inputs of a fixed shape. Datasets with different n require either retraining, or an architecture that takes n as an additional input and is trained across a range of n. The latter is straightforward and is what one would do in practice, but it consumes part of the efficiency advantage claimed in Proposition 10.4.

Model misspecification is untouched. Everything here conditions on the simulator being correct. If the coalescent with the assumed demography is wrong, the generative posterior is confidently wrong, and it is arguably more confidently wrong than ABC, since it does not have a tolerance that inadvertently widens the answer. The check on this is predictive: simulate from the fitted posterior and compare the implied spectra with the observed one.

We compare against rejection ABC. Table 10.5 uses plain rejection ABC. Regression adjustment, ABC MCMC and sequential Monte Carlo variants all improve on it, in some cases substantially, and a full comparison would include them. Our defence is that all of them inherit Proposition 10.1 unchanged, since none of them alters the summary, and Proposition 10.1 is the defect we are addressing.

10.12 Connection to effective sample size

There is a second bridge to the companion literature which is worth recording, because it runs in the opposite direction to everything above: from genetics to computation.

Ewens and Hössjer (2026) show that the inbreeding, variance and eigenvalue effective population sizes are three distinct functionals of a transition kernel. Read as statistics rather than as genetics they are three familiar objects. The inbreeding size $N_e^{(i)} = (2\pi_2)^{-1}$, where π_2 is the probability that two genes descend from a common parent gene, counts identity by descent and is Kish's effective sample size. The variance size $N_e^{(v)} = x(1-x)/(2V)$, for allele frequency x and one generation variance V of its change, is asymptotic variance efficiency. The eigenvalue size $N_e^{(e)} = \frac{1}{2}(1-\lambda_{\max})^{-1}$ is the reciprocal spectral gap, that is the relaxation time, which in Markov chain Monte Carlo governs the integrated autocorrelation time.

Their Table 2, in which $N_e^{(i)}$ over k generations converges to $N_e^{(e)}$, is then the burn in theorem. We reproduce it from the spectral representation of their Equation (42). For the brother sister case $N_m = N_f = 1$ the values at $k = 1, 2, 5, 10, 20, 50$ are 3.0000, 2.7247, 2.6672, 2.6421, 2.6300, 2.6228, converging to $N_e^{(e)} = 2.6180$. These values approach the limit like $1/k$, with excess about $0.24/k$, because each averages over the initial transient. The geometric rate $|\lambda_2/\lambda_1| = 0.382$, which more than halves the error per generation, governs instead the convergence to λ_1 of the ratio of the probabilities of not yet being identical by descent at consecutive generations. For $(N_m, N_f) = (2, 20)$ that rate is 0.078.

Thompson (2013) develops the identity by descent calculus underlying the first of these in detail, and it is the same quantity π_2 that appears in the definition of $N_e^{(i)}$.

The relevance here is methodological. The generative posterior of Section 10.4 produces independent draws by construction, since each draw uses a fresh u, so its effective sample size equals its nominal one. ABC output is independent within an accepted sample but the acceptance rate plays the role of a thinning factor, and sequential ABC variants introduce genuine autocorrelation. Any honest efficiency comparison, including the one in Section 10.10, has to compare effective rather than nominal draws, and the apparatus for doing so is exactly the apparatus Ewens and Hössjer construct for structured populations. Their formulas for dioecious and geographically subdivided cases are, read this way, formulas for effective sample size under structured samplers, which the computational statistics literature lacks.

10.13 Discussion

The chapter makes three claims. Methodologically, the posterior is a transport map, that map is the minimiser of an expected check loss, and it can therefore be fitted from simulations with no summary statistic and no tolerance. Theoretically, Theorem 10.12 makes sufficiency testable: a bottleneck that attains the Bayes risk is a sufficient statistic, and the excess risk at a given bottleneck dimension measures what any summary of that dimension must discard, on a posterior scale given by Theorem 10.16. Empirically, the procedure rediscovers Ewens's 1972 theorem on the model where it holds, and detects and quantifies the failure of Watterson's S on

the infinite sites model, a failure that Proposition 10.25 and Example 10.26 establish exactly.

The programme this opens is the estimation of sufficient statistics in models where none is known. The natural next targets are the ancestral recombination graph, models with selection, and the Λ coalescents with multiple mergers. In each case one increases d until the held out risk stops falling and reads off the learned statistic, with the infinite alleles model of Section 10.8 serving as the check that the procedure works.

A second direction is the one of Section 10.12, in which the effective size formulas of Ewens and Hössjer (2026) are read as effective sample size formulas for structured samplers, and the traffic between the two subjects runs the other way.

10.14 Reproducibility

We record the simulation designs. Scripts are in `code/suff` for the two main network fits of Tables 10.3 and 10.4, and in `code/suff_rev` for every exact calculation and small example in the chapter.

Infinite alleles, Section 10.8. $n = 100$ genes, $M = 12{,}000$ training and 2,500 held out pairs, θ drawn from the Pochhammer prior $p(\theta) \propto \{(\theta+1)(\theta+2)\}^{-1}$ truncated to $[0.2, 25]$ by inverse transform, partitions generated exactly by the Feller coupling (Arratia et al., 1992), in which $B_1 = 1$ and $B_i \sim \text{Bernoulli}(\theta/(\theta+i-1))$ independently for $i = 2, \dots, n$, and the block sizes are the spacings between successive indices i with $B_i = 1$ and $n+1$. Networks as in Section 10.8.2, 700 steps of Adam. The exact posterior uses a 3,000 point geometric grid on $[0.2, 25]$. The capacity and calibration experiments of Sections 10.8.4 to 10.8.6 are a separate set of runs of the same design on the same training data, with their own held out sample of 5,000 pairs drawn after the main one and 20 seeds per configuration, the seeds varying only the training randomness (script `code/suff_rev/second_runs.py`, per seed results and plot caches beside it).

Infinite sites, Section 10.9. $n = 30$ sequences, $M = 30{,}000$ training, 2,000 validation and 4,000 held out pairs, θ log uniform on $[0.5, 30]$. Each replicate simulates the coalescent by holding block sizes for the current lineages, drawing $T_k \sim \text{Exp}\{k(k-1)/2\}$, placing $\text{Poisson}(\theta T_k/2)$ mutations on each of the k branches and recording them against the number of leaves that branch subtends, then merging a uniformly chosen pair. Per site features supplied to φ are $(\log i,\ \log(1+\xi_i))$. Each configuration is trained for 4,000 steps from three random seeds; the loss table reports the mean and range over seeds, and the posterior comparisons use the $d = 4$ fit with the best validation loss over the three seeds. A separate study refits each configuration from 50 seeds on the same data, with the same network and training budget, to put intervals on the gaps between bottlenecks. The exact $\pi(\theta \mid S)$ for the 500 held out comparisons convolves the $n-1$ geometric mass functions of Proposition 10.27 on a 300 point geometric grid in θ, truncating S at 900; on this grid the median for $S = 23$ is 5.96, against 6.00

by adaptive quadrature, a discretisation error well below the differences being measured.

ABC comparison, Section 10.10. 2×10^5 prior draws, S generated exactly from Proposition 10.27 rather than by tree simulation, so that ABC is not penalised for simulation cost relative to the generative method. The observed dataset was simulated by the coalescent at $\theta = 8$ and retained when it gave $S = 23$. Acceptance is $|S_{\mathrm{sim}} - 23| \le \epsilon$. The exact rows of Table 10.5 integrate $\pi(\theta)\mathbb{P}(|S - 23| \le \epsilon \mid \theta)$ by adaptive quadrature.

All experiments run on CPUs, without a GPU. The generative fits take a few minutes at most at these budgets, which is itself part of the argument: the expensive step is the forward simulation, and the generative method uses fewer forward simulations than the ABC comparison by a factor of about seven even before amortisation is counted.

10.15 Exercises

1. Let θ have distribution function F with $\mathbb{E}|\theta| < \infty$, and let $q_u = F^{-1}(u)$. Prove the identity (10.4), and deduce that the u quantile minimises $\mathbb{E}\rho_u(\theta - c)$. *Hint.* Write $\rho_u(\theta - c) - \rho_u(\theta - q_u)$ as $\int_{q_u}^{c} \{\mathbf{1}(\theta \le t) - u\}\,\mathrm{d}t$ and take expectations by Fubini.

2. In Example 10.9 verify that $\mathcal{R}^\star = 1/15$ and that the risk of the map $H(u) = u$ is $1/12$. *Hint.* Show $\mathbb{E}[\rho_u(\theta - \sqrt{u}) \mid Y = 1] = \frac{2}{3}(u - u^{3/2})$ and $\mathbb{E}\rho_u(\theta - u) = u(1-u)/2$ for uniform θ.

3. Let θ be uniform and $Y \mid \theta \sim \mathrm{Binomial}(n, \theta)$. Show that the posterior quantile function given $Y = n$ is $u^{1/(n+1)}$, and explain why a one dimensional bottleneck attains the Bayes risk when the data are the n individual genes. *Hint.* The posterior is $\mathrm{Be}(n+1, 1)$; the count is sufficient by factorisation.

4. Prove that rejection ABC with summary $\bar{y}$ and tolerance ϵ in the Gaussian example of Section 10.7 is exact Bayes for the observation $\bar{y}_{\mathrm{sim}} + V$ with V uniform on $(-\epsilon, \epsilon)$. *Hint.* $\mathbb{P}(|\bar{Y} - s| \le \epsilon \mid \theta) = 2\epsilon$ times the density at s of $\bar{Y} + V$ given θ.

5. For $d_S = 1$, check the coefficient $\epsilon^2/\{2(d_S + 2)\}$ of Proposition 10.2 directly, and compute the leading bias of the ABC posterior mean in the Gaussian example. *Hint.* $\int_{-\epsilon}^{\epsilon} z^2\,\mathrm{d}z = 2\epsilon^3/3$ and $V_1 = 2$; the Gaussian ABC posterior has the form of Section 10.7 with n^{-1} replaced by $n^{-1} + \epsilon^2/3$.

6. Using the Feller coupling, show that $\mathbb{E}K_n = \sum_{i=1}^{n} \theta/(\theta + i - 1)$, and that the maximum likelihood equation for θ from K_n is $K_n = \mathbb{E}K_n$. Solve it numerically for $n = 10$, $K_{10} = 4$. *Hint.* K_n is a sum of independent indicators; the answer is $\hat{\theta} = 1.956$.

7. Derive the generating function $\mathbb{E} z^{S} = \prod_{j=1}^{n-1} j/(j+\theta-\theta z)$ from Proposition 10.27, and check the values $\mathbb{P}(S=0) = 1/5$ and $\mathbb{P}(S=1) = 77/300$ for $n=5$, $\theta = 1$. *Hint.* Differentiate the product once at $z=0$.

8. For $n=3$ derive $\phi_3(\theta) = (\theta+2)/\{2(2\theta+3)\}$ of Example 10.26 directly, and confirm that $\phi_3'(0) = -1/18$ agrees with Proposition 10.25. *Hint.* Use $\mathbb{E} e^{-sT} = r/(r+s)$ and $\mathbb{E}[T e^{-sT}] = r/(r+s)^2$ for $T \sim \mathrm{Exp}(r)$.

9. Show that the pooled statistic $\sum_{i=1}^{n-1} \xi_i \sqrt{p_i}$, with p_i the ith prime, is injective on vectors of nonnegative integers, and hence sufficient for θ under any model for the spectrum. Explain why this does not contradict Table 10.4. *Hint.* Square roots of distinct primes are linearly independent over the rationals (Besicovitch); see Remark 10.15.

10. In Proposition 10.10(i), show that the posterior of $c^{\top}\beta$ depends on y only through the scalar $c^{\top} V X^{\top} W y$, even though $X^{\top} W y$ is p dimensional. *Hint.* Complete the square in the Gaussian exponent to obtain $N(\mu(y), V)$, then use linearity.

11. Compute the excess risk $\mathcal{R}(H) - \mathcal{R}^{\star}$ in Example 10.9 for the map $H(u,y) = u$ restricted to $Y = 1$ by evaluating $\int_0^1 \int_{\sqrt{u}}^{u} (t^2 - u)\, \mathrm{d}t\, \mathrm{d}u$, and compare with the bounds of Theorem 10.16. *Hint.* The answer is $1/60$; the upper bound is $1/30$ and the lower bound is 0.

12. (Computation.) Reproduce the first four rows of Table 10.1 by rejection ABC with 10^5 prior draws, and plot the ABC posterior median against ϵ. *Hint.* Draw $\theta = 2(w-1)/(2-w)$ with $w = 2^{V}$, V uniform, then K_{10} as a sum of independent Bernoulli$(\theta/(\theta+i-1))$ indicators.

10.16 Proofs

Proof of Proposition 10.2. Write $d = d_S$, $f = f(\cdot \mid \theta)$ and $c_d = 1/\{2(d+2)\}$. By Taylor's theorem with the stated uniform bounds, $f(s+z) = f(s) + \nabla f(s)^{\top} z + \tfrac{1}{2} z^{\top} \nabla^2 f(s) z + \tfrac{1}{6} D^3 f(s)[z,z,z] + r(z)$ with $|r(z)| \le C|z|^4$ for a constant C not depending on θ. Integrate over the ball $\{|z| \le \epsilon\}$. The terms of odd degree vanish by symmetry, and $\int_{|z|\le\epsilon} z_i z_j \, \mathrm{d}z = \mathbf{1}(i=j)\, V_d \epsilon^{d+2}/(d+2)$, since $\int_{|z|\le\epsilon} |z|^2 \, \mathrm{d}z = \int_0^{\epsilon} r^2 \, d V_d r^{d-1} \, \mathrm{d}r = d V_d \epsilon^{d+2}/(d+2)$. Hence the acceptance probability is $V_d \epsilon^d \{ f(s) + c_d \epsilon^2 \Delta f(s) \} + O(\epsilon^{d+4})$, uniformly in θ, which is the first display. Integrating against π, and using the uniform bounds to differentiate under the integral, gives $\alpha_\epsilon = V_d \epsilon^d \{ m(s) + c_d \epsilon^2 \Delta m(s) + O(\epsilon^4) \}$ with $\Delta m = \int \pi \Delta f$. The ABC target is the ratio $\pi(\theta)\{ f + c_d \epsilon^2 \Delta f + O(\epsilon^4) \} / \{ m + c_d \epsilon^2 \Delta m + O(\epsilon^4) \}$, and expanding $1/(m+x) = m^{-1}(1 - x/m + O(x^2))$ gives (10.2), since $\pi(\theta) f(s \mid \theta)/m(s) = \pi(\theta \mid s)$. $\square$

Proof of Theorem 10.7. Take $\Theta \subseteq \mathbb{R}$. Let $F_y(t) = \mathbb{P}(\theta \le t \mid Y = y)$ be a regular conditional distribution function, which exists because Θ is a Borel subset of $\mathbb{R}$,

and let $F_y^{-1}(u) = \inf\{t : F_y(t) \geq u\}$, which is jointly measurable in (u, y). For U uniform and independent of Y, the inverse transform argument gives $\mathbb{P}(F_Y^{-1}(U) \leq t \mid Y = y) = \mathbb{P}(U \leq F_y(t)) = F_y(t)$ for almost every y, so $F_Y^{-1}(U)$ has the same conditional law as θ given Y, hence the same joint law with Y. To obtain the almost sure equality $\theta = H(U, Y)$ for the original θ, take $U = F_Y(\theta-) + V\{F_Y(\theta) - F_Y(\theta-)\}$ with V uniform and independent of (θ, Y); then U is uniform, independent of Y, and $F_Y^{-1}(U) = \theta$ almost surely. For general standard Borel Θ, apply a Borel isomorphism to a subset of $\mathbb{R}$ and transport the construction; see Kallenberg (2002, Theorem 6.10). □

Proof of Theorem 10.8. Condition on $Y = y$ and on $U = u$. The inner problem is $\min_{c \in \mathbb{R}} \psi(c)$ with $\psi(c) = \mathbb{E}[\rho_u(\theta - c) \mid Y = y]$, which is finite for almost every y because $\mathbb{E}|\theta| < \infty$. By the identity (10.4), $\psi(c) - \psi(q_u) = \int_{q_u}^{c} (F_y(t) - u)\,\mathrm{d}t$ with $q_u = F_y^{-1}(u)$. The integrand has the sign of $t - q_u$ and, because F_y is continuous and strictly increasing on the interior of the support, it is nonzero on a neighbourhood of every $t \neq q_u$ between q_u and c, so $\psi(c) > \psi(q_u)$ whenever $c \neq q_u$. The objective (10.3) is the average of ψ over the product law of U and Y. Since $(u, y) \mapsto F_y^{-1}(u)$ is measurable, the pointwise minimisers form an admissible H, which is therefore optimal; and any H differing from it on a set of positive measure has strictly larger inner risk on that set and hence strictly larger total risk. So the minimiser is $H^\star(u, y) = F_y^{-1}(u)$, unique up to a null set. □

Proof of Theorem 10.16. Fix y and u, write $F = F_y$, $q_u = F^{-1}(u)$ and $c = g(u, T(y))$, and let $\psi(c) = \mathbb{E}[\rho_u(\theta - c) \mid Y = y]$. By (10.4), $\psi(c) - \psi(q_u) = \int_{q_u}^{c} (F(t) - u)\,\mathrm{d}t$. Since $F(q_u) = u$ and, by hypothesis, $\delta \leq f_y \leq \bar{\delta}$ on the closed interval with endpoints q_u and c, the mean value theorem gives $\delta(t - q_u) \leq F(t) - u \leq \bar{\delta}(t - q_u)$ for $t \geq q_u$ in that interval and the reversed inequalities for $t \leq q_u$, so in either case

$$\frac{\delta}{2}(c - q_u)^2 \;\leq\; \psi(c) - \psi(q_u) \;\leq\; \frac{\bar{\delta}}{2}(c - q_u)^2.$$

Now integrate over $u \sim \mathrm{Unif}(0, 1)$ and over Y. By Theorem 10.8, $\mathbb{E}_Y \mathbb{E}_U \psi(q_U) = \mathcal{R}^\star$, so the middle term integrates to $\mathcal{R}(H) - \mathcal{R}^\star$ and the outer terms to $\frac{\delta}{2}$ and $\frac{\bar{\delta}}{2}$ times $\mathbb{E}_Y \int_0^1 \{g(u, T(Y)) - F_Y^{-1}(u)\}^2\,\mathrm{d}u$. Finally, for probability measures on the line with quantile functions Q_1, Q_2 one has $W_2^2 = \int_0^1 \{Q_1(u) - Q_2(u)\}^2\,\mathrm{d}u$, and $u \mapsto g(u, T(y))$ is by construction the quantile function of $\pi_T(\cdot \mid y)$, giving the claim. □

Proof of Theorem 10.12. By Theorem 10.8 the Bayes risk is attained only at $H^\star(u, y) = F_y^{-1}(u)$, up to a set of (u, y) of measure zero. If $H(u, y) = g(u, T(y))$ also attains it, then $g(u, T(y)) = F_y^{-1}(u)$ for almost every (u, y). By Fubini, outside a P null set of y the identity holds for almost every u, hence on a dense set of u. A distribution function on $\mathbb{R}$ is determined by its quantile function on a dense set,

since $F(t) = \sup\{u : F^{-1}(u) \le t\}$ and F^{-1} is left continuous and nondecreasing, so F_y is a function of $T(y)$ for P almost every y. This is Bayes sufficiency.

Now let $\mathcal{Y}$ be countable, so that P almost every y means every y with $m(y) = P(\{y\}) > 0$. The posterior given $Y = y$ has density $p(y \mid \theta)/m(y)$ with respect to π, and Bayes sufficiency says these densities coincide, as elements of $L^1(\pi)$, for any two points y, y' of positive mass with $T(y) = T(y')$. So there is a π null set $N_{y,y'}$ off which $p(y \mid \theta)\, m(y') = p(y' \mid \theta)\, m(y)$. Let Θ_0 be the complement of the countable union of the $N_{y,y'}$; it has prior probability one. For $\theta \in \Theta_0$ and $T(Y) = t$,

$$\mathbb{P}(Y = y \mid T(Y) = t, \theta) = \frac{p(y \mid \theta)}{\sum_{y' : T(y') = t} p(y' \mid \theta)} = \frac{m(y)}{\sum_{y' : T(y') = t} m(y')},$$

free of θ, which is sufficiency for $\{p(\cdot \mid \theta) : \theta \in \Theta_0\}$; points y of zero marginal mass have $p(y \mid \theta) = 0$ for π almost every θ and can be absorbed in the null set. If π has full support then Θ_0, having null complement, is dense, and if each $\theta \mapsto p(y \mid \theta)$ is continuous the identity $p(y \mid \theta) m(y') = p(y' \mid \theta) m(y)$ extends from Θ_0 to all of Θ, giving sufficiency for the whole family. For general dominated families the same conclusion is the theorem of Halmos and Savage (1949). □

10.17 Notation

symbol	meaning
θ	scaled mutation parameter, $4N_e\mu$ with μ the mutation rate per gene
π, P, m	prior on θ; marginal law of the data; its mass or density
y	the raw data configuration
$S(\cdot)$	a summary statistic; S also denotes the number of segregating sites
ϵ	ABC tolerance
d_S	dimension of the ABC summary
α_ϵ	ABC acceptance probability
U, u	uniform noise variable independent of the data, and its value
$H(u, y)$	the transport map; $H^\star$ its Bayes optimal version
$\rho_u(z)$	check loss $z\{u - \mathbf{1}(z < 0)\}$
$\mathcal{R}$, $\mathcal{R}^\star$, $\mathcal{R}^\star_d$	risk, Bayes risk, best risk with a d dimensional bottleneck
$T_\varphi(y)$	pooled bottleneck statistic $\sum_i \varphi(y_i)$
d, $d^\star$	bottleneck dimension, and its minimal sufficient value
Π, K_n	allelic partition of n genes, and its number of blocks
C_j	number of alleles carried by exactly j genes
B_i	Feller coupling indicator that gene i founds a new allele
ξ_i	number of sites at which i sequences carry the derived allele
T_k	coalescent waiting time while k lineages remain
ℓ_i, L	branch length subtending i leaves, total tree length
a_n, b_n	$\sum_{j=1}^{n-1} j^{-1}$ and $\sum_{j=1}^{n-1} j^{-2}$
π_{T}, D	mean pairwise difference and Tajima's D

The symbol S is overloaded, denoting a generic summary statistic in Sections 10.2 to 10.7 and the number of segregating sites from Section 10.9 onward. The two coincide in the one place where it matters, namely the ABC comparison of Section 10.10, where the summary used is the number of segregating sites.

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CHAPTER 11

Learning to Coalesce

Kingman's coalescent is the historical home of likelihood free inference: simulation is trivial and likelihood evaluation is not. This chapter studies generative Bayesian computation for it, and the story is governed by a sufficiency reduction that exists for Kingman's exchangeable partitions and fails here. We start from the classical fact, implicit in Watterson's generating function, that under the infinite sites model the number of segregating sites is a sum of independent geometrics, $S = \sum_{k=2}^{n} S_k$ with $S_k \sim \mathrm{Geom}\{(k-1)/(k-1+\theta)\}$. This yields Watterson's mean and variance in one line, gives an exactly computable likelihood for S, and couples the coalescent to the Ewens sampling formula: on a common probability space the Ewens block count satisfies $K_n = 1 + \#\{k : S_k \geq 1\}$, so in law the partition records which coalescent levels carry a mutation and the site count records how many. The observed haplotype count has exactly the Ewens law, and all of these statistics carry Fisher information $\theta^{-1} \log n$ to leading order. We then show, with an explicit three sequence example, that S is not sufficient for θ, unlike the Ewens block count, so the summary statistic must be learned rather than derived. We prove a ceiling on what can be learned: any DeepSets encoder built from per site permutation invariant features factors exactly through the site frequency spectrum, which is a strict coarsening of the maximal invariant, so no such architecture recovers linkage information at any depth. The number of distinct haplotypes is a minimal witness that escapes the ceiling, and two four sequence data sets with the same spectrum but different haplotype counts give different posteriors. Experiments at $n = 30$ show that a quantile network matches the exact posterior given S; the spectrum strictly improves on S, cutting the 90% interval for θ from 84.1 to 64.1 at constant population size, and under exponential growth removing 32% of the prior width for the growth rate against 14% for S alone. Appending the haplotype count to S narrows the interval by about 8% relative to the exact posterior given S at low diversity, but appending it to the full spectrum gains at most 1.3%, so in these designs the spectrum already holds most of what the haplotype count adds. Because the likelihood of S under any demography is a one dimensional integral against the tree length law, we build an exact reference posterior for the growth model and show the learned map recovers it to within 4% for θ and 0.21 for the growth rate. Simulation based calibration nonetheless rejects exact uniformity for the learned spectrum maps, and for some trainings of the S only map; the standard recalibration repair flattens the ranks of the S only map, as it must, but cannot reach the residual error that varies with the data, and a small example where the true posterior is known reproduces the phenomenon exactly.

11.1 Introduction

Kingman (1982) defines the n coalescent by a single rule: when k ancestral lineages remain, each of the $\binom{k}{2}$ pairs merges at unit rate. Everything else about the

genealogy of a sample follows. The process is trivial to simulate, requires no parameters, and is the weak limit of essentially every neutral reproduction model, which is why it became the standard prior over genealogies.

It is also the reason population genetics invented likelihood free inference. Once mutations are superimposed on the genealogy, the likelihood of a sample requires integration over the space of trees, and Tavaré et al. (1997) responded by accepting parameter values whose simulated summaries fell close to the observed ones. Approximate Bayesian computation grew from there (Beaumont et al., 2002), and with it a thirty year methodological anxiety about which summaries to use.

Chapter 10 studied generative Bayesian computation (Polson, Ruggeri and Sokolov, 2024) for Kingman's exchangeable partitions, where this anxiety is misplaced. Under the Ewens sampling formula the block count K_n is exactly sufficient for the concentration parameter, so the posterior transport map has two input dimensions no matter how large the sample. One might expect the coalescent to behave the same way, with the number of segregating sites S playing the role of K_n and Watterson's estimator playing the role of the block count. It does not, and the difference is the subject of this chapter.

11.1.1 Plan of the Chapter

The chapter moves from exact distribution theory, through a negative result on sufficiency and a ceiling on learned summaries, to experiments that grade learned posteriors against exact ones.

1. The law of the segregating sites (Section 11.2.3). Mixing the Poisson mutation count over the exponential coalescent holding time produces a geometric, giving $S = \sum_{k=2}^{n} S_k$ with S_k independent and $\mathrm{Geom}\{(k-1)/(k-1+\theta)\}$ distributed. Watterson's mean and variance follow immediately, the likelihood of S is exactly computable, and worked examples at $n = 4$ and $n = 11$ show the law, Watterson's estimator and an exact confidence interval in action.

2. A coupling with the Ewens sampling formula (Section 11.2.5). The same probability sequence $\theta/(\theta+k-1)$ generates both models. On one probability space, $K_n = 1 + \#\{k : S_k \geq 1\}$: in law, Kingman's partition thresholds what the coalescent counts. The observed haplotype partition has the Ewens law exactly, though it is not pathwise the thresholded count. The Fisher informations agree to leading order at $\theta^{-1} \log n$, which explains why partition based and mutation based estimators of θ have the same asymptotic efficiency.

3. Sufficiency fails (Section 11.3). Unlike K_n, the count S is not sufficient for θ, because conditioning on S leaves a θ dependent tilt on the genealogy. A three sequence example makes the failure explicit, and the summary must therefore be learned.

4. A ceiling on learned summaries (Section 11.4). Any encoder of DeepSets form built from per site permutation invariant features collapses exactly to a function of the site frequency spectrum. Since the spectrum is a strict coarsening of the maximal invariant, which is the gene tree, no such architecture recovers linkage information regardless of depth or width. The number of distinct haplotypes lies outside the class, and a four sequence example shows it changes the posterior.
5. Validated generative inference (Sections 11.5 and 11.6). We train quantile networks on prior predictive draws and grade them against exact posteriors: in the constant size model via the geometric law, and in the two parameter growth model via a reference posterior built on the mixed Poisson representation. We then apply rank recalibration to the residual calibration failure of the S only map, report that it flattens the marginal ranks but cannot reach the small error that varies with the data, and explain why with an example in which every posterior is exact.

11.2 The Coalescent and the Infinite Sites Model

This section fixes notation for the genealogy and for the data it generates. Everything later is a statement about these two objects.

11.2.1 Genealogy

Let T_k be the time during which the sample has exactly k ancestral lineages. Under Kingman's coalescent

$$T_k \sim \mathrm{Exp}\Big(\tbinom{k}{2}\Big) = \mathrm{Exp}\Big(\tfrac{k(k-1)}{2}\Big), \qquad k = n, n-1, \ldots, 2, \tag{11.1}$$

independently, and at the end of each holding period a uniformly chosen pair of lineages merges. The topology, meaning the sequence of merging pairs, is independent of the times. Time is measured in units of $2N$ generations for a diploid population of size N (or N for a haploid one), so $\theta = 4N\mu$ in the diploid case. The total branch length is $L_n = \sum_{k=2}^{n} k\, T_k$, with

$$\mathbb{E}[L_n] = \sum_{k=2}^{n} k \cdot \frac{2}{k(k-1)} = 2 \sum_{j=1}^{n-1} \frac{1}{j} =: 2a_1(n) \approx 2 \log n. \tag{11.2}$$

The genealogy carries no parameters, a fact we use repeatedly.

11.2.2 Mutation and the Spectrum

Under the infinite sites model, mutations arrive on the genealogy as a Poisson process of rate $\theta/2$ per unit branch length, and every mutation strikes a previously unmutated site. Writing $X \in \{0,1\}^{n \times S}$ for the resulting matrix of n lineages by S

segregating sites, with a one denoting the derived allele, the observables of interest are

$$\begin{aligned} S &= \text{number of segregating sites,} \\ \xi_i &= \#\{\text{sites with exactly } i \text{ derived alleles}\}, \end{aligned} \tag{11.3}$$

for $i = 1, \dots, n-1$. The vector $\boldsymbol{\xi} = (\xi_1, \dots, \xi_{n-1})$ is the site frequency spectrum, and $S = \sum_i \xi_i$. Writing ℓ_i for the total length of branches subtending exactly i of the n leaves, so that $\sum_i \ell_i = L_n$, the counts ξ_i are conditionally independent with $\xi_i \mid \boldsymbol{\ell} \sim \text{Poisson}(\theta \ell_i / 2)$, and the classical identity $\mathbb{E}[\xi_i] = \theta / i$ holds (Fu, 1995). A third observable, used from Section 11.2.5 on, is the number H of distinct rows of X, the haplotype count.

11.2.3 The Law of the Number of Segregating Sites

The next theorem is the analogue for the coalescent of the Feller coupling for the Ewens formula. Its generating function is due to Watterson (1975) and the explicit mass function to Tavaré (1984); we state it as a decomposition into independent levels, because that is the form the coupling and the information calculations need. The idea is that during each epoch the genealogy has a fixed number of branches of a common length, so the number of mutations in the epoch is a Poisson count mixed over an exponential time, which is geometric.

THEOREM 11.1 (Geometric decomposition). *Under* (11.1) *with infinite sites mutation at rate $\theta/2$, let S_k denote the number of mutations falling on the genealogy while k ancestral lineages remain. Then $S_2, \dots, S_n$ are independent with*

$$\mathbb{P}(S_k = j) = p_k (1 - p_k)^j, \quad j = 0, 1, 2, \dots, \qquad p_k = \frac{k-1}{k-1+\theta}, \tag{11.4}$$

and $S = \sum_{k=2}^{n} S_k$. Consequently

$$\begin{aligned} \mathbb{E}[S] = \theta\, a_1(n), \qquad \operatorname{Var}(S) &= \theta\, a_1(n) + \theta^2 a_2(n), \\ a_r(n) &= \sum_{j=1}^{n-1} j^{-r}. \end{aligned} \tag{11.5}$$

Proof. During the k lineage epoch there are exactly k branches, each of length T_k, whatever the topology, so the total branch length in the epoch is kT_k. Conditionally on $(T_2, \dots, T_n)$ the mutation counts in disjoint epochs are independent Poisson variables with means $\theta k T_k / 2$, by the independent increments of the Poisson process. Hence, for $|z_k| \le 1$,

$$\mathbb{E}\Big[\prod_{k=2}^{n} z_k^{S_k} \;\Big|\; T_2, \dots, T_n\Big] = \prod_{k=2}^{n} \exp\Big\{\tfrac{\theta k T_k}{2}(z_k - 1)\Big\},$$

and since the T_k are independent the unconditional joint generating function is the product over k of $\mathbb{E}[\exp\{\theta k T_k (z_k - 1)/2\}]$. The joint generating function therefore

factorises, which is independence. For the marginal law, with $\lambda_k = k(k-1)/2$ and using $\mathbb{E}[e^{-uT_k}] = \lambda_k/(\lambda_k + u)$ for $u > -\lambda_k$,

$$\mathbb{E}\big[z^{S_k}\big] = \mathbb{E}\Big[\exp\Big\{\tfrac{\theta k T_k}{2}(z-1)\Big\}\Big] = \frac{\lambda_k}{\lambda_k - \theta k(z-1)/2}$$
$$= \frac{k-1}{(k-1) - \theta(z-1)} = \frac{p_k}{1-(1-p_k)z},$$

where the third equality divides numerator and denominator by $k/2$. This is the probability generating function of (11.4). The mean and variance of a geometric on $\{0, 1, \dots\}$ with success probability p are $(1-p)/p$ and $(1-p)/p^2$, so

$$\mathbb{E}[S_k] = \frac{\theta}{k-1}, \qquad \operatorname{Var}(S_k) = \frac{\theta}{k-1}\cdot\frac{k-1+\theta}{k-1} = \frac{\theta}{k-1} + \frac{\theta^2}{(k-1)^2},$$

and summing over $k = 2, \dots, n$, with $j = k-1$ and independence for the variance, gives (11.5). □

Equation (11.5) is Watterson's mean and variance, here obtained without computing a single moment of the tree. The two terms of the variance have a clear genetic reading: θa_1 is the Poisson noise of mutation given the tree, and $\theta^2 a_2$ is the variability of the tree itself, which does not average away as n grows because $a_2(n) \to \pi^2/6$. The proof isolates what makes the constant size case special, and it is worth recording the general statement, since it is what one falls back on otherwise.

PROPOSITION 11.2 (General demography). *Let the population size vary over time, so that the epoch durations T_k are no longer exponential and are in general dependent. Then S_k is a mixed Poisson variable with*

$$\mathbb{E}\big[z^{S_k} \mid T_k\big] = \exp\Big\{\tfrac{\theta k T_k}{2}(z-1)\Big\}, \qquad \mathbb{E}\big[z^{S_k}\big] = \mathcal{L}_{T_k}\Big(\tfrac{\theta k}{2}(1-z)\Big), \tag{11.6}$$

with $\mathcal{L}_{T_k}(u) = \mathbb{E}[e^{-uT_k}]$ the Laplace transform of T_k, and the total satisfies

$$S \mid L_n \sim \operatorname{Poisson}\big(\theta L_n/2\big), \qquad \mathbb{P}_\theta(S = s) = \mathbb{E}\Big[e^{-\theta L_n/2}\,\frac{(\theta L_n/2)^s}{s!}\Big]. \tag{11.7}$$

Theorem 11.1 is the case $T_k \sim \operatorname{Exp}\{k(k-1)/2\}$, where $\mathcal{L}_{T_k}(u) = \lambda_k/(\lambda_k+u)$ is the transform of an exponential and (11.6) *becomes geometric.*

Proof. Conditionally on the genealogy the mutations form a Poisson process of rate $\theta/2$ on a tree whose kth epoch has length kT_k and whose total length is L_n. This gives the conditional generating function in (11.6) and the conditional law in (11.7). Taking expectations over T_k gives the second identity in (11.6), and taking expectations over L_n of the Poisson mass function gives the second identity in (11.7). Substituting the exponential transform at $u = \theta k(1-z)/2$ reproduces the computation in the proof of Theorem 11.1. □

Representation (11.7) is more useful than it looks. Under any demography the likelihood of S is a one dimensional integral against the law of the total tree length, so it can be evaluated by simulating L_n alone, with no need to generate topologies or mutations. Only the geometric structure of Theorem 11.1 is lost when the population size varies, not the tractability of S itself. In the constant size case the likelihood is available in closed form.

COROLLARY 11.3 (Exact likelihood). *The probability generating function of S is $\prod_{k=2}^{n} p_k\{1-(1-p_k)z\}^{-1}$, and for $\theta > 0$ the mass function is*

$$\mathbb{P}_\theta(S=s) = \sum_{k=2}^{n} (-1)^k \binom{n-1}{k-1} \frac{k-1}{\theta+k-1} \Big(\frac{\theta}{\theta+k-1}\Big)^s, \qquad s=0,1,2,\dots \quad (11.8)$$

Hence the exact posterior

$$\pi(\theta \mid S=s) \;\propto\; \mathbb{P}_\theta(S=s)\,\pi(\theta) \quad (11.9)$$

is computable to machine precision for any prior.

Proof. Write $q_k = 1-p_k = \theta/(\theta+k-1)$; these are distinct for $\theta > 0$. By partial fractions,

$$\prod_{k=2}^{n} \frac{1}{1-q_k z} = \sum_{k=2}^{n} \frac{A_k}{1-q_k z}, \qquad A_k = \prod_{j\neq k} \frac{q_k}{q_k - q_j},$$

so the coefficient of z^s in the generating function is $\big(\prod_j p_j\big)\sum_k A_k q_k^s$. Now $q_k - q_j = \theta(j-k)/\{(\theta+k-1)(\theta+j-1)\}$, so $q_k/(q_k-q_j) = (\theta+j-1)/(j-k)$, and

$$\Big(\prod_{j=2}^{n} p_j\Big) A_k = \prod_{j=2}^{n} \frac{j-1}{\theta+j-1} \cdot \prod_{j\neq k} \frac{\theta+j-1}{j-k} = \frac{(n-1)!}{\theta+k-1} \cdot \frac{1}{\prod_{j\neq k}(j-k)}.$$

The last product is $\prod_{j=2}^{k-1}(j-k)\prod_{j=k+1}^{n}(j-k) = (-1)^{k-2}(k-2)!\,(n-k)!$, and $(n-1)!/\{(k-2)!(n-k)!\} = (k-1)\binom{n-1}{k-1}$, which gives (11.8). □

Formula (11.8) is an alternating sum whose binomial coefficients reach $\binom{29}{14} \approx 7.8\times 10^7$ at $n=30$, so it loses digits through cancellation for large n. In computation we use either the convolution recursion $P_k(s) = q_k P_k(s-1) + p_k P_{k-1}(s)$ for the mass function P_k of $S_2+\cdots+S_k$, or inversion of the generating function on 4096 roots of unity by fast Fourier transform. As a check, at $n=30$ the recovered mass function integrates to 1 to eight decimal places and reproduces (11.5) exactly: at $\theta = 10$ the computed mean and variance are 39.6165 and 200.7204, against $\theta a_1 = 39.6165$ and $\theta a_1 + \theta^2 a_2 = 200.7204$ with $a_1 = 3.9617$ and $a_2 = 1.6110$.

EXAMPLE 11.4 (The law of S for four sequences). Take $n=4$ and $\theta = 1$. The three levels have $p_2 = 1/2$, $p_3 = 2/3$, $p_4 = 3/4$, so S is the sum of three independent geometrics with means 1, $1/2$ and $1/3$. The probability of no segregating site is $p_2p_3p_4 = 1/4$. One site arises from exactly one level, so

$$\mathbb{P}(S=1) = p_2p_3p_4\,(q_2+q_3+q_4) = \tfrac{1}{4}\big(\tfrac{1}{2}+\tfrac{1}{3}+\tfrac{1}{4}\big) = \tfrac{13}{48},$$

s	$\mathbb{P}(S = s)$, exact	decimal	simulated
0	1/4	0.2500	0.2514
1	13/48	0.2708	0.2686
2	115/576	0.1997	0.2006
3	865/6912	0.1251	0.1247
4	5971/82944	0.0720	0.0730
5	39193/995328	0.0394	0.0391
6	249355/11943936	0.0209	0.0209
7	1555105/143327232	0.0109	0.0105

TABLE 11.1. Exact law of the number of segregating sites for $n = 4$ and $\theta = 1$, from Theorem 11.1, against frequencies in 2×10^5 simulated coalescents. The Monte Carlo standard error is at most 0.001.

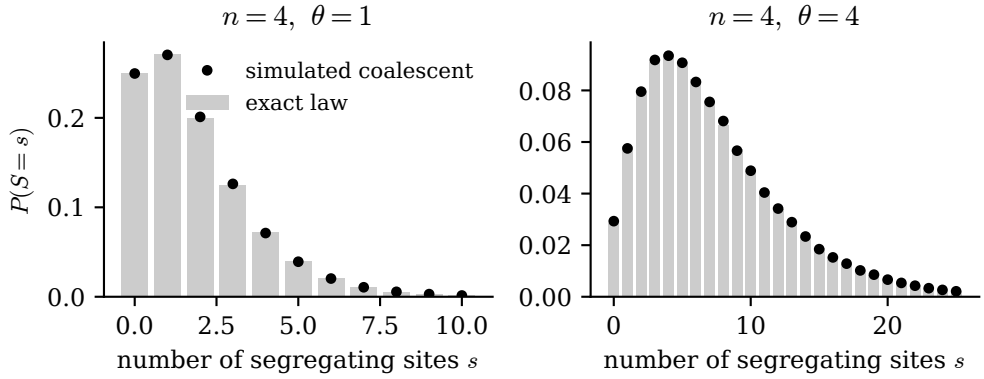


FIGURE 11.1. Exact mass function of S for $n = 4$ (bars) against 10^5 simulated coalescents with branch by branch mutation (points), at $\theta = 1$ and $\theta = 4$. The largest discrepancy is 0.0016.

and in general $\mathbb{P}(S = s) = \frac{1}{4} h_s(\frac{1}{2}, \frac{1}{3}, \frac{1}{4})$, where h_s is the complete homogeneous symmetric polynomial of degree s. Formula (11.8) gives the same numbers: at $s = 1$ its three terms are $\frac{3}{4} - \frac{2}{3} + \frac{3}{16} = \frac{13}{48}$. Table 11.1 lists the first eight probabilities as exact fractions, and compares them with the frequencies in 2×10^5 simulated coalescents in which the tree is built by merging random pairs and mutations are placed branch by branch, with no use of the geometric law. The chi square statistic over the cells $0, \ldots, 7$ and $\{S \geq 8\}$ is 11.5 on eight degrees of freedom, against a five percent critical value of 15.5, and the simulated mean and variance are 1.831 and 3.191 against the exact $a_1(4) = 11/6 = 1.833$ and $a_1 + a_2 = 11/6 + 49/36 = 3.194$. Figure 11.1 shows the same comparison at $\theta = 1$ and $\theta = 4$.

EXAMPLE 11.5 (No segregating sites). The probability of a monomorphic sample is $\mathbb{P}(S = 0) = \prod_{k=2}^{n} p_k = (n-1)!/\{(\theta+1)(\theta+2)\cdots(\theta+n-1)\}$. At $\theta = 1$ this is $1/n$, so among thirty sequences drawn from a population with $\theta = 1$ there is a

one in thirty chance of seeing no variation at all. The same expression is the Ewens probability that all n genes carry one allele, which is the first sign of the coupling in Section 11.2.5.

11.2.4 Watterson's Estimator on a Small Sample

Theorem 11.1 gives more than the moments of Watterson's estimator $\hat{\theta}_W = S/a_1(n)$: it gives an unbiased estimate of its variance and an exact confidence interval. Since $\mathbb{E}[S(S-1)] = \mathrm{Var}(S) + \mathbb{E}[S]^2 - \mathbb{E}[S] = \theta^2\{a_1(n)^2 + a_2(n)\}$, the statistic $S(S-1)/(a_1^2 + a_2)$ is unbiased for θ^2, and

$$\widehat{\mathrm{Var}}(\hat{\theta}_W) = \frac{S}{a_1^2} + \frac{a_2}{a_1^2} \cdot \frac{S(S-1)}{a_1^2 + a_2} \tag{11.10}$$

is unbiased for $\mathrm{Var}(\hat{\theta}_W) = \theta/a_1 + \theta^2 a_2/a_1^2$. Because each S_k is stochastically increasing in θ, so is S, and an exact equal tailed $1-2\alpha$ interval is $[\theta_L, \theta_U]$ with $\mathbb{P}_{\theta_L}(S \geq s) = \alpha$ and $\mathbb{P}_{\theta_U}(S \leq s) = \alpha$, both computed from (11.8).

EXAMPLE 11.6 (A simulated sample of eleven sequences). Kreitman's classic survey of the alcohol dehydrogenase region in *Drosophila melanogaster* sequenced eleven alleles (Kreitman, 1983), and we mimic its size with a clearly labelled simulation: $n = 11$ sequences from the coalescent with $\theta = 4$ (random seed 1983). The data matrix, with rows sorted and the derived allele written as 1, is

```
1111 1 0000 0 1 0 0000 00
1111 1 0000 0 1 0 0000 00
1111 1 0000 0 0 0 0000 00
1111 1 0000 0 0 0 0000 00
1111 1 0000 0 0 0 0000 00
1111 0 0000 0 0 1 0000 00
1111 0 0000 0 0 0 0000 11
0000 0 1111 1 0 0 0000 00
0000 0 1111 1 0 0 0000 00
0000 0 1111 1 0 0 0000 00
0000 0 1111 0 0 0 1111 00
```

with columns grouped by their derived allele count. There are $S = 18$ segregating sites with spectrum $\boldsymbol{\xi} = (7,1,1,4,1,0,4,0,0,0)$, and $H = 6$ distinct haplotypes. With $a_1(11) = 2.92897$ and $a_2(11) = 1.54977$,

$$\hat{\theta}_W = \frac{18}{2.92897} = 6.15, \qquad \widehat{\mathrm{Var}}(\hat{\theta}_W) = 7.56, \qquad \widehat{\mathrm{se}} = 2.75.$$

The normal interval $\hat{\theta}_W \pm 1.645\,\widehat{\mathrm{se}}$ is $(1.62, 10.67)$. The exact 90% interval from the geometric law is $(2.97, 15.05)$: it is shifted upward and is markedly asymmetric, because the law of S is right skewed and its variance grows like θ^2. The maximum likelihood estimate from S alone is 6.49, slightly above $\hat{\theta}_W$. Under the scale invariant prior $\pi(\theta) \propto 1/\theta$ the exact posterior given S has median

6.50 and 90% interval $(2.97, 14.31)$; under an exponential prior with mean 10 the median is 6.89 and the interval $(3.30, 14.11)$. For comparison the mean pairwise difference is $\hat{\theta}_\pi = \binom{11}{2}^{-1} \sum_i i(11-i)\xi_i = 6.65$ and Tajima's D (Tajima, 1989) is 0.37, unremarkable under neutrality. A repeated sampling check with 2×10^4 samples at $\theta = 4$ gives mean $\hat{\theta}_W$ of 4.007, variance 4.267 against the theoretical 4.256, and mean of (11.10) 4.268, confirming unbiasedness.

The example also shows how little eleven sequences say about θ. The exact interval spans a factor of five, and most of that width is the tree term $\theta^2 a_2$, which no amount of additional sequence length removes. Only more sequences, or information about the tree beyond S, can narrow it, and the rest of the chapter is about the second route.

11.2.5 A Coupling with the Ewens Sampling Formula

This section places Kingman's two models of neutral variation on one probability space. Kingman's other model for the same biology is the Ewens sampling formula, whose block count admits the Feller coupling $K_n = \sum_{i=1}^{n} B_i$ with $B_i \sim \text{Bernoulli}\{\theta/(\theta + i - 1)\}$ independent (Ewens, 1972; Kingman, 1977). Comparing this with (11.4), note that the geometric failure probability is

$$1 - p_k = \frac{\theta}{\theta + k - 1}, \tag{11.11}$$

which is exactly the Bernoulli success probability in the Feller coupling. The two models are built from the same sequence.

COROLLARY 11.7 (Bernoulli geometric duality). *On a common probability space carrying $S_2, \ldots, S_n$ as in Theorem 11.1,*

$$K_n \overset{d}{=} 1 + \#\{\, k \in \{2, \ldots, n\} : S_k \geq 1 \,\}, \tag{11.12}$$

where K_n is the number of blocks under the Ewens sampling formula with the same θ.

Proof. $\mathbb{P}(S_k \geq 1) = 1 - p_k = \theta/(\theta + k - 1)$ by (11.11), and the S_k are independent, so $\#\{k \geq 2 : S_k \geq 1\}$ is a sum of independent $\text{Bernoulli}\{\theta/(\theta + k - 1)\}$ variables over $k = 2, \ldots, n$. The Feller coupling gives K_n as the same sum over $i = 1, \ldots, n$, and the $i = 1$ term is $\text{Bernoulli}(1) \equiv 1$. □

The interpretation is clean. Both models decompose the sample along the $n - 1$ coalescent levels, and both attach an independent count to each level with the same parameter. Kingman's partition records only whether a level is marked; the coalescent records how many marks it carries. In law, the partition is the thresholded coalescent. Example 11.5 is the case of no marks: $\mathbb{P}(S = 0) = \mathbb{P}(K_n = 1)$.

There is a second, more literal, link between the two models, and it involves the haplotype count.

PROPOSITION 11.8 (Haplotypes follow the Ewens formula). *Under the infinite sites model on Kingman's coalescent, the partition of the n sequences into identical haplotypes has the law of the Ewens sampling formula with parameter θ. In particular $H \stackrel{d}{=} K_n$, and* $\mathbb{E}[H] = \sum_{i=1}^{n} \theta/(\theta + i - 1)$.

Proof. Two sampled sequences are identical exactly when no mutation falls on the path joining them in the genealogy, since every mutation on that path creates a site at which one carries the derived allele and the other does not, and mutations off the path affect both or neither. Now place the infinite alleles model on the same genealogy and the same mutation events, so that each mutation creates a new allele inherited by its descendants. Two genes then carry the same allele exactly when neither has a mutation on its line of descent below their most recent common ancestor, which is again the condition that the joining path is free of mutations. The two partitions therefore coincide pathwise. The allelic partition of the infinite alleles model on the coalescent has the Ewens law (Ewens, 1972; Kingman, 1982), and the block count of the Ewens formula has mean $\sum_i \theta/(\theta + i - 1)$ by the Feller coupling. □

So H and the thresholded count $K' = 1 + \#\{k : S_k \geq 1\}$ have the same law, but they are different random variables: H is a function of the data, while K' is not, since the data do not reveal in which epoch a mutation arose.

EXAMPLE 11.9 (The coupling for four sequences). Take $n = 4$ and $\theta = 1$. The Ewens law of K_4 is $|s(4,k)|/4!$, where s denotes Stirling numbers of the first kind, giving $6/24$, $11/24$, $6/24$ and $1/24$ for $k = 1, 2, 3, 4$. The thresholded coalescent marks levels $2, 3, 4$ independently with probabilities $1/2$, $1/3$, $1/4$, so $\mathbb{P}(K' = 1) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$ and $\mathbb{P}(K' = 4) = \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{24}$, in agreement. In the 2×10^5 coalescents of Example 11.4 the frequencies of K' are 0.2514, 0.4562, 0.2515, 0.0409 and those of H are 0.2514, 0.4562, 0.2509, 0.0414, against the exact 0.2500, 0.4583, 0.2500, 0.0417. Yet $K' \neq H$ in 13.7% of samples. Table 11.2 shows the joint law: the two agree whenever there is no mutation at all, and otherwise may disagree in either direction. For instance a mutation on lineage c in the three lineage epoch followed by a second mutation on the same lineage in the two lineage epoch gives $K' = 3$ but only $H = 2$, since both derived alleles are carried by one sequence.

The coupling also says how many sites to expect given the partition. Given K', the marked levels carry $1 + \mathrm{Geom}(p_k)$ sites by the memoryless property, with mean $1/p_k$. With only one level marked, the mark sits on level 2, 3 or 4 with probabilities proportional to $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{12}$, so

$$\mathbb{E}[S \mid K' = 2] = \frac{\frac{1}{4} \cdot 2 + \frac{1}{8} \cdot \frac{3}{2} + \frac{1}{12} \cdot \frac{4}{3}}{\frac{11}{24}} = \frac{115}{66} = 1.742,$$

against 1.744 in simulation; similarly $\mathbb{E}[S \mid K' = 3] = 10/3$ and $\mathbb{E}[S \mid K' = 4] = 29/6$.

	$H=1$	$H=2$	$H=3$	$H=4$
$K'=1$	0.2514	0	0	0
$K'=2$	0	0.4081	0.0458	0.0024
$K'=3$	0	0.0467	0.1846	0.0202
$K'=4$	0	0.0014	0.0206	0.0189

TABLE 11.2. Joint frequencies of the thresholded level count $K' = 1 + \#\{k : S_k \geq 1\}$ and the haplotype count H in 2×10^5 coalescents with $n = 4$, $\theta = 1$. Both margins follow the Ewens law, but the variables differ in 13.7% of samples.

In the eleven sequence sample of Example 11.6, mutations fell in seven of the ten epochs, so $K' = 8$, while the data show $H = 6$ haplotypes.

The duality also settles how much is lost by thresholding. Both statistics are sufficient for θ within their own reduced models, so their Fisher informations are computable.

PROPOSITION 11.10 (Information). *Let $I_K(\theta)$ be the Fisher information about θ in the Ewens block count and $I_S(\theta)$ that in the full levelwise vector $(S_2, \dots, S_n)$. Then*

$$I_K(\theta) = \frac{1}{\theta} \sum_{k=2}^{n} \frac{k-1}{(\theta+k-1)^2}, \qquad I_S(\theta) = \frac{1}{\theta} \sum_{k=2}^{n} \frac{1}{\theta+k-1}, \tag{11.13}$$

so $I_K(\theta) \leq I_S(\theta)$ with per term ratio $(k-1)/(\theta+k-1) \uparrow 1$, and both satisfy $I(\theta) = \theta^{-1} \log n + O(1)$ as $n \to \infty$ for fixed θ.

Proof. By the Ewens sampling formula the law of K_n is $\mathbb{P}_\theta(K_n = k) = |s(n,k)|\theta^k / \{\theta(\theta+1)\cdots(\theta+n-1)\}$, a one parameter exponential family with natural parameter $\eta = \log\theta$ and sufficient statistic k. The information in η is therefore $\mathrm{Var}_\theta(K_n)$, and the information in θ is $\mathrm{Var}_\theta(K_n)(d\eta/d\theta)^2 = \theta^{-2}\sum_i q_i(1-q_i)$ with $q_i = \theta/(\theta+i-1)$, using the Feller coupling for the variance. Since $q_i(1-q_i) = \theta(i-1)/(\theta+i-1)^2$, and the $i=1$ term vanishes, the first identity follows.

For the second, write a geometric on $\{0,1,\dots\}$ with mean m as $\mathbb{P}(j) = (1+m)^{-1}\{m/(1+m)\}^j$. The score in m is $j/m - (j+1)/(1+m) = (j-m)/\{m(1+m)\}$, and since $\mathrm{Var}(j) = m(1+m)$ the information in m is $\{m(1+m)\}^{-1}$. Here $m_k = \theta/(k-1)$, so $dm_k/d\theta = 1/(k-1)$, and the information about θ in S_k is $(k-1)^{-2}\{m_k(1+m_k)\}^{-1} = \{\theta(\theta+k-1)\}^{-1}$. Independence makes information additive, which gives the second identity.

The per term ratio of the two sums is $(k-1)/(\theta+k-1) < 1$. For the asymptotics, $\sum_{k=2}^n (\theta+k-1)^{-1} = \log n + O(1)$ by comparison with $a_1(n)$, since the difference of the two sums is bounded by $\theta\, a_2(n) \leq \theta\pi^2/6$. Likewise $\sum_{k=2}^n (k-1)/(\theta+k-1)^2 = \sum_k (\theta+k-1)^{-1} - \theta\sum_k(\theta+k-1)^{-2} = \log n + O(1)$. □

Proposition 11.10 says the partition and the genealogy learn θ at the same leading rate, $\theta^{-1}\log n$, even though the coalescent count has strictly larger variance.

n	θ	$I_K = I_H$	$1/\operatorname{Var}(\hat{\theta}_W)$	I_S	$I_{\mathbf{S}}$	$\theta^{-1}\log n$
4	1	0.660	1.052	1.053	1.083	1.386
4	5	0.023	0.078	0.078	0.087	0.277
30	1	2.383	2.816	2.824	2.995	3.401
30	5	0.215	0.261	0.270	0.367	0.680
500	1	5.150	5.468	5.479	5.793	6.215
500	5	0.724	0.615	0.638	0.903	1.243

TABLE 11.3. Fisher information about θ in the Ewens block count (equivalently the haplotype count), in Watterson's estimator, in the exact law of S, and in the unobserved levelwise vector. I_S is computed from the exact mass function by numerical differentiation.

Larger variance is not less information here, because the mean grows differently; the ordering that does hold, $I_K \le I_{\mathbf{S}}$, is a data processing consequence of the thresholding in (11.12). The levelwise vector is not observed, however, and the observed count S is a coarsening of it, so the question is where S itself sits.

COROLLARY 11.11 (Information in S). *The Fisher information $I_S(\theta)$ in the observed count S satisfies*

$$\frac{a_1(n)^2}{\theta a_1(n) + \theta^2 a_2(n)} \;\le\; I_S(\theta) \;\le\; I_{\mathbf{S}}(\theta),$$

and therefore $I_S(\theta) = \theta^{-1}\log n + O(1)$. By Proposition 11.8 the haplotype count carries exactly $I_K(\theta)$.

Proof. The upper bound holds because S is a function of $(S_2, \dots, S_n)$ and Fisher information cannot increase under a measurable map. The lower bound is the Cramér and Rao inequality applied to the unbiased estimator $\hat{\theta}_W = S/a_1$, whose variance is $\theta/a_1 + \theta^2 a_2/a_1^2$ by (11.5); the family of laws of S is smooth in θ, being a finite signed combination of geometric terms by (11.8). The lower bound equals $\theta^{-1}a_1 - a_2/(1 + \theta a_2/a_1) \ge \theta^{-1}a_1 - a_2$, and the upper bound is at most $\theta^{-1}a_1$, so both are $\theta^{-1}\log n + O(1)$. □

Table 11.3 puts numbers on these bounds. Watterson's estimator is nearly efficient for the law of S, losing at most 4% of I_S in these cases, so little is gained by replacing it with the maximum likelihood estimate from S. The logarithmic asymptotics are slow: even at $n = 500$ the leading term exceeds each of I_K, I_S and $I_{\mathbf{S}}$ by between 7% and 95%. And the ordering between the two observable scalars is not fixed. At small θ the count S is more informative than H, but at $n = 500$ and $\theta = 5$ the haplotype count carries 0.724 against 0.638 for S, because the tree variance $\theta^2 a_2$ that inflates $\operatorname{Var}(S)$ affects a thresholded count much less. This is an early hint that S and H see different aspects of the genealogy.

11.3 Sufficiency Fails

This section shows that the coalescent has no analogue of the block count. For the Ewens model the block count is exactly sufficient, which is what made generative computation trivial in Chapter 10. The obstruction is structural rather than technical: the data are informative about a latent tree whose law, after conditioning on S, still depends on θ.

PROPOSITION 11.12 (S is not sufficient for θ). *Let X denote the full infinite sites data and G the genealogy. Then*

$$p(X \mid \theta) \;=\; \Big(\frac{\theta}{2}\Big)^{S(X)} \int e^{-\theta L_G/2}\, w(X,G)\, \mu(dG), \tag{11.14}$$

where w and the genealogy law μ are free of θ. The conditional law of X given S depends on θ unless the measures $w(X,G)\mu(dG)$, pushed forward to L_G, are proportional across configurations with equal S. For $n = 3$ and $n = 4$ they are not, so S is not sufficient.

Proof. Conditionally on G, the mutations form a Poisson process of rate $\theta/2$ on a tree of total length L_G. The probability that branch b of length t_b receives exactly m_b mutations for every b is $\prod_b e^{-\theta t_b/2}(\theta t_b/2)^{m_b}/m_b! = e^{-\theta L_G/2}(\theta/2)^S \prod_b t_b^{m_b}/m_b!$ with $S = \sum_b m_b$, and summing over the placements compatible with X gives a factor $w(X,G)$ that does not involve θ. Integrating over μ gives (11.14). Sufficiency of S would require the ratio of (11.14) for two configurations X, X' with $S(X) = S(X')$ to be constant in θ. That ratio is

$$\frac{\int e^{-\theta L_G/2} w(X,G)\mu(dG)}{\int e^{-\theta L_G/2} w(X',G)\mu(dG)},$$

the ratio of two Laplace transforms in $\theta/2$ of the image measures of $w(X,G)\mu(dG)$ and $w(X',G)\mu(dG)$ under $G \mapsto L_G$. By uniqueness of the Laplace transform it is constant in θ only if these two measures on $(0,\infty)$ are proportional, that is, only if $\mathbb{E}_\mu[w(X,G) \mid L_G]$ and $\mathbb{E}_\mu[w(X',G) \mid L_G]$ have a constant ratio. Example 11.13 exhibits configurations at $n = 3$ for which they do not, and Example 11.19 does the same at $n = 4$. For larger samples we do not give a general argument, but the same tilt is at work and Section 11.6 detects its consequences at $n = 30$. □

EXAMPLE 11.13 (A minimal failure of sufficiency). Take $n = 3$ and condition on a single segregating site, $S = 1$. Label the first pair to merge a, b and the remaining lineage c. The external branches of a and b have length T_3, that of c has length $T_3 + T_2$, and the internal branch above $\{a, b\}$ has length T_2, so $L_3 = 3T_3 + 2T_2$. The single site is a doubleton, with derived allele in two sequences, exactly when the mutation falls on the internal branch. By (11.14) with $s = 1$,

$$\mathbb{P}_\theta(\xi_2 = 1 \mid S = 1) = \frac{\mathbb{E}\big[T_2 e^{-\theta L_3/2}\big]}{\mathbb{E}\big[L_3 e^{-\theta L_3/2}\big]}.$$

With $T_3 \sim \mathrm{Exp}(3)$ and $T_2 \sim \mathrm{Exp}(1)$ independent, $\mathbb{E}[e^{-3\theta T_3/2}] = (1+\theta/2)^{-1}$, $\mathbb{E}[T_3 e^{-3\theta T_3/2}] = \{3(1+\theta/2)^2\}^{-1}$, $\mathbb{E}[e^{-\theta T_2}] = (1+\theta)^{-1}$ and $\mathbb{E}[T_2 e^{-\theta T_2}] = (1+\theta)^{-2}$. Hence the numerator is $(1+\theta/2)^{-1}(1+\theta)^{-2}$ and the denominator is $(1+\theta/2)^{-2}(1+\theta)^{-1} + 2(1+\theta/2)^{-1}(1+\theta)^{-2}$. Multiplying both by $(1+\theta/2)^2(1+\theta)^2$,

$$\mathbb{P}_\theta(\xi_2 = 1 \mid S = 1) = \frac{1+\theta/2}{(1+\theta)+2(1+\theta/2)} = \frac{2+\theta}{2(3+2\theta)}.$$

This decreases from $1/3$ at $\theta \to 0$, where it is the ratio $\mathbb{E}[\ell_2]/\mathbb{E}[L_3] = 1/3$ of expected branch lengths, to $1/4$ as $\theta \to \infty$. At $\theta = 0.5$, 2 and 8 it equals 0.3125, 0.2857 and 0.2632, and the fractions of doubletons among simulated samples with $S = 1$ are 0.3147, 0.2871 and 0.2632, each within 1.2 standard errors. So whether the one segregating site is shared by two sequences or carried by one is informative about θ beyond the fact that there is one site: a doubleton points to a long internal branch, hence a long tree, hence a smaller θ for the same number of mutations.

The point is worth stating in words. Conditioning on the number of mutations leaves behind a tilt $e^{-\theta L_G/2}$ on the tree, and the configuration X is informative about the tree. So learning where the mutations fell tells us something about θ beyond how many there were. Watterson's estimator, which uses S alone, is unbiased but discards this, which is the source of its inefficiency relative to the full likelihood noted by Felsenstein (1992).

REMARK 11.14 (Contrast). In the Ewens model the analogous tilt is absent because the partition is generated directly, with no latent tree to integrate. This is the precise sense in which Kingman's two models differ for inference, despite Corollary 11.7 placing them on one probability space.

Section 11.6 quantifies the loss at realistic sizes: at $n = 30$ with constant population size, moving from S to the spectrum narrows the 90% posterior interval for θ from 84.1 to 64.1.

11.4 A Ceiling on Learned Summaries

If the summary cannot be derived it must be learned, and the standard device is a permutation invariant encoder of DeepSets type (Zaheer et al., 2017), applied to the multiset of sites. This section establishes in advance what such an encoder can and cannot see.

DEFINITION 11.15. An encoder e is of *per site DeepSets form* if $e(X) = \rho\big(\sum_{j=1}^{S} \phi(x_j)\big)$ for the columns $x_1, \dots, x_S$ of X, with ϕ invariant under permutation of the n lineages.

Per site invariance of ϕ is the natural way to impose exchangeability of lineages, since the labelling of sampled individuals is arbitrary and the same relabelling applies to every site. The sum over sites makes e invariant under permutation of sites as well. The theorem below says these two natural symmetries, imposed site by site, throw away everything except allele frequencies.

THEOREM 11.16 (DeepSets encoders factor through the spectrum). *Every encoder of per site DeepSets form satisfies*

$$e(X) \;=\; \rho\Big(\sum_{i=1}^{n-1} \xi_i \,\tilde{\phi}(i)\Big) \tag{11.15}$$

for some $\tilde{\phi}$, and is therefore a function of the site frequency spectrum $\boldsymbol{\xi}$ alone.

Proof. A function of $x \in \{0,1\}^n$ invariant under all permutations of the n coordinates depends on x only through $|x| = \sum_l x_l$, since any two binary vectors of the same weight are related by a permutation. Hence $\phi(x_j) = \tilde{\phi}(|x_j|)$ where $|x_j|$ is the derived allele count at site j, which lies in $\{1, \ldots, n-1\}$ because site j is segregating. Grouping the sum over sites by that count gives $\sum_j \tilde{\phi}(|x_j|) = \sum_{i=1}^{n-1} \xi_i \tilde{\phi}(i)$, and (11.15) follows. □

The theorem is a statement about architecture, so it holds whatever ρ is and however the network is trained. Its genetic content is that a per site encoder sees each site's allele frequency and nothing about which sequences share which alleles.

COROLLARY 11.17 (The spectrum is a strict coarsening). *Under the infinite sites model without recombination, the maximal invariant of X under relabelling of lineages and permutation of sites is the gene tree, that is, the unlabelled rooted topology with mutation counts on its edges. The spectrum $\boldsymbol{\xi}$ is a strict coarsening of it. Consequently no per site DeepSets encoder, at any depth or width, can recover the information in X that is not already in $\boldsymbol{\xi}$.*

Proof. Under infinite sites without recombination, the sets of sequences carrying the derived allele at the various sites are pairwise nested or disjoint, and the perfect phylogeny they determine is the gene tree. Relabelling lineages and permuting sites preserve these nesting relations and the number of sites with each carrier set, so the gene tree is invariant. Conversely, the gene tree determines X up to an assignment of sequence labels to its leaves and an ordering of its sites, so two matrices with the same gene tree differ by a relabelling and a permutation of sites. That $\boldsymbol{\xi}$ is strictly coarser is seen already at $n = 4$ with two segregating sites, each carrying a single derived allele, so that $\xi_1 = 2$. If both mutations fall on the same external branch the gene tree has one doubly mutated tip; if they fall on different external branches it has two singly mutated tips. The spectra are identical and the gene trees are not. The final claim is Theorem 11.16. □

Corollary 11.17 is a limitation of an architecture class, not of the data, and it is useful precisely because per site DeepSets encoders are the default choice. To show that the ceiling matters one needs a statistic that is computable, invariant, and provably outside the reach of (11.15). The classical choice is the number of distinct haplotypes.

PROPOSITION 11.18 (Haplotype count escapes the spectrum). *The haplotype count $H = H(X)$ is invariant under relabelling of lineages and under permutation of sites, and H is not a function of $\boldsymbol{\xi}$.*

Proof. Invariance is immediate, since permuting rows permutes the multiset of rows and permuting columns applies a fixed bijection to every row. For the second claim take the configuration of Corollary 11.17, with $n = 4$ and $\xi_1 = 2$. If both mutations lie on the same external branch, the rows are $(1, 1)$ for that tip and $(0, 0)$ for the other three, so $H = 2$. If they lie on different external branches, the rows are $(1, 0)$, $(0, 1)$ and $(0, 0)$ twice, so $H = 3$. The spectra coincide. □

Proposition 11.18 identifies H as a minimal witness that the spectrum is lossy: it is not a per site feature, since it depends on which lineages share which mutations, so no encoder of the form (11.15) computes it. The next example shows that the information it carries changes the posterior, not merely the data.

EXAMPLE 11.19 (Same spectrum, different posteriors). Take $n = 4$ and the two data sets of Proposition 11.18: in data set A one sequence carries both derived alleles ($H = 2$); in data set B two sequences carry one each ($H = 3$). Both have $\boldsymbol{\xi} = (2, 0, 0)$. Let $e_1, \dots, e_4$ be the external branch lengths. By the Poisson placement in the proof of Proposition 11.12, up to factors free of θ,

$$p(\mathrm{A} \mid \theta) \propto \theta^2 \, \mathbb{E}\Big[e^{-\theta L_4/2} \sum_j \tfrac{1}{2} e_j^2\Big], \qquad p(\mathrm{B} \mid \theta) \propto \theta^2 \, \mathbb{E}\Big[e^{-\theta L_4/2} \sum_{i<j} e_i e_j\Big],$$

with the same constant of proportionality, so that $p(\mathrm{A} \mid \theta) + p(\mathrm{B} \mid \theta)$ is the probability of the spectrum. Two topologies occur. With probability $2/3$ the tree is a caterpillar with $(e_1, \dots, e_4) = (T_4, T_4, T_4 + T_3, T_4 + T_3 + T_2)$, and with probability $1/3$ it is balanced with $(T_4, T_4, T_4 + T_3, T_4 + T_3)$; in both $L_4 = 4T_4 + 3T_3 + 2T_2$. The expectations reduce to the tilted exponential moments $\mathbb{E}[T_k^m e^{-\theta k T_k/2}] = m!\, \lambda_k / \mu_k^{m+1}$ with $\mu_k = k(k - 1 + \theta)/2$, so the likelihoods are explicit rational functions of θ.

The conditional probability of A given the spectrum is 0.489, 0.457, 0.434, 0.409, 0.388 and 0.373 at $\theta = 0.1$, 0.5, 1, 2, 4, 8, tending to $1/2$ as $\theta \to 0$. Simulation agrees: among coalescents with $\boldsymbol{\xi} = (2, 0, 0)$ the fraction with $H = 2$ is 0.433 at $\theta = 1$ (21708 samples) and 0.389 at $\theta = 4$ (9961 samples). The maximum likelihood estimates are 1.13 from A, 1.30 from B and 1.22 from the spectrum alone, and under a $\mathrm{Ga}(2, 1)$ prior the posterior means are 1.71, 1.80 and 1.76. The data sets agree in every per site summary and disagree about θ. The direction makes sense: a larger θ tilts toward shorter trees, the tilt shrinks the deep epochs most, the external branches become more nearly equal, and two mutations are then more likely to land on different tips.

Capturing linkage more fully requires either features defined on pairs of sites, so that ϕ is no longer per site invariant while e remains invariant overall, or an architecture defined on the inferred gene tree. A pairwise feature already separates

the two data sets of Example 11.19: the number of pairs of identical columns is one in A and zero in B. We take only the scalar step here. What Corollary 11.17 buys is an interpretation of our own results: spectrum based posteriors are the ceiling for the DeepSets class, and Section 11.6.4 measures how much a statistic above the ceiling adds in practice.

11.5 Generative Bayesian Computation

This section recalls the transport map formulation of Chapter 10 in the form used for the experiments, and records the simulator.

11.5.1 The Quantile Map

Let $\vartheta \sim \pi$ be the parameter, $X \sim p(\cdot \mid \vartheta)$ the data, and $Z = e(X)$ an encoding. Generative Bayesian computation (GBC) seeks a map $\mathcal{Q}$ with

$$\vartheta \overset{d}{=} \mathcal{Q}(Z, U) \mid Z, \qquad U \sim \mathrm{Unif}(0,1) \perp\!\!\!\perp Z, \tag{11.16}$$

so that posterior draws are evaluations at fresh uniforms. For scalar ϑ, $\mathcal{Q}(Z, \cdot)$ is the posterior quantile function, characterised as the minimiser of expected check loss,

$$\begin{gathered} \mathcal{Q} = \arg\min_{h} \; \mathbb{E}_{\pi(\vartheta)p(X\mid\vartheta)} \mathbb{E}_U \big[\rho_U\{\vartheta - h(Z,U)\}\big], \\ \rho_u(r) = r\{u - \mathbf{1}(r < 0)\}. \end{gathered} \tag{11.17}$$

The expectation is over the prior predictive, which we can sample from without evaluating any likelihood. That is the entire appeal for the coalescent. Since quantiles commute with increasing maps, we fit the quantile function of $\log \vartheta$ and exponentiate.

11.5.2 Vector Parameters

For $\vartheta = (\vartheta_1, \vartheta_2)$ the check loss applied coordinatewise recovers the two *marginal* posterior quantile functions, not the joint law. The joint is obtained autoregressively,

$$\vartheta_1 = \mathcal{Q}_1(Z, U_1), \qquad \vartheta_2 = \mathcal{Q}_2(Z, \vartheta_1, U_2), \tag{11.18}$$

with $\mathcal{Q}_2$ trained by the same objective on the augmented input. In Section 11.6 we report marginals, which is what (11.17) delivers directly, and we do not evaluate the joint map (11.18).

11.5.3 Algorithm

Algorithm 11.1 GBC for coalescent parameters

1: Input: sample size n, prior π, encoder e, training size N, network h_ϕ.
2: for $r = 1, \dots, N$ do
3: draw $\vartheta^{(r)} \sim \pi$;
4: draw coalescent times $T_k \sim \mathrm{Exp}\{k(k-1)/2\}$ and merge uniformly chosen pairs, accumulating ℓ_i, the branch length subtending i leaves;
5: draw $\xi_i^{(r)} \sim \mathrm{Poisson}(\theta^{(r)} \ell_i / 2)$ for $i = 1, \dots, n-1$ and set $Z^{(r)} = e(\boldsymbol{\xi}^{(r)})$.
6: end for
7: Minimise the empirical check loss over minibatches, drawing $u \sim \mathrm{Unif}(\epsilon, 1-\epsilon)$ afresh at each step.
8: Output: for observed Z, posterior draws $\{h_{\hat{\phi}}(Z, u^{(m)})\}$, sorted in u.

Under exponential growth with $N(t) = N_0 e^{-gt}$ backwards in time, step 4 is modified by the standard time change. The coalescence intensity at time t is proportional to $1/N(t) \propto e^{gt}$, so the coalescent runs at constant rate in the rescaled time $\Lambda(t) = \int_0^t e^{gs} ds = (e^{gt} - 1)/g$. We therefore generate the coalescent in rescaled time and map back by $t = g^{-1} \log(1 + g\Lambda)$. The encoder input is $\log(1 + \xi_i)$, standardised.

11.6 Experiments

The experiments grade learned posteriors against exact ones wherever the theory of Sections 11.2 to 11.4 supplies an exact answer, and measure what the spectrum and the haplotype count add where it does not. Throughout $n = 30$, so the spectrum has 29 cells. Networks have two hidden layers of 128 rectified linear units, trained by Adam on the check loss with a decaying learning rate. As a check on the simulator, at $g = 0$ and $\theta = 8$ the simulated spectrum means over 2×10^4 replicates are 8.047, 4.014, 2.628, 1.567, 0.796 at $i = 1, 2, 3, 5, 10$, against the theoretical θ/i values 8, 4, 2.667, 1.6, 0.8.

11.6.1 Exact Validation: the Posterior Given S

Corollary 11.3 makes the posterior for θ given S exactly computable, so we can grade the learned map rather than merely inspect it. We take $\pi(\theta) = \mathrm{Ga}(3, \text{scale } 3)$, train on $N = 2 \times 10^5$ prior predictive pairs, and compare against quadrature on (11.9).

Table 11.4 shows that the largest discrepancy among the five tabulated quantiles is 0.22, 0.27, 0.16 and 0.21 at $S = 20, 40, 60, 90$. Over a grid of 49 levels spanning $[0.02, 0.98]$ the maximum absolute discrepancies are 0.216, 0.305, 0.257 and 0.752,

S	method	5%	25%	50%	75%	95%
20	exact	3.392	4.760	5.962	7.407	9.992
	GBC	3.410	4.759	5.936	7.370	9.771
40	exact	6.063	8.210	10.038	12.180	15.892
	GBC	6.122	8.206	9.987	12.106	15.620
60	exact	8.268	11.028	13.355	16.048	20.641
	GBC	8.330	10.947	13.368	16.055	20.479
90	exact	10.977	14.437	17.343	20.685	26.316
	GBC	11.144	14.312	17.285	20.471	26.288

TABLE 11.4. Posterior quantiles of θ given the number of segregating sites at $n = 30$ under a $\mathrm{Ga}(3, 3)$ prior. Exact values by inversion of the generating function of Theorem 11.1 followed by quadrature; GBC values from the trained quantile network.

encoder	mean 90% width for θ	coverage at 90%	coverage at 50%
site frequency spectrum	64.09	0.886	0.450
S alone	84.15	0.896	0.497
prior	159.68		

TABLE 11.5. Constant population size, $n = 30$, $\pi(\theta) = \mathrm{Ga}(4, \text{scale } 25)$, $\mathbb{E}[S] = 396$. The spectrum removes 60% of the prior interval width against 47% for S alone, confirming Proposition 11.12 numerically.

mean 0.383. Posterior standard deviations across these cases run from about 2 to 4.7, so the error is between 7 and 16 percent of a posterior standard deviation. Training noise matters at this scale: over four training seeds the grid maximum ranges from 0.06 to 1.40, that is from 3 to 30 percent of a posterior standard deviation, and its largest values sit at the extreme levels 0.02 and 0.98. For every seed it is largest at $S = 90$, where the prior predictive thins in the upper tail.

11.6.2 The Spectrum Beats S at Constant Population Size

Proposition 11.12 says S discards information about θ even when the population size is constant. We test this directly by training two networks on the same 9×10^4 replicates at $g \equiv 0$ with $\pi(\theta) = \mathrm{Ga}(4, \text{scale } 25)$, one encoding the full spectrum and one encoding S alone, and evaluating on 1.2×10^4 fresh replicates.

Table 11.5 is a direct numerical demonstration of Proposition 11.12. Were S sufficient, the two rows would agree up to training error; instead the spectrum delivers an interval 24% narrower. The gap is far larger than the discrepancies in Table 11.4, so it is not an artefact of optimisation. Part of it is miscalibration, since the spectrum row covers 0.886 against a nominal 0.90, but an interval that is 24% narrower would need far worse coverage than this to be explained by overconfidence alone: shrinking a calibrated normal 90% interval by 24% cuts its coverage to about 0.79.

encoder	parameter	mean 90% width	coverage at 90%	coverage at 50%
spectrum	θ	74.83	0.885	0.447
spectrum	g	12.16	0.879	0.466
S alone	θ	95.59	0.903	0.507
S alone	g	15.51	0.901	0.502
prior	θ	159.68		
prior	g	18.00		

TABLE 11.6. Exponential growth, $n = 30$, $\mathbb{E}[S] = 142$. The spectrum removes 32% of the prior width for the growth rate against 14% for S alone.

11.6.3 Exponential Growth

With a growth rate g the model has two parameters and no closed form likelihood. We take $\pi(\theta) = \mathrm{Ga}(4, \text{scale } 25)$ and $g \sim \mathrm{Unif}(0, 20)$, train on 1.2×10^5 replicates, and evaluate on 1.2×10^4 fresh ones. The mean number of segregating sites is 142.

The pattern in Table 11.6 is what the theory predicts. The number of segregating sites is informative about the product of mutation rate and tree length, so it cannot separate a high mutation rate on a short tree from a low rate on a long one; it removes only 14% of the prior width for g. The shape of the spectrum resolves the confounding, because growth skews the genealogy toward starlike topologies and hence toward an excess of singletons, and it removes 32%. The improvement for θ is of similar size, consistent with Table 11.5.

11.6.4 Testing the Ceiling with Haplotype Structure

Proposition 11.18 supplies a statistic outside the DeepSets class, and this section measures what it adds. We track H exactly during simulation by a merge recursion. For each node v of the genealogy let c_v be the number of distinct haplotypes among the leaves below v, and let $\alpha_v \in \{0, 1\}$ record whether some leaf below v carries exactly the sequence of v itself, that is, has no mutation on its path up to v. Leaves have $c = \alpha = 1$. When A and B merge into C, write f_A for the indicator that the branch from A up to C carries no mutation, which has probability $e^{-\theta b_A/2}$ for a branch of length b_A, and similarly f_B. Then

$$c_C = c_A + c_B - \alpha_A \alpha_B f_A f_B, \qquad \alpha_C = \max(\alpha_A f_A,\ \alpha_B f_B), \tag{11.19}$$

because a haplotype below A can equal one below B only if both equal the sequence of C, which requires an unmutated path from a leaf to C on each side, and at most one haplotype on each side has that property. At the root c equals H. Without the flag α the recursion would merge two haplotypes whenever both branches are unmutated, even when neither subtree retains its ancestral type, and it undercounts: at $\theta = 2$ and $n = 30$ it gives a mean of 5.82 against the correct 6.05. In 9.2×10^4 simulated samples the recursion (11.19) agrees with a direct count of distinct rows in every case, and the sample means agree with Proposition 11.8.

prior	method	mean 90% width	coverage at 90%	coverage at 50%	SBC χ^2_9
Ga(2, 4)	exact given S	8.18	0.901	0.495	12.9
	exact given H	9.03	0.902	0.507	16.4
	network, S	8.26	0.900	0.497	23.9, 17.8
	network, S and H	7.55	0.904	0.501	12.5, 8.7
	network, $\boldsymbol{\xi}$	7.14	0.890	0.464	100.7, 91.0
	network, $\boldsymbol{\xi}$ and H	7.05	0.891	0.461	89.7, 112.5
	prior	17.55			
Ga(4, 25)	exact given S	84.75	0.900	0.503	7.8
	exact given H	118.21	0.902	0.508	6.4
	network, S	85.01	0.896	0.497	15.6, 58.4
	network, S and H	80.77	0.904	0.500	15.7, 56.9
	network, $\boldsymbol{\xi}$	62.51	0.877	0.444	224.1, 230.2
	network, $\boldsymbol{\xi}$ and H	62.25	0.877	0.440	275.5, 248.6
	prior	159.68			

TABLE 11.7. Constant population size, $n = 30$, 1.2×10^4 test replicates; priors are Ga(shape, scale); the Monte Carlo standard error of a coverage at 90% is 0.003. Each network row averages two training seeds and the simulation based calibration (SBC) statistic of Section 11.6.7 is given for each seed. The five percent critical value of χ^2_9 is 16.9.

We compare four encoders at constant population size with $\pi(\theta) = \text{Ga}(2, \text{scale } 4)$, training on 8×10^4 replicates and evaluating on 1.2×10^4: S alone, S and H, the spectrum, and the spectrum and H. Here $\mathbb{E}[S] = 8a_1(30) = 31.7$ and, by Proposition 11.8, $\mathbb{E}[H] = 11.75$. The quantile level enters the network as $\frac{1}{2}\log\{u/(1-u)\}$, which resolves the tails better than u itself, and each network is trained for 60 passes with two random seeds; we report the average. Two exact posteriors serve as anchors: the posterior given S from Corollary 11.3, and the posterior given H alone, whose likelihood $\theta^H/\{\theta(\theta+1)\cdots(\theta+n-1)\}$ is the Ewens formula. We repeat the comparison under the Ga(4, scale 25) prior of Table 11.5, where $\mathbb{E}[H] = 25.63$ is close to its maximum of 30, with the same sizes and two seeds.

Table 11.7 separates three effects. First, the haplotype count adds clearly to S. At low diversity, appending H to S narrows the mean interval from 8.26 to 7.55, about 9%, while keeping coverage at nominal and lowering the calibration statistic; the two seed widths for S alone, 8.38 and 8.15, show that training noise is a few percent at most. At high diversity the gain falls to 5%, from 85.01 to 80.77, as H approaches saturation at n and its information about θ shrinks relative to that in S. The exact anchors confirm the direction: H alone is less informative than S alone in both designs, consistent with Table 11.3 at small and moderate θ, yet it is not redundant given S.

Second, the spectrum beats S and H together, by 5% at low diversity and 23% at high diversity, so the allele frequency information captured by the spectrum is

S	parameter	method	5%	25%	50%	75%	95%
40	θ	reference	24.53	34.08	40.76	47.61	58.18
		GBC	24.85	34.02	40.91	48.23	60.30
	g	reference	4.69	10.36	14.40	17.44	19.52
		GBC	4.90	10.29	14.35	17.44	19.52
100	θ	reference	49.43	71.86	88.14	103.13	123.55
		GBC	50.02	72.24	89.19	104.22	125.68
	g	reference	2.62	7.41	11.81	15.88	19.16
		GBC	2.62	7.47	11.84	15.91	19.14
200	θ	reference	71.41	103.96	133.47	166.34	210.03
		GBC	71.11	103.37	133.44	167.90	213.23
	g	reference	0.83	3.29	6.49	10.89	17.21
		GBC	0.82	3.26	6.54	10.97	17.22

TABLE 11.8. Exponential growth model, S only encoder, $n = 30$. Marginal posterior quantiles from the reference posterior built on (11.7) against those from the trained quantile network (one training run, 1.2×10^5 training replicates).

the larger part of what S discards.

Third, and this is the finding that bears on the ceiling, appending H to the spectrum gains only 1.3% at low diversity (7.14 to 7.05) and 0.4% at high diversity (62.51 to 62.25), with coverage unchanged. These gains are comparable to the seed to seed variation under the $\mathrm{Ga}(2, 4)$ prior, which is 0.04 and 0.07 in width, about 0.6% and 1%, for the two spectrum encoders, and 1.30 and 2.06, about 2% and 3%, under the $\mathrm{Ga}(4, 25)$ prior. So although Proposition 11.18 and Example 11.19 prove that H carries information the spectrum cannot, in these designs that information largely overlaps with what the spectrum already provides. The ceiling of Corollary 11.17 is real, but a single haplotype count does not get far above it.

The spectrum encoders also undercover, at 0.88 to 0.89 against a nominal 0.90, which is three to eight Monte Carlo standard errors, and their calibration statistics are large. Their width advantage over the S and H encoder is therefore partly overconfidence. The comparison between the two spectrum rows, which have equal coverage, is unaffected.

11.6.5 Validation Against an Exact Reference Posterior Under Growth

Table 11.6 is validated only by calibration, because the growth model has no closed form likelihood. Proposition 11.2 removes that excuse for the S only encoder: by (11.7) the likelihood of S under any demography is a one dimensional integral against the law of L_n, so a reference posterior can be built by simulating tree lengths alone, with no topology and no mutations. We evaluate $\mathbb{P}_\theta(S = s \mid g) = \mathbb{E}_L[e^{-\theta L/2}(\theta L/2)^s/s!]$ with 2.7×10^8 simulated tree lengths, drawing g from its prior, and integrate against the prior, in closed form for θ since its gamma prior is conjugate to the Poisson factor.

Table 11.8 shows agreement across both parameters and a fivefold range of S. The largest absolute discrepancy for θ is 3.2, at the 95% quantile of the $S = 200$ case, which is 1.5% of the quantile; the largest relative discrepancy is 3.7%, at the 95% quantile of the $S = 40$ case, where the two values differ by 2.13. For the growth rate the largest discrepancy is 0.21 on a prior range of 20. Three further training runs of the same network do no worse, with largest relative discrepancies for θ between 2.3% and 2.4% and largest discrepancies for the growth rate between 0.13 and 0.21. The learned transport map is therefore recovering the correct two parameter posterior and not merely a well calibrated approximation to it, which is what calibration alone could not establish.

Two further remarks. The reference construction extends to any population history for which tree lengths can be simulated, so every S based encoder can now be graded against near truth rather than against itself. It does not extend to the spectrum encoders of Tables 11.5 to 11.7, since ξ is not a function of L_n alone, and those remain validated by calibration only.

11.6.6 Large Samples

All experiments above use $n = 30$. Theorem 11.1 is dimension free and can be checked at any sample size: at $n = 500$, inverting the generating function on 8192 roots of unity returns total mass 1.00000000, mean 54.3266 against Watterson's $\theta a_1(500) = 54.3266$, and variance 159.474 against $\theta a_1 + \theta^2 a_2 = 159.474$, at $\theta = 8$. The exact machinery of Section 11.2.3 therefore scales without modification. What we have not tested is the spectrum encoder at large n, where ξ has $n - 1$ cells and the fixed width network used here would need replacing; see Section 11.7.

11.6.7 Calibration

Coverage tells only part of the calibration story, and this section applies the stronger test. In Tables 11.5 and 11.6 coverage at the 90% level is within about two percentage points of nominal, and at the 50% level the spectrum rows fall three to five points short, but with 1.2×10^4 test replicates the Monte Carlo standard error at the 90% level is 0.003, so the spectrum coverages of 0.886 and 0.885 for θ and 0.879 for g are statistically real, while the S only encoders stay within 0.004 of nominal. Simulation based calibration (Talts et al., 2018) asks for more: if the map were exact, the rank of the true parameter among 99 fitted posterior quantiles would be uniform. Binning 1.2×10^4 replicates into ten cells with expected count 1200 gives chi square statistics on nine degrees of freedom of

$$\begin{aligned} 161.5\ (\text{spectrum},\ \theta), \qquad & 108.6\ (\text{spectrum},\ g), \\ 11.0\ (S,\ \theta), \qquad & 8.5\ (S,\ g) \end{aligned}$$

for the networks of Table 11.6, against a five percent critical value of 16.92; the networks of Table 11.5 give 153.5 for the spectrum and 15.9 for S. Uniformity is rejected decisively for every spectrum map and for no S only map in these two

tables. The test is not oversensitive: the exact posteriors in Table 11.7, scored the same way, give statistics between 6.4 and 16.4 and pass.

We report this because it is the honest result and because the two findings are compatible rather than contradictory. Every 90% interval a practitioner would quote is close to nominal, and the spectrum maps are nonetheless detectably inexact once more than ten thousand replicates are brought to bear. They are also the hardest maps to learn, being the ones where the signal enters through the shape of a 29 dimensional count vector rather than its total, and they fail for both training seeds, with statistics between 80.7 and 208.6 over the two tables. The S only encoders, whose input is one dimensional, are close to the critical value and on either side of it depending on the training run: they pass here, and for a second seed (between 9.2 and 13.1), but a separately trained S only network for the design of Section 11.6.5, the one recalibrated below, gives 33.2 for θ and 8.3 for g on 2×10^4 test replicates, three further training runs of it give 15.5 to 31.0 for θ and 5.9 to 17.8 for g, and the S rows of Table 11.7 range from 15.6 to 58.4.

11.6.8 Rank Recalibration, and What It Cannot Show

The standard remedy for nonuniform ranks is recalibration. If the probability integral transform $V = \hat{F}(\theta \mid Z)$ has marginal distribution function G, then $G(V)$ is uniform by construction, so the corrected quantile map is $h(Z, G^{-1}(\tau))$, with G estimated on a held out calibration set. This is a one line fix requiring no retraining.

It works here, in the marginal sense it is built for. Estimating G on 2×10^4 calibration replicates, with the transform computed on a 400 point grid so that V is effectively continuous, and testing on 2×10^4 fresh replicates, the chi square statistic moves from 33.2 to 14.6 for θ and stays at 8.3 for g. Coverage at the 90% level moves from 0.907 to 0.897 for θ and from 0.899 to 0.901 for g, and the mean 90% width for θ falls from 97.4 to 94.2: the network's intervals for θ were slightly too wide, and a global correction removes that. Over the three further training runs the statistic for θ falls from between 15.5 and 31.0 to between 11.3 and 13.5, and for g it ends between 4.1 and 9.5. The uncorrected critical value is the wrong yardstick after recalibration, because G is itself estimated: for an exact map, with calibration and test sets of 2×10^4 each, the statistic has mean 17.9 and 95th percentile 33.0, so every value after the repair is within the noise the calibration set contributes.

What the repair cannot show is that the map is right for each data set. A global reparameterisation of τ is constant in Z, so it makes the marginal transform uniform whatever the map does at particular values of Z, and a passing statistic after recalibration certifies the marginal and nothing more. A map that is compressed near the median for some values of Z and stretched for others can pass after recalibration while remaining locally wrong; correcting that requires either conditional recalibration, with G estimated as a function of Z, or a map with enough capacity not to need it. For this network the conditional check is available. Coverage at the 90% level within thirds of S is 0.909, 0.900 and 0.914 for θ before recalibration and 0.898, 0.890 and 0.903 after, each with a Monte Carlo

standard error of 0.004: the repair removes the common overcoverage and leaves the spread across thirds, 1.5 points before and 1.3 after, where it was. That spread, between three and four standard errors, is the mild conditional error a global correction cannot reach, and Table 11.8, which compares quantiles at fixed S with the reference posterior, shows that it stays within 3.7% of each quantile. We have not implemented conditional recalibration for the networks.

11.6.9 Calibration When the Truth Is Known

The distinction between marginal and conditional calibration is easy to state and easy to forget, so we close the experiments with an example in which every posterior is known exactly and the distinction can be seen directly.

EXAMPLE 11.20 (Simulation based calibration with exact posteriors). Take $n = 30$ and $\pi(\theta) = \mathrm{Ga}(3, \text{scale } 3)$, and draw $M = 2 \times 10^4$ pairs (θ, S) from the prior predictive, using the geometric law of Theorem 11.1. For each pair compute the probability integral transform $V = F(\theta \mid S)$ under four candidate posteriors, and bin V into ten cells; with continuous V this is the limit of the rank statistic as the number of posterior draws grows.

(i) The exact posterior given S, from Corollary 11.3.

(ii) The prior, reported as if it were the posterior.

(iii) A Poisson gamma approximation that treats S as $\mathrm{Poisson}(\theta a_1)$, ignoring the tree variance $\theta^2 a_2$, giving the posterior $\mathrm{Ga}(3 + S, \text{ rate } 1/3 + a_1)$.

(iv) The approximation (iii) after global rank recalibration, with G estimated on a second set of 2×10^4 replicates.

Table 11.9 and Figure 11.2 give the results. The exact posterior passes, as it must. So does the prior: its transform is $F_\pi(\theta)$ with $\theta \sim \pi$, which is exactly uniform, yet it ignores the data entirely, and its coverage conditional on S is 0.85 in the lowest and highest thirds of S and 1.00 in the middle third. The Poisson gamma approximation is far too narrow, with mean width 4.53 against 8.71 for the exact posterior, and its U shaped histogram fails decisively. Recalibration repairs the marginal histogram, bringing the statistic down to 16.2, just below the critical value, and the overall coverage to 0.898, but only by moving to the levels 0.0004 and 0.9993 of the approximate posterior, and conditionally it is still wrong: coverage is 0.945 when S is small and 0.825 when S is large. The approximation's error grows with S, because the neglected variance $\theta^2 a_2$ grows faster than the retained θa_1, and a correction that does not depend on S cannot follow it.

Example 11.20 is the networks' situation in miniature. Uniform ranks are necessary for a correct posterior but nowhere near sufficient, and a marginal repair can make the histogram look right while leaving the conditional error in place. The diagnostic that matters is calibration conditional on the data, which here is coverage within strata of S, and the corresponding repair must also depend on the data.

posterior	χ^2_9	coverage at 90%	$S < 22$	$22 \le S < 40$	$S \ge 40$	mean 90% width
exact given S	8.7	0.902	0.903	0.904	0.901	8.71
prior	8.5	0.900	0.846	0.998	0.856	16.43
Poisson gamma	11231	0.615	0.697	0.644	0.507	4.53
recalibrated	16.2	0.898	0.945	0.928	0.825	9.02

TABLE 11.9. Simulation based calibration with 2×10^4 prior predictive replicates, $n = 30$, $\pi(\theta) = \text{Ga}(3, 3)$. The middle columns give coverage of the 90% interval within thirds of the observed S, each with Monte Carlo standard error about 0.004. The five percent critical value of χ^2_9 is 16.9.

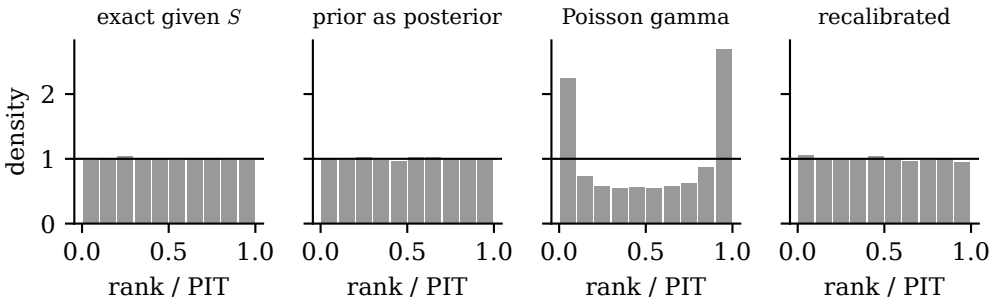


FIGURE 11.2. Histograms of the probability integral transform for the four posteriors of Example 11.20. The first two are uniform, although only the first uses the data; the third is U shaped because the approximation is overconfident; the fourth is flat after recalibration, although its coverage still varies with S.

11.7 Limitations

This section collects what the chapter does not establish.

1. Theorem 11.1 assumes constant population size, no recombination and no selection. The geometric decomposition uses the exponential holding times of (11.1) directly and does not survive a time change, though by Proposition 11.2 the mixed Poisson representation does, which is what Section 11.6.5 exploits.

2. Proposition 11.12 is made explicit by the three sequence calculation of Example 11.13, and Example 11.19 computes exact likelihoods for two four sequence data sets. Beyond these small cases the full data likelihood is not computed, so the gap between spectrum based posteriors and the full data posterior is not measured; methods such as those of Griffiths and Tavaré (1994) would be needed.

3. Proposition 11.18 takes one step past the DeepSets ceiling, not the full step. The haplotype count is a scalar summary of linkage, and in Table 11.7 it narrows the interval given S by about 9% at low diversity and 5% at high

diversity, but narrows the interval given the spectrum by at most 1.3%, comparable to training variation. Encoders with genuine pairwise site features, or architectures on the inferred gene tree, remain unimplemented, and how far above the ceiling the full data posterior lies is unmeasured. This is the largest open gap in the chapter.

4. The reference posterior of Section 11.6.5 validates the S only encoder only. Because ξ is not a function of L_n, the same construction does not grade the spectrum encoders, which carry the chapter's main claims and remain validated by calibration alone.

5. The spectrum encoders undercover, at 0.877 to 0.891 against nominal 0.90 across Tables 11.5, 11.6 and 11.7, so their width advantages over the S based encoders are partly overconfidence. Only comparisons between rows with matching coverage are clean.

6. The exact machinery of Section 11.2.3 is verified at $n = 500$, but every learned encoder is trained at $n = 30$. At the sample sizes of modern sequencing studies the spectrum has thousands of cells and the fixed width network used here would need replacing, most likely by an architecture exploiting the smoothness of $\mathbb{E}[\xi_i] = \theta/i$ in i.

7. Section 11.6.8 applies the standard recalibration fix to the S only network of Section 11.6.5: it repairs the marginal ranks but cannot reach the small conditional error that remains, which Example 11.20 reproduces exactly at a larger scale. We have not applied it to the spectrum maps, whose failures are far larger, and we do not implement conditional recalibration for the networks, so the calibration gap is characterised but still open.

8. Section 11.5 reports marginal posteriors only. The autoregressive construction (11.18) for the joint law of (θ, g) is described but not evaluated, so nothing here speaks to the posterior dependence between mutation rate and growth rate, which is precisely the confounding the spectrum is supposed to resolve.

11.8 Discussion

Chapter 10 found that Kingman's exchangeable partitions hand generative computation an exactly sufficient statistic, reducing posterior inference to a two dimensional transport problem. The natural expectation was that his coalescent would do the same, with segregating sites in place of blocks. Corollary 11.7 shows how close the two models are, placing them on one probability space with the partition as the thresholded coalescent, and Proposition 11.8 shows that the observed haplotype partition is itself an Ewens partition. All of these statistics carry the same leading Fisher information $\theta^{-1} \log n$. Proposition 11.12 shows the models nonetheless part company for inference, because the coalescent hides a genealogy that conditioning on S does not integrate away, and Example 11.13 shows this already with three sequences and one segregating site.

That failure is the reason the field built approximate Bayesian computation on this model and not on the Ewens formula. What generative computation changes is the response to it. Rather than choosing summaries and accepting parameter draws whose summaries fall close to the observed values, one learns a transport map by minimising check loss over prior predictive draws, and obtains the whole posterior quantile function. Theorem 11.16 then supplies something that the summary selection literature could not: a proof of what a given class of learned summaries can see. Per site DeepSets encoders see the spectrum and nothing more, so the choice between hand chosen and learned summaries is, within that class, not a choice at all.

The ceiling is real but, in the designs tested here, low. Proposition 11.18 names a single scalar outside the class, and Example 11.19 shows it moving the posterior for data sets the spectrum cannot tell apart. In Table 11.7 the haplotype count adds substantially to S, but at most 1.3 percent to the full spectrum. The two observations are compatible: Example 11.19 conditions on one spectrum and varies the haplotypes, whereas the experiment averages over the prior predictive, where most of what H says about the tree is already said by the shape of the spectrum. The useful reading is that a scalar is not enough, and that further gains must come from features that see which sequences share which mutations, as in the pairwise statistic noted after Example 11.19, which is where the genuinely linked information lives.

The obvious next step is therefore to go past a scalar. Corollary 11.17 identifies the gene tree as the maximal invariant, which suggests encoding the inferred perfect phylogeny directly and learning on trees rather than on counts. Whether the resulting posteriors close the gap to the full data likelihood, and whether they survive simulation based calibration at the replicate counts used here, are questions our experiments raise and do not answer. Proposition 11.2 already supplies the grading instrument for one half of that programme: since the likelihood of S under any demography is a one dimensional integral against the tree length law, exact reference posteriors are available for arbitrary population histories, and Section 11.6.5 uses one. Extending the same idea past S is the obstacle, because the spectrum is not a functional of tree length alone.

A last word on calibration. Section 11.6.8 reports a repair that passes the marginal test, and we think what it leaves untouched is the more useful finding. Rank recalibration is the reflexive response to a nonuniform simulation based calibration histogram, and it is guaranteed to flatten the histogram, up to the noise of the calibration set, whatever the map does for particular data. For the S only network it did so, removing a common overcoverage, and left the spread of coverage across strata of S where it was, because that error varies with the data rather than across the board. Example 11.20 shows the same thing with every posterior known exactly and a much larger conditional error, and also shows the converse hazard, that a posterior which ignores the data passes the test. Practitioners reaching for the standard fix should check calibration conditional on the data as well, because a flat histogram after recalibration is guaranteed and

therefore says nothing about the conditional regime.

Reproducibility

The exact mass function of Theorem 11.1 is obtained by a 4096 point fast Fourier transform of the generating function, or by the convolution recursion after Corollary 11.3. The coalescent simulator accumulates branch length by leaf count class using a flattened bin count, and is validated against $\mathbb{E}[\xi_i] = \theta/i$ and, at $n = 4$, against the exact law of Table 11.1. Quantile networks are implemented in numpy with hand coded backpropagation of the check loss. Code for the worked examples and for Tables 11.4 to 11.8 is in the directory `code/coal_rev` accompanying the book.

11.9 Exercises

1. Show that $\mathbb{P}(S = 0) = (n-1)!/\{(\theta+1)\cdots(\theta+n-1)\}$, that this equals the Ewens probability $\mathbb{P}(K_n = 1)$, and that it equals $1/n$ when $\theta = 1$. *Hint.* $S = 0$ if and only if every $S_k = 0$; for the Ewens side use the Feller coupling.

2. For $n = 4$ and $\theta = 1$ verify $\mathbb{P}(S = 2) = 115/576$ in two ways: from the complete homogeneous polynomial $h_2(\frac{1}{2}, \frac{1}{3}, \frac{1}{4})$, and from (11.8). *Hint.* $h_2 = \sum_k q_k^2 + \sum_{j<k} q_j q_k = 115/144$.

3. Show that $\mathbb{E}[S(S-1)] = \theta^2\{a_1(n)^2 + a_2(n)\}$ and deduce that (11.10) is unbiased for $\mathrm{Var}(\hat{\theta}_W)$. *Hint.* Use (11.5) and $\mathbb{E}[S(S-1)] = \mathrm{Var}(S) + \mathbb{E}[S]^2 - \mathbb{E}[S]$.

4. Prove that S is stochastically increasing in θ, and hence that the exact interval of Section 11.2.4 has at least the stated coverage. *Hint.* Each S_k is geometric with failure probability $\theta/(\theta+k-1)$, increasing in θ; couple the summands monotonically.

5. For $n = 3$ derive the full law of the spectrum, $\mathbb{P}(\xi_1 = i, \xi_2 = j) = \mathbb{E}\big[\mathrm{Poisson}(i; \theta(3T_3 + T_2)/2)\,\mathrm{Poisson}(j; \theta T_2/2)\big]$, and recover the formula of Example 11.13 for $\mathbb{P}(\xi_2 = 1 \mid S = 1)$. *Hint.* Here $\ell_1 = 3T_3 + T_2$ and $\ell_2 = T_2$; condition on the tree and use the tilted exponential moments.

6. In Example 11.19, show that the conditional probability of data set A given $\boldsymbol{\xi} = (2, 0, 0)$ tends to $1/2$ as $\theta \to 0$. *Hint.* At $\theta = 0$ the tilt disappears; compute $\mathbb{E}[\sum_j e_j^2]$ and $\mathbb{E}[\sum_{i<j} e_i e_j]$ for both topologies from $\mathbb{E}[T_k] = 2/\{k(k-1)\}$ and $\mathbb{E}[T_k^2] = 8/\{k(k-1)\}^2$.

7. Show that H and $K' = 1 + \#\{k : S_k \geq 1\}$ always agree when $n = 2$, and construct a genealogy with mutations for $n = 3$ on which $K' = 3$ but $H = 2$. *Hint.* For $n = 3$ put one mutation on the lone lineage in each of the two epochs.

8. Prove the identity $I_S(\theta) = \theta^{-1} \sum_{k=2}^{n} (\theta + k - 1)^{-1}$ by computing the score of the geometric directly in θ, and check the entry 1.083 of Table 11.3 for $n = 4$, $\theta = 1$. *Hint.* $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}$.

9. Show that a pairwise encoder $\sum_{j<l} \psi(x_j, x_l)$ with ψ invariant under simultaneous permutation of the lineages in both arguments can distinguish the two data sets of Example 11.19, and so is not bound by Theorem 11.16. *Hint.* Take $\psi(x, y) = \mathbf{1}\{x = y\}$.

10. Prove that if a candidate posterior for θ ignores the data and equals the prior, its probability integral transform is exactly uniform under the prior predictive, so simulation based calibration cannot reject it. *Hint.* $F_\pi(\theta)$ is uniform when $\theta \sim \pi$ and F_π is continuous.

11. Derive the rescaled time $\Lambda(t) = (e^{gt} - 1)/g$ for $N(t) = N_0 e^{-gt}$ and show that the expected time to the most recent common ancestor of two sequences is $g^{-1}\mathbb{E}[\log(1 + gE)]$ with $E \sim \mathrm{Exp}(1)$; evaluate it numerically for $g = 1$. *Hint.* The value is $e\,E_1(1) \approx 0.596$, where E_1 is the exponential integral.

12. Using the convolution recursion after Corollary 11.3, compute the exact posterior of θ given $S = 18$ and $n = 11$ under the prior $\pi(\theta) \propto 1/\theta$ and compare its 90% interval with the exact confidence interval of Example 11.6. *Hint.* The answers should be close to $(2.97, 14.31)$ and $(2.97, 15.05)$.

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CHAPTER 12

Open Problems and the Future

Each chapter of this book closes with the problems it leaves open. This chapter collects them, arranges them by the three themes of the book, and states as many as possible as precise mathematical or statistical questions. It then adds the larger questions that the genomic era is now forcing on statistical genetics: samples comparable in size to the population, genealogies inferred at biobank scale, ancient DNA as a time series, the portability of polygenic prediction, and learned inference whose calibration has to be checked rather than assumed. Short calculations show the scale of the difficulties: the slow approach of the singleton fraction to Pitman's discount parameter, the gap between marginal and conditional calibration, and the failure of the coalescent when the sample approaches the size of the population.

12.1 What a century has settled

It is worth beginning with what is no longer in question. Particulate inheritance conserves variance, and that answers Jenkin. The correlations between relatives follow from Mendel's laws applied to many loci, and that answers the biometricians. The Fundamental Theorem, read as Fisher almost certainly meant it and as Ewens has set out, is an identity about the partial change in mean fitness attributable to changes in gene frequency, and Chapter 3 shows exactly what the total change adds to it. The Ewens sampling formula is the law of an exchangeable partition whose predictive rule is affine in the empirical distribution function, and Chapter 6 shows that exchangeability then forces everything else. Under the infinite alleles model the number of alleles is sufficient for the mutation parameter, and under the infinite sites model the number of segregating sites is not (Chapters 10 and 11).

What remains open is, almost without exception, statistical. The mathematics of the classical models is in excellent shape. What we lack is a reliable account of what samples of the size and kind we now collect can tell us about the quantities the classical theory defines. Below, each problem is described in words and then, where it is sharp enough, stated as a displayed question or conjecture precise enough that one could recognise a solution.

12.2 Selection and the additive model

This section collects the problems left by Part I. They concern the empirical content of the Fundamental Theorem, the prior on genetic architecture, and the estimation of quadratic functionals of many small effects.

The measurement that has not been made. The neglected term in the Fundamental Theorem vanishes exactly when the population moves tangentially to the manifold of Hardy Weinberg and linkage equilibrium (Chapter 3). Whether real populations do so on the timescale of selection is an empirical question, and it is answerable: one needs Lindley's Hardy Weinberg disequilibrium coefficient and the linkage disequilibrium spectrum, tracked through time in a population under measurable selection. Long running evolution experiments and the temporal samples that ancient DNA now provides are the natural places to look. If the association coordinates turn out to be stationary, the constant genic environment becomes an observation rather than an assumption. If they move, the size of the remainder can be estimated directly. Of all the problems in this chapter, this is the one most worth doing first.

Write the total rate of change of mean fitness as $\sigma_A^2(t) + R(t)$, where $\sigma_A^2(t)$ is the additive genetic variance in the Malthusian parameter and $R(t)$ is the remainder of Chapter 3, the term that vanishes exactly when the motion is tangential to the manifold of product measures.

QUESTION 12.1. Given genotype samples of sizes $n_0, \dots, n_m$ taken at times $t_0 < \cdots < t_m$ from a population under selection, construct an estimator of the ratio

$$\rho = \frac{\int_{t_0}^{t_m} R(t)\,\mathrm{d}t}{\int_{t_0}^{t_m} \sigma_A^2(t)\,\mathrm{d}t}$$

that is consistent as $\min_i n_i \to \infty$, and determine the sample sizes and spacing needed to distinguish $\rho = 0$ from $|\rho| \geq \rho_0$ with given error probabilities.

The difficulty is one of scale. At a single biallelic locus the departure from Hardy Weinberg proportions is measured by F, and the chi square test has statistic $N\hat{F}^2$ (Exercise 4 of Chapter 1). A one degree of freedom test at level 0.05 has power 0.8 when the noncentrality is 7.85, so detecting $F = 0.1$ needs about 785 genotypes, detecting $F = 0.03$ about 8700, and detecting $F = 0.01$ about 78,500. Departures produced by selection within a generation are of the order of the selection coefficients, so the measurement requires either strong selection or samples of biobank size, repeated through time.

Is the genome sparse? Every estimate in Part I depends on the answer. A horseshoe prior encodes a genetic architecture with a few substantial effects among many negligible ones; a Gaussian prior encodes Fisher's infinitesimal model; the omnigenic model of Boyle, Li and Pritchard suggests that the truth may be dense in a way neither captures. Chapter 5 shows, through an exact identity of marginal

laws, that the choice of prior corresponds to the choice of Pitman's discount σ (written α in Chapter 5), and that the growth of the singleton count identifies it. That makes the sparsity level estimable from the frequency spectrum rather than chosen, but the identification has so far been shown only for the matched species sampling model.

Recall the correspondence of Chapter 5. If the shrinkage weight of a prior in the three parameter beta normal class is $\kappa \sim \mathrm{Be}(a, b)$, then κ has the law of the first stick breaking weight of $\mathrm{GEM}(\sigma, \vartheta)$ with $\sigma = 1 - a$ and $\vartheta = a + b - 1$, and the frequency spectrum intensities agree, $x^{-1} f_\kappa(x) \propto x^{-1-\sigma}(1-x)^{\vartheta+\sigma-1}$. The horseshoe, $a = b = \frac{1}{2}$, corresponds to $\sigma = \frac{1}{2}$ and $\vartheta = 0$.

QUESTION 12.2. Let Π_n be the exchangeable random partition of $\{1, \dots, n\}$ induced by the global local shrinkage prior with $\kappa \sim \mathrm{Be}(a, b)$, constructed as in Chapter 5. Is the exchangeable partition probability function of Π_n equal, for every n, to Pitman's

$$p(n_1, \dots, n_k) = \frac{\prod_{i=1}^{k-1}(\vartheta + i\sigma)}{(\vartheta+1)_{n-1}} \prod_{j=1}^{k} (1-\sigma)_{n_j - 1}, \qquad \sigma = 1 - a,\ \vartheta = a + b - 1,$$

where $(x)_m = x(x+1)\cdots(x+m-1)$? If not, which functionals of the partition (beyond the expected frequency spectrum) do the two laws share?

A positive answer would make the sparsity level a parameter with a sufficient statistic in the data, rather than a modelling choice. Under the two parameter law the proportion of alleles seen exactly j times tends to $\sigma(1-\sigma)_{j-1}/j!$, so for the horseshoe half of all alleles in a large sample are singletons, an eighth are doubletons and a sixteenth tripletons. Example 12.8 shows how slowly this limit is approached.

Functionals, not coefficients. The additive genetic variance is a quadratic functional of the average effects, Robertson's secondary theorem a bilinear one, and under linkage disequilibrium the estimand is $\alpha^\top V \alpha$ rather than a diagonal form (Chapter 4). Posterior intervals for such functionals under shrinkage priors undercover by a few points at moderate sample sizes. The theory for signals of intermediate size suggests that the misses are one sided and that the interval sits low, but the simulations of Chapter 4 show the opposite for the plain horseshoe: its misses lie mostly above the truth (80 of 91 in 600 replicates). A frequentist theory of coverage for quadratic functionals under the horseshoe, analogous to the results of van der Pas et al. (2017) for the coefficients themselves, would make the rate of adaptation a quantity one could report with a calibrated interval.

QUESTION 12.3. In the normal means model $Y_i \sim N(\beta_i, 1)$, $i = 1, \dots, p$, with β in the nearly black class $\ell_0[s_p] = \{\beta : \#\{i : \beta_i \neq 0\} \leq s_p\}$ and $s_p = o(p)$, let C_p be the equal tailed $1 - \gamma$ horseshoe credible interval for $Q(\beta) = \sum_i \beta_i^2$. Characterise the set of $\beta \in \ell_0[s_p]$ on which $\liminf_p \mathbb{P}_\beta\{Q(\beta) \in C_p\} \geq 1 - \gamma$, and decide on which side of $Q(\beta)$ the interval lies, with probability tending to one, on its complement. The

theory of intermediate signals suggests that it lies below, whereas the simulations of Chapter 4 found the misses mostly above the truth for the plain horseshoe, in a design that need not lie in the regime of the theory. Both the set to be characterised and the direction on its complement are open, and the second presupposes that the complement has no sequences with limiting coverage strictly between 0 and $1 - \gamma$.

The need for shrinkage, and the danger of it, are both visible in a small calculation. Take $p = 10{,}000$ coefficients of which 100 equal 3 and the rest are zero, so that $Q = 900$. The plug in estimate $\sum_i Y_i^2$ has expectation $Q + p = 10{,}900$, twelve times too large. The unbiased estimate $\sum_i (Y_i^2 - 1)$ has standard deviation $(2p + 4Q)^{1/2} = 154$, so it is correct on average but noisy. A shrinkage estimator removes most of the noise, and the open question is how much of the signal it removes with it.

Ascertainment is not a shrinkage problem. Measuring fitness only on survivors understates the additive variance severalfold, and no prior repairs it, because the bias is in the estimand rather than in the estimator. Writing the selection mechanism into the likelihood, in the manner of the weighted distributions of Rao that Part II applies to the paintbox, is the obvious route. Chapter 4 simulates truncation at a known fraction and rescales, but writing the mechanism into the likelihood has not been carried out for quantitative genetic designs, and it is the precondition for any empirical test of the Fundamental Theorem in a wild population.

QUESTION 12.4. Let (G, W) be the genotype and fitness of an individual, and suppose (G, W) is observed only with probability $\pi(W)$ for a known or parametrically specified function π. Under what conditions on π is the additive genetic variance in W identified from the law of the observed pairs, and what is the efficiency loss relative to complete observation?

Linkage disequilibrium at biobank scale. The inflation of the plug in additive variance is $\sigma^2 \operatorname{tr}(H^\top H)/N$, where H is the smoother matrix of Chapter 5. The trace equals p for the unregularised projection onto p linearly independent columns, whatever the design, and is smaller for a shrinkage smoother. With real linkage disequilibrium the trace must be computed from an empirical Gram matrix with strong local correlation, and the effective number of parameters is smaller than p by an amount that depends on the population's history. Relating that effective dimension to the inferred genealogy of the sample, which tree sequence methods now make available, would connect Part I to Part III.

QUESTION 12.5. For genotype matrices generated by the coalescent with recombination, with scaled recombination rate ϱ per unit length and p sites per unit length, how does the effective dimension $p_{\mathrm{eff}} = \operatorname{tr}(H^\top H)$ of Chapter 5 behave as $p \to \infty$ and $N \to \infty$, and can it be estimated from an inferred tree sequence without forming the Gram matrix?

12.3 The sampling theory of neutral alleles

The problems left by Part II are about extending exact results beyond the families in which they were proved, and about second order theory for the rare variant end of the frequency spectrum.

Ascertainment and pool size together. Chapter 6 applies two distinct operations to Kingman's paintbox: a Rao weighting for ascertainment, and the two parameter family with negative discount for a finite pool of types. Nothing forbids applying both. The resulting three parameter model would allow ascertainment effort and pool size to be estimated jointly, but its partition probability function has no closed form, so inference must be simulation based, which is exactly the setting of Part III.

QUESTION 12.6. Is the three parameter model obtained by Rao weighting the two parameter paintbox identifiable from the partition of a single sample as $n \to \infty$, and if so at what rate can each parameter be estimated?

Rigidity beyond tail sum weights. Chapter 7 proves that no sampling density acting on the tail sums of the stick breaking construction preserves the Ewens Pitman family when $\sigma > 0$. Ascertainment schemes that depend on the atom index as well as its position are not covered. Whether any of them that charges every interval preserves the family is open, and a negative answer would make the rigidity theorem complete.

CONJECTURE 12.7. *Let $0 < \sigma < 1$ and $\vartheta > -\sigma$. There is no sampling weight, depending measurably on the atom index and the atom's position in the subdivision and charging every interval, that carries* $\mathrm{PD}(\sigma, \vartheta)$ *to* $\mathrm{PD}(\sigma', \vartheta')$ *for any* $(\sigma', \vartheta') \neq (\sigma, \vartheta)$.

Second order theory for the spectrum. The singleton fraction estimates the discount parameter with no normalisation, and its bias under a deterministic envelope is negligible. The same calculation with the random frequencies in place of the envelope, carried to the order at which the Gamma fluctuations and the zeta function correction compete, has not been done. Nor has the second order scale matching between the abundance based estimator and Hill's estimator of the tail index. These are the calculations that would turn rules of thumb into procedures for the rare variant end of the spectrum, which is where the information in very large samples lives.

EXAMPLE 12.8 (How fast does the singleton fraction converge?). Under the two parameter Chinese restaurant process with discount σ and concentration ϑ, the $(n+1)$th gene is a new allele with probability $(\vartheta + \sigma K_n)/(\vartheta + n)$ and joins a given singleton with probability $(1-\sigma)/(\vartheta + n)$. Taking expectations gives two exact linear recursions,

$$\mathbb{E}K_{n+1} = \mathbb{E}K_n + \frac{\vartheta + \sigma \mathbb{E}K_n}{\vartheta + n},$$

$$\mathbb{E}K_{n+1,1} = \mathbb{E}K_{n,1}\Big(1 - \frac{1-\sigma}{\vartheta + n}\Big) + \frac{\vartheta + \sigma \mathbb{E}K_n}{\vartheta + n},$$

where $K_{n,1}$ is the number of singletons. Table 12.1 shows the ratio $\mathbb{E}K_{n,1}/\mathbb{E}K_n$ for $\sigma = \frac{1}{2}$. With $\vartheta = 0$ the ratio is within 0.003 of σ by $n = 100$. With $\vartheta = 10$ it is 0.65 at $n = 100$ and still 0.516 at $n = 10^4$; the excess decays like $c\, n^{-\sigma}$ with $c \approx 1.56$, so a sample of about 6×10^5 genes is needed to bring it below 0.002. The concentration parameter, which is invisible in the first order theory, controls the bias at every sample size one is likely to meet. A Monte Carlo check with 4000 samples of size 100 at $\vartheta = 1$ agrees with the recursions to within simulation error.

TABLE 12.1. Ratio $\mathbb{E}K_{n,1}/\mathbb{E}K_n$ of expected singletons to expected alleles under the two parameter Chinese restaurant process with $\sigma = 1/2$. The limit is $\sigma = 0.5$ in every column.

n	$\vartheta = 0$	$\vartheta = 1$	$\vartheta = 10$
10^2	0.5025	0.5457	0.6506
10^3	0.5003	0.5142	0.5493
10^4	0.5000	0.5044	0.5156
10^5	0.5000	0.5014	0.5049
10^6	0.5000	0.5004	0.5016

QUESTION 12.9. Under PD(σ, ϑ) find the second order expansion

$$\frac{K_{n,1}}{K_n} = \sigma + n^{-\sigma/2}\, Z_n,$$
$$\mathbb{E}\frac{K_{n,1}}{K_n} = \sigma + c_1(\sigma, \vartheta)\, n^{-\sigma} + o(n^{-\sigma}),$$

identifying c_1 and the limit law of Z_n, and use it to give a bias corrected estimator of σ with an interval whose coverage is correct to second order.

Which functionals identify which parameters? The de Finetti limit of the two parameter family is identified by two functionals of the limiting empirical distribution function: the variance of its heights and the growth rate of its jump count (Chapter 6). The Gibbs partitions of Gnedin and Pitman (2006) are the place to start.

QUESTION 12.10. For which families of exchangeable partitions is the directing random measure identified by a finite list of functionals of the limiting empirical distribution function, and is there a principle that assigns one functional to each parameter?

Priors for concentration parameters. Chapters 8 and 9 give two exact routes to Bayesian inference for a concentration parameter: Pólya inverse gamma augmentation, which reaches the transcendental factor directly, and rational conjugacy, which works whenever the Gamma ratio has an integer shift. The two are complementary, and a conjectural meeting point is the half horseshoe, which is a scale mixture of exponentials (Chapter 9 uses it both with residues and, through Proposition 9.12, with a latent exponential scale that needs no residues; combining

that proposition with the augmentation of Chapter 8 has not been carried out here). Two questions remain. The residue expansion of rational conjugacy is exact but ill conditioned, losing about $n \log_{10} 2$ significant digits in a solvable case, or thirty digits at $n = 100$; a stable algorithm for large samples is needed. And the Bernstein and von Mises argument of Chapter 8 shows that for the Ewens richness parameter the Bayes and classical answers agree to order $1/K$; the interesting cases are therefore the ones where K is small or where the shape parameter departs from the logseries limit, and these have not been studied systematically in genetic data.

QUESTION 12.11. Is there an algorithm that evaluates the normalising constant and posterior moments of a Pochhammer posterior after n observations with relative error ϵ in $O(n \log(1/\epsilon))$ operations in fixed precision arithmetic?

12.4 Inference without a likelihood

The problems left by Part III concern what learned summaries can and cannot capture, and how to tell whether a learned posterior can be trusted.

The ancestral recombination graph. With recombination the data are a collection of correlated genealogies and even one dimensional summaries lack closed form laws. The bottleneck test of Chapter 10 applies without change, and its natural output is not another estimate of the recombination rate but the dimension at which the held out risk stops falling: an estimate of how many numbers a recombining sample is worth.

For a model $\{P_\theta\}$ for data X define the *bottleneck dimension* d^* as the smallest d for which some $T : \mathcal{X} \to \mathbb{R}^d$ from a fixed class of encoders, as in Chapter 10, invariant under the symmetries of the model, is sufficient for θ. The class matters: if T may be any measurable map then $d^* \leq 1$ for every model on a countable sample space, because the orbit of X under the symmetries is sufficient and can be coded injectively by a single real number.

QUESTION 12.12. For a sample of n sequences of fixed length under the coalescent with recombination at scaled rate $\varrho > 0$ and infinite sites mutation at scaled rate ϑ, is d^* finite? If it is, how does it grow with n and with the sequence length?

Selection in the genealogy. Tajima's D and its relatives are hand built two dimensional summaries designed to detect departures from the neutral frequency spectrum. A learned two dimensional bottleneck fitted under models with directional or balancing selection is a direct test of how much the classical constructions leave on the table.

Multiple merger coalescents. For species with high fecundity and sweepstakes reproduction the Kingman coalescent is replaced by Λ coalescents, for which the sufficiency question is genuinely open while simulation remains easy. Here the bottleneck dimension is itself the scientific quantity, since it counts the independent

features of the spectrum that distinguish a multiple merger history from a neutral one.

QUESTION 12.13. For the Beta$(2-\alpha, \alpha)$ coalescent with $1 < \alpha < 2$ and infinite sites mutation, is the site frequency spectrum sufficient for (α, ϑ)? If not, what is d^*?

Beyond the permutation invariant ceiling. Chapter 11 proves that any encoder built from per site permutation invariant features factors through the site frequency spectrum, and that the number of distinct haplotypes is a minimal witness that escapes the ceiling. A small case shows what the spectrum loses. With $n = 5$ sequences and two segregating sites, each carried by two sequences, the spectrum is the same whether both mutations sit on the same pair of sequences (haplotypes $11, 11, 00, 00, 00$, so two distinct haplotypes) or on disjoint pairs ($10, 10, 01, 01, 00$, three distinct haplotypes). Both are consistent with a single tree, but they require different trees, and they have different probabilities under the coalescent.

QUESTION 12.14. Under the infinite sites model without recombination, with the sample represented as an $n \times S$ binary matrix modulo permutations of rows and of columns, what is the minimal sufficient statistic for ϑ? Which architectures, built from equivariant layers, can represent it?

Calibration, not just accuracy. Learned posteriors can match exact posteriors to within a few percent and still fail simulation based calibration, and Chapter 11 shows that standard rank recalibration removes the marginal error but cannot show that the conditional error is gone, and a mild conditional residual remains. A theory of conditional calibration for amortised posteriors, with diagnostics that a practitioner can run, is perhaps the most important methodological problem in likelihood free inference today. The next example shows why marginal checks cannot be enough.

EXAMPLE 12.15 (A marginally calibrated posterior that is conditionally wrong). Let $\theta \sim N(0,1)$ and $X \mid \theta \sim N(\theta, 1)$, so the exact posterior is $N(x/2, 1/2)$. Suppose a learned method ignores the data and reports the prior $N(0,1)$ as its posterior. Simulation based calibration (Cook et al., 2006) draws θ from the prior, data from the model, and computes the rank of θ in the reported posterior; here the rank statistic is $\Phi(\theta)$, which is exactly uniform. The method passes the check perfectly, and its 90 percent interval $(-1.645, 1.645)$ covers θ with marginal probability exactly 0.90. Conditionally it fails badly. Given $X = x$ the coverage is $\mathbb{P}(|\theta| < 1.645 \mid x)$ with $\theta \sim N(x/2, 1/2)$, which is 0.98 at $x = 0$, 0.82 at $x = 2$ and 0.58 at $x = 3$; over the 16 percent of data sets with $|X| > 2$ a simulation gives coverage 0.67. Overcoverage near the centre of the data distribution exactly pays for undercoverage in the tails.

QUESTION 12.16. Let $q(\theta \mid x)$ be an amortised posterior and $U = Q(\theta \mid X)$ the probability integral transform of the true parameter under it. Simulation based calibration tests whether U is uniform. Find a test, computable from simulated

pairs (θ_i, X_i) alone, that is consistent against every alternative in which U is not uniform conditionally on X on a set of positive probability, and give its rate against alternatives at distance δ in a suitable metric.

Misspecification. Every method in Part III conditions on the simulator being correct. When it is not, a learned posterior is confidently wrong, arguably more so than approximate Bayesian computation, whose tolerance inadvertently widens the answer. Posterior predictive checks against the observed spectrum are a start. Generalised Bayesian updating and the cut models of modular inference offer more principled options, and their application to coalescent models has hardly begun.

12.5 The larger questions

The problems above are local to the chapters of this book. Several larger ones are reshaping the whole field, and they are less sharply posed; for most of them the first task is to find the right question.

Samples as large as populations. The coalescent is an approximation for a small sample from a large population. When the sample is comparable to the effective population size, as it now is for some human biobanks, multiple and simultaneous mergers appear in the recent past, the frequency spectrum at the rare end departs from the Kingman prediction, and estimates of recent demography and of selection on rare variants are distorted (Bhaskar et al., 2014). The exact Wright Fisher genealogy is tractable numerically, and replacing the coalescent by it, or by a hybrid, is under way.

EXAMPLE 12.17 (The birthday problem in one generation). Trace n lineages back one generation in a Wright Fisher population of $2N$ genes. Each lineage picks its parent uniformly and independently, so this is the birthday problem with $2N$ days. The expected number of pairs that pick the same parent is $\binom{n}{2}/(2N)$, and the probability that no two do is $\prod_{i=1}^{n-1}(1 - i/2N)$. With $2N = 20{,}000$ and $n = 100$ the expected number of merging pairs is 0.25 and the probability of no merger is 0.78, so the Kingman picture of at most one rare merger per generation is reasonable. With $n = 1000$ the expected number is 25, the probability of no merger is 9×10^{-12}, and the sample loses on average 24.6 lineages in the first generation, with simultaneous mergers and occasional triple mergers. With $n = 5000$ about 576 lineages are lost at once. The Kingman coalescent, which allows exactly one pairwise merger at a time, describes none of this, and the errors fall precisely on the recent branches that carry the rare variants.

QUESTION 12.18. Let a sample of n genes be drawn without replacement from a Wright Fisher population of $2N$ genes at stationarity under the infinite alleles model, with $n/(2N) \to c \in (0, 1]$ and $4N\mu \to \vartheta$. What is the limiting law of the allelic partition of the sample, and how does it differ from the Ewens sampling formula in the counts of rare alleles?

Genealogies as data. Methods that infer the ancestral recombination graph of hundreds of thousands of genomes have turned the genealogy from a latent variable into something close to an observation. This changes the statistical questions. Inference about mutation, recombination, demography and selection can be conditioned on an estimated genealogy, but the uncertainty in that estimate has to be propagated, and the estimated genealogies are themselves the output of approximate algorithms whose biases are not well characterised. A Bayesian treatment that is both coherent and feasible at this scale does not yet exist.

Time series of genomes. Ancient DNA has given population genetics what it has rarely had: direct observation of allele frequencies through time. Selection can then be measured as a change in frequency rather than inferred from a pattern of variation at one time. But ancient samples are small, degraded, unevenly spaced, drawn from shifting populations, and subject to strong ascertainment. They are, in the language of Part II, a sample drawn with a nonuniform density from a moving paintbox, and the theory of Chapters 6 and 7 is a beginning, not an answer.

Polygenic adaptation and confounding. Signals of polygenic adaptation for human height, detected by combining many small allele frequency differences, were substantially reduced when recomputed with association effect sizes estimated in a more homogeneous sample (Berg et al., 2019; Sohail et al., 2019). The lesson is that subtle population stratification in the estimated effects aggregates, exactly as the inflation of Chapter 5 aggregates, into a spurious signal. The methodological problem is to estimate a many parameter functional when the errors in the parameters are correlated with the very structure the functional is meant to detect.

Portability of prediction. Polygenic scores built from samples of predominantly European ancestry predict much less accurately in other populations (Martin et al., 2019). Part of the loss is linkage disequilibrium, part is differences in allele frequencies, and part may be genuine differences in effect or in environment. Separating these is a problem in transfer learning with a known genealogical structure, and the hierarchical models of Part I, with population specific local scales sharing a global scale, are a natural starting point.

Learned models of the genome. Large neural networks trained on genomic sequence are now used to predict the functional effects of variants. For population genetics they are potentially a source of prior information: an informative prior on which loci are likely to matter, which is precisely what the local scales of a global local prior need. Using them responsibly requires treating their outputs as priors to be updated, not as answers, and checking the calibration of the resulting posteriors. This is the same discipline that Part III asks of learned posteriors, now applied to learned priors.

12.6 A closing note

Fisher regarded the Fundamental Theorem as one of his deepest results, and Ewens spent much of a long career showing what it does and does not say. The sampling formula that Ewens derived in 1972 became, through Kingman, Ferguson, Antoniak and Pitman, one of the central objects of modern probability and machine learning, far from the electrophoresis gels that motivated it. Both stories show the same pattern: a statement about genes becomes a statement about statistics, and the statistics then returns, years later, to change how the genetics is done.

The thread that runs through this book is that the questions of population genetics are, at bottom, questions of inference: what a sample can tell us about a population, what a prior commits us to, and what is lost when we replace a likelihood by a summary or a simulation. Fisher, Haldane, Wright, Kimura, Ewens and Kingman answered the first of these questions for the models of their day with remarkable completeness. The data we have now are larger and stranger than any of them imagined, and the questions are open again.

12.7 Exercises

1. Show that a one degree of freedom chi square test at level 0.05 has power 0.8 when the noncentrality parameter is 7.85, and deduce the sample sizes quoted after Question 12.1. *Hint.* Under the alternative $N\hat{F}^2$ is approximately noncentral chi square with noncentrality NF^2.

2. Verify that if $\kappa \sim \mathrm{Be}(a, b)$ then $x^{-1} f_\kappa(x)$ is proportional to the intensity $x^{-1-\sigma}(1-x)^{\vartheta+\sigma-1}$ of $\mathrm{PD}(\sigma, \vartheta)$ with $\sigma = 1 - a$ and $\vartheta = a + b - 1$, and that the first stick breaking weight of $\mathrm{GEM}(\sigma, \vartheta)$ is $\mathrm{Be}(1-\sigma, \vartheta+\sigma) = \mathrm{Be}(a, b)$. *Hint.* Compare exponents.

3. Show that the limiting proportion of alleles seen j times under $\mathrm{PD}(\sigma, \vartheta)$ is $\sigma(1-\sigma)_{j-1}/j!$, and check that these proportions sum to one. Evaluate them for $\sigma = \frac{1}{2}$ and $j \le 5$. *Hint.* Use the generating function $1 - (1-z)^\sigma$.

4. In the normal means example after Question 12.3, show that $\mathbb{E} \sum_i Y_i^2 = Q + p$ and $\mathrm{Var} \sum_i (Y_i^2 - 1) = 2p + 4Q$. *Hint.* Y_i^2 is noncentral chi square with one degree of freedom.

5. Derive the two recursions of Example 12.8 and reproduce Table 12.1. Then estimate the constant c of that example numerically for $\vartheta = 1$ and $\vartheta = 10$ from the ratio of expectations. *Hint.* $(\mathbb{E}K_{n,1}/\mathbb{E}K_n - \sigma)\, n^\sigma$ settles near 0.44 and 1.56.

6. In Example 12.15 compute the conditional coverage of the exact 90 percent posterior interval $x/2 \pm 1.645/\sqrt{2}$, and show that it equals 0.90 for every x. Show also that a reported posterior of the form $N(\lambda x, v)$ passes simulation based calibration if and only if $v = (1-\lambda)^2 + \lambda^2$, so that a whole curve of wrong posteriors passes, with the prior ($\lambda = 0$) at one end and the exact

posterior ($\lambda = \frac{1}{2}$) at its minimum. *Hint.* The rank statistic is $\Phi\{(\theta - \lambda X)/\sqrt{v}\}$, and $\theta - \lambda X$ is Gaussian with mean zero and variance $(1-\lambda)^2 + \lambda^2$.

7. Using the birthday calculation of Example 12.17, find the largest n for which the probability of at least one merger in the first generation back is below 0.5 when $2N = 20{,}000$, and compare with $\sqrt{2N \cdot 2\log 2}$. *Hint.* The answer is 166.

8. In Example 12.17 show that the expected number of distinct parents of the n lineages is $2N\{1 - (1 - 1/2N)^n\}$, and deduce the losses of 24.6 lineages at $n = 1000$ and about 576 at $n = 5000$. Show further that the expected loss is $\binom{n}{2}/(2N) - \binom{n}{3}/(2N)^2 + \cdots$, so that it falls short of the expected number of merging pairs by about $\binom{n}{3}/(2N)^2$, and explain why. *Hint.* Write the number of distinct parents as a sum over the $2N$ possible parents of the indicator that at least one lineage picks that parent, and expand $(1 - 1/2N)^n$ by the binomial theorem; a triple merger counts three pairs but loses only two lineages.

9. In Example 12.15 verify the conditional coverages 0.98, 0.82 and 0.58 of the reported interval $(-1.645, 1.645)$ at $x = 0, 2, 3$, show that $\mathbb{P}(|X| > 2) = 0.157$, and compute the coverage over the data sets with $|X| > 2$ by numerical integration, confirming the simulated value 0.67. *Hint.* Marginally $X \sim N(0, 2)$, and given $X = x$ the coverage is $\Phi\{\sqrt{2}(1.645 - x/2)\} - \Phi\{\sqrt{2}(-1.645 - x/2)\}$.

10. Prove the remark before Question 12.12: if the sample space is countable and every P_θ is invariant under a group G acting on it, then the orbit GX is sufficient for θ, and so is a single real number. *Hint.* Points of one orbit have equal probability under every P_θ, so an orbit of positive probability is finite and X is uniform on it given GX; the orbits form a countable set, which can be numbered.

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